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THE MANUAL FOR BRIDGE EVALUATION

American Association of State Highway and Transportation Officials
AASHTO

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Cover photos: **Top Left:** Photo of a demonstration of Olson Instruments Bridge Deck Scanner system on a bridge taken by Larry Olson, Olson Engineering, Inc. **Second Left:** Photo of a through truss bridge taken by Thomas Drda, FHWA. **Third Left:** Photo courtesy of Idaho Department of Transportation. **Bottom Left:** Photo of an inspection of a deck truss bridge using an under bridge inspection truck. Taken by John Thiel, FHWA. **Right:** Leonard P. Zakim Bunker Hill Bridge in Boston, MA. Courtesy of Shay Burrows, FHWA.

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PREFACE

The Manual for Bridge Evaluation (MBE) offers assistance to Bridge Owners at all phases of bridge inspection and evaluation. An abbreviated table of contents follows this preface. Detailed tables of contents precede Sections 1 through 8.

Appendix A includes nine illustrative examples (A1 through A9), previously in the *Guide Manual for Condition Evaluation and Load and Resistance Factor Rating (LRFR) of Highway Bridges*, and two more (A10 and A11) that were added in the Interim Revisions for the Second Edition of this title. All examples are rated using the LRFR method. In addition, Examples A1 and A2 are also rated using the ASR and LFR methods, A4 is also rated using the ASR method, and A11 is also rated using the LFR method. To clarify which rating method is being illustrated, examples using multiple methods are generally divided into Parts A through C and their articles are numbered accordingly as follows:

- Part A, LRFR;
- Part B, ASR and LFR; and
- Part C, example summary.

For ease of reference, the table of contents for Appendix A includes a summary table of the bridge types, rated members, rating live loads, limit states for evaluation, and rating methods, with the starting page number for each example and, in the case of Examples A1, A2, A4, and A11, for each rating method. The typical detailed table of contents follows this summary table.

Appendix A includes numerous citations of other AASHTO bridge publications. To save space, the following shorthand has been adopted:

- "AASHTO Standard Specifications" refers to the current edition of the AASHTO *Standard Specifications for Highway Bridges*, 17th Edition, HB-17,
- "LRFD Design" refers to the current edition of the *AASHTO LRFD Bridge Design Specifications*, Eighth Edition, LRFD-8, and
- "MBE" refers to this publication, *The Manual for Bridge Evaluation*, Third Edition, MBE-3.

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SECTION 1:

INTRODUCTION

1.1—PURPOSE

The purpose of *The Manual for Bridge Evaluation* (MBE) is to serve as a resource for use in developing specific policy and procedures for the inspection and evaluation of existing in-service highway bridges. The MBE also includes the nationally recognized guidance for the load rating of highway bridges.

The National Bridge Inspection Standards (NBIS), as found in the Code of Federal Regulations (23 CFR 650 Subpart C), define the regulations for the inspection and evaluation of the nation's bridges.

The MBE is incorporated by reference in the CFR (23 CFR 650 Subpart C) to be used along with other reference documents such as the American Association of State Highway and Transportation Officials (AASHTO) *Manual for Bridge Element Inspection*, the Federal Highway Administration's (FHWA) *Bridge Inspector's Reference Manual* (BIRM), and the latest National Bridge Inventory (NBI) coding guidance document for the inspection and evaluation of the nation's bridges.

The NBIS have evolved and been improved over the years since their creation in the early 1970s.

The MBE has also evolved and been revised and improved to reflect best practices as determined by research, state departments of transportation (DOTs), and others. In the future as improved practices and research are developed, the MBE will reflect those improvements.

Throughout this Manual there are subsections titled in part "Provisions to Support the NBIS Requirements." These subsections were developed to provide specific guidance and best practices that are considered to be required under the regulations.

1.2—SCOPE

The Manual has been divided into eight Sections, with each Section representing a distinct phase of an overall bridge inspection and evaluation program.

- Section 1—Purpose, scope, applicability, inspection and evaluation quality measures, and definition of general interest terms.
- Section 2—Provisions for proper documentation to be included in a bridge file. The bridge file associated with each bridge provides the foundation against which changes in physical condition can be compared.

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- Section 3—Overview of bridge management systems and their key elements.
- Section 4—Types and frequency of field inspections, as well as specific inspection techniques and procedures.
- Section 5—Various inspection and evaluation testing methods. Conditions at a bridge site or the absence of information from original construction may warrant more elaborate material tests to determine properties for evaluation.
- Section 6—Nationally recognized specification for the load rating of bridges. Includes the Load and Resistance Factor method, the Load Factor method, and the Allowable Stress method.
- Section 7—Provisions for the evaluation of existing bridges for fatigue.
- Section 8—Field-performed load test procedures. Field load testing is a means of supplementing analytical procedures in determining the live-load capacity of a bridge and for improving the confidence in the assumptions.

The successful application of this Manual is directly related to the DOT organizational structure. Such a structure should be both effective and responsive so that the unique characteristics and special problems of individual bridges are considered in developing an appropriate inspection plan and load capacity determination.

1.3—APPLICABILITY

The provisions of this Manual apply to all highway structures that qualify as bridges in accordance with the AASHTO definition of a bridge (see Article 1.5). These provisions may be applied to smaller structures which do not qualify as bridges at the discretion of the DOT.

The NBIS establish minimum requirements for inspection programs and minimum qualifications for bridge inspection personnel. The NBIS apply to all highway bridges that are more than 20 ft in length and located on public roads.

Where conflicts or inconsistencies exist between this Manual and the federal requirements specified in the NBIS, the FHWA coding guidance, or BIRM, the federal requirements shall govern.

1.4—QUALITY

To maintain the accuracy and consistency of inspections and load ratings, bridge inspection programs need to have appropriate quality control (QC) and quality assurance (QA) measures in place. QC procedures are intended to maintain the quality of the bridge inspections, bridge data, scour evaluations, and load ratings, and are usually performed continuously

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within the bridge inspection teams or units performing these functions. QC procedures can vary depending on the structural and scour conditions of a bridge with increased level of review commensurate with increased deterioration of bridge conditions. QA procedures are used to verify the adequacy of the quality control procedures to meet or exceed the standards established by the program manager. QA procedures are usually performed independently of the bridge inspection and load rating teams on a sample of their work.

1.4.1—Provisions to Support the NBIS Requirements

A quality control and quality assurance (QC/QA) program is to include periodic field review of inspection teams, periodic bridge inspection refresher training for program managers and team leaders, QC/QA measures for inventory data, and independent review of inspection reports and computations. The program manager is responsible for developing a QC/QA program that generally conforms to the provisions of Article 1.4. Specific details are to be determined by the program manager.

1.4.2—QC/QA Procedures

Typical quality procedures may include the use of checklists to ensure uniformity and completeness, the review of reports and computations by a person other than the originating individual, and the periodic field review of inspection teams and their work. The documented quality control plan may include:

- Defined QC roles and responsibilities;
- Qualifications for the program manager, bridge inspection personnel, and load rating personnel, including:
  - Education,
  - Certification or registration,
  - Training, and
  - Years and type of experience;
- Procedures for review and validation of inspection reports and data;
- Procedures for documenting important bridge inspection information;
- Procedures for review validation of load rating and scour calculations and data; and
- Procedures for identification and resolution of data issues including errors, omissions, compatibility between items, changes, or any combination thereof.

QA measures include the overall review of the inspection and rating program to ascertain that the

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results meet or exceed the standards established by the program manager. The documented QA plan may include:

- Defined quality assurance roles and responsibilities;
- Frequency parameters for review of districts or units and bridges; and
- Procedures and sampling parameters for selecting bridges to conduct independent review and check of results, including:
    - Condition rating of elements or change in condition rating,
    - Load rating and scour evaluations,
    - Posting status,
    - Deficiency status,
    - Critical findings and the status of any follow-up action, and
    - Location of bridge.

QA measures provide a validation that QC practices are resulting in accurate and thorough inspections, complete bridge files, accurate and complete load ratings and scour evaluations, and qualified inspectors and load raters. Results from QA reviews are used by the program manager to maintain the quality of the program and make improvements where needed.

1.5—DEFINITIONS AND TERMINOLOGY

AASHTO—American Association of State Highway and Transportation Officials.

As-Built Plans—Plans that show the state of the bridge at the end of construction; usually prepared by the Contractor or the resident Engineer.

ASR—Allowable Stress Rating.

Bias—The ratio of mean to nominal value of a random variable.

Bridge—A structure including supports erected over a depression or an obstruction such as water, highway, or railway; having a track or passageway for carrying traffic or other moving loads; and having an opening measured along the center of the roadway of more than 20 ft between undercopings of abutments or spring lines of arches, or extreme ends of openings for multiple boxes. It may also include multiple pipes, where the clear distance between openings is less than half of the smaller contiguous opening.

Bridge Management System (BMS)—A system designed to optimize the use of available resources for the inspection, maintenance, rehabilitation, and replacement of bridges.

Calibration—A process of adjusting the parameters in a new standard to achieve approximately the same reliability as exists in a current standard or specification or to achieve a target reliability index.

Coefficient of Variation—The ratio of the standard deviation to the mean of a random variable.

Collapse—A major change in the geometry of the bridge rendering it unfit for use.

Complex Bridges—Movable, suspension, cable stayed, and other bridges with unusual characteristics.

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Condition Rating—The result of the assessment of the functional capability and the physical condition of bridge components by considering the extent of deterioration and other defects.

Evaluation—An assessment of the performance of an existing bridge.

Exclusion Vehicle—Grandfather provisions in the federal statutes which allow states to retain higher limits than the federal weight limits if such limits were in effect when the applicable federal statutes were enacted. Exclusion vehicles are vehicles routinely permitted on highways of various states under grandfather exclusions to weight laws.

Failure—A condition where a limit state is reached or exceeded. This may or may not involve collapse or other catastrophic occurrences.

FHWA—Federal Highway Administration, U.S. Department of Transportation.

Inventory Rating—Load ratings based on the inventory level allow comparisons with the capacity for new structures and, therefore, results in a live load, which can safely utilize an existing structure for an indefinite period of time.

Inventory Level Rating (LRFR)—Generally corresponds to the rating at the design level of reliability for new bridges in the *AASHTO LRFD Bridge Design Specifications*, but reflects the existing bridge and material conditions with regard to deterioration and loss of section.

LFR—Load Factor Rating.

Limit State—A condition beyond which the bridge or component ceases to satisfy the criteria for which it was designed.

Load Effect—The response (axial force, shear force, bending moment, torque) in a member or an element due to the loading.

Load Factor—A load multiplier accounting for the variability of loads, the lack of accuracy in analysis, and the probability of simultaneous occurrence of different loads.

Load Rating—The determination of the live-load carrying capacity of an existing bridge.

LRFD—Load and Resistance Factor Design.

LRFD Exclusion Limits—Weight and length limits of trucks operating under grandfather exclusions to federal weight laws.

LRFR—Load and Resistance Factor Rating.

Margin of Safety—Defined as R-S, where S is the maximum loading and R is the corresponding resistance (R and S are assumed to be independent random variables).

MUTCD—Manual on Uniform Traffic Control Devices.

National Bridge Inventory (NBI)—The aggregation of structure inventory and appraisal data collected to fulfill the requirements of the National Bridge Inspection Standards.

National Bridge Inspection Standards (NBIS)—Federal regulations establishing requirements for inspection procedures, frequency of inspections, a bridge inspection organization, qualifications of personnel, inspection reports, and preparation and maintenance of bridge inventory records. The NBIS apply to all structures defined as highway bridges located on or over all public roads.

NICET—National Institute for Certification in Engineering Technologies.

Nominal Resistance—Resistance of a component or connection to load effects, based on its geometry, permissible stresses, or specified strength of materials.

Operating Rating (ASR, LFR)—Load ratings based on the operating rating level generally describe the maximum permissible live load to which the structure may be subjected. Allowing unlimited numbers of vehicles to use the bridge at operating level may shorten the life of the bridge.

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Operating Level Rating (LRFR)—Maximum load level to which a structure may be subjected. Generally corresponds to the rating at the operating level of reliability in past load rating practice.

Owner—Agency having jurisdiction over the bridge.

Posting—Signing a bridge for load restriction.

Quality Assurance—The use of sampling and other measures to assure the adequacy of quality control procedures in order to verify or measure the quality level of the entire bridge inspection and load rating program.

Quality Control—Procedures that are intended to maintain the quality of a bridge inspection and load rating at or above a specified level.

RF—Rating Factor.

Reliability Index—A computed quantity defining the relative safety of a structural element or structure expressed as the number of standard deviations that the mean of the margin of safety falls on the safe side.

Resistance Factor—A resistance multiplier accounting for the variability of material properties, structural dimensions and workmanship, and the uncertainty in the prediction of resistance.

Safe Load Capacity—A live load that can safely utilize a bridge repeatedly over the duration of a specified inspection cycle.

Scour Critical Bridge—A bridge whose foundation (or foundations) has been determined to be unstable for the predicted scour conditions.

Service Limit State—Limit state relating to stress, deformation, and cracking.

Serviceability—A term that denotes restrictions on stress, deformation, and crack opening under regular service conditions.

Serviceability Limit States— Collective term for service and fatigue limit states.

Specialized Hauling Vehicle (SHV)—Short wheelbase multi-axle trucks used in the construction, waste management, bulk cargo and commodities hauling industries.

Strength Limit State—Safety limit state relating to strength and stability.

Structure Inventory and Appraisal Sheet (SI&A)—A summary sheet of bridge data required by NBIS.

Target Reliability—A desired level of reliability (safety) in a proposed evaluation.

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SECTION 2: BRIDGE FILES AND DOCUMENTATION

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SECTION 2:

BRIDGE FILES AND DOCUMENTATION

2.1—INTRODUCTION

Maintaining bridges safe for public travel is of utmost importance to transportation officials. In order to ensure public safety, management of in-service highway bridges requires the collection and maintenance of accurate, up-to-date, and comprehensive information for each bridge. The information includes 1) data that does not typically change over the life of a bridge, 2) data that is updated by field inspections, and 3) data that is derived or calculated to assess specific attributes such as scour, vulnerability to extreme events, and safe load-carrying capacity.

Data that does not typically change includes information such as the length and width of the structure and the structure type. Data that is updated as a result of field inspections includes condition information on the structural components and elements, clearances and changes to dead load, and the identification of conditions that introduce a potential risk to the bridge, such as debris accumulation around substructure units and scour or erosion.

Information is organized in a file for each bridge. The bridge file includes a description of the characteristics and conditions of the structure; calculations for determining scour vulnerability (if over water); a determination of the load-carrying capacity, including computations substantiating reduced load limits; and details of any damage and alterations or repairs to the structure.

The information in a bridge file provides a cumulative history of the structure that is useful to review prior to conducting a bridge inspection, rating, or evaluation.

The information in a bridge file may exist electronically, on paper, or in external documents that are appropriately referenced within each bridge file or manual. The external documents may apply to multiple bridges and exist in various locations.

The bridge file provides information that directly relates to requirements of the National Bridge Inspection Standards (Article 2.2) and other supplemental information that bridge owners may find useful in managing bridges (Article 2.3).

2.2—PROVISIONS TO SUPPORT THE NBIS REQUIREMENTS

An inventory of all bridges is to be maintained as required by the National Bridge Inspection Standards (NBIS), in accordance with coding guidance issued by the Federal Highway Administration (FHWA). Certain data is defined by the FHWA that must be collected and retained. This data is typically retained within a computer database. Summaries of the inventory and evaluation data

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may be included in bridge files. The data accurately captures specific, key information including bridge identification, location, structural attributes, dimensions, clearances, condition, and load-carrying capacity.

A bridge file is to be prepared as required by the NBIS and described in this Manual. Specifically:

- Maintain reports on the results of bridge inspections and note any actions taken to address the findings of the inspections,
- Maintain relevant maintenance and inspection data and use it to assist with the assessment of current bridge conditions,
- Document observations and measurements needed to determine the physical and functional condition of the bridge, and
- Identify any changes from initial or previously recorded conditions and other actions needed to ensure that the structure continues to satisfy present structural service requirements.

The format used to document the information in bridge files may vary significantly. If the information is not available in a consolidated inspection report, look for it elsewhere in the inspection file or as referenced to another location. Bridge Owners may make generalized reference to the location of the digital information within their own manuals or other files rather than individually in each bridge file.

Include the following specific bridge information. The level of documentation will vary depending on the complexity, condition, and situation of a structure; however, always include the information that is common to all bridges.

2.2.1—General File Information

Include the following general file information:

- General plan and elevation drawing or a sketch depicting the layout of the bridge, if available;
- Clear and understandable approach to labeling members and elements that enables an assessment of the inspection process and its completeness;
- Inspection and inventory data as defined by the FHWA for reporting to the National Bridge Inventory. This data is often referred to as Structure Inventory and Appraisal information (SI&A);
- Photographs showing a top view of the roadway across the bridge, a side elevation view, and an under view of the main or typical span superstructure configuration. Photographs necessary to show major defects, posting restrictions, and other important features should be included; and
- History of any structural damage.

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2.2.2—Field Inspection Information

Inspection reports provide a chronological record of the date and type of all inspections performed on the bridge. Document each inspection conducted for a bridge and retain in the bridge file. Document the observations and findings from each inspection. Identify the Team Leader responsible for the inspection and report, and the date the inspection was conducted. Observations include a description of conditions of bridge members and the identification of factors that require further review, close monitoring, or additional attention during inspections:

- Brief narrative descriptions of general and component/element conditions with detail to justify a condition rating of 5 or less for NBI items and Condition State 3 or more for elements.
- Sketches or photographs, as appropriate, of elements or members showing typical and deteriorated conditions.
- Sketches documenting remaining section of components with sufficient detail to facilitate determination of the load capacity. These sketches may be part of the load rating documentation.
- Notations of actions taken to address previous inspection findings.

2.2.3—Critical Findings and Actions Taken

Provide a detailed description and photographs of the specific critical finding(s) sufficient to document safety or structural concerns. Identify appropriate immediate actions or follow-up inspections. Include a record of the actions taken to resolve or monitor the critical finding(s).

2.2.4—Waterway Information

Provide channel cross-sections or sketches, soundings, and stream profiles as needed to provide adequate information on the stability of the waterway and allow for adequate assessment of the risk to the structure. Update channel information periodically and when otherwise necessary. Perform a historical comparison to determine the extent of any scour, channel shifting, degradation, or aggradation of the channel. Determine and document a frequency for obtaining and updating these measurements, depending on an assessment of the bridge and stream characteristics. Consider the potential for lateral migration of the stream channel or head-cutting in determining the extent of channel documentation. A single cross-section or sketch at one face may be appropriate for historically stable channels and embankments. Evaluate the need for obtaining cross-sections for pipes and box culverts and structures with small drainage areas on a case-by-case

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basis. The Bridge Inspector's Reference Manual (BIRM) provides additional information on inspecting channels.

2.2.5—Significant Correspondence

Provide correspondence and agreements regarding inspection responsibility, ownership, maintenance responsibilities with other agencies, or other issues that have an impact on the ability to ensure that thorough and timely inspections are completed.

2.2.6—Other Inspection Procedures or Requirements

Provide procedures for specific types of inspections (fracture-critical, underwater, and complex bridges). Address those items that need to be communicated to an inspection team leader to ensure a successful bridge inspection.

Provide procedures to document special access needs, inspection equipment, structural details, inspection methods, and any special qualifications required of inspecting personnel. Overall inspection procedures may exist in a bridge inspection manual that address common aspects of these more complex inspections; however, if there are items unique to that structure that are not covered in the overall inspection procedures, provide written procedures specific to that fracture critical, underwater, and complex bridge inspection.

Document the following items, either in bridge-specific inspection procedures or by referring to general inspection procedures:

1. Identify each of the fracture-critical, complex features, and underwater members, and any elements that need special attention during those inspections, preferably on plan sheets, drawings, or sketches.
2. Describe the inspection method(s), special access needs (under bridge inspection truck, climbing, etc.), special inspection equipment (non-destructive evaluation, a.k.a. NDE), and frequency to be used for the elements.
3. Provide information regarding proximity necessary to details, such as "arm's length" or "hands-on."
4. Provide special qualifications required of inspection personnel by the Program Manager, if any. Section 4 provides additional information on inspection plans for bridges.

2.2.7—Load Rating Documentation

Provide dated load rating results along with the identification of the analyst to determine the safe load-carrying capacity of the bridge and, where necessary, the load limits for posting. Include the load rating results which clearly identify the loads and methodology used in the analysis, a general statement of the results of the

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analysis, identification of members that were found to control the load rating, and any other modifying factors that were assumed in the analysis. Include updates to the calculations as needed to reflect changes in the condition of structural members; changes to the structural configuration, strength of members, or dead load; and changes to the legal live load that may alter the load rating result. If calculations cannot be provided due to a lack of information (missing plans, unknown materials, etc.), provide documentation for justification of determined load rating. If a field load test is used to establish the load-carrying capacity of the bridge, provide reports that describe and document the testing process and results.

2.2.8—Posting Documentation

Provide a summary of posting recommendations and actions taken for the bridge and date of posting.

2.2.9—Scour Assessment

Document the assessment conducted to determine the scour vulnerability of the bridge. Provide a clear reference to an alternate location if documentation is available outside of the inspection file.

2.2.10—Scour Plan of Action

For scour-critical bridges, provide a copy of a plan of action, or a clear reference to the plan of action if the documentation is available in a location other than the inspection file. The plan of action is used to monitor known and potential deficiencies and address or monitor critical scour related findings.

2.3—SUPPLEMENTAL DOCUMENTATION IN BRIDGE FILE

2.3.1—General

Retain the supplemental documentation described in Articles 2.3.2 through 2.3.9 in the bridge file at the discretion of the Program Manager and subject to availability, especially for older bridges. Additional information not listed in this section may be included in the file and may contain information Bridge Owners believe is necessary to operate their Bridge Management Systems. Some of this information may be referenced in the bridge file or references may be provided to outside source locations.

2.3.2—Plans and Drawings

Include one set of final drawings showing the “as-built” condition of the bridge. Construction shop drawings may be included.

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2.3.3—Construction Documentation

Include construction documentation of relevant as-built information regarding the structure, including reference to material certifications and tests performed during construction activities such as pile driving, concrete placement, and prestressing operations.

Retain all pertinent certificates for the type, grade, and quality of materials incorporated in the construction of the bridge, such as steel mill certificates, concrete delivery slips, and other manufacturers' certifications, in accordance with applicable policies and the appropriate statute of limitations.

Reference in the file any reports of nondestructive and laboratory tests of materials incorporated in the bridge during construction.

Retain one complete copy or reference to the special technical specifications under which the bridge was built. Reference the edition and date of the general technical specification in the bridge file.

2.3.4—Original Design Documentation

Retain the original design calculations and documentation for the bridge. Record the AASHTO Bridge Design Specifications version and any project-specific assumptions and criteria.

2.3.5—Unique Considerations

Provide information as necessary on other items that may need to be addressed depending on each unique situation. If appropriate, include special planning considerations such as the following:

- *Special Coordination Procedures*—Coast Guard, security, operations, or other agencies;
- *Natural Safety Concerns*—rattlesnakes, bats, or other wildlife; and
- *Optimum Inspection Periods of the Year*—lake draw down, canal dry time, snow, ice, bird nesting seasons, or other considerations.

Identify special access issues or equipment required to inspect the features (traveler system, climbing, equipment for confined spaces, or remote access camera technology).

Emphasize and highlight special structural details or situations, such as fatigue-prone details, pins and hangers in nonredundant systems, cathodic protection, and weathering or brittle steels.

Document any unusual environmental conditions that may have an effect on the structure, such as salt spray and industrial gases.

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2.3.6—Utilities and Ancillary Attachments

Include information on utilities or ancillary attachments that either 1) are attached to the structure or 2) otherwise affect access to portions of the bridge. The type of connection should be noted. Note a utility in the immediate area, though not fastened to the bridge, e.g., a sewer line crossing the right-of-way and buried in the channel beneath the bridge.

2.3.7—Maintenance and Repair History

Include a chronological record documenting the maintenance and repairs to the bridge since its initial construction. Include details such as significant dates, description of project, contractor, cost, and contract number.

Document the surface protective coatings used, including surface preparation, application methods, and dry-film thickness as well as types of paint, concrete and timber sealants, and other protective membranes. This information may be in the referenced original design file.

2.3.8—Additional Waterway Documentation

For structures over waterways, when available, include a chronological history of major hydraulic events, high-water marks, and scour activity in the bridge file. Channel profiles, mean high-water levels, debris accumulation, and storm surge data are useful information for effectively managing bridges over waterways.

2.3.9—Traffic Data

When the information is available, include the frequency and type of vehicles using the bridge and their historical variations in the bridge file. Vehicle weight data, such as weigh-in-motion (WIM) data, can aid in the determination of bridge-specific load factors when refining LRFR load capacity calculations.

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SECTION 3:

BRIDGE MANAGEMENT SYSTEMS

3.1—INTRODUCTION

As defined by the AASHTO Standing Committee on Highways, Planning Subcommittee on Asset Management, "Transportation asset management is a strategic and systematic process of operating, maintaining, upgrading, and expanding physical assets effectively throughout their lifecycle. It focuses on business and engineering practices for resource allocation and utilization, with the objective of better decision making based upon quality information and well defined objectives."

NCHRP Report 551, *Performance Measures and Targets for Transportation Asset Management*, provides a list of core principles of asset management as follows:

- **Policy-Driven**—Resource allocation decisions are based on a well-defined set of policy goals and objectives. These objectives reflect desired system condition, level of service, and safety provided to customers and are typically tied to economic, community, and environmental goals.
- **Performance-Based**—Policy objectives are translated into system performance measures that are used for both day-to-day and strategic management.
- **Analysis of Options and Tradeoffs**—Decisions on how to allocate resources within and across different assets, programs, and types of investments are based on understanding how different allocations will affect the achievement of policy objectives and what the best options to consider are. The limitations posed by realistic funding constraints also must be reflected in the range of options and tradeoffs considered.
- **Decisions Based on Quality Information**—The merits of different options with respect to an agency's policy goals are evaluated using credible and current data. Decision support tools are applied to help in accessing, analyzing, and tracking these data.
- **Monitoring Provides Clear Accountability and Feedback**—Performance results are monitored and reported for both impacts and effectiveness. Feedback on actual performance may influence agency goals and objectives, as well as future resource allocation and use decisions.

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Transportation agencies must balance limited resources when evaluating individual bridge needs and considering the overall infrastructure needs of an aging highway system. The best action for an individual bridge may not be the best action for an agency's network of bridges when facing funding constraints. Additionally, competing needs from other transportation assets may require adjusting bridge performance targets based on an overall transportation management plan. The goal of asset management is to determine and implement an infrastructure preservation and improvement strategy that best integrates capital and maintenance activities to maximize the net benefit to society.

3.2—OBJECTIVES OF BRIDGE MANAGEMENT SYSTEMS

A bridge management system (BMS) is a tool or collection of tools integrated through a process whose goal is to assist an agency to meet strategic objectives by connecting inventory management and project selection to agency strategic goals through a data driven process. A BMS should meet the needs of both upper management, where it is a strategic planning tool, and technical decision makers, where it is an engineering tool. A BMS helps engineers and decision-makers determine the best fiscally constrained action to take on maintenance programs and short-, medium-, and long-term capital improvement programs. Its purpose is to determine the optimum use of funding by enabling decision-makers to understand the consequences of their actions and strategies. A BMS assists the bridge owner in expending the appropriate level of resources in order to maintain the inventory in an acceptable state of good repair. It also provides essential information to help transportation agencies enhance safety, perform risk assessments, extend the service life of bridges, and serve commerce and the motoring public.

3.3—COMPONENTS OF A BRIDGE MANAGEMENT SYSTEM

A BMS provides three components to support bridge asset management:

- Information Management
- Data Analysis

  Decision Support

3.3.1—Information Management

A BMS requires comprehensive, connected, and well organized relational databases that are capable of

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supporting the various analyses involved in bridge management and reporting this information in a way that can be readily understood by various stakeholders. There are four major types of data required by a BMS:

1. Bridge inventory, general condition ratings, and bridge element ratings;
2. Agency performance measures;
3. Preservation and improvement activity data;
4. Cost data and financial plans.

3.3.1.1—Bridge Inventory, General Condition Ratings, and Bridge Element Ratings

3.3.1.1.1—Bridge Inventory

The National Bridge Inventory (NBI) is data collected by each state Department of Transportation (DOT) and reported to the Federal Highway Administration (FHWA). The data includes inventory, appraisal, and condition information for the nation's highway bridges. NBI data includes component and element level condition data. Bridge owners may also have agency specific bridge inventory data that they use in their BMS.

3.3.1.1.2—General Condition Ratings

In a component level inspection, General Condition Ratings (GCRs) are collected for the deck, superstructure, and substructure for bridge type structures, and a culvert rating for culvert type structures. The GCRs are determined in accordance with the National Bridge Inspection Standards (NBIS) using the 0 (failed condition) to 9 (excellent condition) NBI rating scale to provide an overall assessment of the structure's major components and ensure structural safety of the bridge. NBI appraisal items and condition ratings are often used by state DOTs and the FHWA as bridge performance measures. Bridge owners may also have their own GCRs. For example some state DOTs collect GCRs for each span of the structure. GCRs can be used to categorize bridges into major categories of condition and general need, such as good, fair, and poor, which can be used to do network level bridge management and report on performance measures to stakeholders.

3.3.1.1.3—Bridge Element Ratings

GCRs provide a general assessment of the overall condition of each major component of a bridge, but

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provide limited detail on the type and quantity of deficiencies that may be present. Advanced BMS analyses requires a more detailed condition assessment to predict and prioritize bridge repair, preservation, or replacement actions. To meet the data needs of a modern BMS, AASHTO developed an element level condition assessment system. Bridge elements comprised of National Bridge Elements (NBEs), Bridge Management Elements (BMEs), and Agency-Defined Elements (ADEs) are defined in the *AASHTO Manual for Bridge Element Inspection* (MBEI). The goal of bridge element data is to completely capture the condition of bridges in a simple and effective way that can be standardized across the nation while providing the flexibility to be adapted to both large and small agency settings. Element descriptions consider material composition and where applicable, the presence of protective systems. The condition of each element is reported according to the quantity or percentage of the element rated in four Condition States (CS): CS 1—Good, CS 2—Fair, CS 3—Poor, and CS 4—Severe. The MBEI defines the condition states in objective engineering terms that are intended to provide consistent ratings nationwide. All National Bridge Elements and a select number of Bridge Management Elements on the National Highway System (NHS) are reported to the FHWA to develop bridge condition reports to the United States Congress. Using element level data, DOTs and local bridge owners can better evaluate individual components of a structure, determine and prioritize preservation needs, and estimate cost for projects.

3.3.1.2—Agency Performance Measures

Performance measures are tools used to connect asset management decisions to strategic goals set for the transportation network as well as to communicate the need for and value of investment in the asset. Performance goals are the ideal targets that are set as the high level goal tied directly to the strategic plan and are often set at the executive or legislative level. Performance goals are used to identify funding needs to reach desired levels of service. Performance targets are the predicted value of the measure at a given point in time based upon funding constraints. By comparing performance targets to performance goals, an agency is able to identify performance gaps and can then reevaluate resource allocation.

Performance measures are used by the FHWA to monitor condition trends of the nation's bridges, to provide reports to Congress, and to administer the Federal Aid to Highways Program. Various groups and national publications use performance measures to communicate bridge network conditions and need

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on a uniform national basis. Most state DOTs have their own performance measures that reflect the specific conditions and needs of the state. These are used to communicate program issues to upper management, their state transportation commission, and to their legislature. State specific performance measures are of great importance to owner agencies because they are tailored to the unique needs of that state. Often bridge performance measures use GCRs to categorize bridges as being in good, fair, or poor condition. Bridge owners may use combinations of GCRs and bridge inventory items to set performance measures relevant to their state.

Performance measures using element condition ratings are becoming more prevalent. Bridge elements can be looked at individually, such as the quantity of steel beams in good, fair, poor, or severe condition, or they can be aggregated to provide an overall bridge assessment, such as categorizing the bridges in terms of need (cyclic maintenance, preventive maintenance, rehabilitation, or replacement).

The AASHTOWare BrM software provides a bridge "Health Index" which is a single-number assessment (0–100 scale) of a bridge's condition based on an amalgamation of the individual element health indexes.

The "Health Index" of an element is a single-number assessment of an element's condition based on the weighted, fractional distribution of the quantities across the range of condition states. Index values range from 0 (the worst possible or a completely failed condition) to 100 (the best possible condition).

$$HI_e = C_{e1}y_{e1} + C_{e2}y_{e2} + C_{e3}y_{e3} + C_{e4}y_{e4} \quad \text{(Eq. 3.3.1.3-1)}$$

where:

C_{ei} is the health index coefficient, or weighted value of Condition State i for element e.
y_{ei} is the percentage of element e in Condition State i.

Quantities of an element in CS1 are in good condition and are generally taken to contribute fully to the health index ($C_{e1} = 1$). Quantities of an element in CS4 are in a severe or failed condition and are generally assumed to contribute nothing to the index ($C_{e4} = 0$). Allowing for variable weights in the CS2 and CS3 categories (Fair to Poor) allows the owner to customize the health index into a useable value for that element. For instance, the portions of a protective system in fair condition (CS2) are "substantially effective" while the quantities in poor condition (CS3) have "limited effectiveness." The fair condition (CS2) quantities may

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be given an index value close to that of the "fully effective" portions in good condition because the underlying element is still protected. The quantity in poor condition (CS3), however, may be given a much lower index value to represent the risk of deterioration of the underlying element. By allowing this flexibility, the agency is able to create element level health indexes that correspond to the practices within the agency, and can then be used to develop meaningful performance measures and trigger actions to correct deficiencies at the appropriate times.

The impact of an element on the bridge health index is related to its condition, total quantity, and the weight or importance factor of that element.

$$\boldsymbol{HI} = \frac{\sum_e \boldsymbol{q_e w_e HI_e}}{\sum_e \boldsymbol{q_e w_e}} \qquad \text{(Eq. 3.3.1.3-2)}$$

where:

HI_e is the health index of the element e.
q_e is total quantity of the element e.
w_e is the weight of element e.

Bridge performance measures can also include measures of other variables such as seismic vulnerability, potential scour, substandard load capacity, and functional deficiencies such as low vertical clearance and narrow deck width. Where possible, it is desirable that national and state specific performance measures be related, but there may be good reasons for the measures to be different as long as the differences and the purpose of the measures are communicated clearly.

Best practices in performance measurement include:

- if general condition rating data is used due to limited element history, the measure should be adaptable and meaningful when sufficient data is available to transition to the National Bridge Elements;
- communication should be considered in naming the performance measure;
- the goals should follow the SMART philosophy—Specific, Measurable, Attainable, Realistic, and Timely;
- the measure should be short and simple;
- the measure should be reliable, transparent, and understandable;
- a high priority should be set on preservation and maintenance;

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- the units selected for the performance measure should be carefully considered. For example, creating a condition performance measure using deck area can result in different project prioritization and outcomes than the projects selected if the measure was based upon number of structures;
- the measures should have a direct linkage to the deterioration and the improvement of bridge elements or components to perform the tradeoff analysis or cost/benefit analysis.

3.3.1.3—Preservation and Improvement Action Data

In order to analyze bridges for project selection, the BMS requires information on the action—whether it is preservation or improvement driven—that should be considered in a network and the benefit the action has on a structure. The benefits may vary depending upon the type and condition of the structure that is having the action applied to it, and this complexity must be included in the action data for it to be considered in the analysis. To report bridge needs at both the bridge and network level, an overall category of work type (such as preventative maintenance, rehabilitation, replacement, or mitigation) should be assigned to each action. In addition to maintaining information on possible work activities, historical preservation and improvement data should be stored within the BMS. This information can then be used to evaluate the efficacy of actions, verify assumed benefit information, and influence the timing of future work according to agency policy.

3.3.1.4—Cost Data and Financial Plans

The evaluation of alternative actions and strategies requires knowledge of the costs to perform the work and a knowledge of the available finances. Costs to be considered include the direct cost of the action and the indirect costs associated with a project, such as mobilization and traffic control. Inflation should be included when analyzing the effectiveness of the work performed in future years.

Financial planning information is required to apply fiscal constraints to the analysis. The amount of funds available in the short term and predicted for the long term, the source of the funding, and any other restrictions should be included in the BMS so that it can be incorporated in the analysis. A bridge program does not exist by itself. Financial planning often needs to consider the challenge of planning for prioritization of projects that involve multiple assets and should be

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coordinated with other programs such as pavement, safety, and modernization.

3.3.2—Data Integration

Data integration identifies data driven actions and strategies to optimize the use of limited resources. The optimal decision is the one that minimizes cost and risk over the long term while providing the desired level of service. Because decisions made today on bridge preservation or improvement affect the condition of the bridge system in the future, a BMS should include mechanisms for predicting the effects of today's decisions. Two major prediction tools that are important for BMS operation are bridge deterioration models and bridge-related cost models. The deterioration and cost models provide engineering and economic data to the optimization module, where these inputs, along with agency rules, budgets, and policy data, are analyzed to develop, categorize, and prioritize a selection of projects for maximum benefit within financial constraints.

Data integration is composed of five main components:

1. data analysis;
2. risk assessment;
3. agency rules;
4. cost/benefit analysis; and
5. prioritization and optimization.

3.3.2.1—Data Analysis

A BMS uses data to identify the current and future bridge needs and evaluates system performance. This analysis may involve inventory information, general condition ratings, bridge element ratings, and inspector work recommendations. The BMS should be able to optimize work activities to address the deficiencies and categorize the work types into major categories such as preventive maintenance, rehabilitation, or replacement to report to stakeholders.

Strategic planning requires DOTs to make decisions that are cost-effective over the long term. Assessing future needs based on current condition data is an essential component of BMS data analysis. Bridge deterioration models forecast the probability of element condition states or bridge condition ratings at future points in time, under the assumption that no programmed major preservation, repair, or replacement activity is undertaken. These are typically accompanied by action effectiveness models, which

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forecast bridge condition following completion of preservation activity. Together, these models can be used to predict the rate at which new needs arise, and the effects of alternative policies and funding levels for preservation. When implementing a new bridge management system, deterioration models are typically developed using a process of expert judgment elicitation. Once a few cycles of inspections are completed on a sufficiently large inventory, it becomes possible to develop cross-sectional models, using statistical analysis methods that make year-to-year forecasts of the probabilities of each possible change in condition that might be observed. Such forecasts can be chained year after year to produce an estimate of new needs that are likely to arise each year in a long-range plan. Some deterioration models can consider variables such as climate, traffic volume, and interaction among bridge elements (for example, the effect of paint condition on the underlying steel) to refine the predicted rates of change in condition. Quantifying the uncertain rate of deterioration is important because even long-lived assets can produce near-term preservation needs on a fraction of bridges in the inventory. Methods for developing deterioration models can be found in NCHRP Report 713, *Estimating Life Expectancies of Highway Assets*.

3.3.2.2—Risk Assessment

Risk may be understood as the potential for unplanned adverse events to impact one or more transportation facilities in a way that causes unacceptable transportation system performance according to any or all of the agency's performance objectives. In bridge management, the primary concern is disruption of expected or designed service levels, which may cause injuries or property damage, loss of mobility, and immediate expenditures or long-term excess costs.

The adverse events which may cause these service disruptions may include natural hazards such as earthquakes, landslides, storm surge, high winds, floods, scour, and wildfires; man-made hazards such as overloads and vehicle or vessel collisions, including vehicles containing flammable cargo; and advanced deterioration related to corrosion, section loss, displacement, or fatigue cracking.

Risk assessment evaluates the likelihood and consequence of adverse events. The likelihood of the event includes the probability of the event occurring and may include the vulnerability of the structure to the event. The consequence of the adverse event would quantify the damage to the structure, the impact on the flow of people and goods in the transportation network, and the importance (criticality) of the structure. NCHRP Report 20-07/Task 378 provides a

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risk assessment methodology with worksheets that were developed to assist in estimating likelihood and consequences of service disruption for various types of hazards.

A transportation system is best prepared to reduce the consequence of an extreme event when the system includes redundancy, robustness, and resiliency.

Redundancy is excess capacity in times of emergency. Bridge engineers are familiar with load path redundancy in bridges; however, sound management of the assets on detour and emergency evacuation routes increases a highway system's redundancy.

Robustness is the capacity to cope with stress or uncertainty. Well-maintained assets are generally better able to withstand the stresses of storm events and other disasters than weakened and poorly maintained ones.

Resiliency has been defined in several ways; one definition is "the ability to prepare for, plan for, absorb, recover from, and more successfully adapt to adverse events." Another definition is "the capability of a system to maintain its functions and structure in the face of internal and external change and to degrade gracefully when it must" (Allenby and Fink 2005).

A risk-based BMS should contribute to all three attributes.

A BMS supports risk assessment and strives for risk reduction. A BMS should provide an accurate inventory of bridges with their condition and unit-cost data; identify those bridges at risk for given types of events such as floods, hurricanes, or earthquakes; and prioritize the level of risk. A BMS may have mapping and geographic information system (GIS) capabilities to assist with identifying bridges and prioritizing their coordination with evacuation planning or detour routing. As not all structures have the same criticality or vulnerability, the prioritization can allocate resources–both in terms of dollars and personnel–which may be the limiting factor during an event. A BMS should prioritize critical bridges for mitigation prior to an event, for inspection during critical events, and for repair following an event. A BMS should prioritize bridges for data collection realizing that not all structures have the same criticality or may not have the same level of risk. Sound maintenance practices recommended and tracked by a BMS "harden" assets. For example, bridges with well-maintained wing walls, bank protection, and scour protection are more robust during high water events. Conversely, a BMS can identify bridges where deferring maintenance can allow an undesired condition and lead to an unacceptable vulnerability and risk.

Risk assessment within a BMS can be used to modify and optimize inspection and data collection practices. Bridge inspection and condition data collection can perform two purposes; identify safety

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issues on bridges, and report condition for asset management decisions. NCHRP Report 782, *Proposed Guideline for Reliability-Based Bridge Inspection Practices*, outlines a methodology to develop a risk-based approach for determining the bridge inspection interval. The goal of the methodology is to improve the safety and reliability of bridges by focusing inspection efforts where needed most and optimizing the use of resources.

3.3.2.3—Agency Rules

In order for a BMS to make bridge level decisions consistent with agency practice, agency rules need to be developed. The intent of the rules is to translate agency practices and their effects on bridge, program, and network level recommendations into the system's modeling approach. These rules should be intuitive and reflect agency business practice and policy. Rules may be cyclic or conditional. A cyclic rule is set out as an action and an interval. For example, "Wash every bridge once each year." Conditional rules are set out as action taken as the result of the condition of an element or component. For example, "Replace seals in strip seal expansion joints when quantity in Condition State 2 (Fair) exceeds 20 percent, or quantity in Condition State 3 is greater than 0 percent" is a conditional rule. Conditional rules most often need to be considered concurrently with related elements that could impact how the rules should be applied. For example, a concrete bridge deck may be protected by a wearing surface, or steel beams may be protected by a paint system, or feasible actions such as rehabilitating a deck surface or replacing deck expansion joints may be "constrained" by the poor condition of the bridge's superstructure, which needs to be replaced. Agency rules within the BMS need to provide the recognition of these interdependencies.

Agency rules may provide thresholds for performing work on the entirety of an element when a certain percentage is deteriorated, or there may be rules to address multiple deficiencies at once whenever possible or to defer work for a certain number of years following a project to minimize the impact on the traveling public. Agency rules can be an important tool to reduce computational time by eliminating unrealistic work candidates during network analysis.

Rules may be applied at the bridge, program, or network level. Bridge level rules might reflect different preservation policies for steel versus concrete structures, or prevent replacement recommendations for historic structures. Program level rules may reflect varying performance measure goals or funding constraints, while network rules cover standard agency practice.

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Agency rules may also be used to apply the connection between bridge asset management and the needs and performance measures of the other assets within the transportation system. For example, bridge specific agency rules can time actions to correspond with pavement preservation activities.

3.3.2.4—Cost/Benefit Analysis

With limited resources and competing system needs, cost/benefit analyses are required to determine effective work activities. Cost/benefit analyses can be condition, improvement, or lifecycle driven. Optimization approaches to a BMS recognize that preservation activities including cyclic maintenance, preventive maintenance, repair, and rehabilitation are done to deter deterioration (condition driven benefits) while improvements such as widening, strengthening or mitigation actions respond to user demands or reduce risk. The choice of preservation actions should be predicated on maintaining safety, meeting performance measures, and minimizing agency lifecycle costs while improvements should be based on the benefit to road users or eliminating bridge deficiencies.

3.3.2.4.1—Condition Driven Cost/Benefit Analysis

The most basic asset management program takes condition data, identifies the work activities that could address any deficiencies in condition, and then calculates the cost of the activity based on the quantity to be repaired. In order to compare possible activities, the condition improvement (benefit) is shown as a ratio to the cost. The activity or combination of activities with the highest benefit to cost ratio would be the preferred condition driven action for that structure. The bridge manager needs to take caution when developing a strategy based solely upon condition driven cost/benefit analysis because if condition benefits and costs alone are considered, the alternatives could tend to favor maintenance only to extend life until permanent closure.

3.3.2.4.2—Improvement Cost/Benefit Analysis

Improvement benefits reduce risk or user costs by eliminating, mitigating, or reducing bridge deficiencies. Bridges with low vertical clearance or insufficient load capacity will force a certain volume of truck traffic to be detoured to alternate routes, resulting in increased vehicle operating costs. Bridges having narrow deck width, low vertical clearance, or poor alignment may have a higher occurrence of

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accidents than bridges without these deficiencies. Scour critical bridges or bridges with poor seismic details are at risk during extreme events. Reducing these deficiencies through replacement, widening, strengthening, or mitigation actions provides a benefit to the user. Consideration of improvement benefits is essential in a BMS if functional deficiencies are to be eliminated and risk is to be reduced. AASHTO provides *User and Non-User Benefit Analysis for Highways* (2010) for agencies to help analyze the benefits and costs of highway projects.

3.3.2.4.3—Lifecycle Cost/Benefit Analysis

The direct cost to a highway agency for owning a bridge is not a one-time initial construction cost. Rather, agency costs are a long-term, multi-year investment of a series of expenditures for maintenance, rehabilitation, and replacement. Therefore, a BMS should take a long-term view of the lifecycle of a bridge, reflecting the highway agency's long-term responsibility.

Lifecycle Cost Analysis (LCCA) is an economic evaluation used in the selection of the least cost design and management strategies that impact initial construction and lifecycle agency costs. It compares the differential costs of initial design and lifecycle management alternatives over a specified analysis period to identify the least cost alternative. Design features and management practices similar between the alternatives are not typically included in the analysis. The analysis period should cover a time span that includes at least one major rehabilitation for each alternative being considered. Lifecycle costs can be shown as the discounted present value of the total initial and future agency costs that occur over the analysis period. To perform network level analyses, the assumptions and analysis process of lifecycle costs should be comparable from one structure to another. If lifecycle costs are calculated over a design service life that varies with each type of structure, it is convenient to convert lifecycle costs to equivalent uniform annual costs. If lifecycle costs are calculated over a set period of time for each structure, the remaining value of the structure at the end of the evaluation period should be normalized. LCCA can include user costs attributable to the functional deficiency of a bridge such as a load posting or clearance restriction. It can also include impact to the user resulting from bridge closure or work zone restrictions during construction. Agency costs should not be combined with user costs, since agency costs represent direct impacts to budgets and influence resource allocation decisions. User costs help identify initial construction, preservation, maintenance, and procurement strategies that have the least impact to

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users. The specific elements of agency and user costs considered in a bridge LCCA will depend on the bridge under consideration and the purpose for which the analysis is being made.

When doing LCCA, the use of deterioration models, estimated service life durations for activities, and costs associated with them are needed. BMS software such as AASHTOWare BrM provide deterioration modeling developed from regional and national data and expert elicitation. State agencies typically have historic data that can be used to develop deterioration modeling for bridges in their networks. In addition, construction and maintenance costs from previous projects or studies can be used for developing costs for fix types. There are several resources for doing bridge lifecycle cost analysis, including NCHRP Report 483, *Bridge Lifecycle Cost Analysis*, The National Institute of Standards Technology, Bridge Lifecycle Tool (BLCC) and user's manual, and FHWA's National Bridge Investment Analysis System (NBIAS) and documentation.

3.3.2.5—Prioritization and Optimization

Optimization has become the preferred method for bridge network management. The purpose of optimization at the network level is to select a set of bridge projects in such a way that the total benefit derived from the implementation of the selected projects is maximized (costs and risks are minimized). The ability to establish project priorities and optimally allocate limited funds over a predefined planning horizon, both short- and long-run, is a fundamental part of a BMS.

The system should consider unconstrained, constrained, and required budget cases. If unlimited budgets are available, it is possible to determine the optimum period in which selected alternatives should be scheduled. Where adequate funding is not available to maintain a desired level-of-service, the BMS may calculate the economic consequences of a lower level-of-service and provide an objective means of setting priorities for bridges so that risks are minimized. When a project must be delayed, the BMS can use the deterioration models and cost models to quantify the bridge level effect and the impact on road users, and to determine the new optimal set of actions for the bridge at a later period. By exploring period-by-period project deferrals, multi-year programs can be generated. The system should also be able to indicate the funding required to reach performance goals to identify the financial performance gap.

Modern optimization approaches can take several forms. The differences in optimization approaches tend to be in the specific techniques used and in the

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way that network-level considerations are reflected in the analysis. Two common approaches are:

- Top-Down Approach, where network-level performance measures are addressed first and the results are then used to guide project selection, resource allocation, and scheduling. For the top-down approach, an agency may deteriorate the entire network over the programming time frame, and try to balance the best option for each bridge with the best option for the whole network, increasing or decreasing work on any one bridge to make the most progress toward the agency's performance measures.
- Bottom-Up Approach, where the most cost effective option is identified for each bridge by using condition information and inspector recommendations. The BMS accomplishes this by providing near- and long-term side-by-side comparisons of alternative work candidates and alternate timing of the projects. After bridge level decisions of scope and timing of work are made, the results of the analysis are summarized back up to the network level to determine the level of funding required to address network deficiencies for a given period of time and the optimum projects to schedule subject to budget and other network level constraints.

Agencies will often use a combination of these two methods for optimization.

3.3.2.5.1—Multi-Objective Optimization

Bridge owners often need to consider multiple performance criteria and constraints, such as bridge condition, lifecycle costs, safety, traffic flow disruption, and vulnerability when making decisions and prioritizing projects. They may need to analyze tradeoffs between these performance criteria. NCHRP Report 590, *Multi-Objective Optimization for Bridge Management Systems*, provides a methodology for this decision-making predicated on utility theory. Utility theory is based on the premise that the decision maker's preference structure can be represented by a utility function. Construction of the utility function generally involves three major steps:

- Weighting: This assigns relative weights to each of the multiple criteria based on their relative importance.
- Scaling: This provides a common scale of measurement for performance criteria that have different units. This involves developing a single-

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criterion utility function for each performance criterion.

- Amalgamation: This combines the single-criterion utility functions using the relative weights into a single overall performance measure. The structure of the overall objective function is influenced by the mathematical assumptions made regarding the decision maker's preference structure. The key step in the amalgamation process is the identification of the appropriate functional form for combining the individual utility functions of the various performance criteria into a single quintessential utility value.

The utility functions for each bridge are then used to evaluate alternatives. The increase in utility is the benefit of a project, and therefore the alternative with maximum benefit/cost ratio is the most preferred for a structure. Multi-objective utility functions are then able to be applied using the top-down or bottom-up approaches or a combination of the two.

3.3.3—Decision Support

The function of a BMS is to provide bridge information and data analysis capabilities to improve the decision-making abilities of bridge managers. A BMS should not make decisions. Bridges cannot be managed without the practical, experienced, and knowledgeable input of the Engineer/Manager. A BMS is never used in practice to find one best policy among the possible choices. Instead, managers should use the BMS as a tool to evaluate various policy initiatives, often referred to as "what if" analysis. The available choices may relate to network-level decisions or project-level decisions.

An optimization performed by a BMS is only as valid as its underlying data and assumptions. A BMS may never have all the necessary information in its database. Often the missing information is mostly intangibles, such as engineering experience, local needs, mobility-based traffic capacity improvement, and economic development considerations. A BMS may therefore build in user adjustments at all critical decision areas. To function as decision support, a BMS should assist in compiling and displaying current and predicted condition data and performance measures. Many agencies are making use of executive dashboards to show condition data and performance measures to stakeholders.

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SECTION 4

INSPECTION PROCEDURES

4.1—INTRODUCTION

The first priority of a bridge inspection program is to ensure that bridges are safe for public travel. This requires that inspections be performed by properly trained personnel following proper, systematic inspection procedures to ensure quality and consistency of information across an entire network of bridges.

The collection of accurate, detailed, and comprehensive information is required to operate and manage those bridges. This information is also used within a bridge management system to determine rates of deterioration and routine maintenance needs, as well as priorities of any necessary repairs or rehabilitation.

This section assists the Bridge Owner in developing inspection procedures that directly relate to the requirements of 23 CFR 650 of the National Bridge Inspection Standards (NBIS) (Article 4.2), as well as supplemental inspection practices that Bridge Owners may find useful (Article 4.3).

4.2—PROVISIONS TO SUPPORT THE NBIS REQUIREMENTS

Inspect all bridges, as defined in 23 CFR 650.305, located on public roads in accordance with coding guidance issued by the Federal Highway Administration (FHWA) and the following provisions of this Section.

4.2.1—Bridge Inspection Organization

Ensure that the bridge inspection organization includes one program manager (typically within the state transportation department) who has the responsibility to ensure that all the publically owned bridges are inspected in accordance with the requirements of the NBIS. In addition, provide documentation for the roles and responsibilities for each of the following required sections of the NBIS:

- Policies and Procedures
- Quality Assurance and Quality Control
- Preparation and Maintenance of the Bridge Inventory
- Bridge Inspections and Reports
- Load Ratings

The performance of each section listed above may be delegated to others, such as local Bridge Owners. Define in writing any delegations and

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include any possible actions that may be implemented if the delegated authority does not meet the requirements of the sections.

4.2.2—Qualifications of Personnel

Ensure that specific requirements listed in 23 CFR 650.309 for each of the following personnel are adhered to:

Program Manager: The individual responsible for all bridge inspection activities, including reporting and maintaining an inventory. The program manager provides overall leadership and is available to inspection program staff to provide guidance. The program manager establishes consistent policies and practices for the state or federal agency.

Inspection Team Leader: The individual in charge of an inspection team responsible for planning, preparing, and performing the on-site inspection of the bridge. Ensure that a qualified inspection team leader is at the bridge site at all times during each initial, routine, in-depth, fracture-critical member, and underwater inspection.

Load Rating Manager: The individual responsible for the load ratings. The load rating manager establishes consistent methods, vehicles, etc. to be used and provides guidance as necessary.

Underwater Bridge Inspection Diver: The individual performing the underwater inspection. The underwater bridge inspection diver may also be the inspection team leader if that person meets those requirements as well.

In addition to the requirements in 23 CFR 650.309, document the bridge inspection training and qualifications of each of the above personnel and keep on file. Establish the following:

- Frequency and requirements of periodic bridge inspection refresher training.
- Training and/or experience requirements for inspectors when inspecting complex bridges.

4.2.3—Inspection Types

The types of inspection required for each bridge vary depending on the type and condition of the bridge. For example, a bridge over deep water that has fracture-critical members will require routine, fracture-critical member, and underwater inspections. These may be performed all at the same time. Other types of bridges may only require a routine inspection. It is important to understand each type of inspection in order to ensure that they are applied as required.

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4.2.3.1—Initial Inspection

An initial inspection, also referred to as an inventory inspection, is the first inspection of a new bridge or an existing bridge after a major rehabilitation such as a deck replacement, superstructure replacement, or widening. This inspection serves to record required bridge inventory data, establish baseline conditions, and establish the timeline for other types of inspections normally required such as routine, fracture-critical member, and underwater inspection.

Thoroughly document the initial inspection, performed by persons meeting the required qualifications for each type of inspection. Include an analytical determination of load capacity, as well as a bridge scour evaluation for qualifying bridges over water.

The purpose of the initial inspection is two-fold. First, it is used to provide all National Bridge Inventory (NBI) and bridge element data required by federal and state regulations and all other relevant information normally collected by the Bridge Owner. Second is the determination of baseline structural conditions and the identification and listing of any existing problems or locations in the structure that may experience future problems. The inspection serves as the first routine inspection with provisions as described in Article 4.2.3.2.

4.2.3.2—Routine Inspection

Routine inspections are regularly scheduled inspections consisting of observations for all elements of the bridge and measurements needed to determine the physical and functional condition of the bridge, to identify any changes from initial inspection or previously recorded conditions, and to ensure that the structure continues to satisfy present safety requirements.

Perform a routine inspection to satisfy the requirements of the National Bridge Inspection Standards (NBIS) with respect to maximum inspection frequency and updating of NBI and bridge element data. These inspections are generally conducted from the deck, from ground level and/or water level, and from permanent work platforms and walkways if present. Inspection of underwater portions of the substructure is limited to observations during low-flow periods, and includes probing for evidence of substructure deterioration and undermining as necessary. Special equipment, such as under-bridge inspection equipment, rigging, or staging, is necessary for routine inspection in circumstances where its use provides the only practical means of access.

The areas of the structure to be closely monitored are those determined by previous inspections and/or load rating calculations that are

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critical to the bridge's load-carrying capacity. If, during a routine inspection, there is an area of the structure that requires a closer, more detailed inspection to determine its impact on safety or load-carrying capacity (e.g., a crack in a steel member), then perform and document an additional in-depth inspection of that area.

4.2.3.3—In-Depth Inspection

An in-depth inspection is a close-up, detailed inspection of one or more bridge components located above or below water, using visual or nondestructive techniques as required to identify any deficiencies not readily detectable using routine inspection procedures. Provide traffic control and special equipment, such as under-bridge inspection equipment, staging, and workboats, as necessary to obtain access. Utilize personnel with special skills such as divers and riggers as may be required. When appropriate or necessary to fully ascertain the existence of or the extent of any deficiencies, provide nondestructive field tests, other material tests, or both as may be necessary.

Clearly identify specific portions of the bridge receiving an in-depth inspection in the report, as well as those procedures utilized during the inspection. Document the findings of in-depth inspections in as much detail as needed; these may be significantly more detailed than the information contained within initial and routine inspections. An in-depth inspection of a particular element or detail may be included within other inspection types, such as routine or initial, as described in Articles 4.2.3.1 and 4.2.3.2, respectively.

4.2.3.4—Fracture-Critical Member Inspection

A fracture-critical member (FCM) is a steel member in tension, or with tension elements, whose failure would probably cause a portion of or the entire bridge to collapse. Some examples of steel members that may be considered FCMs include the following:

- One- or two-girder I- or box girder systems
- Suspension systems with eyebar components
- Welded steel pier caps and cross girders
- Two truss systems
- Welded tied arches
- Pin and hanger connections on two- or three-girder systems

The above examples are not a comprehensive list of FCMs.

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Document what types of steel bridge components are considered FCMs. Clearly identify FCM components requiring an FCM inspection in the inspection file.

Perform a hands-on inspection (inspection within arm's length of the component using visual or nondestructive techniques as required) of all FCMs.

4.2.3.5—Underwater Inspection

An underwater inspection is an inspection of the underwater portion of a bridge substructure and surrounding channel which cannot be inspected visually at low water by wading or probing, generally requiring diving or other appropriate techniques.

Document any conditions that require an underwater inspection (such as water depth, turbidity, and current thresholds, if appropriate), as well as requirements for cleaning of submerged bridge elements (e.g. debris, biofouling).

Ensure that a qualified inspection team leader is present during the entire underwater inspection. A separate inspection team leader is not required if the underwater bridge inspection diver meets the qualifications of an inspection team leader.

For bridges determined to require underwater inspection, ensure that the bridge inspection file contains the following information:

- the substructure units requiring underwater inspection
- a description of the underwater elements to be inspected
- the frequency of underwater inspection for the elements
- the procedures to be followed during the underwater inspection

As noted in Article 2.2.6, overall inspection procedures common to underwater inspection may exist or may have been developed in an agency bridge inspection manual, making it unnecessary to include the above bulleted items in each bridge inspection file.

4.2.3.6—Special Inspection

A special inspection is an inspection scheduled at the discretion of the Bridge Owner. It is used to monitor a particular known or suspected deficiency, such as cracks in steel, settlement, or scour. It may also be used to monitor special details or unusual characteristics of a bridge that do not necessarily have defects, such as ultrasonic testing of pins.

A special inspection is not required to be performed by a qualified inspection team leader;

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however, provide a person qualified to perform the specific inspection (for example, qualified nondestructive testing inspector to perform ultrasonic testing of pins). Document the qualifications of typical special inspection personnel such as ultrasonic testing (UT) inspectors and weld inspectors. If a bridge requires unusual special inspection personnel, document their requirements in the bridge file.

Clearly document in the inspection file the location, specific detail to monitor, interval of any special inspection, and how the inspection is to be performed. Perform the special inspection in accordance with the *Bridge Inspector's Reference Manual*, or BIRM (FHWA Publication NHI 12-049) if no specific inspection procedures are identified in the bridge file. Although special inspections are not sufficiently comprehensive to meet NBIS requirements as routine inspections, they may be performed at the same time as the routine inspection.

4.2.3.7—Damage Inspection

A damage inspection is an unscheduled inspection to assess structural damage resulting from environmental events (such as earthquake or flooding) or human actions (such as truck impacts).

The amount of effort expended on this type of inspection can vary significantly depending upon the event and extent of the damage. If major damage has occurred, inspectors may have to evaluate fractured members, determine the extent of section loss, make measurements for misalignment of members, or check for any loss of foundation support.

Ensure that the scope of the damage inspection is sufficient to determine the need for emergency load restrictions or closure of the bridge to traffic, and to assess the level of effort necessary to implement a repair.

Inspect all damaged bridge elements in accordance with the visual inspection method as described in the BIRM. If a significant defect is suspected through visual inspection, use of advanced inspection methods as described by the BIRM may be necessary.

This inspection may be supplemented by a timely in-depth inspection to obtain additional data needed to refine load analyses, determine repair quantities/methodologies, or adjust load restrictions.

For scour-critical bridges, follow the scour plan of action (POA) included in the bridge file.

4.2.4—Inspection Interval

Inspection intervals vary depending on the type of inspection and condition of the bridge. However, there are minimum intervals required by the NBIS for several inspection types. The following sub-

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articles list the minimum requirements for each inspection type. Document the criteria to be used in determining when a routine, FCM, complex, or underwater inspection is to be placed on a shorter interval (less than 24 months). These criteria can be engineering judgment but shall also contain other data driven requirements such as load ratings or condition ratings. It is the responsibility of each state or federal agency to establish its own criteria.

4.2.4.1—Initial Inspection Interval

Initial inspections are not recurrent and thus do not have intervals. However, upon completion of new bridge construction or upon completion of major bridge rehabilitation and, once all travel lanes are open to traffic, complete the initial inspection and update all NBI and element level data within the following timeframe:

- for state or federally owned bridges: within 90 days
- for all other bridges: within 180 days

4.2.4.2—Routine Inspection Interval

Ensure that routine inspections are performed at intervals not to exceed 24 months. However, certain bridges may be inspected at longer intervals, not to exceed 48 months. Ensure that bridges identified for inspection on an interval longer than 24 months meet the specific written criteria approved by the FHWA. The criteria may include past inspection findings, age, condition, material type, load rating capacity, and posting status.

The requirements for a 48-month inspection interval policy are described in the FHWA Technical Advisory T 5140.21 dated September 16, 1988. This document is available at http://www.fhwa.dot.gov/bridge/nbis/t514021.cfm.

4.2.4.3—In-Depth Inspection Interval

In-depth inspections are scheduled and performed at the discretion of the Bridge Owner. Therefore, there are no minimum inspection intervals specified in regulation.

4.2.4.4—Fracture-Critical Member Inspection Interval

Perform fracture-critical member inspections at intervals not to exceed 24 months, in accordance with the NBIS. There is no allowance in the NBIS for FCMs to be inspected at longer intervals than 24 months.

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4.2.4.5—Underwater Inspection Interval

Perform underwater inspections at intervals not exceeding 60 months. However, certain bridges may be inspected at longer intervals, not exceeding 72 months. Bridges identified for underwater inspection on an interval longer than 60 months should meet specific written criteria approved by the FHWA. The criteria may include other items such as past inspection findings, age, condition, scour, or scour history.

Guidance for developing a 72-month underwater inspection interval policy can be found in the American Society of Civil Engineers (ASCE) Manuals of Practice 101, *Underwater Investigations Standard Practice Manual*, and FHWA-DP-80-1, *Underwater Inspection of Bridges*. The following NBI rating attributes are also suggested:

- The substructure should be in at least Good to Fair condition.
- NBI item 60 should have a rating of 5 or better. If the substructure elements are unprotected steel or unwrapped wood and are in an aggressive environment such as salt water or fast currents they should not be considered for a 72-month inspection.
- The channel should be stable with NBI item 61 of 7 or better. The structure should not have stream stability or scour issues and should be a known foundation type.
- NBI item 113 should have a rating of 4, 5, 7, 8, or 9.

4.2.4.6—Special Inspection Interval

Special inspections are scheduled and performed at the discretion of the Bridge Owner, and thus do not have minimum inspection intervals. Determine the inspection interval for a particular bridge considering the type and significance of the defect(s) requiring inspection, and then inspect the bridge at this interval.

A special inspection may occur coincident with a routine or other type of inspection, in the interest of efficiency.

4.2.4.7—Damage Inspection Interval

As they are event-driven, damage inspections do not have any inspection intervals. If the inspection reveals defects prompting continued, recurrent monitoring over time, then special inspections may be scheduled as a result of the damage inspection.

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4.2.5—Inspection Procedures

4.2.5.1—General

Inspect bridges according to the procedures listed in Article 4.2.5 to meet the requirements of 23 CFR 650 (NBIS). These procedures constitute general procedures for each type of inspection.

Develop bridge-specific procedures to facilitate the inspection of unique components or conditions not covered here. Include bridge-specific inspection procedures in the bridge file.

Record NBI data, including condition ratings, and bridge element data in accordance with the FHWA *Recording and Coding Guide for the Structure Inventory and Appraisal of the Nation's Bridges*, the AASHTO *Manual for Bridge Element Inspection*, and any supplemental state coding guides or manuals.

Use the BIRM as a principal inspection reference. It provides methods and techniques for the inspection of different bridge materials and components. Ensure that each inspector has a thorough knowledge of this manual.

Ensure that the inspection team leader reviews previous inspection reports for any defects, access equipment needs, etc., in order to properly prepare for the inspection. Ensure that the bridge file is reviewed for any required specific inspection procedures.

Ensure that all condition ratings are supported by narrative that appropriately justifies and documents the rating.

Clearly identify significant deficiencies, and include supporting photographic documentation. If there are a number of similar defects, a representative picture may be used.

Develop written procedures for the evaluation and subsequent re-evaluation of bridge load capacity ratings. Incorporate in these procedures the need to review and/or revise the load rating due to changes in the physical condition of the bridge, as well as policy-driven changes, such as changes in subject live load configurations or changes in the load rating methods themselves. A revised load rating may not be necessary depending on the exact nature of the circumstance/change(s); however, clearly state in the bridge file what specific inspection(s) were evaluated when the load rating has been reviewed.

Develop written procedures for the evaluation of scour at bridges, and the determination of scour susceptibility. Develop and document a scour plan of action (POA) in the bridge files for all bridges evaluated and determined to be scour-critical. Additional information on scour POAs is available at http://www.fhwa.dot.gov/engineering/hydraulics/bridgehyd/poa.cfm.

Develop procedures for bridge scour monitoring and assessment during and after flood events and

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include in the scour POA. Scour monitoring may be performed using different levels of effort depending on the degree of risk. Monitoring is described in the FHWA guidance manuals *Evaluating Scour at Bridges* (HEC-18) and *Bridge Scour and Stream Instability Countermeasures* (HEC-23). These publications can be found at https://www.fhwa.dot.gov/engineering/hydraulics/pubs/hif12003.pdf and https://www.fhwa.dot.gov/engineering/hydraulics/pubs/09111/09111.pdf, respectively.

Develop and establish the definition of a critical inspection finding (CIF). Develop procedures regarding:

- Immediate public safety actions to be taken upon finding a CIF.
- Initial CIF notification of Bridge Owners and the FHWA.
- Ensuring that CIFs are addressed in a timely manner.
- Follow-up actions to be taken with all concerned parties, until CIFs are eliminated at the bridge.

4.2.5.2—Initial Inspection Procedures

Ensure that a qualified inspection team leader is present during the entire initial inspection.

Prior to the initial inspection, review all plans and relevant information in order to encode NBI data, as well as bridge element quantities to the maximum extent possible.

Perform the initial inspection in accordance with the routine inspection procedures in order to collect outstanding NBI and element level data, as well as to identify existing deficiencies and encode condition ratings.

Ensure that baseline bridge inventory photographs are taken during the initial inspection, following the protocol developed by the Bridge Owner.

4.2.5.3—Routine Inspection Procedures

Ensure that a qualified inspection team leader is present during the entire routine inspection.

Perform an inspection of each bridge element in accordance with the visual inspection method as described in the BIRM. If a significant defect is suspected through visual inspection, use of advanced inspection methods as described by the BIRM may be performed, as warranted.

Determine all condition ratings and bridge element quantity and condition state quantity distribution data as necessary, following the FHWA and AASHTO guides and manuals previously described, and appropriate state guidance documents.

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Evaluate all height and weight restriction posting at the bridge site, to ensure that proper signage and notifications are in place. Note changes, either suspected or confirmed, to either weight or height posting values.

For scour-critical bridges, follow the scour POA included in the bridge file with regard to any items to be evaluated during the routine inspection.

Report any CIFs discovered, following established CIF procedures.

4.2.5.4—In-Depth Inspection Procedures

Ensure that a qualified inspection team leader is present during the entire in-depth inspection.

Perform an in-depth inspection as necessary to provide information or data using visual or advanced techniques and to identify any deficiencies not readily detectable using routine inspection procedures.

Provide traffic control and special equipment, such as under-bridge inspection equipment, staging, and workboats, as necessary to obtain access to areas not accessible during the routine inspection. Provide personnel with special skills such as divers and riggers, as necessary.

Perform in-depth inspections, at a minimum, using visual inspection methods, per the BIRM. Use advanced inspection methods, such as nondestructive field tests or material tests, whenever they are required to fully ascertain the existence of or the extent of a deficiency.

Clearly identify specific portions of the bridge receiving in-depth inspection in the report, as well as the procedures utilized during the inspection. Document findings of in-depth inspections to the level of detail which provides the information needed, which may be significantly more detailed than the information obtained from initial and routine inspections.

4.2.5.5—Fracture-Critical Member Inspection Procedures

An FCM inspection requires a hands-on inspection of the entire fracture-critical member(s) or member component(s) shown in the inspection file, which may include nondestructive testing or nondestructive evaluation.

Perform FCM inspections in accordance with FHWA-IP-86-26, *Inspection of Fracture Critical Bridge Members*, as well as with guidance provided in the BIRM.

Ensure that a qualified inspection team leader is present during the entire FCM inspection.

Review the inspection file prior to the inspection.

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Record the FCM inspection as such in the bridge inventory, with NBI Items 92a and 93a updated accordingly.

4.2.5.6—Underwater Inspection Procedures

Ensure that a qualified inspection team leader is present during the entire underwater inspection. A separate inspection team leader is not required if the underwater bridge inspection diver meets the qualifications of an inspection team leader.

Ensure that an appropriate level of pre-inspection preparation occurs, as with above-water routine inspections.

Reference the bridge inspection file prior to the inspection.

Perform the underwater inspection in accordance with methods and techniques contained in the BIRM, as well as state requirements, including requirements for evaluation of the waterway adjacent to the bridge.

Follow the scour POA as it pertains to requirements during an underwater inspection.

Record the underwater inspection as such in the bridge inventory, with NBI Items 92b and 93b updated accordingly.

4.2.5.7—Special Inspection Procedures

Ensure that a person(s) qualified to perform the specific inspection, as indicated in standard policies or the bridge file, performs the special inspection.

If no specific inspection procedures are identified in the bridge file, then perform the inspection in accordance with the BIRM.

Record the special inspection as such in the bridge inventory, with NBI Items 92c and 93c updated accordingly.

4.2.5.8—Damage Inspection Procedures

Ensure that the scope of the damage inspection is sufficient to determine the need for emergency load restrictions or closure of the bridge to traffic, and to assess the level of effort necessary to affect a repair.

Inspect all damaged bridge elements in accordance with the visual inspection method as described in the BIRM. If a significant defect is suspected through visual inspection, inspect using advanced inspection methods as described in the BIRM.

Supplement this inspection, as necessary, with a timely in-depth inspection to obtain additional data needed to refine load analyses, determine repair quantities/methodologies, or adjust load restrictions.

For scour-critical bridges, follow the scour POA requirements in the bridge file.

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4.2.5.9—Complex Bridge Inspection Procedures

A complex bridge inspection is an inspection of a complex bridge, including all movable, suspension, cable-stayed, and other bridges with unusual characteristics.

Determine and document which bridges are to be classified as complex bridges.

Develop written procedures for complex bridge inspections, indicating any special requirements for the inspection team leader or specific bridge inspectors (i.e. mechanical or electrical systems inspectors). Provide at a minimum, an inspection team leader that meets the requirements referred to in Article 4.2.2.

If necessary, document in the bridge inspection file any additional specialized inspection procedures to be followed, and additional/specialized training and experience required of inspectors.

Complex bridges are a bridge classification, and not an inspection type. Perform inspections at intervals not to exceed 24 months.

4.2.6—Collection of Element Level Data

Federal regulations (MAP-21) require element level data to be collected and transmitted to the FHWA for highway bridges on the National Highway System (NHS).

In addition to federally mandated minimum requirements, Bridge Owners may record additional element level data for bridges on the NHS to suit individual needs. Collection of element level data for bridges off the NHS is at the Bridge Owner's discretion.

Develop and implement procedures to collect element level data for bridges on the NHS in accordance with the AASHTO *Manual for Bridge Element Inspection*.

Develop and implement standardized Element Level inspections of NHS bridges, following procedures identified in the AASHTO *Manual for Bridge Element Inspection*.

4.3—NONREGULATORY INSPECTION PRACTICES

4.3.1—General

Bridge inspection is fundamental to public safety and serves as the foundation for a comprehensive bridge asset management system. Accordingly, it is imperative that inspections be accomplished in a safe, systematic manner, following established practices using appropriate equipment and inspection techniques.

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Reference and follow the inspection procedures and techniques in the FHWA *Bridge Inspector's Reference Manual*, or BIRM (NHI-12-049), during ongoing bridge safety inspections. The following are recommended practices and techniques that may be incorporated into a bridge safety inspection program in addition to the regulatory information previously provided in Article 4.2.

4.3.2—Safety

Develop a safety program for inspection personnel with information concerning their safety and health, including the proper operation of inspection tools and equipment. Safety of both the inspection team members and the public is paramount. This safety program shall embody applicable state and federal legislation governing safety and health in the bridge inspection work environment.

4.3.2.1—Safety for Inspectors

Wear personal protective clothing at all times, including hard hats, vests, safety glasses, and appropriate footwear during inspections. Practice proper hearing, sight, and face protection methods when using manual and power tools. Keep all equipment, safety devices, and machinery in the best possible operating condition.

Operate inspection vehicles in accordance with the operating manuals provided by the manufacturer. Train for and implement emergency procedures in the event of equipment or vehicle failure.

Wear safety belts, lanyards, harnesses, and other personal safety equipment in accordance with applicable standards. Maintain lifelines, belts, lanyards, and other equipment in good repair. Discard worn or damaged equipment. Keep safety equipment clean and away from potentially harmful chemicals such as gasoline, dye penetrant, and oil.

Employ proper safety precautions when entering confined spaces such as the interior of a box girder. Air quality testing, air ventilation, and/or the use of self-contained breathing apparatus (SCBA) should be employed as necessary. Use good judgment and common sense in addition to established safety programs. Recognize that each bridge site is unique, and take appropriate caution in situations where unusual working conditions may exist. Note that some spaces may be classified as "permit required" confined spaces, and require permit conditions to be satisfied prior to working in that space.

Periodically take training in first aid and CPR. Take training in fall protection, HAZCOM/global harmonized system, and defensive driving as necessary.

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Take respirator training and respirator fit testing.

Some bridge inspection activities may require removal of, or exposure to, bridge coatings containing heavy metals. Develop a procedure to facilitate safety when working in these conditions, eliminating the presence of airborne heavy metals, such as lead, to the maximum extent practicable. Incorporate safety training, employee air quality monitoring, and perhaps even blood lead level (BLL) blood testing and monitoring for inspectors working in these environments.

Provide formal vision testing (including color-blindness) and hearing testing as necessary for inspection staff.

4.3.2.2—Public Safety

In the interest of public safety, employ proper procedures for traffic control and work zone protection during the inspection of a bridge. Use the MUTCD, supplemented by state and local authorities, as a guide for such procedures. To the maximum extent practicable, perform inspections so as to minimize disruption to traffic flow.

Coordinate any work in which personnel or equipment will be working or stationed within 25 ft of any railroad track in advance with the appropriate authority or company. This authority will determine if a flagman needs to be present and whether any special protective safety measures need to be implemented. Some railroads require completion of a formal railroad safety awareness course prior to working in the railroad right-of-way.

4.3.3—Planning, Scheduling, and Equipment

4.3.3.1—Planning

The key to the successful performance of a safe and effective bridge inspection is proper advance planning and preparation. Develop an inspection plan based on a review of the bridge record and a preinspection site visit as necessary. The following items are to be considered:

- Determine the type of inspection required.
- Determine if specialized inspectors or knowledge is necessary.
- Determine the number of personnel and type of equipment and tools necessary to perform the inspection.
- Determine which members or locations are noted in previous inspections or maintenance records as having existing defects or areas of concern.

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- Determine whether the bridge is subject to height or weight capacity restriction and plan access equipment needs accordingly.
- Estimate the duration of the inspection and the scheduled work hours.
- Establish coordination with or notification of other agencies or the public as needed.
- Assemble field-recording forms and prepare appropriate predrafted sketches of typical details.
- Determine the extent of underwater inspection required, as well as any vulnerability to scour. Identify special needs such as diving inspection or scour evaluation.
- Decide whether nondestructive or other specialized testing is appropriate.
- Determine whether the structure contains members or details requiring special attention, such as fracture-critical members, fatigue-prone details, and non-redundant members.
- Determine whether there are structures nearby that are also scheduled for inspection and that require a similar crew with similar tools and equipment.
- Determine special access requirements, including equipment placement locations, traffic control, and railroad coordination.

Where appropriate, consult with the local highway maintenance superintendent or foreman regarding the bridges to be inspected. This person observes the bridges at all times of the year under a variety of conditions, and may point out peculiarities that may not be apparent at the time of the investigation. Stream action during periods of high water and position of expansion joints at times of very high and low ambient temperatures are examples of conditions observed by local maintenance personnel that may not be seen by the inspector.

4.3.3.2—Scheduling

As practicable, schedule bridge inspections during periods of the year offering conditions most suitable to promote efficiency and thorough inspections. Substructures of bridges over streams or rivers can best be inspected at times of low water. Inspect structures requiring climbing using rope access techniques when high winds or temperature extremes are not prevalent. Inspect the bearings, joints, or other elements during temperature extremes when trouble from thermal movement is suspected.

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To minimize risks to inspector health while maximizing inspection productivity, schedule inspections to avoid temperature extremes. Where environmental risks to inspector/public health and safety can be demonstrated to be significant, a variance can be requested from FHWA for a one-time reassignment of the date of next inspection for a bridge or group of bridges. Document the variance request with sufficient detail to warrant the request being approved.

4.3.3.3—Equipment

Bridge inspection equipment consists of those items used to provide access to the bridge component(s) being inspected, as well as those items used to perform actual inspection tasks. Include in the bridge record all equipment requirements necessary to perform the inspection.

4.3.3.3.1—Access Methods and Equipment

The variation in types of structures to be inspected requires that bridge inspectors gain access to structural elements to perform an inspection using a broad range of techniques and equipment. The methods and equipment used to access bridge members include:

- Ladders,
- Mechanical lift vehicles,
- Under-bridge crane vehicles,
- Power lift staging,
- Rigging and scaffolds,
- Boats,
- Rope access climbing, and
- Underwater diving equipment.

In selecting the use of access equipment, consider the following items:

- The degree of proximity to bridge components necessary to successfully accomplish the inspection, and the difficulty in accessing those components. Select the best equipment/method to accomplish the task at hand.
- The ability of the soil, pavement, and/or bridge structures to safely support the access equipment.
- The need for traffic control or lane closure, depending on the type and location of the equipment, as well as the impact to the motoring public. Use the MUTCD and/or state and local requirements as a guide in planning such measures.

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- The need for special care to prevent accidents if utilities are present.
- The need for permits, flagmen, and other special considerations for bridges over railroads.

Use experienced, trained personnel to plan and execute inspections involving specialized access equipment.

4.3.3.3.2—Inspection Methods and Equipment

The inspection methods and equipment to be employed will depend on the type of inspection as described in Article 4.2.3, information gained from a review of previous inspection reports and bridge plans, and any information regarding specific needs or requirements expressed by the Bridge Owner. Ensure that basic inspection equipment, including personal protective equipment, cameras, and hand tools for cleaning, hammer sounding, and measuring, are readily available for all inspection work. Assemble additional equipment, such as waders and nondestructive testing supplies, as necessary to facilitate an efficient, thorough inspection. In planning the inspection, a pre-inspection site visit by the team leader may be helpful.

If plans are available, the pre-inspection should be done plans-in-hand, to allow preliminary verification of structure configuration and details. The pre-inspection determines the means of access, and discloses areas of potential concern that will require close attention and/or specialized equipment during subsequent inspections. Additionally, it forms the basis for decisions on timing, weather conditions, traffic controls, railroad or agency coordination, and utility de-energizing.

4.3.4—Inspection Forms and Reports

4.3.4.1—Field Notebooks

Prepare inspection forms and reports for field use in a systematic manner that contains sketches, relevant photographs, and room for notes. For routine inspection of smaller bridges, these materials may be assembled within a simple file for each bridge. For larger or more complex bridges, this information is often assembled in an inspection notebook or binder, providing the inspector with quick access to information in a logical sequence.

Document information and data collected in the field in a clear and detailed fashion. Ensure that notes and sketches can be fully interpreted at a later date. Take photographs in the field to illustrate defects or provide references. Clearly annotate with captions in the field notes.

Ensure that all inspectors produce uniform documentation, easily understood by all inspection

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teams and office personnel; and use standardized abbreviations, legends, and systematic numbering of bridge components. Use consistent nomenclature to describe bridge components.

Use sketches and photographs to supplement the written notes concerning the location and physical characteristics of deficiencies. Use photographs and sketches to define the areas and extent of deterioration. Simple elevation and section sketches of deteriorated members permit dimensioning of defects without resorting to lengthy written notes.

4.3.4.2—Reports

Bridge inspection reports document the findings of a field inspection. Include in the inspection report a description of the condition of the bridge components inspected, identification of any notable deficiencies, and recommendations for follow up as necessary. Ensure that reports accurately note location and extent of all distress and deterioration so that future inspectors can readily make a comparison of conditions. Ensure that bridge inspection reports include the source(s) of all information contained in the report, including the names of the inspectors and the date of the inspection, as well as any other sources of data.

Ensure that inspectors keep in mind when assembling a report that funds may be allocated or repairs designed based on the quality and accuracy of the information presented. Bridge inspection data is also used in determining the safe load capacity of a bridge, which ties to posting levels and permits. Reports are also a legal record, which may form an important element in some future litigation.

Write reports in a manner so as to be easily understood by all stakeholders in the inspection process, from bridge engineers to maintenance staff. Avoid the use of language that is confusing or not readily understood.

Write recommendations for repair and maintenance in a manner to aid in prioritizing follow-up actions.

4.3.5—Inspection Techniques

Defects found throughout the structure require a thorough investigation to determine and evaluate their cause. The cause of most defects will be readily evident; however, it may take considerable time and effort to determine the cause of some defects and to fully assess their seriousness.

One very important aspect of the inspection deals with collaboration. The bridge inspector is constantly faced with evaluating damage and deterioration relating to structural, geotechnical, hydraulic, and roadway engineering issues. Utilize technical resources and assistance as necessary, and

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collaborate with colleagues and other engineering professionals when faced with a difficult inspection question or challenge.

Field inspections include, and perhaps even begin with, a visualization of overall bridge alignment, noting any signs of settlement. A sighting conducted along the bridge rails or parapets can quickly reveal signs of bridge settlement or misalignment that may pose a serious safety concern.

If possible, observe bridges during passage of heavy loads to determine if there is any excessive noise, vibration, or deflection. If detected, further investigation should be made to determine the cause. Careful measurement of line, grade, and length may be required for this evaluation. The seriousness of the condition can then be appraised and corrective action taken as required.

Identify possible fire hazards at the bridge, including accumulation of debris such as drift, weeds, brush, and garbage. Report the storage of combustible material under or near a bridge, in control houses on movable bridges, or in storage sheds in the vicinity of the bridge. Exercise care when inspecting bridges being used as shelter by transients.

In addition to the inspection techniques presented in the BIRM, utilize the inspection techniques described in Articles 4.3.5.1 through 4.3.5.13 below to perform the inspection. Utilize special consideration beyond what is presented here as necessary if unusual or unique bridges or portions are encountered. Document and define these considerations in the inspection plan for the bridge.

4.3.5.1—Field Measurements

Make field measurements to provide baseline data on existing bridge components; to track changes which may occur over time, such as crack width and length; and to provide data for use in designing repairs or rehabilitation measures.

Measurements may be required on bridges for which no plans are available or to verify data shown on plans. Make measurements with sufficient precision to serve the purpose for which they are intended.

The limits of accuracy in Table 4.3.5.1-1 are generally ample for field measurement of overall member dimensions.

Table 4.3.5.1-1—Field Measurement Limits of Accuracy

Timber Members	Nearest $^1/_4$ in.
Concrete Members	Nearest $^1/_2$ in.
Asphalt Surfacing	Nearest $^1/_2$ in.
Steel Shapes	Accuracy necessary to identify the section
Span Lengths	Nearest 0.1 ft

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When plans are available for a bridge to be load rated, dimensions and member types and sizes will normally be taken from the plans. However, many of the plans for older structures are not as-built plans, nor do they reflect all changes made to the bridge. Check during field inspections to ensure that the plans truly represent the structure before they are used in structural calculations. Give special attention to checking for possible changes in dead load, such as a change in the type of decking, additional overlays, new utilities, or some combination thereof.

Element-level inspection requires quantification of bridge elements, typically in terms of total number, total linear length, or total surface area, based on the specific element quantification scheme. Element quantities can often be determined from bridge plans; however, field verify quantities (as well as accuracy of the elements themselves selected) as necessary to ensure accuracy.

Make and record measurements sufficient to track changes in expansion joint opening, crack size, or rocker-bearing position. Make measurements to monitor suspected or observed structure misalignment. In these cases, the inspector records the information in field notes to serve as a datum for future readings. Keep a log of the readings in the inspection file and update with the readings after each inspection cycle. Permanent markings may be made directly on the structural components to aid in the future identification and tracking of such deficiencies.

Include the field measurement of spalls with length, width, and depth. Document the size and location of erosion or scour voids.

When assessing damage or deterioration to steel members, exposed reinforcement, or prestressing, identify the remaining section of steel (as well as the original section, where necessary) rather than a visual estimation of percentage section loss.

4.3.5.2—Cleaning

It is good inspection practice to clean selected areas to allow close inspection for corrosion, damage, deterioration, or other hidden defects. Debris, vegetation, marine growth, and other items can accumulate and obscure problem areas.

On steel structures, particularly on fracture-critical members, it may be necessary to remove cracked or peeling paint for proper inspection. Remove coatings adjacent to suspected or known fatigue cracking to ensure that it is indeed the base metal, and not simply the coating, which is cracked. Steel components exhibiting impacted crevice corrosion (pack rust) or corrosive delamination require chipping with a hammer or other means to remove corrosion down to the base metal in order to measure the remaining section of sound metal. Make provisions to recoat such areas exposed during the

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inspection, which are critical to the structural integrity of the bridge.

When dealing with impacted crevice corrosion, it is inadvisable to estimate corrosion depth from the thickness of corrosion bloom for many reasons. The corrosion thickness varies with environmental conditions and the existing corrosion. At the time of inspection there could be new deterioration on top of a previously deteriorated and cleaned area. Rather, the thickness of sound, remaining section under the pack rust should be measured and recorded.

On concrete structures, efflorescence and debris may cover heavily corroded reinforcing, cracks, or other deterioration. Vegetation and debris on piles can obscure defects such as cracks or spalls.

Loose, delaminated concrete in areas of spalling and incipient spalling can hide underlying, corroded reinforcement. As such, carefully remove representative areas of such delaminated concrete, and determine the condition of underlying concrete and reinforcing steel.

Timber structures are particularly susceptible to termites and decay in areas where debris causes a wet/dry condition. Give particular attention to cleaning and carefully inspect such areas, especially when they are present near end grain.

4.3.5.3—Condition Rating Inspection of Bridges

Develop and implement standardized practices for condition rating inspections of bridges, following procedures identified in the *Recording and Coding Guide for the Structure Inventory and Appraisal of the Nation's Bridges* (FHWA, 1995). The Coding Guide provides guidance for the uniform recording of numerical condition ratings for the bridge deck, superstructure, and substructure, as well as channel and culvert, as they may exist.

In addition to these overall condition ratings, Bridge Owners may record condition ratings for individual bridge components. For example, individual condition ratings for the bearings, beams or girders, diaphragms, paint system, or other elements may be recorded in addition to the overall condition rating for NBI item 59.

Rather than selecting condition ratings manually, Bridge Owners may have the ability to automatically generate condition rating data directly from element level inspection data by means of data translator software.

Ascertain and follow the procedures for condition rating inspection of bridges.

4.3.5.4—Critical Inspection Findings

Promptly bring to the attention of the Bridge Owner or responsible agency any critical structural

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and safety-related deficiencies found during the field inspection or evaluation of a bridge. Implement standard procedures for addressing critical inspection findings, including:

- Rapid evaluation of the deficiencies found.
- Immediate public safety response procedures, including emergency traffic control and information reporting steps.
- Notification of public safety officials, including the Bridge Owner, FHWA officials, and the public.
- Rapid implementation of corrective or protective actions, including bridge closure or weight restriction posting as necessary.
- A tracking system to ensure adequate follow-up actions and reporting.
- Provisions for identifying other bridges with similar structural details for follow-up inspections.

Develop CIF procedures, working in cooperation with the FHWA Division Office.

4.3.5.5—Decks

This Article includes discussion covering the inspection of decks constructed of reinforced concrete, prestressed concrete, steel, and timber. In addition to the information presented here, consult the BIRM for comprehensive deck inspection techniques.

Many decks were designed to act in a composite manner under live load with the supporting superstructure members. Check to see that composite decks are acting as intended by the designer. Movement between the bottom of the deck and top flange of supporting members or the loss of camber may be indicative of a reduction in composite action.

4.3.5.5.1—Concrete Decks

Check concrete decks for cracking, leaching, scaling, potholing, spalling, and other evidence of deterioration, as indicated in the BIRM. Evaluate each deficiency to determine its effect on the structure, as well as the need to restore any loss of structural integrity and maintain a smooth riding surface. Evidence of deterioration in the reinforcing steel is to be examined closely to determine its extent. Decks treated with deicing salts or which are located in a salt air environment are likely to be affected.

The extent of spalling and delamination can be determined by tapping lightly with a hammer or by dragging a chain across the deck in the vicinity of

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the spall. A hollow sound indicates a separation or fracture plane in the concrete beneath the surface. Map and record the hollow sounding areas. These and other nondestructive field test methods are discussed in Section 5.

Always examine the underside of the deck slab for indications of deterioration or distress. Report and document any loose concrete that is a safety hazard that could fall and harm an individual under the bridge. Develop procedures for removing incipient falling debris in a timely manner.

Note any evidence of water passing through cracks in the slab. When permanent stay-in-place forms have been used in construction of the deck, the inspector may recommend that some panels at random locations be removed to check the condition of the slab. The inspector should pay special attention to low lying elevations of the deck as moisture and chlorides typically concentrate in these locations.

Asphaltic or other types of wearing surfaces on a deck may hide deterioration in the deck until it is advanced. Examine the surfacing very carefully for evidence of deterioration in the deck or the wearing surface. Such defects may show as cracking or breaking up of the surfacing. In areas where deck deterioration is suspected, the inspector may recommend the removal of small sections of the wearing surface for a more thorough investigation. In addition to this destructive evaluation, alternative technologies are available for the evaluation of these covered decks, including Ground Penetrating Radar and infrared scanning. Many Bridge Owners report positive success in using these methods to detect damage or deterioration in such covered decks.

Examine concrete decks for rutting and wear that may result in reduced skid resistance. Concrete containing certain varieties of limestone aggregate is especially susceptible to wear and the polishing action of tires. Request and perform skid resistance tests as necessary to determine the need for remedial action to restore the surface skid resistance.

4.3.5.5.2—Prestressed Concrete Deck Panels

This Article covers precast prestressed concrete deck slabs, with or without composite action. The slab units may or may not be covered with a wearing surface. Not included in this discussion are those precast panels used as stay-in-place forms for cast-in-place concrete decks.

As with conventionally reinforced concrete, check the surfaces of prestressed concrete deck panels for cracking, leaching, scaling, potholing, spalling, and other evidence of deterioration, as per guidance provided in the BIRM. Make notations of the location and extent of damage for comparison with previous reports, as well as for a basis of comparison with future inspections.

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Examine the ends of slab units for evidence of deterioration or failure in the anchorage zone, and also the joints between adjacent slab units for spalling and for intrusion of foreign material.

At locations where the slab units are covered by a wearing surface of asphalt concrete or other material, defects will tend to be hidden from view. This requires very close inspection for cracking, lifting, or loss of bond of the wearing surface, as well as a close inspection of the underside of the slabs.

Note evidence of cracking, spalling, water leakage through cracks, or separation at the joints between slabs during inspection of the slab underside. Examine areas where the slab units bear on the girders for cracking and spalling of concrete in the deck slabs or on the edges of concrete girders.

Examine the neoprene or fabric shims between slabs and girders for tearing, bulging, or loosening.

Check to see that nuts or bolt heads on slab anchoring bolts are tight.

Check the slab units under passage of heavy loads to see that keys or other connecting devices between adjacent slab units are functioning properly. Slab units that are transversely post-tensioned in order to act in unison may together yield a reduced deck capacity, should individual slab units be found to be working independently from one another.

4.3.5.5.3—Steel Decks

The inspector should evaluate the steel deck section and differentiate structural members from members acting only as wearing surfaces.

Check open grid decks carefully for broken tie-down welds. Document any fatigue cracking of primary, secondary, and tertiary bars found. Check for wear in the bar top surfaces in the wheel lines, which reduces skid resistance.

Closed grid decks are either filled full depth or partial depth with concrete. Check closed grid decks for the same defects as open grid decks. In addition, these decks are susceptible to corrosive deterioration of the grid elements embedded in concrete. Corrosive delamination and resulting expansion of the steel bars can fracture the tie-down welds or distort the supporting structure. Examine the concrete fill-wearing surface for spalling or scaling, which exposes the grid. Where the grid is visible, check for evidence of water ponding which can cause a traffic hazard and promote further concrete deterioration, corrosion of the grid, or both. Check the underside of the filled grid for evidence of water leakage and corrosion of grid elements.

Corrugated metal pan decks consist of a corrugated sheet metal structural element with portland cement concrete, asphalt concrete, or gravel fill, which forms the wearing surface. Check this type of deck for evidence of rust-through of the bottom corrugations, due to trapped water. This type

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of deck is usually attached to the stringers with plug welds, which are not directly observable. Vertical movement of the deck under the passage of live load may indicate plug weld failure. Examine the fill material of the wearing surface for cracks or rutting. Open cracks in the wearing surface will allow rust-through of the deck elements to occur at an accelerated rate.

Orthotropic steel plate decks consist of a flat steel plate with a series of stiffening web elements. A wearing surface is bonded to the top of the steel plate. On some structures, the steel plate is itself a flange element of a box girder section. Check for de-bonding of the overlay, rust-through or cracks in the steel plate, and for the development of fatigue cracks in the web elements or connecting welds. Check the connection between the orthotropic plate deck and supporting members where visible, and note any evidence of live load movement.

4.3.5.5.4—Timber Decks

Examine timber decks for decay and evidence of localized distress, especially at their contact surfaces where they bear on underlying stringers, and between layers of planking or laminated members. Note any looseness that may have developed from inadequate nailing or bolting, or where spikes have worked loose. Observation under passing traffic will reveal looseness or excessive deflection in the members. Consult the BIRM for specific techniques for inspecting timber decks.

4.3.5.5.5—Expansion Joints

Expansion joints are important components of a bridge. They accommodate thermal expansion of the deck and superstructure and also prohibit water leakage through the joint openings. They have a limited design life and are thus often found to exhibit significant deterioration, allowing water leakage through the joint openings, which, in turn, can cause significant damage to underlying bridge components. The amount of damage found in bridges caused by water transmission through failed expansion joints is significant.

Check expansion joints for freedom of expansion. The clear opening of the joint should provide for adequate expansion of the adjacent superstructure elements, considering the span lengths and temperature at the time of inspection. The inspector should measure expansion joint openings and ambient temperature at easily identifiable locations so that a record of joint movement over time can be established. Inspect for solid objects (non-compressible), which can become wedged in the joint and prevent joint contraction.

On joints without armoring, inspect for proper joint alignment, the presence and condition of any

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joint sealant material, and for evidence of spalls or "D" cracking in the slab edges, which would prevent proper sealing of the joint.

Inspect armored joints without sealant material, such as sliding plate dams or finger joints, both above and below deck, evaluating the condition of the supports. Record any horizontal or vertical misalignment of the joint elements and check them again at future inspections. Where drainage troughs are provided, check for a build-up of debris that prevents proper drainage and causes spillover onto the superstructure and substructure components or impedes joint movement. Also evaluate the condition and effectiveness of the troughs themselves.

Sealed armored joints such as strip seals or compression seals should be checked for deterioration, including tears, separations, sagging, protrusions, or embedment of foreign material. Ultraviolet degradation of the seal material is evidenced by hardening and brittleness of the surface and by the appearance of pattern cracking. Check the underside of all sealed deck joints for evidence of active joint leakage, shown by water staining of the underlying structural elements. Clearly mark areas of water staining on drawings or in the field notes so that joint repairs or seal replacement can take place.

Reinforced elastomeric joints are composed of various combinations of steel supports and sealant material. Inspect for missing anchor bolt covers, separation of joint elements, and audible or visual evidence of loose joint panels under traffic. Repair loose joint panels because the bolt failure is progressive and may result in one of the joint panels breaking loose under traffic.

Elastomeric plug joints utilize an elastomeric concrete "plug" as the seal, which is pliant enough to accommodate minor thermal movements of the deck. While simple in concept, the elastomeric plug should be evaluated for de-bonding from adjacent deck and approach roadway elements, as well as tearing, delamination, or degradation of the plug material itself.

Modular joints are composed of single or multiple support systems working together to accommodate large bridge movements. Inspect for surface damage to seals and separation beams. Examine the underside for evidence of leakage and also for unusual noise, which may indicate fractured welds or bolts.

Consult the BIRM for additional inspection techniques for expansion joints.

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4.3.5.5.6—Railings, Sidewalks, and Curbs

Evaluate bridge railing and parapets for structural condition and as to adequacy of geometry and structural capacity. Be familiar with the railing requirements of the Bridge Owner, as well as NBI item 36, as presented in the Coding Guide.

Determine and document whether the structural elements, especially fracture-critical members such as eyebars and hangers on through-truss bridges are separated from traffic by an adequate vehicular railing system to prevent vehicle impact from causing major structural damage and to protect the vehicle.

Inspect reinforced concrete parapets and curb-line barriers for evidence of impact damage or rotation. Record areas of collision damage or movement. On precast parapet elements, check for evidence of anchorage failure. Sound exposed anchor bolts with a hammer. Check for separations of the base of the precast element from the deck, or evidence of active water leakage between the parapet and the deck.

Inspect post and beam railing systems for collision damage and deterioration of the various elements, along with post bases for any loss of anchorage. The exposed side of vehicular railing should be smooth and continuous.

Inspect sidewalk areas for structural defects and pedestrian safety items such as tripping hazards, ponding of water or ice, and the walking surface, which should not be slippery in wet weather**.** Tripping hazards constitute a significant legal issue for many Bridge Owners, and thus should be communicated and dealt with promptly.

Examine the type, condition, and alignment of the curbs. Check curbs also to see that they are properly anchored.

4.3.5.5.7—Drainage

Properly functioning drainage systems are very important to the health of the bridge. Examine the bridge drainage system for adequacy and functionality as well as condition.

Check that gratings over the scupper or drains are intact. Promptly report broken or missing grates that are a traffic hazard. Report and document clogged scuppers and downspouts.

Note drainage through open joints, cracks, or spalls in the curbs or parapets, or other unintended routes.

Check that the bridge drainage travels through the down spouting and is adequately terminated in drainage facilities or splash blocks. Record any areas of erosion or undermining caused by downspout outfalls. Water ponding on the bridge deck due to clogged scuppers can accelerate freeze-thaw deterioration of the deck and poses a hazard to the

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traveling public. Establish a procedure for reporting and clearing clogged bridge drainage systems.

4.3.5.5.8—Lighting

Inspect lighting standards and supports for proper anchorage and fatigue damage. Report any missing or broken luminaires, exposed wiring, or missing junction box covers.

4.3.5.5.9—Deck Overlays

Assess the condition of the deck overlay using techniques presented in the BIRM. Carefully examine the condition of the overlay at the curblines, joints, and scuppers. Report the extent of surface deterioration as well as the overlay thickness.

4.3.5.6—Superstructure

This Article includes discussion covering inspection of superstructure components composed of reinforced concrete, structural steel, or timber, including bearings, connection devices, and protective coatings. In addition to the information presented here, consult the BIRM for comprehensive superstructure inspection techniques.

Check girders over a traveled way for any damage resulting from being struck by overheight loads passing under the bridge. Some Bridge Owners incorporate the remeasurement of minimum vertical underclearance (MVU) as a standard part of their routine inspection practice using data collection equipment such as hand-held laser distance measurement units to quickly collect MVU data during the inspection.

If feasible, note any excessive vibration or deflection as truckloads move across the superstructure.

4.3.5.6.1—Steel Beams, Girders, and Box Sections

Evaluate steel beams, girders, and box sections as to whether or not they are fracture-critical members (FCM) or contain fatigue-prone details, as defined in the *AASHTO LRFD Bridge Design Specifications*. More information on fatigue-prone details and FCMs may be found in Articles 4.2.3.4, 4.3.7, and 4.3.7.1, respectively. Document in the bridge record in a complete listing of all FCMs and the type and location of various fatigue-prone details found on the structure.

Inspect structural steel members for loss of section due to corrosion. Where a build-up of rust scale is present, a visual observation is usually not sufficient to evaluate section loss. Hand scrape areas

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of rust scale to base metal and measure the remaining section using calipers, ultrasonic thickness meters, or other appropriate method. Take sufficient measurements to allow the evaluation of the effect of the losses on member capacity.

Check members for out-of-plane bending in webs or connection plates and compression flanges for buckling.

Check the tension zone of members for cracking near erection or tack welds and at other fatigue-prone details.

Enter and inspect box members from within where accessible. Check enclosed members for water intrusion. Access points to enclosed box members should be closed or screened to prevent entry of birds, rodents, and other animals. Check for collection of debris, bird/animal waste, and other deleterious materials.

Check for fatigue cracks, which typically begin near weld terminations of stiffeners and gusset plates due to secondary stresses or out-of-plane bending. Carefully document any evidence of cracking for evaluation and appropriate follow-up, as necessary. Documentation on cracks should include their location, length, and width, as well as any changes in crack size since the previous inspection, as applicable. Written documentation should be supplemented with photographs and/or sketches to provide additional detail.

On FCMs, perform periodic inspections at a level of effort sufficient to detect very small cracks. Nondestructive testing equipment, including ultrasonic testing, magnetic particle testing, eddy current testing, and dye penetrant testing, are commonly used by Bridge Owners as an integral part of their FCM inspection process.

When evaluating suspected fatigue damage, remove coatings in the affected areas to determine if the damage is to the coating or in underlying metal. Determine coating composition prior to the start of such work. Only trained personnel using appropriate methods and equipment should remove coatings containing heavy metals.

Inspect uncoated weathering steel structures for:

- Details or conditions that promote continuous wetting of the uncoated steel.
- Bridge geometrics that result in salt spray (marine or traffic generated) reaching the uncoated steel.
- Pitting of the surface of the steel indicating unacceptable degradation of the steel.

4.3.5.6.2—Reinforced Concrete Beams, Girders, and Box Sections

All reinforced concrete superstructures should be inspected for cracking. Carefully note the locations of the cracks and their sizes for future

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reference and comparison. An effort should be made to determine the probable cause of the cracking, such as shrinkage, overstress, settlement of substructure, or possible chemical action.

Check stems of members for abnormal cracking and any disintegration of the concrete, especially over bearings. Diagonal cracks radiating from the bearings toward the center of span indicate stress caused by shear. Vertical cracks extending upward from the girder soffit near centerline of span indicate stress in tension. High edge pressure at the bearings may cause spalling in the adjacent areas of the girders.

Inspect the soffit of the lower slab in box girder structures as well as the outside face of the girders for significant cracking. Note any offset at hinges that might indicate problems with hinge bearing. An abnormal offset may require further exploration to determine the cause and severity of the condition. Examine the inside of box girders for cracks and to see that the drains are open and functioning properly. Note that concrete box girders may have timber falsework left in place from the construction process, which may significantly hinder internal inspection of the girder cells. Check the diaphragms for cracks.

If there are earthquake restrainer mechanisms at abutments, bents, or hinges, the inspection should cover close examination of these elements for damage due to corrosion or stress. Note vertical, lateral, and longitudinal movements relative to the substructure.

4.3.5.6.3—Prestressed Concrete Beams, Girders, and Box Sections

Prestressed concrete girders should be examined for alignment, cracking, and deterioration of the concrete, per the BIRM. Check for cracking or spalling in the area around the bearings, and at cast-in-place diaphragms where creep and camber of the girders may have had an effect. Carefully note the location of any cracks and their sizes for future reference and comparison. Evidence of rust staining at cracks can mean possible damage to prestressing steel, and should warrant additional investigation.

Check pretensioned box sections during the passage of heavy loads to see whether any unit is acting independently of the others. Such independent action could indicate spreading of the girders or failure of the longitudinal key between girders.

Sound the concrete cover over prestressed strands and note all delamination of concrete. Delamination may indicate prestressed strands have begun to corrode. Since corroded strands expand from their original size, the concrete cover delaminates. On bridges with underpassing traffic, the exterior faces and the soffits of all types of

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prestressed girders should be examined. Note and evaluate spalling, cracking, or damage to prestressing steel.

Inspections of earthquake restrainer mechanisms and for earthquake damage should be conducted as outlined in Article 4.3.5.6.2.

4.3.5.6.4—Timber Systems

Examine timber stringers for splitting, cracking, checking, and excessive deflection. Evaluate stringers for vertical/diagonal cracking, indicative of potential overstress. Look for crushing and evidence of decay where they bear on the bent caps or abutment seats, and at their top edge where the deck is supported. Stringers should be kept clear of dirt accumulations to help prevent decay from starting and to help prevent its acceleration once it has started.

In order to evaluate the capacity of existing timber structures:

- Record the beam size, spacing, and span length.
- Record the type of beam: rough-sawn, dressed, nail-laminated, or glue-laminated.
- Note beams that have been cut or notched at the bearing areas, and to what extent. Horizontal shear capacity is controlled by beam depth.
- Estimate the age of the timber.
- Estimate or measure the moisture content of the timber.
- Identify the species and grade of the lumber. Check if original and repair construction records for material delivery slips are available. Where no information is available, the inspector must use judgment based on factors including local experience, visual appearance, odor, and cross grain. Where more exact information is required, obtain a sample for testing by a laboratory.
- Check the bridging between the timber stringers to see that it is tight and functioning properly.
- Check timber connections for loose or missing fasteners.
- Consider borings to determine the integrity of larger members and log stringers, as timber superstructure members can decay from the inside out without any noticeable exterior indications of decay.

4.3.5.6.5—Floor Systems

Truss and deck girder structures are constructed with a system of stringers, floor beams, and if present, brackets to transmit the live load from the

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deck to the main load-carrying members (girders or trusses). Dependent upon the Bridge Owner's policy, transverse floor beams, brackets, or both can be fracture-critical members depending on the framing used. Clearly indicate in the bridge record whether or not the floor system contains FCMs.

Inspect stringers, floor beams, and connections for cracks and section loss due to corrosion. Floor beams and connections often exhibit significant corrosion due to leakage through the deck joints.

4.3.5.6.6—Trusses

The examination of a truss will normally begin with sighting along the roadway rail or curb and along the truss chord members to determine any misalignment, either vertical or horizontal. Check alignment of trusses carefully for any sag that may indicate partial failure in joints or improper adjustments of the steel verticals or counters. Investigate any deviation from the normal alignment fully to determine its cause. Check each of the truss members and their connections.

Examine steel compression members to see if they are straight with no kinks or bows. Also, check compression members to see that their connections are intact. Eccentricity in the connecting details has a great influence on the strength of the member and therefore warrants a close check.

Identify steel tension members in trusses as to whether or not they are fracture-critical members. Inspect all fracture-critical members closely in accordance with the provisions of Articles 4.2.5.5 and 4.3.7.

When a tension member consists of more than one component, each component should be checked to see that it is being properly loaded. Check counter members to see that they are in proper adjustment and not carelessly tightened, thus throwing abnormal stresses into the counters or other members. Carefully check looped rod tension members found in old trusses for abnormal cracking where the loop is formed and examine eyebar members for cracks in the eyes.

Inspect gusset plate member connections closely according to the provisions of Article 4.3.5.6. Confirm that gusset plate dimensions and connection details match those shown in the bridge record plans and shop drawings. Record any differences found, and all gusset plate dimensions and connection information if record plans or shop drawings are not available. Field inspections of gusset plates need to focus on corrosion, distortion, and connections. Section losses can occur along gusset plate areas that trap debris or hold water, usually along the top of the bottom chord. Distortion in the gusset plate can be from original construction or can be caused by overstressing of the plate due to overloads, inadequate thickness/bracing, forces

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associated with pack rust between plates, or traffic impact.

Visual inspections may not detect or accurately quantify corrosion on gusset plates. Use of ultrasonic thickness gauges and calipers are recommended for determining any reduced thickness due to section loss. Out-of-plane distortion can be determined by the use of a straight edge. Check for individual broken or loose rivets or bolts. Inspect for slipped surfaces around the individual bolts and rivets and for any cracking in the gusset plate at bolt and rivet holes.

Evaluate truss and bracing members for traffic damage. Portal bracing usually is the most restrictive overhead clearance and consequently is most susceptible to damage from over-height vehicles.

Check all upper and lower lateral-bracing members for damage and observe if they are properly adjusted and functioning satisfactorily. In old bridges, an appraisal of the lateral and sway bracing should be made to determine its adequacy. This appraisal will normally be a judgment of the Engineer based on observation of transverse vibration or movement of the structure under traffic.

Check the conditions of the pins at the connections and verify that the nuts and keys are in place. Also, verify that spacers on the pins are holding eyebars and looped rods in their proper position. Nondestructive examination of the pins via ultrasonic inspection may be warranted to determine notching (wear) or cracking of the pin.

Check rivets and bolts to verify that none are loose, worn, or sheared.

Examine all timber members for checks, splits, and decay. Decay is most often found at the joints where there are contact surfaces, daps in the timbers where moisture can enter, and around holes through which truss rod bolts are fitted. End panel joints are likely areas for decay because of the dirt and debris, which tends to accumulate on the bridge seat.

Check for any evidence of crushing at the ends of compression chords and diagonal members.

Check all splice points for soundness in the connections. All bolts should be checked to see that they are tight and in good condition.

Investigate roofs and sides of covered bridges for adequacy of protecting the structural members from the elements.

Report any fire hazards that exist and need correction to safeguard the structure.

4.3.5.6.7—Cables

Inspect wire rope cables for breakage, fraying, and surface pitting and cable terminations to detect fretting fatigue due to flexure.

Inspect saddles, socket assemblies, and connections for cracking and evidence of internal corrosion. Where severe surface deterioration or

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wire breakage is present, a more detailed inspection of the cable, such as spreading with wedges or nondestructive testing techniques, is required to determine the extent of loss.

Long runs of cable should be observed for excessive vibration due to the passage of trucks or wind. Give special attention to cable in the vicinity of saddles and at low points.

Closely examine cable hangers for cracked wires at the socket attachment.

Enter cable anchorages and examine the wire terminations for loss of section and the presence of moisture.

4.3.5.6.8—Diaphragms and Cross-Frames

Check the condition of diaphragms and cross-frames on steel multi-girder bridges, particularly at the points of attachment to the main structural elements.

Welded attachments and gusset plates in the tensile zones of girders are fatigue sensitive and may induce out-of-plane bending in girder webs. Check for cracking or distortion in the diaphragms/cross-frames and the girder web.

Check riveted or bolted connection points for evidence of prying and soundness of the fasteners.

Evaluate section loss in the members due to corrosion.

4.3.5.6.9—Lateral Bracing, Portals, and Sway Frames

Check lateral bracing and sway frame connection plates for fatigue cracking due to wind or live load induced vibrations.

Remove build-up of debris at gussets to examine for loss of section.

Note any lateral brace or sway frame that vibrates excessively due to wind or live load passage.

Examine truss portal members for collision damage or misalignment. Measure the vertical clearance to knee braces or other portal connections and record the actual minimum clearance.

4.3.5.6.10—Rivets, Bolts, and Welded Connections

Connections between structural members are either welded or mechanically fastened using rivets or bolts. Bolted connections are either designed to act in bearing (load transferred through the bolts) or in friction where the bolts clamp the joined pieces together, relying on friction to transfer the load. The inspector should be familiar with the types of connections present on each bridge. The details of these connections are normally part of the bridge record.

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Check friction type, high-strength bolted connections to verify that all their bolts are fully tightened. As necessary, where direct tension indicator (DTI) washers are used, check for correct tension using appropriate thickness gauges. Look for signs of rubbing or broken paint or corrosion around the bolts. For example, the presence of primer dust and fretting corrosion stains near the connection is an indication of abrasion caused by slipping of the joint. Sound suspect bolt heads with a hammer for audible sounds of distress and observe any movement of the bolts when struck. This is especially true on stringer/floor beam connections; rivets with rust dribbling from behind a rivet head usually mean that the shank is broken and the head will break off when struck with a hammer. Document the number and locations of loose bolts.

Check riveted and bearing type high-strength bolted connections in shear for condition and loose elements and record severe loss to the heads of rivets.

Hammer sound rivets and bolts which act in tension for the presence of distress or movement. Report missing or unsound rivets or bolts in such a connection and make follow-up repairs to avoid the possibility of a progressive failure of the connection.

Check welded connections for the development of fatigue cracking, which occurs most commonly at weld terminations and returns. Examine the weld for fine cracks, which frequently exhibit rust staining. Where such areas are visually detected, microscopic or nondestructive tests can be performed to confirm and define the cracks present (see Section 5). Fracture-critical members must receive prompt attention when weld cracks are detected.

For guidance on inspection of truss gusset plate connections, see Article 4.3.5.6.6.

4.3.5.6.11—Pins and Hangers

Pin and hanger assemblies are frequently used to allow thermal movement of adjacent spans. This movement is provided for by longitudinal translation of the upper pin past the lower pin, causing rotation of the hanger. See Figure 4.3.5.6.11-1 illustrating the many parts that make up one type of pin and hanger assembly.

When present on trusses or two-girder systems, a pin and hanger assembly is an FCM. On multi-girder systems, the hanger may not be an FCM if sufficient cross-framing is present to redistribute the load to adjacent members without causing progressive failure. Bridge Owners should determine which bridges have pin and hanger assemblies that are FCMs.

Pin and hanger assemblies are usually located next to an open joint and are therefore vulnerable to corrosion. This corrosion can cause binding of the components, which places unanticipated torsional stresses on the pins and bending stresses in the

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hangers. Fatigue cracking can also develop along the entire length of the hanger assembly. Evaluate the entire pin and hanger assembly for evidence of corrosion-induced binding and inspect the hangers to detect evidence of misalignment or bowing, as well as fatigue cracking.

Some pin and hanger assemblies are built with a limited distance between the end of the pin and the hanger plate. Impacted corrosion build-up between the elements can develop enough force to move the hanger laterally to a point where the bearing area is insufficient, causing the pin to shear or the hanger fall off the pin. Cap plates may not be strong enough to restrain this movement. Check the retainer nuts or cap plates to ensure that they are adequately secured. Carefully check all welds on pin and hanger assemblies and evaluate the assemblies to detect rust build-up between the elements and evidence of lateral movement along the pin.

Pin and hanger assembly failure mechanisms include fracture of the hanger, fracture or shear in the pin, or the aforementioned movement of the hanger. The pins are frequently obscured from direct view. Check for evidence of fracture or distress such as displacement of connected elements or fretting. Where the end of the pin is exposed, such as with threaded nuts, ultrasonic testing may be used to check for cracks in the pins parallel to the tested face of the pins. On those pins that are covered by cap plates, a program should be established to routinely remove the cap plates and ultrasonically test the pins, consistent with the testing program established for pins.

Pin and hanger assemblies at fixed connections usually are provided with a restrainer or thrust plate to prevent longitudinal movement. Check that this restrainer is not subject to flexure or distortion.

Properly trained personnel should conduct ultrasonic testing of pins.

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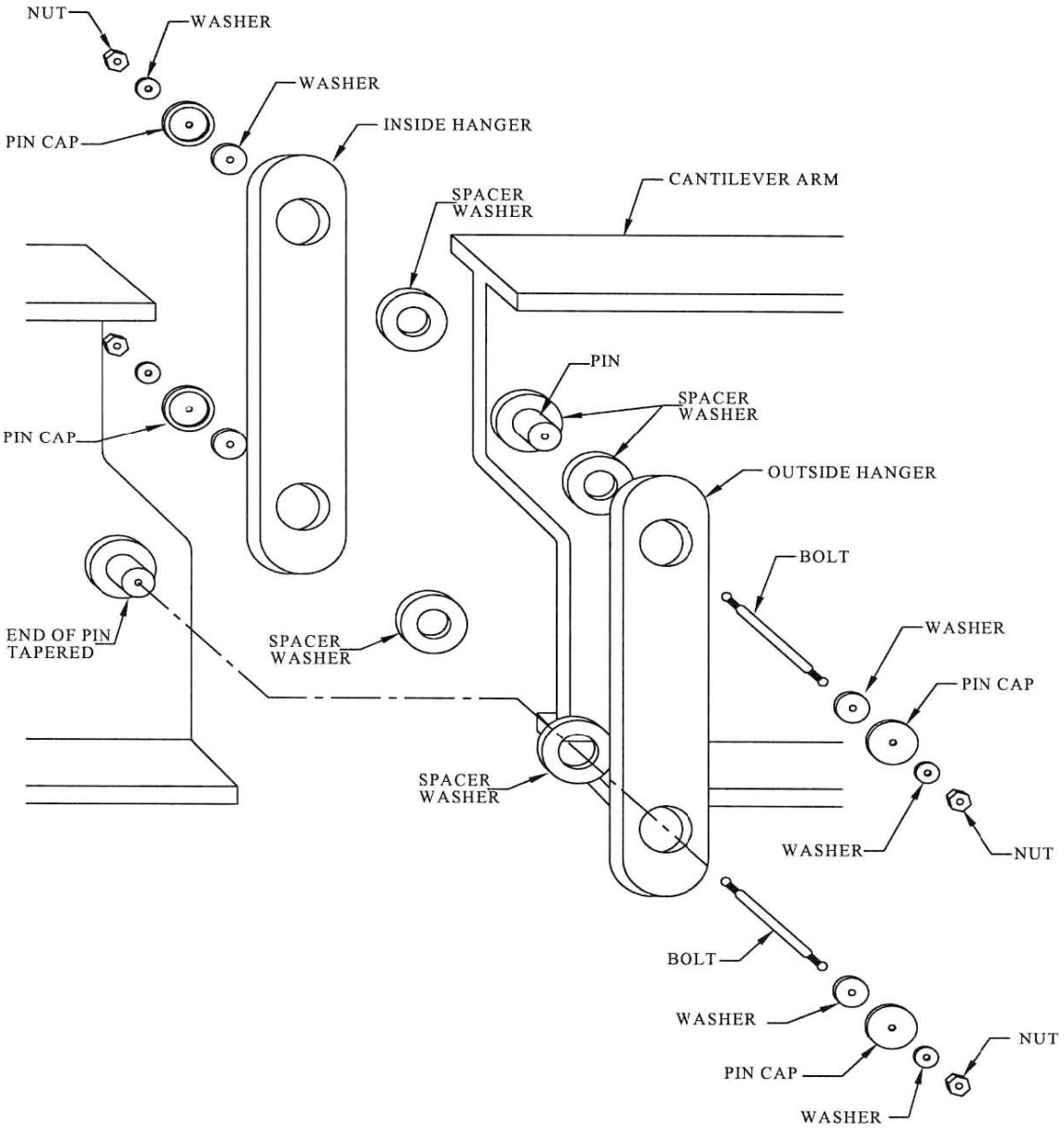

Figure 4.3.5.6.11-1—Pin and Hanger Assembly

4.3.5.6.12—Bearings

Examine bearing devices to determine that they are functioning properly. Small changes in other portions of the structure, such as pier or abutment settlement, may be reflected in the bearings.

Check bearings for post-tensioned concrete box spans for bearing loss, misalignment, or tearing of the bearing material as shortening of the structure may occur over time.

Bearings and lateral shear keys are subject to binding and damage from creep in bridges with a relatively high skew, carefully examine for any such defects.

Check expansion bearings to see that they can move freely and are clear of all foreign material.

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Rollers and rockers should bear evenly for their full length and should be in proper position relative to the temperature at the time of the inspection. Check lubricated-type bearings to see that they are being properly lubricated.

Inspect anchor bolts for corrosion, damage, and to see that nuts are secure. See that anchor bolt nuts are properly set on expansion bearings to allow normal movement. Missing anchor nuts, sheared anchor bolts, or both should be identified and reported.

Note the physical condition of the elastomeric bearings pads and any abnormal flattening, bulging, or splitting which may indicate overloading or excessive unevenness of loading. Pads should also be in the proper position relative to both the girder and abutment seat.

Sharp skewed and curved girder bridges may not have bearings that permit multi-rotation and adequate movement. In such instances, expect uneven wear of the bearing components. Check the substructure in the vicinity of such bearings for possible distress.

Examine pot, disc, and spherical bearings and note any instances of extruded or deformed elastomer, polyether urethane, or PTFE (polytetrafluorethylene); damaged seals or rings; or cracked steel.

Examine grout pads and pedestals for cracks, spalls, or deterioration.

Carefully examine bearings, keys, and earthquake restrainer mechanisms after unusual occurrences such as heavy traffic damage, earthquakes, and impact from debris during flood periods.

Examine the concrete at pier caps and abutment seats adjacent to bearings for cracks and spalls.

Evaluate bearings, if feasible, while under passage of heavy moving loads to detect rattles. The probable cause of such movement should be identified and noted.

4.3.5.6.13—Paint

Provide in the bridge file a record of the paint systems present, the dates of application, and the nature of surface preparation used prior to the last application.

Most Bridge Owners have one or more paint systems. Have a copy of the as-installed paint specification available.

Make an overall judgment as to the condition of the paint based on the condition of the majority of surfaces, not on localized areas of corrosion. The painted surfaces should be free of rust, pitting, chalking, crazing, or generalized rust staining. Report individual areas of more severe corrosion for touchup painting.

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Examine the condition of the paint and document the extent of corrosion. Check carefully around bolt and rivet heads. Truss chord and panel joint connection details are particularly susceptible to corrosion, especially where contaminants from the roadway surface such as deicing salts may be deposited on the steel. It is difficult to inspect many of the areas around connection details for condition of paint and to determine if any corrosion is beginning. These areas should not be overlooked, however, as they frequently are the spots where the corrosion will first start. Look for the deformation in riveted or bolted multi-plate sections where moisture may have entered and corroded the contact surfaces of the plates, causing them to be pushed apart.

Investigate cracks on painted surfaces, which may indicate a crack in the underlying material. This is especially true if rust staining is present.

4.3.5.6.14—Utilities

Bridges are frequently used to support utilities such as water supply, sanitary sewer, gas, electric, and telephone. Most commonly these are suspended between beams or girders below the deck. In most jurisdictions, the utility and the supports are owned, installed, and maintained by the utility company. In certain cases such as lighting circuits, the owner agency may be the same as the Bridge Owner.

Describe in the bridge record a clear description of the utilities present on the bridge, the owner of the utility, the agency responsible for maintaining the utility, the date of installation or modification of the utility encroachment, and a party to notify in case any defects are uncovered by the inspection.

Be familiar with the type of utility present and the nature of hazards that may be present during the inspection.

Utilities are frequently retrofitted on to bridges. Inspect the nature and type of the retrofitted support system for the presence of improperly welded connections, which may be fatigue sensitive or which may result in overloading secondary bridge elements.

Failures in the utilities can introduce several different types of problems:

- Structural deterioration may occur as a result of liquid carrying pipes leaking onto the superstructure or substructure elements. They may also cause a build-up of ice during cold weather periods.
- Utilities on bridges over waterways may cause restriction in the hydraulic capacity or navigational clearance of the structure.
- Leaks in gas or sewer lines can cause fire and health risks. The risk of fire or explosion in an

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enclosed area or adjacent to a major structural element should be evaluated.

- Electric short circuits can cause a danger to the inspector or the general public.
- Utilities on bridges over roadways or railroads may cause restrictions in vertical and/or horizontal clearances to the structures.

4.3.5.6.15—Arches

This Article covers steel, timber, concrete, and masonry arch bridge superstructures and long-span concrete arch culverts. Since arches are compression members, carefully note any cracking in the arch ring as indicative of improper loading or movement of supports.

Inspect elements of steel and timber arches as generally covered for steel and timber members in the BIRM, as well as Articles 4.3.5.6.1 and 4.3.5.6.4, respectively.

The concrete in the arch ring and in the elements supporting the deck is to be inspected as generally covered in the BIRM, as well as Article 4.3.5.6.2. Note any cracking, spalling, or other deterioration and compare this with previous inspection reports.

Check masonry arches or masonry-faced concrete arches for mortar cracks, vegetation, water seepage through the cracks, loose or missing stones or blocks, weathering, and spalled or split blocks and stones.

Inspect concrete arch culverts as described for concrete box culverts in Article 4.3.5.9. Pay special attention to the footing area for evidences of undermining, settlement, or outward movement; and to the soffit of the arch ring, roughly one third of the distance outward between crown and springline. Longitudinal cracks in this area of the soffit indicate shear or flexure problems.

4.3.5.7—Substructure

An inspection of the bridge substructure is generally comprised of the examination and recording of damage, deterioration, movement, erosion, and, if in water, evidence of scour. This Article covers the inspection of the substructure elements themselves, while the scour inspection is addressed in Article 4.3.5.8. In addition to the information presented here, consult the BIRM for comprehensive substructure inspection techniques.

4.3.5.7.1—Abutments

Inspect all exposed abutment and wingwall surfaces to detect damage, deterioration, and potential overstress, as per the BIRM.

The horizontal surfaces of the tops of abutments are particularly vulnerable to attack from deicing

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salts. In some areas, corrosion of reinforcing steel near the surface can result in cracking, spalling, and discoloration of the concrete.

Examine devices installed to restrain the structure against seismic events for evidence of corrosion; broken strands; missing bolts, nuts, or cable clamps; and for proper adjustment. Check for evidence of horizontal or vertical movement of the superstructure relative to the abutment.

Structural steel partially encased in concrete should be closely inspected at the concrete interface for corrosive deterioration and for movement relative to the concrete surface.

Check stone masonry for cracking in the mortar joints to see that the pointing is in good condition. Check the stone masonry for erosion, cavities, cracking, and other signs of deterioration of the stones.

Check abutments for evidence of rotation, lateral or longitudinal shifting, or settlement of foundations as compared to previous records. Such movement is usually evidenced by the opening or closing of cracks or joints, by bearings being off center or at a changed angle, or by changes in measured clearances between ends of girders and the abutment backwall. Perform evaluations after an earthquake has occurred in the vicinity.

Evaluate abutment foundations for exposure due to erosion or scour, as well as structural damage and deterioration. Inspect exposed portions of foundation piling, if any, for damage and deterioration, and document the exposed heights.

Evaluate abutment drains and weep holes for proper function. Seepage of water at cracks or joints away from the weep holes may indicate an accumulation of water in the fill behind the abutment due to malfunction of the weep holes.

If in water, the submerged portion of abutments and wingwalls should be evaluated for local scour in accordance with Article 4.3.5.8.1.

4.3.5.7.2—Retaining Walls

Inspect retaining walls using techniques presented in the BIRM. Additionally:

Examine retaining wall footings for soil settlement, as well as erosion and scour, which could lessen capacity of the wall.

Examine stone masonry for the existence and severity of cracks, as well as deterioration of the masonry and mortar.

The exposed ends of headers of concrete crib walls should be closely examined for cracks that could indicate possible future loss of the interlocking feature and failure of the wall.

Check wall faces, tops, and joints for bulging or settlement since the last inspection. Cracks in the slope behind a wall can indicate settlement of the

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foundation and rotation of the wall. Bulges in the faces of sheet pile walls or mechanically stabilized earth walls can indicate failure of individual anchors, which may result in potentially reduced capacity.

Inspect any exposed piling, whether exposed as a design feature of the wall (sheet pile and soldier pile walls) or by adverse action (scour, erosion, or settlement), as described in the applicable portions of Article 4.3.5.8.1.

If the retaining wall is adjacent to water, examine the submerged portion for scour in accordance with Article 4.3.5.8.1.

4.3.5.7.3—Piers and Bents

Examine all exposed concrete and stone masonry for the existence and severity of cracks and any deterioration of the concrete, masonry, or mortar. Areas of special vulnerability are the water line and splash zones, the ground line, and locations where the concrete is exposed to roadway drainage, including the tops of piers or bents. Examine bearing seats, grout pads, and pedestals for cracks, spalls, or other deterioration.

Check steel piers and bents for corrosion, especially at joints and splices. Cable connections, bolts, and rivets are especially vulnerable to corrosion.

Check all bents and piers for lateral movement, tilt, or settlement, particularly after periods of high water, storms, or earthquakes. Observe bent members during passage of heavy loads to determine whether movements are unusual or as expected.

Note any material deposited against a bent or pier that was not provided for in the original design, as horizontal instability could result from such loads.

Check timber piles for decay, especially in areas where they are alternately wet and dry. The most likely place for this condition to be found is at the ground line or tidal zone in coastal areas. When decay at the groundline is suspected, the soil may be removed from around the pile to a depth of approximately one foot and the timber probed or bored.

For some soil types, where there is a perched water table, decay can be one to two feet below ground while there is no evidence of decay at the ground line. Where appropriate, dig below the ground line around all timber posts, piles, or sills; or angle the timber boring device below ground to check for decay; or both. Fill holes made for testing with treated wooden plugs or silicone sealant in order to prevent subsequent decay. The necessity of such borings will vary greatly from area to area because of climatic variations, the type of wood used for piling, and the preservative treatment that has been used on the timber. Although piles may

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appear sound on the outer surface, they may contain advanced interior decay. Creosote treated piles, for example, may exhibit severe decay in the core area where the treatment has not penetrated, even though the outside surface shows no evidence of deterioration. Sounding with a hammer may reveal an unsound pile. Where decay is suspected or encountered, a three pound hammer used to sound the timber has proven to be effective in getting the proper tonal quality to indicate decay.

In addition to the above, special attention should be given to the following:

- Contact surfaces of timber when exploring for decay.
- Areas where vegetation or debris may have accumulated.
- Tops of the piles, where the cap bears.
- Areas where bracing members are fastened.
- Areas which are checked or split.

Examine timber members for decay, cracks, checking, and any evidence of overstress. Further information on the inspection of timber members is found in Article 4.3.5.6.4.

Examine steel and concrete piles throughout their length for physical damage, including in the splash zone and below the water surface, for corrosion and deterioration.

Inspect all submerged steel piles for deterioration and loss of section. Special attention should be given to exposed piles in or near salt water. Corrosion of exposed steel piles may be more active at the terminus of concrete encasements on partially encased structural steel members, at the waterline or tide affected zone, and at the mud line. Coastal streams may be brackish for several miles upstream due to tidal effects and are considered a potentially corrosive environment until confirmed otherwise.

Observe pile caps under heavy loads to detect unusual movement or any excessive deflection. Steel and timber caps should be observed for any rotational movement resulting from eccentric connections. Bracing members must be checked to see that they are adequate, sound, and securely fastened.

Evaluate pier foundations for exposure due to erosion or scour, as well as structural damage and deterioration. Exposed portions of foundation piling, if any, should be inspected for damage and deterioration, and their exposure heights should be documented. Additionally examine footings situated in above water locations mined, for exposure and potential undermining caused by soil settlement or wind erosion.

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Inspect piers and bents located in or adjacent to water for evidence of scour as described in Article 4.3.8.1.

4.3.5.7.4—Bridge Stability and Movements

Establish the baseline condition of the structure during the initial inspection. This acts as the basis for the future determination of movement.

Check for transverse movement by sighting along the top of railing, edge of deck, or along a girder. Similarly, an inspector can check for differential vertical movements by sighting along the top of railing or edge of deck. On large structures or structures on complex alignment, it may be necessary to use a level or transit to detect movement. Differential settlement between one side of a bridge and the other may also require checking with a level.

The use of a transit is suggested for checking bents, piers, and faces of abutments and retaining walls for rotational movements or tilt. A plumb bob may be used where heights are not great or where only a preliminary determination is desired.

Vertical movement in the superstructure is usually evidence of foundation settlement or rotation of the abutments or piers. Lateral or longitudinal sliding is caused by high water, ice pressure, earthquake, or other application of horizontal forces. Small, relatively equal movements should be noted, but usually are of little consequence. Investigate large or differential movements to further determine the probable cause with a view toward corrective measures being taken.

Examine bearing rockers, rollers, and hanger elements for movement or inclination that is not consistent with the temperature. Compare with notes from previous inspections to see if movements or inclinations are signs of settlement or shifting of foundations.

Inspect joints at abutments, bents, piers, and hinges. Jamming, unusually large openings, and elevation differentials on opposite sides of the joint constitute evidence of substructure movement or bearing failure.

Check abutment backwalls and ends of beams for cracking, spalling, or improper clearances. Causes could be the rotation or sliding of the abutment, or pressure from the roadway pavement against the back of the abutment.

Inspect abutments, wingwalls, and retaining walls to detect distortion, unusual cracking, or changes in joint widths or inclination. This damage could have been caused by settlement or a change in pressure against the walls. Look for cracks, slipouts, or seepage in the earth slopes in front of or behind the walls, as well as for imbalanced post-construction embankment exerting pressure against these walls.

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4.3.5.8—Scour and Waterway Inspections

Scour is the leading cause of bridge failure worldwide. Although local scour inspections may likely be conducted in conjunction with the inspection of the bridge substructure, it is fitting that bridge scour inspection methods be discussed separately, to highlight their significance to public safety. In addition to the guidance provided here, guidance contained within the BIRM as well as FHWA *Technical Advisory—Evaluating Scour at Bridges*, T5140.23 is followed during bridge scour inspections.

4.3.5.8.1—Substructure/Culvert Scour Inspection

Scour inspection is often accomplished by wading or probing methods when substructure elements are located in relatively shallow, slow moving water. Bridge sites with more aggressive hydraulic conditions warrant underwater diving inspections. Develop policies to establish the hydraulic and environmental thresholds between wading and diving inspections.

Investigate bridge abutments and piers for evidence of scour-induced exposure or undermining of footings. Typical evidence of abutment scour for spill-through abutments is an observable instability of the slope protection due to the removal of material at the toe of slope.

Give particular attention to foundations on spread footings where scour or erosion is more critical than for foundations on piles. However, be aware that scour and undermining of pile foundations can also occur. Inspect any exposed piling in undermined areas to determine the number of exposed piles and their exposure heights. The vertical support capacity of the piles normally should not be greatly affected unless the scour is excessively severe, but the horizontal stability may be in question.

Include measurements from a fixed reference point to determine the current elevation of the channel bottom during the inspection of a scour-susceptible bridge. Compare the channel elevation to the key substructure elevations in order to determine the current embedment of the support and identify any potential stability situations. Box culvert bridges are also susceptible to scour and undermining, both at their upstream and downstream ends, as well as undermining and failure of protective items such as concrete aprons and grouted riprap at the culvert ends.

When local scour is found at substructure units and culverts, record the approximate volume and location of the scour void. When undermining is found, measure the length, height, and penetration of the undermined area under the foundation. Record

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the number of exposed piles, if applicable, as well as typical and maximum piling exposure heights. Document evidence of scour-induced damage to piles and foundation, as it may exist.

When limited visibility, debris accumulations, or severe hydraulic or environmental conditions exist onsite, advanced inspection techniques such as underwater imaging sonar may be employed to evaluate local scour conditions without human intervention underwater.

When serious local scour is found, a more macroscopic view of the bridge is warranted, to ascertain any bridge settlement that may exist as a result.

When erosion has occurred on one face of the abutment only, leaving solid material on the opposite face, horizontal instability may result. Horizontal instability may also result from soil or rock fills piled against abutments or on slopes retained by wingwalls.

Evaluate riprap that has been placed as a countermeasure against pier scour for stability. Verify that the material being observed as riprap is actually riprap, as opposed to naturally occurring rock or material deposited at the pier by the stream that may not be providing adequate protection. The key to making the evaluation is the shape of the material. Angular rock is typically specified for riprap while material deposited by a stream is usually rounded.

4.3.5.8.2—Underwater Inspection

Underwater bridge inspections may be conducted as routine, in-depth, emergency, or special inspection types, and the focus may involve the inspection of substructure, scour, and/or waterway elements.

Underwater inspectors should be qualified as divers in accordance with OSHA 29 CFR Part 1910, Subpart T—Commercial Diving Operations, as well as other state regulations related to commercial diving safety. Refer to the BIRM for additional information regarding diving qualifications for underwater inspectors.

Determine and document conditions warranting an underwater inspection (such as water depth, turbidity, and current thresholds), as well as requirements for cleaning of submerged bridge elements (e.g., debris, biofouling).

Determine level of inspection intensity to be undertaken during underwater inspections (Level I, II, or III) as described in the BIRM. Consider the use of acoustical imaging technology in situations where water depth, high flow, turbidity, or other conditions could affect diver ability to determine localized conditions and scour effects. Use of this technology in the presence of submerged debris and other hazards provides safety benefits in lieu of deployment of a diver for Level I, II, or III

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inspections. Acoustical imaging technology could be a particularly effective technique for newer structures or where previous underwater inspections (Level I, II, or III) indicate no substructure element deficiencies. An appropriate mix of inspection techniques involving diving and nondestructive testing may be appropriate. States should develop policies and procedures for underwater inspection techniques; document these procedures; and include a schedule for Level I, II, and III inspections or Acoustical Imaging Technology, as appropriate.

Determine and document whether the evaluation of the waterway (as per Article 4.3.5.8.3) is to be conducted in conjunction with the underwater inspection, as opposed to the routine inspection.

For scour-critical bridges, follow the scour plan of action (POA) included in the bridge file as it pertains to underwater inspection.

4.3.5.8.3—Inspection of Waterways

Assess the adequacy of the waterway opening under the structure. When assessing the adequacy of the waterway opening, the inspector should bear in mind the potential for debris build-up during periods of high flow and the hazard posed by ice jamming under the bridge during winter and early spring periods.

Develop and revise a channel cross-section record for the structure as significant changes occur. This provides an invaluable record of the tendency toward scour, channel shifting, degradation, or aggradation. Observe evidence of materials mining in the waterway. A study of these characteristics can help predict when protection of pier and abutment footings may be required to avoid or minimize future problems.

Check existing bank protection and other protective devices, such as groins and guide banks (spur dikes), to see if they are sound and functioning properly. Determine if changes in the channel have caused the present protection to be inadequate and if it may be advisable to place more protection or to revise the existing protection.

See that the waterway is not obstructed and affords free flow of water. Obstructions such as debris or vegetation growth may contribute to scour and may present a fire hazard to the structure. Watch for sand and gravel bars deposited in the channel that may direct stream flow in such a manner as to cause harmful scour at piers and abutments.

Check areas upstream and downstream of the bridge to see if the bridge and its approaches are causing any problems or potential problems. Items to look for include possible flooding from inadequate openings at the structure, erosion of banks or levees from improper location, or skew of the piers or abutments. Upstream and downstream

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channel cross-sections may be needed in locations with shifting channels (banks eroding, channel migrating, or streambed degrading). Additionally, record evidence of overtopping of the bridge by floods.

4.3.5.8.4—Channel Protection Features—Dolphins and Fenders

Dolphins and fenders are used to protect substructure units from impacts by floating debris or maneuvering vessels. The term "dolphin" refers to a stand-alone unit placed upstream or downstream from the pier. The term "fender" refers to a protective unit or cover that is placed around the pier or abutment face and frequently attached to the substructure.

Piles used in dolphins or fenders are inspected per the BIRM, and as described in Article 4.3.5.8.3.

Inspect steel piles, frame members, fasteners, and cables for corrosion damage, particularly in the splash zone. Since both dolphins and fenders may suffer frequent hits and abrasion, the inspection should include a close examination for damage from these sources.

Examine timber piles and other timber members for decay, insect damage, marine organisms, abrasion, and structural damage. Check at the water line for weathering of material (see Article 4.3.5.8.3). Note whether protective treatment needs patching or replacement. Examine cable ties and bolts for corrosion. Examine catwalks and their fastenings for decay and other damage.

Examine concrete members for spalling, cracking, corrosion of the reinforcing steel, and damage from abrasion or collisions. For concrete surfaces that have a protective treatment, indicate the condition of the treatment and the need for patching or replacement.

Examine rubber elements for missing parts, deterioration, cracking, and other damage to elements or fastening devices. Examine pneumatic and hydraulic elements for damage and to see if they are functioning properly under impact.

Check lighting devices on dolphins or fenders for corrosion, broken or missing lenses, and to see whether the lights are functioning correctly. Examine wiring, conduits, and fastening devices for corrosion, breaks, or loose connections.

4.3.5.9—Box Culverts as Bridges

This Article covers reinforced concrete single- or multiple-cell box culverts which are classified as bridges in accordance with the AASHTO definition of a bridge (see Section 1). Much of the material is also applicable to concrete arch culverts and to reinforced concrete structures constructed either without a bottom slab or with a bottom slab not

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rigidly connected to the sidewalls. In addition to guidance contained within the BIRM, the following techniques are presented:

Check the culvert for outward evidence of settlement or other movement by sighting for a sag in the profile of the roadway overhead, sag of the culvert floor or in the underside of the top slab, differential movement at joints in the box, and for rotation of the wingwalls at the ends of the box.

Evaluate the sidewalls, floor slab, and any footings for abrasion, cracking, or other deterioration of the concrete surfaces. Check for leakage of water through the expansion joints and for any undermining of the structure at the outlet due to scour. Check for accumulations of debris, particularly at the inlet and immediately upstream from the inlet, which could block the entrance. Note whether vegetation, debris, or trees are interfering with proper flow through the culvert. Note excessive accumulations of fill in the culvert. Check for erosion channels in the roadway embankment and in the banks of the waterway, which could affect the performance or structural integrity of the culvert.

Inspect the underside of the ceiling slab to detect cracks and spalls. Note the location and size for comparison with previous and subsequent reports. Cracks transverse to the direction of the span may indicate shear or tension stresses due to loadings in excess of those the structure can safely carry. Longitudinal cracks usually indicate differential settlement along the barrel of the box.

Check masonry facing, if present, for mortar cracks, loose or missing stones or blocks, weathering, and spalled or split blocks and stones.

4.3.5.10—Corrugated Metal Plate Structures

Corrugated Metal Plate (CMP) Structures, also known as “flexible culverts,” depend on interaction with the backfill soil for their stability and ability to carry loads. The CMP arch is a compression ring with little bending resistance.

Inspect the shape of the CMP arch and compare this to the as-built shape. Note any flattening of the top arch elements or sides, and all changes from the as-built condition or previous inspection.

Check the base of the CMP arch for differential settlement or undermining.

Inspect the backfill material at the outlet for evidence of material being removed from underneath and alongside of the structure due to water infiltrating the material from the inlet. Coring or test pits may be required to determine the extent of loss of backfill material.

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Check the entire length of the barrel of the CMP arch for misalignment of plate elements, leakage at seams, and dents or other local defects.

Check all CMP structures for cracks and distortion, especially at bolt locations.

CMP structures should be checked for partial or full concrete headwalls at the inlet to which the structures are anchored. In the absence of headwalls, check for evidence of an upward displacement of the inlet. For those installations with an inlet end mitered to the embankment slope, check for evidence of the edges folding inward.

4.3.5.11—Approaches

This Article provides guidance concerning the inspection of those approach elements situated immediately adjacent to the bridge itself. While not technically a part of the bridge, the inspection of these approach elements is nevertheless important to the safety of the motoring public.

4.3.5.11.1—Approach Slabs

Check approach slabs for cracking, unevenness, settlement, or roughness. Existence of one or more of these defects may cause vehicles coming onto the bridge to induce undesirable impact stresses in the structure. Cracking or unevenness in a concrete approach slab may indicate a void under the slab from fill settlement or erosion.

Some concrete approach slabs are designed to function as a beam, with the capacity to take live loads without relying on full support from underlying fill material. Other approach slabs must rely on support from underlying fill to function properly and may not safely support all traffic loads when undermined. The inspector thus should determine the capacity of the slab when erosion of underlying fills is found.

Examine expansion joints between the approach slabs and the abutment backwall or bridge deck, with deficiencies documented, as per Article 4.3.5.5.5. Additionally, some bridges exhibit relief joints between the approach pavement and exterior ends of the approach slabs, which are designed to accommodate local thermal movement of the approach pavement adjacent to the bridge. When inspecting relief joints, determine whether or not there is adequate clearance to provide for this movement. If the joint was intended to be sealed, determine if the seal is adequate to prevent leakage.

4.3.5.11.2—Approach Pavement

The Bridge Owner develops the procedures for consistently inspecting approach pavements adjacent to bridges, including the appropriate length of approach pavement at each end of the bridge to be inspected as part of the bridge inspection process.

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Approach pavements may be categorized as either rigid (typically portland cement concrete) or flexible (typically asphaltic concrete).

Check rigid pavements for cracking, unevenness, settlement, or roughness in the vicinity of the bridge, as well as the smoothness of the bridge transition. Existence of one or more of these defects may cause vehicles coming onto the bridge to induce undesirable impact stresses in the structure. Cracking or unevenness in a concrete approach slab may indicate a void under the slab from fill settlement or erosion.

Check flexible pavements for rutting, cracking, and evidence of deterioration or disintegration.

4.3.5.11.3—Drainage

The approach roadway drainage should be directed away from the bridge. Check that:

- Roadway drainage facilities adjacent to the bridge are functioning and
- Runoff flows into the drainage facilities and does not pond in the roadway or shoulder areas and does not erode the approach fill.

Settlement of the approach pavement or fill can significantly alter the roadway's profiles and cross slope and redirect water away from the drainage facilities.

4.3.5.11.4—Traffic Safety Features

This Article covers the inspection of traffic safety features situated immediately adjacent to the bridge, such as steel rail or wire cable approach guide rail (guardrail, slope-faced (New Jersey shape concrete barriers, and impact attenuation devices, as well as the transition of the approach guide rail to the bridge railing or parapet. Ensure that inspectors are familiar with current agency standards for approach guide rail types, installation heights, and any minimum clearances. Check each approach guide rail assembly as to its conformance to current standards. The NBI Coding Guide, item 36 also provides guidance for the evaluation of these features.

Check the guide rail condition for collision damage, cracks, rust, or component breakage. Check that connections between rails and posts are secure and tight. Check the alignment of the rail. Note all areas of settlement or frost heave. The posts, made of wood, concrete, or steel, must be firmly embedded in the ground. Promptly report posts, which have been hit by vehicles and displaced horizontally. Check wood posts for rot or insect damage, especially at the ground line. Check the slope beyond the guide rail posts for settlement or erosion that may reduce the

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embedment of the posts. Check guide rail approach ends and connections to the bridge parapet or railing for conformance to current standards.

Check impact attenuation devices adjacent to bridge elements for evidence of damage due to impact, and that the energy absorbing elements such as water or sand filled tubes have not ruptured. Check that cables and anchorages are secure and undamaged.

On structures over highways, review the adequacy and condition of traffic safety devices for both the upper and lower roadways.

4.3.5.11.5—Embankment Slopes

Inspect approach embankment slopes for evidence of excessive erosion, settlement, and undermining of pavements, approach slabs, curbing, or guide railing. Also check for slope failure in the vicinity of abutments, which could constitute a significant safety hazard to the traveling public.

4.3.5.12—Encroachments

Encroachments at or adjacent to a bridge site are man-made or natural elements which restrict the clearance under a bridge or, in some areas, over the bridge. Signs and sign structures, utilities, dense vegetation, rock outcroppings, and debris are examples of encroachments that reduce the horizontal and vertical clearances for the passage of vehicles. The encroachment of waterways is discussed in Article 4.3.5.8 and the inspection of utilities carried by the bridge is described in Article 4.3.5.6.14.

The inspector should note if the encroachment is located where there is a possibility that it may be hit and damaged by traffic. Check the horizontal and vertical clearances by field measurements, particularly after repaving projects.

Note the aesthetic effect encroachments may have on the bridge. This item must be considered in permitting encroachments to remain on a bridge. The general appearance of the vicinity around the structure will be a factor in making this determination.

4.3.5.13—Signs

Establish policies to facilitate the uniform signage posting for weight limit restriction, speed restriction, reduced vertical clearance, or bridge closure notification.

Inspect bridge signage to ensure that signs are in their proper place. This inspection includes signs at or on the structure and any necessary advance warning signs, as required by policy. Check also to see that the lettering is clear and legible and that signs are in generally good physical condition.

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Inspections that occur during the colder months of the year must account for summer foliage in assessing sign visibility. Additionally, for bridges that are weight restriction posted, record any truck traffic using the bridge in excess of posted weight limits and notify the Bridge Owner.

Document any revision made which will alter the vertical clearances, such as addition of surfacing to the roadway that will necessitate secondary measurement of the clearances and correction of the signs and records to reflect the change.

For bridges over navigable channels, check to see that the required navigational signs for water traffic are in place and in good condition. The inspector should be familiar with the regulations of the United States Coast Guard to the extent necessary for making these determinations. Examine navigational lights, if appropriate, to see that they are properly installed in their intended positions and functioning. Inspect the aerial obstruction lights on high bridges to see if they are functioning.

Inspect sign-framing members including the connections and anchor bolts for structural integrity. Connections used in sign-framing members may be fatigue prone and should be inspected in accordance with Article 4.3.7. Pay careful attention to epoxy anchors in tension, used to secure sign structures to bridges, and note any evidence of movement in the anchorages.

The parties responsible for replacing missing or damaged signs and for removal of vegetation and otherwise restoring sign visibility should be designated. Report sign deficiencies appropriately.

4.3.6—Complex Bridge Inspections

The inspection of complex bridges requires considerable planning, scheduling, and equipment, and may also require the utilization of inspectors with specialized training and expertise. As described in Article 4.2.5.9, each state should determine and document which bridges are to be classified as complex bridges.

Develop a specialized inspection plan for each complex bridge, which reflects the unique characteristics of the structure. For each bridge classified as complex, the inspection plan, as well as the bridge file, describes the inspection approach, equipment to be used, and the training and experience required of the inspection team. Where specialized inspection methods are required to evaluate special elements, document in the inspection plan and the bridge file a description of these methods. Identify in the inspection plan any safety considerations and safety procedures to be followed during the inspection.

Some complex bridge types are briefly described below.

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4.3.6.1—Movable Bridges

The most common types of movable bridges are the swing span, vertical lift span, and bascule span (single or double leaf). Movable bridges and their inspections are described in detail in the AASHTO *Movable Bridge Inspection, Evaluation, and Maintenance Manual* (2017).

Movable bridges employ electrical and mechanical systems as well as structural components and they require inspection staff with specialized equipment and training in the evaluation of these systems.

4.3.6.2—Suspension Bridges

Suspension bridges include spans incorporating cable-suspended and eyebar-chain suspension systems.

For cable suspension systems, examine the main suspension cables to see that their protective covering or coating is in good condition and protects the steel from corrosion. Give special attention to the cable areas adjacent to the cable bands, at the saddles over the towers, and at the anchorages.

Place emphasis on checking the condition of caulking at cable band locations on suspension bridge main cables, when it exists.

Examine the bands holding the suspenders to the main suspension cable to see that no slippage has occurred and that all bolts appear to be tight.

Check anchorages for corrosion and to see that there is adequate protection against moisture entering or collecting where it may cause corrosion. Give special attention to steel anchor bars embedded in concrete at the interface of the steel and the concrete.

Utilize nondestructive testing as necessary in evaluating the condition of cables (see Section 5).

Inspect stiffening trusses, floor system, towers, and cable bands in detail.

Eyebar suspension systems that have flat steel bars fabricated into a chain, with each link member consisting of two or more eyebars, connected by pins are considered fracture-critical unless evaluation indicates otherwise.

Eyebars used in a chain suspension span are very similar to those in a truss. The same type of inspection should be used on a suspension chain as that used on the truss chord.

- Carefully inspect the area around the eye and the shank for cracking.
- Examine the spacers on the pins at the end of each eyebar to be sure they are holding the eyebars in their proper position.

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- Observe the eyebars under live load to assure that the load is distributed evenly to each member of the link.
- Examine closely spaced eyebars at the pin for corrosion build-up (pack rust) between each member.
- Look for weld repairs.
- Inspect pins, pin nuts, pin caps, through bolts, and other similar components very carefully.

4.3.6.3—Cable-Stayed Bridges

Cable-stayed bridges consist of concrete or steel box girders or trusses supported by cables originating from a tall tower or towers. These cables may be fracture-critical elements and inspection is paramount. Cable inspection procedures should address cable enclosures, anchorages, and damping systems.

Each cable-stayed bridge should have an inspection manual prepared by the designer that provides a comprehensive set of special procedures for use in conducting inspections. The manual will usually describe the various components of the bridge, the design requirements, and construction techniques used. The manual will also outline the inspection procedures to be followed for each element and will include recommended maintenance procedures.

The inspection of the other structural elements is done in accordance with appropriate Articles of this Section.

4.3.6.4—Tied Arch Bridges

Tied arch bridges tie the ends of the arch ribs to the lower chord in such a manner that horizontal forces generated within the arch ribs as they are loaded are restrained by the lower chord, acting in tension like a bowstring. Vertical ties transmit vertical dead and live load forces from the deck up to the arch.

Tied arch bridges are considered to be fracture-critical and require careful planning and inspection, particularly at the connections between the vertical ties and the arch ribs and at the connections of the arch ribs with the deck.

4.3.6.5—Prestressed Concrete Segmental Bridges

Prestressed concrete segmental bridges may be made up of cast-in-place segments or precast segments. The inspection of the superstructure of a segmentally cast-in-place or precast bridge is much the same as that for prestressed concrete bridges.

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The inspection of substructure, bearings, deck, and expansion joints is carried out in accordance with the applicable discussions in Article 4.3. Closely examine the deck surface for longitudinal cracks at the edge of the exterior girder web. Cracking could have resulted from heavy loads on the overhang or by casting or curing methods that resulted in transverse bowing of individual units and resultant cracking during stressing.

Pay particular attention to the profile of the roadway surface, done by sighting the top of railing or edge of deck. Humps or sags of an entire span length are evidences of long-term creep of tendons or concrete not anticipated in the original design.

Localized sags or humps are indications of problems deserving closer inspection to see if there has been a failure of prestressing units or their anchorages. Such an inspection requires entry into the box sections and examination of the interior anchorages. Prestressing tendons may also be susceptible to corrosion due to issues with moisture trapped within the tendon ducts. This requires specialized expertise and equipment to evaluate.

Closely examine the areas around the keys in the girder stems and the slabs for cracks, particularly at interlocking corners.

Inspect the deck soffit for cracks and spalls and for evidences of water leakage through cracks or joints.

While inside the box, check the underside of the deck at joints between segments under passage of heavy loads. Differential movements indicate improper functioning of keys in the girder stems, or possible failure of the bearings under an end unit at its support. Differential movement between segments will also show up as cracks in the wearing surface on the deck.

4.3.7—Fatigue-Prone Details and Fracture-Critical Members

Fatigue cracks may occur in steel members and connection details at locations of stress concentration, where the rigidity of the member changes. Connection details, damaged components, and corrosion-notched sections are examples of such locations. Fracture-critical members (FCMs) are steel tension members or steel tension components of members whose failure would be expected to result in a partial or full collapse of the bridge. Both fatigue-prone details and fracture-critical members require close attention during inspections, and often require the use of nondestructive testing methods and equipment.

4.3.7.1—Fatigue-Prone Details

Various connection details have been identified and assigned a fatigue stress category, indicative of

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their relative fatigue life. (See LRFD Design Table 6.6.1.2.3-1.) Generally, Category E′ details have the shortest fatigue life and are the most prone to fatigue cracking. The susceptibility of the detail to cracking decreases from Category E′ to Category A. Many of the problems associated with these details are related to weld terminations and weld defects. Welds made in the field, including tack welds, are especially susceptible to fatigue cracking.

Ensure that bridge inspectors are trained to identify fatigue-prone details. Give a close visual inspection to all locations prone to fatigue cracking as described in Article 4.3.5.6.1. Frequency of such inspections is based on the category of the detail, the size and number of repetitions of trucks loads, and other related factors. The inspection of fatigue-prone details may include nondestructive testing. (See Section 5.)

4.3.7.2—Fracture-Critical Members

Give special attention to steel bridges with the following structural characteristics or components during the inspections:

- One- or two-girder I- or box girder systems.
- Suspension systems with eyebar components.
- Steel pier caps and cross girders.
- Two-truss systems.
- Welded tied arches.
- Pin and hanger connections on two- or three-girder systems.

The above is not a comprehensive list of FCMs or fracture-critical bridge structure types and does not serve as a checklist.

Tension components of a bridge member consist of components of a tension member and those portions of a flexural member that are subject to tension stress. Any attachment having a length in the direction of the tension stress greater than 4 in. (10 cm) that is welded to the tension area of a component of a FCM is considered part of the tension component, and is therefore be considered fracture-critical.

FCMs have all or part of their cross section in tension. Most cracks in steel members occur in the tension zones, generally at a flaw or defect in the base material. Frequently the crack is a result of fatigue occurring near a weld, a material flaw, changes in member cross section, or some combination thereof.

After the crack occurs, failure of the member could be sudden and may lead to the collapse of the bridge. For this reason, bridges with FCMs should receive special attention during inspections. The NBIS also specifies specialized reporting

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requirements for bridges classified as being fracture critical.

Review the previous inspection for any previous defects, access equipment needs, or other information in order to properly prepare for the inspection. Review the bridge file for any required specific FCM inspection procedures.

Perform a hands-on inspection for all FCMs in accordance with the BIRM's physical inspection methods. Give special attention to the tension zones, connections, and fatigue sensitive details. Note and evaluate all cracks in the steel for any changes. Perform BIRM advanced inspection methods for any new crack or suspected changes to existing cracks to determine the extent of any new or changed crack. Bring to the attention of the Bridge Owner any new or changed cracks.

Any findings considered CIFs cause the state's CIF procedures to begin.

4.3.8—Data Collection for Load Rating

Bridge evaluation involves not only the inspection of a bridge to assess its physical condition and functional capability, but also analysis and calculations for determining its load capacity rating and for reviewing overload permit applications. The scope of the inspection should be sufficient to provide the data necessary for load capacity evaluation of primary members and connections. The re-evaluation of in-service bridges for load capacity is required to the extent that changed structural conditions would affect any previously recorded ratings. The load ratings used in conjunction with the inspection findings will assist in determining the need for posting, strengthening, or closure of the bridge to traffic.

4.3.8.1—Field Data Collection

Before load rating a bridge, collect current condition and loading data for the bridge. The quality and the availability of data will have a direct influence on the accuracy and reliability of the load rating results. Where certain data is unavailable or unknown, this Manual provides guidance on arriving at suitable estimated values.

The following important items of data required for load rating are obtained from field inspection and from available bridge records. Where feasible, verify all important plan data used in the field at the time of inspection.

Geometric Data

- Span length/member lengths
- Support conditions/continuity/overhangs
- Bridge skew at each bearing

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- Girder/truss/floor beam spacing
- Roadway, traffic lane, and sidewalk widths

Member and Condition Data

- Member types and actual member sizes
- Material grade and specifications
- Reinforcing/prestressing/post-tensioning data
- Material losses due to deterioration
- Condition ratings/flagged conditions
- Elements and defects (NHS bridges)
- Presence of fatigue-sensitive details
- Presence of fracture-critical members and connections

Loading and Traffic Data

- Actual wearing surface thickness, if present
- Non-structural attachments and utilities
- Depth of fill, soil type, and condition (buried structures)
- Number and positioning of traffic lanes on the bridge
- Pedestrian traffic intensity
- ADTT or traffic volume and composition
- Posted load limit, if any
- Posted speed limit, if any
- Roadway surface conditions at approaches and on bridge
- Roadway condition/bumps at deck joints

4.3.8.2—Observation under Traffic

Observe bridges during passage of heavy loads to determine if there is any excessive noise, vibration, or deflection. If detected, further investigation should be made until the cause is determined. Investigate a bridge that exhibits a permanent sag in its profile further to determine a likely cause, such as overloads.

Bearings are designed to move freely about their pins or bearings and, if feasible, should be inspected carefully under passage of heavy loads to confirm that their movement is not being restrained.

Many decks were designed to act compositely under live load with the supporting superstructure members. The inspector should check to see that composite decks are acting as intended by the designer. Movement between the bottom of the deck

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passage of heavy loads may be indicative of a breakdown in the composite action.

Observations under traffic will reveal looseness or excessive deflection of timber decks and stringers. Check the bridging between the timber stringers to see that it is tight and functioning properly.

4.3.8.3—Loading Determination

Dead load effects of the superstructure are computed through detailed calculations of the existing dead loads. To this end, the evaluator should utilize all available bridge records. Determine in the field all pertinent information when as-built information is incomplete or unavailable. Dead and super-imposed dead loads are accurately estimated by undertaking detailed measurements of the structure. Measure overlay thickness and depth of fill during each inspection. Verify weight of utilities present and their distribution during inspection.

The live loading depends on the number of traffic lanes carried by the bridge. The actual number of lanes in service may be less than the maximum number of lanes that could be accommodated by the bridge. Record the clear width of roadway and sidewalks and position of lanes on the bridge. Observations regarding travel speed, apparent violations of load postings when present, and nature of pedestrian traffic would also assist the evaluator during the load rating process.

Traffic Data—The expected loading during the evaluation exposure period is affected by the truck traffic at the site. Data may be available from recent traffic surveys including *ADT*, *ADTT*, and truck-load data measurements. Advice should be sought from the Bridge Owner/Traffic Division regarding available traffic data.

Dynamic Load Allowance (Impact)—The main parameters affecting dynamic load allowance are the bridge approach, bumps, and other pavement roughness. Check approach pavement condition for cracking, unevenness, settlement, or roughness. Existence of one or more of these defects may cause vehicles coming onto the bridge to induce undesirable dynamic stresses in the structure. The inspector should assess the condition of the deck overlay and report the condition of the overlay, deck joints, and approaches.

The inspector should record all the parameters necessary to determine the strength of primary members and connections, in accordance with Article 4.3.

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SECTION 5

MATERIAL TESTING

5.1—GENERAL

This Section describes the more common testing procedures for assessing the strength and condition of materials and structural components of bridges. New testing procedures are evolving rapidly as a result of improved technology. Material testing should be performed by properly trained personnel.

C5.1

This Section defines the types of nondestructive field tests and provides guidance on when to use them. In addition, guidelines are provided for sampling bridge materials and using related laboratory tests. Source material included FHWA *Manual on the Inspection of Fracture Critical Bridge Members*; NCHRP Report 312, *Condition Surveys of Concrete Bridge Components*; NCHRP Report 206, *Detection and Repair of Fatigue Damage in Welded Highway Bridges*; NCHRP Report 242, *Ultrasonic Measurement of Weld Flaw Size*; FHWA *Training Course on Nondestructive Testing*; NCHRP Project 10-30, *Nondestructive Methods for Field Inspection of Embedded or Encased High Strength Steel Rods and Cables*; various ASTM specifications; and state manuals.

Properly trained personnel should perform the testing described in this Section. The American Society for Nondestructive Testing has programs for certifying technicians at various skill levels which may be used as a guide in establishing minimum levels of competency for test personnel.

5.2—FIELD TESTS

Numerous field test procedures are available for concrete, steel, and timber structures. Many of these procedures are nondestructive, while others result in some removal or damage of the material.

5.2.1—Concrete Field Tests

Typical field test procedures for concrete bridge components are described below. A comparison of the test methods in terms of their capability of detecting defects in concrete components is shown in Table 5.2.1-1. This table should be used as a guide in selecting an appropriate field test method for concrete components.

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Table 5.2.1-1—Capability of Investigating Techniques for Detecting Defects in Concrete Structures in Field Use

Capability of Defect Detection[a]						
Method Based on	Cracking	Scaling	Corrosion	Wear and Abrasion	Chemical Attack	Voids in Grout
Strength	N	N	P	N	P	N
Sonic	F	N	G[b]	N	N	N
Ultrasonic	G	N	F	N	P	N
Magnetic	N	N	F	N	N	N
Electrical	N	N	G	N	N	N
Nuclear	N	N	F	N	N	N
Thermography	N	G[b]	G[c]	N	N	N
Radar	N	G[b]	G[c]	N	N	N
Radiography	F	N	F	N	N	F

[a] G = Good; F = Fair; P = Poor; N = Not suitable

[b] Beneath bituminous surfacings

[c] Detects delaminations

5.2.1.1—Strength Methods

Rebound and penetration tests measure the hardness of concrete and are used to predict the strength of concrete. The Schmidt Hammer is probably the most commonly used device of this type. It consists of a plunger and a spring-loaded mass that strikes the free end of a plunger, which is in contact with the concrete, and rebounds. The extent of rebound gives an indication of the strength of the concrete at the surface position tested. The measurement is influenced by the finish of the concrete, age, and other factors. As an inspection technique, the hammer may be used to compare the quality of the concrete in different parts of the concrete bridge components. It should be remembered that only the surface of the concrete is being checked and the strength values are relative. This test is covered in ASTM C805, "Standard Test Method for Rebound Number for Hardened Concrete." Actual strength must be determined by other means.

The relative compressive strength of concrete can also be determined by the Windsor Probe. The Windsor Probe is a commercial test system that utilizes procedures outlined in ASTM C803, "Standard Test Method for Penetration Resistance of Hardened Concrete." This device drives a steel probe into the concrete using a constant amount of energy supplied by a precise powder charge. The lengths of the probes projecting from the concrete are measured. A normal result is based on the average of three measurements. This test and the Schmidt Hammer are considered usable only with relatively new concrete, e.g., less than one year old.

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5.2.1.2—Sonic Methods

Mechanical sonic pulse-velocity methods have been used for concrete for many years. Hammer blows create the impulse and the time of travel of this sonic pulse between pickups placed on the concrete is measured. The time of travel is related to the modulus of elasticity and, hence, the strength. This technique can be effective, but is tedious and can be applied to small areas only. The procedure is capable of detecting differences between areas of sound and unsound concrete and is frequently used to detect delaminations or other fractures. The technique is impractical in evaluating large surface areas, such as concrete decks. However, on vertical surfaces there is currently no alternative that is practical and reliable.

Chain drags, sounding rods, or even hammers are frequently used for detecting delaminations on horizontal surfaces, such as decks or tops of piers. The chain drag can be used to quickly traverse a large area with reasonable accuracy in determining areas of delamination provided the inspector has experience in detecting hollow sounds. Chain-drag surveys of asphalt-covered decks are not totally accurate but they are quick and inexpensive and may be used as an initial test to determine the need for more thorough investigations.

The practice for measuring delaminations in concrete bridge decks is discussed in ASTM D4580.

Portable, automated acoustic methods have been developed for bridge decks. The instrument consists of three components: a tapping device, a sonic receiver, and a signal interpreter. The instrument is moved across a deck as acoustic signals are generated, propagated through the concrete, received, and interpreted electronically. The output is used to generate a plan of the deck indicating delaminated areas. The accuracy decreases when used on an asphalt-covered deck.

5.2.1.3—Ultrasonic Techniques

Ultrasonic devices are normally used by measuring the velocity in concrete of a pulse generated by a piezoelectric transducer. The pulse velocity depends on the composition and maturity of the concrete and its elastic properties. The relationship to strength depends on several other properties and is best determined experimentally.

The recommended procedure is the direct transmission method which has the transmission and receiving probes in line on opposite sides of a concrete thickness. Caution should be used in comparing results from indirect transmission tests with calibration or tests from direct transmission techniques.

There appear to be reasonably good correlations between pulse velocity and compressive strength provided the system has been calibrated with cores of the particular concrete being evaluated. The concrete strength can be predicted within about 20 percent of the calibration curve established for the particular concrete being investigated. It is not possible to predict the strength of concrete without calibration with the particular concrete in question.

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The presence of steel parallel to the line of transmission provides a path along which the pulse can travel more rapidly. Corrections can be made for this situation but detailed information on the reinforcement is needed. It is generally desirable to choose path lengths that avoid the influence of reinforcing steel.

Open cracks or voids may also affect the ultrasonic pulse. The path of the pulse will thus travel around any cavity in the concrete and the time of transmission of the pulse is lengthened. Large cracks and voids may be detected by this means. Narrow cracks will transmit the pulse through points of contact, and small voids will increase the path length only a small amount and may not be distinguishable from the normal variability of the measurements.

Ultrasonic techniques can, with proper experience and training, provide excellent information regarding the condition of the concrete. However, the method is complex and requires some skill to obtain usable results. The technique is not normally used in routine bridge evaluation.

5.2.1.4—Magnetic Methods

The principal application of magnetic methods in testing of concrete bridge components is in determining the position of the reinforcement. Magnetic methods are not techniques for detecting defects or deterioration directly but the fact that inadequate cover is often associated with corrosion-induced deterioration indicates that a method for locating the reinforcing bars can be important in corrosion control.

Several portable, battery-operated magnetic devices known as cover meters or pachometers have been designed to detect the position of reinforcement and measure the depth of cover. The devices generate a magnetic field between the two poles of a probe and the intensity of the magnetic field is proportional to the cube of the distance from the pole faces. When a reinforcing bar is present, the magnetic field is distorted and the degree of distortion is a function of the bar diameter and its distance from the probe.

In general, the cover meters can measure cover within 0.25 in. in the range of 0 to 3 in. The instruments give satisfactory results in lightly reinforced members but, in heavily reinforced members or where large steel members are nearby, it is not possible to obtain reliable results. In addition, some reports indicate that epoxy coatings distort readings.

5.2.1.5—Electrical Methods

Electrical methods for inspection of concrete bridge components include resistance and potential measurements. Electrical resistance has been used for measuring the permeability of bridge deck seal coats. The procedure has been published as a standard test in ASTM D3633 and involves measuring the resistance between the reinforcing steel and a wet sponge on the concrete surface.

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Corrosion of reinforcement produces a corrosion cell caused by differences in electrical potential. This difference in electrical potential can be detected by placing a copper–copper sulfate half-cell on the surface of the concrete and measuring the potential differences between the half-cell and steel reinforcement. It is generally agreed that the half-cell potential measurements can be interpreted as follows:

- Less negative than –0.20 volts indicates a 90 percent probability of no corrosion,
- Between –0.20 and –0.35 volts, corrosion activity is uncertain, and
- More negative than –0.35 volts is indicative of greater than 90 percent probability that corrosion is occurring.

If positive readings are obtained, it usually means that insufficient moisture is available in the concrete and the readings are not valid. These tests do not indicate the rate of corrosion and the measurements only manifest the potential for corrosion at the time of measurement.

Although most commonly used with bridge decks, the half-cell has been used with other bridge components, such as bents, to determine active corrosion.

5.2.1.6—Nuclear Methods

The main use of nuclear methods is to measure the moisture content in concrete by neutron absorption and scattering techniques. These moisture measurements are then used to determine if corrosion of reinforcement is likely to occur. A direct measurement of the rate of corrosion would be more useful to the bridge inspector and, hence, the nuclear methods are more research-oriented than operational.

5.2.1.7—Thermography

Infrared thermography has been found to be a useful supplemental test in detecting delaminations in concrete bridge decks. The method could be used for other concrete bridge components exposed to direct sunlight. Thermography works on the principle that as the concrete heats and cools, there is a substantial thermal gradient within the concrete because concrete is a poor conductor of heat. Delaminations and other discontinuities interrupt the heat transfer through the concrete, and these discontinuities cause a higher surface temperature during periods of heating than the surrounding concrete and the reverse situation during periods of cooling. The differences in surface temperature can be measured using sensitive infrared detection systems. The equipment can record and identify areas of delamination and correlations can indicate depth of delamination below the surface by the differences in surface temperature.

The test method for detecting delaminations in bridge decks using infrared thermography is discussed in ASTM D4788.

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5.2.1.8—Radar

Ground-penetrating radar has been used to detect deterioration of bridge decks. These investigations are carried out by low-power, high-frequency pulsed radar. The radar picks up any discontinuity such as air to asphalt, asphalt to concrete, or cracks in concrete. The ability to measure the thickness of asphalt covering is an important benefit. The radar method also has important potential for examining the condition of the top flange of box beams that are otherwise inaccessible. More than a little experience is necessary for proper interpretation of the data.

5.2.1.9—Radiography

Gamma radiation will penetrate concrete and therefore can be used to investigate concrete by exposing photograph film to radiation. A source of radiation is placed on one side of the concrete and a film is attached to the other side. Steel impedes the transmission and an image shows up on the developed film as lighter than the surrounding concrete. Void areas show up as darker images. The inspector then can get a reasonable idea of the concrete steel reinforcement pattern and the location and extent of defects in the concrete mass.

Radiography can be carried out only by licensed firms that can handle radioactive isotopes. Radiography of concrete is expensive and limited applications of the technique are likely to be used in bridge inspection.

5.2.1.10—Endoscopes

Endoscopes consist of rigid or flexible viewing tubes that can be inserted into holes drilled into concrete bridge components. Light can be provided by glass fibers from an external source. In the rigid tubes, viewing is provided through reflecting prisms and, in the flexible tubes, a fiber optics system is used. These scopes allow close examination of parts of the structure which could not be otherwise viewed. The inside of a box girder or a hollow post-tensioning duct are two examples. Some equipment is available with attachments for a camera or television monitor. Although this is a viewing instrument, some destruction of material is necessary for its proper use with concrete.

5.2.2—Steel Field Tests

Typical field test procedures for detecting defects in steel bridge components are described below.

A general summary of the relative capabilities of the steel test methods is given in Table 5.2.2-1. This table should be used as a guide in selecting an appropriate field test method for steel components.

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Table 5.2.2-1—Capability of Nondestructive Examination Techniques for Detecting Defects in Steel Structures in Field Use

Method Based on		Capability of Defect Detection[a]									
		Minute Surface Cracks	Deeper Surface Cracks	Internal Cracks	Fatigue Cracks	Internal Voids	Porosity and Slag in Welds	Thickness	Stress Corrosion	Blistering	Corrosion Pits
Radiography		N	F[b]	F[b]	P	G	G	F	F	P	G
Magnetic Particle (A.C.)	Wet	G	G	N	G	N	N	N	G	N	N
	Dry	F	G	N	G	N	N	N	F	N	P
Eddy Current		F	G	N	N	N	P	P	N	N	N
Dye Penetrants		F	G	N	G	N	N	N	G	N	F
Ultrasonics[c]		P	G	G	G	G	F	G	F	F	P

[a] G = Good; F = Fair; P = Poor; N = Not suitable

[b] If beam is parallel to cracks

[c] Capability varies with equipment and operating mode

5.2.2.1—Radiography

Nondestructive examination by use of X-rays depends on the fact that X-radiation, produced either by a commercial X-ray machine or by radioactive decay of a radioisotope, will be absorbed by a material in proportion to the thickness of the part examined and the atomic number. Thus, if a defective piece of material is examined by this method, the X-ray absorption at the region of the defect will be different (usually less) than sound material next to this region. The X-radiation coming through the part is recorded on a film or fluorescent screen; the image is usually darker in the area where the defect is located. The X-ray image on film provides a permanent record of the defect and also shows the size and shape of the defect in two dimensions. It does not show its position in depth in the part.

It follows from this description that defects such as slag inclusions or porosity in welds or castings are easily detected by this method. Planar defects such as cracks are also detectable but only if oriented approximately parallel to the axis of the X-ray beam. Cracks or planar defects perpendicular to the X-ray beam axis will not change the X-ray absorption significantly and thus will be undetected. Intermediate orientations will produce varying degrees of defect detectability.

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Advantages of this method of nondestructive examination are the permanent record that normally results, the ability to determine internal defect size and shape (and thus defect nature), and its almost universal acceptance in codes and by the engineering profession in general. The prime disadvantages to this method are its inability to locate the depth of the defect, its inability to locate poorly oriented planar defects, and the need to use, in general, large or hazardous equipment. It may also be difficult to apply in some field locations. One special consideration with this method which makes it particularly attractive is the fact that the resulting film is, in fact, a photograph of the part, and thus is immediately geometrically relatable to the part examined. No secondary analysis of the data is necessary.

5.2.2.2—Magnetic Particle Examination

This method of inspection, like the dye penetrant examination, is limited to surface or near-surface defects. An additional limitation placed on the process is the fact that only magnetic materials may be examined. In the shop application of the method, the part to be examined is placed in a magnetic field and fine powdered iron is sprayed (in suspension) or blown on it. If the magnetic field is undisturbed by any surface or subsurface discontinuities, the iron powder aligns itself with the field in a uniform film. If a discontinuity (such as a crack) disturbs the field, a concentration of magnetic lines of force will occur and, thus, a concentration of iron powder. This concentration will show the presence of the crack during visual inspection. In order to detect the crack, it must be aligned transverse or nearly transverse to the magnetic field. For this reason, the magnetic field must either be aligned perpendicular to the expected direction of defect formation or must be varied in direction. For shop tests, this is usually accomplished by sequentially magnetizing the part in a large circular coil to produce a longitudinal magnetic field and passing current through the part to produce a circular magnetic field.

Field applications use permanent magnets, electromagnets, portable coils, conductive cables, or electrode prods. Prods require current on the same level as welding current. If prods are not held and maintained in proper contact, electrical arcing may occur and cause component damage. Prods should not be used on fatigue prone components or highly loaded members. Instead, alternative field application methods should be used. Electromagnetic yokes are generally preferred on these components. If the electromagnet or other portable magnetizing equipment is moved about the part or structure to be examined, defects at any orientation may be detected.

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The advantages to this method are its relative portability, the minimum skills required to operate it, and its ability to detect even tight cracks. Of course, it is limited in the materials that it may be applied to and the type of defects it may detect. Again, in some applications, it has the additional limitation that it leaves the part in the magnetized condition. Although this is not normally a problem, it may interfere with some subsequent operations, such as welding. It is possible to demagnetize the area examined by this method, but this is time consuming and adds to the cost.

5.2.2.3—Eddy Current Examination

This method operates very similarly to magnetic particle inspection but the defect is detected by a perturbation in the electrical, not magnetic, field in the material examined. In this technique, a coil carrying alternating current produces eddy currents in a conductor nearby. The conductor eddy currents, in turn, create impedance in the exciting or, if desired, a separate search coil. The impedance produced depends on the nature of the conductor and the exciting coil, the magnitude and frequency of the current, and the presence or absence of discontinuities in the conductor. The method is therefore instrumented such that a coil is scanned over the surface of the area to be examined and defects produce a characteristic change in impedance as read from a dial or meter (output can be put on a chart if desired).

This method has been given only limited application for several reasons, most important of which has been that generally only simple geometries can be examined. Complex geometries change the impedance readings in themselves and thus limit the usefulness of the procedure. Again, as with magnetic particle examination, only conductors can be examined.

There is some potential for this method. Defects in depth can be detected or, with suitable frequency control, examination may be limited to the surface. Defect size can also be estimated from the response of the area examined. It is insensitive to many surface conditions (for example, paint) which limit other methods. This method appears to need further development, however, to be generally applicable. Certainly the geometry sensitivity of the method is a real disadvantage.

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5.2.2.4—Dye Penetrant Examination

The dye penetrant method of inspection is probably the most commonly employed shop and field method of defect detection. Although it is limited entirely to defects that penetrate the surface of the structure, it is inexpensive, easily applied, and easily interpreted.

The method itself is simple: The surface of the part to be examined is cleaned, usually mechanically, with a chemical degreasing agent, or both. A fluid is placed on the surface to be examined, often with an aerosol spray, and allowed to penetrate cracks or surface defects by capillary attraction or other surface wetting phenomena. After a period of time, usually minutes, the penetrant is removed and a second solution is sprayed on the surface. The second coating, called a developer, usually dries to a chalky powder and remains unchanged in the regions where no defect exists. In the location of a crack, the penetrant seeps from the crack where it is trapped and stains the developer. For this reason, brightly-colored (often red) penetrants are used. The red penetrant stains on the white chalky developer indicate the presence of a crack or other defect when visually inspected by the examiner. Modifications of the system include penetrants of different viscosity to detect different size cracks, wet rather than dry developers, and penetrants that fluoresce under ultraviolet light to make smaller defects visible.

The principal advantages of the method are the ease with which the tests are conducted, the minimal skills required, and the low cost. Tests are not time consuming and may be made frequently during other operations (for example, to determine if a defect being removed by grinding is completely eliminated). It must be considered the most portable of all methods.

The principal disadvantage is that only surface defects can be detected. This places a limitation on the usefulness of the method for the defect depth determination and “code” approval of most structures. However, from the practical shop viewpoint, many defects that occur during construction (for example, weld cracks) are detectable if dye penetrant is used at intermediate stages in the construction. Thus, defects that are later buried can be detected and repaired before they are hidden from view. Use of dye penetrant during fabrication may prevent later rejection when ultrasonic or X-ray examination is used. The more sophisticated dye penetrant methods using ultraviolet light are rarely used in field applications.

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5.2.2.5—Ultrasonic Examination

Ultrasonic testing relies on the wave properties of sound in materials to detect internal flaws. High-frequency sound waves in the form of mechanical vibrations are applied to the part to be tested and the waves, passing through the material, strike either a defect or, eventually, an external surface. The sound vibrations are then reflected and the nature of the return signal indicates the location and type of reflecting surface. Normal instrumentation includes a sound wave generator and pick-up device (usually combined in one unit) and a display screen on which the initial and the reflected pulse are displayed. Display instrumentation permits an estimation of the position (in depth) of the defect, the nature of the defect and, by moving the detection portion of the unit (called the search unit) along the part to be examined, the size of the defect. The test sensitivity is influenced by a great number of testing variables, such as sound frequency, design of the search unit, instrumentation, electronic processing of the return signal, and the skill of the operator. Typically, results of the examination are listed in a form prepared by the operator based on his observations of the display screen.

The major advantages of this system of nondestructive examination are its portability, sensitivity, and ability to detect the location of cracks or defects in depth. On the other hand, the major fault of the system is that, until very recent times, no permanent record of the defect was produced. It is now possible to make photographic records of the display and equipment is now available to permit the storage of field data in a format suitable for subsequent computer processing and reporting. Another characteristic of the system often cited as a difficulty is the sensitivity of the method. It is possible to see too much; e.g., grain size in metals and minor defects not observable by other methods. The system cannot detect surface defects very well. The dependency of the method on operator skill must also be considered an unfavorable factor.

More research has been undertaken to modify this method and make it more widely applicable than most of the others, so advances in technology are more likely in this field.

5.2.3—Timber Field Tests

Typical field test procedures for detecting defects and deterioration in timber bridge components are described below.

A summary of the capabilities of each of the test methods for detecting defects and deterioration in timber components is given in Table 5.2.3-1. This table should be used as a guide in selecting an appropriate field test procedure for timber components.

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Table 5.2.3-1—Capability of Investigative Techniques for Detecting Defects in Timber Structures in Field Use

Capability of Defect Detection[a]					
Method Based on	Surface Decay and Rot	Internal Decay and Voids	Weathering	Chemical Attack	Abrasion and Wear
Penetration	G	G	F	F	N
Electrical	F	F	N	N	N
Ultrasonics	N	G	G	N	N

[a] G = Good; F = Fair; P = Poor; N = Not suitable

5.2.3.1—Penetration Methods

Any probe, such as a knife, ice pick, nail, or brace and bit, can be used to test for internal decay or vermin infestation. The ease with which a member can be penetrated is then a measure of its soundness. Only a qualitative assessment is obtained because the pressure on the instrument is neither controlled nor measured.

Although the procedure is rather crude, it is rapid and an overall assessment of the condition of a structure can be obtained quickly. The use of a probe is much more satisfactory than attempting to identify a hollow member by sounding because the load on the member affects the response, and may lead to erroneous conclusions.

An increment borer, which consists of a sharpened hollow tube, usually about $^1/_4$-in. (6-mm) internal diameter, can also be used to penetrate the wood. The borer is superior to a nail or ice pick because it gives a more accurate record of the depth of decay or infestation. It also allows samples to be removed from the interior of the member for detailed examination or testing for such items as moisture content and preservative penetration, or to be cultured for positive evidence of decay fungi.

5.2.3.2—Electrical Methods

The main application of electrical methods is to measure the moisture content of timber. There are several electrical techniques available for measuring moisture content.

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Resistance meters are based on a direct current measurement of electrical resistance between point or blade electrodes pushed into the timber. The resistance is related to the moisture content, which is displayed on a calibrated scale. The results are affected by the species of timber and correction factors must be applied. Resistance moisture meters are light, compact, and inexpensive, but the major disadvantage is that they measure the moisture content of the surface layers unless special deep probes are used. Readings over 30 percent moisture content are not reliable and contamination by some chemicals, such as salt, affects the readings.

Capacitance meters are based on an alternating current measure of the dielectric constant of wood, which is proportional to its moisture content. The results are a function of the relative density of the wood and correction factors must be applied. The meters measure primarily surface moisture content and, on lumber thicker than 2 in. (50 mm), do not respond to internal moisture adequately. Capacitance meters have a wider range (0 to at least 35 percent moisture content) than resistance meters and are less affected by the presence of chemicals.

Radio frequency power-loss meters operate in the frequency range 0 to 25 MHz and are based on an alternating current measurement of the impedance (combined effect of resistance) and capacitance of timber. They need to be calibrated for wood species and density. The meters use plate-type electrodes and the field penetrates about $^{3}/_{4}$ in. (20 mm) but the surface layers have the predominant effect. The cost of the meters is similar to that of capacitance-type meters, being higher than that of simple resistance types.

Electrical resistance measurements are also the basis of an instrument designed to detect internal rot. The device consists of a resistance probe, which is inserted to various depths in a hole $^{3}/_{32}$ in. (2.4 mm) in diameter. A marked change in electrical resistance is an indication of decay. Although the device effectively detects rot, it is susceptible to false indications of decay in apparently sound wood.

5.2.3.3—Ultrasonic Techniques

The same ultrasonic pulse-velocity equipment and techniques described in Article 5.2.1.3 for application to concrete members can also be used for the in situ testing of timber structures, both above and below the water surface.

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Pulse-velocity measurements relate to the elastic properties of the wood and are, therefore, sensitive to the direction of the grain. However, pulse-velocity measurements have been found to follow similar trends to strength changes caused by fluctuations in density and local defects. Consequently, the strength and stiffness properties of the timber can be assessed. The ultrasonic method can also be used to identify internal decay and hollow areas, as well as internal knots, checks, and shakes. Because a discontinuity, such as a crack or a hollow area caused by decay, reflects part of the sound wave and changes the velocity of the transmitted wave, the technique is most sensitive to detecting defects that are oriented perpendicularly to the pulse. For this reason, the direct transmission mode with transducers on opposite faces of the member is generally the most useful configuration. However, in some situations, it may be necessary to investigate other relative positions of the transducers in order to produce a maximum response. To simplify interpretation of the results, it is common practice to compare the pulse velocity from a suspected area of deterioration with that from an area known to be sound (measured using the same transducer configuration), thereby eliminating the need to measure the density of the timber. In all cases, a good contact between the transducer and the surface of the timber is essential. A light grease or glycerol is suitable for the coupling medium. Bentonite paste has also been found satisfactory.

5.3—MATERIAL SAMPLING

Tests which require the removal of material from the structure should be used only when a particular piece of information is desired, and only when the results can provide something useful in the overall evaluation of the bridge.

A few common material sampling standards are shown in Table 5.3-1. Samples should be removed from those areas of a bridge subjected to low stress levels as determined by the Engineer. An adequate number of samples should be obtained to provide results representative of the entire structure being evaluated. Normally, a minimum of three samples would be required.

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Additional guidance on repairing areas of bridge members from which material was removed for testing may be found in the *AASHTO Maintenance Manual for Roadways and Bridges*; NCHRP Report 271, *Guidelines for Evaluation and Repair of Damaged Steel Bridge Members*; and NCHRP Report 280, *Guidelines for Evaluation and Repair of Prestressed Concrete Bridge Members*.

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Table 5.3-1—Standard ASTM and AASHTO Methods for Material Sampling

Designation[a]	Title
C42/T 24	Standard Method of Obtaining and Testing Drilled Cores and Sawed Beams of Concrete
T 260	Sampling and Testing for Chloride Ion in Concrete and Concrete Raw Materials
C823	Standard Practice for Examination and Sampling of Hardened Concrete in Constructions
A610	Standard Test Methods for Sampling Ferroalloys for Size
A673	Standard Specification for Sampling Procedure for Impact Testing of Structural Steel (Charpy Test)
A370	Standard Test Methods and Definitions for Mechanical Testing of Steel Products

[a] ASTM test methods are designated A or C. AASHTO test methods are designated T

The removal of material from a structure will leave a hole or void in one or more members. Repairs can be readily made to concrete, masonry, and timber members. Repairs to steel members may be much more complex, particularly if welding is used, and should be carried out by experienced personnel. Care should be taken to minimize any residual stress resulting from the repair.

5.4—LABORATORY TESTS

To supplement field tests and observations, there are many laboratory tests which have been standardized and used routinely in the evaluation of materials used in bridges. Tables 5.4-1, 5.5-1, and 5.5-2 list the ASTM and AASHTO standards governing the laboratory testing of concrete, steel, and timber components, respectively.

Laboratory tests should be conducted by testing laboratories familiar with the AASHTO, ASTM, and Bridge Owner standards to be employed.

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Table 5.4-1—Standard ASTM and AASHTO Test Methods for Concrete for Use in the Laboratory

Designation[a]	Title
C39/T 22	Standard Method of Test for Compressive Strength of Cylindrical Concrete Specimens
C1084/ T 178	Test Method for Cement Content of Hardened Portland Cement Concrete
C174/T 148	Method of Measuring Thickness of Concrete Elements using Drilled Concrete Cores
C457	Standard Test Method for Microscopical Determination of the Air-Void System in Hardened Concrete
C469	Standard Test Method for Static Modulus of Elasticity and Poisson's Ratio of Concrete in Compression
C496	Standard Test Method for Splitting Tensile Strength of Cylindrical Concrete Specimens
C617/T 231	Standard Practice for Capping Cylindrical Concrete Specimens
C642	Standard Test Method for Density, Absorption, and Voids in Hardened Concrete
C666/T 161	Standard Test Method for Resistance of Concrete to Rapid Freezing and Thawing
C856	Standard Practice for Petrographic Examination of Hardened Concrete
T 259	Method of Test for Resistance of Concrete to Chloride Ion Penetration[b]
T 260	Method of Sampling and Testing for Total Chloride Ion in Concrete and Concrete Raw Materials
T 277	Standard Method of Test for Electrical Indication of Concrete's Ability to Resist Chloride Ion Penetration

[a] ASTM test methods are designated C. AASHTO test methods are designated T

[b] Corrosion threshold is about 1.3 to 2.0 lbs of chloride per yd^3

5.5—INTERPRETATION AND EVALUATION OF TEST RESULTS

Field and laboratory test results must be interpreted and evaluated by a person experienced in such activity. If the same test has been previously run on material from this structure, the test results should be compared, differences noted, and then evaluated. When more than one type of test is used to measure the same material property, the individual test results should be compared and differences explained.

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Care must be exercised in the interpretation and evaluation of field and laboratory test results.

Several issues may play a part in the evaluation, for instance:

- Was sampling done properly? (Location, size, number to adequately represent the member being tested.)
- Do the results confirm expectations? Any surprises?
- Is there a pattern or consistency to the results of the group of tests or to previous test results?

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- Was the test performed by an experienced individual or firm? (The reliability factor)
- Do the results indicate incipient failure, the need for immediate repairs, or for weight-limit posting? (If so, must verify data.)
- Are other tests or inspections needed to verify results, to investigate other members in the same structure for like defects, or to look into the possibility of there being companion-type defects in the same member?
- Is there likelihood that other structures on the system have experienced like problems—or that there may be similar structures where the problem is as yet undiscovered?

Table 5.5-1—Standard ASTM and AASHTO Test Methods for Steel for Use in the Laboratory

Designation[a]	Title
A370/T 244	Standard Test Methods and Definitions for Mechanical Testing of Steel Products
E3	Standard Guide for Preparation of Metallographic Specimens
E8/T 68	Standard Method of Test for Tension Testing of Metallic Materials
E10/T 70	Standard Test Method for Brinell Hardness of Metallic Materials
E92	Standard Test Method for Vickers Hardness of Metallic Materials
E103	Standard Practice for Rapid Indentation Hardness Testing of Metallic Materials
E110	Standard Test Method for Rockwell and Brinell Hardness of Metallic Materials by Portable Hardness Testers
E112	Standard Test Methods for Determining Average Grain Size
E340	Standard Practice for Macroetching Metals and Alloys
E384	Standard Test Method for Microindentation Hardness of Materials
E407	Standard Practice for Microetching Metals and Alloys
E883	Standard Guide for Reflected-Light Photomicrography

[a] ASTM test methods are designated A or E. AASHTO test methods are designated T

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Table 5.5-2—Standard ASTM Test Methods for Timber for Use in the Laboratory

Designation[a]	Title
D143	Standard Test Methods for Testing Small Clear Specimens of Timber
D198	Standard Test Methods of Static tests of Lumber in Structural Sizes
D1860	Standard Test Method for Moisture and Creosote-Type Preservation in Wood[a]
D4442	Test Methods for Moisture Content of Wood
D2017 (Withdrawn)	Method for Accelerated Laboratory Test of Natural Decay Resistance of Woods
D2085 (Withdrawn)	Test Methods for Chloride for Calculating Pentachlorophenol in Solutions for Wood (Lime Ignition Method)
D2395	Standard Test Methods for Density and Relative Density of Wood and Wood-Base Materials
D2915	Standard Practice for Sampling and Data Analysis for Structural Wood and Wood Based Products
D3345	Standard Test Method for Laboratory Evaluation of Wood for Resistance to Termites

[a] Substantially the same as AWPA-A6

5.6—TESTING REPORTS

It is important that all field and laboratory tests be documented in writing and become part of the bridge file. Where instrumentation is used in the conduct of the test, the report should include the type of equipment, the manufacturer, and the serial number; a copy of the most recent calibration certificate; and the name of the trained operator.

For laboratory tests, the results should be submitted in a formal report using the laboratory letterhead, signed by a responsible official of the laboratory.

5.7—REFERENCES

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LOAD RATING

6.1—SCOPE

This Section sets forth criteria for the load rating and posting of existing bridges and provides a choice of load rating methods. Part A incorporates provisions specific to the Load and Resistance Factor Rating (LRFR) method developed to provide uniform reliability in bridge load ratings, load postings, and permit decisions. Part B provides safety criteria and procedures for the Allowable Stress and Load Factor methods of evaluation. No preference is placed on any rating method. Any of the three methods identified above may be used to establish live load capacities and load limits for purposes of load posting.

Bridge Owners should implement standardized procedures for determining the load rating of bridges based on this Manual.

Bridge load rating provides a basis for determining the safe load capacity of a bridge. Load rating requires engineering judgment in determining a rating value that is applicable to maintaining the safe use of the bridge and arriving at posting and permit decisions.

The specific load ratings are used in identifying the need for load posting or bridge strengthening and in making overweight-vehicle permit decisions. Load ratings are routinely reported to the NBI for national bridge administration and are also used in local bridge management systems.

This Section is intended for use in evaluating the types of highway bridges commonly in use in the United States that are subjected primarily to permanent loads and vehicular loads. Methods for the evaluation of existing bridges for extreme events such as earthquake, vessel collision, wind, flood, ice, or fire are not included herein. Rating of long-span bridges, movable bridges, and other complex bridges may involve additional considerations and loadings not specifically addressed in this Section and the rating procedures should be augmented with additional evaluation criteria where required.

6.1.1—Assumptions

The load rating of a bridge is based on existing structural conditions, material properties, loads, and traffic conditions at the bridge site. To maintain this capacity, the bridge is assumed to be subject to inspections at regular intervals, not to exceed the maximum interval cited in Article 4.2.4. Changes in existing structural conditions, material properties, loads, or site traffic conditions could require re-evaluation.

The procedures for computing load rating of concrete bridges are based on the assumptions that materials and construction are of good quality and there is no loss of material design strength, or, when warranted, the material

C6.1

Load and Resistance Factor Rating provisions in Part A of this Section have been carried over and updated from the 2003 AASHTO *Guide Manual for Condition Evaluation and Load and Resistance Factor Rating (LRFR) of Highway Bridges*. Allowable Stress and Load Factor rating procedures given in Section 6 of the 1994 AASHTO *Manual for Condition Evaluation of Bridges* have been incorporated in Part B. This Manual replaces both these documents and will serve as a single standard for bridge evaluation.

Load rating of bridges by nondestructive load testing is discussed in Section 8 of this Manual.

Primary focus of this Section is the assessment of the safety of bridges for live loads (including overloads) and fatigue. Extreme events have a very low probability of occurrence but impart very high-magnitude forces on a structure. Hydraulic considerations (scour/ice/debris), wind loads, temperature effects, collisions, and the effects of creep and shrinkage are generally not considered in the load rating of bridges, but should be considered under unique circumstance and structure types. The vulnerability to extreme events is an important bridge design consideration but it holds even greater significance in the overall safety assessment of existing bridges. It is important that Bridge Owners and evaluators recognize the vulnerabilities to these other failure modes so that a comprehensive safety assurance program may be developed for in-service bridges on a consistent and rational basis.

C6.1.1

Load rating of a bridge should not be undertaken without a recent thorough field inspection, which:

- Provides the condition data and other critical noncondition data necessary for evaluation,
- Minimizes the possibility of the evaluator making a gross error in assessing the capacity of a component or connection, and
- Improves bridge safety through early discovery of deterioration or signs of distress that could signal impending failure.

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As necessary, size, number, and relative location of bolts and rivets through tension members should be determined and recorded so that the net area of the section can be calculated. Also, in addition to the physical condition, threaded members such as truss rods at turnbuckles should be checked to see if the rod has been upset (check the diameter at the smallest diameter of the rod) so that the net area will be properly calculated. This information will normally be taken from plans when they are available, but should be determined in the field otherwise. Any misalignment, bends, or kinks in compression members should be measured carefully. Such defects will have a great effect on the load-carrying capability of a member and may be the controlling factor in the load-carrying capacity of the entire structure. Also, examine the connections of compression members carefully to see if they are detailed such that eccentricities are introduced which must be considered in the structural analysis.

The effective area of members to be used in the calculations shall be the gross area less that portion which has deteriorated due to decay or corrosion. The effective area should be adjusted for rivet or bolt holes in accordance with the *AASHTO LRFD Bridge Design Specifications* or *Standard Specifications for Highway Bridges*.

6.1.3—Evaluation Methods

This Manual provides analytical and empirical methods for evaluating the safe maximum live load capacity of bridges or for assessing their safety under a particular loading condition. Empirical methods are load ratings by load testing. Only the specific analytical method, Load and Resistance Factor Rating of bridges, is discussed in Part A of Section 6. Other analytical methods are discussed in Part B of Section 6, and load testing is discussed in Section 8.

C6.1.3

Load testing may be used as an alternative method to directly assess the load capacity of a bridge when analytical methods of evaluation are not applicable or need verification.

Safety assessment of a bridge using structural reliability methods may be used in special cases where the uncertainty in load or resistance is significantly different from that assumed in this Manual.

(Reference: NCHRP Report 454, *Calibration of Load Factors for LRFR Bridge Evaluation*.)

6.1.4—Bridges with Unknown Structural Components

For bridges where necessary details, such as reinforcement in a concrete bridge, are not available from plans or field measurements, a physical inspection of the bridge by a qualified inspector and evaluation by a qualified engineer may be sufficient to establish an approximate load rating based on rational criteria. Load tests may be helpful in establishing the safe load capacity for such structures.

A concrete bridge or concrete bridge length culvert with unknown details need not be posted for restricted loading if it has been carrying normal traffic for an appreciable period and shows no distress. The bridge shall be inspected regularly to verify satisfactory performance. A bridge may also be load tested to determine its capacity.

C6.1.4

Knowledge of the live load used in the original design, the current condition of the structure, and live load history may be used to provide a basis for assigning a safe load capacity. Bridge Owners may consider nondestructive proof load tests to establish a safe load capacity for such bridge.

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the bridge. Special considerations of these nonredundant components may be required in load rating the structure.

6.1.8—Qualifications and Responsibilities

A registered Professional Engineer shall be charged with the overall responsibility for bridge-capacity evaluation. The engineering expertise necessary to properly evaluate a bridge varies widely with the complexity of the bridge. A multi-disciplinary approach that utilizes the specialized knowledge and skills of other engineers may be needed in special situations for inspection and office evaluation.

6.1.9—Documentation of Load Rating

The load rating should be adequately documented, including all background information such as field inspection reports, material and load test data, all supporting computations, and a clear statement of all assumptions used in calculating the load rating. If a computer model was used, the input data file should be retained for future use.

in this area are provided in NCHRP Report 406, *Redundancy in Highway Bridge Superstructures.*

C6.1.8

Engineer qualifications are also subject to requirements specific to a state or Bridge Owner.

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PART A—LOAD AND RESISTANCE FACTOR RATING

6A.1—INTRODUCTION

6A.1.1—General

The load and resistance factor rating procedures of Part A provide a methodology for load rating a bridge consistent with the load and resistance factor design philosophy of the *AASHTO LRFD Bridge Design Specifications*.

C6A.1.1

A structure, designed and checked by the load and resistance factor method using HL-93 loading, may not require load rating calculations to determine the inventory or operating rating until changes to the structure occur that would reduce the inventory rating below the design load level. At the discretion of the owner, the inventory or operating ratings for the design load for these structures may be assigned based on the design loading. The HL-93 rating factors would generally be: Inventory 1.0, Operating 1.3.

Bridges will have adequate capacity for all AASHTO legal loads and state legal loads that fall within the exclusion limits described in the *AASHTO LRFD Bridge Design Specifications* if the design and assigned ratings are based on HL-93. Load rating calculations would be necessary if the force effects from state legal loads or permit loads exceed those from the design loading.

The LRFD and LRFR specifications are used for the design and load rating of modern bridge structures. Generally, the level and detail of analysis of a designed and checked bridge is much more comprehensive than a bridge load rating. Therefore, the inventory and operating rating may be conservatively determined from the design loading. See Article 6A.1.5.1, Design Load Rating.

The operating rating factor is determined by multiplying inventory rating by the ratio of the inventory live load factor (1.75) to the operating level live load factor (1.35). These live load factors for the Strength I limit state are provided in Table 6A.4.2.2-1.

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6A.1.2—Scope

Part A provides procedures for the rating of bridges using the load and resistance factor philosophy. Procedures are presented for load rating bridges for the LRFD design loading, AASHTO and state legal loads, and overweight permit loads. These procedures are consistent in philosophy and approach of the *AASHTO LRFD Bridge Design Specifications*. The methodology is presented in a format using load and resistance factors that have been calibrated based on structural reliability theory to achieve a minimum target reliability for the strength limit state. Guidance is provided on service limit states that are applicable to bridge load rating.

Specific provisions for the evaluation of horizontally curved steel-girder bridges are included in Article 6A.6.

C6A.1.2

The service limit states are not calibrated based on reliability theory to achieve a target reliability but are based on past practice. Part A provides guidance to incorporate these traditional service limit states into the evaluation.

6A.1.3—Philosophy

Bridge design and rating, though similar in overall approach, differ in important aspects. Bridge ratings generally require the Engineer to consider a wider range of variables than is typical in bridge design. Design may adopt a conservative reliability index and impose checks to ensure serviceability and durability without incurring a major cost impact. In rating, the added cost of overly conservative evaluation standards can be prohibitive as load restrictions, rehabilitation, and replacement become increasingly necessary.

The rating procedures presented herein recognize a balance between safety and economics. In most cases, a lower target reliability than design has been chosen for load rating at the strength limit state. Application of serviceability limit states to rating is done on a more selective basis than is prescribed for design in the *AASHTO LRFD Bridge Design Specifications*.

C6A.1.3

The term "evaluation criteria" denotes safety and serviceability standards adopted for assessing existing bridges.

LRFD calibration reported a target LRFD reliability index β of 3.5. The LRFD design criteria based on this index were derived for a severe traffic-loading case (including the presence of 5000*ADTT*). The LRFR procedures in Part A adopt a reduced target reliability index of approximately 2.5 calibrated to past AASHTO operating level load rating. This value was chosen to reflect the reduced exposure period, consideration of site realities, and the economic considerations of rating vs. design.

6A.1.4—Application of *AASHTO LRFD Bridge Design Specifications*

This Section of the Manual is consistent with the current *AASHTO LRFD Bridge Design Specifications*. Where this Section of the Manual is silent, the current *AASHTO LRFD Bridge Design Specifications* shall govern. Where appropriate, reference is made herein to specific articles in the *AASHTO LRFD Bridge Design Specifications*.

Where the behavior of a member under traffic is not consistent with that predicted by the governing specifications, as evidenced by a lack of visible signs of distress or excessive deformation or cases where there is evidence of distress even though the specification does not predict such distress, deviation from the governing specifications based on the known behavior of the member under traffic may be used and shall be fully documented. Material sampling, instrumentation, and load tests may be helpful in establishing the load capacity for such members.

C6A.1.4

Judgment must be exercised in evaluating a structure, and in some cases the evaluation criteria may be adjusted based on site conditions and/or structure conditions as recorded in the most recent inspection report. All data used in the decision to adjust the evaluation criteria shall be fully documented.

Nearly all existing bridges have been designed in accordance with the *AASHTO Standard Specifications for Highway Bridges*, most according to older editions of the specifications. The LRFD Specifications do not provide guidance on older bridge types that use materials and details no longer in common use. However, the *AASHTO LRFD Bridge Design Specifications* incorporate the state-of-the-art in design and analysis methods, loadings, and strength of materials.

Specifications are calibrated documents in which the loads, load factors, and design methods are part of the whole and should not be separated. Combining factors

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Strength is the primary limit state for load rating; service limit states are selectively applied. The results of the load rating for legal loads could be used as a basis for decision making related to load posting or bridge strengthening.

6A.1.5.3—Permit Load Rating

Permit load rating checks the safety and serviceability of bridges in the review of permit applications for the passage of vehicles above the legally established weight limitations. This is a third level rating that should be applied only to bridges having sufficient capacity for AASHTO legal loads. Calibrated load factors by permit type and traffic conditions at the site are specified for checking the load effects induced by the passage of the overweight truck. Guidance is also provided on the serviceability criteria that should be checked when reviewing permit applications.

6A.2—LOADS FOR EVALUATION

6A.2.1—General

Article 6A.2 describes the loads to be used in determining the load effects in the load rating equation provided in Article 6A.4.2. In general, only permanent loads and vehicular loads are considered to be of consequence in load rating. Environmental loads such as wind, ice, temperature, stream flow, and earthquake are usually not considered in rating except when unusual conditions warrant their inclusion.

6A.2.2—Permanent Loads and Load Factors

The load rating of bridges shall consider all permanent loads. Permanent loads include dead loads and locked-in force effects from the construction process.

6A.2.2.1—Dead Loads: *DC* and *DW*

The dead load effects on the structure shall be computed in accordance with the conditions existing at the time of analysis. Dead loads should be based on dimensions shown on the plans and verified with field measurements. Where present, utilities, attachments, and thickness of wearing surface should be field verified at the time of inspection. Minimum unit weights of materials used in computing dead loads should be in accordance with LRFD Design Table 3.5.1-1, in the absence of more precise information.

6A.2.2.2—Permanent Loads Other Than Dead Loads: *P*

Secondary effects from post-tensioning shall be considered as permanent loads.

C6A.2.2

Allowance for future wearing surface need not be provided in evaluation.

C6A.2.2.1

Care should be exercised in estimating the weight of concrete decks because significant variations of deck thickness have been found. Wearing surface thicknesses are also highly variable. Multiple measurements at curbs and roadway centerline should be used to determine an average wearing surface thickness.

C6A.2.2.2

In continuous post-tensioned bridges, secondary moments are introduced as the member is stressed.

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6A.2.2.3—Load Factors

Load factors for permanent loads are as given in Table 6A.4.2.2-1. If the wearing surface thickness is field measured, γ_{DW} may be taken as 1.25.

A load factor of 1.0 shall be applied to the secondary effects from post-tensioning, cited in Article 6A.2.2.2 ($\gamma_p = 1.0$).

6A.2.3—Transient Loads

6A.2.3.1—Vehicular Live Loads (Gravity Loads): *LL*

The nominal live loads to be used in the evaluation of bridges are selected based on the purpose and intended use of the evaluation results. Live load models for load rating include:

Design Load: HL-93 Design Load per LRFD Design Specifications

Legal Loads:
1. AASHTO legal loads, as specified in Article 6A.4.4.2.1a.
2. The Notional Rating Load as specified in Article 6A.4.4.2.1b or state legal loads.

Permit Load: Actual Permit Truck

Load factors for vehicular live loads appropriate for use in load rating are as specified in Articles 6A.4.3.2.2, 6A.4.4.2.3, and 6A.4.5.4.2.

State legal loads having only minor variations from the AASHTO legal loads should be evaluated using the same procedures and factors specified for AASHTO trucks in this Manual.

State legal loads significantly heavier than the AASHTO legal loads should be load rated using load factors specified for routine permits in this Manual, if the span has sufficient capacity for AASHTO legal loads.

C6A.2.3.1

The evaluation of bridge components to include the effects of longitudinal braking forces, specified in LRFD Design Article 3.6.4 in combination with dead- and live load effects, should be done only where the evaluator has concerns about the longitudinal stability of the structure.

Bridges that do not satisfy the HL-93 design load check should be evaluated for legal loads in accordance with the provisions of Article 6A.4.4 to determine the need for load posting or strengthening. Legal loads for rating given in Article 6A.4.4.2.1a that model routine commercial traffic are the same family of three AASHTO trucks (Type 3, Type 3S2, and Type 3-3) used in current and previous AASHTO evaluation Manuals. The single-unit legal load models given in Article 6A.4.4.2.1b represent the increasing presence of Formula B multi-axle specialized hauling vehicles in the traffic stream in many states.

6A.2.3.2—Application of Vehicular Live Load

The number of traffic lanes to be loaded and the transverse placement of wheel lines shall be in conformance with the *AASHTO LRFD Bridge Design Specifications* and the following:

- Roadway widths from 18 to 20 ft shall have two traffic lanes, each equal to one half the roadway width.
- Roadway widths less than 18 ft shall carry one traffic lane only.

C6A.2.3.2

In the past, a distance as little as 1 ft between wheel load and edge of the roadway was used for rating by some agencies. This deviation from design is considered overly conservative and especially affected the rating of exterior stringers. The design of exterior stringers in many older bridges, especially those designed prior to 1957, may not have included a minimum live load distribution to the outside stringers.

In design it is required to place lanes starting at the face of the barrier, including the shoulder areas. Many existing truss/arch and two-girder bridges have wide shoulders, so moving the wheel load within 2 ft of the barrier could significantly increase load distribution to the truss/arch/girder and result in lower ratings. Similarly, for multi-girder bridges with large overhangs, establishing live

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- The center of any wheel load shall not be closer than 2.0 ft from the edge of a traffic lane or face of the curb.
- The distance between adjacent wheel lines of passing trucks shall not be less than 4.0 ft.
- The standard gauge width, distance between the wheels of a truck shall be taken to be 6.0 ft unless noted otherwise.

Utilizing the number and transverse placement of lanes in accordance with *AASHTO LRFD Bridge Design Specifications* may not be consistent with the actual usage of the bridge as defined by the striped lanes, and could result in conservative load ratings for bridge types such as trusses, two-girder bridges, arches, and exterior girders, where live load distribution factors are established using the lever rule method. Upon the approval of the bridge owner, an alternate load rating of the bridge for normal operating conditions or current usage may be performed by placing truck loads only within the striped lanes. When load rating a structure based on the existing striped lanes, the transverse positioning of the truck should include placing the wheel load anywhere within the lane, including on the lane stripe. This alternate load rating may be performed for all live load models. Placement of striped lanes on the bridge should be field verified and documented in the inspection report.

load distribution (using the lever rule method or rigid deck distribution method) by moving the wheel load within 2 ft of the barrier significantly increases the load distribution to the exterior girder and results in lower ratings. Moving the wheel load within 2 ft of the barrier is consistent with design but not consistent with how the bridge is loaded under normal service conditions. Ratings per the design code lane placement will provide ratings for the maximum likely live loading for the bridge, whereas the ratings with wheel loads within striped lanes will provide ratings under current use conditions. Placing loads within striped lanes is also particularly relevant to permit reviews, where permit trucks are required to operate in the traffic lanes. It is recognized that on occasion a random vehicle may enter the roadway areas outside of the striped lanes. The probability that this will occur simultaneously with full live loads in the striped lanes is quite low. In cases where the striped lanes are less than 10 ft wide, the load should be centered in the lane so that the distance to the wheel load from the barrier would be less than 2 ft.

6A.2.3.3—Dynamic Load Allowance: *IM*

The dynamic load allowance for evaluation shall be as specified in Articles 6A.4.3.3, 6A.4.4.3, and 6A.4.5.5.

C6A.2.3.3

In the AASHTO Standard Specifications, the dynamic load allowance was termed impact.

Part A allows the use of reduced dynamic load allowance for load rating under certain conditions as discussed in Article C6A.4.4.3.

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6A.2.3.4—Pedestrian Live Loads: *PL*

Pedestrian loads on sidewalks need not be considered simultaneously with vehicular loads when load rating a bridge unless the Engineer has reason to expect that significant pedestrian loading will coincide with maximum vehicular loading. Pedestrian loads considered simultaneously with vehicular loads in calculations for load ratings shall be the probable maximum loads anticipated, but in no case should the loading exceed the value specified in LRFD Design Article 3.6.1.6.

6A.2.3.5—Wind Loads: *WL* and *WS*

Wind loads need not be considered unless special circumstances justify otherwise.

C6A.2.3.5

Wind loads are not normally considered in load rating. However, the effects of wind on special structures such as movable bridges, long-span bridges, and other high-level bridges should be considered in accordance with applicable standards.

6A.2.3.6—Temperature Effects: *TG* and *TU*

Temperature effects need not be considered in calculating load ratings for nonsegmental bridge components that have been provided with well-distributed steel reinforcement to control thermal cracking.

C6A.2.3.6

Where temperature effects are considered, a reduced long-term modulus of elasticity for concrete may be used in the analysis.

Temperature gradient, *TG,* may be considered when evaluating segmental bridges.

6A.2.3.7—Earthquake Effects: *EQ*

Earthquake effects need not be considered in calculating load ratings.

C6A.2.3.7

In regions prone to seismic activity, the safety of bridges under earthquake loads may be evaluated in accordance with the provisions of *Seismic Retrofitting Manual for Highway Bridges, FHWA-RD94-052*, May 1995.

6A.2.3.8—Creep and Shrinkage: *CR* and *SH*

Creep and shrinkage effects do not need to be considered in calculating load ratings where there is well-distributed reinforcement to control cracking in nonsegmental, nonprestressed components.

6A.3—STRUCTURAL ANALYSIS

6A.3.1—General

Methods of structural analysis suitable for the evaluation of bridges shall be as described in Section 4 of the *AASHTO LRFD Bridge Design Specifications* and in this Section.

C6A.3.1

Evaluation seeks to verify adequate performance of existing bridges with an appropriate level of effort. Within a given evaluation procedure, the evaluator has the option of using simplified methods that tend to be somewhat conservative or pursue a more refined approach for improved accuracy. It is recommended that wherever feasible, simplified evaluation procedures should be first applied before resorting to higher level evaluation methods. Refined approaches to capacity evaluation of existing bridges can be economically justified where increased capacity is required to achieve a desired safe load capacity or permit load capability.

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6A.3.2—Approximate Methods of Structural Analysis

Except as specified herein, approximate methods of distribution analysis as described in LRFD Design Article 4.6.2 may be used for evaluating existing straight bridges.

For steel box-girder bridges, the provisions of LRFD Design Article 6.11.1.1 shall apply in determining the applicability of approximate analysis methods.

Approximate analysis of horizontally curved steel bridges may be used provided that the Engineer ascertains that approximate analysis methods are appropriate according to the provisions of LRFD Design Article 4.6.2.2.4. The effects of curvature may be ignored in the determination of the major-axis bending moments in horizontally curved steel I- and box-girder bridges provided that the appropriate conditions specified in LRFD Design Articles 4.6.1.2.4b and 4.6.1.2.4c, respectively, are satisfied.

The multiple presence factor of 1.2 which is included in the LRFD distribution factors for single-lane loadings should not be used when checking fatigue or special permit loads. Adjustments in distributions to account for traffic volume provided in the *AASHTO LRFD Bridge Design Specifications* should also not be factored into the evaluation distribution factors.

C6A.3.2

The live load distribution formulas provided in the *AASHTO LRFD Bridge Design Specifications* were developed for common bridge types and dimensions, for the HS family of trucks. Their validity has been verified for parameter variations within specific ranges as indicated in the tables of LRFD Design Article 4.6.2. The live load distribution formulas can also be applied to the AASHTO family of legal trucks, and permit vehicles whose overall width is comparable to the width of the design truck. If the bridge or loading parameters fall outside these specified ranges, the accuracy is reduced or the formulas may not be applicable. In such cases, refined methods of analysis should be considered.

Applying a multi-lane distribution factor to a loading involving a heavy permit truck only in one lane can be overly conservative. Permit load rating procedures provided in Article 6A.4.5 should be applied to the review of permits. The live load factors for permit loads given in Table 6A.4.5.4.2a-1 have been derived for the possibility of simultaneous presence of nonpermit trucks on the bridge when the permit vehicle crosses the span.

Engineers using the LRFD live load distribution formulas may find distributions for multi-lane loadings now reduced on the average by some ten percent compared to distributions computed with simplified *S*/over approximations of the AASHTO *Standard Specifications for Highway Bridges*. However, the reduction in the distributions for single-lane loading computed by the *AASHTO LRFD Bridge Design Specifications* and compared to the *S*/over formulas will be much greater and differ by 30–40 percent or more. The distributions for single lane are important when checking special permits or fatigue life estimates, which both use single-lane distributions.

Unusual wheel configurations and wider gauge widths may be characteristic of certain permit vehicles. The AASHTO LRFD distribution factors were developed using the HS-20 truck model that has a standard 6-ft gauge width. Sensitivity studies of the load distribution factor to several different truck parameters indicate that most parameters such as gross weight, number of axles, and axle spacings have only a small effect on the load distribution factor for flexure. It was found that the single most important parameter is gauge width. The distribution factor is generally lower for increased gauge widths.

Exterior girders of existing bridges may have been designed for less capacity than the interior girders. Additionally, they may also be subjected to increased deterioration due to their increased environmental exposure. Approximate methods of analysis for exterior girders are often less reliable than interior girders due to the structural participation of curbs and parapets. The level of structural participation could vary from bridge to bridge. Field testing (load testing) procedures described in Section 8 of this Manual may be employed to verify the behavior of exterior girders.

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Prestressed concrete adjacent box-beam and slab bridges built prior to 1970 may not have sufficient transverse post-tensioning (LRFD Design Article C4.6.2.2.1 requires a minimum prestress of 0.25 ksi) to act as a unit. These bridges should be analyzed using the *S*/*D* method of live load distribution provided in the *AASHTO LRFD Bridge Design Specifications*.

Analysis of segmental bridges is covered in LRFD Design Article 4.6.2.9.

6A.3.3—Refined Methods of Analysis

Bridges that exhibit insufficient load capacity when analyzed by approximate methods, and bridges or loading conditions for which accurate live load distribution formulas are not readily available may be analyzed by refined methods of analysis as described in LRFD Design Article 4.6.3.

As specified in LRFD Design Article 4.6.3.3.2, analysis of bridges curved in plan should be performed using refined methods of analysis, unless the Engineer ascertains that approximate methods of analysis are appropriate.

C6A.3.3

Some cases where refined analysis methods would be considered appropriate include:

- girder spacings and span lengths outside the range of LRFD-distribution formulas,
- varying skews at supports,
- curved bridges,
- low-rated bridges, and
- permit loads with nonstandard gage widths and large variations in axle configurations.

Many older bridges have parapets, railings, and curbs that are interrupted by open joints. The stiffness contribution of these elements to bridge response should be verified by load testing, if they are to be included in a refined analysis.

Most analytical models are based on linear response, where load effect is proportional to the load applied. Conversely, the resistance models used for design and evaluation assume nonlinear response at the strength limit state. The rationale for this inconsistency is found in the "lower bound theorem", which states that for a structure that behaves in a ductile manner the collapse load computed on the basis of an assumed equilibrium moment diagram is less than or equal to the true ultimate collapse load. Restated in simpler terms, the theorem implies that as long as the requirements of ductility and equilibrium are satisfied, the exact distribution of internal force effects is not required at the strength limit state. The lower bound theorem does not apply in cases where buckling may occur prior to yielding and redistribution of force effects.

Evaluation of the fatigue and service limit states is concerned with nonductile failure modes and service level loads where there is little likelihood of load redistribution. Hence, the lower bound theorem does not apply to these limit states. Analytical procedures that underestimate the load effects in some locations and overestimate the effects in others, while acceptable at the strength limit state may result in significant inaccuracies for the fatigue and service limit states. Refined analysis procedures that can properly model the relative stiffnesses of all bridge components assume added significance when evaluating bridges for nonstrength-related criteria. Use of refined analytical

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methods could significantly influence the repair/rehabilitation strategy or posting load that may be governed by service or fatigue criteria.

When a refined method of analysis is used, a table of distribution factors for extreme force effects in each span should be provided in the load rating report to aid in future load ratings.

6A.3.4—Analysis by Field Testing

C6A.3.4

Bridges may be evaluated by field testing (load testing) if the evaluator feels that analytical approaches do not accurately represent the true behavior and load distribution of the structure and its components. Procedures for load testing are described in Section 8 of this Manual.

One important use of diagnostic load tests is to confirm the precise nature of load distribution to the main load-carrying members of a bridge and to the individual components of a multi-component member.

6A.4—LOAD-RATING PROCEDURES

6A.4.1—Introduction

C6A.4.1

Three load-rating procedures that are consistent with the load and resistance factor philosophy have been provided in Article 6A.4 for the load capacity evaluation of in-service bridges:

- design load rating (first level evaluation)
- legal load rating (second level evaluation)
- permit load rating (third level evaluation)

The load-rating procedures are structured to be performed in a sequential manner, as needed, starting with the design load rating (see flowchart in Appendix A6A). Load rating for AASHTO legal loads is required only when a bridge fails ($RF < 1$) the design load rating at the operating level. Similarly, only bridges that pass the load rating for AASHTO legal loads should be evaluated for overweight permits.

Each procedure is geared to a specific live load model with specially calibrated load factors aimed at maintaining a uniform and acceptable level of reliability in all evaluations.

Load factors for evaluation may be taken from Articles 6A.4.3, 6A.4.4, and 6A.4.5, as applicable. Where adequate information on the traffic is available, site-, route-, or region-specific load factors may be developed. If accepted by the Owner, these load factors may be used in lieu of the values given in this Manual.

Weigh-In-Motion (WIM) data collected at a specific site, along a specific route, or around a specific region may be used to perform a load calibration to determine site-, route-, or region-specific load factors. Depending on the traffic pattern and truck counts, these load factors may be higher or lower than those listed in this Manual.

The load rating is generally expressed as a rating factor for a particular live load model, using the general load-rating equation provided in Article 6A.4.2.

The live load factors given in Articles 6A.4.3.2, 6A.4.4.2, and 6A.4.5.4 are established assuming that demands based on the two or more lane load distribution factor will always govern over the demands based on single-lane distribution factor for design loads, legal loads, and routine permits. However, in the unusual case where the single-lane distribution factor is greater than the two-lane distribution factor, the higher distribution factor shall be used.

Calibration of LRFR load factors was done using the simplified live load distribution factors of LRFD Design. During this calibration, it was assumed that the demands based on the two or more lane load scenario will govern over the demands based on the single lane loaded case for multi-lane bridges for legal and routine permit trucks. However, there may be a few instances where the single lane distribution factor (based on the simplified equations given in LRFD Design, the lever rule method, or the rigid cross-section distribution of exterior girder) may exceed the two-lane distribution factor. In these situations, to be conservative, use of the higher live load distribution factor with the load factors in this Manual is recommended.

The multiple presence factor listed in LRFD Design does not apply for permit trucks. Therefore, the multiple presence factor should not be considered when establishing the load demands for routine permit trucks, when the single-lane distribution governs.

The multiple presence factor (MPF, m) of 1.2 incorporated within the single-lane distribution factor equation accounts for the presence of a vehicle heavier than the design or rating truck in the lane. In other words, MPF

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listed within LRFD Design is applicable for design loadings (HL-93) and legal trucks only, not for permit trucks. As a result, the MPF for permit trucks was not considered during the LRFR calibration of permit loads. The MPF should not be considered when establishing the load demand of either routine or special permit trucks. It is important to note that when comparing the simplified live load distribution factors to establish the demand of routine permit trucks, the (MPF, m) of 1.2 included within the simplified single-lane distribution factor equation should be removed.

Article 6A.4.5.4.2c addresses the use of refined analysis for permits and provides an adjustment to the load factors when using refined analysis methods (such as 3-D analysis). Again, it is not necessary to incorporate the (MPF, m) for permit trucks during evaluation.

6A.4.2—General Load-Rating Equation

6A.4.2.1—General

The following general expression shall be used in determining the load rating of each component and connection subjected to a single force effect (i.e., axial force, flexure, or shear):

$$RF = \frac{C - (\gamma_{DC})(DC) - (\gamma_{DW})(DW) \pm (\gamma_P)(P)}{(\gamma_{LL})(LL + IM)} \quad (6A.4.2.1\text{-}1)$$

For the strength limit states:

$$C = \varphi_c \varphi_s \varphi R_n \quad (6A.4.2.1\text{-}2)$$

Where the following lower limit shall apply:

$$\varphi_c \varphi_s \geq 0.85 \quad (6A.4.2.1\text{-}3)$$

For the service limit states:

$$C = f_R \quad (6A.4.2.1\text{-}4)$$

where:

RF = Rating factor

C = Capacity

f_R = Allowable stress specified in the LRFD code

R_n = Nominal member resistance (as inspected)

DC = Dead load effect due to structural components and attachments

C6A.4.2.1

It should be noted that load modifiers, η, relating to ductility, redundancy, and operational importance contained in Article 1.3.2.1 of the *AASHTO LRFD Bridge Design Specifications* are not included in the general load-rating equation. In load rating, ductility is considered in conjunction with redundancy and incorporated in the system factor, φ_s. Operational importance is not included as a factor in the LRFR load rating provisions.

The load rating of a deteriorated bridge should be based on a recent thorough field inspection. Only sound material should be considered in determining the nominal resistance of the deteriorated section. Load ratings may also be calculated using as-built member properties to serve as a baseline for comparative purposes.

Resistance factor, φ, has the same value for new design and for load rating. Also, φ = 1.0 for all nonstrength limit states. For condition factors, see Article 6A.4.2.3. For system factors, see Article 6A.4.2.4.

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DW = Dead load effect due to wearing surface and utilities

P = Permanent loads other than dead loads

LL = Live load effect

IM = Dynamic load allowance

γ_{DC} = LRFD load factor for structural components and attachments

γ_{DW} = LRFD load factor for wearing surfaces and utilities

γ_P = LRFD load factor for permanent loads other than dead loads = 1.0

γ_{LL} = Evaluation live load factor

φ_c = Condition factor

φ_s = System factor

φ = LRFD resistance factor

The load rating shall be carried out at each applicable limit state and load effect with the lowest value determining the controlling rating factor. Limit states and load factors for load rating shall be selected from Table 6A.4.2.2-1.

Components subjected to combined load effects shall be load rated considering the interaction of load effects (e.g., axial-bending interaction or shear-bending interaction), as provided in this Manual under the sections on resistance of structures.

Secondary effects from prestressing of continuous spans and locked-in force effects from the construction process should be included as permanent loads other than dead loads, P (see Articles 6A.2.2.2. and 6A.2.2.3).

6A.4.2.2—Limit States

Strength is the primary limit state for load rating; service and fatigue limit states are selectively applied in accordance with the provisions of this Manual. Applicable limit states are summarized in Table 6A.4.2.2-1.

C6A.4.2.2

Service limit states that are relevant to load rating are discussed under the articles on resistance of structures (see Articles 6A.5, 6A.6, and 6A.7).

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Table 6A.4.2.2-1—Limit States and Load Factors for Load Rating

Bridge Type	Limit State*	Dead Load γ_{DC}	Dead Load γ_{DW}	Design Load Inventory γ_{LL}	Design Load Operating γ_{LL}	Legal Load γ_{LL}	Permit Load γ_{LL}
Steel	Strength I	1.25	1.50	1.75	1.35	Tables 6A.4.4.2.3a-1 and 6A.4.4.2.3b-1	—
	Strength II	1.25	1.50	—	—	—	Table 6A.4.5.4.2a-1
	Service II	1.00	1.00	1.30	1.00	1.30	1.00
	Fatigue	0.00	0.00	0.75	—	—	—
Reinforced Concrete	Strength I	1.25	1.50	1.75	1.35	Tables 6A.4.4.2.3a-1 and 6A.4.4.2.3b-1	—
	Strength II	1.25	1.50	—	—	—	Table 6A.4.5.4.2a-1
	Service I	1.00	1.00	—	—	—	1.00
Prestressed Concrete	Strength I	1.25	1.50	1.75	1.35	Tables 6A.4.4.2.3a-1 and 6A.4.4.2.3b-1	—
	Strength II	1.25	1.50	—	—	—	Table 6A.4.5.4.2a-1
	Service III	1.00	1.00	0.80	—	1.00	—
	Service I	1.00	1.00	—	—	—	1.00
Wood	Strength I	1.25	1.50	1.75	1.35	Tables 6A.4.4.2.3a-1 and 6A.4.4.2.3b-1	—
	Strength II	1.25	1.50	—	—	—	Table 6A.4.5.4.2a-1

* Defined in the *AASHTO LRFD Bridge Design Specifications*

Notes:

- Shaded cells of the table indicate optional checks.
- Service I is used to check the 0.9 F_y stress limit in reinforcing steel.
- Load factor for DW at the strength limit state may be taken as 1.25 where thickness has been field measured.
- Fatigue limit state is checked using the LRFD fatigue truck (see Article 6A.6.4.1).

6A.4.2.3—Condition Factor: φ_c

Use of condition factors as presented below may be considered optional based on an agency's load-rating practice.

The condition factor provides a reduction to account for the increased uncertainty in the resistance of deteriorated members and the likely increased future deterioration of these members during the period between inspection cycles.

Table 6A.4.2.3-1—Condition Factor: φ_c

Structural Condition of Member	φ_c
Good or Satisfactory	1.00
Fair	0.95
Poor	0.85

C6A.4.2.3

The uncertainties associated with the resistance of an existing intact member are at least equal to that of a new member in the design stage. Once the member experiences deterioration and begins to degrade, the uncertainties and resistance variabilities are greatly increased (scatter is larger).

Additionally, it has been observed that deteriorated members are generally prone to an increased rate of future deterioration when compared to intact members. Part of φ_c relates to possible further section losses prior to the next inspection and evaluation.

Improved inspections will reduce, but not totally eliminate, the increased scatter or resistance variability in deteriorated members. Improved inspection and field measurements will reduce the uncertainties inherent in identifying the true extent of deterioration for use in calculating the nominal member resistance. If section properties are obtained accurately, by actual field measurement of losses rather than by an estimated percentage of losses, the values specified for φ_c in Table 6A.4.2.3-1 may be increased by 0.05 ($\varphi_c \leq 1.0$).

The condition factor, φ_c, tied to the structural condition of the member, accounts for the member deterioration due to natural causes (e.g., atmospheric corrosion). Damage caused by accidents is specifically not considered here.

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If condition information is collected and recorded in the form of NBI condition ratings only (not as element level data), then the following approximate conversion may be applied in selecting φ_c.

Table C6A.4.2.3-1—Approximate Conversion in Selecting φ_c

Superstructure Condition Rating (SI & A Item 59)	Equivalent Member Structural Condition
6 or higher	Good or Satisfactory
5	Fair
4 or lower	Poor

6A.4.2.4—System Factor: φ_s

System factors are multipliers applied to the nominal resistance to reflect the level of redundancy of the complete superstructure system. Bridges that are less redundant will have their factored member capacities reduced, and, accordingly, will have lower ratings.

System factors that correspond to the load factor modifiers in the *AASHTO LRFD Bridge Design Specifications* should be used. The system factors in Table 6A.4.2.4-1 are more conservative than the LRFD design values and may be used at the discretion of the evaluator until they are modified in the *AASHTO LRFD Bridge Design Specifications*; however, when rating nonredundant superstructures for legal loads using the generalized load factors given in Article 6A.4.4.2.3, the system factors from Table 6A.4.2.4-1 shall be used to maintain an adequate level of system safety.

The system factor for riveted and bolted gusset plates and their connections for all force effects shall be taken as 0.90.

Table 6A.4.2.4-1—System Factor: φ_s for Flexural and Axial Effects

Superstructure Type	φ_s
Welded Members in Two-Girder/Truss/Arch Bridges	0.85
Riveted Members in Two-Girder/Truss/Arch Bridges	0.90
Multiple Eyebar Members in Truss Bridges	0.90
Three-Girder Bridges with Girder Spacing 6 ft	0.85
Four-Girder Bridges with Girder Spacing ≤4 ft	0.95
All Other Girder Bridges and Slab Bridges	1.00
Floorbeams with Spacing >12 ft and Noncontinuous Stringers	0.85
Redundant Stringer Subsystems between Floorbeams	1.00

C6A.4.2.4

Structural members of a bridge do not behave independently, but interact with other members to form one structural system. Bridge redundancy is the capability of a bridge structural system to carry loads after damage to or the failure of one or more of its members. Internal redundancy and structural redundancy that exists as a result of continuity are neglected when classifying a member as nonredundant.

If Table 6A.4.2.4-1 is used, the system factors are used to maintain an adequate level of system safety. Nonredundant bridges are penalized by requiring their members to provide higher safety levels than those of similar bridges with redundant configurations. The aim of φ_s is to add a reserve capacity such that the overall system reliability is increased from approximately an operating level (for redundant systems) to a more realistic target for nonredundant systems corresponding to inventory levels.

If the Engineer can demonstrate the presence of adequate redundancy in a superstructure system (Reference: NCHRP Report 406), then φ_s may be taken as 1.0. In some instances, the level of redundancy may be sufficient to utilize a value of φ_s greater than 1.0, but in no instance should φ_s be taken as greater than 1.2.

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The single safe load capacity produced by the procedures presented in this Manual considers redundancy and bridge condition in the load-rating process. The load and resistance factors have been calibrated to provide uniform levels of reliability and permit the introduction of bridge- and site-specific data in a rational and consistent format. It provides a level of reliability corresponding to the operating level reliability for redundant bridges in good condition. The capacity of nonredundant bridges and deteriorated bridges should be reduced during the load-rating process by using system factors and condition factors. The safe load capacity may approach or exceed the equivalent of operating rating for redundant bridges in good condition on low traffic routes, and may fall to the equivalent of inventory levels or below for heavily deteriorated, nonredundant bridges on high traffic routes.

6A.4.4.2—Live Loads and Load Factors

6A.4.4.2.1—Live Loads

C6A.4.4.2.1

6A.4.4.2.1a—Routine Commercial Traffic

C6A.4.4.2.1a

The AASHTO legal vehicles and lane-type load models shown in Figures D6A-1 through D6A-5 shall be used for load rating bridges for routine legal commercial traffic.

For all span lengths the critical load effects shall be taken as the larger of the following:

- For all load effects, AASHTO legal vehicles (Type 3, Type 3S2, Type 3-3; applied separately) or state legal loads.
- For negative moments and reactions at interior supports, a lane load of 0.2 klf combined with two AASHTO Type 3-3 vehicles or state legal loads multiplied by 0.75 heading in the same direction separated by 30 ft.

Take the largest of Type 3, Type 3S2, Type 3-3 vehicles, or state legal loads plus lane loads. The lane load model is common to all truck types. In addition, for span lengths greater than 200 ft, critical load effects shall be created by:

- AASHTO Type 3-3 or state legal load multiplied by 0.75 and combined with a lane load of 0.2 klf.

Dynamic load allowance shall be applied to the AASHTO legal vehicles and state legal loads but not the lane loads. If the *ADTT* is less than 500, the lane load may be excluded and the 0.75 factor changed to 1.0 if, in the Engineer's judgment, it is warranted.

Usually bridges are load rated for all three AASHTO trucks and lane loads to determine the governing loading and governing load rating. A safe load capacity in tons may be computed for each vehicle type (see Article 6A.4.4.4). When the lane type, load model governs the load rating, the equivalent truck weight for use in calculating a safe load capacity for the bridge shall be taken as 80 kips.

AASHTO legal vehicles, designated as Type 3, Type 3S2, and Type 3-3 are sufficiently representative of average truck configurations in use today, and are used as vehicle models for load rating. These vehicles are also suitable for bridge posting purposes. Load ratings may also be performed for state legal loads that have only minor variations from the AASHTO legal loads using the live load factors provided in Table 6A.4.4.2.3a-1 for the AASHTO vehicles. It is unnecessary to place more than one vehicle in a lane for spans up to 200 ft because the load factors provided have been modeled for this possibility.

The federal bridge formula (Reference: TRB Special Report 225, *Truck Weight Limits Issues and Options*, 1990) restricts truck weights on interstate highways through (a) a total, or gross, vehicle weight limit of 80 kips; (b) limits on axle loads (20 kips for single axles, 34 kips for tandem axles); and (c) a bridge formula that specifies the maximum allowable weight on any group of consecutive axles based on the number of axles in the group and the distance from first to the last axles. Grandfather provisions in the federal statutes allow states to retain higher limits than these if such limits were in effect when the applicable federal statutes were first enacted.

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The objective of producing new *AASHTO LRFD Bridge Design Specifications* that will yield designs having uniform reliability required as its basis a new live load model with a consistent bias when compared with the exclusion vehicles. The model consisting of either the HS-20 truck plus the uniform lane load or the tandem plus the uniform lane load (designated as HL-93 loading) resulted in a tight clustering of data around a 1.0 bias factor for all force effects over all span lengths. This combination load was therefore found to be an adequate basis for a notional design load in the *AASHTO LRFD Bridge Design Specifications*.

While this notional design load provides a convenient and uniform basis for design and screening of existing bridges against new bridge safety standards, it has certain limitations when applied to evaluation. The notional design load bears no resemblance or correlation to legal truck limits on the roads and poses practical difficulties when applied to load rating and load posting of existing bridges.

A characteristic of the AASHTO family of legal loads (Type 3, Type 3S2, and Type 3-3) is that the group satisfies the federal bridge formula. The AASHTO legal loads model three portions of the bridge formula which control short, medium, and long spans. Therefore, the combined use of these three AASHTO legal loads results in uniform bias over all span lengths, as achieved with the HL-93 notional load model (see Figure C6A-1). These vehicles are presently widely used for load rating and load posting purposes. These AASHTO vehicles model many of the configurations of present truck traffic. They are appropriate for use as rating vehicles as they satisfy the major aim of providing uniform reliability over all span lengths. They are also widely used as truck symbols on load posting signs. Additionally, these vehicles are familiar to engineers and provide continuity with current practice.

6A.4.4.2.1b—Specialized Hauling Vehicles

The notional rating load (NRL) shown in Figure D6A-6, which envelopes the load effects of the Formula B specialized hauling vehicle configurations (see Figure D6A-7) weighing up to 80 kips, should be used for legal load ratings.

C6A.4.4.2.1b

The vehicles referred to as specialized hauling vehicles (SHV) are legal single-unit, short-wheelbase, multiple-axle trucks commonly used in the construction, waste management, bulk cargo, and commodities hauling industries.

Trucks weighing up to 80 kips are typically allowed unrestricted operation and are generally considered "legal" provided they meet weight guidelines of Federal Bridge Formula B (Formula B). In the past, the maximum legal weight for short-wheelbase trucks was usually controlled by Formula B rather than by the 80 kips gross weight limit. Since the adoption of the AASHTO family of three legal loads, the trucking industry has introduced specialized single-unit trucks with closely spaced multiple axles that make it possible for these short-wheelbase trucks to carry the maximum load of up to 80,000 lb and still meet Formula B. The AASHTO family of three legal loads selected at the time to closely match the Formula B in the short, medium, and long truck length ranges do not represent these newer axle configurations. These SHV trucks cause force effects that exceed the stresses induced

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by HS-20 in bridges by up to 22 percent and by the Type 3, 3S2, or 3-3 posting vehicles by over 50 percent, in certain cases. The shorter bridge spans are most sensitive to the newer SHV axle configurations.

The NRL represents a single load model that will envelop the load effects on simple and continuous span bridges of the worst possible Formula B single-unit truck configurations with multiple axles up to 80 kips. It is called "notional" because it is not intended to represent any particular truck. Vehicles considered to be representative of the newer Formula B configurations were investigated through the analysis of weigh-in-motion data and other truck and survey data obtained from the states (refer to NCHRP Project 12-63 Final Report). Bridges that rate for the NRL loading will have adequate load capacity for all legal Formula B truck configurations up to 80 kips. Bridges that do not rate for the NRL loading should be investigated to determine posting needs using the single-unit posting loads SU4, SU5, SU6, and SU7, specified in Article 6A.8.2. These SU trucks were developed to model the extreme loading effects of single-unit SHVs with four or more axles.

The Federal tandem axle weight limit on the Interstate System is 34,000 pounds. Although tandem axles are generally defined as an axle group consisting of two axles, CFR Title 23 658.5 defines tandem axle in terms of the distance between the outer axles. That is, two or more consecutive axles whose centers may be spaced more than 40 in. and not more than 96 in. apart.

By this definition, the total weight on three axles spaced at 8 ft between the outer axles would be limited to 34,000 pounds. Spacings even slightly above 8 ft would be governed by Formula B with a maximum weight of 42,000 pounds.

The SHVs referenced in Article 6A.8.2 are load models for rating and posting that show three axle groups with a spacing of 8 ft between the outer axles and a total weight on the three axles of 42,000 lb. This is not in strict compliance with the tandem axle definition in the CFR. However, these load models provide analysis efficiency in load ratings by serving as envelope vehicles and are not meant as examples of real trucks, where the actual axle spacings may be slightly different.

In context of the previous paragraph, for all practical purposes, the SHVs are compliant with Formula B.

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In the NRL loading, axles that do not contribute to the maximum load effect under consideration should be neglected. For instance, axles that do not contribute to the maximum positive moments need to be neglected or they will contribute to bending in the opposite (negative) direction. This requirement may only affect certain continuous bridges, usually with short span lengths. The drive axle spacing of 6 ft may also be increased up to 14 ft to maximize load effects. Increasing the drive axle spacing to 14 ft could result in a slight increase in moments, again in continuous span bridges.

It is unnecessary to consider more than one NRL loading per lane. Load ratings may also be performed for state legal loads that have only minor variations from the AASHTO legal loads using the live load factors provided in Tables 6A.4.4.2.3a-1 and 6A.4.4.2.3b-1.

6A.4.4.2.2—Live Load Factors

C6A.4.4.2.2

The LRFR provisions provide generalized live load factors for load rating that have been calibrated to provide a uniform and acceptable level of reliability. Load factors appropriate for use with the AASHTO and state legal vehicles are defined based on the traffic data available for the site.

Traffic conditions at bridge sites are usually characterized by traffic volume. The *ADTT* at a site is usually known or can be estimated. Generalized load factors are representative of bridges nationwide with similar traffic volumes.

FHWA requires an *ADTT* to be recorded on the Structural Inventory and Appraisal (SI&A) form for all bridges. In cases where site traffic conditions are unavailable or unknown, worst-case traffic conditions should be assumed.

The HS-20 truck may be substituted in place of the three AASHTO legal trucks for load rating purposes. This does not mean that the HS-20 is the worst loading. The SHVs and exclusion vehicles are more severe than HS-20.

Live load varies from site to site. More refined site-specific load factors appropriate for a specific bridge site may be estimated if more detailed traffic and truck load data are available for the site. *ADTT* and truck loads through weigh-in-motion measurements recorded over a period of time allow the estimation of site-specific load factors that are characteristic of a particular bridge site.

6A.4.4.2.3—Generalized Live Load Factors: γ_L

C6A.4.4.2.3

6A.4.4.2.3a—Generalized Live Load Factors for Routine Commercial Traffic

C6A.4.4.2.3a

Generalized live load factors for the Strength I limit state are specified in Table 6A.4.4.2.3a-1 for routine commercial traffic on structures other than buried structures. If in the Engineer's judgment, an increase in the live load factor is warranted due to conditions or situations not accounted for in this Manual when determining the safe legal load, the Engineer may increase the factors in Table 6A.4.4.2.3a-1, not to exceed the value of the factor multiplied by 1.3.

Service limit states that are relevant to legal load rating are discussed under the articles on resistance of structures (see Articles 6A.5, 6A.6, and 6A.7).

The generalized live load factors are intended for AASHTO legal loads and state legal loads that have only minor variations from the AASHTO legal loads. Legal loads of a given jurisdiction having gross vehicle rates that are significantly greater than the AASHTO legal loads should preferably be load rated using load factors provided for routine permits in this Manual.

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Among the variables used in evaluation, the uncertainty associated with live loads is generally the greatest. It is, therefore, a logical candidate for closer scrutiny. Much of the total uncertainty about bridge loads represents site-to-site uncertainty rather than inherent randomness in the truck-loading process itself. In design, conservative load factors are assigned to encompass all likely site-to-site variabilities in loads to maintain a uniform and satisfactory reliability level. In evaluation, much of the conservatism associated with loads can be eliminated by obtaining site-specific information. The reduction in uncertainty could result in reduced load factors for evaluation. However, if site investigation shows greater overloads, the load factor may be increased rather than reduced.

For a specific bridge with a low-load rating using generalized load factors, further investigation of site-specific loading could result in improved load rating. In many cases, assessing the site-specific loading will require additional load data collection. Advances in weigh-in-motion technology have significantly lowered the cost of collecting load and traffic data. The cost of additional data collection should be weighed against the potential benefit that may result from improved load ratings.

Permit vehicles should be removed from the stream, if possible, when estimating statistical parameters. WIM data on trucks should be unbiased and should capture any seasonal, weekly, or daily fluctuations. The data collection period should be sufficient to capture at least 400 trucks in the upper 20 percent of the weight sample for the site. Additional guidance on determining site-specific load factors can be found in the NCHRP Report 454.

Alternate Approach to Deriving Site-Specific Load Factors from WIM Data

The commentary above on site specific live load factors describes a simplified procedure for calculating load factors using statistics for the heaviest 20 percent of the truck weight spectra to model the maximum load effects expected on typical bridges. It assumes that the heaviest trucks follow a Normal distribution and that 1 in 15 trucks will cross the bridge side-by-side. Studies performed in NCHRP 12-76 have shown that these simplifying assumptions may not be valid in all cases. NCHRP Project 12-76 has proposed a more consistent approach for using WIM data for live load modeling, which takes into consideration the actual distribution of the truck traffic data, including the actual configurations and the actual percentage of side-by-side crossings.

Calculating Maximum Load Effect, L_{max}

The estimation of the maximum load effect, L_{max}, expected over a 5-year bridge evaluation period can be executed through a variety of methods. The one implemented herein is based on the assumption that the tail end of the histogram of the maximum load effect over a given return period approaches a Gumbel distribution as the return period increases. The method assumes that the WIM

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data is assembled over a sufficiently long period of time to ensure that the data is representative of the tail end of the truck weight histograms. Equations in closed form for statistical projections can be utilized provided the tail of the load effect histogram for the original population of trucks approaches the tail end of a Normal distribution. A Normal distribution fit can usually be obtained for the top five percent of data points.

The process begins by assembling the measured load effects histograms (moment effect or shear force effect) for single lane events and two-lane events for a suite of simple and continuous spans. Then calculate cumulative distribution function for each load effect and obtain the standard deviate of the cumulative function. A plot is made of the upper five percent of the values of the normal deviate versus the load effect, X. The slope, m, and intercept, n, of the best fit regression line provides the statistics for the normal distribution that best fits the tail end of the distribution.

Calculation of L_{max} for each span using equations in closed form for statistical projections can be performed as follows:

- The mean of Normal that best fits the tail end of the distribution: $\mu_{event} = -n/m$.
- The standard deviation of the best fit Normal: $\sigma_{event} = 1/m$.
- Let n_{day} = total number of trucks per day
- For 5 years: $N = n_{day} \times 365 \times 5$
- The most probable value, u_N, for the Gumbel distribution that models the maximum value in 5 years L_{max} is given as:

$$u_N = \mu_{event} + \sigma_{event} \times \left[\sqrt{2\ln(N)} - \frac{\ln(\ln(N)) + \ln(4\pi)}{2\sqrt{2\ln(N)}} \right] \quad \text{(C6A.4.4.2.3a-3)}$$

- The dispersion coefficient for the Gumbel distribution that models the maximum load effect L_{max} is given as:

$$a_N = \frac{\sqrt{2\ln(N)}}{s_{event}} \quad \text{(C6A.4.4.2.3a-4)}$$

- The mean value of L_{max} is given as:

$$L_{max} = \mu_{max} = u_N + \frac{0.577216}{\alpha_N} \quad \text{(C6A.4.4.2.3a-5)}$$

The next step in the derivation of live load factors applies the projected maximum load effect, L_{max}, from the WIM data in Equations 30 and 34 contained in NCHRP Report 454, *Calibration of Load Factors for LRFR Evaluation.* The general expressions for site-specific live load factors for the Strength I limit state, following the same format as the derivation of the LRFR load factors are as follows:

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$$RT = RF \times W \qquad (6A.4.4.4\text{-}1)$$

where:

RT = Rating in tons for truck used in computing live load effect

W = Weight in tons of truck used in computing live load effect

When the lane-type load model (see Figures D6A-4 and D6A-5) governs the load rating, the equivalent truck weight, W, for use in calculating a safe load capacity for the bridge shall be taken as 80 kips.

6A.4.5—Permit Load Rating

6A.4.5.1—Background

Bridge Owners usually have established procedures and regulations which allow the passage of vehicles above the legally established weight limitations on the highway system. These procedures involve the issuance of a permit which describes the features of the vehicle and/or its load and, in most jurisdictions, which specifies the allowable route or routes of travel.

Permits are issued by states on a single-trip, multiple-trip, or annual basis. Routine or annual permits are usually valid for unlimited trips over a period of time, not to exceed one year, for vehicles of a given configuration within specified gross and axle weight limits. Special permits are usually valid for a single trip only, for a limited number of trips, or for a vehicle of specified configuration, axle weights, and gross weight. Special permit vehicles are usually heavier than those vehicles issued annual permits. Depending upon the authorization, these permit vehicles may be allowed to mix with normal traffic or may be required to be escorted in a manner which controls their speed, lane position, the presence of other vehicles on the bridge, or some combination thereof.

C6A.4.5.1

To assure that permit restrictions and conditions are met and to warn the other traffic, special escort vehicles may be needed or required by state law. Traffic safety needs should always be considered.

6A.4.5.2—Purpose

Article 6A.4.5 provides procedures for checking bridges to determine the load effects induced by the overweight permit loads and their capacity to safely carry these overloads. Permit load rating should be used only if the bridge has a rating factor greater than 1.0 when evaluated for AASHTO legal loads.

C6A.4.5.2

Permit vehicles should be rated by using load-rating procedures given in Article 6A.4.5, with load factors selected based on the permit type, loading condition, and site traffic data. The live load to be used in the load-rating equation for permit decisions shall be the actual permit vehicle weight and axle configuration.

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The factors recommended for evaluating permit loads are calibrated with the assumptions that the bridge, as a minimum, can safely carry AASHTO legal loads, as indicated by the evaluation procedures given in Article 6A.4.4. This requirement is especially evident when using reduced live load factors for permits based on a small likelihood that there will be multiple presence of more than one heavy vehicle on the span at one time. Such multiple presence situations are considered in the calibration of the checking equations of both the *AASHTO LRFD Bridge Design Specifications* and the evaluation procedures given in this Manual.

6A.4.5.3—Permit Types

6A.4.5.3.1—Routine (Annual) Permits

Routine permits are usually valid for unlimited trips over a period of time, not to exceed one year. The permit vehicles may mix in the traffic stream and move at normal speeds without any movement restrictions. Some permits may be restricted to specified routes.

6A.4.5.3.2—Special (Limited Crossing) Permits

C6A.4.5.3.2

Special permits are usually valid for a single trip only or for a limited number of trips. These permit vehicles are usually heavier than those vehicles issued routine permits.

Single-trip permits are good for only one trip during a specified period of time (typically 3–5 days). Multiple-trip permits grant permission to transport overweight shipments during a 30–90 day period.

Single-trip permits for excessively heavy loads may have certain conditions and restrictions imposed to reduce the load effect, including, but not limited to:

- Requiring the use of escorts to restrict all other traffic from the bridge being crossed.
- Requiring the permit vehicle to be in a certain position on the bridge (e.g., in the center or to one side) to reduce the loading on critical components.
- Requiring crossing at crawl speed (<10 mph) to reduce dynamic load allowance.

An upper limit of 100 special permit crossings was used for calibration purposes in this Manual. Permits operating at a higher frequency should be evaluated as routine permits.

6A.4.5.4—Live Load and Load Factors

6A.4.5.4.1—Live Load

C6A.4.5.4.1

The live load to be used in the evaluation for permit decisions shall be the actual permit truck or the vehicle producing the highest load effect in a class of permit vehicles operating under a single permit. The loading shall consider the truck weight, its axle configuration and distribution of loads to the axles, designated lane position, and any speed restrictions associated with the issuance of the permit.

Service limit states that are relevant to permit load rating are discussed under the articles on resistance of structures (see Articles 6A.5, 6A.6, and 6A.7).

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For spans up to 200 ft, only the permit vehicle shall be considered present in the lane. For spans between 200 and 300 ft, and when checking negative moments in continuous span bridges, an additional lane load shall be applied to simulate closely following vehicles. The lane load shall be taken as 0.2 klf in each lane. The lane load may be superimposed on top of the permit vehicle (for ease of analysis) and is applied to those portions of the span(s) where the loading effects add to the permit load effects.

6A.4.5.4.2—Load Factors

Table 6A.4.5.4.2a-1 specifies live load factors for permit load rating that are calibrated to provide a uniform and acceptable level of reliability. Load factors are defined based on the permit type, loading condition, and site traffic data.

Permit load factors given in Table 6A.4.5.4.2a-1 for the Strength II limit state are intended for spans having a rating factor greater than 1.0 when evaluated for AASHTO legal loads. Permit load factors are not intended for use in load-rating bridges for legal loads except as allowed under Article 6A.2.3.1.

6A.4.5.4.2a—Routine (Annual) Permits

The live load factors given in Table 6A.4.5.2a-1 for evaluating routine permits shall be applied to a given permit vehicle or to the maximum load effects of all permit vehicles allowed to operate under a single-routine permit. A multi-lane loaded distribution factor shall be used to account for the likelihood of the permits being present alongside other heavy vehicles while crossing a bridge.

C6A.4.5.4.2

The target reliability index set for the calibration of permit load factors is $\beta_{target} = 2.5$ with the goal of achieving reliability index values for all conditions that remain above a minimum $\beta_{min} = 1.50$ and keeping a minimum live load factor of $\gamma_L = 1.1$. The minimum reliability index always governed the recalibration, which raised the average beta values to be mostly greater than 2.5.

Reliability index values for permit checking, multiple presence probabilities, and estimation of the load effect of the random trucks that will cross a bridge alongside a permit truck are calculated using the actual live load effects of trucks recorded by WIM from several sites throughout the U.S. Results of this study may be found in the Final Report for NCHRP 20-07 Task 285.

C6A.4.5.4.2a

In order to better reflect the load effects from the different truck types, the routine permits are categorized based on a combination of their gross vehicle weights (in kips) and their front to rear axle lengths (in ft). The load factors for routine permits can be reduced for the cases where the permit truck's gross vehicle weight and load effect is high to reflect the lower probability of having a random truck of equal or higher weight and load effect crossing alongside the permit truck as given in Table 6A.4.5.4.2a-1. These checks should be performed with the multilane AASHTO LRFD load distribution factors.

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Table 6A.4.5.4.2a-1—Permit Load Factors: γ_L

Permit Type	Frequency	Loading Condition	*DF*[a]	*ADTT* (one direction)	Load Factor by Permit Weight Ratio[b]		
					GVW / AL < 2.0 (kip/ft)	2.0 < GVW / AL < 3.0 (kip/ft)	GVW / AL > 3.0 (kip/ft)
Routine or Annual	Unlimited Crossings	Mix with traffic (other vehicles may be on the bridge)	Two or more lanes	>5,000	1.4	1.35	1.30
				=1,000	1.35	1.25	1.20
				<100	1.30	1.20	1.15
	Unlimited Crossings (Reinforced Concrete Box Culverts)[c]	Mix with traffic (other vehicles may be on the bridge)	One lane	All *ADTT*s	1.40		
					All Weights		
Special or Limited Crossing	Single-Trip	Escorted with no other vehicles on the bridge	One lane	N/A	1.10		
	Single-Trip	Mix with traffic (other vehicles may be on the bridge)	One lane	All *ADTT*s	1.20		
	Multiple Trips (less than 100 crossings)	Mix with traffic (other vehicles may be on the bridge)	One lane	All *ADTT*s	1.40		

Notes:

[a] *DF* = LRFD-distribution factor. When one-lane distribution factor is used, the built-in multiple presence factor should be divided out.

[b] Permit Weight Ratio = GVW/AL;. GVW = Gross Vehicle Weight; AL = Front axle to rear axle length; Use only axles on the bridge.

[c] Refer to Article 6A.5.12.

The live load distribution analysis for routine permits is done using LRFD two-lane distribution factors which assume the simultaneous side-by-side presence of two equally heavy vehicles in each lane. This condition is too conservative for permit load analysis. The live load factors herein were derived to account for the possibility of simultaneous presence of nonpermit heavy trucks on the bridge when the permit vehicle crosses the span. Thus, the load factors are higher for spans with higher *ADTT*s and lower for heavier permits. The live load factors in Table 6A.4.5.2a-1 for routine permits must be applied together with the upper limit of permit weights operating under a single permit and the corresponding two-lane distribution factor.

The live load factors in Table 6A.4.5.2a-1 should be used for interpolation between various *ADTT*s and weight limits.

6A.4.5.4.2b—Special (Limited-Crossing) Permits

Special permits shall be evaluated using the live load factors given in Table 6A.4.5.4.2a-1. These factors shall be applied to the load effects induced by a permit load of magnitude and dimensions specified in the permit application. The live load factors given in Article 6A.4.5.4 for special permits shall only be used for spans having a

C6A.4.5.4.2b

For special permits that are valid for a limited number of trips (below 100 crossings), the probability of simultaneous presence of heavy vehicles alongside the permit vehicle is small. The calibration of these live load factors reflects some contribution from vehicles in adjacent lanes.

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rating factor of 1.0 or higher for AASHTO legal loads or the design load.

A one-lane distribution factor shall be used for special permit review. Such a distribution factor shall be based on tabulated LRFD distribution factors without including any built-in, multiple presence factor, statistical methods where applicable, or refined analysis.

If the agency expects that the special permit will be used with a frequency greater than 100 crossings, then the permit shall be treated as a routine permit.

The live load distribution shall be based on only a single-lane loaded condition. If tabulated LRFD one-lane distribution factors are used, any built-in multiple presence factor (such as a value of 1.2) should be divided out.

Using a live load factor $\gamma_L = 1.10$ for escorted special permit loads will provide average reliability index values greater than the target $\beta_{target} = 2.5$ when the single-lane AASHTO LRFD load distribution factors are used after removing the multiple presence factor $MP = 1.2$.

For special permits mixed with traffic, a live load factor $\gamma_L = 1.40$ leads to an average conditional reliability index of $\beta_{average} = 3.00$ with a minimum value of 1.50. The conditional reliability analysis assumes that a heavy random truck will always cross alongside the permit truck.

6A.4.5.4.2c—Permit Checks Using Refined Analysis

C6A.4.5.4.2c

When routine permit checks are evaluated using a refined analysis, the load factors as given in Table 6A.4.5.4.2a-1 shall be increased (by adding) 0.10 and applied on the two permit trucks placed in adjacent lanes.

When escorted special permits with no other vehicles on the bridge are evaluated using a refined analysis, $\gamma_L = 1.1$ should be applied to the escorted vehicle.

If the permit check with no traffic restriction is to be executed using a rigorous analysis, it is important for the rating engineer to know what live load factor to apply on the permit truck and what truck load and what live load factors to apply to a random truck in the adjacent lane.

In the case of routine permits, the expected number of such permit crossings is unknown so a conservative approach to dealing with the possibility of multiple presence is adopted.

Escorted special permits traveling over bridges at crawl speed (*IM* is neglected) should be checked with the load factor given in Table 6A.4.5.4.2a-1 increased by a factor of 1.05 to satisfy the minimum value of $\beta_{min} = 1.50$.

When special permits mixed with traffic are evaluated using a refined analysis, a live load factor $\gamma_L = 1.0$ shall be applied on the permit truck while a $\gamma_L = 1.10$ shall be applied on the governing AASHTO or state legal truck placed in the adjacent lane.

The calibration of load factors for refined analysis for multi-girder bridges accounts for the conservatism and the variability of the AASHTO LRFD load distribution factor compared to those obtained from refined structural analyses.

6A.4.5.5—Dynamic Load Allowance: *IM*

The dynamic load allowance to be applied for permit load rating shall be as specified in Article 6A.4.4.3 for legal loads, except that for slow moving (≤10 mph) permit vehicles the dynamic load allowance may be eliminated.

6A.4.5.6—Exterior Beams

C6A.4.5.6

Permit load factors given in Table 6A.4.5.4.2a-1 are applicable to both interior and exterior beam ratings. Distribution of live load to exterior beams as defined in LRFD Design Article 4.6.2.2.2d shall apply with the following modifications:

In LRFD, live load distribution to the exterior beams for bridges with diaphragms or cross-frames must be checked by an additional investigation that assumes rigid body behavior of the section, per LRFD Design Article 4.6.2.2.2d.

- For special permits, use a one-lane loaded condition only. Where a one-lane loaded condition is assumed, the LRFD multiple presence factor need not be applied

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(the built-in multiple presence factor in the LRFD one-lane distribution factor should be divided out).

- For routine permits, a multi-lane loaded condition shall be assumed. Permit trucks of equal weights shall be assumed to be present in each lane in determining the governing distribution factor.

6A.4.5.7—Continuous Spans

Closely spaced heavy axles can cause uplift in end spans of continuous bridges. During permit reviews, uplift in continuous span bridges and its effect upon bearings should be considered.

C6A.4.5.7

When the upward *LL* reaction reduces the total reaction to less than ten percent of normal downward *DL* reaction, uplift may be considered to occur. Unless the uplift is counteracted (by weights or tie-downs), the vehicle should not be permitted on the bridge.

6A.5—CONCRETE STRUCTURES

6A.5.1—Scope

The provisions of Article 6A.5 apply to the evaluation of concrete bridge components reinforced with steel bars and/or prestressing strands or bars. The provisions of Article 6A.5 combine and unify the requirements for reinforced and prestressed concrete.

Provisions for the rating of segmental concrete bridges using the LRFR methodology are given in Article 6A.5.11.

6A.5.2—Materials

6A.5.2.1—Concrete

When the compressive strength of concrete, f'_c, is unknown and the concrete is in satisfactory condition, f'_c for reinforced concrete superstructure components may be taken as given in Table 6A.5.2.1-1 by considering the date of construction.

C6A.5.2.1

Cores may also be taken where the initial load capacity based on design concrete strength is considered inadequate. Concrete strength may have little effect on the capacity of flexural members. However, in the case of compression members, the axial capacity increase may be as large as the concrete strength increase.

Table 6A.5.2.1-1—Minimum Compressive Strength of Concrete by Year of Construction

Year of Construction	Compressive Strength, f'_c, ksi
Prior to 1959	2.5
1959 and Later	3.0

For prestressed concrete components, the compressive strengths shown above may be increased by 25 percent.

Where the quality of the concrete is uncertain, cores should be taken for mechanical property testing. Where mechanical properties have been established by testing, the nominal value for strength is typically taken as the mean of the test values minus 1.65 standard deviations to provide a 95 percent confidence limit. Average test values should not be used for evaluation. Guidance on material sampling for bridge evaluation is provided in Article 5.3.

6A.5.2.2—Reinforcing Steel

Yield strengths for reinforcing steels are specified in Table 6A.5.2.2-1. Yield strengths of unknown reinforcing steel may be estimated by considering the date of

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construction. Where practical, specimens of unknown steel should be obtained for testing to ascertain more accurate mechanical properties.

Table 6A.5.2.2-1—Yield Strength of Reinforcing Steel

Type of Reinforcing Steel	Yield Strength, f_y, ksi
Unknown steel constructed prior to 1954	33.0
Structural grade	36.0
Billet or intermediate grade, Grade 40, and unknown steel constructed during or after 1954	40.0
Rail or hard grade, Grade 50	50.0
Grade 60	60.0

6A.5.2.3—Prestressing Steel

Where the tensile strength of the prestressing strand is unknown, the values specified in Table 6A.5.2.3-1 based on the date of construction may be used.

C6A.5.2.3

Stress-relieved strands should be assumed when the strand type is unknown.

Table 6A.5.2.3-1—Tensile Strength of Prestressing Strand

Year of Construction	Tensile Strength, f_{pu}, ksi
Prior to 1963	232.0
1963 and Later	250.0

Yield strengths for prestressing steel are specified in Table 6A.5.4.2.2b-1.

Stress limits for stress-relieved strand are $0.70f_{pu}$ immediately prior to transfer, and $0.80f_{py}$ at service limit state after all losses.

6A.5.3—Resistance Factors

Resistance factors, φ, for concrete members, for the strength limit state, shall be taken as specified in LRFD Design Article 5.5.4.2.

C6A.5.3

For service limit states, φ = 1.0.

6A.5.4—Limit States

The applicable limit states and their load combinations for the evaluation of concrete members are specified for the various rating procedures. The load combinations, and the load factors which comprise them, are specified in Table 6A.4.2.2-1 and in these Articles.

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6A.5.4.1—Design-Load Rating

The Strength I load combinations shall be checked for reinforced concrete components. The Strength I and Service III load combinations shall be checked for prestressed concrete components.

C6A.5.4.1

Service III need not be checked for HL-93 at the operating level as Service III is a design-level check for crack control in prestressed components.

The Service I load combination of the *AASHTO LRFD Bridge Design Specifications* need not be checked for reinforced concrete bridges, as it pertains to the distribution of reinforcement to control crack widths in reinforced concrete beams. Distribution of reinforcement for crack control is a design criterion that is not relevant to evaluation. In LRFD, Service I is also used to check compression in prestressed concrete bridges. This check may govern at prestress transfer, but usually will not govern live load capacity under service conditions.

Most prestressed designs are designed for no cracking under full-service loads. Fatigue is not a concern until cracking is initiated. Hence, prestressed components need not be routinely checked for fatigue.

Rating factors for applicable limit states computed during design-load rating will aid in identifying vulnerable limit states for further evaluation and future inspections.

6A.5.4.2—Legal Load Rating and Permit Load Rating

Load ratings for legal loads and permit loads shall be based on satisfying the requirements for the strength limit and service limit states, guided by considerations presented in these articles.

6A.5.4.2.1—Strength Limit State

Concrete bridge components shall be load rated for the Strength I load combination for legal loads, and for Strength II load combination for permit loads.

6A.5.4.2.2—Service Limit State

6A.5.4.2.2a—Legal Load Rating

Load rating of prestressed concrete bridges based on satisfying limiting concrete tensile stresses under service loads at the Service III limit state is considered optional, except for segmentally constructed bridges. A live load factor of 1.0 is recommended for legal loads when using this check for rating purposes.

C6A.5.4.2.2a

These provisions for evaluation of prestressed concrete bridges permit, but do not encourage, the past practice of limiting concrete tensile stresses at service load. In design, limiting the tensile stresses of fully prestressed concrete members based on uncracked section properties is considered appropriate. This check of the Service III load combination may be appropriate for prestressed concrete bridges that exhibit cracking under normal traffic.

Service limit states are mandatory for the rating of segmental concrete bridges, as specified in Article 6A.5.11.5.

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6A.5.7—Evaluation for Flexural and Axial Force Effects

Members such as arches and beam-columns that are subjected to a combination of axial load and moment shall be evaluated by considering the effect on load capacity of the interaction of axial and bending load effects. Rating factors should be obtained based on both the moment capacity and axial capacity.

C6A.5.7

The use of interaction diagrams as capacity evaluation aids is recommended. The interaction diagram represents all possible combinations of axial loads and bending moments that could produce failure of a particular section in its current condition. The intersection of the line representing dead load and live load eccentricities with the interaction curve provides a convenient method for evaluating load capacity (see Appendix G6A).

6A.5.8—Evaluation for Shear

The shear capacity of existing reinforced and prestressed concrete bridge members should be evaluated for permit loads. In-service concrete bridges that show no visible signs of shear distress need not be checked for shear when rating for the design load or legal loads.

When using the Modified Compression Field Theory (MCFT) for the evaluation of concrete shear resistance, the longitudinal reinforcement should be checked for the increased tension caused by shear, in accordance with LRFD Design Article 5.7.3.6.3.

Whenever the shear failure plane crosses multiple stirrup zones, as shown in Figure 6A.5.8-1, the capacity due to shear reinforcement may be established using the average shear reinforcement area per unit length (A_v/S) existing within the shear failure plane. The average A_v/S can be established using Eq. 6A.5.8-1.

C6A.5.8

Design provisions based on the MCFT are incorporated in the LRFD Design specifications. The MCFT is capable of giving more accurate predictions of the shear response of existing reinforced and prestressed concrete bridge members, with and without web reinforcement. In lieu of the more detailed analysis outlined in the LRFD Design specifications, a simplified analysis that assumes $\beta = 2.0$ and $\theta = 45°$ may be first attempted for reinforced concrete sections and standard prestressed concrete sections with transverse reinforcement. The expressions for shear strength then become essentially identical to those traditionally used for evaluating shear resistance. Where necessary, a more accurate evaluation using MCFT may be performed.

Live load shear for existing bridge girders using the LRFD Design specifications could be higher than the shear obtained from the AASHTO Standard Specifications due to higher live load, higher live load distribution factors for shear, and the higher dynamic load allowance. On the other hand, LRFD Design specifications may yield higher shear resistance for prestressed concrete sections at high-shear locations. MCFT uses the variable angle (θ) truss model to determine shear resistance. Higher prestress levels give flatter θ angles. Flatter θ angles could give higher shear resistances except at regions with high moment and shear.

Prestressed concrete shear capacities are load dependent, which means computing the shear capacity involves an iterative process when using the current AASHTO MCFT. Multiple locations, preferably at 0.05 points, need to be checked for shear. Location where shear is highest may not be critical because the corresponding moment may be quite low. Typically, locations near the 0.25 point could be critical because of relatively high levels of both shear and moment. Also contributing to the need for checking multiple locations along the beam is the fact that the stirrup spacings are typically not constant, but vary.

Unlike flexural failures, shear failures occur over an inclined plane and a shear crack typically intersects a number of stirrups. The length of failure along the longitudinal axis of the member is approximately $d_v \cot\theta$. Each of the stirrups intersected by this crack participate in resisting the applied shear.

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$$\left(\frac{A_v}{S}\right)_{avg} = \frac{\sum\left(\frac{A_v}{S}\right)_i a_i}{d_v Cot(\theta)} \qquad (6A.5.8\text{-}1)$$

where:

A_v = area of shear reinforcement
S = spacing of shear stirrups
a_i = horizontal distance of shear plane crossing the stirrup zone i
d_v = effective shear depth
θ = angle inclination of shear failure plane

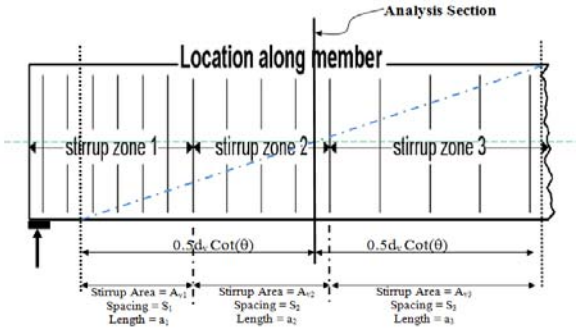

Figure 6A.5.8-1

For bridge girders where dead loads and live loads are generally applied above mid depth of the girders, the shear failure plane may be conservatively assumed to cross the analysis point at mid-depth of the section.

6A.5.9—Secondary Effects from Prestressing

Secondary effects from post-tensioning shall be considered as permanent loads with load factors as cited in Article 6A.2.2.3.

6A.5.10—Temperature, Creep, and Shrinkage Effects

Typically, temperature, creep, and shrinkage effects need not be considered in calculating load ratings for components that have been provided with well-distributed steel reinforcement to control cracking.

These effects may need to be considered in the strength evaluation of long span, framed, and arch bridges.

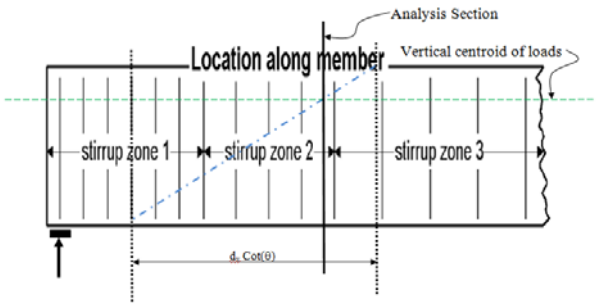

Figure C6A.5.8-1

The relationship between the location of the analysis section and longitudinal zone of stirrups that resist the shear at that section is a function of the vertical position of the load applied to the member, including its self-weight. Ideally, a shear crack inclined at an angle θ intersects the vertical centroid of the applied load as shown in Figure C6A.5.8-1. However, since establishing the vertical centroid requires additional resources and is difficult to implement within software, it is recommended to assume that the shear failure plane intersects the section at mid depth of the member, which will yield conservative capacity.

C6A.5.9

Reactions are produced at the supports in continuous spans under post-tensioning loads, giving rise to secondary moments in the girders. The secondary moments are combined with the primary moments to provide the total moment effect of the post-tensioning.

C6A.5.10

Temperature, creep, and shrinkage are primarily strain-inducing effects. As long as the section is ductile, such changes in strain are not expected to cause failure.

Where temperature cracks are evident and analysis is considered warranted, temperature effects due to time-dependent fluctuations in effective bridge temperature may be treated as long-term loads, with a long-term modulus of elasticity of concrete reduced to one-third of its normal value.

The temperature loading (*T*) could be significant in superstructures that are framed into bents and abutments with no hinges. Uniform temperature loading (*TU*) could induce a significantly large tension in the superstructure girders, which would result in reduction in shear capacity of reinforced concrete girders. Temperature gradient

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loading (*TG*) could induce significantly higher bending moments in framed structures.

Bearings' becoming nonfunctional generally leads to thermal forces being applied onto the bridge elements that were not designed for such loads. Keeping bearings in good working order could prevent temperature and shrinkage forces from occurring.

6A.5.11—Rating of Segmental Concrete Bridges

6A.5.11.1—Scope

This Article incorporates provisions specific to the rating of segmental concrete bridges.

6A.5.11.2—General Rating Requirements

The load-rating capacity of post-tensioned concrete segmental bridges shall be checked in the longitudinal and transverse directions.

C6A.5.11.2

It is possible for transverse effects in a typical segmental box section to govern a capacity or load rating for a bridge. This can be a consequence of the flexural capacity of the top slab at the root of the cantilever wing or interior portion. Such sections are normally governed by serviceability considerations, such as limiting tensile stresses (Service III). Consequently, examination of transverse effects is necessary for a complete load rating.

6A.5.11.3—Application of Vehicular Live Load

For the transverse operating load ratings of the top slab of segmental concrete box girders, the factor of 1.20 specified in LRFD Design Table 3.6.1.1.2-1 for one loaded lane shall be limited to a maximum of 1.00.

C6A.5.11.3

The notional design load of LRFD Design Article 3.6.1.2 was normalized assuming that the governing load condition is two lanes loaded. The value in LRFD Design Table 3.6.1.1.2-1 for one lane loaded reflects the probability of a single heavy truck exceeding the effect of two or more fully correlated heavy side-by-side trucks. The transverse design of the top slab of segmental bridges is governed by axle loads. The amplification of individual axle loads for the single-lane condition is not appropriate. Maximum credible axle loads are less uncertain than maximum credible vehicle loads as axle loads are limited by the bending resistance of vehicle axles.

6A.5.11.4—Design-Load Rating

The Strength I and both the Service I and the Service III limit states shall be checked for the design-load rating of segmental concrete bridges. For operating rating of the design load at the service limit state, the number of live load lanes may be taken as the number of striped lanes. However, loads shall be positioned so as to create maximum effects, for example, on shoulders if necessary. For segmental concrete bridges, the Service III limit state specifically includes the principal tensile stress check of LRFD Design Article 5.9.2.3.3.

C6A.5.11.4

If the Owner anticipates adding additional striped lanes in the near future, the ultimate number of striped lanes should be used. The principal tensile stress check is necessary to verify the adequacy of webs of segmental box girder bridges for longitudinal shear and torsion.

The use of the number of striped lanes is an attempt to "calibrate" the service limit states and distinguishes the operating rating (where the number of striped lanes is used) from the inventory rating (where the number of design lanes is appropriately used). The lesser load effects resulting from the use of striped lanes for the operating rating acknowledges a lower target reliability index for operating as opposed to inventory. If the Owner chooses to use the number of striped lanes in the rating analysis, this assumption should be clearly noted in the rating report.

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The strength limit states are calibrated to achieve target reliabilities, β_T, of 3.5 and 2.5 for inventory and operating evaluation levels, respectively. While the use of the number of striped lanes results in lower reliability for ratings at the service limit states than the number of design lanes, the resultant increment in β_T is unknown. However, a brief study of existing bridges suggests that the use of the number of striped lanes results in adequate operating ratings at the service limit states for well-performing segmental box girder bridges, which is not the case when using the number of design lanes.

6A.5.11.5—Service Limit State

6A.5.11.5.1—Legal Load Rating

Both the Service I and Service III limit states are mandatory for legal load rating of segmental concrete box girder bridges. For these service limit state checks, the number of live load lanes may be taken as the number of striped lanes. However, the loads shall be positioned so as to create maximum effects, for example, on shoulders if necessary. For segmental concrete box girder bridges, the Service III limit state specifically includes the principal tensile stress check of LRFD Design Article 5.9.2.3.3.

C6A.5.11.5.1

See C6A.5.11.4.

6A.5.11.5.2—Permit Load Rating

Both the Service I and Service III limit states are mandatory for permit load rating of segmental concrete box girder bridges. For these service limit state checks, the number of live load lanes may be taken as the number of striped lanes. However, loads shall be positioned so as to create maximum effects, for example, on shoulders if necessary. For segmental concrete box girder bridges, the Service III limit state specifically includes the principal tensile stress check of LRFD Design Article 5.9.2.3.3.

C6A.5.11.5.2

See C6A.5.11.4.

6A.5.11.6—System Factors: φ_s

System factors for longitudinal flexure of post-tensioned segmental concrete box girder bridges are given in Table 6A.5.11.6-1.

C6A.5.11.6

In the context of post-tensioned segmental box girders, the system factor must account for a few significant and important aspects different than other types of bridges. In particular, for a post-tensioned segmental bridge, the system factor, φ_s, must properly and appropriately account for:

- longitudinally continuous versus simply supported spans,
- inherent integrity afforded by the closed continuum of the box section,
- multiple-tendon load paths,
- number of webs per box, and
- types of details and their post-tensioning.

Results of research, load-rating analysis, studies of performance of existing bridges and application of principles underlying LRFR, in particular NCHRP Report 406,

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Redundancy in Highway Bridge Superstructures, were used to address the above needs and establish the system factors summarized in Table 6A.5.11.6-1.

Table 6A.5.11.6-1—System Factors for Post-Tensioned Segmental Concrete Box Girder Bridges

Bridge Type	Span Type	# of Hinges to Failure	System Factors (φ_s) No. of Tendons per Web[a] 1/web	2/web	3/web	4/web
Precast Balanced Cantilever Type A Joints	Interior Span	3	0.90	1.05	1.15	1.20
	End or Hinge Span	2	0.85	1.00	1.10	1.15
	Statically Determinate	1	n/a	0.90	1.00	1.10
Precast Span-by-Span Type A Joints	Interior Span	3	n/a	1.00	1.10	1.20
	End or Hinge Span	2	n/a	0.95	1.05	1.15
	Statically Determinate	1	n/a	n/a	1.00	1.10
Precast Span-by-Span Type B Joints	Interior Span	3	n/a	1.00	1.10	1.20
	End or Hinge Span	2	n/a	0.95	1.05	1.15
	Statically Determinate	1	n/a	n/a	1.00	1.10
Cast-in-Place Balanced Cantilever	Interior Span	3	0.90	1.05	1.15	1.20
	End or Hinge Span	2	0.85	1.00	1.10	1.15
	Statically Determinate	1	n/a	0.90	1.00	1.10

[a] For box girder bridges with three or more webs, table values may be increased by 0.10.

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For longitudinal shear and torsion, transverse flexure, and punching shear of segmental concrete box girder bridges, the system factor shall be taken as 1.0.

System factors for longitudinal flexure in segmental bridges given in Table 6A.5.11.6-1 were selected based on the following reasoning: for flexural conditions, the values of 0.85 (one tendon per web) and 1.00 (two tendons per web) for internal tendons in precast segmental cantilever bridges stem from examination and knowledge of existing bridges—some in Florida, but also others elsewhere—in which only one tendon per web was provided passing through the bottom (tension face) of some end-span segments. This borders upon "condition" but is not strictly a function of it.

After studies of existing bridges and adoption of "multiple tendon paths" as a policy by the Florida Department of Transportation in its *New Directions for Post-Tensioned Bridges*, it was realized that the idea of providing at least two tendons per web and then applying a system factor of 1.00 to the section capacity calculation offered a conservative and comfortable solution. Although a larger system factor might be well justified, 1.00 would certainly be a minimum for such situations. The same could not be held for only one (intact) tendon per web on the tension face. Therefore, a value of 0.85 was chosen. This is judgmental, but is based on observations and experience.

There is a first-generation span-by-span bridge in the Florida Keys with only two external tendons per web in some spans. The fact that this bridge has been performing satisfactorily provides confidence to adopt 1.00 as the lowest possible system factor relating to multiple tendon paths when applied to continuous (interior) spans using external tendons. There is much less confidence and comfort in providing only one external tendon per web—even if, theoretically, this were sufficient to satisfy structural design requirements. In fact, there is no known case of only one external tendon per web. This consideration led to the insertion of "n/a" in Table 6A.5.11.6-1 (meaning "not applicable" or "not allowed") and the choice of 0.85 as the "bottom line" if such a case were found to exist.

Based on the approach in NCHRP Report 406 and studies of its application to segmental bridges, it is considered that system factors for the design of simple and continuous segmental bridges could be 1.10 and 1.20, respectively; with the potential for even greater values for system factors for rating pending the results of yet further studies and research. For the time being, it is certainly safe to adopt at least these values for flexural rating purposes. Considering the need to address multiple tendon paths and that under the *New Directions for Post-Tensioned Bridges*, a minimum of four external tendons per web is recommended for span-by-span construction, it is considered appropriate to apply the 1.10 and 1.20 values to the case of simple and continuous spans of these bridges. It then follows that because precast segmental cantilever bridges usually contain more than 4 cantilever tendons per web then these same values can be safely applied to ratings for cantilever construction.

Longitudinal continuity is recognized through the simple concept of the number of plastic hinges needed to form a collapse mechanism: this is one hinge for a simple

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span or statically determinate structure, two hinges for the end span of a continuous unit, and three hinges for an interior span or monolithic portal frame. The same applies whether a bridge is built using span-by-span or balanced cantilever construction. The significance of the distinction between simple and continuous spans really refers to the difference between statically determinate and indeterminate structures. The possibility of a statically determinate cantilever bridge (in other words, cantilevers with a suspended drop-in span) is treated like a simple-span bridge. For an interior span or statically indeterminate structure, the system factor is set at 1.20, but for an end span or statically determinate bridge, the system factor is 1.10 for two-web boxes with at least four tendons per web. For longitudinal flexure, an enhancement of 0.10 is added to the system factor for boxes with three or more webs.

System factors for intermediate conditions (for example, to account for three tendons per web) were selected by interpolation.

For longitudinal shear and shear torsion, the system factor is taken as 1.00 for the strength limit state for all circumstances.

With transverse post-tensioning of the deck slab, a segmental box is simply a prestressed concrete structure. Therefore, the system factor for transverse flexure of 1.00 is appropriate, regardless of the spacing of tendons; likewise for the local detail of a transverse beam support to an expansion joint device, although the possibility of having only one tendon in the effective section is recognized by reducing the system factor to 0.90.

For local details involving local shear and/or strut-and-tie action or analysis where the resistance is provided by local post-tensioning tendons or bars, a system factor of 1.00 is considered appropriate for two or more tendons. A reduced factor of 0.90 should be used where only one tendon or bar provides the resistance.

6A.5.11.7—Evaluation for Shear and Torsion

For post-tensioned segmental bridges, longitudinal shear and torsion capacity shall be evaluated for design load, legal load, and permit load rating. Refer to Article 5.7 of the *AASHTO LRFD Bridge Design Specifications* for guidance. The shear and torsion for a closed box section shall be determined in accordance with Article 5.7 of the *AASHTO LRFD Bridge Design Specifications*, or otherwise be determined from first principles.

6A.5.12—Rating of Reinforced Concrete Box Culverts

6A.5.12.1—Scope

C6A.5.12.1

This Article incorporates provisions specific to the load rating of single- or multiple-cell reinforced concrete box culverts. The procedures described herein specifically

It is common practice for most of the culvert-specific variables to be taken directly from the construction documents or standard plans. These include culvert

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apply to load rating in-service cast-in-place and precast reinforced concrete box culverts with drained backfill conditions. The intended purpose of this Article is the load rating of box culverts designed using the LRFD methodology and it provides a load rating method that is consistent with that design approach. This Article does not specifically address load rating for alternative culvert structures with respect to shape (circular or elliptical pipe, arch, etc.), material type (aluminum, plastic, steel, etc.), or drainage.

dimensions, material properties, reinforcing schedules, and installation methods. The data from the construction documents should be confirmed during a visual inspection of the culvert, and any discrepancies from the construction documents should be addressed.

It is assumed that culverts that are being load rated will have had a recent visual inspection to establish the condition rating of the culvert. Any discrepancies from plans, any distress, or any deterioration that may affect structural capacity should be noted and considered in the load rating analysis.

In these load-rating provisions for box culverts, the loads are assumed to be carried internally by the box culvert. Flexible pipes and corrugated steel structures are assumed to carry loads with soil–structure interaction (not covered by these provisions).

6A.5.12.2—General Rating Requirements

Culvert ratings should recognize that these structures experience several loadings that are not applicable to most bridge superstructures, including vertical and horizontal soil loads and live load surcharge.

Culvert structural members shall be evaluated for flexure, shear, and axial thrust. Load ratings at several critical sections for the culvert structure must be calculated for each load effect in order to establish the lowest load rating.

Water load on interior walls may be neglected in load rating calculations.

C6A.5.12.2

Typically, flexure in the top and bottom slabs is the controlling case for load ratings. Top slabs with less than 2 ft of fill, designed for moment (LRFD Design Article A5.14.4.1), can be considered adequate for shear.

Loadings for culvert load rating analyses consist of:

- Vertical dead loads representing the self-weight of the structure and the weight of the soil on top of the structure,
- Vertical live loads on top of the culvert,
- A live load surcharge that provides an allowance for trucks which are approaching but not directly above the culvert, and
- Lateral earth pressure consisting of a trapezoidal load placed on the outside walls of the culvert.

6A.5.12.3—Structural Analysis of Box Culverts

The analysis of box culverts may be based on any rational method acceptable to the Owner and consistent with how the culvert is reinforced.

C6A.5.12.3

The analysis of precast culverts is commonly based on the equivalent strip method assuming a frame fixed against lateral movement at the base and free to side-sway at the top. Depending on how the culvert corners are reinforced, top slabs of cast-in-place culverts may be modeled as simple or continuous spans pinned at one corner with rollers at the interior supports or at the other corner.

The force demands on the structure are determined by analytical modeling. Repeatability of the modeling procedure is important to maintain reliability and confidence in the load ratings. Though various levels of sophistication in culvert modeling are possible, simplification of the demand modeling approach will result in more consistent and repeatable load ratings. A common yet conservative approach is to use a two-dimensional frame model by taking a 1.0-ft wide slice normal to the culvert flowline. It provides a quick, conservative, repeatable load rating. The following assumptions can be made in the two-dimensional frame analysis:

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- A 1.0-ft section of the culvert may be analyzed as a frame.
- The frame has a pinned support at one end and pin-roller support at the opposite end.
- Gross section properties are used for strength ratings.
- Supporting soil pressures are uniform over the length of the bottom slab. Vertical loading is balanced by bottom slab loads with no reaction in the supports.
- No hydrostatic pressure (water) exists inside the culvert.
- Supporting soils are fully drained, i.e., no hydrostatic pressure outside the culvert.

This simplified modeling approach tends to produce conservative force demands, particularly in the bottom slab. Alternate methods of restraint for the bottom slab include 1) vertical springs (linear-elastic or nonlinear soil springs), placed at a minimum of at the tenth points along the bottom slab and a horizontal restraint at one corner, or 2) beam on elastic foundation and a horizontal restraint at one corner. These methods account for the effect of differential settlement in the foundation and allow more natural distributions of the live load across the bottom slab and permit more precise estimates of force effects.

Potential failure modes for culvert structural members are bending moment (or flexure), shear, and axial thrust. Force demands at several critical sections for this frame must be calculated in order to establish the lowest load rating for a culvert structure. The critical sections will typically be either near mid-span or at a corner of the culvert structure.

Where needed to verify higher load capacity, analytical models of increasing complexity and sophistication may be used, including finite element models that consider soil–structure interaction effects.

6A.5.12.4—Load Rating Equation for Box Culverts

The load rating of culverts shall be carried out for each load effect using the following rating factor expression, with the lowest value determining the controlling rating factor. Limit states and load factors for load rating shall be selected from Table 6A.5.12.5-1.

$$RF = \frac{C \pm \gamma_{DC} DC \pm \gamma_{DW} DW \pm \gamma_{EV} EV \pm \gamma_{EH} EH \pm \gamma_{ES} ES}{(\gamma_{LL})(LL + IM) \pm (\gamma_{LS})(LS)} \quad (6A.5.12.4\text{-}1)$$

In which, for the strength limit states:

$$C = \varphi_c \varphi_s \varphi R_n \quad (6A.5.12.4\text{-}2)$$

where:

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RF = rating factor
C = capacity
R_n = nominal member resistance (as inspected)
DC = dead load effect due to structural components and attachments
DW = dead load effect due to wearing surface and utilities
EV = vertical earth pressure
EH = horizontal earth pressure
ES = uniform earth surcharge
LL = live load effect
IM = dynamic load allowance
LS = live load surcharge
γ_{DC} = LRFD load factor for structural components and attachments
γ_{DW} = LRFD load factor for wearing surfaces and utilities
γ_{EV} = LRFD load factor for vertical earth pressure
γ_{EH} = LRFD load factor for horizontal earth pressure
γ_{ES} = LRFD load factor for earth surcharge
γ_{LL} = evaluation live load factor
γ_{LS} = LRFD load factor for live load surcharge
φ_c = condition factor
φ_s = system factor
φ = LRFD resistance factor

The product of φ_c and φ_s shall not be taken less than 0.85.

Components subjected to combined load effects shall be load rated considering the interaction of load effects.

6A.5.12.5—Limit States

Concrete box culverts shall be load rated for the Strength I load combination for the design and legal loads, and for Strength II load combination for permit loads.

The applicable loads and their load combinations for the evaluations are specified in Table 6A.5.12.5-1 and in Articles 6A.5.12.6 through 6A.5.12.10.

Service limit state for crack width control need not be checked when load rating reinforced concrete box culverts.

C6A.5.12.5

Maximum and minimum load factors for different loads should be combined to produce the largest load effect. Unbalanced lateral earth pressure should also be checked. The load cases should be selected to generate the maximum moment, shear, and axial demands on the whole culvert. Ratings should consider moment, shear, and thrust demands for all of the critical sections for each load case.

It would be prudent to also perform an evaluation of the culvert under permanent loads only as the earthfill depth may have been increased since the original construction.

6A.5.12.6—Resistance Factors

Resistance factors, φ, for concrete members for the strength limit state shall be taken as specified in LRFD Design Article 12.5.5.

6A.5.12.7—Condition Factor: φ_c

Use of condition factors as presented in Table 6A.4.2.3-1 may be considered optional based on an agency's load-rating practice.

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6A.5.12.8—System Factor: φ_s

System factor for strength limit states for concrete box culverts shall be taken as 1.0.

6A.5.12.9—Materials

The load rater should use best available information on material properties available for the constructed culvert. In the absence of project-specific information from design plans or as-built drawings, default material properties as shown in Article 6A.5.2 may be taken, considering the year of construction.

C6A.5.12.9

It is customary to rely on construction documents, applicable standard culvert plans, or both to identify the materials used in the original construction. In certain cases, as dictated by field conditions or calculated load capacities, it may be appropriate to obtain samples of the actual culvert materials and determine material properties based on laboratory tests.

6A.5.12.10—Loads for Evaluation

6A.5.12.10.1—Dead Loads

Where specific data is unavailable, the unit weights of materials may be taken as shown in LRFD Design Table 3.5.1-1.

6A.5.12.10.2—Earth Pressure

6A.5.12.10.2a—Vertical Earth Pressure: EV

The unit weight of the soil may be taken as shown in LRFD Design Table 3.5.1-1.

C6A.5.12.10.2a

EV represents the dead load due to the earth fill.

Modification of earth loads for soil–structure interaction may be used in accordance with LRFD Design Article 12.11.2.2.1.

6A.5.12.10.2b—Lateral Earth Pressure: EH

Lateral earth pressure shall be assumed to be linearly proportional to the depth of the soil based on the at-rest pressure coefficient as shown in LRFD Design Article 3.11.5.2.

A reduced earth pressure should be used when calculating maximum positive moment in the top slab of culverts.

C6A.5.12.10.2b

EH represents lateral earth pressure from fill up to the top of the culvert. A minimum equivalent fluid pressure of 60 lb/ft^2 of depth may be assumed for determining lateral earth pressure.

Horizontal earth pressures can create top slab moments with the opposite sense of vertical live loads, so a reduced lateral load case is used to generate the maximum moment demands in the top and bottom slab midspans.

LRFD Design Article 3.11.7 recommends a 50 percent reduction in earth pressure where earth pressure may reduce effects caused by other loads but this reduction need not be combined with the minimum load factor.

6A.5.12.10.2c—Uniform Surcharge Loads: ES

The fill above the top slab of the culvert shall be considered an earth surcharge and a constant horizontal earth pressure shall be applied in addition to the basic earth pressure. The uniform horizontal pressure due to earth surcharge should be based on the at-rest pressure coefficient as shown in LRFD Design Article 3.11.5.2.

A reduced earth pressure should be used when calculating maximum positive moment in the top slab of culverts.

C6A.5.12.10.2c

ES represents a constant lateral earth pressure from fill above the top slab of the culvert.

LRFD Design Article 3.11.7 recommends a 50 percent reduction in earth pressure where earth pressure may reduce effects caused by other loads but this reduction need not be combined with the minimum load factor.

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6A.5.12.10.3—Live Loads

Culvert load ratings for the design live load, legal loads, and permit loads shall be based on satisfying the requirements for the strength limit states, guided by considerations presented in this Article and elsewhere in this Manual. Load factors for load rating shall be selected from Table 6A.5.12.5-1.

For load rating top slabs of box culverts for the design load, only the axle loads of the design truck or the design tandem, without the lane load, shall be applied.

For traffic traveling parallel to the span, box culverts shall be load rated for a single loaded lane with the following multiple presence factors:

- *Design Load*—LRFD single-lane multiple presence factor shall be applied to the load.
- *Legal Load and Permit Load*—Only the single-lane loaded condition needs to be checked for legal load and permit load ratings, even when the culvert carries multiple lanes. The 1.2 single-lane multiple presence factor should not be applied to this loading.

  A single legal load factor of 2.00 shall be specified for all traffic volumes. For routine as well as special permits, utilize the load factors provided in Table 6A.4.5.4.2a-1, without applying the multiple presence factor.

C6A.5.12.10.3

A single lane with a multiple presence factor of 1.2 on the live load and using the live load distribution widths in Article 4.6.2.10 will also verify adequate capacity for multiple loaded lanes with MPF, m = 1.0 or less for design loads when the traffic direction is parallel to the span (LRFD Design Article C12.11.2.1).

The LRFR legal load factors recommended for bridge superstructures have been calibrated for two-lane loaded conditions. For culverts, where the single-lane loading governs, a higher legal load factor of 2.00 is recommended for β_{target} = 2.5 to account for the greater likelihood of heavier truck weights in one-lane situations. It should also be noted that the 2.00 legal load factor is applied without the 1.2 LRFD multiple presence factor for single-lane loading.

A single legal load factor of 2.00 is specified for all AASHTO legal loads for all traffic volumes as one-lane load factors are not significantly influenced by traffic volume as side-by-side presence is not a factor.

6A.5.12.10.3a—Live Load Distribution

Distribution of wheel loads for culverts with less than 2.0 ft of fill shall be taken as specified in LRFD Design Article 4.6.2.10. Distribution of wheel loads to culverts with 2.0 ft or more of cover shall be as specified in LRFD Design Article 3.6.1.2.6. Single-span culverts with depth of fill more than 8 ft need not be load rated for live loads as the live load effects are negligible; for multiple-span culverts, the effects of live load may be neglected where the depth of fill exceeds the distance between faces of end walls. The vertical live load should be applied as a moving load across the top of the culvert structure.

Culverts with deep fills should be evaluated for the effects of permanent loads only.

C6A.5.12.10.3a

For culverts with depth of fill greater than 8 ft, the live loads will constitute a negligible portion of the overall loading. The capacity of the culvert should be checked for permanent loads only for the possible ultimate demand obtained by the maximum and minimum load factors.

Box culverts are normally analyzed as two-dimensional frames. Equivalent strip widths defined in LRFD Design Article 4.6.2.10 for box culverts with depth of fill less than 2 ft are used to simplify the analysis. Distribution length parallel to the span may be conservatively neglected in most load ratings.

For earth fills of 2 ft or more, the tire contact area for distribution purposes may be taken as 20 in. wide × 10 in. long, for a wheel of one or two tires (LRFD Design Article A3.6.1.2.5). For other truck loads, the tire area may be calculated following the provisions of LRFD Design Article C3.6.1.2.5. Lane loads are distributed only transversely.

LRFD Design Article 3.6.1.2.6 states that wheel loads may be considered to be uniformly distributed over a rectangular area with sides equal to the dimensions of the tire contact area and increased by 1.15 times the depth of fill in select granular backfill. Where such areas from multiple wheels overlap, the total load should be uniformly distributed over the area but the total width of distribution shall not

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exceed the total width of the supporting slab. By comparison, the AASHTO *Standard Specifications for Highway Bridges* (2002) used to treat the wheel loads as point loads and distribute them over an area equal to a square with dimensions of 1.75 times the depth of fill. The LRFD method yields higher live load forces than were calculated with the *Standard Specifications* for box culverts. The difference is due to a difference in the way the effects of vehicle loading are transferred through the pavement/subgrade to the underlying culvert.

6A.5.12.10.3b—Dynamic Load Allowance: IM

The dynamic load allowance for culverts shall be taken as given in LRFD Design Article 3.6.2.2 except that, for slow moving (≤10 mph) permit vehicles, the dynamic load allowance may be eliminated.

C6A.5.12.10.3b

Dynamic load allowance, *IM*, for culverts is reduced based on the minimum depth of fill but is not less than zero. *IM* is considered for fills up to 8 ft and is only applied to the truck or tandem loads (LRFD Design Article 3.6.2.1).

6A.5.12.10.3c—Live Load Surcharge: LS

Traffic loads on backfills shall be treated as a uniform earth surcharge on the retained soil. A live load surcharge should be applied as a lateral soil pressure. At the barrel wall, the equivalent height of soil for highway loadings shall be determined based on LRFD Design Table 3.11.6.4-1.

C6A.5.12.10.3c

This loading accounts for a vehicle axle positioned just off the end of the culvert which creates additional soil pressure on the retained soil.

This new model for live load surcharge has had a significant increase in lateral load from vehicle loading compared to that previously required by the Standard Specifications. An updated treatment of live load surcharge in LRFD using a new live load model was needed to reflect the current weight and size of vehicular traffic.

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Table 6A.5.12.5-1—Limit States and Load Factors for Culvert Load Rating

Bridge Type	Limit State	*DC*		*DW*		Design Load[a]		Legal Load[b]	Permit[b] Load	*LS*		*EH*[c]		*EV*		*ES*[d]	
		Max	Min	Max	Min	Inv.	Opr.			Max	Min	Max	Min	Max	Min	Max	Min
		γ_{DC}	γ_{DC}	γ_{DW}	γ_{DW}	γ_{LL}	γ_{LL}	γ_{LL}	γ_{LL}	γ_{LS}	γ_{LS}	γ_{EH}	γ_{EH}	γ_{EV}	γ_{EV}	γ_{ES}	γ_{ES}
Reinforced Concrete Box Culvert	Strength I	1.25	0.90	1.50	0.65	1.75	1.35	2.00	—	Same as LF for Design/ Legal Loads	0.00	1.35	0.90	1.30	0.90	1.50	0.75
	Strength II	1.25	0.90	1.50	0.65	—	—	—	Table 6A. 4.5.4.2a-1	Same as LF for Permit Loads	0.00	1.35	0.90	1.30	0.90	1.50	0.75

Notes:

[a] In addition to the load factor, use the 1.2 multiple presence factor for single-lane loading

[b] Multiple presence factor is not included and is not required for single-lane loading

[c] Use a 50 percent reduction to *EH* for rating positive moment in top slabs; need not be combined with the minimum load factor

[d] Use a 50 percent reduction to *ES* for rating positive moment in top slabs; need not be combined with the minimum load factor. Water loads on interior walls are neglected.

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6A.6—STEEL STRUCTURES

6A.6.1—Scope

The provisions of Article 6A.6 shall apply to the evaluation of steel and wrought-iron components of bridges. The provisions of Article 6A.6 apply to components of straight or horizontally curved I-girder bridges and straight or horizontally curved single or multiple closed-box or tub girder bridges.

C6A.6.1

LRFD Design Article 6.10 provides a unified approach for consideration of combined major-axis bending and flange lateral bending from any source in I-sections. In load rating, flange lateral bending effects from wind and deck placement need not be considered.

Bridges containing both straight and curved segments are to be treated as horizontally curved bridges.

6A.6.2—Materials

6A.6.2.1—Structural Steels

The minimum mechanical properties of structural steel given in Table 6A.6.2.1-1 may be assumed based on the year of construction of the bridge when the specification and grade of steel are unknown.

Table 6A.6.2.1-1—Minimum Mechanical Properties of Structural Steel by Year of Construction

Year of Construction	Minimum Yield Point or Minimum Yield Strength, F_y, ksi	Minimum Tensile Strength, F_u, ksi
Prior to 1905	26	52
1905 to 1936	30	60
1936 to 1963	33	66
After 1963	36	66

Where it is possible to identify the designation (AASHTO or ASTM) and grade of the steel from available records, it is possible to determine the minimum yield and tensile strengths to be used for evaluation by reviewing the designation specification.

In cases where the initial evaluation suggests load capacity inadequacies, or there is doubt about the nature and quality of a particular material, the mechanical properties can be verified by testing. Mechanical properties of the material should be determined based on coupon tests. The nominal values for yield and tensile strength are typically taken as the mean test value minus 1.65 standard deviation to provide a 95 percent confidence limit. Average test values should not be used for evaluation. Guidance on material sampling for bridge evaluation is provided in Article 5.3.

Actual values of yield and ultimate tensile stresses reported on mill certificates should not be used for evaluation. Instead, the strength used should be the guaranteed minimum value as specified for the grade of steel shown. The resistance factors account for the fact that the mean strength of the actual material supplied usually exceeds the minimum specified strength.

C6A.6.2.1

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Mechanical properties of eyebars, high-strength eyebars, forged eyebars, and cables vary depending on manufacturer and year of construction. When information from records is not available, microstructural and chemical analyses and hardness testing are helpful in classifying the material. In the absence of material tests, the Engineer should carefully investigate the material properties using manufacturer's data and compilation of older steel properties before establishing the yield point and tensile strength to be used in load rating the bridge.

6A.6.2.2—Pins

If the material designation for pins is unknown, the yield strength may be selected from Table 6A.6.2.2-1, based on the year of construction.

Table 6A.6.2.2-1—Minimum Yield Point of Pins by Year of Construction

Year of Construction	Minimum Yield Point, F_y, ksi
Prior to 1905	25.5
1905 through 1935	30
1936 through 1963	33
After 1963	36

6A.6.2.3—Wrought Iron

When the material designation is unknown for wrought iron, the minimum tensile strength, F_u, should be taken as 48 ksi and the minimum yield point, F_y, should be taken as 26 ksi.

Where practical, coupon tests should be performed to confirm the minimum mechanical properties used in the evaluation.

6A.6.3—Resistance Factors

Except as specified herein, resistance factors, φ, for steel members, for the strength limit state, shall be taken as specified in LRFD Design Article 6.5.4.2.

If the year of construction is prior to 1991, the resistance factor for axial compression for steel, φ_c, shall be taken as 0.90 for built-up compression members, unless it can be established that the member has not been fabricated from universal mill plate, in which case φ_c may be taken as 0.95.

For load rating of main truss member gusset plates, the resistance factors shall be taken as follows:

- Gusset plate compression $\varphi_{cg} = 0.95$
- Gusset plate chord splices $\varphi_{cs} = 0.85$
- Gusset plate shear yielding $\varphi_{vy} = 1.00$
- Gusset plate block shear rupture $\varphi_{bs} = 1.00$
- Gusset plate shear fracture $\varphi_{vu} = 0.80$
- Tension, fracture in net section $\varphi_u = 0.80$
- Tension, yielding in gross section $\varphi_y = 0.95$
- A325 and A490 bolts in shear $\varphi_s = 0.80$
- A307 bolts in shear $\varphi_s = 0.75$
- Fasteners bearing on material $\varphi_{bb} = 0.80$

C6A.6.3

For service limit states, φ = 1.0.

Users of this Manual and the LRFD Bridge Design Specifications will note some differences in the specified resistance factors for main truss member gusset plates. The differences are due to the fact that a higher acceptable level of reliability can be tolerated more readily in design than in rating. In addition, the determination of the resistance factors in the two specifications was based on different dead-to-live load ratios in order to provide more lenient factors for use in rating. The resistance factors are based on the findings from NCHRP Project 12-84 (Ocel, 2013), which did not obtain sufficient data for all possible gusset-plate modes of failure to justify a difference in some of the factors that are provided in the two specifications.

The resistance factor, φ_c, for built-up members subject to axial compression is reduced from 0.95 to 0.90 if the year of construction is prior to 1991 to appropriately reflect the fact that steel built-up compression members may have been fabricated from universal mill plate. Such columns are contained in the data band of lowest strength reflected by SSRC Column Category 3P. Since only one column curve based on SSRC Column Category 2P is used for all columns, earlier versions of LRFD Design specified a

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lower φ_c of 0.90 to reflect the larger data spread and coefficient of variation found in the data for all column categories. Since the production of universal mill plates was discontinued around 1990, and changes in steelmaking practice since that time have resulted in materials of higher quality and much better defined properties, φ_c was raised from 0.90 to 0.95 in the seventh edition of LRFD Design.

6A.6.4—Limit States

The applicable limit states and their load combinations for the evaluation of structural steel and wrought iron members are specified for the various rating procedures. The load combinations, and the load factors which comprise them, are specified in Table 6A.4.2.2-1 and in these Articles.

6A.6.4.1—Design-Load Rating

Strength I and Service II load combinations shall be checked for the design loading. Live load factors shall be taken as tabulated in Table 6A.4.2.2-1.

In situations where fatigue-prone details are present (category C or lower) a rating factor for infinite fatigue life should be computed. Members that do not satisfy the infinite fatigue life check may be evaluated for remaining fatigue life using procedures given in Section 7. This is an optional requirement.

C6A.6.4.1

Rating factors for applicable strength, service, and fatigue limit states computed during the design load rating will aid in identifying vulnerable limit states for further evaluation and future inspections.

6A.6.4.2—Legal Load Rating and Permit Load Rating

Ratings for legal loads and permit loads shall be based on satisfying the requirements for the strength and service limit states, guided by the considerations discussed in this Article.

6A.6.4.2.1—Strength Limit State

Steel bridge components shall be load rated for the Strength I load combination for legal loads, and for Strength II load combination for permit loads.

C6A.6.4.2.1

Load factors for the Strength Limit state are given in Table 6A.4.4.2.3a-1 and Table 6A.4.5.4.2a-1.

6A.6.4.2.2—Service Limit State

Service II load combination check, in conjunction with the service limit state control of permanent deflection of LRFD Design Article 6.10.4.2 and 6.11.4, shall apply to flexural members of all section types. Live load factors shall be taken as tabulated in Table 6A.4.2.2-1. The flange stresses in bending shall not exceed the limiting stresses specified in LRFD Design Article 6.10.4.2.2 for composite and noncomposite sections.

f_R in Eq. 6A.4.2.1-4 shall be taken as:

f_R = 0.95 F_{yf} for composite sections, including negative flexural regions of continuous spans

C6A.6.4.2.2

The reduced load factors for Service II, compared to load-factor design and rating, reflect a more liberalized approach to applying Service II checks for evaluation versus design. Load Factor design and evaluation procedures require the service behavior of steel bridges to be checked for an overload taken as 5/3 times the design load. Serviceability checks for evaluation need not be as stringent as in new designs as there is less uncertainty in traffic loads and the exposure period is reduced. During an overweight permit review, the actual truck weight is available for evaluation. Also, past performance of the bridge under traffic conditions is known and is available to guide the evaluation.

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f_R = 0.80 $R_h F_{yf}$ for noncomposite sections

where:

F_{yf} = Yield stress

The inclusion of the f_ℓ term in LRFD Design Eqs. 6.10.4.2.2-2 and 6.10.4.2.2-3 may be considered optional for straight girder bridges, at the discretion of the Owner.

The f_ℓ term, determined as specified in LRFD Design Article 6.10.1.6, shall be considered when load rating horizontally curved bridges.

Some Bridge Owners have restricted legal loads by posting bridges to control permanent deformations that might result from very heavy unauthorized or illegal overloads. It is not considered likely that unauthorized or illegal loads will obey posted load restrictions.

It is important to note that the live load factors for Service II limit state were not established through reliability-based calibration, but were selected based on engineering judgment and expert opinion. The level of reliability represented by this serviceability check is unknown.

In regions of negative flexure in straight continuous-span I-girder bridges meeting the restrictions specified in LRFD Design Article B6.2, higher load ratings at the service limit state may be achieved by considering the service limit state moment redistribution procedures given in Appendix B to LRFD Design Section 6. For sections in negative flexure that meet the requirements of LRFD Design Article 6.10.4.2.1, the concrete may be considered effective in tension for computing flexural stresses. In such cases the increased susceptibility to web bend buckling should be checked. The appropriate value of D_c to be used at the service limit state should be as specified in LRFD Design Article D6.3.1.

For existing straight bridges as permitted in Article 6A.3.2, the Service II limit state check should be done using the LRFD distribution analysis methods as described in LRFD Design Article 4.6.2. The Strength II limit state check for special permits uses the one-lane distribution factor with the multiple presence factor divided out, and reduced load factors established through reliability-based calibration (see Table 6A.4.5.4.2a-1). This would lead to different methods of live load distribution analyses for the Strength II and Service II limit states for special permit loads, with the Service II distribution method being more conservative when the two-lane distribution is applied. As the load factor prescribed for Service II limit state check for permit loads was based on fitting to load-factor operating level serviceability rating, the built-in conservatism in the distribution analysis is considered appropriate. Escorted special permits operating with no other vehicles on the bridge may be analyzed using one-lane distribution factors for Service II. For existing structures that are curved in plan, either approximate analysis methods or refined analysis methods may be used for the Service II limit state check according to the guidelines of Articles 6A.3.2 and 6A.3.3.

The stress limitation of $0.8F_{yf}$ for the negative moment region of composite spans with longitudinal reinforcement has been found to be conservative. The Autostress design method places no restriction on the maximum stress due to negative moment at overload. Continuous span bridges are allowed to redistribute moments and respond to subsequent overloads in an elastic manner. This can also be applied to the rating of existing bridges.

Use of discontinuous cross-frame or diaphragm lines in straight bridges having skews exceeding 20° may warrant investigation for lateral bending stresses. In the evaluation

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of such bridges where flange lateral bending effects may be significant, it would provide additional conservatism for control of permanent deformation to consider the f_ℓ term in the load rating equation. The determination of f_ℓ due to horizontal curvature is addressed in LRFD Design Article 4.6.1.2.4b. The f_ℓ term may be included in the load rating analysis by adding to the other appropriate component major-axis bending stresses.

6A.6.5—Effects of Deterioration on Load Rating

A deteriorated structure may behave differently than the structure as originally designed and different failure modes may govern its load capacity. Corrosion is the major cause of deterioration in steel bridges. Effects of corrosion include section loss, unintended fixities, movements and pressures, and reduced fatigue resistance.

C6A.6.5

Tension Members with Section Losses Due to Corrosion

Corrosion loss of metals can be uniform and evenly distributed or it can be localized. Uniform reduction in the cross-sectional area of a tension member causes a proportional reduction in the capacity of the member. Since localized corrosion results in irregular localized reductions in area, a simplified approach to evaluating the effects of localized corrosion is to consider the yielding of the reduced net area as the governing limit state. Due to their self-stabilizing nature, stress concentrations and eccentricities induced by asymmetrical deterioration may be neglected when estimating the tension strength of members with moderate deterioration.

For eyebars and pin plates, the critical section is located at the pin hole normal to the applied stress. In evaluating eyebars with significant section loss in the head, the yielding of the reduced net section in the head should be checked as it may be a governing limit state.

Deterioration of lacing bars and batten plates in built-up tension members may affect the load sharing among the main tension elements at service loads. At ultimate load, yielding will result in load redistribution among the tension elements and the effect on capacity is less significant.

Compression Members with Section Losses Due to Corrosion

Uniform Corrosion

Local Effects—The susceptibility of members with reduced plate thickness to local buckling should be evaluated with respect to the limiting width/thickness ratios specified in LRFD Design Article 6.9.4.2. If these values are exceeded, AISC *LRFD Manual of Steel Construction* may be used to evaluate the local residual compressive capacity.

Overall Effects—Most compression members encountered in bridges are in the intermediate length range and have a box-shape or H-shape cross section. Moderate uniform corrosion of these sections has very little effect on the radius of gyration. The reduction of compressive resistance for short and intermediate length members, for moderate deterioration, is proportional to the reduction in cross-sectional area.

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Localized Corrosion

Deterioration at the ends of fixed-end compression members may result in a change in the end restraint conditions and reduce its buckling strength. Localized corrosion along the member can cause changes in the moment of inertia. Asymmetric deterioration can induce load eccentricities. The effects of eccentricities can be estimated using the eccentricity ratio ec/r^2, where e is the load eccentricity in the member caused by localized section loss, c is the distance from the neutral axis to the extreme fiber in compression of the original section, and r is the radius of gyration of the original section. Effects of eccentricity may be neglected for eccentricity ratios under 0.25.

Built-Up Members with Deteriorated Lacing Bars/Batten Plates

The main function of lacing bars and batten plates is to resist the shear forces that result from buckling of the member about an axis perpendicular to the open web. They also provide lateral bracing for the main components of the built-up member. Localized buckling of a main component can result because of loss of lateral bracing from the deterioration of the lacing bars. The slenderness ratio of each component shape between connectors and the nominal compressive resistance of built-up members should be evaluated as specified in LRFD Design Article 6.9.4.3.

Corrosion of lacing bars and batten plates reduces the shear resistance of the built-up member and, therefore, a reduction in its overall buckling strength may result. Approximate analytical solutions for the buckling resistance of built-up members with deteriorated lacing and batten plates can be formulated using a reduced effective modulus of elasticity of the member, given in NCHRP Report 333. It has been determined that moderate deterioration of up to about 25 percent loss of the original cross-section of lacing bars and batten plates has very little effect on the overall member capacity, as long as the resistance to local failure is satisfactory.

Flexural Members with Section Losses Due to Corrosion

Uniform Corrosion

The reduction in bending resistance of laterally supported beams with stiff webs will be proportional to the reduction in section modulus of the corroded cross-section compared to the original cross-section. Either the elastic or plastic section modulus shall be used, as appropriate. Local and overall beam stability may be affected by corrosion losses in the compression flange.

The reduction in web thickness will reduce shear resistance and bearing capacity due to both section loss and web buckling. When evaluating the effects of web losses, failure modes due to buckling and out-of-plane movement that did not control their original design may govern. The loss in shear resistance and bearing capacity is linear up to the point where buckling occurs.

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Localized Corrosion

Small web holes due to localized losses not near a bearing or concentrated load may be neglected. All other web holes should be analytically investigated to assess their effect.

A conservative approach to the evaluation of tension and compression flanges with highly localized losses is to assume the flange is an independent member loaded in tension or compression. When the beam is evaluated with respect to its plastic moment capacity, the plastic section modulus for the deteriorated beam may be used for both localized and uniform losses.

Main Truss Member Gusset Plates

The resistance of gusset plates may be reduced if section loss due to corrosion is present at certain locations coinciding with the failure planes assumed in applying the resistance equations specified in Article 6A.6.12.6.

For evaluating the tension resistance, only the section loss that intersects the Whitmore section must be accounted for when calculating the resistance. The section loss may be smeared uniformly over the entire Whitmore section.

For evaluating the shear resistance, the use of the remaining area across a failure plane is sufficient for determining the resistance regardless of whether or not multilayered gusset plates are present or the corrosion is localized, is asymmetric about the connection work point, or affects only one gusset plate.

For evaluating the compressive resistance, the actual area remaining in the partial shear plane defined in Article 6A.6.12.6.6 is to be considered. When evaluating the compressive resistance according to Article 6A.6.12.6.7, an equivalent plate thickness should be defined for the Whitmore section based on a projection upon the Whitmore section of all cross-sectional area loss occurring between the Whitmore section and the adjoining members in the direction of the member, as shown in Figure C6A.6.5-1. In this case, a smeared uniform plate thickness must be derived for L_{total} considering the isolated section loss occurring over $L_{corrosion}$. These methods were found to be conservative as reported in Ocel (2013).

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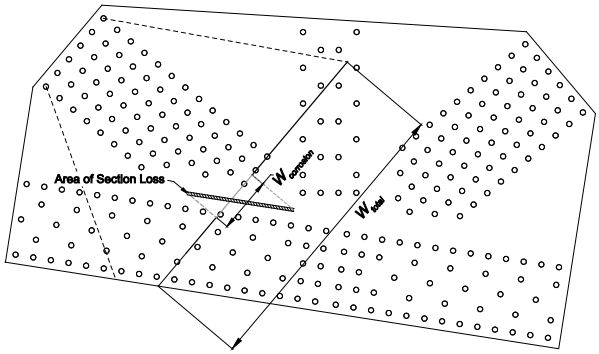

Figure C6A.6.5-1—Section-Loss Band Projected upon the Whitmore Section to Determine an Equivalent Average Plate Thickness for the Compressive Resistance Evaluation

6A.6.6—Tension Members

Members and splices subjected to axial tension shall be investigated for yielding on the gross section and fracture on the net section as specified in LRFD Design Article 6.8.2.

6A.6.6.1—Links and Hangers

The following provisions are given for the evaluation of pin-connected tension members other than eyebars:

1. The net section through the pin hole transverse to the axis of the member shall be 40 percent greater than the net section of the main member.

2. The net section back of the pin hole parallel to the axis of the member shall not be less than the net section of the main member away from the pin hole.

3. In the event that the net section at the pin does not conform to 1) or 2) above, the net section of the member shall be reduced proportionately for rating purposes.

6A.6.6.2—Eyebars

The following provisions are given for the evaluation of eyebars:

1. The section of the head through the pin hole transverse to the axis of the member shall be 35 percent greater than the section of the body.

2. The section of the head beyond the pin hole taken in the longitudinal direction shall not be less than 75 percent of the section of the body away from the pin hole.

3. In the event that the section at the pin does not conform to 1) or 2) above, the section of the body used

C6A.6.6.1

Design of pin and hanger connections assumes free rotation at the pin. Accumulation of dirt and corrosion developed between the elements of the pin and hanger assembly could result in unintended partial or complete fixity of the pin and hanger connection. Very large in-plane bending stresses in the hangers and torsional stresses in the pins could be expected from rotational fixity. The fatigue life of the hangers could also be reduced. Build-up of corrosion products between the hangers and web or gusset plates could cause out-of-plane bending in the hangers. Failure modes not routinely considered in the original design may need to be considered during evaluation.

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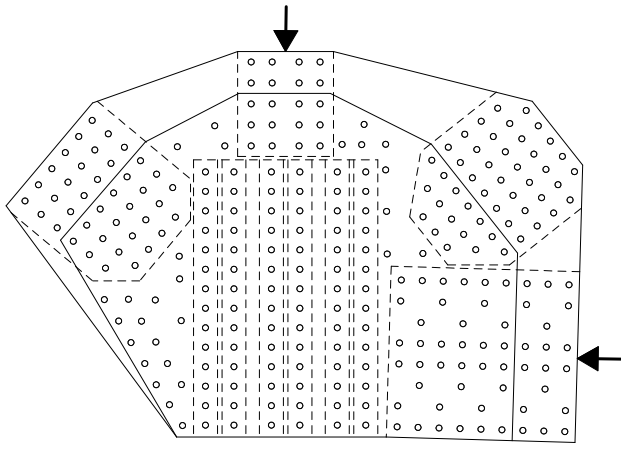

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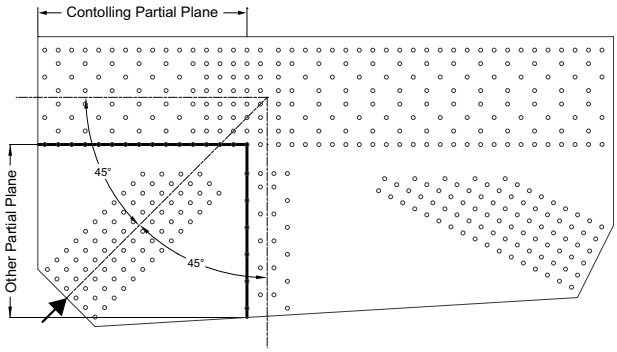

Figure 6A.6.12.6.6-1—Example of a Controlling Partial Shear Plane that Parallels the Chamfered End of the Compression Member Since that Member Frames In at an Angle of 45 Degrees to Both the Chord and the Vertical

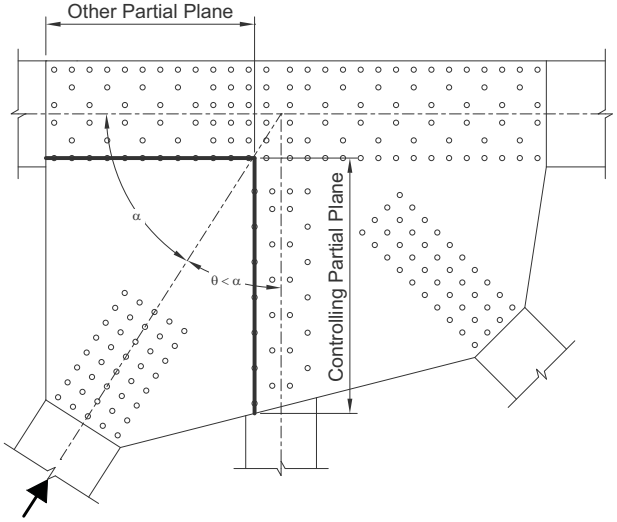

Figure 6A.6.12.6.6-2—Example of a Controlling Partial Shear Plane on the Side of a Compression Member without a Chamfered End that Has the Smaller Framing Angle between That Member and the Other Adjoining Members (i.e., θ < α)

6A.6.12.6.7—Gusset Plate Compressive Resistance

The factored compressive resistance, P_r, of gusset plates at the strength limit state shall be taken as:

$$P_r = \varphi_{cg} P_n \qquad (6A.6.12.6.7\text{-}1)$$

in which:

P_n = nominal compressive resistance of a Whitmore section determined from Eq. 6A.6.12.6.7-2 or 6A.6.12.6.7-3, as applicable (kips):

- If $\frac{P_e}{P_o} \geq 0.44$, then:

C6A.6.12.6.7

Gusset plate zones in the vicinity of compression members are to be load rated for plate stability at the strength limit state. Experimental testing and finite element simulations performed as part of NCHRP Project 12-84 (Ocel, 2013) and by others (Yamamoto et al., 1988; Hafner et al., 2012) have found that truss gusset plates subject to compression always buckle in a sidesway mode in which the end of the compression member framing into the gusset plate moves out-of-plane. The buckling resistance is dependent on the chamfering of the member, the framing angles of the members entering the gusset, and the standoff distance of the compression member relative to the surrounding members; i.e. the distance, L_{mid}. An example connection showing a typical chamfered member end and member framing angle is provided in Figure C6A.6.12.6.7-1. The research found that the compressive resistance of gusset plates with large L_{mid} distances was

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$$P_n = \left[0.658^{\left(\frac{P_o}{P_e}\right)}\right] P_o \qquad \text{(6A.6.12.6.7-2)}$$

- If $\frac{P_e}{P_o} < 0.44$, then:

$$P_n = 0.877 P_e \qquad \text{(6A.6.12.6.7-3)}$$

In which:

P_e = elastic critical buckling resistance (kips)

$$= \frac{3.29E}{\left(\frac{L_{mid}}{t_g}\right)^2} A_g \qquad \text{(6A.6.12.6.7-4)}$$

where:

φ_{cg} = resistance factor for gusset plate compression specified in Article 6A.6.3

P_o = equivalent nominal yield resistance = $F_y A_g$ (kips), in which:

F_y = specified minimum yield strength (ksi)

A_g = gross cross-sectional area of the Whitmore section determined based on 30 degree dispersion angles, as shown in Figure 6A.6.12.6.7-1 (in.^{2}). The Whitmore section shall not be reduced if the section intersects adjoining member bolt lines

E = modulus of elasticity (ksi)

L_{mid} = distance from the middle of the Whitmore section to the nearest member fastener line in the direction of the member, as shown in Figure 6A.6.12.6.7-1 (in.)

t_g = gusset plate thickness (in.)

reasonably predicted using modified column buckling equations and Whitmore section analysis. When the members were heavily chamfered, reducing the L_{mid} distance, the buckling of the plate was initiated by shear yielding on the partial shear plane adjoining the compression member causing a destabilizing effect, as discussed in Article C6A.6.12.6.6.

Eq. 6A.6.12.6.7-4 is derived by substituting plate properties into column buckling formulas along with an effective length factor of 0.5 that was found to be relevant for a wide variety of gusset plate geometries (Ocel, 2013).

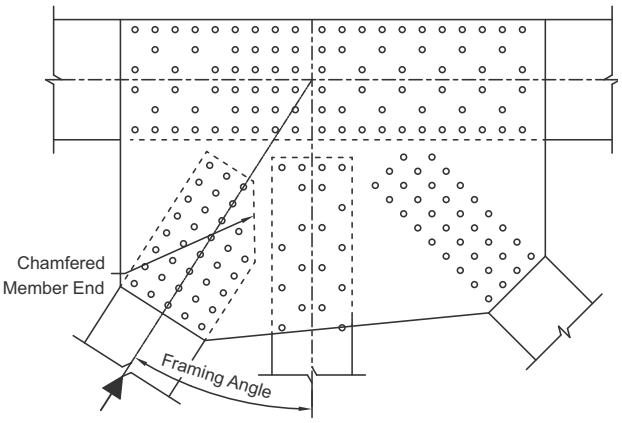

Figure C6A.6.12.6.7-1—Example Connection Showing a Typical Chamfered Member End and Member Framing Angle

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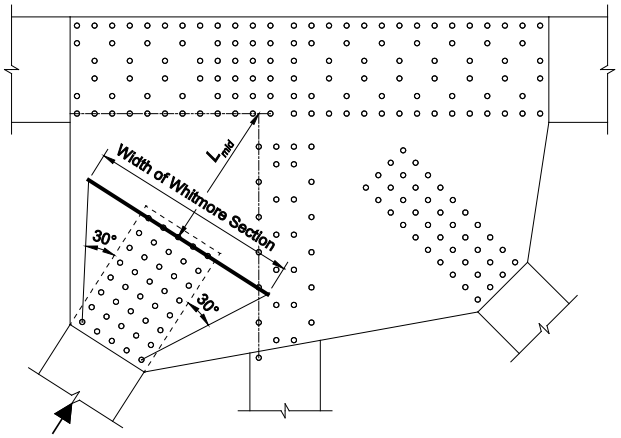

Figure 6A.6.12.6.7-1—Example Connection Showing the Whitmore Section for a Compression Member Derived From 30 Degree Dispersion Angles and the Distance L_{mid}

The provisions of this Article shall not be applied to compression chord splices.

6A.6.12.6.8—Gusset Plate Tensile Resistance

The factored tensile resistance, P_r, of gusset plates at the strength limit state shall be taken as the smallest factored resistance in tension based on block shear rupture, yielding on the Whitmore section, and net section fracture on the Whitmore section.

The factored block shear rupture resistance shall be taken as:

$$P_r = \varphi_{bs} R_p (0.58 F_u A_{vn} + F_u A_{tn}) \le \varphi_{bs} R_p (0.58 F_y A_{vg} + F_u A_{tn}) \quad (6A.6.12.6.8\text{-}1)$$

where:

φ_{bs} = resistance factor for gusset plate block shear rupture specified in Article 6A.6.3

R_p = reduction factor for holes taken equal to 0.90 for bolt holes punched full size and 1.0 for bolt holes drilled full size or subpunched and reamed to size

F_u = specified minimum tensile strength of the connected material (ksi)

A_{vn} = net area along the plane resisting shear stress (in.^{2})

A_{tn} = net area along the plane resisting tension stress (in.^{2})

F_y = specified minimum yield strength of the connected material (ksi)

A_{vg} = gross area along the plane resisting shear stress (in.^{2})

C6A.6.12.6.8

A conservative model has been adopted to predict the block shear rupture resistance in which the resistance to rupture along the shear plane is added to the resistance to rupture on the tensile plane. Block shear is a rupture or tearing phenomenon and not a yielding phenomenon. However, gross yielding along the shear plane can occur when tearing on the tensile plane commences if $0.58F_uA_{vn}$ exceeds $0.58F_yA_{vg}$. Therefore, Eq. 6A.6.12.6.8-1 limits the term $0.58F_uA_{vn}$ to not exceed $0.58F_yA_{vg}$. Eq. 6A.6.12.6.8-1 is consistent with the philosophy for tension members where the gross area is used for yielding and the net area is used for rupture.

The reduction factor, R_p, conservatively accounts for the reduced rupture resistance in the vicinity of holes that are punched full size (Brown et al., 2007). No reduction in the net section fracture resistance is required for holes that are drilled full size or subpunched and reamed to size.

The net area, A_n, is the product of the plate thickness and its smallest net width. The width of each standard hole is to be taken as the nominal diameter of the hole. The width of oversize and slotted holes, where permitted, is to be taken as the nominal diameter or width of the hole. The net width is to be determined for each chain of holes extending across the member or element along any transverse, diagonal, or zigzag line.

The net width for each chain is to be determined by subtracting from the width of the element the sum of the widths of all holes in the chain and adding the quantity $s^2/4g$ for each space between consecutive holes in the chain, where:

s = pitch of any two consecutive holes (in.)

g = gauge of the same two holes (in.)

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The factored resistances for yielding on the Whitmore section and net section fracture on the Whitmore section shall be determined from Eqs. 6A.6.12.6.8-2 and 6A.6.12.6.8-3, respectively.

$$P_r = \varphi_y F_y A_g \quad (6A.6.12.6.8\text{-}2)$$

$$P_r = \varphi_u F_u A_n R_p U \quad (6A.6.12.6.8\text{-}3)$$

where:

φ_y = resistance factor for yielding of tension members specified in Article 6A.6.3

F_y = specified minimum yield strength of the gusset plate (ksi)

A_g = gross cross-sectional area of the effective Whitmore section determined based on 30 degree dispersion angles, as shown in Figure 6A.6.12.6.8-1 (in.^{2}). The Whitmore section shall not be reduced if the section intersects adjoining member bolt lines

φ_u = resistance factor for fracture of tension members specified in Article 6A.6.3

F_u = specified minimum tensile strength of the gusset plate (ksi)

A_n = net cross-sectional area of the effective Whitmore section determined based on 30 degree dispersion angles, as shown in Figure 6A.6.12.6.8-1 (in.^{2}). The Whitmore section shall not be reduced if the section intersects adjoining member bolt lines

R_p = reduction factor for holes taken equal to 0.90 for bolt holes punched full size and 1.0 for bolt holes drilled full size or subpunched and reamed to size

U = reduction factor to account for shear lag; taken as 1.0 for gusset plates

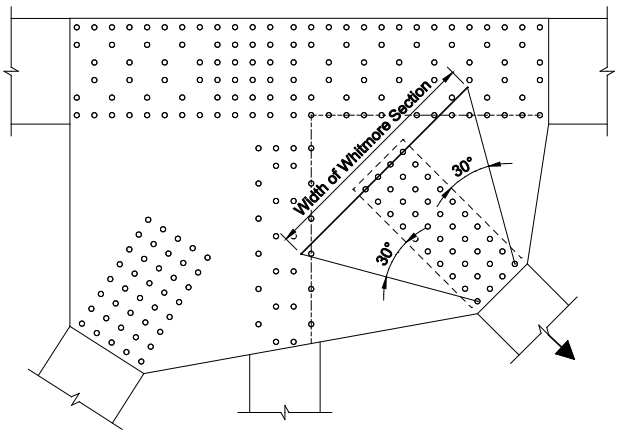

Figure 6A.6.12.6.8-1—Example Connection Showing the Whitmore Section for a Tension Member Derived from 30 Degree Dispersion Angles

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The provisions of this Article shall not be applied to tension chord splices.

6A.6.12.6.9—Chord Splices

Gusset plates that splice two chord sections together shall be checked using a section analysis considering the relative eccentricities between all plates crossing the splice and the loads on the spliced plane.

For compression chord splices, the factored compressive resistance, P_r, of the spliced section at the strength limit state shall be taken as:

$$P_r = \varphi_{cs} F_{cr} \left(\frac{S_g A_g}{S_g + e_p A_g} \right) \quad \text{(6A.6.12.6.9-1)}$$

in which:

F_{cr} = stress in the spliced section at the limit of usable resistance (ksi); shall be taken as the specified minimum yield strength of the gusset plate when the following equation is satisfied:

$$\frac{KL_{\text{splice}}\sqrt{12}}{t_g} < 25 \quad \text{(6A.6.12.6.9-2)}$$

where:

φ_{cs} = resistance factor for gusset plate chord splices specified in Article 6A.6.3

S_g = gross section modulus of all plates in the cross-section intersecting the spliced plane (in.^{3})

A_g = gross area of all plates in the cross-section intersecting the spliced plane (in.^{2})

e_p = distance between the centroid of the cross-section and the resultant force perpendicular to the spliced plane (in.)

K = effective column length factor taken as 0.50 for chord splices

L_{splice} = center-to-center distance between the first lines of fasteners in the adjoining chords as shown in Figure 6A.6.12.6.9-1 (in.)

t_g = gusset plate thickness (in.)

C6A.6.12.6.9

This Article is only intended to cover the load rating of chord splices that occur within the gusset plates. For gusset plates also serving the role of a chord splice, the forces from all members framing into the connection must be considered. The chord splice forces are the resolved axial forces acting on each side of the spliced section, as illustrated in Figure C6A.6.12.6.9-1. Generally, a difference in the resolved forces on the two sides of a chord splice arises due to the use of envelope forces, i.e. forces due to nonconcurrent loads. Where envelope forces are used, the resolved forces should include consideration of the concurrence or nonconcurrence of forces to avoid potentially unconservative reductions in the connection force. The chord splice should be investigated for the larger of the two resolved forces on either side of the splice.

For chord splices that are in full compression under the loads being considered and that are detailed with milled ends in full contact bearing at the splice, resolved forces may be adjusted to account for the transfer of a portion of the compressive load though end bearing as long as the capacity in bearing is verified. The portion of the compressive load that may be transferred in bearing in such cases is specified in LRFD Design Article 6.13.6.1.3.

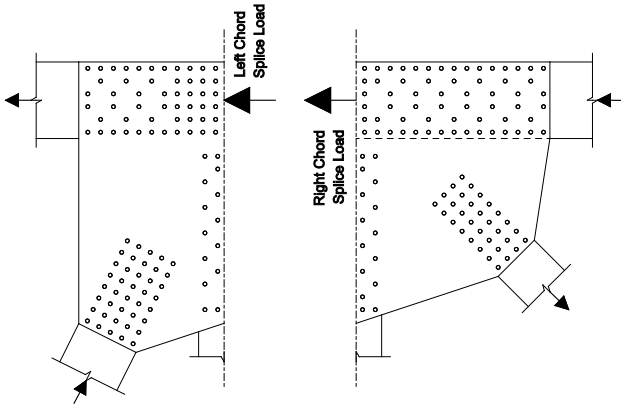

Figure C6A.6.12.6.9-1—Example Connection Showing the Resolution of the Member Forces into Forces Acting on Each Side of a Chord Splice

The resistance equations in this Article assume the gusset and splice plates behave as one combined spliced

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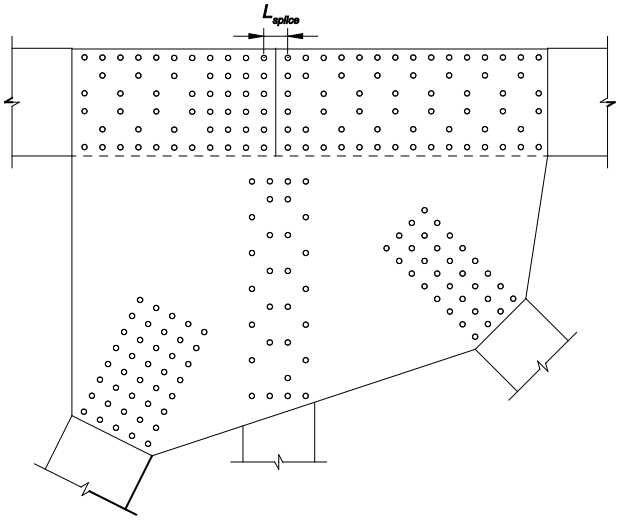

Figure 6A.6.12.6.9-1—Example Connection Showing the Chord Splice Parameter, L splice

For tension chord splices, the factored tensile resistance, P_r, of the spliced section at the strength limit state shall be taken as the lesser of the values given by Eqs. 6A.6.12.6.9-3 and 6A.6.12.6.9-4.

$$P_r = \varphi_{cs} F_y \left(\frac{S_g A_g}{S_g + e_p A_g} \right) \quad \text{(6A.6.12.6.9-3)}$$

$$P_r = \varphi_{cs} F_u \left(\frac{S_n A_n}{S_n + e_p A_n} \right) \quad \text{(6A.6.12.6.9-4)}$$

where:

φ_{cs} = resistance factor for gusset plate chord splices specified in Article 6A.6.3

F_y = specified minimum yield strength of the gusset plate (ksi)

S_g = gross section modulus of all plates in the cross-section intersecting the spliced plane (in.^{3})

A_g = gross area of all plates in the cross-section intersecting the spliced plane (in.^{2})

section to resist the applied axial load and eccentric bending that occurs due to the fact that the resultant forces on the section are offset from the centroid of the combined section, as illustrated in Figure C6A.6.12.6.9-2. The combined spliced section is treated as a beam and the factored resistance at the strength limit state is determined assuming the stress in the combined section at the limit of usable resistance is equal to the specified minimum yield strength of the gusset plate if the slenderness limit for the spliced section given by Eq. 6A.6.12.6.9-2 is met, which will typically be the case. If not, the Engineer will need to derive a reduced value of F_{cr} to account for possible elastic buckling of the gusset plate within the splice.

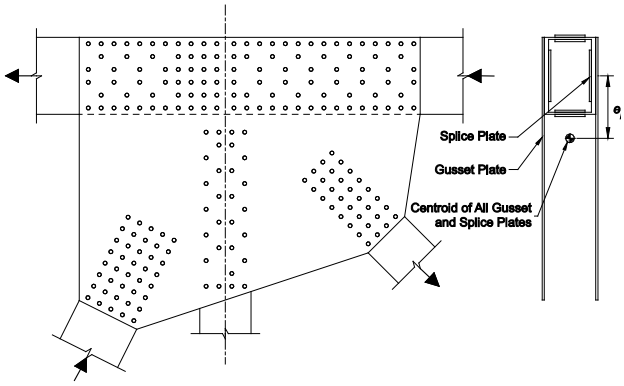

Figure C6A.6.12.6.9-2—Illustration of the Combined Spliced Section at a Chord Splice

The Whitmore section check specified in Article 6A.6.12.6.7 is not considered applicable for the load rating of a compression chord splice.

The yielding and net section fracture checks on the Whitmore section specified in Article 6A.6.12.6.8 are not considered applicable for the load rating of a tension chord splice.

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F_u = specified minimum tensile strength of the gusset plate (ksi)

S_n = net section modulus of all plates in the cross section intersecting the spliced plane (in.^{3})

A_n = net area of all plates in the cross-section intersecting the spliced plane (in.^{2})

e_p = distance between the centroid of the cross-section and the resultant force perpendicular to the spliced plane (in.)

Tension chord splice members shall also be checked for block shear rupture as specified in Article 6A.6.12.6.8.

6A.6.12.6.10—Edge Slenderness

Gusset plates shall not be load rated on the basis of edge slenderness.

C6A.6.12.6.10

NCHRP Project 12-84 (Ocel, 2013) found no direct correlation between the buckling resistance of the gusset plate and the free edge slenderness. In addition, merely adding stiffeners to just the free edges will not provide any appreciable increase in the compressive resistance of the plate. However, properly stiffening the free edges, as discussed below, could suppress plate buckling.

Since gusset plate buckling was always observed to occur in a sway mode, either a diaphragm must be added between the two gussets, preferably also connected to the chord, to stiffen against sway; or else stiffening elements must be placed along the free edges such that their full out-of-plane yield moment resistance can be developed at the planes that would bend if sway occurs. These requirements do not apply if the free edges are merely being stiffened without relying on an increase in buckling resistance. In this case, there are no criteria specified for sizing of the edge stiffeners, but the traditional practice of using angles with leg thicknesses of 0.50 in. has generally proven adequate to reduce deformations of the free edges during fabrication, erection, and service.

The effect of proper edge stiffening on the compressive resistance of the gusset plate was examined experimentally and analytically in NCHRP Project 12-84 (Ocel, 2013). The increase in compressive resistance was highly dependent upon the configuration of the connection and was found to vary from 6 percent to 45 percent. Generally, connections using chamfered members that allowed for very closely spaced member arrangements experienced little increase in compressive resistance. Connections that had large spans of free plate between the compression members and the surrounding members experienced the largest increase in compressive resistance. That is, properly stiffened free edges tend to suppress buckling as predicted by the Whitmore section analysis specified in Article 6A.6.12.6.7 in gusset plates with large L_{mid} distances. However, proper edge stiffening will likely not suppress the buckling resulting from partial plane shear yielding in cases with small L_{mid} distances. Therefore, in such cases, the resistance calculated according to the provisions of Article 6A.6.12.6.6 would be considered to represent the upper

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bound of compressive buckling resistance with properly stiffened free edges, unless a refined simulation analysis indicates otherwise. A refined simulation analysis, which is permitted according to the provisions of Article 6A.6.12.6.11, may be used to better quantify the increase in compressive resistance offered by properly stiffened free edges.

6A.6.12.6.11—Refined Analysis

A refined simulation analysis using the finite element method may be employed to determine the nominal resistance of a gusset plate connection at the strength limit state in lieu of satisfying the requirements specified in Articles 6A.6.12.6.6 through 6A.6.12.6.9. The nominal resistance obtained from the refined simulation analysis shall be multiplied by 0.90 in order to obtain the factored resistance of the connection.

If a load rating conducted in accordance with Articles 6A.6.12.6.6 through 6A.6.12.6.9 indicates an unacceptable load rating and the limiting capacity is based on compression (e.g., Partial Shear, Whitmore) or a deteriorated condition, then a more refined analysis may be performed. Any more rigorous analysis must be consistent with a rational application of established engineering principles.

C6A.6.12.6.11

A refined simulation analysis does not consider the variability of material properties and fabrication tolerances assumed in the AASHTO LRFR calibration. As a result, to be consistent with the philosophy of the AASHTO LRFR specifications, the 0.90 reduction factor was developed as a partial φ factor accounting for these two issues. This value assumes the simulation analysis is accurate enough such that there is no variation in the professional factor and was calibrated to provide a target reliability index of 3.5. The reduction factor specified in Article 6A.6.12.6.1 is also to be considered.

The necessary fidelity of the model is dependent on the failure mode under investigation. For instance, simple planar shell finite element models of single gusset plates have been successfully used to identify the nominal shear resistance of gusset plate connections. These models included nonlinear material properties with strain hardening, and member loads were applied as surface tractions at fastener locations. However, additional modeling effort is required to predict the nominal compressive buckling resistance of a gusset plate.

Considering the following list of model attributes, NCHRP Project 12-84 researchers were able to attain model predictions within 9 percent of experimental values for a three-dimensional, two-panel truss system isolated out of an entire bridge where the connection of interest was located in the center between two panels (Ocel, 2013). Model symmetry was not used because the sway buckling mode would not be captured. The following list, which is not considered exhaustive, summarizes other important attributes of this model:

- The gusset plate, splice plates, and the members for a distance of two member depths away from the gusset plate edge were modeled with shell elements. The truss was represented with beam elements at all other locations;
- The shell elements were able to capture nonlinear geometric and material effects. Nonlinear material properties considered strain hardening;
- Each fastener was represented with a line element with deformable, nonlinear material properties;
- The mesh contained initial imperfections on all compression members with a maximum out-of-plane magnitude limited by the smaller of: 1) the longest free edge length divided by 150; 2) 0.1 times the gap between the end of the compression member and the

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next adjoining member; or 3) 100 percent of the gusset plate thickness; and

- The model was proportionally loaded until failure. Typically, buckling can be identified when the analysis no longer converges to a solution. Shear failures are more difficult to identify but typically occur when the plate exhibits load/displacement softening or when a strain threshold is exceeded, after which the analysis predictions become unrealistic.

Because the basic compression checks comprise empirical fit of a wide range of conditions, significant improvements in accuracy can be provided by explicitly considering the flow of forces through the plate and the capacities of the sections resisting those forces. Refined modeling approaches based on a first principles analytical approach utilizing fundamental steel design theory may be used. Examples of other approaches are illustrated in Figures C6A.6.12.6.11-1 and C6A.6.12.6.11-2. The 0.90 reduction factor required for the application toward the results of a finite element analysis should be applied to the basic corner check of Figure C6A.6.12.6.11-1, but not to the truncated Whitmore section of Figure C6A.6.12.6.11-2 as it was shown to be comparable in accuracy to the full Whitmore section for which there is no reduction.

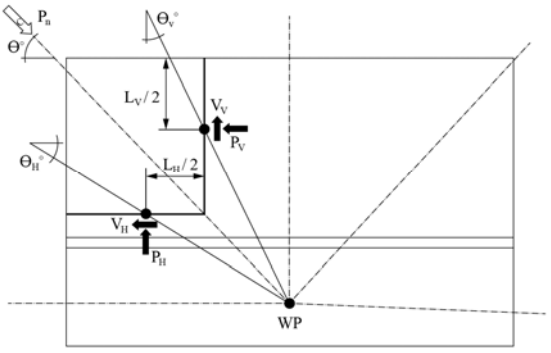

Figure C6A.6.12.6.11-1—Basic Corner Check

In this approach the following assumptions and constraints are made:

- Failure surfaces represent minimum section that includes all member fasteners.
- Forces act at centroid of respective section surfaces.
- Surfaces can carry no moment.
- Combination and normal and shear forces limited by von Mises stress criterion.
- Resultant of each section forces pass through nodal work point.
- Resultant of all section forces must align with member.

Subject to the limitations of other checks, this approach provides a more accurate estimate of capacity when compared to the partial shear check. Since this

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method is generally conservative, it can be further refined by removing certain constraints. For example, it is not essential for the resultants of the section forces to pass through the work point, nor is it necessary for the failure sections to carry no moment. Provided that there is adequate capacity in other areas of the gusset plate, these constraints can be eliminated. If they are eliminated, the other sections of the plate must be evaluated for the corresponding demands. All other checks, i.e., horizontal shear, block shear, etc. still apply. Refer to the WJE reference for examples demonstrating this approach.

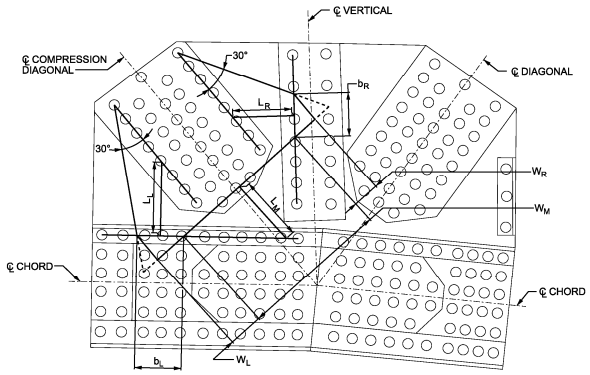

Figure C6A.12.6.11-2—Truncated Whitmore Section

- The Truncated Whitmore Section Method was developed at the Georgia Institute of Technology by Dr. Donald White, et al., as a part of the NCHRP 12-84 Project. Illustrative examples of its application are found in Appendix I, Section 5 of the Final Report, and designated Method 2.
- In utilizing the Truncated Whitmore Section Method, the equations found in Article 6A.6.12.6.7 are applicable, except that:

  a) L_{mid} is replaced by the lengths L_M, L_L, and L_R in calculating the nominal resistance of the widths W_M, W_L, and W_R.

  b) The constant value 3.29 in Eq. 6A.6.12.6.7-4 is to be replaced with the value 6.71 in Warren trusses with a vertical framing into the joint in addition to the diagonal members and 4.25 for all other gusset and joint geometries for calculation of the nominal resistance of the width, W_M, and by 6.71 for calculation of the nominal resistance of the widths W_L and W_R. These coefficients are due to calibration differences between Method 1 (Partial Shear Plane Method) and Method 2.

  c) When computing the nominal compression resistance, P_n, the tributary portions of the gusset gross cross-sectional area within the base dimensional widths W_L and W_R are to be reduced 10 percent.

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d) The total gusset plate compressive resistance is taken as the sum of the contributions from W_M, W_L, and W_R.

- In cases where the Whitmore section is not truncated by the adjacent fastener lines, as shown in Figure C6A.6.12.6.11-2, L_L and L_R are zero and the Truncated Whitmore Section Method reduces to the traditional Whitmore section method with W_M equal to the Whitmore Section width defined in Article 6A6.12.6.7 and $L_M = L_{mid}$.
- When using the Truncated Whitmore Section Method (Method 2), no Partial Plane Shear strength check is required.

6A.7—WOOD STRUCTURES

6A.7.1—Scope

The provisions of Article 6A.7 apply to the evaluation of wood bridges constructed of sawn lumber or glued laminated timber.

6A.7.2—Materials

The reference design values for existing timber bridge components in satisfactory condition may be taken as given in LRFD Design Articles 8.4.1.1.4 and 8.4.1.2.3 and adjusted for actual conditions of use in accordance with LRFD Design Article 8.4.4. To obtain values for species and grades not included in the LRFD articles, a direct conversion of Allowable Stress Design Values in the *National Design Specification for Wood Construction*, 2005 Edition may be performed.

C6A.7.2

The material and member properties based on as-built information may need to be adjusted for field conditions such as weathering or decay. The Engineer's judgment and experience are required in assessing actual member resistance.

Southern Pine and Douglas Fir are the more common types of timber used in bridge construction. Plans and other relevant contract documents should be reviewed to determine the species and grade of wood. When the type of timber is unknown, field identification and grading may be done based on visual appearance, grade marks, local experience, and grade description requirements. Sampling for testing may be done where more exact information is required.

6A.7.3—Resistance Factors

Resistance factors (φ) for the strength limit state shall be taken as given in LRFD Design Article 8.5.2.2.

C6A.7.3

Some older timber bridges may not have the roadway deck continuously attached to the beams. The resistance of beams not continuously braced in the lateral direction should be reduced in accordance with LRFD provisions (LRFD Design Article 8.6.2).

6A.7.4—Limit States

The applicable limit states for the evaluation of wood bridges shall be taken as specified in Table 6A.4.2.2-1 and in these Articles.

C6A.7.4

Deflection control on timber components as specified in LRFD Design Article 2.5.2.6.2 may be applied to evaluation if the bridge superstructure was observed to exhibit excessive flexing under normal traffic. This is an optional requirement.

6A.7.4.1—Design-Load Rating

Rating factors for the design-load rating shall be based on the Strength I load combination.

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6A.7.4.2—Legal Load Rating and Permit Load Rating

Wood bridge components shall be load rated for the Strength I load combination for legal loads, and for Strength II load combination for permit loads.

6A.7.5—Dynamic Load Allowance

Dynamic load allowance need not be applied to wood components (LRFD Design Article 3.6.2.3).

6A.7.6—Evaluation of Critical Connections

Critical connections of timber bridges shall be evaluated for shear at the strength limit state.

C6A.7.6

External connections of nonredundant members are considered critical connections. Split rings and shear plates may be concealed between wood members. These significantly increase the shear strength of bolted connections. Available records should be consulted to verify their presence. Sometimes a probe may be used to locate them.

6A.8—POSTING OF BRIDGES

6A.8.1—General

Weight limitations for the posted structure should conform to local regulations or policy, using the guidelines in this Manual. Bridge posting should not be confused with bridge-load rating. Bridge inspection and rating are engineering-related activities, whereas bridge posting is a policy decision. If state legal loads exceed the calculated load capacity of the bridge, the bridge must be posted; however, the bridge may be posted at a lower level.

Bridges not capable of carrying a minimum gross live load weight of three tons must be closed. A Bridge Owner may close a structure at any higher posting threshold. When deciding whether to close or post a bridge, the Owner should consider the character of traffic, the likelihood of overweight vehicles, and the enforceability of weight posting.

C6A.8.1

Field experience and tests on reinforced concrete bridges (T-beam and slab bridges) have shown that there is considerable reserve capacity beyond the computed value, and that such spans show considerable distress (e.g., cracking, spalling, deflections, etc.) before severe damage and collapse actually occur.

6A.8.2—Posting Loads

When the maximum legal load under state law exceeds the safe load capacity of a bridge, restrictive load posting shall be required. Though there is variation among the states with respect to the type of signs preferred for posting bridges, most states use either a single weight-limit sign or a three-vehicle combination sign. In any case, the posting signs shall conform to the *Manual on Uniform Traffic Control Devices* (MUTCD).

The live load to be used for posting considerations should be any of the typical AASHTO legal loads given below or state legal loads:

1. Type 3, Type 3S2, Type 3-3, or lane loads (shown in Figures D6A-1 thru D6A-5), for routine single and combination commercial vehicles, and

C6A.8.2

The wide variety of vehicle types cannot be effectively controlled by any single-posting load. A single-posting load based on a short truck model would be too restrictive for longer truck combinations, particularly for short-span bridges. A single-posting load based on a longer combination would be too liberal for almost any span combination.

The three vehicles: Type 3, 3S2, and 3-3 adequately model short vehicles and combination vehicles in general use in the United States. The four single-unit posting trucks SU4, SU5, SU6, and SU7 model the short wheelbase, multi-axle Specialized Hauling Vehicles (SHVs) that are becoming increasingly more common. These SU trucks were developed to model the extreme loading effects of single-unit SHVs with 4 or more axles.

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2. A single Type SU4, Type SU5, Type SU6, or Type SU7 (shown in Figure D6A-7) for single-unit specialized hauling vehicles.

The rating factors obtained for the AASHTO posting vehicles and lane type loads are used in Article 6A.8.3 to develop safe posting loads for single and combination vehicles.

6A.8.3—Posting Analysis

The decision to load post a bridge should be made by the Bridge Owner based on the general procedures as set forth in this Manual and established practices of the Bridge Owner. The following guidelines may be of assistance to authorities responsible for establishing posting weight limits.

When the rating factor, *RF,* calculated for each legal truck (AASHTO vehicle) is greater than 1.0, the bridge need not be posted.

When for any legal truck the *RF* is between 0.3 and 1.0, the following equation should be used to establish the safe posting load for that vehicle type:

$$\text{Safe Posting Load} = \frac{W}{0.7}\left[\left(RF\right) - 0.3\right] \qquad (6A.8.3\text{-}1)$$

where:

For bridges that do not rate for the NRL loading, a posting analysis should be performed to resolve posting requirements for single-unit multi-axle trucks. While a single envelope notional rating load NRL can provide considerable simplification of load-rating computations, additional legal loads for posting are needed to give more accurate posting values. Certain multi-axle Formula B configurations that cause the highest load effects appear to be common only in some states, and they should not lead to reduced postings in all states. Further, some states may have specific rules that prohibit certain Formula B configurations.

Setting weight limits for posting often requires the evaluator to determine safe load capacities for legal truck types that operate within a given state, in accordance with state posting practices. The four single-unit Formula B legal loads shown in Figure D6A-7 include the worst 4-axle (SU4), worst 5-axle (SU5), worst 6-axle (SU6), and worst 7-axle (SU7) trucks (7-axle is also representative of 8-axle trucks) identified in the NCHRP 12-63 study. This series of loads affords the evaluator the flexibility of selecting only posting loads that model commercial Formula B trucks in a particular state or jurisdiction.

The more compact four- and five-axle trucks (SU4 and SU5) that produce the highest moment or shear per unit weight of truck will often govern the posting value (result in the lowest weight limit). States that post bridges for a single tonnage for all legal single-unit trucks may consider it desirable to reduce the number of new posting loads that need to be evaluated. Here it would be appropriate to use truck SU5 as a single representative posting load for the series of Formula B truck configurations with 5 to 8 axles. This simplification will introduce added conservatism in posting, especially for short span bridges. It should be noted that situations could arise where a bridge may have a $RF > 1.0$ for SU5 but may have a $RF < 1.0$ for SU6 or SU7. Here the SU5 load model is being utilized to determine a single posting load for a bridge for trucks with six or seven axles, even though the bridge has adequate capacity for SU5.

C6A.8.3

The safe load capacity for an existing bridge established using load rating procedures provided in this Manual represents an upper bound for posting loads. It reflects superstructure redundancy, traffic characteristics, and condition of the bridge so that further consideration of these factors during posting would not be necessary.

The lower limit of $RF = 0.3$ at which the bridge must be closed was derived based on several factors which change the uncertainties of the safety of posted bridges compared to unposted situations. The rating factor of 0.3 may also in some cases be similar to existing bridge closing levels based on inventory levels of stress. The posting graph in Figure 6A.8.3-1 provides posting loads which drop off more quickly than does the rating factor. This causes a conservative selection of posting loads relative to the numerically calculated rating factor and is intended to cover the following variables:

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RF = Legal load rating factor

W = Weight of rating vehicle

When the RF for any vehicle type falls below 0.3, then that vehicle type should not be allowed on the span.

Where the RF is governed by the lane load shown in Figures D6A-4 and D6A-5, then the value of W in Eq. 6A.8.3-1 shall be taken as 80 kips. When States use their own legal loads which are different from the AASHTO legal loads, Eq. 6A.8.3-1 may be used for the posting load, but the gross weight of the State's legal vehicle shall be substituted in the posting equation.

A Bridge Owner may close a structure at any posting threshold but bridges not capable of carrying a minimum gross live load weight of three tons must be closed.

- The statistical distribution of gross vehicle weights will be markedly different for a posted structure with a greater percentage of vehicles at or exceeding the posted limit compared to numbers exceeding the legal limit on an unposted bridge. An allowance for potential overloads is contained in the posting curve presented herein. Any overload allowance or safety margin should not be used as a justification for subverting legal posted signs.
- The dynamic load allowance increases as the gross weight of a vehicle decreases and this increase is reflected in the posting curve.
- The distribution of gross vehicle weight to individual axles may change as the gross legal weight decreases. A vehicle could satisfy both the posted gross and the individual axle combination limits and still cause a load effect in excess of that assumed in the rating factor calculation which uses a standard axle distribution. This acute load distribution on the axles has been incorporated in the posting curve.

The reliability level inherent in the posting curve is raised at the lower posting loads to achieve reliability targets closer to design inventory levels rather than the evaluation or operating reliability characteristic of other practices in this Manual.

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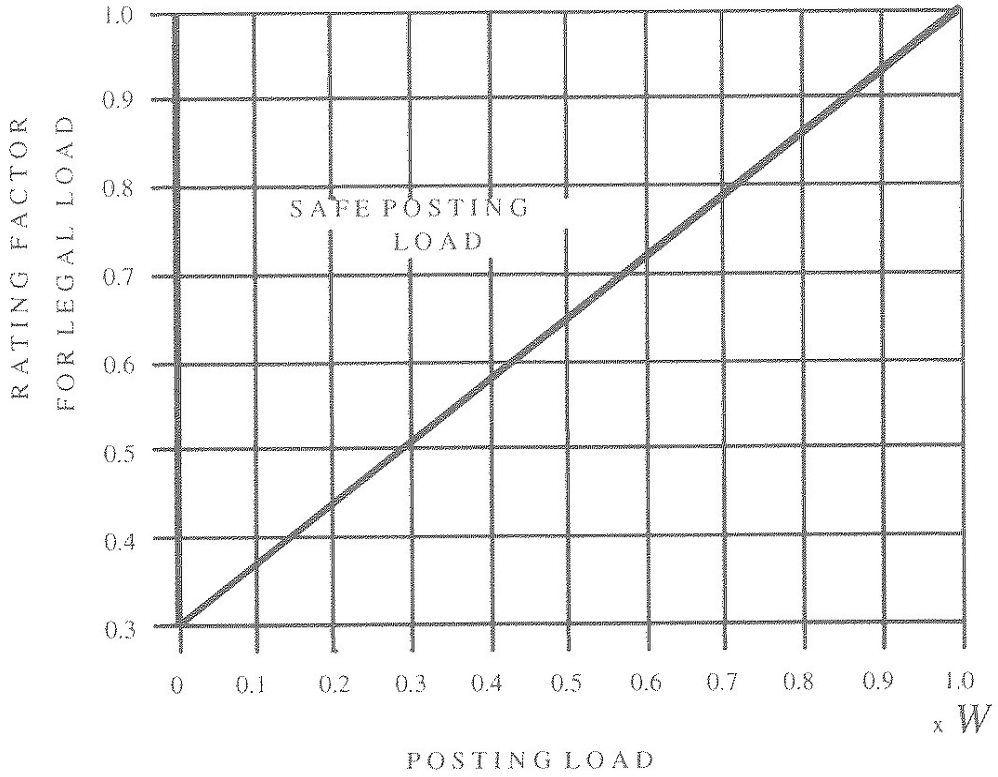

Figure 6A.8.3-1—Calculation of Posting Load

6A.8.4—Regulatory Signs

Regulatory signing shall conform to the requirements of the *Manual on Uniform Traffic Control Devices* (MUTCD) or other governing regulations, and shall be established in accordance with the requirements of the agency having authority over the highway.

When a decision is made to close a bridge, signs and properly designed, structurally sound traffic barriers shall be erected to provide adequate warning and protection to the traveling public. If pedestrian travel across the bridge is also restricted, adequate measures to prevent pedestrian use of the bridge shall be installed. Signs and barriers shall meet or exceed the requirements of local laws and the applicable sections of the MUTCD. Bridge closure signs and barriers shall be inspected periodically to ensure their continued effectiveness.

6A.8.5—Speed Limits

In some cases, lower speed limits will reduce impact loads to the extent that lowering the weight limit may not be required. Consideration of a speed posting will depend upon alignment, general location, volume, and type of

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traffic. A speed posting should not be considered as a basis for increasing the weight limit in areas where enforcement will be difficult and frequent violations can be anticipated.

6A.9—SPECIAL TOPICS

6A.9.1—Evaluation of Unreinforced Masonry Arches

6A.9.1.1—General

The predominant type of unreinforced masonry bridge is the filled spandrel arch. Materials may be unreinforced concrete, brick, and ashlar or rubble stone masonry. Mortar used to bind the individual masonry units should be classified in accordance with ASTM C270.

The total load-carrying capacity of an unreinforced masonry arch should be evaluated by the Allowable Stress method (Article 6B.5.2.6) based on limitation of the tensile and compressive stresses developed in the extreme fiber when axial and bending stresses are combined, and on failure modes due to instability.

6A.9.1.2—Method of Analysis

Internal stresses of masonry arches are usually analyzed by regarding the arch as an elastic redundant structure. When evaluating masonry arches, three types of failures are generally investigated: 1) overturning of two adjacent masonry units of the arch, 2) sliding or shear failure, and 3) compressive failure of the masonry.

There may be instances in which the capacity of the arch based on approximate analysis methods may be inadequate or the behavior of the arch under traffic is not consistent with that predicted by evaluation. In these situations load tests or more refined analysis may be helpful in establishing a more accurate safe load capacity.

C6A.9.1.2

Failure due to crushing of the masonry material is not common. In classical arch analysis, the stability of the arch masonry units is ensured by keeping the line of resistance (or the resultant of the moment and thrust at a given point) within the middle third of the arch ring (or within the kern). Keeping the resultant within the kern will ensure that no part of the arch is subjected to tension.

Classical analysis of filled arches tends to greatly under-estimate their true capacity. The filled arch is a very complex structure composed of both the arch ring and the surrounding fill. A rigorous solution to establish the load capacity of masonry arches should consider the soil-structure interaction including the effects of lateral earth pressure. Classical arch analysis neglects the effects of lateral earth pressure on arch behavior. In filled arches the passive restraint of the fill is sufficient to greatly limit the distortion of the arch under live load. A large portion of the composite stiffness of the arch and fill is due to the restraint of the fill.

A number of simple empirical methods and computer-based analysis methods have been developed to assess masonry arch bridges in the United Kingdom, where a significant portion of the bridge stock is said to consist of masonry arches. Details of these methods are contained in *The Assessment of Highway Bridges and Structures BD 21/97 & BA 16/93, Department of Transport, UK.*

6A.9.1.3—Allowable Stresses in Masonry

The allowable stresses in masonry materials shall be as specified in Article 6B.5.2.6 of this Manual.

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6A.9.2—Historic Bridges

Most states have undertaken historic bridge surveys to identify which of their bridges that were built more than 50 years ago are historic. Historic bridge survey information is generally maintained by the state Department of Transportation, and it may be in a master database and/or may have been entered into the state's BMS database. This information is frequently part of the bridge record, and it offers guidance on why the bridge is noteworthy. The survey data may also contain useful information about original design details.

Historic bridges are defined as those that meet the National Register of Historic Places' criteria for evaluation. The criteria establish a measure of consideration to evaluate which bridges have the significance and integrity to be determined historic and thus worthy of preservation. Many types of bridges, from stone arch and metal truss bridges to early continuous stringer and prestressed beam bridges have been determined to be historic for their technological significance. Other bridges are historic because they are located in historic districts or are associated with historic transportation routes, such as rail lines or parkways.

Historic bridges, like all other National Register-listed or eligible resources, are affected by federal laws intended to strengthen the governmental commitment to preservation. This means that all work needs to be done in compliance with the applicable federal, and often state, regulations and procedures. They require consideration of the historic significance of the bridge when developing maintenance, repair, and/or rehabilitation methodologies. The goal is to avoid having an adverse effect on the historic bridge. Guidance on how to develop successful approaches for working on historic bridges can be found in *The Secretary of the Interior's Standards for Rehabilitation* and *The Secretary of the Interior's Standards for the Treatment of Historic Properties 1992*. Both offer approaches for considering ways to upgrade structures while maintaining their historic fabric and significance, and they are available from the National Park Service Preservation Assistance Division or the state historic preservation office.

Because historic bridges require demonstrated consideration of ways to avoid adverse effects, evaluations should be complete, encompassing the relevant parts of this Manual. Nondestructive testing methods should be considered to verify components and system performance. Repair rather than replacement of original elements should be considered, and any replacement should be in kind where feasible. Strengthening should be done in a manner that is respectful to the historic bridge.

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APPENDIX A6A—LOAD AND RESISTANCE FACTOR RATING FLOW CHART

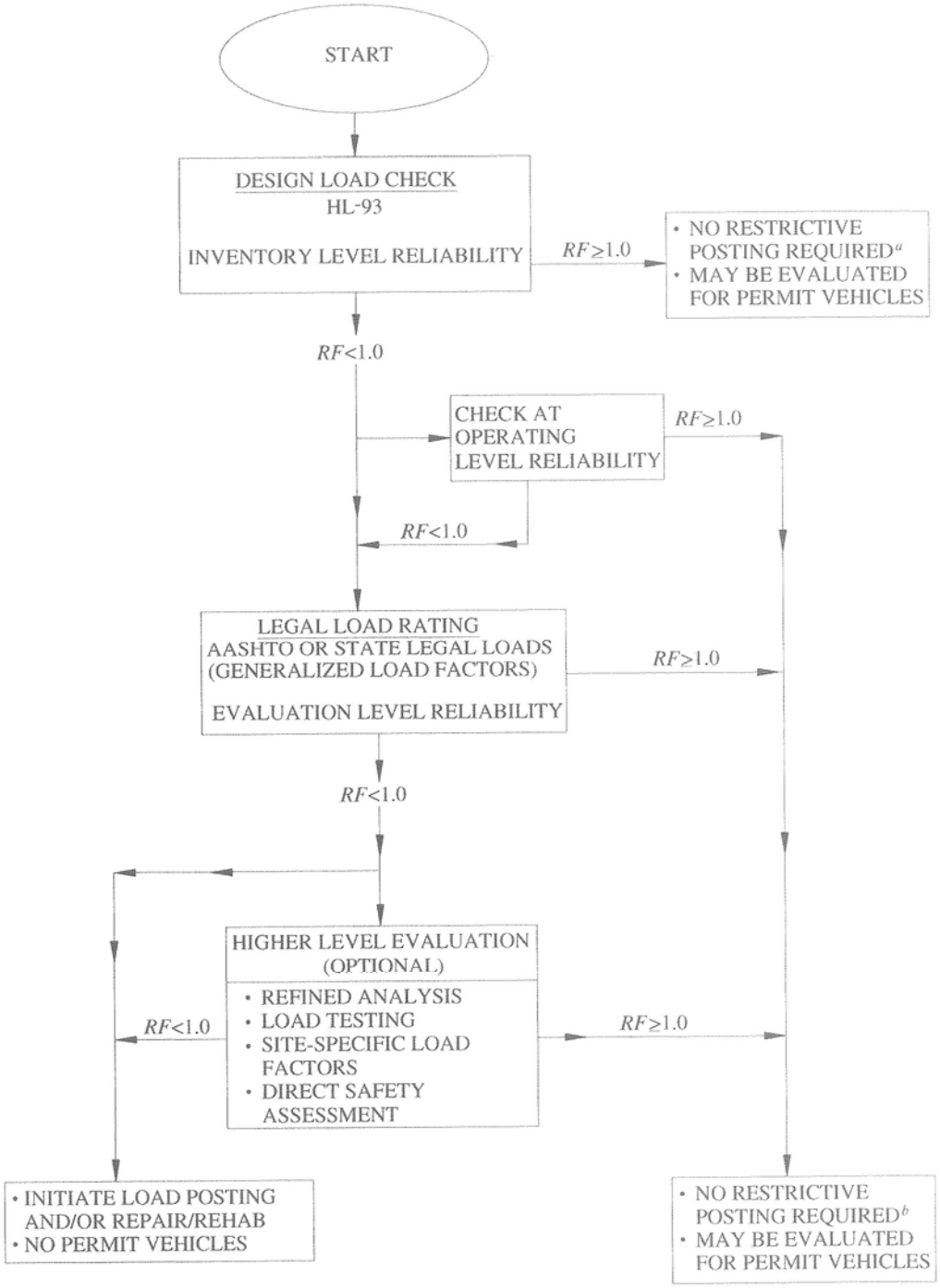

Figure A6A-1—Load and Resistance Factor Rating Flow Chart

[a] For routinely permitted commercial traffic on highways of various states under grandfather exclusions to federal weight laws

[b] For legal loads that comply with federal weight limits and Formula B

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APPENDIX B6A—LIMIT STATES AND LOAD FACTORS FOR LOAD RATING

Table B6A-1—Limit States and Load Factors for Load Rating (6A.4.2.2-1)

Bridge Type	Limit State*	Dead Load *DC*	Dead Load *DW*	Design Load Inventory *LL*	Design Load Operating *LL*	Legal Load *LL*	Permit Load *LL*
Steel	Strength I	1.25	1.50	1.75	1.35	Tables 6A.4.4.2.3a-1 and 6A.4.4.2.3b-1	—
	Strength II	1.25	1.50	—	—	—	Table 6A.4.5.4.2a-1
	Service II	1.00	1.00	1.30	1.00	1.30	1.00
	Fatigue	0.00	0.00	0.80	—	—	—
Reinforced Concrete	Strength I	1.25	1.50	1.75	1.35	Tables 6A.4.4.2.3a-1 and 6A.4.4.2.3b-1	—
	Strength II	1.25	1.50	—	—	—	Table 6A.4.5.4.2a-1
	Service I	1.00	1.00	—	—	—	1.00
Prestressed Concrete	Strength I	1.25	1.50	1.75	1.35	Tables 6A.4.4.2.3a-1 and 6A.4.4.2.3b-1	—
	Strength II	1.25	1.50	—	—	—	Table 6A.4.5.4.2a-1
	Service III	1.00	1.00	0.80	—	1.00	—
	Service I	1.00	1.00	—	—	—	1.00
Wood	Strength I	1.25	1.50	1.75	1.35	Tables 6A.4.4.2.3a-1 and 6A.4.4.2.3b-1	—
	Strength II	1.25	1.50	—	—	–	Table 6A.4.5.4.2a-1

* Defined in the *AASHTO LRFD Bridge Design Specifications*.

Shaded cells of the table indicate optional checks.

Service I is used to check the $0.9F_y$ stress limit in reinforcing steel.

Load factor for *DW* at the strength limit state may be taken as 1.25 where thickness has been field measured.

Fatigue limit state is checked using the LRFD fatigue truck (see Article 6A.6.4.1).

Table B6A-2—Generalized Live Load Factors for Legal Loads: γ_L for Routine Commercial Traffic (6A.4.4.2.3a-1)

Traffic Volume (one direction)	Load Factor
Unknown	1.45
ADTT ≥ 5,000	1.45
ADTT = 1,000	1.30

Note: Linear interpolation is permitted for other *ADTT* values between 1,000 and 5,000.

Table B6A-3—Generalized Live Load Factors, γ_L for Specialized Hauling Vehicles (6A.4.4.2.3b-1)

Traffic Volume (one direction)	Load Factor
Unknown	1.45
ADTT ≥ 5,000	1.45
ADTT = 1,000	1.30

Note: Linear interpolation is permitted for other *ADTT* values between 1,000 and 5,000.

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Table B6A-4—Permit Load Factors: γ_L (6A.4.5.4.2a-1)

Permit Type	Frequency	Loading Condition	*DF*[a]	*ADTT* (one direction)	Load Factor by Permit Weight Ratio[b]		
					GVW / AL < 2.0 (kip/ft)	2.0 < GVW/AL < 3.0 (kip/ft)	GVW/AL > 3.0 (kip/ft)
Routine or Annual	Unlimited Crossings	Mix with traffic (other vehicles may be on the bridge)	Two or more lanes	>5,000	1.40	1.35	1.30
				=1,000	1.35	1.25	1.20
				<100	1.30	1.20	1.15
					All Weights		
Special or Limited Crossing	Single-Trip	Escorted with no other vehicles on the bridge	One lane	N/A	1.10		
	Single-Trip	Mix with traffic (other vehicles may be on the bridge)	One lane	All *ADTT*s	1.20		
	Multiple Trips (less than 100 crossings	Mix with traffic (other vehicles may be on the bridge)	One lane	All *ADTT*s	1.40		

Notes:

[a] *DF* = LRFD distribution factor. When one-lane distribution factor is used, the built-in multiple presence factor should be divided out.

[b] Permit Weight Ratio = GVW/AL where GVW = Gross Vehicle Weight and AL = Front axle to rear axle length. Use only axles on the bridge.

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APPENDIX C6A—LRFD DESIGN LIVE LOAD (HL-93) (LRFD DESIGN ARTICLE 3.6.1)

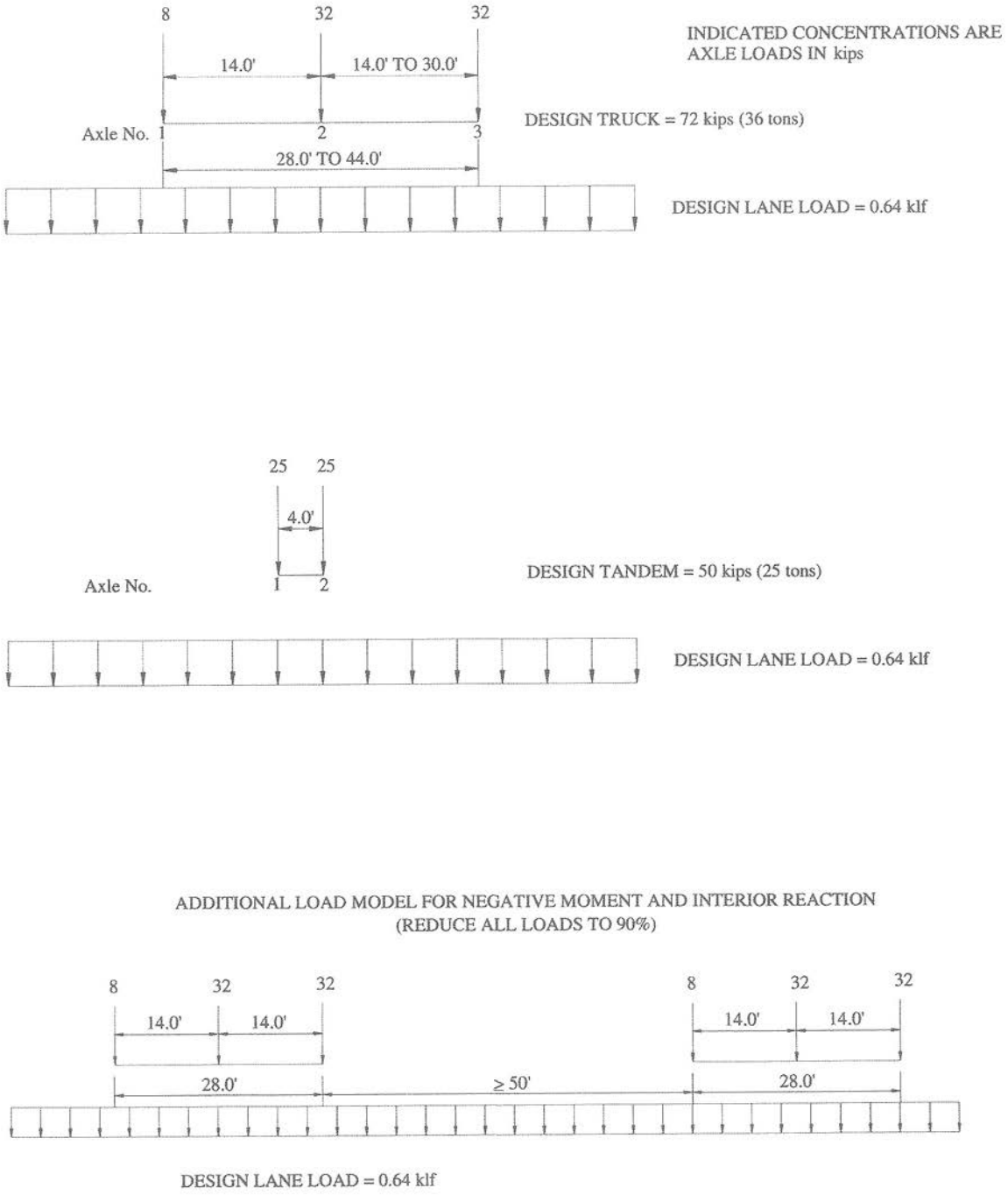

Figure C6A-1—LRFD Design Live Load (HL-93)

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APPENDIX D6A—AASHTO LEGAL LOADS

a. *AASHTO Trucks*—Apply for all span lengths and load effects.

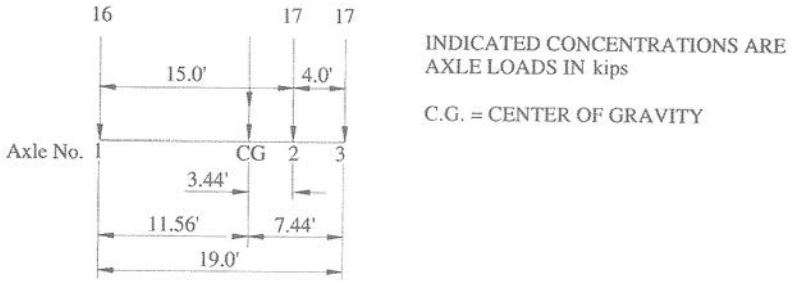

Figure D6A-1—Type 3 Unit; Weight = 50 kips (25 tons)

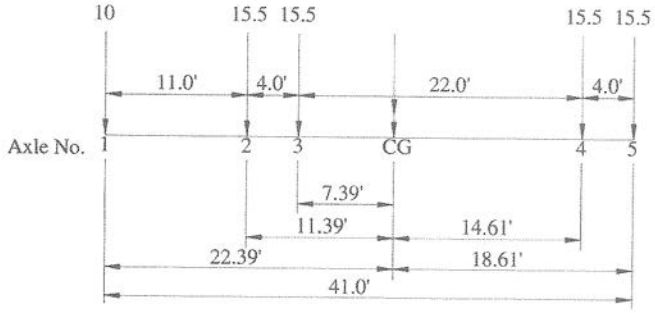

Figure D6A-2—Type 3S2 Unit; Weight = 72 kips (36 tons)

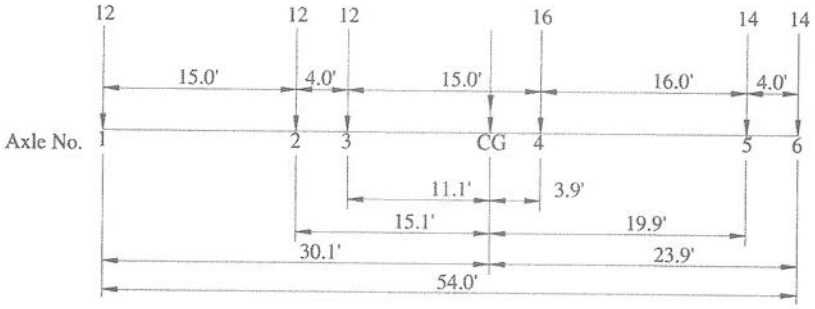

Figure D6A-3—Type 3-3 Unit; Weight = 80 kips (40 tons)

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b. *Lane-Type Legal Load Model*—Apply for spans greater than 200 ft and all load effects.

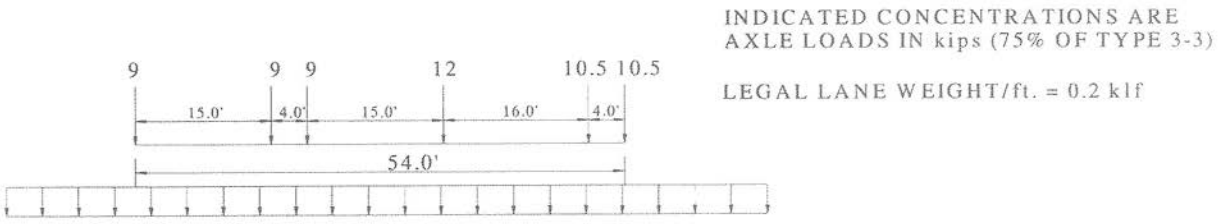


Figure D6A-4—Lane-Type Loading for Spans Greater than 200 ft

c. *Lane-Type Legal Load Mode*l—Apply for negative moment and interior reaction for all span lengths.

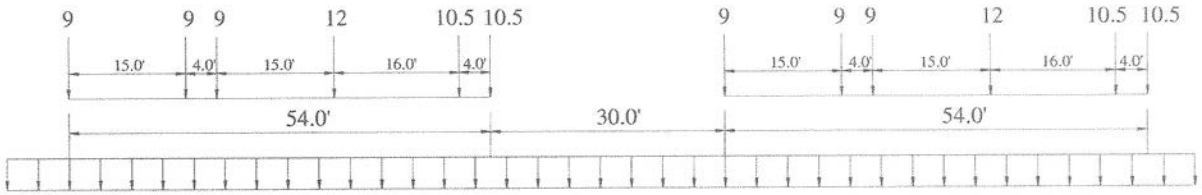
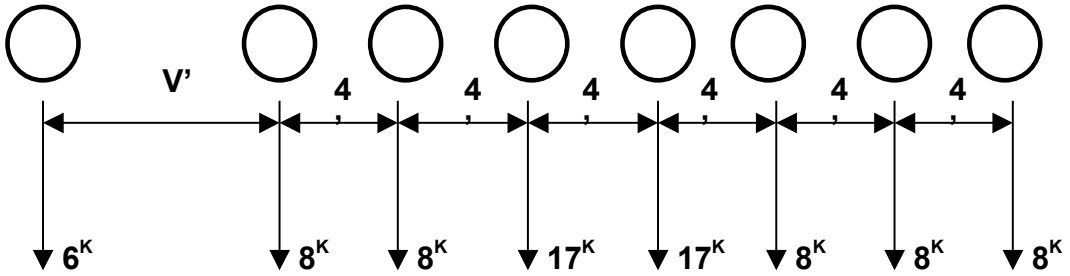


Figure D6A-5—Lane-Type Loading for Negative Moment and Interior Reaction

d. *Notional Rating Load*—Apply for all span lengths and load effects.



V=VARIABLE DRIVE AXLE SPACING—6'0" TO 14'0". SPACING TO BE USED IS THAT WHICH PRODUCES MAXIMUM LOAD EFFECTS.

AXLES THAT DO NOT CONTRIBUTE TO THE MAXIMUM LOAD EFFECTS UNDER CONSIDERATION SHALL BE NEGLECTED.

MAXIMUM GVW = 80 KIPS

AXLE GAGE WIDTH = 6'0"

Figure D6A-6—Notional Rating Load (NRL) for Single-Unit SHVs that Meet Federal Bridge Formula B

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e. Single-Unit Bridge Posting Loads

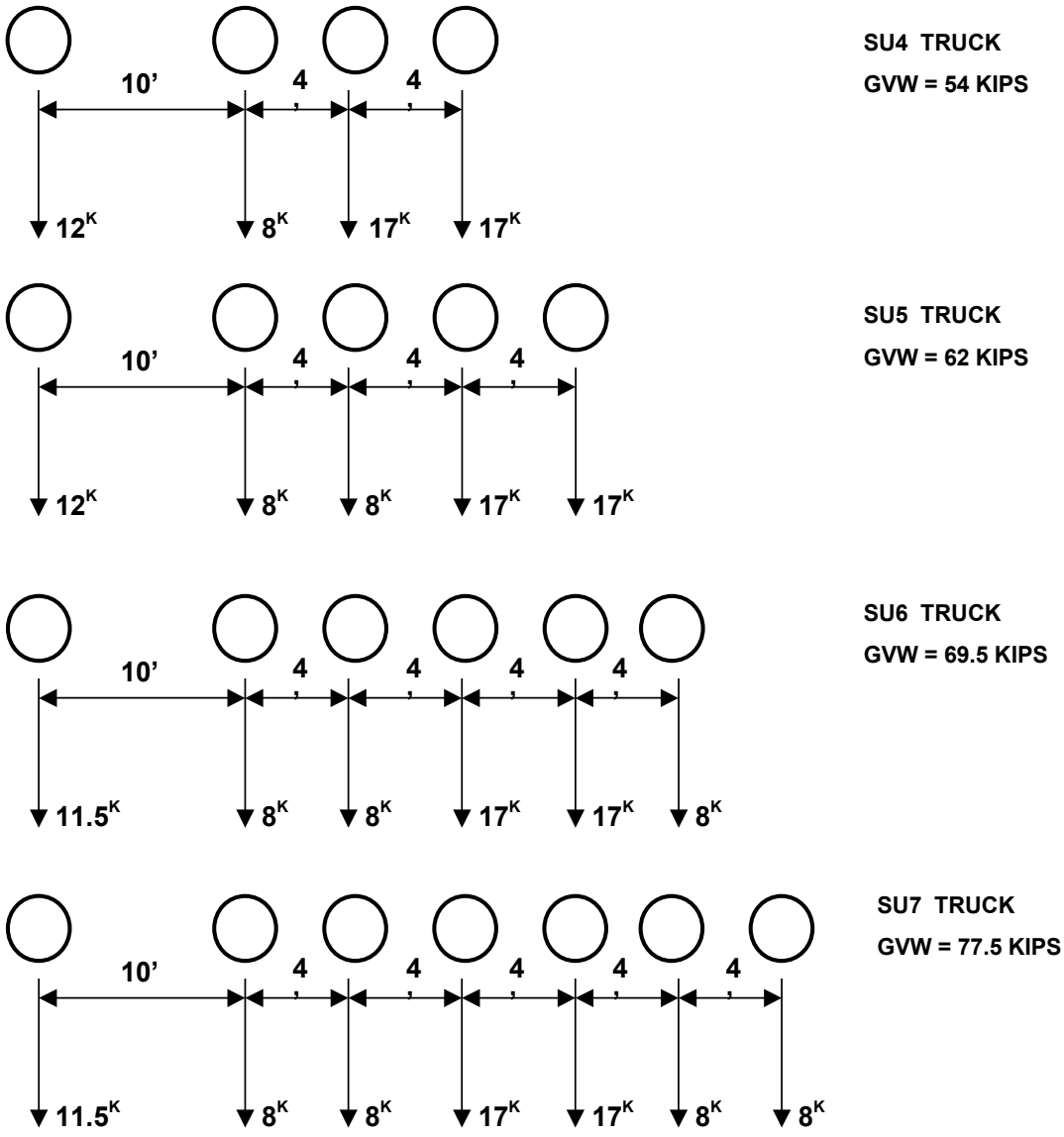

Figure D6A-7—Bridge Posting Loads for Single-Unit SHVs that Meet Federal Bridge Formula B

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APPENDIX E6A—LIVE LOAD MOMENTS ON LONGITUDINAL STRINGERS OR GIRDERS (SIMPLE SPAN)

Table E6A-1—Live Load Moments in kip ft per Lane with 33 percent *IM*

Span, ft	AASHTO Legal Loads				Design Load HL-93
	3	3-S2	3-3	Lane	
20	183.3	167.0	150.8		296.0
21	194.4	177.2	160.1		315.7
22	205.6	187.5	169.2		335.6
23	216.8	200.0	178.5		355.6
24	228.0	213.6	187.8		375.8
25	239.1	227.1	197.1		396.2
26	250.6	240.7	206.2		416.7
27	261.7	254.3	215.5		437.3
28	272.9	267.9	224.8		458.2
29	284.1	281.7	234.1		479.2
30	300.3	295.3	243.4		500.3
32	333.3	322.4	270.0		543.1
34	366.0	349.8	298.7		586.5
36	399.0	376.9	327.4		630.5
38	432.0	404.1	356.4		675.2
40	465.0	431.5	385.2		722.0
42	498.0	458.6	414.2		781.2
44	531.2	486.0	443.2		843.5
46	564.2	513.1	471.9		906.4
48	597.2	540.5	500.9		970.0
50	630.4	587.3	530.1		1034.0
52	663.4	634.1	570.0		1099.0
54	696.4	681.2	615.3		1164.0
56	729.6	728.3	660.5		1230.0
58	762.9	775.1	705.7		1297.0
60	795.9	822.5	750.9		1364.0
70	961.6	1058.7	990.1		1711.0
80	1127.6	1295.7	1255.3		2073.0
90	1293.6	1533.2	1520.7		2451.0
100	1459.5	1771.3	1786.2		2846.0
120	1791.8	2248.0	2317.7		3682.0
140	2124.0	2725.2	2849.1		4582.0
160	2456.5	3202.9	3380.6		5546.0
180	2788.7	3680.6	3912.3		6574.0
200	3121.2	4158.9	4444.3	4333.2	7665.0
250	3952.2	5354.6	5773.8	5892.8	10672.0
300	4783.2	6550.5	7103.5	7577.6	14077.0

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Table E6A-2—Live Load Moments in kip ft per Lane with 33 percent *IM*

Span, ft	Specialized Hauling Vehicles				NRL	Design Load HL-93
	SU4	SU5	SU6	SU7		
20	213.3	223.4	234.3	234.3	234.3	296.0
21	227.2	240.2	253.5	253.5	253.5	315.7
22	241.3	256.7	272.9	272.9	272.9	335.6
23	255.1	273.4	292.1	293.1	293.1	355.6
24	269.2	289.9	311.5	314.9	314.9	375.8
25	283.0	306.7	330.6	336.6	336.6	396.2
26	296.9	323.2	350.1	358.6	360.4	416.7
27	311.0	339.9	369.2	380.4	385.2	437.3
28	328.2	356.4	388.6	402.2	409.6	458.2
29	346.1	373.2	407.8	424.0	434.4	479.2
30	363.9	389.7	427.1	445.8	458.9	500.3
32	399.5	422.9	465.8	489.4	508.1	543.1
34	435.2	457.5	504.3	533.3	557.3	586.5
36	471.1	498.2	546.4	577.0	608.9	630.5
38	506.7	539.2	592.4	626.7	661.8	675.2
40	542.5	579.9	638.4	678.3	715.0	722.0
42	578.3	620.8	684.4	729.6	768.2	781.2
44	614.2	661.5	730.7	781.2	821.1	843.5
46	649.8	702.5	776.7	832.8	874.3	906.4
48	685.7	743.5	822.7	884.2	927.5	970.0
50	721.7	784.4	868.8	935.8	980.6	1034.0
52	757.6	825.4	915.0	987.4	1033.7	1099.0
54	793.2	866.4	961.1	1038.7	1086.9	1164.0
56	829.1	907.3	1007.3	1090.3	1140.1	1230.0
58	865.0	948.6	1053.4	1141.9	1193.3	1297.0
60	900.8	989.5	1099.4	1193.4	1246.2	1364.0
70	1080.1	1194.9	1330.3	1451.0	1512.1	1711.0
80	1259.5	1400.5	1561.2	1708.7	1777.9	2073.0
90	1438.8	1606.1	1792.0	1966.3	2043.8	2451.0
100	1618.3	1812.0	2022.9	2224.0	2309.7	2846.0
120	1977.2	2223.8	2484.8	2739.3	2841.5	3682.0
140	2336.3	2635.7	2946.9	3254.6	3373.4	4582.0
160	2695.1	3047.7	3408.9	3770.0	3905.4	5546.0
180	3054.2	3459.7	3871.1	4285.4	4437.4	6574.0
200	3413.3	3871.9	4333.1	4800.8	4969.3	7665.0
250	4311.1	4902.4	5488.4	6089.3	6299.1	10672.0
300	5208.5	5932.9	6643.9	7377.5	7629.1	14077.0

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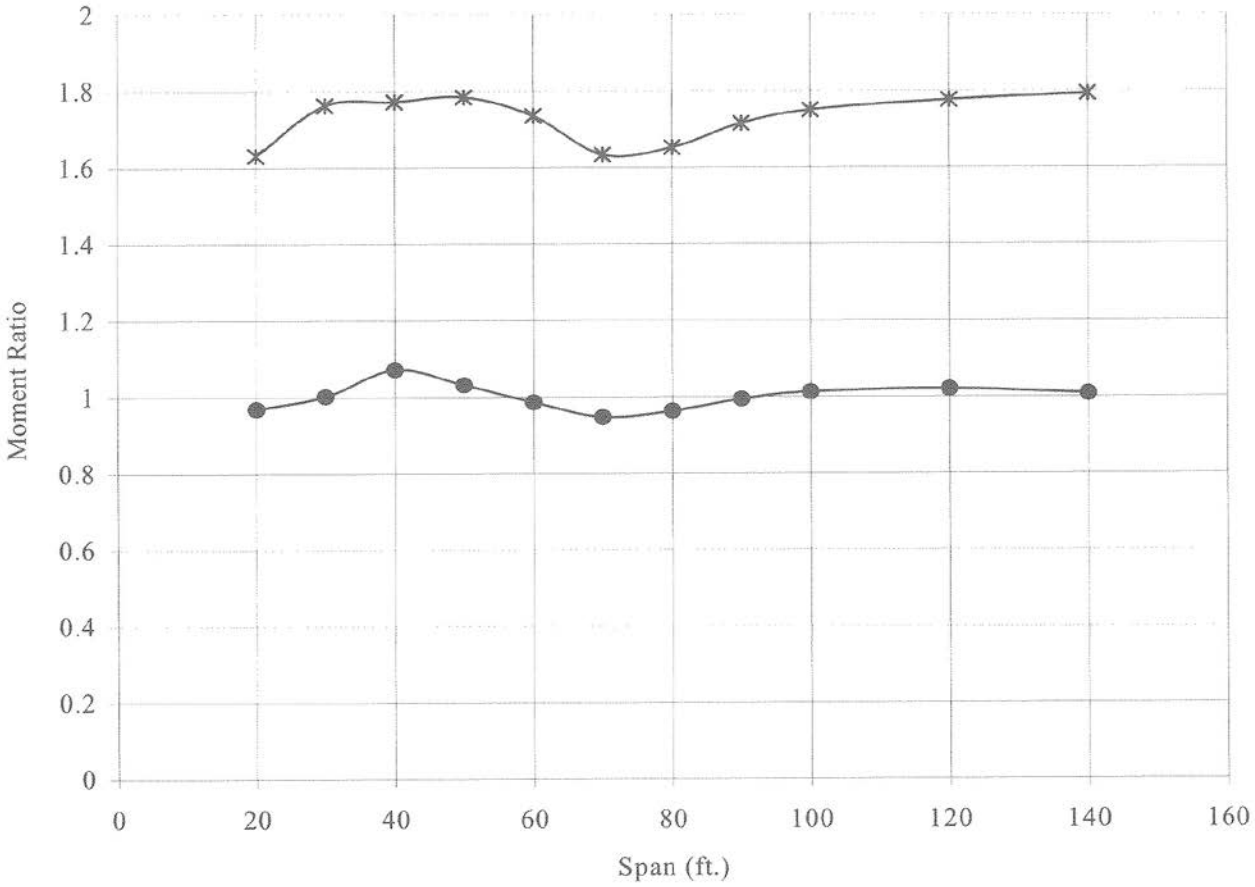
APPENDIX F6A—VARIATION IN MOMENT RATIO WITH SPAN LENGTH

$$\text{Moment Ratio} = \frac{\text{Simple Span Maximum Moment Caused by the Exclusion Vehicle Population}}{\text{Simple Span Maximum Moment Caused by Each Load Model}}$$



$$\text{—}*\text{—} = \frac{\text{Exclusion Vehicle Population}}{\text{AASHTO Legal Loads}}$$

$$\text{—}\bullet\text{—} = \frac{\text{Exclusion Vehicle Population}}{\text{HL}-93}$$

Figure F6A-1—Variation in Moment Ratio with Span Length

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APPENDIX G6A—RATING OF CONCRETE COMPONENTS FOR COMPRESSION PLUS BENDING

Steps for Obtaining Rating Factors (see Figure G6A-1)

1. Develop the interaction diagram, by computer or manual methods, using as-inspected section properties.
2. Point **A** represents the factored dead load moment and thrust.
3. Using the factored live load moment and thrust for the rating live load, compute the live load eccentricity ($e_1 = M_{LL}/P_{LL}$).
4. Continue from Point **A** with the live load eccentricity to the intersection with the interaction diagram.
5. Read the ultimate moment and axial capacities from the diagram.
6. $$\text{Moment } RF = \frac{\text{Moment Capacity} - \text{Factored } M_{DL}}{\text{Factored } M_{LL+IM}}$$

$$\text{Axial } RF = \frac{\text{Axial Capacity} - \text{Factored } P_{DL}}{\text{Factored } P_{LL+IM}}$$

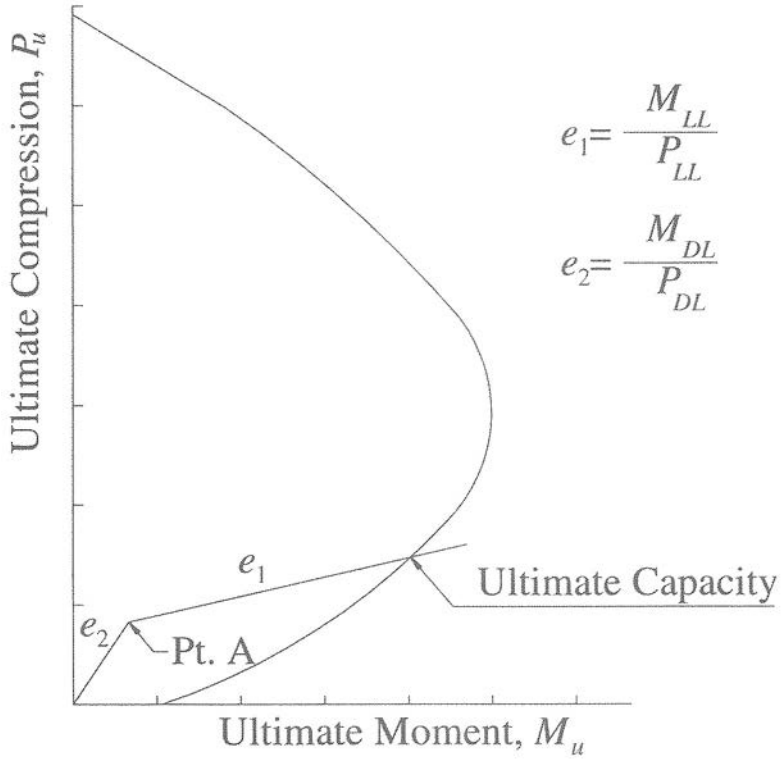

Figure G6A-1—Axial Plus Bending Interaction Diagram for Concrete Structures

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APPENDIX H6A—RATING OF STEEL MEMBERS FOR COMPRESSION PLUS BENDING

Combined Axial and Flexural Strength I Limit State for Steel Moment Magnification—Beam Columns

LRFD Design Articles 6.9.2.2 and 4.5.3.2.2b

$P_u = \gamma_D P_{DL} + (RF)\gamma_L P_{LL+IM}$

$M_u = \delta_b \left[\gamma_D M_{DL} + (RF)\gamma_L M_{LL+IM}\right]$

δ_b = Moment or stress magnifier for braced mode deflection

If $\frac{P_u}{P_r} > 0.2$ and $M_{uy} = 0$ then:

$\frac{P_u}{P_r} + \frac{8.0}{9.0}\frac{M_{ux}}{M_{rx}} \le 1.0$ for rating the correct *RF* will make this an equality.

LRFD Design Eq. 6.9.2.2-2

Substituting:

$$\frac{\gamma_D P_{DL} + RF \times \gamma_L \times P_{LL+IM}}{P_r} + \frac{8}{9M_r}\left(\delta_b \gamma_D M_{DL} + \delta_b \times RF \times \gamma_L M_{LL+IM}\right) = 1.0$$

$$\gamma_D\left[\frac{P_{DL}}{P_r} + \frac{8}{9}\delta_b\left(\frac{M_{DL}}{M_r}\right)\right] + RF \times \gamma_L\left[\frac{P_{LL+IM}}{P_r} + \frac{8}{9}\delta_b\left(\frac{M_{LL+IM}}{M_r}\right)\right] = 1.0$$

$$RF = \frac{1 - \gamma_D\left[\frac{P_{DL}}{P_r} + \frac{8}{9}\delta_b\left(\frac{M_{DL}}{M_r}\right)\right]}{\gamma_L\left[\frac{P_{LL+IM}}{P_r} + \frac{8}{9}\delta_b\left(\frac{M_{LL+IM}}{M_r}\right)\right]}$$

where:

$$\delta_b = \frac{C_m}{1 - \frac{P_u}{\varphi P_e}} \ge 1.0$$

$$= \frac{C_m}{1 - \frac{\gamma_D P_{DL} + RF \times \gamma_L P_{LL+IM}}{\varphi P_e}}$$

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Moment magnifier may be approximated by assuming *RF* = 1.0.

$$\delta_b = \frac{C_m}{1 - \frac{\gamma_D P_{DL} + \gamma_L P_{LL+IM}}{\varphi P_e}}$$

An iterative analysis could be used for improved accuracy.

If $\frac{P_u}{P_r} < 0.2$ then:

$\frac{P_u}{2.0P_r} + \frac{M_u}{M_r} \le 1.0$ for rating the correct *RF* will make this an equality. LRFD Design Eq. 6.9.2.2-1

$$P_u = \gamma_D P_{DL} + (RF)\gamma_L P_{LL+IM}$$

$$M_u = \delta_b \left[\gamma_D M_{DL} + (RF)\gamma_L M_{LL+IM}\right]$$

$$\frac{\gamma_D P_{DL} + RF\gamma_L P_{LL+IM}}{2P_r} + \frac{\delta_b \left[\gamma_D M_{DL} + RF \times \gamma_L \times M_{LL+IM}\right]}{M_r} = 1.0$$

$$\gamma_D \left[\frac{1}{2}\frac{P_{DL}}{P_r} + \delta_b \left(\frac{M_{DL}}{M_r}\right)\right] + RF \times \gamma_L \left[\frac{P_{LL+IM}}{2P_r} + \delta_b \left(\frac{M_{LL+IM}}{M_r}\right)\right] = 1.0$$

$$RF = \frac{1 - \gamma_D \left[\frac{P_{DL}}{2P_r} + \delta_b \left(\frac{M_{DL}}{M_r}\right)\right]}{\gamma_L \left[\frac{P_{LL+IM}}{2P_r} + \delta_b \left(\frac{M_{LL+IM}}{M_r}\right)\right]}$$

where: (for *RF* = 1.0)

$$\delta_b = \frac{C_m}{1 - \frac{\gamma_D P_{DL} + \gamma_L P_{LL+IM}}{\varphi P_e}}$$

An iterative analysis could be used for improved accuracy.

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APPENDIX I6A—RATING OF STEEL COMPRESSION MEMBERS WITH ECCENTRIC CONNECTIONS (SECANT FORMULA METHOD)

In compression members with unsymmetrical sections (such as truss chords) the gravity axis of the section may not coincide with the working lines, resulting in an eccentric connection. Compression members having equal end eccentricities are conveniently analyzed using the secant formula. The LRFD Design Specifications, like most modern codes, does not utilize the secant formula, but provides an interaction equation for the design of members with combined axial loads and concurrent moments. Rating compression members using an interaction equation is somewhat tedious, as an iterative approach may be required to establish the governing rating.

As an alternative to analyzing axial compression members with eccentric connections as combined compression-flexure members (LRFD Design Article 6.9.2.2), an axial load magnification factor may be applied to rate the member as a concentrically loaded member with an equivalent load. The secant formula is used to include the first and second order bending effects to produce a magnified axial load (dead and live) that would produce a constant stress over the cross-section equal to the peak stress in an eccentric member. This approach is applicable to members assumed to be pinned at the ends and without lateral loads on the member. Pin-connected compression chord members in truss bridges are a common example of this type.

The axial load magnification factor is given by:

$$\delta_A = \left[1 + \frac{eA}{S}\sec\left(\frac{L}{2}\sqrt{\frac{P_u}{EI}}\right)\right] \tag{I6A-1}$$

e = Eccentricity of connection from working line of member

A = Area of member

S = Section modulus of the member about the axis of bending caused by the eccentric connection for the extreme fiber of the member in the direction of the eccentricity

L = Length of the member between connections

P_u = Factored axial load (dead load + live load)

E = Modulus of elasticity

I = Moment of inertia of the member about the axis of bending caused by the eccentric connection

Any set of consistent units may be used.

Generally, end eccentricities may be neglected if ec/r^2 is less than 0.25. The LRFD Design beam-column equation with the moment magnification approach could also be used to evaluate compression members with only end eccentricities and no transverse loading. This process is a more lengthy approach as the beam-column method is a general approach applicable to a variety of situations. Limited investigation of the LRFD Design method vs. secant formula method indicates that the secant formula is simpler to use and would give comparable results. The following example shows the impact on load rating when the end eccentricity is increased from 0 in. to 1 in.

Example rating using axial load magnification:

Section based on member in Appendix A, Example 6 but with the pins assumed to be 1 in. eccentric in the negative y coordinate. Member forces calculated assuming centerline of pin to be concentric with center of gravity of top chord.

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e = 1 in.

A = 55.3 in.^{2}

S_x *bottom* = 376.0 in.^{3}

L = 300 in.

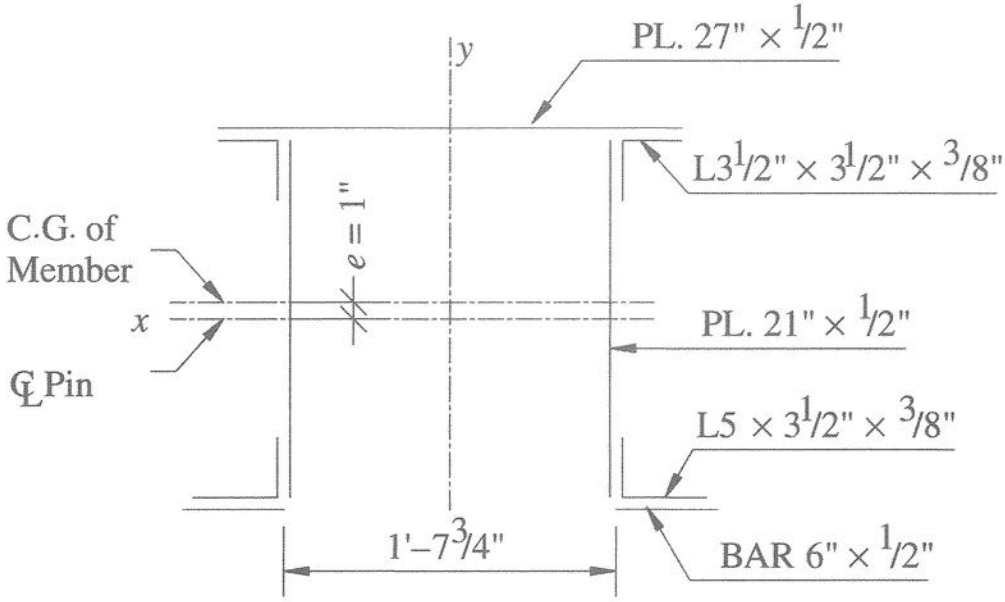
E = 29,000 ksi

I_x = 5,716.8 in.^{4}

P_{DC} = 558.1 kips

P_{DW} = 39.4 kips

P_{LL+IM} = 231.1 kips



$$P_u = 1.75 \times 231.1 + 1.25 \times 558.1 + 1.25 \times 39.4 = 1151.3 \text{ kips}$$

$$\delta_A = \left[1 + \frac{1 \times 55.3}{376.0} \sec\left(\frac{300}{2}\sqrt{\frac{1151.3}{29000 \times 5716.8}}\right)\right]$$

$$\delta_A = 1.159$$

$$RF = \frac{0.85 \times 1.0 \times 0.9 \times 1906.6 - 1.25 \times 1.159 \times 558.1 - 1.25 \times 1.159 \times 39.4}{1.75 \times 1.159 \times 231.1}$$

$$RF = 1.26$$

$\left(RF = 1.76 \text{ for } e = 0\right)$

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PART B—ALLOWABLE STRESS RATING AND LOAD FACTOR RATING

6B.1—GENERAL

Section 6, Part B of this Manual provides a choice of load rating methods. Load ratings at operating and inventory levels using the allowable stress method can be calculated and may be especially useful for comparison with past practices. Similarly, load ratings at operating and inventory levels based on the load factor method can also be calculated. Each of these rating methods is presented below.

6B.1.1—Application of Standard Design Specifications

For all matters not covered by this Manual, the current applicable AASHTO *Standard Specifications for Highway Bridges* (AASHTO Standard Specifications) should be used as a guide. However, there may be instances in which the behavior of a member under traffic is not consistent with that predicted by the controlling specification. In this situation, deviations from the controlling specifications based on the known behavior of the member under traffic

C6B.1

Bridge engineers have recognized that for the same bridge conditions a wide range of ratings may arise, depending on the rating method selected. Historically, several approaches have been used in rating bridges including inventory and operating rating levels and the use of allowable stress and load factor methods of analysis.

In recent years, methods have been developed to provide more uniform safety margins for structures in terms of a reliability index. For bridge evaluation, the load and resistance factor rating (LRFR) method contained in this Manual provides uniform reliability in bridge load ratings and load postings. See Section 6, Part A, for more information on LRFR.

A structure, designed and checked by the load factor method, using HS-20 loading (or greater HS load, e.g. HS-25), may not require calculations to determine the inventory or operating rating for the design loading until changes to the structure occur that would reduce the inventory rating below the design load level. At the discretion of the owner, for a bridge having an HS-20 (or greater HS load, e.g. HS-25) design live load, the inventory or operating rating for these structures may be assigned based on the design loading.

The HS-20 (or greater HS load, e.g. HS-25) design load rating factors would generally be: inventory 1.0, operating 1.67. Load ratings for state legal loads or permit loads would be necessary if the force effects from state legal loads or permit loads exceed those of the design loading. Effects from specialized hauling vehicles, some state legal loads, and permit loads can exceed force effects of HS-20.

The LFD and LFR Specifications have been used for the design and load rating of modern bridge structures. Generally, the level and detail of analysis of a designed and checked bridge is much more comprehensive than a bridge load rating. Therefore, the inventory and operating rating may be conservatively determined from the design loading.

The operating rating factor is determined by multiplying the inventory rating by the ratio of the inventory live load factor (2.17) to the operating level live load factor (1.3). These live load factors are provided in Article 6B.5.3.

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may be used and should be fully documented. Diagnostic load tests may be helpful in establishing the safe load capacity for such members.

For ease of use and where appropriate, reference is made to specific articles in the AASHTO *Standard Specifications for Highway Bridges.*

6B.2—RATING LEVELS

Each highway bridge should be load rated at two levels, inventory and operating levels.

6B.2.1—Inventory Rating Level

The inventory rating level generally corresponds to the customary design level of stresses but reflects the existing bridge and material conditions with regard to deterioration and loss of section. Load ratings based on the inventory level allow comparisons with the capacity for new structures and, therefore, results in a live load, which can safely utilize an existing structure for an indefinite period of time.

6B.2.2—Operating Rating Level

Load ratings based on the operating rating level generally describe the maximum permissible live load to which the structure may be subjected. Allowing unlimited numbers of vehicles to use the bridge at operating level may shorten the life of the bridge.

6B.3—RATING METHODS

In the load rating of bridge members, two methods for checking the capacity of the members are provided in Section 6, Part B—the allowable stress method and load factor method.

6B.3.1—Allowable Stress: *AS*

The allowable or working stress method constitutes a traditional specification to provide structural safety. The actual loadings are combined to produce a maximum stress in a member, which is not to exceed the allowable or working stress. The latter is found by taking the limiting stress of the material and applying an appropriate factor of safety.

6B.3.2—Load Factor: *LF*

The load factor method is based on analyzing a structure subject to multiples of the actual loads (factored loads). Different factors are applied to each type of load, which reflect the uncertainty inherent in the load calculations. The rating is determined such that the effect of the factored loads does not exceed the strength of the member.

C6B.3

In addition to the two methods described in Part B, the LRFR method may be used. See Section 6, Part A, for more information on LRFR.

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6B.4—RATING EQUATION

6B.4.1—General

The following general expression should be used in determining the load rating of the structure:

$$RF = \frac{C - A_1 D}{A_2 L(1+I)} \qquad (6B.4.1\text{-}1)$$

where:

RF = The rating factor for the live load carrying capacity. The rating factor multiplied by the rating vehicle in tons gives the rating of the structure (see Eq. 6B.4.1-2)

C = The capacity of the member (see Article 6B.5)

D = The dead load effect on the member (see Article 6B.6.1). For composite members, the dead load effect on the noncomposite section and the dead load effect on the composite section need to be evaluated when the allowable stress method is used

L = The live load effect on the member (see Article 6B.6.2)

I = The impact factor to be used with the live load effect (see Article 6B.6.4)

A_1 = Factor for dead loads (see Articles 6B.4.2 and 6B.4.3)

A_2 = Factor for live load (see Articles 6B.4.2 and 6B.4.3)

In the equation above, "load effect" is the effect of the applied loads on the member. Typical "load effects" used by engineers are axial force, vertical shear force, bending moment, axial stress, shear stress, and bending stresses. Once the "load effect" to be evaluated is selected by the Engineer, the "capacity" of a member to resist such a load effect may be determined (see Article 6B.5).

$$RT = (RF)W \qquad (6B.4.1\text{-}2)$$

where:

RT = Bridge member rating (tons)

W = Weight of nominal truck used in determining the live load effect, L (tons)

The rating of a bridge is controlled by the member with the lowest rating in tons.

C6B.4.1

The rating equation may be used regardless of the method (allowable stress or load factor) used to evaluate a member's capacity. The application of the basic rating equation to steel, concrete, and timber bridges is illustrated in Appendix A (load rating examples).

For example, at the maximum moment section of a girder, the bending stress may be selected as the "load effect" to be evaluated. The capacity of the girder would be determined based on the maximum stress which the girder cross-section could safely carry at the rating level desired. The dead load effect would be the theoretical bending stress due to dead loads at the section being evaluated. The live load bending stress would be computed based on the truck configuration or lane load selected for the rating and AASHTO impact and distribution factors. Appropriate factors (A_1 and A_2) would be selected and RF determined.

The RF would then be multiplied by the total weight (tons) of the nominal truck used in establishing the live load effect, L. Thus, the final rating for a bridge member will be expressed in tons.

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6B.4.2—Allowable Stress

For the allowable stress method, $A_1 = 1.0$ and $A_2 = 1.0$ in the general rating equation.

The capacity, C, depends on the rating level desired, with the higher value for C used for the Operating level. The determination of the nominal capacity of a member is discussed in Article 6B.5.2.

6B.4.3—Load Factor

For the load factor method, $A_1 = 1.3$ and A_2 varies depending on the rating level desired. For inventory level, $A_2 = 2.17$ and for operating level, $A_2 = 1.3$.

The nominal capacity, C, is the same regardless of the rating level desired (see Article 6B.5.3).

6B.5—NOMINAL CAPACITY: *C*

6B.5.1—General

The nominal capacity to be used in the rating equation depends on the structural materials, the rating method, and rating level used. Nominal capacities based on the Allowable Stress method are discussed in Article 6B.5.2 and those based on the Load Factor method are discussed in Article 6B.5.3.

The Bridge Owner is responsible for selecting the rating method. The method used should be identified for future reference.

6B.5.2—Allowable Stress Method

In the allowable stress method, the capacity of a member is based on the rating level evaluated: inventory level-allowable stress, or operating level-allowable stress. The properties to be used for determining the allowable stress capacity for different materials follow. For convenience, the tables provide, where appropriate, the inventory, operating, and yield stress values. Allowable stress and strength formulas should be those provided herein or those contained in the AASHTO Standard Specifications. When situations arise that are not covered by these specifications, then rational strength of material formulae should be used consistent with data and plans verified in the field investigation. Deviations from the AASHTO Standard Specifications should be fully documented.

When the bridge materials or construction are unknown, the allowable stresses should be fixed by the Engineer, based on field investigations and/or material testing conducted in accordance with Section 5, and should be substituted for the basic stresses given herein.

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6B.5.2.1—Structural Steel

The allowable unit stresses used for determining safe load capacity depend on the type of steel used in the structural members. When nonspecification metals are encountered, coupon testing may be used to determine a nominal yield point. When information on specifications of the steel is not available, allowable stresses should be taken from the applicable "Date Built" column of Tables 6B.5.2.1-1 and 6B.5.2.1-2.

Table 6B.5.2.1-1 gives allowable inventory stresses and Table 6B.5.2.1-2 gives the allowable operating stresses for structural steel. The nominal yield stress, F_y, is also shown in Tables 6B.5.2.1-1 and 6B.5.2.1-2. Tables 6B.5.2.1-3 and 6B.5.2.1-4 give the allowable inventory and operating Stresses for bolts and rivets. For compression members, the effective length, KL, may be determined in accordance with the AASHTO Standard Specifications or taken as follows:

KL = 75 percent of the total length of a column having riveted end connections

= 87.5 percent of the total length of a column having pinned end connections

The modulus of elasticity, E, for steel should be 29,000,000 lb/in.^{2}

If the investigation of shear and stiffener spacing is desirable, such investigation may be based on the AASHTO Standard Specifications.

C6B.5.2.1

When nonspecification materials are encountered, standard coupon testing procedures may be used to establish the nominal yield point. To provide a 95 percent confidence limit, the nominal yield point would typically be the mean coupon test value minus 1.65 standard deviations.

Mechanical properties of eyebars, high-strength eyebars, forged eyebars, and cables vary depending on manufacturer and year of construction. In the absence of material tests, the Engineer should carefully investigate the material properties using manufacturer's data and compilations of older steel properties before establishing the yield and allowable stresses to be used in load rating the bridge.

The formulas for the allowable bending stress in partially supported or unsupported compression flanges of beams and girders, given in Tables 6B.5.2.1-1 and 6B.5.2.1-2 are based on the corresponding formula given in Table 10.32.1A of the Allowable Stress Design portion of the AASHTO Standard Specifications. The equation in Table 6B.5.2.1-1 is to be used for an inventory rating and the equation in Table 6B.5.2.1-2 is to be used for an operating rating.

The previously used formulas are inelastic parabolic formulas which treat the lateral torsional buckling of a beam as flexural buckling of the compression flange. This is a very conservative approach for beams with short unbraced lengths. The flexural capacity is reduced for any unbraced length greater than zero. This does not reflect the true behavior of a beam. A beam may reach M_p with unbraced lengths much greater than zero. In addition, the formula neglects the St. Venant torsional stiffness of the cross-sections. This is a significant contribution to the latera1 torsional buckling resistance of rolled shapes, particularly older "I" shapes. The previous formulas must also be limited to the values of I/b listed. This limit is the slenderness ratio when the estimated buckling stress is equal to half the yield strength or 0.275 F_y in terms of an allowable stress. Many floor stringers will have unbraced lengths beyond this limit. If the formulas are used beyond these limits, negative values of the allowable stress can result.

The new formulas have no upper limit which allows the determination of allowable stresses for all unbraced lengths. In addition, the influence of the moment gradient upon buckling capacity is considered using the modifier, C_b, in the new formulas.

The specification formulas are based on the exact formulations of the lateral torsional buckling of beams. They are currently used in the AISC LRFD Specifications and other specifications throughout the world. They are also being used to design and rate steel bridges by the load factor method. Figures 6B.5.2.1-1 and 6B.5.2.1-2 show a comparison between the specification formulas and the previous specification formulas for two sections. Figure 6B.5.2.1-1 compares results for a $W18 \times 46$ rolled section. The new specification gives a much higher capacity than the previous specification. The difference is due to the inclusion of the St. Venant torsional stiffness, J, in the proposed specification. Figure 6B.5.2.1-2 shows a similar comparison for a plate-girder section. The section,

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labeled section 3, has 1.5 × 16 in. flanges and a $^{5}/_{16}$ × 94 in. web. The previous specification equation gives higher values than the new specification for large unbraced lengths. The previous specification is unconservative in this range. Both graphs show that, for small unsupported lengths, the new specification gives higher allowable stress values. The higher values result from the fact that there is an immediate reduction in capacity versus unsupported length in the previous specification.

Tables 6B.5.2.1-3 and 6B.5.2.1-4 contain the allowable inventory and operating stresses for low-carbon steel bolts, rivets, and high-strength bolts. For high-strength bolts (Table 6B.5.2.1-4), the values for inventory rating correspond to the allowable stress design values in the AASHTO Standard Specifications (Tables 10.32.3B and 10.32.3C). The values for the operating rating correspond to the inventory rating values multiplied by the ratio 0.75/0.55. The corresponding values for low-carbon steel bolts (ASTM A307) in Table 6B.5.2.1-3 are based on the values given in Table 10.32.3A of the Standard Specifications.

Specifications and guidance for determining the capacity of gusset plates can be found in Appendix L6B. allowable inventory and operating stresses for fasteners used in gusset plates can be found in Tables 6B.5.2.1-3 and 6B.5.2.1-4.

Guidance on considering the effects of deterioration on load rating of steel structures can be found in Article C6A.6.5.

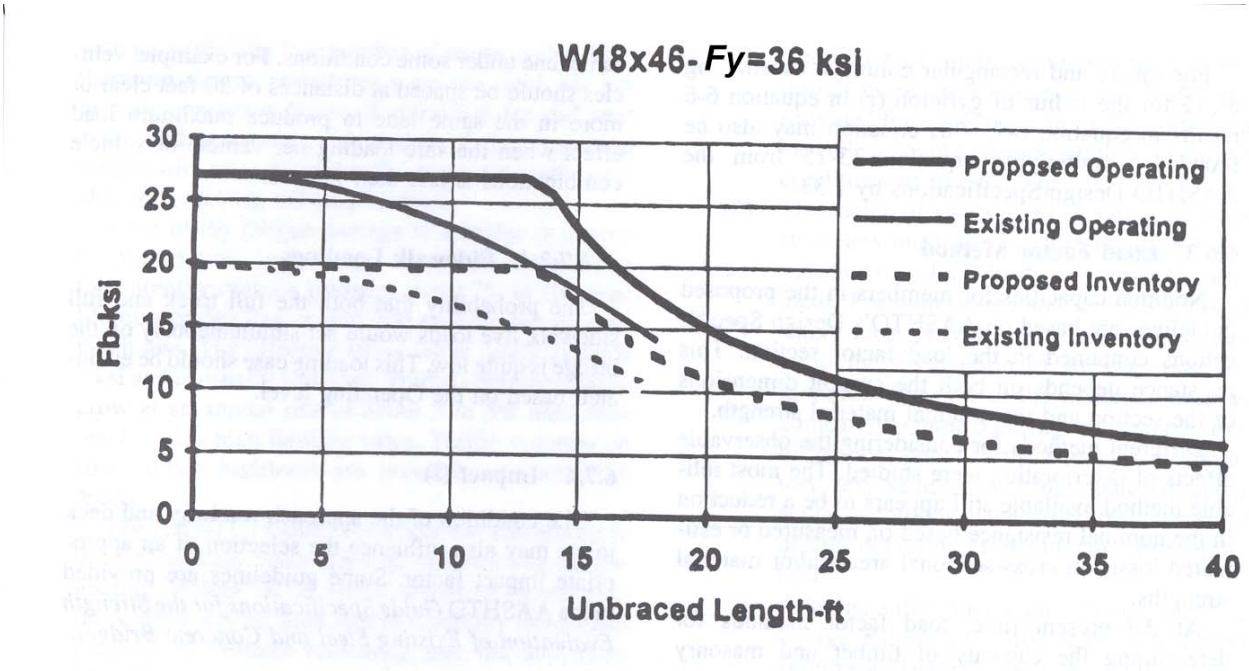

Figure 6B.5.2.1-1—Allowable Bending Stresses in Beams with Unsupported Compression Flanges: *W*18 × 46 – F_y = 36 ksi

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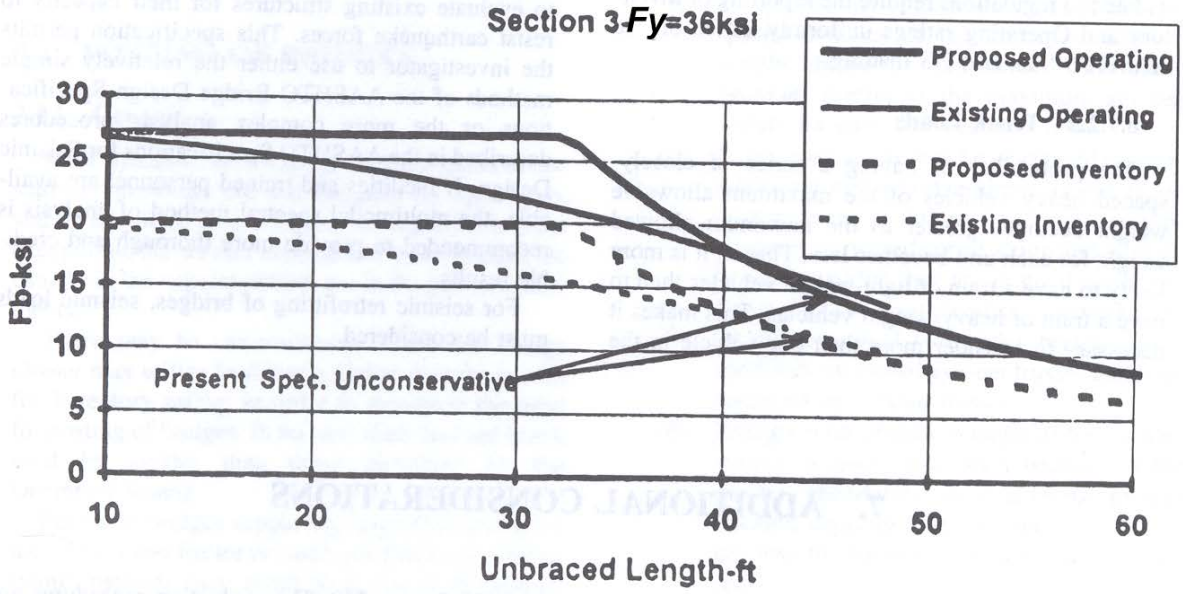

Figure 6B.5.2.1-2—Allowable Bending Stresses in Beams with Unsupported Compression Flanges: Section 3 – F_y = 36 ksi

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Table 6B.5.2.1-1—Inventory Rating Allowable Stresses, psi

		DATE BUILT-STEEL UNKNOWN								
		Prior to 1905	1905 to 1936	1936 to 1963	After 1963	Carbon Steel	Silicon Steel Over 2 in. to 4 in. incl.	Nickel Steel	$1^1/_8$ in. and under	Over $1^1/_8$ in. to 2 in. incl.
AASHTO Designation[a]						M 94 (1961)	M 95 (1961)	M 96 (1961)		
ASTM Designation[a]						A7 (1967)	A94 (1966)	A8 (1961)	A94	A94
Minimum Tensile Strength	F_u	52,000	60,000			60,000	70,000	90,000	75,000	72,000
Minimum Yield Point	F_y	26,000	30,000	33,000	36,000	33,000	45,000	55,000	50,000	47,000
Axial tension in members with no holes for high-strength bolts or rivets. Use net section when member has any open holes larger than $1^1/_4$-in. diameter, such as perforations.	$0.55F_y$	14,000	16,000	18,000	20,000	18,000	24,000	30,000	27,000	25,000
	$0.46F_y$	NOT APPLICABLE								
Axial tension in members with holes for high-strength bolts or rivets and tension in extreme fiber of rolled shapes, girders, and built-up sections subject to bending.	Gross* Section $0.55F_y$ (use whichever is smaller)	14,000	16,000	18,000	20,000	18,000	24,000	30,000	27,000	25,000
• When the area of holes deducted for high-strength bolts or rivets is more than 15 percent of the gross area, that area in excess of 15 percent shall be deducted from the gross area in determining stress on the gross section. In determining gross section, any open holes larger than $1^1/_4$-in. diameter, such as perforations, shall be deducted.	Net Section $0.50F_u$	26,000	30,000	30,000	30,000	30,000	35,000	45,000	37,000	36,000
	Net Section $0.46F_u$	NOT APPLICABLE								
Axial tension in members without holes. Axial compression, gross section: stiffeners of plate girders. Compression in splice material, gross section.	$0.55F_y$	14,000	16,000	18,000	20,000	18,000	24,000	30,000	27,000	25,000
Compression in extreme fibers of rolled shapes, girders, and built-up sections, subject to bending, gross section, when compression flange is:	$0.55F_y$	14,000	16,000	18,000	20,000	18,000	24,000	30,000	27,000	25,000
(A) Supported laterally its full length by embedment in concrete										
(B) Partially supported or unsupported[b]										

$$F_b = \frac{91\times10^6 C_{1s}}{(F.S.)\,S_{xc}}\left(\frac{1_{yc}}{l}\right)\sqrt{0.772\frac{J}{1_{yc}}+9.87\left(\frac{d}{l}\right)^2} \le 0.55F_y$$

C_b = $1.75 + 1.05\,(M_1/M_2) + 0.3\,(M_1/M_2)2 \le 2.3$ where M_1 is the smaller and M_2 is the larger end moment in the unbraced segment of the beams: M_1/M_2 is positive when the moments cause reverse curvature and negative when bent in single curvature.

= 1.0 for unbraced cantilevers and for members where the moment within a significant portion of the unbraced segment is greater than or equal to the larger of the segment end moments.

$F.S.$ = factor of safety at inventory level = 1.82

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Table 6B.5.2.1-1—Inventory Rating Allowable Stresses, psi (continued)

			$1^1/_2$ in. max	$^1/_2$ in. max	Over $2^1/_2$ in. to 4 in. incl.	$^3/_4$ in. and under	To $2^1/_2$ in. incl. (A514) All thick (A517)	Over 4 in. to 5 in. incl. (A588) Over $^3/_4$ in. to $1^1/_2$ in. incl.
AASHTO Designation[a]								
ASTM Designation[a]			A572	A572	A514	A242, A440, A441	A514/A517	A242, A440, A441, A588
Minimum Tensile Strength		F_u	60,000	80,000		70,000	115,000	67,000
Minimum Yield Point		F_y	45,000	65,000	90,000	50,000	100,000	46,000
Axial tension in members with no holes for high-strength bolts or rivets. Use net section when member has any open holes larger than $1^1/_4$-in. diameter, such as perforations.		$0.55F_y$	25,000	36,000	N.A.	27,000	55,000	25,000
		$0.46F_y$	NOT APPLICABLE		48,300	53,000	53,000	N.A.
Axial tension in members with holes for high-strength bolts or rivets and tension in extreme fiber of rolled shapes, girders, and built-up sections subject to bending • When the area of holes deducted for high-strength bolts or rivets is more than 15 percent of the gross area, that area in excess of 15 percent shall be deducted from the gross area in determining stress on the gross section. In determining gross section, any open holes larger than $1^1/_4$-in. diameter, such as perforations, shall be deducted.	use whichever is smaller	Gross* Section $0.55F_y$	25,000	36,000	49,000	27,000	55,000	25,000
		Net Section $0.50F_u$	30,000	40,000	N.A.	35,000	N.A.	33,500
		Net Section $0.46F_u$	NOT APPLICABLE		48,300	N.A.	53,000	N.A.
Axial tension in members without holes. Axial compression, gross section: stiffeners of plate girders. Compression in splice material, gross section.		$0.55F_y$	25,000	36,000	49,000	27,000	55,000	25,000

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Table 6B.5.2.1-1—Inventory Rating Allowable Stresses, psi (continued)

		$1\frac{1}{2}$ in. max	$\frac{1}{2}$ in. max	Over $2\frac{1}{2}$ in. to 4 in. incl.	$\frac{3}{4}$ in. and under	To $2\frac{1}{2}$ in. incl. (A514) All thick (A517)	Over 4 in. to 5 in. incl. (A588) Over $\frac{3}{4}$ in. to $1\frac{1}{2}$ in. incl.
Compression in extreme fibers of rolled shapes, girders, and built-up sections, subject to bending, gross section, when compression flange is:	$0.55F_y$	25,000	36,000	49,000	27,000	55,000	25,000
(A) Supported laterally its full length by embedment in concrete							
(B) Partially supported or unsupported[b]							
Compression in concentrically loaded columns[c]							
with $C_c = \sqrt{\frac{2\pi^2 E}{F_y}}$		112.8	93.8	79.8	107.0	75.7	111.6
$F_a = \frac{F_y}{F.S.}\left[1 - \frac{\left(\frac{KL}{r}\right)^2 F_y}{4\pi^2 E}\right]$ when $\frac{KL}{r} \le C_c$		$21{,}230 - 0.83\left(\frac{KL}{r}\right)^2$	$30{,}660 - 1.74\left(\frac{KL}{r}\right)^2$	$42{,}450 - 3.34\left(\frac{KL}{r}\right)^2$	$23{,}580 - 1.03\left(\frac{KL}{r}\right)^2$	$47{,}170 - 4.12\left(\frac{KL}{r}\right)^2$	$21{,}700 - 0.87\left(\frac{KL}{r}\right)^2$
$F_a = \frac{\pi^2 E}{F.S.\left(\frac{KL}{r}\right)^2} = \frac{135{,}008{,}740}{\left(\frac{KL}{r}\right)^2}$ when $KL \ge C_c$ with $F.S. = 2.12$							
F_b = $\frac{91 x 10^6 C_{1s}}{(F.S.) S_{xc}}\left(\frac{1_{yc}}{1}\right)\sqrt{0.772\frac{J}{1_{yc}} + 9.87\left(\frac{d}{1}\right)^2} \le 0.55F_y$							

C_b = $1.75 + 1.05\,(M_1/M_2) + 0.3\,(M_1/M_2)^2 \le 2.3$ where M_1 is the smaller and M_2 is the larger end moment in the unbraced segment of the beams: M_1/M_2 is positive when the moments cause reverse curvature and negative when bent in single curvature.

= 1.0 for unbraced cantilevers and for members where the moment within a significant portion of the unbraced segment is greater than or equal to the larger of the segment end moments.

F.S. = factor of safety at inventory level = 1.82

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Table 6B.5.2.1-1—Inventory Rating Allowable Stresses, psi (continued)

		1 1/2 in. max	1/2 in. max	Over 2 1/2 in. to 4 in. incl.	3/4 in. and under	To 2 1/2 in. incl. (A514) All thick (A517)	Over 4 in. to 5 in. incl. (A588) Over 3/4 in. to 1 1/2 in. incl.
Shear in girder webs, gross section		15,000	22,000	30,000	17,000	30,000	15,000
Bearing on milled stiffeners and other steel parts in contact. Stress in extreme fiber of pins	$0.80F_y$	37,000	52,000	72,000	40,000	80,000	37,000
Bearing on pins not subject to rotation		37,000	52,000	72,000	40,000	80,000	37,000
Bearing on pins subject to rotation (such as rockers and hinges)		18,000	26,000	36,000	20,000	40,000	18,000
Shear in pins	$0.40F_y$	18,000	26,000	36,000	20,000	40,000	18,000
Bearing on power-driven rivets and high-strength bolts (or as limited by allowable bearing on the fasteners)	$1.35F_u$	81,000	108,000	142,000	94,500	155,000	90,500

Table 6B.5.2.1-1—Inventory Rating Allowable Stresses, psi (continued)

		1 1/2 in. max	1 in. max	Over 5 in. to 8 in. incl. (A588) Over 1 1/2 in. to 4 in. incl.	Over 4 in. to 8 in. incl.
AASHTO Designation[a]					M 188 (1983)
ASTM Designation[a]		A572	A572	A242, A440, A441, A588, A572	A441 (1985)
Minimum Tensile Strength	F_u	70,000	75,000	63,000	60,000
Minimum Yield Point	F_y	55,000	60,000	42,000	40,000
Axial tension in members with no holes for high-strength bolts or rivets. Use net section when member has any open holes larger than 1 1/4-in. diameter, such as perforations.	$0.55F_y$	30,000	33,000	23,000	22,000
	$0.46F_y$	NOT APPLICABLE			

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Table 6B.5.2.1-1—Inventory Rating Allowable Stresses, psi (continued)

			$1^1/_2$ in. max	1 in. max	Over 5 in. to 8 in. incl. (A588) Over $1^1/_2$ in. to 4 in. incl.	Over 4 in. to 8 in. incl.
Axial tension in members with holes for high strength bolts or rivets and tension in extreme fiber of rolled shapes, girders, and built-up sections subject to bending • When the area of holes deducted for high-strength bolts or rivets is more than 15 percent of the gross area, that area in excess of 15 percent shall be deducted from the gross area in determining stress on the gross section. In determining gross section, any open holes larger than $1^1/_4$-in. diameter, such as perforations, shall be deducted.	use whichever is smaller	Gross* Section $0.55F_y$	30,000	33,000	23,000	22,000
		Net Section $0.50F_u$	35,000	37,500	31,500	30,000
		Net Section $0.46\ F_u$				
Axial tension in members without holes. Axial compression, gross section: stiffeners of plate girders. Compression in splice material, gross section.		$0.55F_y$	30,000	33,000	23,000	22,000
Compression in extreme fibers of rolled shapes, girders, and built-up sections, subject to bending, gross section, when compression flange is: (A) Supported laterally its full length by embedment in concrete (B) Partially supported or unsupported[b]		$0.55F_y$	30,000	33,000	23,000	

$$F_b = \frac{91\,x\,10^6 C_{1s}}{(F.S.)\,S_{xc}}\left(\frac{1_{yc}}{1}\right)\sqrt{0.772\frac{J}{1_{yc}}+9.87\left(\frac{d}{1}\right)^2} \le 0.55F_y$$

C_b = $1.75 + 1.05\,(M_1/M_2) + 0.3\,(M_1/M_2)^2 \le 2.3$ where M_1 is the smaller and M_2 is the larger end moment in the unbraced segment of the beams: M_1/M_2 is positive when the moments cause reverse curvature and negative when bent in single curvature.

C_b = 1.0 for unbraced cantilevers and for members where the moment within a significant portion of the unbraced segment is greater than or equal to the larger of the segment end moments.

F.S. = factor of safety at Inventory Level = 1.82

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Table 6B.5.2.1-1—Inventory Rating Allowable Stresses, psi (continued)

		$1^1/_2$ in. max	1 in. max	8 in. incl. (A588) Over $1^1/_2$ in. to 4 in. incl.	Over 4 in. to 8 in. incl.
Compression in concentrically loaded columns[c]					
with $C_c = \sqrt{\frac{2\pi^2 E}{F_y}}$		102.0	97.7	116.7	
$F_a = \frac{F_y}{F.S.}\left[1 - \frac{\left(\frac{KL}{r}\right)^2 F_y}{4\pi^2 E}\right]$ when $\frac{KL}{r} \le C_c$		$25{,}940 - 1.25\left(\frac{KL}{r}\right)^2$	$28{,}300 - 1.48\left(\frac{KL}{r}\right)^2$	$19{,}810 - 0.73\left(\frac{KL}{r}\right)^2$	
$F_a = \frac{\pi^2 E}{F.S.\left(\frac{KL}{r}\right)^2} = \frac{135{,}008{,}740}{\left(\frac{KL}{r}\right)^2}$ when $KL \ge C_c$ with $F.S. = 2.12$					
Shear in girder webs, gross section		18,000	20,000	14,000	
Bearing on milled stiffeners and other steel parts in contact. Stress in extreme fiber of pins	$0.80F_y$	44,000	48,000	34,000	32,000
Bearing on pins not subject to rotation		44,000	48,000	34,000	32,000
Bearing on pins subject to rotation (such as rockers and hinges)		22,000	24,000	17,000	16,000
Shear in pins	$0.40F_y$	22,000	24,000	17,000	
Bearing on power-driven rivets and high-strength bolts (or as limited by allowable bearing on the fasteners)	$1.35F_u$	94,500	101,000	85,000	81,000

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Table 6B.5.2.1-2—Operating Rating Allowable Stress, psi

		DATE BUILT-STEEL UNKNOWN						
		Prior to 1905	1905 to 1936	1936 to 1963	After 1963	Carbon Steel	Silicon Steel Over 2 in. to 4 in. incl.	Nickel Steel
AASHTO Designation[a]						M 94 (1961)	M 95 (1961)	M 96 (1961)
ASTM Designation[a]						A7 (1967)	A4 (1966)	A8 (1961)
Minimum Tensile Strength	F_u	52,000	60,000			60,000	70,000	90,000
Minimum Yield Point	F_y	26,000	30,000	33,000	36,000	33,000	45,000	55,000
Axial tension in members with no holes for high-strength bolts or rivets.	$0.75F_y$	19,500	22,500	24,500	27,000	24,500		
Use net section when member has any open holes larger than $1^1/_4$-in. diameter, such as perforations.	$0.60F_u$	NOT APPLICABLE					33,500	41,000
Axial tension in members with holes for high-strength bolts or rivets and tension in extreme fiber of rolled shapes, girders, and built-up sections subject to bending • When the area of holes deducted for high-strength bolts or rivets is more than 15 percent of the gross area, that area in excess of 15 percent shall be deducted from the gross area in determining stress on the gross section. In determining gross section, any open holes larger than $1^1/_4$-in. diameter, such as perforations, shall be deducted.	use whichever is smaller	Gross* Section $0.75F_y$	19,500	22,500	24,500	27,000	24,500	33,500	41,000
		Net Section $0.67F_u$	35,000	40,000	40,000	40,000	40,000	46,500	60,000
		Net Section $0.60F_u$	NOT APPLICABLE						
Axial tension in members without holes. Axial compression, gross section: stiffeners of plate girders. Compression in splice material, gross section.	$0.75F_y$	19,500	22,500	24,500	27,000	24,500	33,500	41,000

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Table 6B.5.2.1-2—Operating Rating Allowable Stress, psi (continued)

		DATE BUILT-STEEL UNKNOWN						
		Prior to 1905	1905 to 1936	1936 to 1963	After 1963	Carbon Steel	Silicon Steel Over 2 in. to 4 in. incl.	Nickel Steel
Compression in extreme fibers of rolled shapes, girders, and built-up sections, subject to bending, gross section, when compression flange is: (A) Supported laterally its full length by embedment in concrete (B) Partially supported or unsupported[b]	$0.75F_y$	19,500	22,500	24,500	27,000	24,500	33,500	41,000
Compression in concentrically loaded columns[c]								
with $C_c = \sqrt{\frac{2\pi^2 E}{F_y}}$ when $\frac{KL}{r} \le C_c$		148.4	138.1	131.7	126.1	131.7	112.8	102.0
$F_a = \frac{F_y}{F.S.}\left[1 - \frac{\left(\frac{KL}{r}\right)^2 F_y}{4\pi^2 E}\right]$ when $\frac{KL}{r} \ge C_c$		$15{,}290 - 0.35\left(\frac{KL}{r}\right)^2$	$17{,}650 - 0.46\left(\frac{KL}{r}\right)^2$	$19{,}410 - 0.56\left(\frac{KL}{r}\right)^2$	$21{,}180 - 0.67\left(\frac{KL}{r}\right)^2$	$19{,}410 - 0.56\left(\frac{KL}{r}\right)^2$	$26{,}470 - 1.04\left(\frac{KL}{r}\right)^2$	$32{,}350 - 1.55\left(\frac{KL}{r}\right)^2$
$F_a = \frac{\pi^2 E}{F.S.\left(\frac{KL}{r}\right)^2} = \frac{168{,}363{,}840}{\left(\frac{KL}{r}\right)^2}$ with $F.S. = 1.70$								

$$F_b = \frac{91 x 10^6 C_b}{(F.S.) S_{xc}}\left(\frac{1_{yc}}{1}\right)\sqrt{0.772\frac{J}{1_{yc}} + 9.87\left(\frac{d}{1}\right)^2} \le 0.75F_y$$

C_b = $0.75 + 1.05\,(M_1/M_2) + 0.3\,(M_1/M_2)^2 \le 2.3$ where M_1 is the smaller and M_2 is the larger end moment in the unbraced segment of the beams: M_1/M_1 is positive when the moments cause reverse curvature and negative when bent in single curvature.

= 1.0 for unbraced cantilevers and for members where the moment within a significant portion of the unbraced segment is greater than or equal to the larger of the segment end moments.

F.S. = factor of safety at inventory level = 1.34

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Table 6B.5.2.1-2—Operating Rating Allowable Stress, psi (continued)

		DATE BUILT-STEEL UNKNOWN						
		Prior to 1905	1905 to 1936	1936 to 1963	After 1963	Carbon Steel	Silicon Steel Over 2 in. to 4 in. incl.	Nickel Steel
Shear in girder webs, gross section	$0.45F_y$	11,500	13,500	15,000	16,000	15,000	20,000	24,500
Bearing on milled stiffeners and other steel parts in contact. Stress in extreme fiber of pins	$0.90F_y$	23,000	27,000	29,500	32,000	29,500	40,500	49,500
Bearing on pins not subject to rotation	$0.90F_y$	23,000	27,000	29,500	32,000	29,500	40,500	49,500
Bearing on pins subject to rotation (such as rockers and hinges)	$0.55F_y$	14,000	16,500	18,000	19,500	18,000	24,500	30,000
Shear in pins	$0.55F_y$	14,000	16,500	18,000	19,500	18,000	24,500	30,000
Bearing on power-driven rivets and high-strength bolts (or as limited by allowable bearing on the fasteners)	$1.85F_u$	96,000	111,000	111,000	111,000	111,000	129,500	166,500

[a] Number in parentheses represents the last year these specifications were printed.

[b] For the use of larger C_b values, see Structural Stability Research Council Guide to Stability Design Criteria for Metal Structures, Third Edition, p. 135. If cover plates are used, the allowable static stress at the point of theoretical cutoff shall be as determined by the formula.

ℓ = length of unsupported flange between lateral connections, knee braces, or other points of support, in.

ℓ_{yc} = moment of inertia of compression flange about the vertical axis in the plane of the web, in.^{4}

d = depth of girder, in.

J = $\frac{\left[(bt^3)_c + (bt^3)_t + Dt_w^3\right]}{3}$, in.^{4}, where b and t represent the flange width and thickness of the compression and tension flange, D is the web depth, and t_w is the web thickness.

S_{xc} = section modulus with respect to the compression flange, in.^{3}

[c] E = modulus of elasticity of steel

r = governing radius of gyration

L = actual unbraced length

K = effective length factor

Note: The formulae do not apply to members with variable moment of inertia.

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Table 6B.5.2.1-2—Operating Rating Allowable Stress, psi (continued)

			8 in. and Under	$1^1/_8$ in. and Under	Over $1^1/_8$ in. to 2 in. incl.	$1^1/_2$ in. max	$^1/_2$ in. max	Over $2^1/_2$ in. to 4 in. incl.	$^3/_4$ in. and under 4 in. and under (A588)
AASHTO Designation[a]									
ASTM Designation[a]			A36	A94	A94	A572	A572	A514	A242, A440, A441, A588, A572
Minimum Tensile Strength		F_u	58,000	75,000	72,000	60,000	80,000	105,000	70,000
Minimum Yield Point		F_y	36,000	50,000	47,000	45,000	65,000	90,000	50,000
Axial tension in members with no holes for high-strength bolts or rivets. Use net section when member has any open holes larger than $1^1/_4$-in. diameter, such as perforations.		$0.75F_y$	27,000	37,500	35,000	33,500	48,500	N.A.	37,500
		$0.60F_u$		NOT APPLICABLE				63,000	N.A.
Axial tension in members with holes for high-strength bolts or rivets and tension in extreme fiber of rolled shapes, girders, and built-up sections subject to bending • When the area of holes deducted for high-strength bolts or rivets is more than 15 percent of the gross area, that area in excess of 15 percent shall be deducted from the gross area in determining stress on the gross section. In determining gross section, any open holes larger than $1^1/_4$-in. diameter, such as perforations, shall be deducted.	use whichever is smaller	Gross* Section $0.75F_y$	27,000	37,500	35,000	33,500	48,500	67,500	37,500
		Net Section $0.67F_u$	38,000	50,000	48,000	40,000	53,000	N.A.	46,500
		Net Section $0.60F_u$			NOT APPLICABLE			63,000	N.A.
Axial tension in members without holes. Axial compression, gross section: stiffeners of plate girders. Compression in splice material, gross section.		$0.75F_y$	27,000	37,500	35,000	33,500	48,500	67,500	37,500

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Table 6B.5.2.1-2—Operating Rating Allowable Stress, psi (continued)

		8 in. and Under	$1^1/_8$ in. and Under	Over $1^1/_8$ in. to 2 in. incl.	$1^1/_2$ in. max	$^1/_2$ in. max	Over $2^1/_2$ in. to 4 in. incl.	$^3/_4$ in. and under 4 in. and under (A588)
Compression in extreme fibers of rolled shapes, girders and built-up sections, subject to bending, gross section, when compression flange is: (A) Supported laterally its full length by embedment in concrete (B) Partially supported or unsupported[b] $F_b = \frac{91\,x\,10^6 C_b}{(F.S.)\,S_{xc}}\left(\frac{l_{yc}}{l}\right)\sqrt{0.772\frac{J}{l_{yc}} + 9.87\left(\frac{d}{l}\right)^2} \le 0.75F_y$	$0.75F_y$	27,000	37,500	35,000	33,500	48,500	67,500	37,500
C_b = $1.75 + 1.05\,(M_1/M_2) + 0.3\,(M_1/M_2)^2 \le 2.3$ where M_1 is the smaller and M_2 is the larger end moment in the unbraced segment of the beams; M_1/M_2 is positive when the moments cause reverse curvature and negative when bent in single curvature. = 1.0 for unbraced cantilevers and for members where the moment within a significant portion of the unbraced segment is greater than or equal to the larger of the segment end moments. F.S. = factor of safety at operating level = 1.34								
Compression in concentrically loaded columns[c]								
with $C_c = \sqrt{\frac{2\pi^2 E}{F_y}}$ when $\frac{KL}{r} \le C_c$			126.1	107.0	110.4	112.8	93.8	79.8
$F_a = \frac{F_y}{F.S.}\left[1 - \frac{\left(\frac{KL}{r}\right)^2 F_y}{4\pi^2 E}\right]$ when $\frac{KL}{r} \ge C_c$			$21{,}180 - 0.67\left(\frac{KL}{r}\right)^2$	$29{,}410 - 1.28\left(\frac{KL}{r}\right)^2$	$27{,}650 - 1.13\left(\frac{KL}{r}\right)^2$	$26{,}470 - 1.04\left(\frac{KL}{r}\right)^2$	$38{,}240 - 2.17\left(\frac{KL}{r}\right)^2$	$52{,}940 - 4.16\left(\frac{KL}{r}\right)^2$
$F_a = \frac{\pi^2 E}{F.S.\left(\frac{KL}{r}\right)^2} = \frac{168{,}363{,}840}{\left(\frac{KL}{r}\right)^2}$ with $F.S. = 1.70$								
Shear in girder webs, gross section	$0.45F_y$	16,000	22,500	21,000	20,000	29,000	40,500	
Bearing on milled stiffeners and other steel parts in contact. Stress in extreme fiber of pins	$0.90F_y$	32,000	45,000	42,000	40,500	58,500	81,000	
Bearing on pins not subject to rotation	$0.90F_y$	32,000	45,000	42,000	40,500	58,500	81,000	
Bearing on pins subject to rotation (such as rockers and hinges)	$0.55F_y$	19,500	27,500	25,500	24,500	35,500	49,500	
Shear in pins	$0.55F_y$	19,500	27,500	25,500	24,500	35,500	49,500	
Bearing on power-driven rivets and high-strength bolts (or as limited by allowable bearing on the fasteners)	$1.85F_u$	107,000	138,500	133,000	171,000	148,000	194,000	

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Table 6B.5.2.1-2—Operating Rating Allowable Stress, psi (continued)

			To $2^1/_2$ in. incl (A514) All thick (A517)	Over 4 in. to 5 in. incl. (A588) Over $^3/_4$ in. to $1^1/_2$ in. incl.	$1^1/_2$ in. max	1 in. max	Over 5 in. to 8 in. incl. (A588) Over $1^1/_2$ in. to 4 in. incl.	Over 4 in. to 8 in. incl.
AASHTO Designation[a]								
ASTM Designation[a]			A514–A517	A242, A440, A441, A588	A572	A572	A242, A440, A441, A588, A572	A441
Minimum Tensile Strength		F_u	115,000	67,000	70,000	75,000	63,000	60,000
Minimum Yield Point		F_y	100,000	46,000	55,000	60,000	42,000	40,000
Axial tension in members with no holes for high-strength bolts or rivets. Use net section when member has any open holes larger than $1^1/_4$-in. diameter, such as perforations.		$0.75F_y$ $0.60F_u$	75,000 69,000	34,500	41,000	45,000 NOT APPLICABLE		30,000
Axial tension in members with holes for high-strength bolts or rivets and tension in extreme fiber of rolled shapes, girders, and built-up sections subject to bending. • When the area of holes deducted for high-strength bolts or rivets is more than 15 percent of the gross area, that area in excess of 15 percent shall be deducted from the gross area in determining stress on the gross section. In determining gross section, any open holes larger than $1^1/_4$-in. diameter, such as perforations, shall be deducted.	use whichever is smaller	Gross* Section $0.75F_y$	75,000	34,500	41,000	45,000	31,500	30,000
		Net Section $0.67F_u$	N.A.	44,500	46,500	50,000	42,000	40,000
		Net Section $0.60F_u$	69,000	NOT APPLICABLE				
Axial tension in members without holes. Axial compression, gross section: stiffeners of plate girders. Compression in splice material, gross section.		$0.75F_y$	75,000	34,500	41,000	45,000	31,500	30,000

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Table 6B.5.2.1-2—Operating Rating Allowable Stress, psi (continued)

		To $2^1/_2$ in. incl (A514) All thick (A517)	Over 4 in. to 5 in. incl. (A588) Over $^3/_4$ in. to $1^1/_2$ in. incl.	$1^1/_2$ in. max	1 in. max	Over 5 in. to 8 in. incl. (A588) Over $1^1/_2$ in. to 4 in. incl.	Over 4 in. to 8 in. incl.
Compression in extreme fibers of rolled shapes, girders and built-up sections, subject to bending, gross section, when compression flange is: (A) Supported laterally its full length by embedment in concrete (B) Partially supported or unsupported[b]	$0.75F_y$	75,000	34,500	41,000	45,000	31,500	

$$F_b = \frac{91\times10^6 C_b}{(F.S.)S_{xc}}\left(\frac{1_{yc}}{1}\right)\sqrt{0.772\frac{J}{1_{yc}}+9.87\left(\frac{d}{1}\right)^2} \le 0.75F_y$$

C_b = $1.75 + 1.05\,(M_1/M_2) + 0.3\,(M_1/M_2)^2 \le 2.3$ where M_1 is the smaller and M_2 is the larger end moment in the unbraced segment of the beams; M_1/M_2 is positive when the moments cause reverse curvature and negative when bent in single curvature.

C_b = 1.0 for unbraced cantilevers and for members where the moment within a significant portion of the unbraced segment is greater than or equal to the larger of the segment end moments.

$F.S.$ = factor of safety at inventory level = 1.34

Table 6B.5.2.1-2—Operating Rating Allowable Stress, psi (continued)

	To $2^1/_2$ in. incl. (A511) All thick (A 517)	Over 4in. to 5in. incl. (A588) Over $^3/_4$ in. to $1^1/_2$ in. incl.	$1^1/_2$ in. max	1 in. max	Over 5 in. to 8 in. incl. (A588) Over $1^1/_2$ in. to 4 in. incl.	Over 4 in. to 8 in. incl.
Compression in concentrically loaded columns[c]						
with $C_c = \sqrt{\frac{2\pi^2 E}{F_y}}$	75.7	111.6	102.0	97.7	116.7	
when $\frac{KL}{r} \le C_c$						

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Table 6B.5.2.1-2—Operating Rating Allowable Stress, psi (continued)

		To $2^1/_2$ in. incl. (A511) All thick (A 517)	Over 4in. to 5in. incl. (A588) Over $^3/_4$ in. to $1^1/_2$ in. incl.	$1^1/_2$ in. max	1 in. max	Over 5 in. to 8 in. incl. (A588) Over $1^1/_2$ in. to 4 in. incl.	Over 4 in. to 8 in. incl.
$F_a = \frac{F_y}{F.S.}\left[1 - \frac{\left(\frac{KL}{r}\right)^2 F_y}{4\pi^2 E}\right]$ when $\frac{KL}{r} \geq C_c$ $F_a = \frac{\pi^2 E}{F.S.\left(\frac{KL}{r}\right)^2} = \frac{168{,}363{,}840}{\left(\frac{KL}{r}\right)^2}$ with $F.S. = 1.70$		58,820 – $5.14\left(\frac{KL}{r}\right)^2$	27,060 – $1.09\left(\frac{KL}{r}\right)^2$	32,350 – $1.55\left(\frac{KL}{r}\right)^2$	35,290 – $1.85\left(\frac{KL}{r}\right)^2$	24,710 – $0.91\left(\frac{KL}{r}\right)^2$	
Shear in girder webs, gross section	$0.45F_y$	45,000	20,500	24,500	27,000	18,500	18,000
Bearing on milled stiffeners and other steel parts in contact. Stress in extreme fiber of pins	$0.90F_y$	90,000	41,000	49,500	54,000	37,500	36,000
Bearing on pins not subject to rotation	$0.90F_y$	90,000	41,000	49,500	54,000	37,500	36,000
Bearing on pins subject to rotation (such as rockers and hinges)	$0.55F_y$	55,000	25,000	30,000	33,000	23,000	22,000
Shear in pins	$0.55F_y$	55,000	25,000	30,000	33,000	23,000	22,000
Bearing on power-driven rivets and high-strength bolts (or as limited by allowable bearing on the fasteners)	$1.85F_u$	213,000	124,000	129,500	138,500	116,500	111,000

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Table 6B.5.2.1-3—Allowable Inventory and Operating Stresses for Low Carbon Steel Bolts and Power-Driven Rivets, psi

Type of Fastener	Rating Level	Tension	Bearing	Shear Bearing Type Connection
(A) Low Carbon Steel Bolts: Turned Bolts (ASTM A307) and Ribbed Bolts[a, b]	INV	18,000	20,000	11,000[c]
	OPR	24,500	27,000	15,000[c]
(B) Power-Driven Rivets (rivets driven by pneumatically or electrically operated hammers are considered power driven) Structural Steel Rivet (ASTM A502 Grade 1 or ASTM A141)	INV	—	40,000	13,500
	OPR	—	54,500	18,000
Structural Steel Rivet (High Strength) (ASTM A502 Grade 2)	INV	—	40,000	20,000
	OPR	—	54,500	27,000

[a] The AASHTO Standard Specifications indicate that ASTM A307 bolts shall not be used in connections subject to fatigue.

[b] Based on nominal diameter of bolt.

[c] Threads permitted in the shear plane.

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Table 6B.5.2.1-4—Allowable Inventory and Operating Stresses for High-Strength Bolts, ksi[a]

Load Condition	Hole Type	Rating Level	AASHTO M 164[d] (ASTM A325) Bolts	AASHTO M 253 (ASTM A490) Bolts
Applied Tension T	Standard, oversize, or slotted	INV OPR	38[e] 52[e]	47 64
Shear F_v: Slip-critical connection[b]	Standard	INV OPR	15[e] 20[e]	19 26
	Oversize or short-slotted loaded in any direction	INV OPR	13[e] 18[e]	16 22
	Long-slotted Load transverse	INV OPR	11[e] 15[e]	13 18
	Long-slotted Load parallel	INV OPR	9[e] 12[e]	11 15
Shear F_v: Bearing type connection[c] Threads in any shear plane	Standard or slotted	INV OPR	19[e] 26[e]	24 33
No threads in shear plane	Standard or slotted	INV OPR	24[e] 33[e]	30 41
Bearing f_p on connected material	Standard, oversize, or short-slotted loaded in any direction	INV	$\frac{0.5L_cF_u}{d} \le F_u$ [f]	
		OPR	$\frac{0.7L_cF_u}{d} \le 1.4F_u$ [f]	
	Long-slotted Load parallel	INV	$\frac{0.5L_cF_u}{d} \le F_u$ [f]	
		OPR	$\frac{0.7L_cF_u}{d} \le 1.4F_u$ [f]	
	Long-slotted Load transverse	INV	$\frac{0.4L_cF_u}{d} \le 0.8F_u$ [f]	
		OPR	$\frac{0.55L_cF_u}{d} \le 1.1F_u$ [f]	

[a] The tabulated stresses, except for bearing stress, apply to the nominal area of bolts used in any grade of steel.

[b] Applicable for contact surfaces with clean mill scale (slip coefficient 0.33).

[c] In bearing-type connections whose length between extreme fasteners in each of the spliced parts measured parallel to the line of an axial force exceeds 50 in. (1.27 m), tabulated value shall be reduced by 20 percent.

[d] AASHTO M 164 (ASTM A325) and AASHTO M 253 (ASTM A490) high-strength bolts are available in three types, designated as types 1, 2, or 3.

[e] The tensile strength of M 164 (A325) bolts decreases for diameters greater than 1 in. The values listed are for bolts up to 1-in. diameter. The values shall be multiplied by 0.875 for diameters greater than 1 in.

[f] L_c is equal to the clear distance between the holes or between the hole and the edge of the material in the direction of applied bearing force, in.; F_u is the specified minimum tensile strength of the connected material; d is the nominal diameter of the bolt, in.

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6B.5.2.1.1—Combined Stresses

The allowable combined stresses for steel compression members may be calculated by the provisions of AASHTO Standard Specifications as modified below or by the procedure contained in Appendix K6B.

In using the AASHTO Standard Specifications (Article 10.36), the allowable compressive axial stress, $F_{a,}$ and the allowable compressive bending stresses, F_{bx} and $F_{by,}$ should be based on Tables 6B.5.2.1-1 and 6B.5.2.1-2. The safety factor *F.S.* to be used in computing the Euler buckling stress F'_e should be as follows:

F.S. = 2.12 at inventory level

= 1.70 at operating level

6B.5.2.1.2—Batten Plate Compression Members

To allow for the reduced strength of batten plate compression members, the actual length of the member shall be multiplied by the following factor to obtain the adjusted value of L/r to be substituted in the compression member formulae discussed in Articles 6B.5.2.1 and 6B.5.2.1.1.

Actual L/r	Factor Spacing Center-to-Center of Batten Plates			
	Up to 2*d*	4*d*	6*d*	10*d*
40	1.3	2.0	2.8	4.5
80	1.1	1.3	1.7	2.3
120	1.0	1.2	1.3	1.8
160	1.0	1.1	1.2	1.5
200	1.0	1.0	1.1	1.3

d = depth of member perpendicular to battens

For compression members having a solid plate on one side and batten plates on the other, the foregoing factors shall be reduced 50 percent.

Adjusted L/r (batten plate both sides) = Actual L/r × factor

Adjusted L/r (batten plate one side)
= Actual L/r × [1 + $^1/_2$(factor – 1)]

6B.5.2.2—Wrought Iron

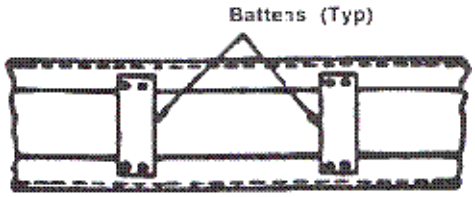
Allowable maximum unit stress in wrought iron for tension and bending:

Operating = 20,000 psi

Inventory = 14,600 psi

C6B.5.2.1.2

Built-up compression members are generally connected across their open sides. Typical connections include stay plates in combination with single or double lacing, perforated cover plates, and battens. This Article covers the use of batten plates only, when used as shown below:



C6B.5.2.2

Allowable maximum unit stresses in wrought iron for tension and bending at the Inventory level should be between 10,000 psi and 14,000 psi, depending on material test results.

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Where possible, coupon tests should be performed to confirm material properties used in the rating.

6B.5.2.3—Reinforcing Steel

The following are the allowable unit stresses in tension for reinforcing steel. These will ordinarily be used without reduction when the condition of the steel is unknown.

Table 6B.5.2.3-1—Allowable Unit Stresses for Reinforcing Steel

	Stresses (psi)		
	Inventory Rating	Operating Rating	Yield
Structural or unknown grade prior to 1954	18,000	25,000	33,000
Structural Grade	20,000	27,000	36,000
Grade 40 billet, intermediate, or unknown grade (after 1954)	20,000	28,000	40,000
Grade 50 rail or hard	20,000	32,500	50,000
Grade 60	24,000	36,000	60,000

6B.5.2.4—Concrete

Unit stresses in concrete may be determined in accordance with the Service Load Design Method of the AASHTO Standard Specifications (Article 8.15) or be based on the articles below. When the ultimate strength, f'_c, of the concrete is unknown and the concrete is in satisfactory condition, f'_c may be determined from Table 6B.5.2.4-1.

Table 6B.5.2.4-1—Allowable Unit Stresses for Concrete

Year Built	f'_c (psi)
Prior to 1959	2,500
1959 and later	3,000

For prestressed concrete components, the compressive strengths shown above may be increased by 25 percent.

6B.5.2.4.1—Bending

The following maximum allowable bending unit stresses in concrete in lb/in.^{2} may be used:

C6B.5.2.4

Some guidance on the ultimate strength, f'_c, of concrete may be obtained from compression testing of cores removed from the structure. (See Article 5.3.)

Guidance on considering the effects of deterioration on the load rating of concrete structures can be found in Article C6A.5.5.

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Table 6B.5.2.4.1-1—Compression Due to Bending, f'_c

f'_c (psi)	Compression Due to Bending, f'_c (psi)		n
	Inventory Level	Operating Level	
2,000–2,400	800	1,200	15
2,500–2,900	1,000	1,500	12
3,000–3,900	1,200	1,900	10
4,000–4,900	1,600	2,400	8
5,000 or more	2,000	3,000	6

The value of n may be varied according to Table 6B.5.2.4.1-1.

6B.5.2.4.2—Columns

The determination of the capacity of a compression member based on the AASHTO Standard Specifications (Article 8.15.4) results in an inventory level capacity. The following simplified approach establishes the maximum operating level capacity:

Maximum safe axial load in columns at operating rating:

$$P = f_c A_g + f_s A_s \qquad \text{(6B.5.2.4.2-1)}$$

where:

P = Allowable axial load on column

f_c = Allowable unit stress of concrete taken from Eq. 6B.5.2.4.2-2 or 6B.5.2.4.2-3

A_g = Gross area of column

f_s = Allowable stress of steel = $0.55f_y$

f_y = Yield strength of reinforcing steel

A_s = Area of longitudinal reinforcing steel

Compression, short columns, in which L/D is 12 or less:

$$f_c = 0.3f'_c \qquad \text{(6B.5.2.4.2-2)}$$

Compression, long columns, in which L/D is greater than 12:

$$f_c = 0.3f'_c(1.3 - 0.03L/D) \qquad \text{(6B.5.2.4.2-3)}$$

L = Unsupported length of column

D = Least dimension of column

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6B.5.2.4.3—Shear (Diagonal Tension)

The Inventory level shear strength should be determined in accordance with the Service Load Design method of the AASHTO Standard Specifications (Article 8.15.5).

The Operating level shear strength in beams showing no diagonal tension cracking may be found as follows:

(Total Unit Shear) = (Shear Taken by Steel)
+ (Shear Taken by Concrete)
or:

$$v = v_s + v_c \qquad \text{(6B.5.2.4.3-1)}$$

The allowable shear stress carried by the concrete, v_c, may be taken as $1.3\sqrt{f'_c}$, and a more detailed calculation of the allowable shear stress can be made using:

$$v_c = 1.25\sqrt{f'_c} + 1,600\rho_w(Vd/M) \le 2.3\sqrt{f'_c} \qquad \text{(6B.5.2.4.3-2)}$$

where:

d = Distance from extreme compression fiber to centroid of tension reinforcement

ρ_w = Reinforcement ratio = $A_s/(b_w d)$

b_w = Width of the web

M is the moment acting simultaneously with the shear force, V, at the section being considered. The quantity Vd/M shall not be taken greater than 1.0.

Where severe diagonal tension cracking has occurred, v_c should be considered as zero and all shear stress should be taken by the reinforcing steel.

6B.5.2.5—Prestressed Concrete

Rating of prestressed concrete members should be based on the criteria presented under Article 6B.5.3.3.

C6B.5.2.5

As in design, the rating of prestressed concrete members is a combination of strength (load factor method) and serviceability requirements (allowable stress method). The criteria for rating prestress concrete members are presented under the load factor method in Article 6B.5.3.3.

6B.5.2.6—Masonry

Stone, concrete, and clay brick masonry structures should be evaluated using the allowable stress rating method. Mortar used to bind the individual masonry units should be classified in accordance with ASTM C270.

The allowable inventory level compressive stresses for masonry assemblies are shown in Table 6B.5.6-1. These are minimum values and may be used in the absence of more reliable data such as the results of a prism test conducted in accordance with ASTM E447. The condition of the masonry unit and mortar should be considered when assigning an allowable stress.

C6B.5.2.6

The allowable stresses for evaluating masonry structures are based on the ACI empirical method. (See ACI 530-05.) These values are conservative and constitute a lower bound for allowable masonry stresses. The Engineer may use the more rigorous approach in ACI 530-05 as an alternative.

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Allowable operating level stresses for masonry are not included in this Manual. Masonry components should be evaluated at the inventory level.

Reinforced masonry construction may be evaluated using the allowable unit stresses for reinforcing steel. Article 6B.5.2.3 and an appropriate allowable stress in the masonry.

Table 6B.5.2.6-1—Allowable Inventory Compressive Stresses for Evaluation of Masonry

	Allowable Inventory Compressive Stresses Gross Cross-Sectional Area, psi	
Construction: Compressive Strength of Unit, Gross Area, psi	Type M or S Mortar[a]	Type N Mortar[a]
Solid masonry of brick and other solid units of clay or shale; sand-lime or concrete brick:		
8,000 or greater	350	300
4,500	225	200
2,400	160	140
1,500	115	100
Grouted masonry, of clay or shale; sand-lime or concrete:		
4,500 or greater	225	200
2,400	160	140
1,500	115	100
Solid masonry of solid concrete masonry units:		
3,000 or greater	225	200
2,000	160	140
1,200	115	100
Masonry of hollow load-bearing units:		
2,000 or greater	140	120
1,500	115	100
1,000	75	70
700	60	55
Stone ashlar masonry:		
Granite	720	640
Limestone or marble	450	400
Sandstone or cast stone	360	320
Rubble stone masonry		
Coarse, rough, or random	120	100

[a] Mortar is classified in accordance with ASTM C270

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6B.5.2.7—Timber

Determining allowable stresses for timber in existing bridges will require sound judgment on the part of the Engineer making the field investigation.

(1) Inventory Stress

The inventory unit stresses should be equal to the allowable stresses for stress-grade lumber given in the AASHTO Standard Specifications.

Allowable inventory unit stresses for timber columns should be in accordance with the applicable provisions of the AASHTO Standard Specifications.

(2) Operating Stress

The maximum allowable Operating unit stresses should not exceed 1.33 times the allowable stresses for stress-grade lumber given in the current AASHTO Standard Specifications. Reduction from the maximum allowable stress will depend upon the grade and condition of the timber and should be determined at the time of the inspection.

Allowable operating stress in lb/in.^{2} of cross-sectional area of simple solid columns should be determined by the following formulas but the allowable operating stress should not exceed 1.33 times the values for compression parallel to grain given in the design stress table of the AASHTO Standard Specifications.

$$\frac{P}{A} = \frac{4.8E}{(1/r)^2} \qquad \text{(6B.5.2.7-1)}$$

where:

P = Total load, lb

A = Cross-sectional area, in.^{2}

E = Modulus of elasticity

ℓ = Unsupported overall length between points of lateral support of simple columns, in.

r = Least radius of gyration of the section, in.

For columns of square or rectangular cross-section, this formula becomes:

$$\frac{P}{A} = \frac{0.40E}{(1/d)^2} \qquad \text{(6B.5.2.7-2)}$$

where:

d = Dimension of the narrowest face, in.

The above formula applies to long columns with ℓ/d over 11, but not greater than 50.

For short columns, ℓ/d not over 11, use the allowable design unit stress in compression parallel to grain times 1.33 for the grade of timber used.

C6B.5.2.7

The material and member properties based on as-built information may need to be adjusted for field conditions such as weathering or decay. The Engineer's judgment and experience are required in assessing actual member resistance.

Eq. 6B.5.2.7-1 is based on the Euler long-column formula with two adjustments as follows. First, E is reduced by dividing by 2.74. This corresponds to a safety factor of 1.66 for solid timber members according to the National Design Specifications for Wood Construction (2005). Then the Euler allowable stress is multiplied by 1.33 to provide an operating level allowable stress as shown in Eq. 6B.5.2.7-1.

For square and rectangular columns, substituting $d/\sqrt{12}$ for the radius of gyration, r, in Eq. 6B.5.2.7-1 results in Eq. 6B.5.2.7-2.

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6B.5.3—Load Factor Method

Nominal capacity of structural steel, reinforced concrete and prestressed concrete should be the same as specified in the load factor sections of the AASHTO Standard Specifications. Nominal strength calculations should take into consideration the observable effects of deterioration, such as loss of concrete or steel-sectional area, loss of composite action or corrosion.

Allowable fatigue strength should be checked based on the AASHTO Standard Specifications. Special structural or operational conditions and policies of the Bridge Owner may also influence the determination of fatigue strength.

C6B.5.3

Nominal capacities for members in the proposed guidelines are based on AASHTO's Standard Specifications contained in the load factor section. This resistance depends on both the current dimensions of the section and the nominal material strength.

Different methods for considering the observable effects of deterioration were studied. The most reliable method available still appears to be a reduction in the nominal resistance based on measured or estimated losses in cross-sectional area and/or material strengths.

At the present time, load factor methods for determining the capacity of timber and masonry structural elements are not available.

6B.5.3.1—Structural Steel

The yield stresses used for determining ratings should depend on the type of steel used in the structural members. When nonspecification metals are encountered, coupon testing may be used to determine yield characteristics. The nominal yield value should be substituted in strength formulas and is typically taken as the mean test value minus 1.65 standard deviations. When specifications of the steel are not available, yield strengths should be taken from the applicable "date built" column of Tables 6B.5.2.1-1 to 6B.5.2.1-4.

The capacity of structural steel members should be based on the load factor requirements stated in the AASHTO Standard Specifications. The capacity, C, for typical steel bridge members is summarized in Appendix L6B. For beams, the overload limitations of Article 10.57 of the AASHTO Standard Specifications should also be considered.

Except as specified in Appendix L6B.2.6.1, the Operating rating for welds, bolts, and rivets should be determined using the maximum strengths from Table 6B.5.3.1-1.

The Operating rating for friction joint fasteners (ASTM A325 bolts) should be determined using a stress of 21 ksi. A_1 and A_2 should be taken as 1.0 in the basic rating equation.

Where rivets carrying loads pass through undeveloped fillers 0.25 in. or more in thickness in axially loaded connections, refer to Article 6A.6.12.5.1 and AASHTO LRFD Design Article 6.13.6.1.4 for a potential capacity reduction factor.

C6B.5.3.1

Guidance on considering the effects of deterioration on load rating of steel structures can be found in Article C6A.6.5.

Specifications and guidance for determining the capacity of gusset plates can be found in Appendix L6B.

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Table 6B.5.3.1-1—Design Strength of Connectors

Type of Fastener	Strength (φF)
Groove Weld[a]	$1.00F_y$
Fillet Weld[b]	$0.45F_u$
Low-Carbon Steel Bolts	
ASTM A307	
Tension	30 ksi
Shear on Bolt with	
Threads in Shear Plane[d]	18 ksi
Power-Driven Rivets[f]	
ASTM A502	
φF_{vu} Shear—Grade 1	32.0 ksi
φF_{vu} Shear—Grade 2	43.0 ksi
ASTM A141 or Carbon Steel	
φF_{vu} Shear	32.0 ksi
Rivets[f]	
Unknown Rivet Type and Origin φF_{vu}	27.0 ksi
High Strength Bolts	
AASHTO M 164	
(ASTM A325)	
Applied Static Tension[c]	68 ksi
Shear on Bolt with	
Threads in Shear Plane[c,d,e]	35 ksi
AASHTO M 253	
(ASTM A490)	
Applied Static Tension	85 ksi
Shear on Bolt with	
Threads in Shear Plane[d,c]	43 ksi

[a] F_y = yield point of connected material.

[b] F_u = minimum strength of the welding rod metal.

[c] The tensile strength of M 164 (A325) bolts decreases for diameters greater than 1 inch. The design values listed are for bolts up to 1 inch in diameter. The design values shall be multiplied by 0.875 for diameters greater than 1 inch.

[d] For bolts, the tabulated values shall be reduced by 20 percent in bearing-type connections whose length between extreme fasteners in each of the spliced parts measured parallel to the line of axial force exceeds 50 inches. For flange splices, the 50-inch length is to be measured between the extreme bolts on only one side of the connection.

[e] If material thickness or joint details preclude threads in the shear plane, multiply values by 1.25.

[f] For rivets, tabulated values shall be reduced by a reduction factor for length in bearing-type connections. When the length (L) between extreme fasteners in each of the spliced parts measured parallel to the line of axial force is less than or equal to 50 inches, the reduction factor shall be determined from the following variable reduction equation *1-(0.25L/50)*. Tabulated values shall be reduced by 25 percent in bearing-type connections whose length between extreme fasteners in each of the spliced parts measured parallel to the line of axial force exceeds 50 inches. For flange and other splices, the 50-inch length is to be measured between the extreme rivets on only one side of the connection.

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6B.5.3.2—Reinforced Concrete

The following are the yield stresses for reinforcing steel.

Reinforcing Steel	Yield Point, F_y (psi)
Unknown steel (prior to 1954)	33,000
Structural Grade	36,000
Billet or Intermediate Grade and Unknown after 1954 (Grade 40)	40,000
Rail or Hard Grade (Grade 50)	50,000
Grade 60	60,000

The capacity of concrete members should be based on the strength requirements stated in AASHTO Standard Specifications (Article 8.16). Appendix L6B contains formulas for the capacity, *C*, of typical reinforced concrete members. The area of tension steel at yield to be used in computing the ultimate moment capacity of flexural members should not exceed that available in the section or 75 percent of the reinforcement required for balanced conditions.

C6B.5.3.2

Guidance on considering the effects of deterioration on the load rating of concrete structures can be found in Article C6A.5.5.

6B.5.3.3—Prestressed Concrete

The rating of prestressed concrete members at both inventory and operating level should be established in accordance with the strength requirements of the AASHTO Standard Specifications. Additionally at inventory level, the rating must consider the allowable stresses at service load as specified in the AASHTO Standard Specifications. In situations of unusual design with wide dispersion of the tendons, the operating rating might further be controlled by stresses not to exceed 0.90 of the yield point stress in the prestressing steel nearest the extreme tension fiber of the member.

A summary of the strength and allowable stress rating equations is presented at the end of this Section. More stringent allowable stress values may be established by the Bridge Owner.

Typically, prestressed concrete members used in bridge structures will meet the minimum reinforcement requirements of Article 9.18.2.1 of the AASHTO Standard Specifications. While there is no reduction in the flexural strength of the member in the event that these provisions are not satisfied, an Owner, as part of the flexural rating, may choose to limit live loads to those that preserve the relationship between φM_n and M_{cr} that is prescribed for a new design. The use of this option necessitates an adjustment to the value of the nominal moment capacity φM_n, used in the flexural strength rating equations. Thus when $\varphi M_n < 1.2M_{cr}$, the nominal moment capacity becomes $(k)(\varphi)(M_n)$, where k is the larger of:

$$k = \frac{\varphi M_n}{1.2M_{cr}}$$

C6B.5.3.3

In the design of prestressed concrete members, both the strength at ultimate and the allowable stress criteria at the transfer and in-service conditions must be satisfied. The strength design is based on factored loads and the flexural capacity of the section computed in accordance with Article 9.17 of the AASHTO Standard Specifications.

The limitation on the maximum stress of pre-stressing steel at the operating level to 0.90 of the yield point stress is not a design requirement, but should be used to ensure sufficient reserve ductility in the prestressing steel.

Reactions are produced at the supports in continuous spans under post-tensioning loads, giving rise to secondary moments in the girders. The secondary moments are combined with the primary moments to provide the total moment effect of the post-tensioning.

Guidance on considering the effects of deterioration on the load rating of concrete structures can be found in Article C6A.5.5.

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or:

$$k = \frac{\varphi M_n}{1.33 M_u}$$

Rating Equations

Inventory Rating

$$RF = \frac{6\sqrt{f_c'} - (F_d + F_p + F_s)}{F_1} \text{ Concrete Tension}$$

$$RF = \frac{0.6 f_c' - (F_d + F_p + F_s)}{F_1} \text{ Concrete Compression}$$

$$RF = \frac{0.4 f_c' - \frac{1}{2}(F_d + F_p + F_s)}{F_1} \text{ Concrete Compression}$$

$$RF = \frac{0.8 f_y^* - (F_d + F_p + F_s)}{F_1} \text{ Prestressing Steel Tension}$$

$$RF = \frac{\varphi R_n - (1.3D + S)}{2.17 L(1 + I)} \text{ Flexural and Shear Strength}$$

Operating Rating

$$RF = \varphi R_n - \frac{(1.3D + S)}{1.3 L(1 + I)} \text{ Flexural and Shear Strength}$$

$$RF = \frac{0.9 f_y^* - (F_d + F_p + F_s)}{F_1} \text{ Prestressing Steel Tension}$$

where:

RF = Rating factor

f'_c = Concrete compressive strength

$6\sqrt{f_c'}$ = Allowable concrete tensile strength. A factor of $3\sqrt{f_c'}$ may be applicable, or this allowable stress may be zero, as provided by Article 9.15 of the AASHTO Standard Specifications.

F_d = Unfactored dead loss stress

F_p = Unfactored stress due to prestress force after all losses

F_s = Unfactored stress due to secondary prestress forces

F_1 = Unfactored live load stress including impact

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6B.6.2.1—Wheel Loads (Deck)

In general, stresses in the deck do not control the load rating except in special cases. The calculation of bending moments in the deck should be in accordance with the AASHTO Standard Specifications. Wheel loads should be in accordance with the current AASHTO Standard Specifications.

6B.6.2.2—Truck Loads

The live or moving loads to be applied on the deck for determining the rating should be the standard AASHTO "HS" loading.

The number of traffic lanes to be loaded, and the transverse placement of wheel lines should be in conformance with the current AASHTO Standard Specifications and the following:

Roadway widths from 18 to 20 ft should have two design lanes, each equal to one-half the roadway width. Live loadings should be centered in these lanes. Roadway widths less than 18 ft should carry one traffic lane only.

When conditions of traffic movements and volume would warrant it, fewer traffic lanes than specified by AASHTO may be considered.

Utilizing the number and transverse placement of lanes in accordance with the AASHTO Standard Specifications may not be consistent with the actual usage of the bridge as defined by the striped lanes, and could result in conservative load ratings for bridge types such as trusses, two-girder bridges, arches, and exterior girders, where live load distribution factors are established using the lever rule method. Upon the approval of the Bridge Owner, an alternate load rating of the bridge for normal operating conditions or current usage may be performed by placing truck loads only within the striped lanes. When load rating a structure based on the existing striped lanes, the transverse positioning of the truck should include placing the wheel load anywhere within the lane, including on the lane stripe. This alternate load rating may be performed for all live load models. Placement of striped lanes on the bridge should be field verified and documented in the inspection report.

C6B.6.2.2

The probability of having a series of closely spaced heavy vehicles of the maximum allowable weight becomes greater as the maximum allowed weight for each unit becomes less. That is, it is more likely to have a train of light-weight vehicles than to have a train of heavy-weight vehicles. This makes it necessary to consider more than one vehicle in the same lane under some conditions. For example, vehicles should be spaced at distances of 30 ft clear or more in the same lane to produce maximum load effect when the safe loading per vehicle or vehicle combinations is less than 12 tons.

In design it is required to place lanes starting at the face of the barrier, including the shoulder areas. Many existing truss/arch and two-girder bridges have wide shoulders, so moving the wheel load within 2 ft of the barrier could significantly increase load distribution to the truss/arch/girder and result in lower ratings. Similarly, for multi-girder bridges with large overhangs, establishing live load distribution (using the lever rule method) by moving the wheel load within 2 ft of the barrier significantly increases the load distribution to the exterior girder and results in lower ratings. Moving the wheel load within 2 ft of the barrier is consistent with design but not consistent with how the bridge is loaded under normal service conditions. Ratings per the design code lane placement will provide ratings for the maximum likely live loading for the bridge whereas the ratings with wheel loads within striped lanes will provide ratings under current use conditions. Placing loads within striped lanes is also particularly relevant to permit reviews where permit trucks are required to operate in the traffic lanes. It is recognized that on occasion a random vehicle may enter the roadway areas outside of the striped lanes. The probability that this will occur simultaneously with full live loads in the striped lanes is quite low. In cases where the striped lanes are less than 10 ft wide, the load should be centered in the lane so that the distance to the wheel load from the barrier would be less than 2 ft.

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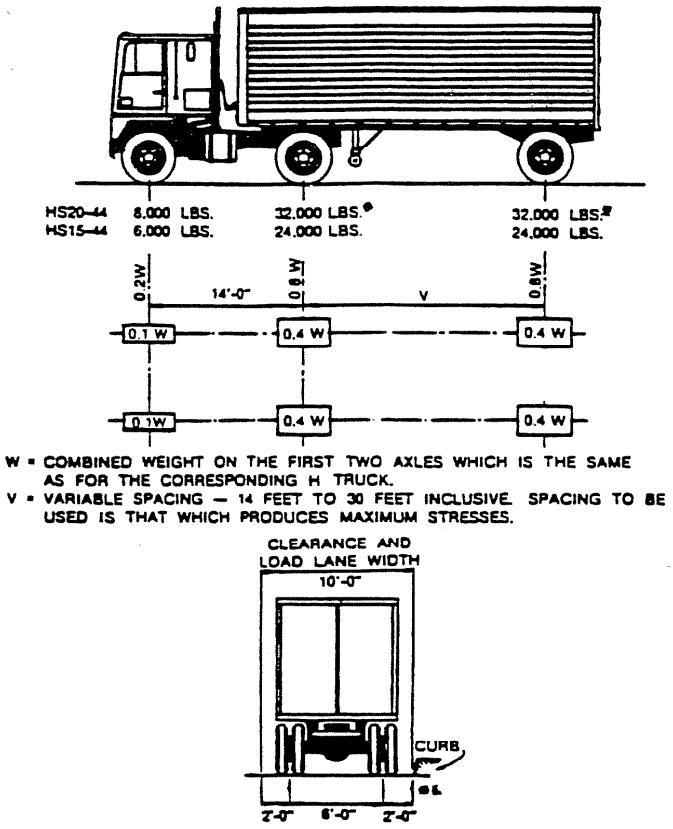
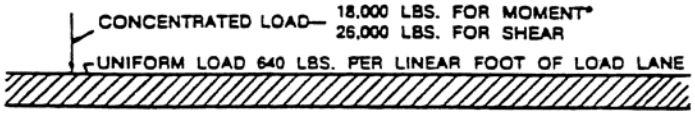


*In the design of timber floors and orthotropic steel decks (excluding transverse beams) for HS20 loading, one-axle load of 24,000 pounds or two-axle loads of 16,000 pounds each, spaced 4 feet apart may be used, whichever produces the greater stress, instead of the 32,000-pound axle shown.

Figure 6B.6.2-1—Standard HS Truck



H20–44 LOADING
HS20–44 LOADING

*For the determination of maximum negative moment in continuous spans, the lane load shown shall be modified by the addition of a second, equal weight concentrated load placed in one other span in the series in such position to produce the maximum effect.

Figure 6B.6.2-2—Standard Lane Load

6B.6.2.3—Lane Loads

The Bridge Owner may use the standard AASHTO HS lane load for all span lengths where it may result in load effects which are greater than those produced by the AASHTO standard HS truck loading.

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6B.6.2.4—Sidewalk Loadings

Sidewalk loadings used in calculations for safe load capacity ratings should be the probable maximum loads anticipated. Because of site variations, the determination of loading to be used will require engineering judgment, but in no case should it exceed the value given in AASHTO Standard Specifications.

The operating level should be considered when full truck and sidewalk live loads act simultaneously on the bridge.

C6B.6.2.4

The probability that both the full truck and full sidewalk live loads would act simultaneously on the bridge is quite low. This loading case should be evaluated based on the operating level.

6B.6.2.5—Live Load Effects: *L*

Live load moments in longitudinal stringers and girders may be calculated using Tables C6B-1 and C6B-2 for live load moments produced by the HS-20 load.

Live load moments in the intermediate and end floor beams of trusses and through girders may be calculated by using Tables D6B-1, D6B-2, E6B-1, and E6B-2. The tables, along with the moment formulas on the same sheets, provide a convenient means of computing the live load moments based on the HS-20 load.

Live loads in truss members can be calculated by using the formulas for maximum shear and moments given in Appendices F6B through J6B. Using these formulas will give the maximum live load stresses for the HS-20 truck. Note that the formulas are valid only when used within the given limits. Modifications of the formulas may be required under loadings not meeting these limits. Such modifications may be found necessary when the structure or panels are too short to permit the entire load to be on the structure with the load positioned to produce the maximum shear or moment.

6B.6.3—Distribution of Loads

The fraction of live load transferred to a single member should be selected in accordance with the current AASHTO Standard Specifications. These values represent a possible combination of diverse circumstances. The option exists to substitute field-measured values, analytically calculated values, or those determined from advanced structural analysis methods based on the properties of the existing structure. Loadings should be placed in positions causing the maximum response in the components being evaluated.

6B.6.4—Impact: *I*

Impact should be added to the live load used for rating in accordance with the current AASHTO Standard Specifications. However, specification impact may be reduced when conditions of alignment, enforced speed posting, and similar situations require a vehicle to substantially reduce speed in crossing the structure.

C6B.6.4

The condition of the approach roadway and deck joints may also influence the selection of an appropriate impact factor. Some guidelines are provided in Article C6A.4.4.3.

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6B.6.5—Deflection

Live load deflection limitations should not be considered in load rating except in special cases.

6B.6.6—Longitudinal Loads

The rating of the bridge members to include the effects of longitudinal loads in combination with dead and live load effects should be done at the operating level. Where longitudinal stability is considered inadequate, the structure may be posted for restricted speed. In addition, longitudinal loads should be used in the evaluation of the adequacy of the substructure elements.

6B.6.7—Environmental Loads

The rating of the bridge members to include the effects of environmental loads in combination with dead and live load effects should be done at the operating level.

6B.6.7.1—Wind

Lateral loads due to wind normally need not be considered in load rating.

However, the effects of wind on special structures such as movable bridges, suspension bridges, and other high-level structures should be evaluated.

6B.6.7.2—Earthquake

Earthquake loads should not be considered in calculating load ratings or in determining live load restrictions.

To evaluate the resistance of the structure to seismic forces, the methods described in Division I-A, Seismic Design of the AASHTO Standard Specifications, may be used.

C6B.6.7.2

Bridge Maintenance Engineers may be called upon to evaluate existing structures for their capacity to resist earthquake forces. This specification permits the investigator to use either the relatively simple methods of the AASHTO *Standard Specifications for Highway Bridges* or the more complex analysis procedures described in the *AASHTO Guide Specifications for LRFD Seismic Bridge Design*. If facilities and trained personnel are available, the multimodal spectral method of analysis is recommended to provide more thorough and credible results.

For seismic retrofitting of bridges, seismic loads must be considered.

6B.6.7.3—Temperature, Creep, and Shrinkage

Typically, temperature, creep, and shrinkage effects need not be considered in calculating load ratings for components that have been provided with well-distributed steel reinforcement to control cracking.

These effects may need to be considered in the strength evaluation of long-span, framed, and arch bridges.

C6B.6.7.3

Temperature, creep, and shrinkage are primarily strain-inducing effects. As long as the section is ductile, such changes in strain are not expected to cause failure.

Where temperature cracks are evident and analysis is considered warranted, temperature effects due to time-dependent fluctuations in effective bridge temperature may be treated as long-term loads, with a long-term modulus of elasticity of concrete reduced to one-third of its normal value.

The temperature loading, *T*, could be significant in superstructures that are framed into bents and abutments with no hinges. Uniform temperature loading, *TU*, could induce a significantly large tension in the superstructure

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girders, which would result in reduction in shear capacity of reinforced concrete girders. Temperature gradient loading, *TG*, could induce significantly higher bending moments in framed structures.

Bearings' becoming nonfunctional generally leads to thermal forces being applied onto bridge elements that were not designed for such loads. Keeping bearings in good working order could prevent temperature and shrinkage forces from occurring.

6B.6.7.4—Stream Flow

Forces caused by water movements should not be considered in calculating the load rating. However, remedial action should be considered if these forces are especially critical to the structure's stability.

6B.6.7.5—Ice Pressure

Forces caused by ice pressure should be considered in the evaluation of substructure elements in those regions where such effects can be significant. If these forces are especially important, then corrective action should be recommended.

6B.6.7.6—Permanent Loads Other Than Dead Loads

Secondary effects from post-tensioning shall be considered as permanent loads.

C6B.6.7.6

In continuous post-tensioned bridges, secondary moments are introduced as the member is stressed.

6B.7—POSTING OF BRIDGES

6B.7.1—General

Weight limitations for the posted structure should conform to local regulations or policy within the limits established by this Manual. A bridge should be capable of carrying a minimum gross live load weight of three tons at inventory or operating level. When deciding whether to close or post a bridge, the owner may particularly want to consider the volume of traffic, the character of traffic, the likelihood of overweight vehicles, and the enforceability of weight posting. A Bridge Owner may close a structure at any posting threshold, but bridges not capable of carrying a minimum gross live load weight of three tons must be closed.

The total load on any member caused by dead load, live load, and such other loads deemed applicable to the structure, should not exceed the member capacity as set forth in this Manual or in the rating report. When it becomes necessary to reduce the allowable live loads in order to conform to the capacity of a structure, such a reduction should be based on the assumption that each axle load maintains a proportional relation to the total load of the vehicle or vehicle combination.

C6B.7.1

Most structures which require weight limits below statutory limits are old and designed for light loads, and/or are weak as a result of damage. With some exceptions, the weaker elements of older bridges are usually in the superstructure, not in the piers or abutments.

There may be circumstances where the Bridge Owner may utilize load levels higher than those used for inventory rating, in order to minimize the need for posting of bridges. In no case shall the load levels used be greater than those permitted by the operating rating.

For those bridges supporting large dead loads, the use of the Load Factor or Load and Resistance Factor rating methods may result in a live load capacity greater than that determined based on the allowable stress rating method.

Bridges which use a load level above the inventory level should be subject to more frequent, competent inspections. Several factors may influence the selection of the load level. For instance:

1. The factor of safety commonly used in the design or inventory level rating may have provided for an increase in traffic volume, a variable amount of deterioration, and extreme conditions of live loading.

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2. The factor of safety used in rating existing structures must provide for unbalanced distribution of vehicle loads, and possible overloads. For both design and rating, factors of safety must provide for lack of knowledge as to the distribution of stresses, possible minimum strength of the materials used as compared to quoted average values, possible differences between the strength of laboratory test samples and the material under actual conditions in the structure, and normal defects occurring in manufacture or fabrication.
3. A higher safety factor for a bridge carrying a large volume of traffic may be desirable as compared with the safety factor for a structure carrying few vehicles, especially if the former includes a high percentage of heavy loads.
4. The probability of having a series of closely spaced vehicles of the maximum allowed weight should be considered. This effect becomes greater as the maximum allowed weight for each unit becomes less.
5. Lower load levels may be warranted for nonredundant metal bridge elements due to the consequences of failure. Exceptions may be elements of riveted construction and all floor beams, provided they are in good condition. Examples of nonredundant elements are welded or rolled two-girder bridges, truss members, or pinned eyebar trusses and truss members on welded trusses.
6. Bridges with extensive material losses may warrant a lower load level because of the greater uncertainty in evaluating present strength capacity. This is especially true if the loss in material is in a highly stressed area.
7. Sites for which it is suspected that there are frequent truck overloads should be considered for lower load levels unless enforcement methods are put in place.
8. The ratio of dead load to live load may have an influence on the selection of appropriate load level. Structures with high ratios of dead to live load and for which there are no visible signs of distress may be considered for the higher load levels.

6B.7.2—Posting Loads

The live load to be used in the rating Eq. 6B.4.1-1 for posting considerations should be any of the three typical legal loads shown in Figure 6B7.2-1, any of the four single-unit legal loads shown in Figure 6B7.2-2 or state legal loads. For spans over 200 feet in length, the selected legal load should be spaced with 30 feet clear distance between vehicles to simulate a train of vehicles in one lane and a single vehicle load should be applied in the adjacent lanes(s). When the maximum legal load under state law exceeds the safe load capacity of a bridge, restrictive posting shall be required.

C6B.7.2

Trucks weighing up to 80,000 lb are typically allowed unrestricted operation and are generally considered "legal" provided they meet weight guidelines of Federal Bridge Formula B (Formula B). In the past, the maximum legal weight for short wheelbase trucks was usually determined by Formula B rather than by the 80,000-lb gross weight limit. Since the adoption of the AASHTO family of three legal loads, the trucking industry has introduced specialized single-unit trucks with closely spaced multiple axles that make it possible for these short wheelbase trucks to carry the maximum load of up to 80,000 lb and still meet Formula B. The current AASHTO legal loads selected at the time to closely match the Formula B in the short,

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medium, and long truck length ranges do not represent these newer axle configurations. These specialized hauling vehicles cause force effects that exceed the stresses induced by HS-20 by up to 22 percent and by the Type 3, 3S2, or 3-3 posting vehicles by over 50 percent in certain cases. The shorter spans are most sensitive to axle configurations.

The Notional Rating Load, *NRL*, shown in Figure 6B7.2-3 may be used as a screening load model for single-unit trucks that meet Formula B. Bridges that result in $RF \geq 1.0$ for the *NRL* loading will have adequate load capacity for all legal single-unit Formula B truck configurations up to 80,000 lb.

The *NRL* loading represents a single load model that will envelop the load effects on simple and continuous span bridges of the worst possible Formula B single-unit truck configurations up to 80,000 lb. It is called "notional" because it is not intended to represent any particular truck. Vehicles considered to be representative of the newer Formula B configurations were obtained through the analysis of weigh-in-motion data and other truck and survey data obtained from the states. The single *NRL* load model with a maximum gross weight of 80,000 lb produces moments and shears that exceed the load effects for a series of 3- to 8-axle single-unit trucks allowed to operate under current federal weight laws (NCHRP Report 575).

In the *NRL* loading, axles that do not contribute to the maximum load effect under consideration shall be neglected. For instance, axles that do not contribute to the maximum positive moments need to be neglected or they will contribute to bending in the opposite (negative) direction. This requirement may only affect certain continuous bridges, usually with short span lengths. The drive axle spacing of 6 ft may also be increased up to 14 ft to maximize load effects. Increasing the drive axle spacing to 14 ft could result in a slight increase in moments for continuous bridges.

For bridges with $RF < 1.0$ for the *NRL* loading, a posting analysis should be performed to resolve posting requirements for single-unit multi-axle trucks. While a single envelope *NRL* loading can provide considerable simplification of load-rating computations, additional legal loads for posting are needed to give more accurate posting values. Certain multi-axle Formula B configurations that cause the highest load effects appear to be common only in some states, and they should not lead to reduced postings in all states.

Setting weight limits for posting often requires the evaluator to determine safe load capacities for legal truck types that operate within a given state, in accordance with state posting practices. The four single-unit Formula B legal loads shown in Figure 6B.7.2-2 include the worst 4-axle (SU4), worst 5-axle (SU5), worst 6-axle (SU6), and worst 7-axle (SU7) trucks (7-axle is also representative of 8-axle trucks) identified in the NCHRP 12-63 study. This series of loads affords the evaluator the flexibility of selecting only posting loads that model commercial Formula B trucks in a particular state or jurisdiction.

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The more compact four- and five-axle trucks that produce the highest moment or shear per unit weight of truck will often govern the posting value (result in the lowest weight limit). States that post bridges for a single tonnage for all single-unit trucks may consider it desirable to reduce the number of new posting loads that need to be evaluated. Here it would be appropriate to use truck SU5 as a single representative posting load for the series of Formula B truck configurations with 5 to 8 axles. This simplification will introduce added conservatism in posting, especially for short-span bridges. It should be noted that situations could arise where a bridge may have a $RF > 1.0$ for SU5 but may not rate ($RF < 1.0$) for SU6 or SU7. Here the SU5 load model is being utilized to determine a single posting load for a bridge that has adequate capacity for SU5 but not for the heavier trucks.

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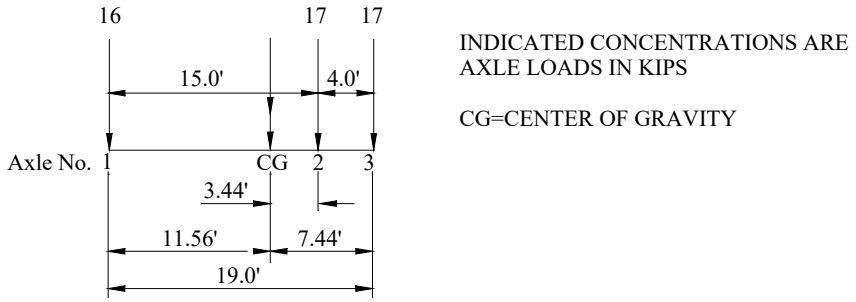
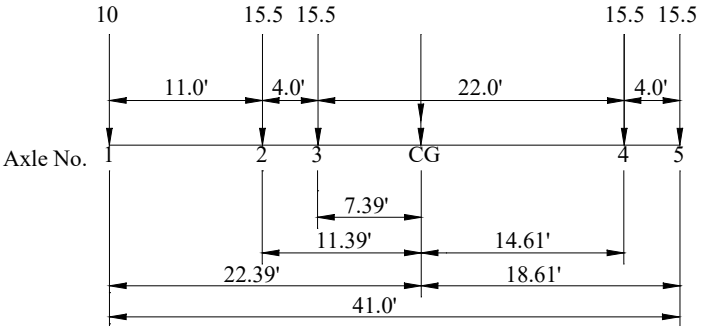
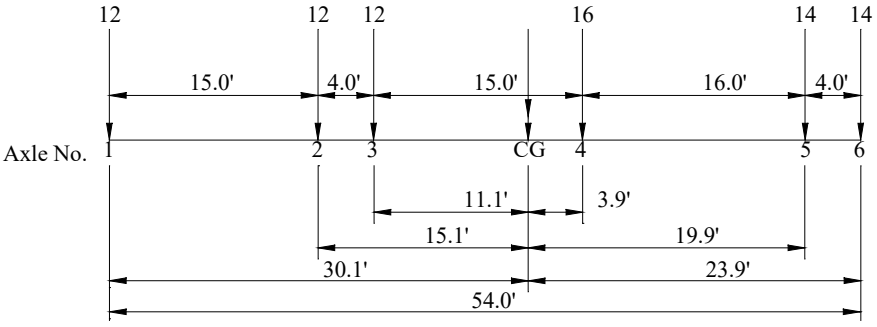


TYPE 3 UNIT WEIGHT = 50 KIPS (25 TONS)



TYPE 3S2 UNIT WEIGHT = 72 KIPS (36 TONS)



TYPE 3-3 UNIT WEIGHT = 80 KIPS (40 TONS)

Figure 6B.7.2-1—Typical Legal Loads Used for Posting

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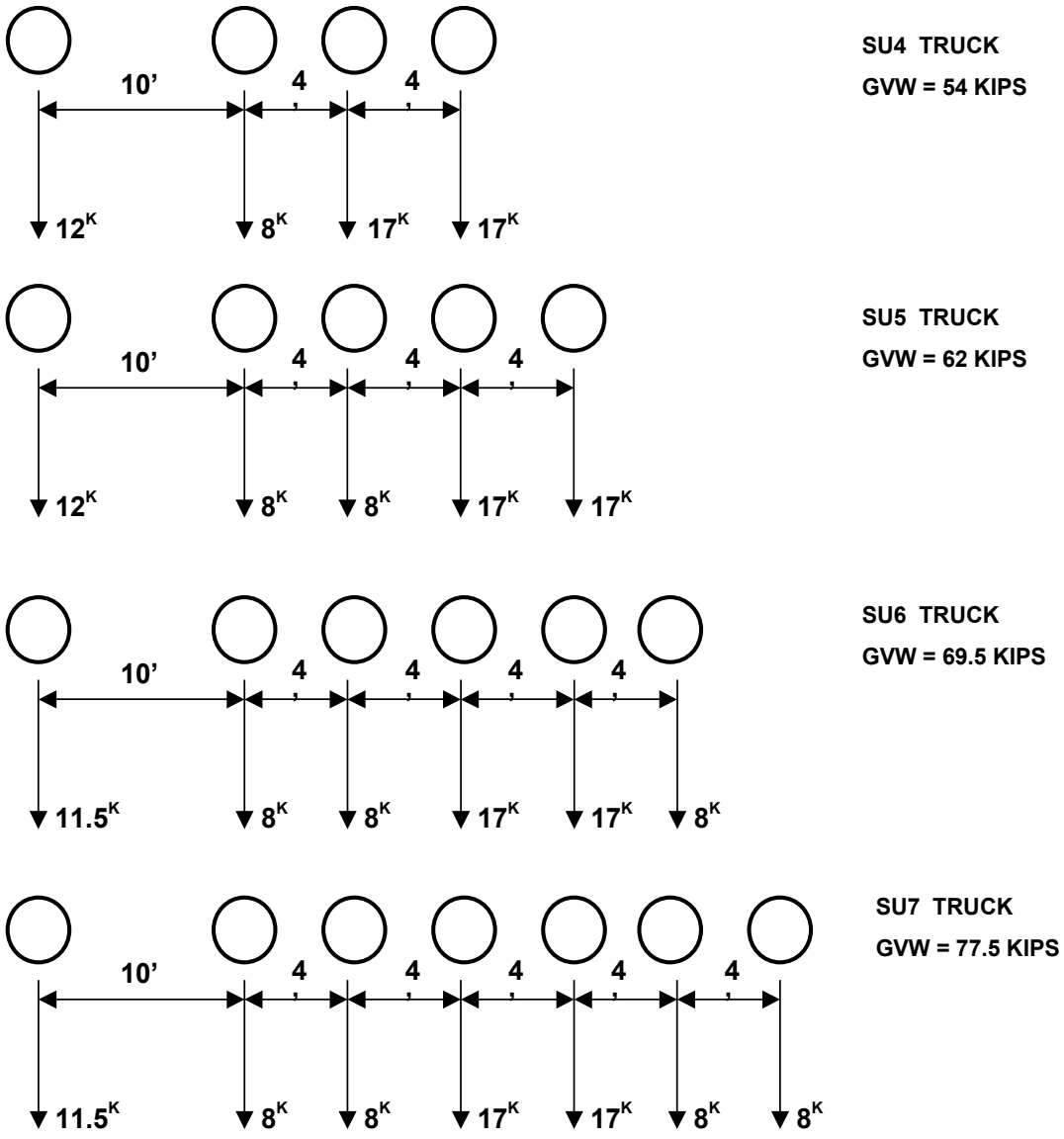

Figure 6B.7.2-2—Bridge Posting Loads for Single Unit Trucks that Meet Formula B

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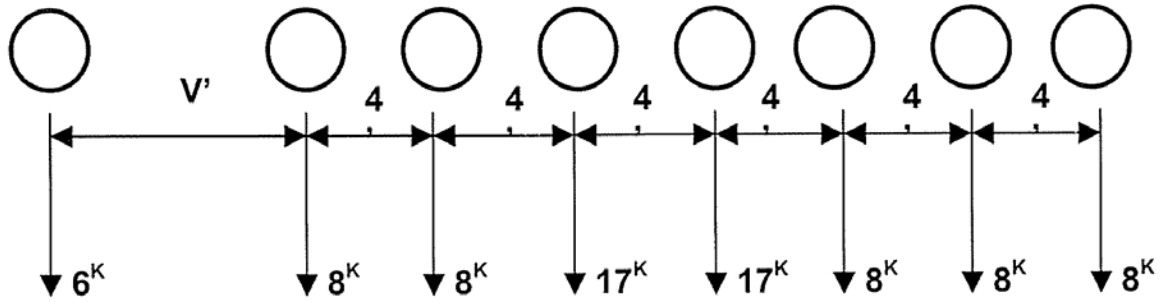

V = VARIABLE DRIVE AXLE SPACING — 6'0" TO 14'-0". SPACING TO BE USED IS THAT WHICH PRODUCES MAXIMUM LOAD EFFECTS.

AXLES THAT DO NOT CONTRIBUTE TO THE MAXIMUM LOAD EFFECT UNDER CONSIDERATION SHALL BE NEGLECTED.

MAXIMUM GVW = 80 KIPS

AXLE GAGE WIDTH = 6'-0"

Figure 6B.7.2-3—Notional Rating Load (NRL) for Single-Unit SHVs that Meet Federal Bridge Formula

6B.7.3—Posting Analysis

The determination of the need to load-post a bridge should be made by the Bridge Owner based on the general procedures in Section 6, Part B, and established practices of the Bridge Owner. When the maximum legal load under state law exceeds the safe load capacity of a bridge calculated at the operating level, restrictive posting shall be required.

6B.7.4—Regulatory Signs

Regulatory signing shall conform to the requirements of the *Manual on Uniform Traffic Control Devices* (MUTCD) or other governing regulations, and shall be established in accordance with the requirements of the agency having authority over the highway.

When a decision is made to close a bridge, signs and properly designed, structurally sound traffic barriers shall be erected to provide adequate warning and protection to the traveling public. If pedestrian travel across the bridge is also restricted, adequate measures to prevent pedestrian use of the bridge shall be installed. Signs and barriers shall meet or exceed the requirements of local laws and the applicable sections of the MUTCD. Bridge closure signs and barriers shall be inspected periodically to ensure their continued effectiveness.

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6B.7.5—Speed Limits

In some cases, lower speed limits will reduce impact loads to the extent that lowering the weight limit may not be required. Consideration of a speed posting will depend upon alignment, general location, volume, and type of traffic. A speed posting should not be considered as a basis for increasing the weight limit in areas where enforcement will be difficult and frequent violations can be anticipated.

6B.8—PERMITS

6B.8.1—General

Bridge Owners usually have established procedures which allow oversized/overweight vehicles to travel on the highway system. These procedures involve the issuance of a permit which describes the features of the vehicle and/or its load and, in most jurisdictions, will specify the allowable route or routes of travel. Generally speaking, permits should not be approved in situations where the load or the hauling vehicle can be reduced to conform to the size and weight limitations of local regulations.

Most Bridge Owners have methods for checking bridges to determine the effects which would be caused by the passage of vehicles above the legally established weight limitations. One approach is to check permit vehicles by the general methods of Section 6, Part B.

The live load to be used in the rating Eq. 6B.4.1-1 for permit decisions should be the actual vehicle size, weight, and type using the highway, together with an impact factor dependent on local conditions. The actual loading used may vary from time to time and from state to state in accordance with local laws and regulations.

The operating level may be used for evaluating special permits for heavier than normal vehicles. Bridges which have members theoretically stressed to near the operating level stress should be inspected more frequently than other structures.

6B.8.2—Routine Permits

Routine permit vehicles are expected to mix in the random traffic stream and move at normal times and speeds. The maximum load effects of all permit vehicles allowed to move on a routine basis should be evaluated. The structural component with the lowest permit load rating on the route system should determine whether a vehicle should be issued a permit.

For routine permits, it is usually necessary to calculate load effects by assuming that a permit vehicle may occur on the bridge alongside another heavy vehicle.

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6B.8.3—Controlled Permits

Special or controlled permits are usually valid for a single trip only. These permit vehicles are usually heavier than those vehicles issued routine permits for unlimited trips. Depending on the authorization, these special vehicles may be allowed to mix with random traffic or may be required to be escorted in a manner which controls speed, lane position, or both.

6B.8.4—Escorted Permits

If a special permit vehicle is escorted, then the loading for that permit vehicle may be applied in a designated lane position. Impact values may be reduced if speed control is ensured. If the escort control is able to ensure that no other trucks will be on the bridge simultaneously with the permit vehicle, then other live loads need not be applied.

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APPENDIX A6B

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APPENDIX B6B

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APPENDIX C6B—LIVE LOAD MOMENTS ON LONGITUDINAL STRINGERS OR GIRDERS

Table C6B-1—Live Load Moments on Longitudinal Stringers or Girders for Routine Commercial Traffic

Live Load Moments in ft-kips per Wheel Line										
Type of Loading (without Impact)					Span,	Type of Loading (with Impact)				
H-15	**HS-20**	3	3S2	3-3	ft c/c	H-15	**HS-20**	3	3S2	3-3
15.0	20.0	10.6	9.7	10.0	5	19.5	26.0	13.8	12.6	13.0
18.0	24.0	12.8	11.6	12.0	6	23.4	31.2	16.6	15.1	15.6
21.0	28.0	15.2	13.8	14.0	7	27.3	36.4	19.7	18.0	18.2
24.0	32.0	19.1	17.4	16.0	8	31.2	41.6	24.9	22.7	20.8
27.0	36.0	23.1	21.1	19.1	9	35.1	46.8	30.1	27.4	24.8
30.0	40.0	27.2	24.8	22.4	10	39.0	52.0	35.4	32.2	29.1
33.0	44.0	31.3	28.5	25.8	11	42.9	57.2	40.7	37.1	33.5
36.0	48.0	35.4	32.2	29.2	12	46.8	62.4	46.0	42.0	37.9
39.0	52.0	39.6	36.1	32.6	13	50.7	67.6	51.4	46.9	42.3
42.0	56.0	43.7	39.9	36.0	14	54.6	72.8	56.8	51.8	46.8
45.0	60.0	47.9	43.7	39.4	15	58.5	78.0	62.2	56.8	51.3
48.0	64.0	52.1	47.5	42.9	16	62.4	83.2	67.7	61.7	55.7
51.0	68.0	56.3	51.3	46.3	17	66.3	88.4	73.1	66.7	60.2
54.0	72.0	60.4	55.1	49.8	18	70.2	93.6	78.6	71.6	64.7
57.0	76.0	64.6	58.9	53.2	19	74.1	98.8	84.0	76.6	69.2
60.0	80.0	68.9	62.8	56.7	20	78.0	104.0	89.5	81.6	73.7
63.0	84.0	73.1	66.6	60.2	21	81.9	109.2	95.0	86.6	78.2
66.0	88.0	77.3	70.5	63.6	22	85.8	114.4	100.5	91.6	82.7
69.0	92.0	81.5	75.2	67.1	23	89.7	119.6	105.9	97.7	87.2
72.0	96.3	85.7	80.3	70.6	24	93.6	125.2	111.4	104.4	91.8
75.0	103.7	89.9	85.4	74.1	25	97.5	134.8	116.9	111.0	96.3
78.0	111.1	94.2	90.5	77.5	26	101.4	144.4	122.4	117.7	100.8
81.3	118.5	98.4	95.6	81.0	27	105.7	154.1	127.9	124.3	105.3
85.1	126.0	102.6	100.7	84.5	28	110.6	163.8	133.4	131.0	109.8
88.8	133.5	106.8	105.9	88.0	29	115.4	173.6	138.9	137.6	114.4
92.5	141.0	112.9	111.0	91.5	30	120.2	183.3	146.8	144.3	118.9
99.8	156.2	125.3	121.2	101.5	32	130.0	203.1	162.9	157.6	132.0
107.4	171.8	137.6	131.5	112.3	34	139.6	223.3	178.9	170.9	146.0
114.8	189.4	150.0	141.7	123.1	36	149.2	246.2	195.0	184.2	160.1
122.3	207.1	162.4	151.9	134.0	38	159.0	269.2	211.1	197.5	174.1
129.7	224.9	174.8	162.2	144.8	40	168.6	292.4	227.3	210.8	188.3
137.2	242.7	187.2	172.4	155.7	42	178.3	315.3	243.3	224.0	202.3
144.7	260.4	199.7	182.7	166.6	44	187.5	337.5	258.7	236.7	215.8
152.1	278.3	212.1	192.9	177.4	46	196.6	359.6	274.1	249.3	229.3
159.6	296.1	224.5	203.2	188.3	48	205.7	381.7	289.4	261.9	242.8
167.1	314.0	237.0	220.8	199.3	50	214.8	403.8	304.7	283.9	256.2
174.6	331.8	249.4	238.4	214.3	52	223.9	425.5	319.9	305.8	274.8
182.0	349.7	261.8	256.1	231.3	54	232.8	447.3	335.0	327.6	295.9
189.5	367.6	274.3	273.8	248.3	56	241.8	469.1	350.1	349.4	316.9
198.8	385.4	286.8	291.4	265.3	58	253.1	490.6	365.1	371.1	337.7
209.2*	403.3	299.2	309.2	282.3	60	265.8*	512.2	380.1	392.7	358.5
265.1*	492.8	361.5	398.0	372.2	70	333.1*	619.0	454.2	500.1	467.6
327.0*	582.4	423.9	487.1	471.9	80	406.8*	724.5	527.3	605.9	587.0
394.9*	672.2	486.3	576.4	571.7	90	486.7*	828.8	599.4	710.5	704.6
468.8*	762.0	548.7	665.9	671.5	100	572.9*	931.2	670.7	813.9	820.7
634.5*	941.6	673.6	845.1	871.3	120	764.0*	1,133.7	811.1	1,017.5	1,049.1
824.2*	1,121.4	798.5	1,024.5	1,071.1	140	979.8*	1,333.3	949.2	1,217.8	1,273.2
1,038.0*	1,384.0*	923.5	1,204.1	1,270.9	160	1,220.1*	1,626.2*	1,085.5	1,415.3	1,493.9
1,275.8*	1,701.0*	1,048.4	1,383.7	1,470.8	180	1,484.9*	1,980.0*	1,222.3	1,610.6	1,712.0
1,537.5*	2,050.0*	1,173.4	1,563.5	1,670.8	200	1,774.0*	2,365.7*	1,353.9	1,804.0	1,927.8
2,296.9*	3,062.5*	1,485.8	2,013.0	2,170.6	250	2,603.1*	3,469.8*	1,683.9	2,281.4	2,460.0
3,206.2*	4,275.0*	1,798.2	2,462.6	2,670.5	300	3,583.5*	4,779.4*	2,009.8	2,752.4	2,984.7

* Based on standard lane loading. All other values are based on standard truck loading.

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Table C6B-2—Live Load Moments on Longitudinal Stringers or Girders for Specialized Hauling Vehicles

Live Load Moments in ft-kips per Wheel Line												
Type of Loading (without Impact)						Span, ft c/c	Type of Loading (with Impact)					
HS-20	NRL	SU4	SU5	SU6	SU7		HS-20	NRL	SU4	SU5	SU6	SU7
20.0	10.6	10.6	10.6	10.6	10.6	5	26.0	13.8	13.8	13.8	13.8	13.8
24.0	12.8	12.8	12.8	12.8	12.8	6	31.2	16.6	16.6	16.6	16.6	16.6
28.0	15.2	15.2	15.2	15.2	15.2	7	36.4	19.8	19.8	19.8	19.8	19.8
32.0	19.1	19.1	19.1	19.1	19.1	8	41.6	24.8	24.8	24.8	24.8	24.8
36.0	23.1	23.1	23.1	23.1	23.1	9	46.8	30.0	30.0	30.0	30.0	30.0
40.0	27.9	27.9	27.9	27.9	27.9	10	52.0	36.3	36.3	36.3	36.3	36.3
44.0	33.1	33.1	33.1	33.1	33.1	11	57.2	43.0	43.0	43.0	43.0	43.0
48.0	38.3	38.3	38.3	38.3	38.3	12	62.4	49.8	49.8	49.8	49.8	49.8
52.0	43.5	43.5	43.5	43.5	43.5	13	67.6	56.6	56.6	56.6	56.6	56.6
56.0	48.8	48.8	48.8	48.8	48.8	14	72.8	63.4	63.4	63.4	63.4	63.4
60.0	54.4	54.0	54.0	54.4	54.4	15	78.0	70.7	70.2	70.2	70.7	70.7
64.0	60.6	59.2	59.2	60.6	60.6	16	83.2	78.8	77.0	77.0	78.8	78.8
68.0	66.7	64.5	65.3	66.7	66.7	17	88.4	86.7	83.9	84.9	86.7	86.7
72.0	73.6	69.7	71.5	73.6	73.6	18	93.6	95.7	90.6	93.0	95.7	95.7
76.0	80.8	74.9	77.8	80.8	80.8	19	98.8	105.0	97.4	101.1	105.0	105.0
80.0	88.1	80.2	84.0	88.1	88.1	20	104.0	114.5	104.3	109.2	114.5	114.5
84.0	95.3	85.4	90.3	95.3	95.3	21	109.2	123.9	111.0	117.4	123.9	123.9
88.0	102.6	90.7	96.5	102.6	102.6	22	114.4	133.4	117.9	125.5	133.4	133.4
92.0	110.2	95.9	102.8	109.8	110.2	23	119.6	143.3	124.7	133.6	142.7	143.3
96.3	118.4	101.2	109.0	117.1	118.4	24	125.2	153.9	131.6	141.7	152.2	153.9
103.7	126.6	106.4	115.3	124.3	126.6	25	134.8	164.5	138.3	149.9	161.6	164.5
111.1	135.5	111.6	121.5	131.6	134.8	26	144.4	176.2	145.1	158.0	171.1	175.2
118.5	144.8	116.9	127.8	138.8	143.0	27	154.1	188.2	152.0	166.1	180.4	185.9
126.0	154.0	123.4	134.0	146.1	151.2	28	163.8	200.2	160.4	174.2	189.9	196.6
133.5	163.3	130.1	140.3	153.3	159.4	29	173.6	212.3	169.1	182.4	199.3	207.2
141.0	172.5	136.8	146.5	160.6	167.6	30	183.3	224.3	177.8	190.5	208.7	217.9
156.2	191.0	150.2	159.0	175.1	184.0	32	203.1	248.3	195.3	206.7	227.6	239.2
171.8	209.5	163.6	172.0	189.6	200.5	34	223.3	272.4	212.7	223.6	246.5	260.7
189.4	228.9	177.1	187.3	205.4	216.9	36	246.2	297.6	230.2	243.5	267.0	282.0
207.1	248.8	190.5	202.7	222.7	235.6	38	269.2	323.4	247.7	263.5	289.5	306.3
224.9	268.8	204.0	218.0	240.0	255.0	40	292.4	349.4	265.1	283.4	312.0	331.5
242.7	288.8	217.4	233.4	257.3	274.3	42	315.4	375.3	282.5	303.3	334.3	356.4
260.4	308.7	230.9	248.7	274.7	293.7	44	337.4	400.0	299.2	322.3	356.0	380.6
278.3	328.7	244.3	264.1	292.0	313.1	46	359.7	424.8	315.7	341.3	377.4	404.6
296.1	348.7	257.8	279.5	309.3	332.4	48	381.7	449.5	332.3	360.3	398.7	428.5
314.0	368.7	271.3	294.9	326.6	351.8	50	403.7	474.0	348.8	379.2	419.9	452.3
331.8	388.6	284.8	310.3	344.0	371.2	52	425.5	498.4	365.3	398.0	441.2	476.1
349.7	408.6	298.2	325.7	361.3	390.5	54	447.4	522.7	381.5	416.7	462.2	499.6
367.6	428.6	311.7	341.1	378.7	409.9	56	469.1	547.0	397.8	435.3	483.3	523.1
385.4	448.6	325.2	356.6	396.0	429.3	58	490.7	571.2	414.1	454.0	504.2	546.6
403.3	468.5	338.7	372.0	413.3	448.7	60	512.2	595.1	430.2	472.5	525.0	569.9
492.8	568.5	406.1	449.2	500.1	545.5	70	619.2	714.2	510.2	564.4	628.3	685.4
582.5	668.4	473.5	526.5	586.9	642.4	80	724.5	831.4	589.0	654.9	730.0	799.0
672.2	768.4	540.9	603.8	673.7	739.2	90	828.5	947.0	666.7	744.2	830.4	911.1
762.0	868.3	608.4	681.2	760.5	836.1	100	931.3	1,061.3	743.6	832.6	929.5	1,021.9
941.6	1,068.3	743.3	836.0	934.2	1,029.8	120	1,133.8	1,286.3	895.0	1,006.6	1,124.8	1,240.0
1,121.4	1,268.2	878.3	990.9	1,107.9	1,223.6	140	1,333.0	1,507.5	1,044.0	1,177.8	1,316.9	1,454.4
1,384.0*	1,468.2	1,013.2	1,145.8	1,281.6	1,417.3	160	1,626.8*	1,725.8	1,191.0	1,346.8	1,506.4	1,665.9
1,701.0*	1,668.2	1,148.2	1,300.7	1,455.3	1,611.1	180	1,979.9*	1,941.7	1,336.4	1,513.9	1,693.9	1,875.2
2,050.0*	1,868.2	1,283.2	1,455.6	1,629.0	1,804.8	200	2,365.4*	2,155.6	1,480.6	1,679.5	1,879.6	2,082.5
3,062.5*	2,368.1	1,620.7	1,843.0	2,063.3	2,289.2	250	3,470.8*	2,683.8	1,836.8	2,088.7	2,338.4	2,594.4
4,275.0*	2,868.1	1,958.1	2,230.4	2,497.7	2,773.5	300	4,777.9*	3,205.5	2,188.5	2,492.8	2,791.5	3,099.8

* Based on standard loading. All other values based on standard truck loading.

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APPENDIX D6B—STRINGER LIVE LOAD REACTIONS ON TRANSVERSE FLOOR BEAMS AND CAPS (INTERMEDIATE TRANSVERSE BEAMS) (SIMPLE SPAN ONLY)

Table D6B-1—Live Load Reactions, *R*, in kips per Wheel Line, No Impact, for Routine Commercial Traffic

Stringer Span, ft	Live Load Reactions, *R*, in kips per Wheel Line, No Impact				
	Type of Loading				
	Type 3	Type 3S2	Type 3-3	H-15	**HS-20**
10	13.6	12.4	11.2	12.0	16.0
11	13.9	12.7	11.5	12.0	16.0
12	14.2	13.1	11.7	12.0	16.0
13	14.4	13.7	11.9	12.0	16.0
14	14.6	14.2	12.0	12.0	16.0
15	14.8	14.6	12.2	12.2	17.3
16	15.3	15.0	12.3	12.4	18.5
17	15.8	15.4	12.7	12.5	19.5
18	16.4	15.6	13.3	12.7	20.4
19	16.8	15.9	13.7	12.8	21.3
20	17.2	16.1	14.2	12.9	22.0
21	17.6	16.3	14.5	13.0	22.7
22	18.0	16.5	14.9	13.1	23.3
23	18.3	16.7	15.2	13.2	23.8
24	18.5	16.9	15.5	13.3	24.3
25	18.8	17.0	15.7	13.4	24.8
26	19.0	17.5	16.2	13.4	25.2
27	19.3	18.2	16.8	13.5	25.6
28	19.5	18.8	17.5	13.5	26.0
29	19.7	19.4	18.0	13.6	26.3
30	19.9	20.1	18.8	13.6	26.7

One-Lane Loading $M = \dfrac{(L-3)^2 R}{2L}$

*Two-Lane Roadway over 18 ft $M = \left(L - 9 + \dfrac{2.25}{L}\right) R$

*Wheel Line/Truss: $\begin{cases} \text{One-Lane Loading} = \left(1 + \dfrac{W-9}{C}\right) \\ \text{Two-Lane Loading} = \left(1 + \dfrac{W-18}{C}\right) 2 \end{cases}$

where:

M = Moment in transverse beam

R = Reaction (tabular value)

L = Span of transverse beam

W = Width of roadway

C = Spacing, center-to-center of trusses

All values based on standard truck loadings.

* Based on 9-ft lane width.

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Table D6B-2—Live Load Reactions, *R*, in kips per Wheel Line, No Impact, for Specialized Hauling Vehicles

	Live Load Reactions, *R*, in kips per Wheel Line, No Impact					
	Type of Loading					
Stringer Span, ft	SU4	SU5	SU6	SU7	NRL	HS-20
10	16.0	16.8	17.6	17.6	17.6	16.0
11	16.5	17.5	18.6	18.6	18.6	16.0
12	16.8	18.2	19.5	19.5	19.5	16.0
13	17.2	18.7	20.2	20.5	20.2	16.0
14	17.4	19.1	20.9	21.4	20.9	16.0
15	18.1	19.5	21.4	22.2	21.4	17.3
16	18.6	19.9	21.9	22.9	21.9	18.5
17	19.1	20.2	22.3	23.5	22.3	19.5
18	19.6	20.4	22.7	24.0	23.0	20.4
19	19.9	21.0	23.3	24.8	23.7	21.3
20	20.3	21.5	23.9	25.5	24.3	22.0
21	20.6	22.0	24.4	26.1	24.8	22.7
22	20.9	22.4	24.9	26.7	25.3	23.3
23	21.2	22.7	25.3	27.2	26.0	23.8
24	21.4	23.1	25.7	27.7	26.6	24.3
25	21.6	23.4	26.1	28.1	27.1	24.8
26	21.8	23.7	26.4	28.5	27.6	25.2
27	22.0	24.0	26.7	28.9	28.1	25.6
28	22.2	24.2	27.0	29.3	28.5	26.0
29	22.4	24.4	27.3	29.6	28.9	26.3
30	22.5	24.7	27.5	29.9	29.3	26.7

All values based on standard truck loadings.

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APPENDIX E6B—STRINGER LIVE LOAD REACTIONS ON TRANSVERSE FLOOR BEAMS AND CAPS (END TRANSVERSE BEAMS) (SIMPLE SPAN ONLY)

Table E6B-1—Live Load Reactions, *R*, in kips per Wheel Line, No Impact, for Routine Commercial Traffic

	Live Load Reactions, *R*, in kips per Wheel Line, No Impact				
	Type of Loading				
Stringer Span, ft	Type 3	Type 3S2	Type 3-3	H-15	**HS-20**
10	13.6	12.4	11.2	12.0	16.0
11	13.9	12.7	11.5	12.0	16.0
12	14.2	12.9	11.7	12.0	16.0
13	14.4	13.1	11.9	12.0	16.0
14	14.6	13.3	12.0	12.0	16.0
15	14.7	13.4	12.1	12.2	17.1
16	14.9	13.9	12.3	12.4	18.0
17	15.0	14.3	12.4	12.5	18.9
18	15.1	14.6	12.4	12.7	19.6
19	15.2	14.9	12.5	12.8	20.2
20	15.7	15.2	12.6	12.9	20.8
21	16.1	15.5	13.1	13.0	21.3
22	16.6	15.7	13.5	13.1	21.8
23	16.9	15.9	13.8	13.2	22.2
24	17.3	16.1	14.2	13.3	22.6
25	17.6	16.3	14.5	13.4	23.0
26	17.9	16.4	14.8	13.4	23.4
27	18.1	16.6	15.0	13.5	23.7
28	18.4	16.7	15.3	13.5	24.0
29	18.6	16.8	15.5	13.6	24.4
30	18.8	17.0	15.7	13.6	24.8

All values based on standard truck loadings.

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Table E6B-2—Live Load Reactions, *R*, in kips per Wheel Line, No Impact, for Specialized Hauling Vehicles

Stringer Span, ft	Live Load Reactions, *R*, in kips per Wheel Line, No Impact Type of Loading					
	SU4	SU5	SU6	SU7	NRL	HS-20
10	14.4	14.4	14.4	14.4	14.4	16.0
11	14.9	15.0	15.0	15.0	14.9	16.0
12	15.5	15.4	15.5	15.5	15.5	16.0
13	15.8	16.0	16.1	16.1	16.1	16.0
14	16.2	16.6	16.7	16.7	16.9	16.0
15	16.6	17.3	17.3	17.4	17.4	17.1
16	16.8	17.8	17.8	17.8	17.8	18.0
17	17.1	18.1	18.2	18.2	18.4	18.9
18	17.1	18.5	18.5	18.7	18.9	19.6
19	17.8	18.9	19.0	19.0	19.4	20.2
20	18.0	19.3	19.3	19.3	20.1	20.8
21	18.6	19.5	19.5	19.5	20.3	21.3
22	19.0	19.6	19.6	19.6	20.7	21.8
23	19.2	20.2	20.2	20.2	21.0	22.2
24	19.8	20.6	20.7	20.7	21.5	22.6
25	20.0	21.2	21.2	21.2	21.9	23.0
26	20.1	21.4	21.4	21.4	21.9	23.4
27	17.2	21.7	21.7	21.7	22.2	23.7
28	20.6	22.1	22.1	22.2	22.6	24.0
29	20.7	22.5	22.4	22.4	22.8	24.4
30	21.2	22.6	22.6	22.6	23.1	24.8

All values based on standard truck loadings.

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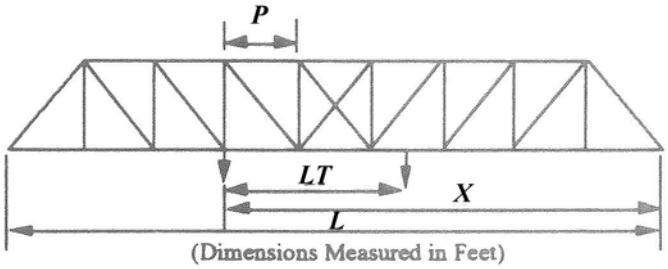
APPENDIX F6B—FORMULAS FOR MAXIMUM SHEAR[a] AT ANY PANEL POINT (NO IMPACT INCLUDED) (SIMPLE SPAN ONLY)

Type Load[b]	*LT*	Min. *X*	Formula	Use for Truss with No. Panels	(1)	(2)
3	19 ft	19 ft	$V = \frac{25(X - 7.44)}{L}$	All	3	Rt
3S2	41 ft	41 ft	$V = \frac{36(X - 18.61)}{L}$	5 or more	5	Rt
		30 ft	$V = \frac{36(X - 11.39)}{L} - \frac{55}{P}$	3, 4	2	Lt
		26 ft	$V = \frac{36(X - 7.39)}{L} - \frac{106}{P}$	2	3	Lt
3-3	54 ft	54 ft	$V = \frac{40(X - 23.9)}{L}$	6 or more	6	Rt
		50 ft	$V = \frac{40(X - 19.9)}{L} - \frac{28}{P}$	4, 5	5	Rt
		35 ft	$V = \frac{40(X - 11.1)}{L} - \frac{138}{P}$	3	3	Lt
		34 ft	$V = \frac{40(X - 3.9)}{L} - \frac{252}{P}$	2	4	Rt



where:

L = Length of truss

LT = Length of truck

P = Length of panel

X = Distance from panel point to end of truss

V = Shear at panel point in kips per wheel line

(1) = Axle no. at panel point

(2) = Truck facing

[a] Applicable when entire truck is on span.

[b] See Appendix H6B for shear resulting from H and HS load types.

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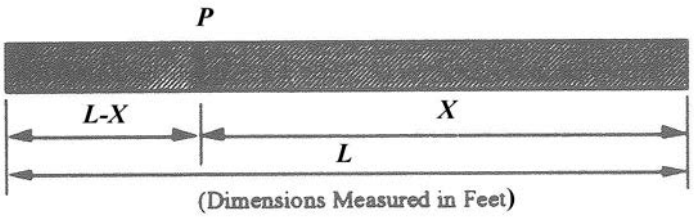
APPENDIX G6B—FORMULAS FOR MAXIMUM SHEAR AT ANY POINT ON SPAN (NO IMPACT INCLUDED) (SIMPLE SPANS ONLY)

Type Load	$\frac{L-X}{L}$	Formula for Maximum Shear[a]	Length of Truck	Minimum	
				$L-X$	X [b]
3	0–0.500	$V=\frac{25(X-7.44)}{L}$	19 ft	0	19 ft
3S2	0–0.500	$V=\frac{36(X-18.61)}{L}$	41 ft	0	41 ft
3-3	0–0.500	$V=\frac{40(X-23.90)}{L}$	54 ft	0	54 ft



where:

V = Shear at a point P which is $L-X$ distance from end of span in kips per wheel line

[a] These formulas are applicable only when dimension X exceeds total length of truck.

[b] For spans where dimension X is less than the minimum, the maximum shears are to be determined from statics.

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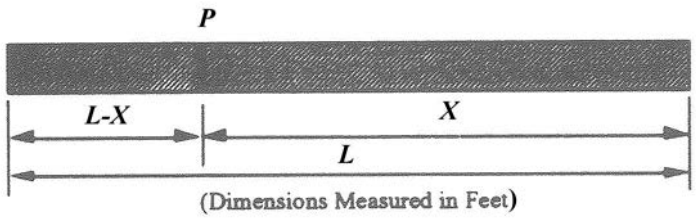
APPENDIX H6B—FORMULAS FOR MAXIMUM SHEAR AT ANY POINT ON SPAN (NO IMPACT INCLUDED) (SIMPLE SPANS ONLY)

Type Load	$\frac{L-X}{L}$	Use for Girder Lengths	Formula for Maximum Shear[a]	Minimum	
				$L-X$	X
HS-20	0–0.500	Under 42 ft	$V=\frac{36(X-4.67)}{L}-4$	14	14
		42 ft to 120 ft[b]	$V=\frac{36(X-9.33)}{L}$	0	28
HS-15	0–0.500	Under 42 ft	$V=\frac{27(X-4.67)}{L}-3$	14	14
		42 ft to 120 ft[b]	$V=\frac{27(X-9.33)}{L}$	0	28
H-20	0–0.500	To 35 ft[b]	$V=\frac{20(X-2.8)}{L}$	0	14
H-15	0–0.500	To 35 ft[b]	$V=\frac{15(X-2.8)}{L}$	0	14

[a] All values based on standard truck loadings

[b] Truck loading does *not* govern shear beyond the lengths specified; use lane loading.



where:

V = Shear to left of point P in kips per wheel line

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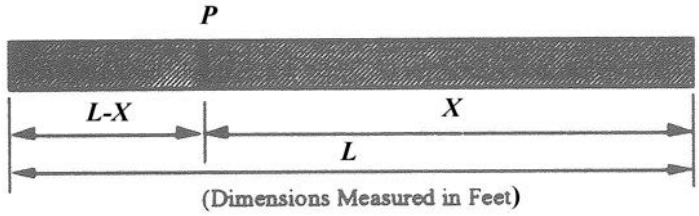
APPENDIX I6B—FORMULAS FOR MOMENT SHEAR AT ANY POINT ON SPAN (NO IMPACT INCLUDED) (SIMPLE SPANS ONLY)

Type Load	$\frac{L-X}{L}$	Formula for Maximum Moment at P	Minimum $L-X$	Minimum X	(1)	(2)
3	0–0.340	$25(X-7.44)\frac{(L-X)}{L}$	0	19.0	3	Rt
	0.340–0.500	$25(X-3.44)\frac{(L-X)}{L}-34$	4.0	15.0	2	Rt
3S2	0–0.211	$36(X-18.61)\frac{(L-X)}{L}$	0	41.0	5	Rt
	0.211–0.354	$36(X-11.39)\frac{L-X}{L}-55$	11.0	30.0	2	Lt
	0.354–0.500	$36(X-7.39)\frac{(L-X)}{L}-106$	15.0	26.0	3	Lt
3-3	0–0.175	$40(X-23.9)\frac{(L-X)}{L}$	0	54.0	6	Rt
	0.175–0.3125	$40(X-19.9)\frac{(L-X)}{L}-28$	4.0	50.0	5	Rt
	0.3125–0.396	$40(X-11.10)\frac{(L-X)}{L}-138$	19.0	35.0	3	Lt
	0.396–0.500	$40(X-3.9)\frac{(L-X)}{L}-252$	20.0	34.0	4	Rt

(1) Axle No. at P

(2) Truck facing



Moments in ft-kips per wheel line at a distance $L - X$ from end of span.

Formulas are applicable when entire truck is on span.

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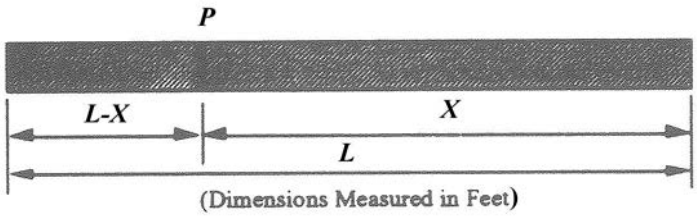
APPENDIX J6B—FORMULAS FOR MAXIMUM MOMENT AT ANY POINT ON SPAN (NO IMPACT INCLUDED) (SIMPLE SPANS ONLY)

Type Load	$\frac{L-X}{L}$	Formula for Maximum Moment at P	Minimum		Max L[a]
			$L-X$	X	
HS-20	0–0.333	$\frac{36(L-X)(X-9.33)}{L}$	0	28	—
	0.333–0.500	$\frac{36(L-X)(X-4.67)}{L}-56$	14	14	144.5
HS-15	0–0.333	$\frac{27(L-X)(X-9.33)}{L}$	0	28	—
	0.333–0.500	$\frac{27(L-X)(X-4.67)}{L}-42$	14	14	144.5
H-20	0–0.500	$\frac{20(L-X)(X-2.8)}{L}$	0	14	56
H-15	0–0.500	$\frac{15(L-X)(X-2.8)}{L}$	0	14	56



Moments in ft-kips per wheel line.

These formulas are applicable when all loads are on the span

[a] Span lengths greater than this value are controlled by lane loading

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APPENDIX K6B—FORMULAS FOR STEEL COLUMNS[a]

The allowable combined stresses for steel compression members may be calculated either by the provisions of AASHTO Standard Specifications or from the following relationship. The permissible average unit stress for steel columns shall be:

$$f_s = \frac{\dfrac{f_y}{\eta}}{1 + \left(0.25 + \dfrac{e_g c}{r^2}\right) B \text{ cosec } \Phi} = \frac{P}{A} \qquad \text{(K6B-1)}^{b}$$

P = load parallel to the axis of the member (lb)

A = gross cross-sectional area of column (in.^{2})

f_y = yield point or yield strength (see Tables 6B.5.2.1-1 and 6B.5.2.1-2)

η = factor of safety based on yield point or yield strength

= 1.82 at inventory level

= 1.48 at operating level

c = distance from neutral axis to the extreme fiber in compression

r = radius of gyration in the plane of bending

$\Phi = \dfrac{L}{r}\sqrt{\dfrac{\eta f_s}{E}}$ (rad)

L = effective length of the columns

= 75 percent of the total length of a column having riveted end connections

= 87.5 percent of the total length of a column having pinned end connections

E = modulus of elasticity of steel

= 29,000,000 lb/in.^{2}

$B = \sqrt{\alpha^2 - 2\alpha \cos \Phi + 1}$

$$\alpha = \frac{\dfrac{e_s c}{r^2} + 0.25}{\dfrac{e_g c}{r^2} + 0.25}$$

When e_g and e_s lie on the same side of the column axis, α is positive; when on opposite sides, α is negative.

e_g = eccentricity of applied load at the end of column having the greater computed moment (in.)

e_s = eccentricity at opposite end

[a] Refer also to the column formulas given in Tables 6B.5.2.1-1 and 6B.5.2.1-2.

[b] When the radius of gyration perpendicular to the plane of bending is less than r, the column shall be investigated for the case of a long column concentrically loaded, having a greater value of L/r.

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For values of $^{L}/_{r}$ equal to or less than:

$$\left(\cos^{-1}\alpha\right)\left[\frac{E\left(1+0.25+\frac{e_g c}{r^2}\right)}{f_y}\right]^{\frac{1}{2}} \tag{K6B-2}$$

the permissible f_s shall be determined from the formula:

$$f_s = \frac{\frac{f_y}{\eta}}{1+0.25+\frac{e_g c}{r^2}} \tag{K6B-3}$$

For $\alpha = -1$ with values of $^{L}/_{r}$ greater than determined by Formula B, the permissible f_s shall be determined by the Euler formula:

$$f_s = \frac{\pi^2 E}{\eta\left(\frac{L}{r}\right)^2} \tag{K6B-4}$$

When the values of end moments are not computed but considered negligible in amount, α shall be assumed equal to +1.

α shall be assumed equal to +1 for a member subject to bending stresses induced by the components of externally applied loads acting perpendicular to its axis. For this case, the formula becomes:

$$f_s = \frac{\frac{f_y}{\eta} - \frac{M_c}{I}}{1+\left[0.25+\left(e_g+d\right)\frac{c}{r^2}\right]\sec\frac{1}{2}\Phi} \tag{K6B-5}$$

d = deflection due to transverse components of externally applied loads (in.)

I = moment of inertia of section about an axis perpendicular to the plane of bending (in.^{4})

M = moment due to the transverse components of externally applied load (in.-lb)

Note: The value of 0.25 in the above formulas provides for inherent crookedness and unknown eccentricity.

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APPENDIX L6B—FORMULAS FOR THE CAPACITY, *C*, OF TYPICAL BRIDGE COMPONENTS BASED ON THE LOAD FACTOR METHOD

L6B.1—GENERAL

When using the load factor method, the capacity, *C*, in the basic load rating Eq. 6B.4.1-1 is based on procedures in the latest edition of AASHTO's *Standard Specifications for Highway Bridges* (AASHTO Standard Specifications). This Appendix summarizes the capacity determination for typical bridge members of steel, reinforced concrete, or prestressed concrete. For more conditions not covered in this Appendix, the AASHTO Standard Specifications should be used.

The formulas shown below have been taken from the AASHTO Standard Specifications. All equation and article numbers cited below refer to this Specification, except for a few article numbers that refer to Appendix L6B itself. The notation used in the formulas is as defined in the AASHTO Standard Specifications.

L6B.2—CAPACITY OF STEEL MEMBERS (PART D, STRENGTH DESIGN METHOD)

L6B.2.1—Sections in Bending

The capacities specified in L6B.2.1.1 and L6B.2.1.2 are applicable to compact rolled or welded beams and girders, satisfying the applicable cross-sectional limitations, which are rolled or fabricated from steels with a specified minimum yield strength between 33,000 and 50,000 psi. The capacities specified in L6B.2.1.3 through L6B.2.1.5 are applicable to noncompact rolled, riveted, or welded beams and girders satisfying the applicable cross-sectional limitations, which are rolled or fabricated from steels with a minimum specified yield strength between 33,000 and 100,000 psi. The equations found in L6B.2.1.1 through L6B.2.1.5 are not applicable to hybrid girders.

L6B.2.1.1—Compact, Braced, Noncomposite

$$C = F_y Z \tag{10-92}$$

L6B.2.1.2—Compact, Composite

Positive Moment Sections

For composite positive moment sections satisfying the cross-sectional limitations specified in Article 10.50.1.1.2:

In simple spans or in continuous spans with compact noncomposite negative-moment pier sections:

$C = M_u$

where M_u is determined according to Eq. 10-129b or Eq. 10-129c, as applicable, in Article 10.50.1.1.2. For steel with F_y = 33,000 psi, $\beta = 0.9$ in Article 10.50.1.1.2.

In continuous spans with noncompact noncomposite or composite negative-moment pier sections:

Tension and Compression Flange

$C = F_y$

Alternatively, *C* may be taken as M_u, where M_u is determined according to Eq. 10-129d in Article 10.50.1.1.2.

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According to the preceding requirements, the capacity of a composite positive moment section satisfying the cross-sectional limitations for a compact section specified in Article 10.50.1.1.2 will be at or just below the full plastic moment capacity, M_p, in simple spans and in continuous spans with compact pier sections. In this case, the dead and live load moments are to be used in the basic load-rating equation to compute a rating factor for the section. In continuous spans with noncompact pier sections, the capacity of a compact composite positive moment section will typically be taken equal to the yield stress, F_y. In this case, the dead and live load stresses in each flange are to be used in the basic load rating equation to compute a rating factor for each flange. In either case, however, the web slenderness requirement for the positive moment section given by Eq. 10-129 is to be checked using the depth of the web in compression at the plastic moment, D_{cp}. The elastic depth of the web in compression, D_c, is not to be used in checking the web slenderness requirement for these sections.

Negative Moment Sections

For composite negative moment sections satisfying the cross-sectional limitations specified in Article 10.50.2.1:

$C = M_u$

where M_u is determined according to the provisions of Article 10.50.2.1.

L6B.2.1.3—Noncompact, Noncomposite

The lesser of:

$$C = F_y S_{xt} \tag{10-98}$$

or if Eq. 10-101 is satisfied:

$$C = F_y S_{xt} \tag{10-99}$$

where:

$$F_{cr} = \left(4{,}400\frac{t}{b}\right)^2 \le F_y$$

R_b shall be calculated from the provisions of Article 10.48.4.1 with F_{cr} substituted for the term M_r/S_{xc} when Eq. 10-103b applies.

If Eq. 10-101 is not satisfied:

$C = F_{cr} S_{xc} Rb \le M_u$

where M_u is determined according to the provisions of Article 10.48.4.1.

L6B.2.1.4—Noncompact, Composite, Positive Moment Section

Tension Flange

$C = F_y$

Compression Flange

$C = F_y R_b$

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When R_b is determined from Eq. 10-103b, F_y shall be substituted for the term M_r/S_{xc} and A_{fc} shall be taken as the effective combined transformed area of the top flange and concrete deck that yields D_c calculated in accordance with Article 10.50(b). The resulting R_b factor shall be distributed to the top flange and concrete deck in proportion to their relative stiffness.

Since D_c is a function of the dead-to-live load stress ratio according to the provisions of Article 10.50(b), an iterative procedure may be required to determine the rating factor for the compression flange.

L6B.2.1.5—Noncompact, Composite, Negative Moment Section

Tension Flange

$C = F_y$

Compression Flange

If Eq. 10-101 is satisfied:

$C = F_{cr}R_b$

where:

$$F_{cr} = \left(4,400\frac{t}{b}\right)^2 \le F_y$$

R_b shall be calculated from the provisions of Article 10.48.4.1 with F_{cr} substituted for the term M_r/S_{xc} when Eq. 10-103b applies.

If Eq. 10-101 is not satisfied:

$$C = F_{cr}R_b \le M_u / S_{xc}$$

where M_u and S_{xc} are determined according to the provisions of Article 10.48.4.1.

D_c of the composite section consisting of the steel section plus the longitudinal reinforcement may conservatively be used in lieu of D_c calculated according to the provisions of Article 10.50(b).

L6B.2.2—Sections in Shear

$C = V_u$ (10-113 or 10-114)

where V_u is found in accordance with Article 10.48.8.1.

L6B.2.3—Sections in Shear and Bending (Article 10.48.8.2)

For sections subject to combined shear and bending where the shear capacity is governed by Eq. 10-114 for stiffened girders, the load rating shall be determined according to the following procedure. For composite noncompact sections, replace the moments M_D and $M_{L(1+I)}$ with the corresponding stresses f_D and $f_{L(1+I)}$ and the maximum bending strength M_u of the section with the maximum bending strength, F_u, of the compression or tension flange, as applicable, in the following equations.

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STEP 1: Determine the initial load rating factors for shear and bending moment ignoring moment–shear interaction:

Initial Shear Rating Factor

$$RF_{Vi} = \frac{V_u - A_1 V_D}{A_2 V_{L(1+I)}}$$

Initial Moment Rating Factor

$$RF_{Mi} = \frac{M_u - A_1 M_D}{A_2 M_{L(1+I)}}$$

where:

M_u is found as described above for sections in bending

V_u is found as for sections in shear

M_D is the dead load bending moment

V_D is the dead load shear

$M_{L(1+I)}$ is the maximum live load plus impact bending moment

$V_{L(1+I)}$ is the maximum live load plus impact shear

For composite noncompact sections, the initial moment rating factor shall be taken as the smaller of the rating factors determined separately for the compression and tension flange.

STEP 2: Determine the initial controlling rating factor ignoring moment–shear interaction:

Initial Controlling Rating Factor

RF = minimum of $\left(RF_{Vi}, RF_{Mi}\right)$ from STEP 1

STEP 3: Determine the factored moment and shear using the initial controlling rating factor from STEP 2 as follows:

$$V = A_1 V_D + RF \times A_2 \times V_{L(1+I)}$$
$$M = A_1 M_D + RF \times A_2 \times M_{L(1+I)}$$

STEP 4: Determine the final controlling rating factor as follows:

Final Controlling Rating Factor

RF = minimum of $\left(RF_{Vi}, RF_{M}\right)$ determined from one of the following four cases:

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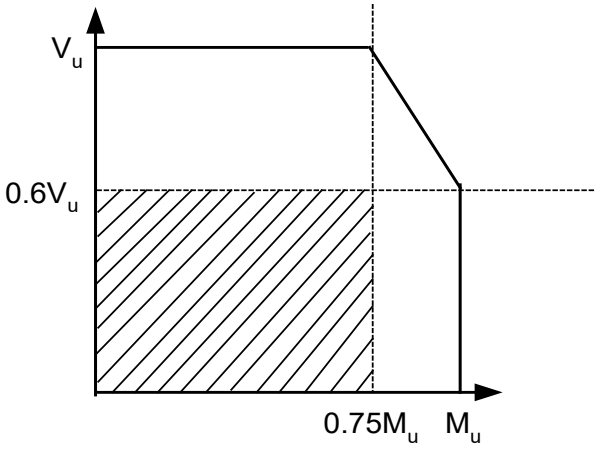
CASE A:

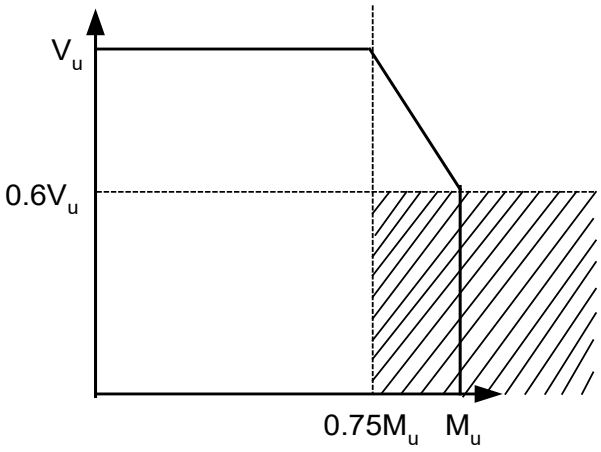
If $V \le 0.6V_u$ and $M \le 0.75M_u$ then:



$RF_V = RF_{Vi}$ and $RF_M = RF_{Mi}$

CASE B:

If $V \le 0.6V_u$ and $M > 0.75M_u$ then:



Reduced Shear Rating Factor

$$RF_V = \frac{V_{Limit} - A_1 V_D}{A_2 V_{L(1+i)}}$$

Moment Rating Factor

$$RF_M = \frac{M_u - A_1 M_D}{A_2 M_{L(1+I)}}$$

where:

$V_{Limit} = 0.6V_u \ge CV_p$

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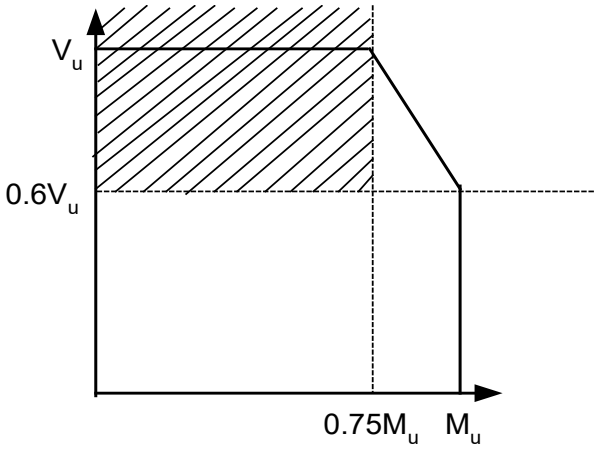
CASE C:

If $V > 0.6V_u$ and $M \le 0.75M_u$ then:



Shear Rating Factor

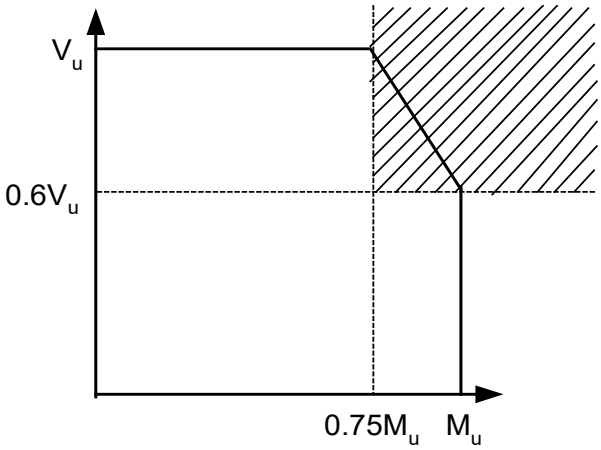
$$RF_V = \frac{V_u - A_1 V_D}{A_2 V_{L(1+I)}}$$

Reduced Moment Rating Factor

$$RF_M = \frac{0.75M_u - A_1 M_D}{A_2 M_{L(1+i)}}$$

CASE D:

Otherwise:



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Moment–Shear Rating Factor

$$RF_M = RF_V = RF_{M-V} = \frac{2.2V_uM_u - A_1V_DM_u - 1.6A_1M_DV_u}{A_2V_{L(1+i)}M_u + 1.6A_2M_{L(1+i)}V_u}$$

if:

$$RF_{M-V} \geq \frac{CV_p - A_1V_D}{A_2V_{L(1+i)}} \Rightarrow RF_M = RF_V = RF_{M-V}$$

Otherwise:

$$RF_V = \frac{CV_p - A_1V_D}{A_2V_{L(1+i)}};\ RF_M = \frac{M_u - A_1M_D}{A_2M_{L(1+i)}}$$

STEP 5: If the final controlling rating factor is different than the initial controlling rating factor, STEPS 2–4 can be repeated (using the final controlling rating factor as the initial controlling rating factor) only if a more accurate rating factor is justified.

STEP 6: When CASE B, C, or D controls the rating and a higher rating is desired for moment and/or shear, STEPS 2–5 may be repeated using sets of concurrent factored live load plus impact moments and shears to determine the final controlling rating factor.

In lieu of investigating numerous combinations of concurrent moments and shears, it is recommended that the rating be repeated using: i) the maximum factored live load plus impact moment in conjunction with a percentage (less than 100 percent) of the maximum factored live load plus impact shear, and ii) the maximum factored live load plus impact shear in conjunction with a percentage (less than 100 percent) of the maximum factored live load plus impact moment.

The final controlling rating factor is the lesser of the factors obtained using i) and ii). If the resulting final controlling rating factor is affected by moment–shear interaction, it must not exceed the initial rating factor for the controlling action.

In lieu of a more rigorous analysis, the determination of the appropriate percentage to be applied should be based on rational engineering judgment. The percentage that is applied should not reduce the maximum factored live load plus impact moment or shear, as applicable, below the actual concurrent factored live load plus impact moment or shear.

Example #1

Load Factor Design

Inventory Rating $\left(A_1 = 1.3; A_2 = 2.17\right)$

Composite Noncompact Section

Assume the following:

$V_u = 411.7$ kips

$V_D = 100$ kips

$V_{L(1+I)} = 90$ kips

$V_p = 700$ kips

$f_D = 20$ ksi

$f_{L(1+I)} = 10.05$ ksi

$F_u = 50$ ksi

$C = 0.42$

$$RF_{Vi} = \frac{V_u - A_1V_D}{A_2V_{L(1+I)}} = \frac{411.7 - 1.3(100)}{2.17(90)} = 1.44$$

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Try second iteration:

$$\left(1.3f_D + 2.17 * RF * f_{L(1+I)}\right) = \left[1.3(20) + 2.17(0.90)(10.05)\right] = 45.6 \text{ ksi} > 37.5 \text{ ksi} \left(= 0.75F_u\right)$$

$$\left(1.3V_D + 2.17 * RF * V_{L(1+I)}\right) = \left[1.3(100) + 2.17(0.90)(90)\right] = 305.8 \text{ k} > 247.0 \text{ k} \left(= 0.6V_u\right)$$

Therefore:

$$RF_M = RF_V = RF_{M-V} = \frac{2.2V_uF_u - A_1V_DF_u - 1.6A_1f_DV_u}{A_2V_{L(1+I)}F_u + 1.6A_2f_{L(1+I)}V_u} = 0.90 \text{ (converged)}$$

Example #2

Load Factor Design

Inventory Rating $\left(A_1 = 1.3;\ A_2 = 2.17\right)$

Composite Noncompact Section

Assume the following:

$V_u = 411.7$ kips $\quad f_D = 18$ ksi

$V_D = 30$ kips $\quad f_{L(1+I)} = 9.86$ ksi

$V_{L(1+I)} = 60$ kips $\quad F_u = 48$ ksi

$V_p = 600$ kips $\quad C = 0.383$

$$RF_{Vi} = \frac{V_u - A_1V_D}{A_2V_{L(1+I)}} = \frac{411.7 - 1.3(30)}{2.17(60)} = 2.87$$

$$RF_{Mi} = \frac{F_u - A_1f_D}{A_2f_{L(1+I)}} = \frac{48 - 1.3(18)}{2.17(9.86)} = 1.15$$

$\therefore\ RF = RF_{Mi} = 1.15$

$$\left(1.3f_D + 2.17 * RF * f_{L(1+I)}\right) = \left[1.3(18) + 2.17(1.15)(9.86)\right] = 48.0 \text{ ksi} > 36.0 \text{ ksi} \left(= 0.75F_u\right)$$

$$\left(1.3V_D + 2.17 * RF * V_{L(1+I)}\right) = \left[1.3(30) + 2.17(1.15)(60)\right] = 188.7 \text{ k} < 247.0 \text{ k} \left(= 0.6V_u\right)$$

$$V_{Limit} = 0.6V_u \geq CV_p = 247.0 \text{ kips} > (0.383)(600) = 230 \text{ kips}$$

Therefore:

$$RF_M = \frac{F_u - 1.3f_D}{2.17f_{L(1+I)}} = \frac{48 - 1.3(18)}{2.17(9.86)} = 1.15$$

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$$RF_V = \frac{V_{Limit} - 1.3V_D}{2.17V_{L(1+I)}} = \frac{247.0 - 1.3(30)}{2.17(60)} = 1.60$$

$\therefore RF = RF_M = 1.15$ (Case B controls) (converged by inspection)

Example #3

Load Factor Design

Inventory Rating $(A_1 = 1.3;\ A_2 = 2.17)$

Composite Noncompact Section

Assume the following:

$V_u = 411.7$ kips $f_D = 5$ ksi

$V_D = 60$ kips $f_{L(1+I)} = 6$ ksi

$V_{L(1+I)} = 90$ kips $F_u = 48$ ksi

$V_p = 700$ kips $C = 0.353$

$$RF_{Vi} = \frac{V_u - A_1V_D}{A_2V_{L(1+I)}} = \frac{411.7 - 1.3(60)}{2.17(90)} = 1.71$$

$$RF_{Mi} = \frac{F_u - A_1f_D}{A_2f_{L(1+I)}} = \frac{48 - 1.3(5)}{2.17(6)} = 3.19$$

$\therefore RF = RF_{Vi} = 1.71$

$$\left(1.3f_D + 2.17 * RF * f_{L(L+I)}\right) = \left[1.3(5) + 2.17(1.71)(6)\right] = 29.0 \text{ ksi} < 36.0 \text{ ksi} \left(= 0.75F_u\right)$$

$$\left(1.3V + 2.17 * RF * V\right) = \left[1.3(60) + 2.17(1.71)(90)\right] = 411.7 \text{ k} > 247.0 \text{ k} \left(= 0.6V\right)$$

Therefore:

$$RF_M = \frac{0.75F_u - 1.3f_D}{2.17f_{L(1+I)}} = \frac{0.75(48) - 1.3(5)}{2.17(6)} = 2.27$$

$$RF_V = \frac{V_u - 1.3V_D}{2.17V_{L(1+I)}} = \frac{411.7 - 1.3(60)}{2.17(90)} = 1.71$$

$\therefore RF = RF_V = 1.71$ (Case C controls) (converged by inspection)

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Example #4

Load Factor Design
Inventory Rating $(A_1 = 1.3;\ A_2 = 2.17)$
Composite Noncompact Section

Assume the following:

$V_u = 411.7$ kips $\quad f_D = 5$ ksi

$V_D = 30$ kips $\quad f_{L(1+I)} = 6$ ksi

$V_{L(1+I)} = 60$ kips $\quad F_u = 48$ ksi

$V_p = 700$ kips $\quad C = 0.353$

$$RF_{Vi} = \frac{V_u - A_1 V_D}{A_2 V_{L(1+I)}} = \frac{411.7 - 1.3(30)}{2.17(60)} = 2.87$$

$$RF_{Mi} = \frac{F_u - A_1 f_D}{A_2 f_{L(1+I)}} = \frac{48 - 1.3(5)}{2.17(6)} = 3.19$$

$\therefore\ RF = RF_{Vi} = 2.87$

$$\left(1.3 f_D + 2.17 * RF * f_{L(1+I)}\right) = \left[1.3(5) + 2.17(2.87)(6)\right] = 44.0 \text{ ksi} > 36.0 \text{ ksi} \left(= 0.75 F_u\right)$$

$$\left(1.3 V_D + 2.17 * RF * V_{L(1+I)}\right) = \left[1.3(30) + 2.17(2.87)(60)\right] = 411.7 \text{ k} > 247.0 \text{ k} \left(= 0.6 V_u\right)$$

Therefore:

$$RF_M = RF_V = RF_{M-V} = \frac{2.2 V_u F_u - A_1 V_D F_u - 1.6 A_1 f_D V_u}{A_2 V_{L(1+I)} F_u + 1.6 A_2 f_{L(1+I)} V_u}$$

$$= \frac{2.2(411.7)(48) - 1.3(30)(48) - 1.6(1.3)(5)(411.7)}{2.17(60)(48) + 1.6(2.17)(6)(411.7)} = 2.52$$

Continuing:

$$\frac{C V_p - A_1 V_D}{A_2 V_{L(1+I)}} = \frac{0.353(700) - 1.3(30)}{2.17(60)} = 1.60 < RF_{M-V} = 2.52$$

$\therefore\ RF = RF_{M-V} = 2.52$ (Case D1 controls)

Try a second iteration:

$$\left(1.3 f_D + 2.17 * RF * f_{L(1+I)}\right) = \left[1.3(5) + 2.17(2.52)(6)\right] = 39.3 \text{ ksi} > 36.0 \text{ ksi} \left(= 0.75 F_u\right)$$

$$\left(1.3 V_D + 2.17 * RF * V_{L(1+I)}\right) = \left[1.3(30) + 2.17(2.52)(60)\right] = 367.1 \text{ k} > 247.0 \text{ k} \left(= 0.6 V_u\right)$$

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Therefore:

$$RF_M = RF_V = RF_{M-V} = \frac{2.2V_uF_u - A_1V_DF_u - 1.6A_1f_DV_u}{A_2V_{L(1+I)}F_u + 1.6A_2f_{L(1+I)}V_u} = 2.52 \text{ (converged)}$$

Example #5

Load Factor Design
Inventory Rating $(A_1 = 1.3;\ A_2 = 2.17)$
Composite Noncompact Section

Assume the following:

$V_u = 411.7$ kips $\quad f_D = 20$ ksi

$V_D = 70$ kips $\qquad f_{L(1+I)} = 10$ ksi

$V_{L(1+I)} = 90$ kips $\quad F_u = 50$ ksi

$V_p = 700$ kips $\qquad C = 0.42$

$$RF_{Vi} = \frac{V_u - A_1V_D}{A_2V_{L(1+I)}} = \frac{411.7 - 1.3(70)}{2.17(90)} = 1.64$$

$$RF_{Mi} = \frac{F_u - A_1f_D}{A_2f_{L(1+I)}} = \frac{50 - 1.3(20)}{2.17(10)} = 1.11$$

$$\therefore RF = RF_{Mi} = 1.11$$

$$\left(1.3f_D + 2.17 * RF * f_{L(1+I)}\right) = \left[1.3(20) + 2.17(1.11)(10)\right] = 50.0 \text{ ksi} > 37.5 \text{ ksi} \left(= 0.75F_u\right)$$

$$\left(1.3V_D + 2.17 * RF * V_{L(1+I)}\right) = \left[1.3(70) + 2.17(1.11)(90)\right] = 307.0 \text{ k} > 247.0 \text{ k} \left(= 0.6V_u\right)$$

Therefore:

$$RF_M = RF_V = RF_{M-V} = \frac{2.2V_uF_u - A_1V_DF_u - 1.6A_1f_DV_u}{A_2V_{L(1+I)}F_u + 1.6A_2f_{L(1+I)}V_u}$$

$$= \frac{2.2(411.7)(50) - 1.3(70)(50) - 1.6(1.3)(20)(411.7)}{2.17(90)(50) + 1.6(2.17)(10)(411.7)} = 0.98$$

Continuing:

$$\frac{CV_p - A_1V_D}{A_2V_{L(1+I)}} = \frac{0.42(700) - 1.3(70)}{2.17(90)} = 1.04 > RF_{M-V} = 0.98$$

Therefore:

$$RF_V = 1.04$$

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$$RF_M = \frac{F_u - 1.3f_D}{2.17f_{L(1+I)}} = \frac{50 - 1.3(20)}{2.17(10)} = 1.11$$

$\therefore RF = RF_V = 1.04$ (Case D2 controls)

Try second iteration:

$$\left(1.3f_D + 2.17 * RF * f_{L(1+I)}\right) = \left[1.3(20) + 2.17(1.04)(10)\right] = 48.6 \text{ ksi} > 37.5 \text{ ksi} \left(= 0.75F_u\right)$$

$$\left(1.3V_D + 2.17 * RF * V_{L(1+I)}\right) = \left[1.3(70) + 2.17(1.04)(90)\right] = 294.0 \text{ k} > 247.0 \text{ k} \left(= 0.6V_u\right) \text{ (converged)}$$

Table L6B.2.3-1—Summary of Load Rating Results

	A_t = 1.3		A_2 = 2.17		Inventory						First Iteration:							
								Step 1		Step 2	Step 3				Step 4			
Example	V_D	V_L	V_n	CV_p	f_D	f_L	F_p	RF_V	RF_M	***RF**i*	V	$0.6V_n$	f	$0.75F_u$	Case	RF_V	RF_M	**Min *RF***
1	100	90	411.7	294.0	20	10.05	50	1.44	1.10	**1.10**	344.9	247.0	50	38	D1	0.90	0.90	**0.90**
2	30	60	411.7	230.0	18	9.86	48	1.15	1.15	**1.15**	188.7	247.0	48	36	B	1.60	1.15	**1.15**
3	60	90	411.7	247.0	5	6	48	3.19	3.19	**1.71**	411.7	247.0	29	36	C	1.71	2.27	**1.71**
4	30	60	411.7	247.0	5	6	48	3.19	3.19	**2.87**	411.7	247.0	44	36	D1	2.52	2.52	**2.52**
5	70	90	411.7	294.0	20	10	50	1.11	1.11	**1.11**	307.0	247.0	50	38	D2	1.04	1.11	**1.04**
											Second Iteration: (second iteration was not needed)							
								0.90	0.90	**0.90**	305.3	247.0	46	38	D1	0.90	0.90	**0.90**
								1.60	1.15	**1.15**	188.7	247.0	48	36	B	1.60	1.15	**1.15**
								1.71	2.27	**1.71**	411.7	247.0	29	36	C	1.71	2.27	**1.71**
								2.52	2.52	**2.52**	367.1	247.0	39	36	D1	2.52	2.52	**2.52**
								1.04	1.11	**1.04**	294.0	247.0	49	38	D2	1.04	1.11	**1.04**

L6B.2.4—Compression Members

L6B.2.4.1—Concentrically Loaded Members

$$C = 0.85 A_s F_{cr} \tag{10-150}$$

where F_{cr} is found in accordance with Article 10.54.1.1.

L6B.2.4.2—Combined Axial Load and Bending

Interaction Eqs. 10-155 and 10-156 must be satisfied by factored axial force, *P*, and factored axial moment, *M*. See Article 10.54.2.

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L6B.2.5—Capacity Based on Overload Provisions of Article 10.57

Note $A_1 = 1.0$ and $A_2 = 1.67$ in the basic rating Eq. 6B.4.1-1 when making this check.

L6B.2.5.1—Noncomposite Beams

$$C = 0.8F_yS \quad \text{(Article 10.57.1)}$$

L6B.2.5.2—Composite Beams

$$C = 0.95F_y \quad \text{(Article 10.57.2)}$$

L6B.2.5.3—Web Compressive Stress

$$C = F_{cr} \quad \text{(Article 10.57)}$$

where F_{cr} is found in accordance with Eq. 10-173.

Since D_c is a function of the dead-to-live load stress ratio according to the provisions of Article 10.50(b), an iterative procedure may be necessary to determine the rating factor at composite positive moment sections. At composite negative moment sections, D_c of the composite section consisting of the steel section plus the longitudinal reinforcement may conservatively be used in lieu of D_c calculated according to the provisions of Article 10.50(b).

L6B.2.6—Gusset Plates

Main truss member gusset plates shall be load rated for shear, compression, or tension, or some combination thereof, occurring in the vicinity of each connected member. Articles L6B.2.6.1 through L6B.2.6.8 outline the necessary checks for performing a load factor rating for these gusset plates and their connections, which are based on the research performed under NCHRP Project 12-84 (Ocel, 2013) that only considered LRFR. These provisions supersede the 2009 FHWA guidelines for gusset plate load ratings. All of the resistance factors used in this Article have not been rigorously determined considering the base HS-20 live load model used for load factor rating. Additional reductions are not required in LFR based on the DL/LL ratio. Future research looking into the live load variability for truss systems may justify the use of lower resistance factors. An example gusset plate rating by LFR can be found in Appendix A.

L6B.2.6.1—Fasteners

Fasteners in bolted and riveted gusset plate connections shall be evaluated to prevent fastener shear and plate bearing failures.

The shear capacity of one fastener shall be taken as:

$$C = (\phi F)mA \quad \text{(L6B.2.6.1-1)}$$

where:

ϕF = shear capacity per fastener area of one fastener specified in Table L6B.2.6.1-1 (ksi)

m = number of shear planes

A = cross-sectional area of one fastener (in.^{2}); for rivets, use the undriven diameter to calculate the area

When bolts carrying loads pass through undeveloped fillers 0.25 in. or more in thickness in axially loaded connections, the bolt shear capacity shall be reduced by:

$$R = \left[\frac{(1+\gamma)}{(1+2\gamma)}\right] \quad \text{(L6B.2.6.1-2)}$$

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where:

γ = A_f/A_p where:

A_f = sum of the area of the fillers on both sides of the connected plate (in.^{2})

A_p = smaller of either the connected plate area or the sum of the splice plate areas on both sides of the connected plate (in.^{2}); for chord splices, when considering the gusset plate(s), only the portion of the gusset plate(s) that overlaps the connected plate shall be considered in the calculation of the splice plate areas

Alternatively, if fillers are extended beyond the connected parts and connected with enough bolts to develop the force in the fillers, the fillers may be considered developed. For rivets, the undeveloped filler plate reduction factor, R_3, shall be considered as specified in Article 6A.6.12.5.1.

Table L6B.2.6.1-1—Shear Capacity Per Fastener Area

	ϕF (ksi) [a]
A307	18
A325—threads included in shear plane	35
A325—threads excluded from shear plane	43
A490—threads included in shear plane	43
A490—threads excluded from shear plane	53
Rivets	See Table 6A.6.12.5.1-1

a Tabulated values shall be reduced by 20 percent in bearing-type connections whose length between extreme fasteners in each of the spliced parts measured parallel to the line of axial force exceeds 50 in.

The bearing capacity of the connected material at standard, oversize, short-slotted holes or long-slotted holes parallel to the applied force shall be taken as:

$$C = 0.9L_c tF_u \le 1.8dtF_u \quad \text{(10-166b)}$$

where:

L_c = clear distance between the holes or between the hole and the edge of the material in the direction of the applied bearing force (in.)

t = thickness of the gusset plate (in.)

d = nominal diameter of the fastener (in.)

F_u = specified minimum tensile strength of the gusset plate given in AASHTO Standard Specifications Table 10.2A (ksi)

For determining the bearing capacity of long-slotted holes loaded perpendicular to the applied force, refer to AASHTO Standard Specifications Article 10.56.1.3.

The operating rating for friction joint high-strength bolts should be determined according to the provisions of Article 6B.5.3.1.

L6B.2.6.2—Multilayered Gusset and Splice Plates

Where multilayered gusset and splice plates are used, the resistances of the individual plates may be added together in determining the overall resistance provided that enough fasteners are present to develop the force in the layered gusset and splice plates.

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L6B.2.6.3—Gusset Plate Shear Resistance

Gusset plates shall be load rated for shear yielding and shear rupture on relevant partial and full shear failure plane widths.

Yielding:

$$C = \varphi_{vy}(0.58)F_yA_g\Omega \quad \text{(L6B.2.6.3–1)}$$

Rupture:

$$C = \varphi_{vu}(0.58)F_uA_n \quad \text{(L6B.2.6.3–2)}$$

where:

φ_{vy} = resistance factor for gusset plate shear yielding taken as 1.00

F_y = specified minimum yield strength of the gusset plate given in AASHTO Standard Specifications Table 10.2A (ksi)

A_g = gross area of the plate resisting shear (in.^{2})

Ω = shear reduction factor for gusset plates taken as 0.88

φ_{vu} = resistance factor for gusset plate shear rupture taken as 0.85

F_u = specified minimum tensile strength of the gusset plate given in AASHTO Standard Specifications Table 10.2A (ksi)

A_n = net area of the plate resisting shear (in.^{2})

Partial shear planes shall only be checked around compression members, and only shear yielding on partial shear planes shall be checked. The partial shear plane length shall be taken along adjoining member fastener lines between plate edges and other fastener lines. The following partial shear planes, as applicable, shall be evaluated to determine which shear plane controls:

- The plane that parallels the chamfered end of the compression member, as shown in Figure L6B.2.6.3-1;
- The plane on the side of the compression member that has the smaller framing angle between that member and the other adjoining members, as shown in Figure L6B.2.6.3-2; and
- The plane with the least cross-sectional shear area if the member end is not chamfered and the framing angle is equal on both sides of the compression member.

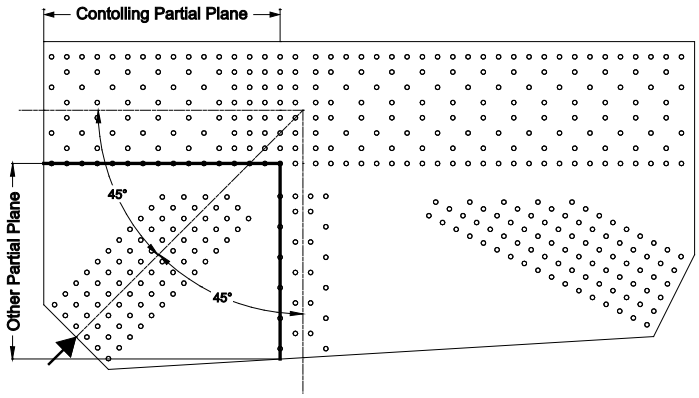

Figure L6B.2.6.3-1—Example of a Controlling Partial Shear Plane that Parallels the Chamfered End of the Compression Member Since That Member Frames in at an Angle of 45 Degrees to Both the Chord and the Vertical

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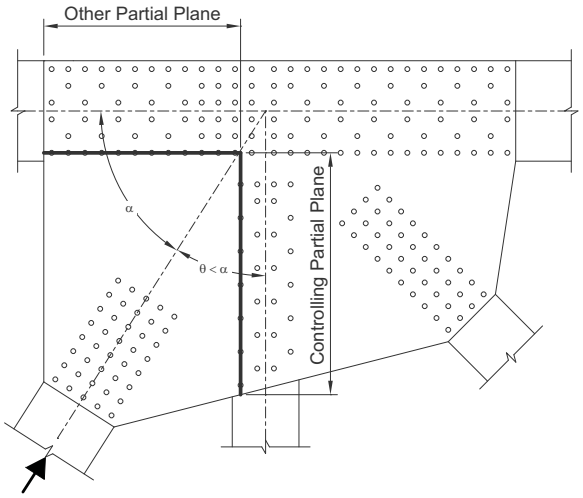

Figure L6B.2.6.3-2—Example of a Controlling Partial Shear Plane on the Side of a Compression Member without a Chamfered End that Has the Smaller Framing Angle between that Member and the Other Adjoining Members (i.e. $\theta < \alpha$)

L6B.2.6.4—Gusset Plate Compressive Resistance

Gusset plate zones in the vicinity of compression members shall be load rated for plate stability. The compressive capacity may be taken as the compressive capacity of a Whitmore section. The buckling capacity is dependent on the chamfering of the member, the framing angles of the members entering the gusset, and the standoff distance of the compression member relative to the surrounding members; i.e. the distance, L_c. An example connection showing a typical chamfered member end and member framing angle is provided in Figure C6A.6.12.6.7-1.

The provisions of this Article shall not be applied for the load rating of compression chord splices.

The compressive capacity of a Whitmore section shall be taken as:

$$C = \varphi_{cg}\,(0.85) A_s F_{cr} \qquad \text{(L6B.2.6.4-1)}$$

in which:

- If $\dfrac{KL_c\sqrt{12}}{t} \le \sqrt{\dfrac{2\pi^2 E}{F_y}}$, then:

$$F_{cr} = F_y\left[1 - \frac{F_y}{4\pi^2 E}12\left(\frac{KL_c}{t}\right)^2\right]$$

- If $\dfrac{KL_c\sqrt{12}}{t} > \sqrt{\dfrac{2\pi^2 E}{F_y}}$, then:

$$F_{cr} = \frac{\pi^2 E}{12\left(\dfrac{KL_c}{t}\right)^2}$$

where:

φ_{cg} = resistance factor for gusset plate compression taken as 1.00
A_s = gross cross-sectional area of the Whitmore section determined based on 30 degree dispersion angles, as shown in Figure L6B.2.6.4-1 (in.^{2}). The Whitmore section shall not be reduced if the section intersects adjoining member bolt lines.

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K = effective length factor in the plane of buckling taken as 0.50 for gusset plates

L_c = distance from the middle of the Whitmore section to the nearest member fastener line in the direction of the member, as shown in Figure L6B.2.6.4-1 (in.)

t = gusset plate thickness (in.)

E = modulus of elasticity of gusset plate (ksi)

F_y = specified minimum yield strength of the gusset plate (ksi)

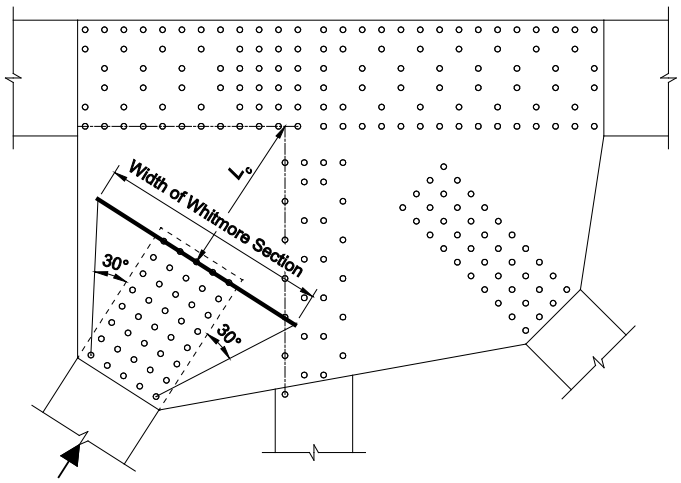

Figure L6B.2.6.4-1—Example Connection Showing the Whitmore Section for a Compression Member Derived from 30 Degree Dispersion Angles and the Distance L_c

L6B.2.6.5—Gusset Plate Tensile Resistance

Gusset plate zones in the vicinity of tension members shall be rated for yielding on the effective area of the Whitmore section and for block shear rupture.

The Whitmore section check shall not be applied for the load rating of tension chord splices.

Yielding:

$$C = \varphi_y F_y A_e \quad \text{(L6B.2.6.5-1)}$$

in which:

A_e = effective cross-sectional area of the Whitmore section determined based on 30 degree dispersion angles, as shown in Figure L6B.2.6.5-1 (in.^{2}). The Whitmore section shall not be reduced if the section intersects adjoining member bolt lines.

$$= A_n + \beta A_g \leq A_g \quad \text{(L6B.2.6.5-2)}$$

where:

φ_y = resistance factor for yielding of tension members taken as 1.00

F_y = yield strength of the plate specified in AASHTO Standard Specifications Table 10.2A (ksi)

A_n = net section of the member (in.^{2})

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β = 0.0 for AASHTO M 270 Grade 100/100W steels, or when holes exceed $1^1/_4$ in. in diameter

= 0.15 for all other steels and when holes are less than or equal to $1^1/_4$ in. in diameter

Block Shear Rupture:

The block shear rupture capacity shall be taken as:

$$C = \varphi_{bs}R_p(0.58F_uA_{vn} + F_uA_{tn}) \le \varphi_{bs}R_p(0.58F_yA_{vg} + F_uA_{tn}) \quad \text{(L6B.2.6.5-3)}$$

where:

φ_{bs} = resistance factor for block shear rupture taken as 0.85

R_p = reduction factor for holes taken equal to 0.90 for bolt holes punched full size and 1.0 for bolt holes drilled full size or subpunched and reamed to size

F_u = tensile strength of the plate specified in AASHTO Standard Specifications Table 10.2A (ksi)

A_{vn} = net area along the plane resisting shear stress (in.^{2})

A_{tn} = net area along the plane resisting tension stress (in.^{2})

F_y = yield strength of the plate specified in AASHTO Standard Specifications Table 10.2A (ksi)

A_{vg} = gross area along the plane resisting shear stress (in.^{2})

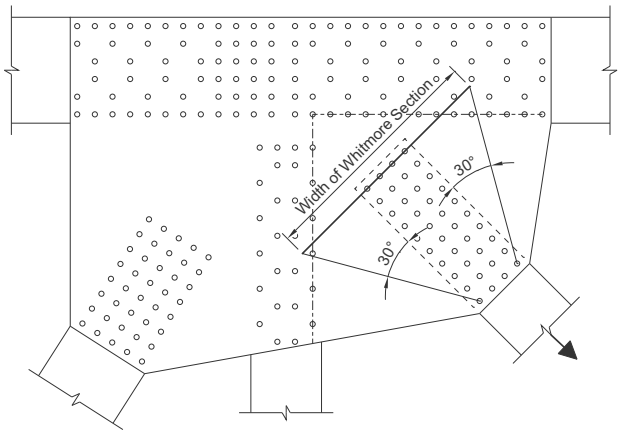

Figure L6B.2.6.5-1—Example Connection Showing the Whitmore Section for a Tension Member Derived from 30 Degree Dispersion Angles

L6B.2.6.6—Chord Splices

Gusset plates that splice two chord sections together shall be checked using a section analysis considering the relative eccentricities between all plates crossing the splice and the loads on the spliced plane. This Article is only intended to cover the load rating of chord splices that occur within the gusset plates. For gusset plates also serving the role of a chord splice, the forces from all members framing into the connection must be considered. The chord splice forces are the resolved axial forces acting on each side of the spliced section, as illustrated in Figure C6A.6.12.6.9-1. Generally, a difference in the resolved forces on the two sides of a chord splice arises due to the use of envelope forces, i.e. forces due to nonconcurrent loads. Where envelope forces are used, the resolved forces should include consideration of the concurrence or nonconcurrence of forces to avoid potentially unconservative reductions in the connection force. The chord splice should be investigated for the larger of the two resolved forces on either side of the splice.

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For chord splices that are in full compression under the loads being considered and that are detailed with milled ends in full contact bearing at the splice, resolved forces may be adjusted to account for the transfer of a portion of the compressive load though end bearing as long as the capacity in bearing is verified. The portion of the compressive load that may be transferred in bearing in such cases is specified in AASHTO *Standard Specifications for Highway Bridges,* Article 10.18.3.

The capacity equations in this Article assume that the gusset and splice plates behave as one combined spliced section to resist the applied axial load and eccentric bending that occurs due to the fact that the resultant forces on the section are offset from the centroid of the combined section, as illustrated in Figure C6A.6.12.6.9-2 (substitute e for e_p in the figure). The combined spliced section is treated as a beam and the capacity is determined assuming the stress in the combined section at the limit of usable capacity is equal to the specified minimum yield strength of the gusset plate if the slenderness limit for the spliced section given below is met, which will typically be the case.

The application of the Whitmore section check specified in Article L6B.2.6.4 is not considered applicable for the load rating of a compression chord splice.

For compression chord splices, the compressive capacity of the spliced section shall be taken as:

$$C = \varphi_{cs} F_{cr} \left(\frac{S_g A_g}{S_g + eA_g} \right) \qquad \text{(L6B.2.6.6-1)}$$

in which:

F_{cr} = stress in the spliced section at the usable limit of capacity (ksi). F_{cr} shall be taken as the specified minimum yield strength of the gusset plate when the following equation is satisfied:

$$\frac{KL_{splice}\sqrt{12}}{t_g} < 25 \qquad \text{(L6B.2.6.6-2)}$$

(Note: if the preceding equation is not satisfied, the Engineer will need to derive a reduced value of F_{cr} to account for possible elastic buckling of the gusset plate within the splice.)

where:

φ_{cs} = resistance factor for gusset plate chord splices taken as 1.00

S_g = gross section modulus of all plates in the cross-section intersecting the spliced plane (in.^{3})

A_g = gross area of all plates in the cross-section intersecting the spliced plane (in.^{2})

e = distance between the centroid of the cross-section and the resultant force perpendicular to the spliced plane (in.)

K = effective column length factor taken as 0.50 for chord splices

L_{splice} = center-to-center distance between the first lines of fasteners in the adjoining chords as shown in Figure L6B.2.6.6-1 (in.)

t_g = gusset plate thickness (in.)

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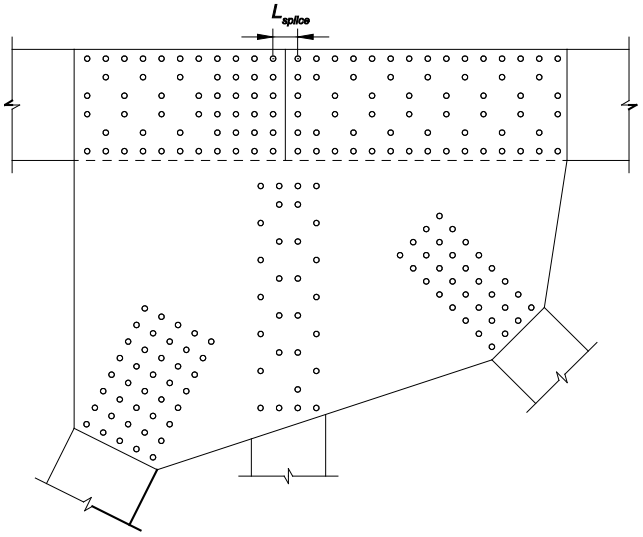

Figure L6B.2.6.6-1—Example Connection Showing Chord Splice Parameter, L_{splice}

For tension chord splices, the tensile capacity of the spliced section shall be taken as the lesser of the values given by the following equations:

$$C = \varphi_{cs} F_y \left(\frac{S_g A_g}{S_g + eA_g} \right)$$

$$C = \varphi_{cs} F_u \left(\frac{S_n A_n}{S_n + eA_n} \right)$$

where:

φ_{cs} = resistance factor for gusset plate chord splices taken as 1.00

F_y = specified minimum yield strength of the gusset plate (ksi)

S_g = gross section modulus of all plates in the cross-section intersecting the spliced plane (in.^{3})

A_g = gross area of all plates in the cross-section intersecting the spliced plane (in.^{2})

e = distance between the centroid of the cross-section and the resultant force perpendicular to the spliced plane (in.)

F_u = specified minimum tensile strength of the gusset plate (ksi)

S_n = net section modulus of all plates in the cross-section intersecting the spliced plane (in.^{3})

A_n = net area of all plates in the cross-section intersecting the spliced plane (in.^{2})

The yielding check on the effective area of the Whitmore section specified in Article L6B.2.6.5 is not considered applicable for the load rating of a tension chord splice; however, tension chord splice members shall be checked for block shear rupture as specified in Article L6B.2.6.5.

L6B.2.6.7—Edge Slenderness

Gusset plates should not be load rated based on any edge slenderness criteria. Refer to Article C6A.6.12.6.10 for additional guidance.

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L6B.2.6.8—Refined Analysis

A refined simulation analysis using the finite element method may be employed to determine the nominal resistance of a gusset-plate connection at the strength limit state in lieu of satisfying the requirements specified in Articles L6B.2.6.3 through L6B.2.6.6. Refer to Article C6A.6.12.6.11 for a list of some suggested model attributes, which is not considered exhaustive.

L6B.3—REINFORCED CONCRETE MEMBERS (AASHTO *STANDARD SPECIFICATIONS FOR HIGHWAY BRIDGES*, 17TH EDITION, 8.16)

L6B.3.1—Sections in Bending

L6B.3.1.1—Rectangular Sections with Tension Reinforcement Only

$$C = \varphi M_n = \varphi\left[A_s f_y \left(d - \frac{a}{2} \right) \right] \tag{8-16}$$

where:

$$a = \frac{A_s f_y}{0.85 f'_c b} \tag{8-17}$$

L6B.3.1.2—Tee Section (Flanged) with Tension Reinforcement Only

L6B.3.1.2.1—Compression Zone within Flange Area

$C = \varphi M_n$ as for Article L6B.3.1.1 above.

L6B.3.1.2.2—Compression Zone Includes Both Flange Area and a Portion of the Web

$$C = \varphi M \tag{8-19}$$

where M_n is found in accordance with AASHTO *Standard Specifications for Highway Bridges*, 17th Edition, Article 8.16.3.3.2.

L6B.3.2—Sections in Compression

See Article 8.16.4 of the AASHTO *Standard Specifications for Highway Bridges*, 17th Edition.

L6B.3.3—Sections in Shear

$$C = \phi V_n \tag{8-46}$$

See Article 8.16.6 of the AASHTO *Standard Specifications for Highway Bridges*, 17th Edition, for the procedure for computing φV_n.

L6B.4—PRESTRESSED CONCRETE MEMBERS (SECTION 9)

L6B.4.1—Sections in Bending

L6B.4.1.1—Rectangular Sections without Nonprestressed Reinforcement

$$C = \varphi M_n = \varphi\left[A_s^* f_{su}^* d \left(1 - 0.6 \frac{\rho^* f_{su}^*}{f_c'} \right) \right] \tag{9-13}$$

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L6B.4.1.2—Tee (Flanged) Sections without Nonprestressed Reinforcement

L6B.4.1.2.1—Compression Zone within Flange Area

$C = \varphi M_n$ as for Rectangular Sections; see Article L6B.4.1.1 above.

L6B.4.1.2.2—Compression Zone Includes Flange Area and Part of Web

$$C = \varphi M_n \tag{9-14}$$

See Article 9.17.3 of the AASHTO *Standard Specifications for Highway Bridges*, 17th Edition, for the evaluation of this equation.

L6B.4.2—Sections in Shear

$$C = \varphi V_n \tag{9-26}$$

See Article 9.20 of the AASHTO *Standard Specifications for Highway Bridges*, 17th Edition, for the procedure for computing φV_n.

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SECTION 7

FATIGUE EVALUATION OF STEEL BRIDGES

7.1—LOAD-INDUCED VERSUS DISTORTION-INDUCED FATIGUE

Fatigue damage has been traditionally categorized as either load-induced or distortion-induced fatigue damage.

Load-induced fatigue is that due to the in-plane stresses in the steel plates that comprise bridge member cross-sections. These in-plane stresses are those typically calculated by designers during bridge design or evaluation.

Distortion-induced fatigue is due to secondary stresses in the steel plates that comprise bridge member cross-sections. These stresses, which are typically caused by out-of-plane forces, can only be calculated with very refined methods of analysis, far beyond the scope of a typical bridge design or evaluation. These secondary stresses are minimized through proper detailing.

C7.1

The previous most comprehensive codification of fatigue evaluation of steel bridges, the *Guide Specifications for Fatigue Evaluation of Existing Steel Bridges* (AASHTO, 1990), explicitly considered only load-induced fatigue damage. The Guide Specifications referenced NCHRP Report 299 for considering "fatigue due to secondary bending stresses that are not normally calculated," NCHRP (1987). Additional fatigue evaluation recommendations were provided in NCHRP Report 721 for both load-induced and distortion-induced bending, NCHRP (2012).

These "plates" may be the individual plates which comprise a built-up welded, bolted, or riveted plate girder, or may be the flanges, webs, or other elements of rolled shapes.

The traditional approximate methods of analysis utilizing lateral live-load distribution factors have encouraged bridge designers to discount the secondary stresses induced in bridge members due to the interaction of longitudinal and transverse members, both main and secondary members.

Detailing to minimize the potential for distortion-induced fatigue, such as connecting transverse connection plates for diaphragms and floorbeams to both the compression and tension flanges of girders, is specified in LRFD Design Article 6.6.1.3.

7.2—LOAD-INDUCED FATIGUE DAMAGE EVALUATION

7.2.1—Application

Two levels of fatigue evaluation are specified for load-induced fatigue: the infinite-life check of Article 7.2.4 and the finite-life calculations of Article 7.2.5. Only bridge details which fail the infinite-life check are subject to the more complex finite-life fatigue evaluation.

Except for the case of riveted connections and tack weld details specified below, the list of detail categories to be considered for load-induced fatigue-damage evaluation and the illustrative examples of these categories are shown in LRFD Design Table 6.6.1.2.3-1.

Except as specified herein, the base metal at net sections of riveted connections shall be evaluated based upon the requirements of Category C, given in LRFD Design Table 6.6.1.2.3-1, instead of Category D as specified for new designs. The exception is for riveted

C7.2.1

The initial infinite-life check should be made with the simplest, least refined stress-range estimate. If the detail passes the check, no further refinement is required. The stress-range estimate for the infinite-life check should be refined before the more complex procedures of the finite-life fatigue evaluation are considered.

For new design, the base metal at net sections of riveted connections is specified to be Category D. This represents the first cracking of a riveted member, which is highly redundant internally. Category C more accurately represents cracking that has propagated to a

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members of poor physical condition, such as with missing rivets or indications of punched holes, in which case Category D shall be used.

Tack welds may be evaluated based upon the requirements of Category C, given in LRFD Design Table 6.6.1.2.3-1.

As uncertainty is reduced from the evaluation by more refined analysis or site-specific data, the increased certainty is reflected in lower partial load factors, summarized in Table 7.2.2.1-1 and described in Articles 7.2.2.1 and 7.2.2.2.

If cracks have been visually detected, then the fatigue life evaluation procedure specified herein should be used with caution. Generally, upon visual detection of load-induced fatigue cracking, the majority of the fatigue life has been exhausted and either retrofitting measures should be initiated, or a fracture mechanics approach can be used to evaluate the fatigue crack damage.

critical size. This increase in fatigue life for evaluation purposes is appropriate due to the redundancy of riveted members.

Tack welds were frequently left in place in riveted connections. The tack welds were used to hold the members in place, prior to placement of the rivets. Tack welds in this context are typically less than 2 in. in length. The strength of tack welds was found to conform to fatigue Category C based on laboratory testing.

The partial load factors specified in Article 7.2 were adapted from the *Guide Specifications for Fatigue Evaluation of Existing Steel Bridges* (AASHTO, 1990).

7.2.2—Estimating Stress Ranges

C7.2.2

The effective stress range shall be estimated as:

$$(\Delta f)_{eff} = R_p R_s \Delta f \qquad (7.2.2\text{-}1)$$

where:

R_p = The multiple presence factor, calculated as described in Article 7.2.2.1 for calculated stress ranges, or 1.0 for measured stress ranges

R_s = The stress-range estimate partial load factor, calculated as $R_{sa}R_{st}$, unless otherwise specified, summarized in Table 7.2.2.1-1, and

Δf = Measured effective stress range; or factored calculated stress range due to the passage of the fatigue truck as specified in LRFD Design Article 3.6.1.4 for Fatigue II Load Combination, or the calculated stress range due to a fatigue truck determined by a truck survey or weigh-in-motion study.

The calculated stress range is due to a single truck in a single lane on the bridge.

The multiple presence factor takes into account the effect of trucks present simultaneously in multiple lanes instead of a single lane loading. When using measured stress ranges, the multiple presence factor should not be used in the equation, as the effects of multiple presence are already reflected in the measured stress ranges.

The load factor is 0.80 for live load specified for the Fatigue II limit state (finite load-induced fatigue life) in LRFD Design Table 3.4.1-1.

7.2.2.1—Calculating Estimated Stress Ranges

C7.2.2.1

The multiple presence factor, R_p, shall be calculated as:

For Longitudinal Members:

$$R_p = 0.988 + 6.87\times10^{-5}\,(\mathrm{L}) + 4.01\times10^{-6}\,[ADTT]_{PRESENT} + 0.0107/(n_L) \geq 1.0 \qquad (7.2.2.1\text{-}1)$$

The limits used in developing the equation are noted as follows: $2 \leq n_L \leq 4$; $[ADTT]_{PRESENT} < 8{,}000$ for n_L=2; 11,000 for n_L=3, and 13,000 for n_L=4, and 30 ft < L < 220 ft. These are the ranges used in the analysis, based on the WIM data available. Use of these

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For Transverse Members:

$R_p = 1.0$

where:

L = span length in feet,

$[ADTT]_{PRESENT}$ = Present average number of trucks per day for all directions of truck traffic including all lanes on the bridge, and

n_L = number of striped lanes.

Two sources of uncertainty are present in the calculation of effective stress range at a particular fatigue detail:

- Uncertainty associated with analysis, represented by the analysis partial load factor, R_{sa}, and
- Uncertainty associated with assumed effective truck weight, represented by the truck-weight partial load factor, R_{st}.

equations may be justified outside of these ranges, but are not based on experimental evidence. The multiple presence factor is applicable to longitudinal parallel members only.

Table 7.2.2.1-1—Partial Load Factors: R_{sa}, R_{st}, and R_s

Fatigue-Life Evaluation Methods	Analysis Partial Load Factor, R_{sa}	Truck-Weight Partial Load Factor, R_{st}	Stress-Range Estimate Partial Load Factor, R_s[a]
For Evaluation or Minimum Fatigue Life			
Stress range by simplified analysis, and truck weight per LRFD Design, Article 3.6.1.4	1.0	1.0	1.0
Stress range by simplified analysis, and truck weight estimated through weigh-in-motion study	1.0	0.95	0.95
Stress range by refined analysis, and truck weight per LRFD Design, Article 3.6.1.4	0.95	1.0	0.95
Stress range by refined analysis, and truck weight by weigh-in-motion study	0.95	0.95	0.90
Stress range by field-measured strains	N/A	N/A	0.85
For Mean Fatigue Life			
All methods	N/A	N/A	1.00

[a] In general, $R_s = R_{sa}R_{st}$

7.2.2.1.1—For Determination of Evaluation or Minimum Fatigue Life

In the calculation of effective stress range for the determination of evaluation or minimum fatigue life, the

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stress-range estimate partial load factor shall be taken as the product of the analysis partial load factor and the truck-weight partial load factor:

$$R_s = R_{sa} R_{st} \quad (7.2.2.1.1\text{-}1)$$

If the effective stress range is calculated through refined methods of analysis, as defined in LRFD Design Article 4.6.3:

$$R_{sa} = 0.95 \quad (7.2.2.1.1\text{-}2)$$

otherwise:

$$R_{sa} = 1.0 \quad (7.2.2.1.1\text{-}3)$$

If the effective truck weight is estimated through a weight-in-motion study at, or near, the bridge:

$$R_{st} = 0.95 \quad (7.2.2.1.1\text{-}4)$$

otherwise:

$$R_{st} = 1.0 \quad (7.2.2.1.1\text{-}5)$$

7.2.2.1.2—For Determination of Mean Fatigue Life

In the calculation of effective stress range for the determination of mean fatigue life, the stress-range estimate partial load factor shall be taken as 1.0.

7.2.2.2—Measuring Estimated Stress Ranges

The effective stress range may be estimated through field measurements of strains at the fatigue-prone detail under consideration and under typical traffic conditions. The effective stress range shall be computed as the cube root of the weighted sum of the cubes of the measured stress ranges, as given in:

$$\left(\Delta f\right)_{eff} = R_s \left(\Sigma \gamma_i \Delta f_i^3\right)^{\frac{1}{3}} \quad (7.2.2.2\text{-}1)$$

where:

γ_i = Percentage of cycles at a particular stress range and

Δf_i = The particular stress range in a measured stress range histogram of magnitude greater than 0.45 of the constant-amplitude fatigue threshold of the fatigue prone detail under consideration, i.e. $> \Delta F_{TH}/2.2$.

C7.2.2.2

Field measurements of strains represent the most accurate means to estimate effective stress ranges at fatigue-prone details.

The *AASHTO LRFD Bridge Design Specifications* assume that the maximum stress range is 2.2 times the effective stress range. It is unlikely that the maximum stress range during the service life of the bridge will be captured during a limited field measurement session; therefore means to extrapolate from the measured stress range histogram to the maximum stress range must be used.

The lower portion of field measured stress range histograms must be truncated in order to avoid underestimating the effective stress range.

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7.2.2.2.1—For Determination of Evaluation or Minimum Fatigue Life

Where field-measured strains are used to generate an effective stress range, R_s, for the determination of evaluation or minimum fatigue life, the stress-range estimate partial load factor shall be taken as 0.85.

7.2.2.2.2—For Determination of Mean Fatigue Life

Where field-measured strains are used to generate an effective stress range, R_s, for the determination of mean fatigue life, the stress-range estimate partial load factor shall be taken as 1.0.

7.2.3—Determining Fatigue-Prone Details

Bridge details are only considered prone to load-induced fatigue damage if they experience a net tensile stress. Thus, fatigue damage need only be evaluated if, at the detail under evaluation:

$$2.2(\Delta f)_{tension} > f_{dead\text{-}load\ compression} \quad (7.2.3\text{-}1)$$

where:

$(\Delta f)_{tension}$ = Tensile portion of the effective stress range as specified in Article 7.2.2, and

$f_{dead\text{-}load\ compression}$ = Unfactored compressive stress at the detail due to dead load

C7.2.3

The multiplier of two in the equation represents the assumed relationship between maximum stress range and effective stress range, as specified in the *AASHTO LRFD Bridge Design Specifications*.

When measured stress ranges are used to evaluate fatigue life, the multiplier of two in the equation should be reconsidered based upon the discussion of Article C7.2.2.2.

If the effective truck weight is significantly less than 54 kips, a multiplier more than two should be considered. Similarly, for a measured effective truck weight greater than 54 kips, a multiplier less than two would be appropriate.

7.2.4—Infinite-Life Check

If:

$$(\Delta f)_{max} \le (\Delta F)_{TH} \quad (7.2.4\text{-}1)$$

then:

$$Y = \infty \quad (7.2.4\text{-}2)$$

where:

$(\Delta f)_{max}$ = The maximum stress range expected at the fatigue-prone detail, which may be taken as:

C7.2.4

Theoretically, a fatigue-prone detail will experience infinite life if all of the stress ranges are less than the constant-amplitude fatigue threshold; in other words, if the maximum stress range is less than the threshold.

When measured stress ranges are used to evaluate fatigue life, the multiplier of 2.2 in the equation for $(\Delta f)_{max}$ should be reconsidered based upon the discussion in Article C7.2.2.2.

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- R_p times the factored calculated stress range due to the passage of the fatigue truck as specified in LRFD Design Article 3.6.1.4; for Fatigue I Load Combination

- $2.2(\Delta f)_{eff}$; for calculated stress range due to a fatigue truck determined by a truck survey or weigh-in-motion study with R_s=1.0

- Larger of maximum (Δf_i), $2.2(\Delta f)_{eff}$, or other suitable value; for measured stress ranges with R_s=1.0

$(\Delta F)_{TH}$ = The constant-amplitude fatigue threshold given in LRFD Design Table 6.6.1.2.5-3

Otherwise, the total fatigue life shall be estimated as specified in Article 7.2.5.

The load factor is 1.75 for live load specified for the Fatigue I limit state (infinite load-induced fatigue life) in LRFD Design Table 3.4.1-1.

When measured stress ranges are used to evaluate fatigue life, the maximum stress range should be taken as the larger value of 2.2 times the field measured effective stress range or the field measured maximum stress range, unless another suitable value is justified.

7.2.5—Estimating Finite Fatigue Life

7.2.5.1—General

Four levels of finite fatigue life may be estimated:

- The minimum expected fatigue life (which equals the conservative design fatigue life),
- Evaluation 1 fatigue life (which equals a conservative fatigue life for evaluation),
- Evaluation 2 fatigue life (which equals a less conservative fatigue life for evaluation), and
- The mean fatigue life (which equals the statistically most likely fatigue life).

The total finite fatigue life of a fatigue-prone detail, in years, shall be determined as:

$$Y = \frac{\log\left[\dfrac{R_R A}{365n\left[(ADTT)_{SL}\right]_{PRESENT}\left[(\Delta f)_{eff}\right]^3} g(1+g)^{a-1}+1\right]}{\log(1+g)} \quad (7.2.5.1\text{-}1)$$

C7.2.5.1

Much scatter, or variability, exists in experimentally derived fatigue lives. For design, a conservative fatigue resistance of two standard deviations shifted below the mean fatigue resistance or life is assumed. This corresponds to the minimum expected finite fatigue life of this Article. Limiting actual usable fatigue life to this design fatigue life is most cautious and can be costly. As such, means of estimating the two evaluation fatigue lives and the mean finite fatigue life are also included to aid the evaluator in the decision making. Evaluation 1 is equivalent to the evaluation life in the previous specification, while Evaluation 2 fatigue life provides an additional choice for the user midway between Evaluation 1 and the mean fatigue life values.

Recent research has made it possible to obtain a closed-form solution for the total finite fatigue life using an estimated traffic growth rate and the present $(ADTT)_{SL}$. The estimated annual traffic growth rate can be obtained using available information in the agency's bridge inventory or the NBI. For cases with zero traffic growth, a very small value of g should be selected (less than 0.01 percent) for use in the expression for Y. Evaluators should be cautious in estimating traffic growth practically. For example an assumed $ADTT$ in a single lane of 10,000 trucks per day corresponds to a truck passing every 8.6 seconds, around the clock 24/7. Clearly this would be an unrealistic assumption for many bridges.

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where:

R_R = Resistance factor specified for evaluation, minimum, or mean fatigue life as given in Table 7.2.5.1-1

A = Detail-category constant given in LRFD Design Table 6.6.1.2.5-1

n = Number of stress-range cycles per truck passage estimated according to Article 7.2.5.2

g = Estimated annual traffic-growth rate, percent, expressed as a decimal; i.e., 5 percent = 0.05

a = Present age of the detail in years

$[(ADTT)_{SL}]_{PRESENT}$ = Present average number of trucks per day in a single lane

$(\Delta f)_{eff}$ = The effective stress range as specified in Article 7.2.2

The resistance factors for fatigue life, specified in Table 7.2.5.1-1, represent the variability of the fatigue life of the various detail categories, A through E′. The minimum life, Evaluation 1 Life and Evaluation 2 Life fatigue-life curves are shifted from the mean fatigue-life S-N curves in log-log space. Scatter of the fatigue lives at given stress range values from controlled laboratory testing provides statistical information on fatigue behavior of bridge details under cyclic loading. Accordingly, the probability of failure associated with each level of fatigue life approaches 2 percent, 16 percent, 33 percent and 50 percent for the minimum Evaluation 1, Evaluation 2 and mean fatigue lives, respectively. Typically, the minimum life or Evaluation 1 Life is used to evaluate the fatigue serviceability. If concerns are encountered regarding the computed fatigue serviceability, then the serviceability index can be revised according to Article 7.2.7.2.

Table 7.2.5.1-1—Resistance Factor for Evaluation, Minimum or Mean Fatigue Life, R_R

Detail Category (from Table 6.6.1.2.5-1 of the *LRFD Specifications*)	R_R			
	Minimum Life	Evaluation 1 Life	Evaluation 2 Life	Mean Life
A	1.0	1.5	2.2	2.9
B	1.0	1.3	1.7	2.0
B’	1.0	1.3	1.6	1.9
C	1.0	1.3	1.7	2.1
C’	1.0	1.3	1.7	2.1
D	1.0	1.3	1.7	2.0
E	1.0	1.2	1.4	1.6
E’	1.0	1.3	1.6	1.9

7.2.5.2—Estimating the Number of Cycles per Truck Passage

The number of stress-range cycles per truck passage may be estimated (in order of increasing apparent accuracy and complexity):

- Through the use of LRFD Design Table 6.6.1.2.5-2,
- Through the use of influence lines, or
- By field measurements.

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7.2.6—Fatigue Serviceability Index

7.2.6.1—Calculating the Fatigue Serviceability Index

The fatigue serviceability index shall be calculated as:

$$Q = \left(\frac{Y - a}{N}\right) GRI \qquad (7.2.6.1\text{-}1)$$

where:

Y = Calculated fatigue life, as given in Section 7.2.5.1

a = Present age of detail, in years

N = Greater of Y or 100 years

G = Load Path Factor, as given in Table 7.2.6.1-1

R = Redundancy Factor, as given in Table 7.2.6.1-2

I = Importance Factor, as given in Table 7.2.6.1-3

Table 7.2.6.1-1—Load Path Factor, *G*

Number of Load Path Members	*G*
1 or 2 members	0.8
3 members	0.9
4 or more members	1

Table 7.2.6.1-2—Redundancy Factor, *R*

Type of Span	*R*
Simple	0.9
Continuous	1

Table 7.2.6.1-3—Importance Factor, *I*

Structure or Location	Importance Factor, *I*
Interstate Highway Main Arterial State Route Other Critical Route	0.90
Secondary Arterial Urban Areas	0.95
Rural Roads Low *ADTT* Routes	1.00

C7.2.6.1

The fatigue serviceability index is a dimensionless relative measure of the performance of a structural detail, at a particular location in the structure, with respect to the overall fatigue resistance of the member. The numerical values for Q vary between 1.0 and 0.

The load path, redundancy, and importance factors are risk factors that modify the fatigue serviceability index. They reduce the index from its base value, i.e. based on fatigue resistance alone, to a reduced value that reflects greater consequences from the lack of ability to redistribute the load (load path factor), lack of redundancy (redundancy factor), or use of the structure (importance factor). The net effect of a reduction in the index will be to move the composite index value to a lower value that may initiate a lower fatigue rating. These risk factors are similar to the ductility, redundancy, and operational classification factors in the *AASHTO LRFD Bridge Design Specifications*. Improved quantification with time will possibly modify these factors.

The number of members that carry load when a fatigue truck is placed on the bridge is used to select the load path factor; e.g., two members for a two-girder bridge and for a typical truss structure; four or more members for a multi-beam or multi-girder bridge; etc. For floorbeams, consider the number of floorbeam members loaded by the fatigue truck. For diaphragms and secondary members, use $G = 1$.

The redundancy factor is to be applied for the specific member or detail being considered. Redundant subsystems need not be penalized when supported by a non-redundant main system.

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7.2.6.2—Recommended Actions Based on Fatigue Serviceability Index

The fatigue serviceability index may be used as a guide for actions that may be undertaken based on fatigue ratings and assessments. Bridge owners should develop a uniform methodology for recommended actions based upon the fatigue serviceability index.

C7.2.6.2

In the recommended actions provided, it is expected that based upon increasing risk, the inspection frequency of the bridge shall be increased on a case-by-case assessment by the bridge owner. Actions may include increased inspection frequency, reassessment as given in Article 7.2.7, or retrofit.

7.2.7—Strategies to Increase Fatigue Serviceability Index

7.2.7.1—General

If the fatigue serviceability index is deemed unacceptable, the strategies of Articles 7.2.7.2 and 7.2.7.3 may be applied to enhance the fatigue serviceability index.

C7.2.7.1

Retrofit, increased inspection, or load-restriction decisions should be made based upon the evaluation fatigue life unless the physical condition or fabrication quality of the bridge is poor. In general, it is uneconomical to limit the useful fatigue life of in-service bridges to the minimum (design) fatigue life.

If the estimated fatigue serviceability index based upon the evaluation fatigue life is deemed unacceptable, a fatigue life approaching the mean fatigue life can be used for evaluation purposes if the additional risk of fatigue cracking is acceptable.

7.2.7.2—Recalculate Fatigue Serviceability Index

7.2.7.2.1—Through Accepting Greater Risk

In general, Evaluation 1 Life of Article 7.2.5 is used in determining the fatigue serviceability index of a bridge detail according to Article 7.2.6. If the evaluator is willing to accept greater risk of fatigue cracking due to:

- Long satisfactory cyclic performance of the detail to date,
- A high degree of redundancy, and/or
- Increased inspection effort, (e.g., decreased inspection interval),
- Some combination of the above,

the fatigue serviceability index may be determined using a fatigue life approaching the mean fatigue life of Article 7.2.5.

C7.2.7.2.1

Greater risk of fatigue cracking can be realized by using either the Evaluation 2 or mean fatigue life values to compute the fatigue serviceability index.

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7.2.7.2.2—Through More Accurate Data

The calculated fatigue serviceability index may be refined by using more accurate data as input to the fatigue-life estimate. Sources of improvement of the estimate include:

- Field measurement of stress ranges at the fatigue-prone detail under construction
- 3-D finite element analysis for stresses at the fatigue-prone detail under consideration
- Weigh-in-motion data of truck weights at or near the bridge site,
- Site-specific data on average daily truck traffic (*ADTT*) at or near the bridge site

This strategy is based upon achieving a better estimate of the fatigue life.

7.2.7.2.3—Through Truncated Fatigue Life Distribution

When a negative fatigue serviceability index is obtained according to Article 7.2.6, the detail's fatigue serviceability index may be updated using the equations below for mean, evaluation, and minimum lives, provided a field inspection finds no evidence of fatigue cracking at the detail.

$$Y'_{mean} = 2.19 Y_{mean} e^{0.73\Phi^{-1}[0.18(1-P)+P]-0.27} \quad (7.2.7.2.3\text{-}1)$$

$$Y'_{eval2} = 2.19 Y_{mean} e^{0.73\Phi^{-1}[0.12(1-P)+P]-0.27} \quad (7.2.7.2.3\text{-}2)$$

$$Y'_{eval1} = 2.19 Y_{mean} e^{0.73\Phi^{-1}[0.074(1-P)+P]-0.27} \quad (7.2.7.2.3\text{-}3)$$

$$Y'_{minimum} = 2.19 Y_{mean} e^{0.73\Phi^{-1}[0.039(1-P)+P]-0.27} \quad (7.2.7.2.3\text{-}4)$$

where:

Y'_{mean} = updated mean life in years

Y_{mean} = mean life in years without updating based on no detection of cracking at detail in question

Y'_{eval1} = updated Evaluation 1 Life in years

Y'_{eval2} = updated Evaluation 2 Life in years

$Y'_{minimum}$ = updated minimum life in years

Φ^{-1} = inverse of the standard normal variable's cumulative probability function (Table 7.2.7.2-1)

C7.2.7.2.3

The fatigue life of a structural detail is modeled using a log normally distributed random variable, as shown in Figure C7.2.7.2.3-1. When the estimated life using Article 7.2.5 is smaller than the present age, the remaining life becomes negative as illustrated.

In this situation, if field inspection finds no evidence of cracking, the estimated life is an overly-conservative estimate. The low tail of the total life distribution is truncated up to the present life. The eliminated probability, *P*, is computed, and the resulting probability density function is divided by $(1 - P)$ to ensure that the total probability under the distribution curve is still 1.0 as shown in Figure C7.2.7.2.3-2. Then the updated life is determined to maintain the same reliability level for fatigue life distribution. Functions Φ (.) and Φ^{-1}(.) are commonly available in commercial spreadsheet programs.

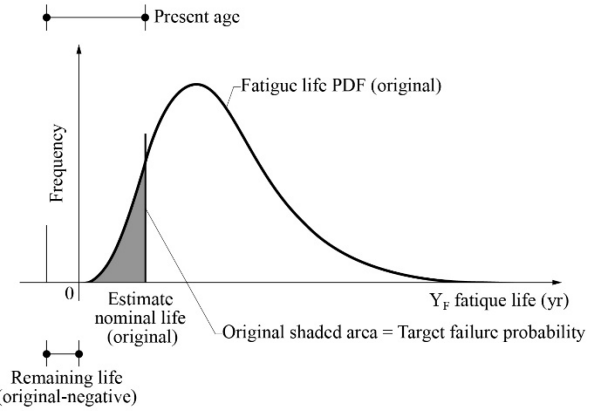

Figure C7.2.7.2.3-1—Probability Density Function of Fatigue Life and Estimated Life as a Value on Horizontal Axis

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P = probability of fatigue life being shorter than current age before updating based on no crack found

$$= \Phi\left[\frac{Ln\left(\frac{a}{2.19Y_{mean}}\right)+0.27}{0.73}\right] \qquad (7.2.7.2.3\text{-}5)$$

where:

a = present age in years

Φ = standard normal variable's cumulative probability function (Table 7.2.7.2-1)

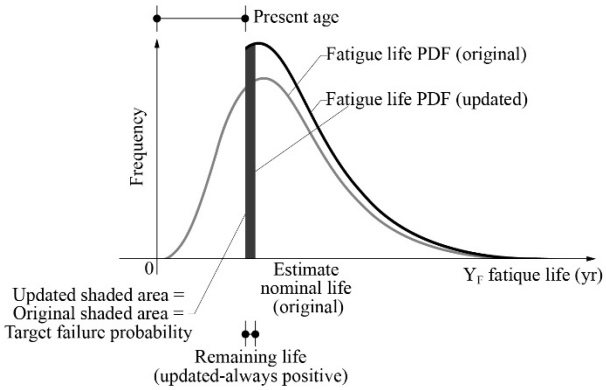

Figure C7.2.7.2.3-2—Truncated Probability Density Function of Fatigue Life and Updated Life as a Value on the Horizontal Axis

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Table 7.2.7.2-1—Cumulative Distribution Function Φ(x) for Standard Normal Variable x

x	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857

7.2.7.3—Retrofit

If the recalculated fatigue serviceability index is not ultimately acceptable, the actual fatigue serviceability index may be increased by retrofitting the critical details to improve the detail category, and thus increase the fatigue serviceability index. This strategy increases the actual fatigue serviceability index when further enhancement of the calculated fatigue serviceability index, through improved input, is no longer possible or practical.

C7.2.7.3

In certain cases, Owners may wish to institute more intensive inspections, in lieu of more costly retrofits, to assure adequate safety. Restricting traffic to increase the fatigue serviceability index is generally not considered cost effective. If the fatigue serviceability index is deemed inadequate, the appropriate option to increase the fatigue serviceability index should be determined based upon the economics of the particular situation.

7.3—DISTORTION-INDUCED FATIGUE EVALUATION

Distortion-induced fatigue is typically caused by the out-of-plane deformation of the web plate, which causes fatigue crack formation on details that are prone to such cracking under cyclic loading. The cracks tend to form in the member web at locations where there is a geometrical discontinuity, such as a vertical gap between a stiffener or connection plate and the girder flange or a horizontal gap between a gusset plate and a connection plate. Distortion-induced fatigue is a stiffness problem (more precisely the lack thereof) versus a load problem.

C7.3

Often, distortion-induced fatigue cracks initiate after relatively few stress-range cycles at fatigue-prone details. However, depending upon the magnitude of the out-of-plane distortion and the geometry of the web gap detail, the crack growth may be slow and a significant period of time may be required before they become large enough to be detected visually. Therefore, existing bridges should not be assumed to be insensitive to distortion-induced cracking if fatigue cracks do not appear after a short period of time. Experience has shown that in some cases,

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cracking may not be evident for 10 years after the beginning of service.

7.3.1—Methods to Assess Distortion-Induced Cracking

Out-of-plane distortions caused by truck loading must be accommodated by the regions that contain unsupported web gaps. Fatigue cracks caused by out-of-plane distortions should be repaired when they develop.

C7.3.1

Typically, smaller web gaps are subject to higher distortion-induced stresses than larger web gaps provided the same demand for the out-of-plane distortion. The demand for out-of-plane distortion is determined by the global behavior of the structural system. Accurate quantification of the stress field in an unsupported web gap detail can be very difficult, even for finite element modeling or field measurement of strains and/or local deformations. This is especially the case when the dimension of the web gap is comparable to the thicknesses of the surrounding plates and the sizes of the connecting welds, resulting in high stress gradients across the web gap.

Even very small distortions can cause high local stresses that may induce fatigue cracking. Often, the fatigue cracks grow in a plane that is parallel to the primary stresses of the member and will slow down or even stop as the web gap becomes more flexible due to the presence of the crack. However, it is possible that the crack may turn and become perpendicular to the primary stress of the member, leading to more rapid crack growth. Therefore, distortion-induced fatigue cracks should be repaired.

7.3.2—Retrofit Options for Distortion-Induced Fatigue Cracking

Retrofit should be considered if distortion-induced cracking has been detected. Two primary retrofit methods are available: softening or stiffening. The softening approach is used to increase the overall flexibility of the detail in question to accommodate the out-of-plane deformations without further cracking. The stiffening approach is used to minimize the local distortion by providing a positive load path for the forces that tend to cause the distortion. In either case, a hole should be drilled at the tip of each crack.

C7.3.2

In the softening retrofit, the flexibility of the detail in question in increased. Drilling holes to eliminate the tip of distortion-induced fatigue cracks will typically increase the local flexibility somewhat. However, the primary method used to increase the flexibility is to increase the size of the web gap. This can be effective since the out-of-plane bending stresses are related to the inverse of the square of the web gap length. One critical issue for this approach is to avoid an excessive increase of out-of-plane deformation resulting from the web gap enlargement. Removal of portions of a stiffener or other plates to increase the size of the web gap will also require removal of the connecting weld in those regions to provide a smooth, flush surface. Non-destructive inspection should be conducted to ensure that no undesirable gouges, notches, or discontinuities remain.

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In the stiffening retrofit, the stiffness of the detail in question is increased to minimize the out-of-plane distortion. Commonly, this will require the addition of a WT section, or a double- or single-angle section. Drilling retrofit holes to eliminate the tip of any distortion-induced fatigue cracks should be done prior to installation of the retrofit connection element. Typically, the installation of a retrofit element will increase the stiffness and significantly decrease the out-of-plane deformation at the detail. However, the force effect of the retrofit on the primary and secondary members should be considered. One critical issue for this approach is to size the retrofit connection of sufficient thickness and strength for the loading forces to be generated at the new connection.

7.4—FRACTURE-CONTROL FOR OLDER BRIDGES

Bridges fabricated prior to the adoption of AASHTO's *Guide Specifications for Fracture-Critical Nonredundant Steel Bridge Members* (1978) may have lower fracture toughness levels than are currently deemed acceptable. Destructive material testing of bridges fabricated prior to 1978 to ascertain actual toughness levels may be justified. Decisions on fatigue evaluations of a bridge can be made based upon the information from these tests.

C7.4

The fatigue life of a steel bridge detail generally consists of crack initiation and stable crack propagation. The propagation stage continues until the crack reaches a critical length associated with unstable, rapid crack extension, namely fracture. An exception is constraint-induced fracture, where very little or no crack growth occurs prior to fracture.

Fracture toughness reflects the tolerance of the steel for a crack prior to fracture. Fracture of steel bridges is governed by the total stress, including the dead-load stress, and not just the live-load stress range as is the case with fatigue. Older bridges with satisfactory performance histories likely have adequate fracture toughness for the maximum total stresses that they have experienced. However, propagating fatigue cracks in bridges of questionable fracture toughness are very serious, and may warrant immediate bridge closure.

7.5—ALTERNATE ANALYSIS METHODS

Alternative analysis techniques, such as fracture mechanics and hot-spot stress analysis, may be used to predict the finite fatigue life of a detail. The estimate for finite life obtained from these methods should be used in place of Y in Article 7.2.6 to determine the fatigue serviceability index.

C7.5

These analyses may be helpful in assessing cases where S-N test data from appropriately sized connections are not available. Hot-spot stress fatigue design has been used in certain industries to evaluate structures with complex geometries where nominal stress is not easily defined and where weld toe cracking is the most likely mode of failure. More detailed information on the hot-spot analysis method is provided in Niemi et al. (2006). Fracture mechanics, on the other hand, have also been used in certain industries as a "fitness for purpose" type of assessment to establish a suitable design life for members with certain known flaw sizes. Barsom and Rolfe (1999) provide detailed examples illustrating the use of this method. When using alternative analysis techniques, efforts should be made to use a level of safety comparable with those levels prescribed in Article 7.2.6 for minimum, evaluation, or mean fatigue life.

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SECTION 8:

NONDESTRUCTIVE LOAD TESTING

8.1—INTRODUCTION

8.1.1—General

Load testing is the observation and measurement of the response of a bridge subjected to controlled and predetermined loadings without causing changes in the elastic response of the structure. Load tests can be used to verify both component and system performance under a known live load and provide an alternative evaluation methodology to analytically computing the load rating of a bridge.

Literally thousands of bridges have been load tested over the last 50 years in various countries. In some countries, load tests are used to verify the performance of new bridges compared to design predictions. The aim of this Section is to emphasize the use of load testing as part of bridge load-rating procedures.

8.1.2—Classification of Load Tests

Basically, two types of load tests are available for bridge evaluation: diagnostic tests and proof tests. Diagnostic tests are performed to determine certain response characteristics of the bridge, its response to loads, the distribution of loads; or to validate analytical procedures or mathematical models. Proof tests are used to establish the maximum safe load capacity of a bridge, where the bridge behavior is within the linear-elastic range.

Load testing may be further classified as static load tests and dynamic load tests. A static load test is conducted using stationary loads to avoid bridge vibrations. The intensity and position of the load may be changed during the test. A dynamic load test is conducted with time-varying loads or moving loads that excite vibrations in the bridge. Dynamic tests may be performed to measure modes of vibration, frequencies, dynamic load allowance, and to obtain load history and stress ranges for fatigue evaluation. Diagnostic load tests may be either static or dynamic tests. Proof load tests are mostly performed as static tests.

8.2—FACTORS WHICH INFLUENCE THE LOAD-CARRYING CAPACITY OF BRIDGES

8.2.1—General

The actual performance of most bridges is more favorable than conventional theory dictates. When a structure's computed theoretical safe load capacity or remaining fatigue life is less than desirable, it may be beneficial to the Bridge Owner to take advantage of some of the bridge's inherent extra capacity that may have been ignored in conventional calculations.

C8.1.1

The procedures outlined in this Section for the nondestructive load testing of bridges were developed in NCHRP Project 12-28(13)A and reported in *NCHRP Research Results Digest*, November 1998—Number 234, "Manual for Bridge Rating Through Load Testing," and include certain modifications necessary to ensure consistency with the load and resistance factor load-rating procedures presented in this Manual.

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Several factors not considered in routine design and evaluation could affect the actual behavior of bridges. Load testing is an effective methodology to identify and benefit from the presence of certain load capacity enhancing factors as outlined below.

8.2.2—Unintended Composite Action

Field tests have shown that a noncomposite deck can participate in composite action with the girders in carrying live load, provided the horizontal shear force does not exceed the limiting bond strength between the concrete deck slab and steel girder flanges. However, as test loads are increased and approach the maximum capacity of the bridge, slippage can take place and composite action can be lost, resulting in a sudden increase in main member stresses. Thus, it is important that for noncomposite steel bridges, load test behavior and stress values taken at working loads or lower not be arbitrarily extrapolated to higher load levels. The unintended composite action contributes to both the strength of a girder bridge and its ability to distribute loads transversely. Advantage can be taken of unintended composite action in fatigue evaluation computations provided there is no observed slippage between the deck and stringer flange under normal traffic.

8.2.3—Unintended Continuity/Fixity

Simply supported bridges are assumed to be supported on idealized rollers that do not carry any moment. However, tests have shown that there can be significant end moments attributable to the continuity provided by the deck slab at stringer-to-floorbeam connections and to frozen bearings. Frozen bearings could also result in unintended arching action in the girders to reduce the applied moments at midspan by a significant margin. For load-rating purposes, it may not be justified to extrapolate the results of a load test done at moderate-load levels when such restraints are detected during the test. It is quite possible that the enhanced behavior attributable to unintended continuity and frozen bearings would not be present at extreme load levels.

8.2.4—Participation of Secondary Members

Secondary bridge members are those members which are not directly in the load path of a structure, such as: diaphragms, cross-frames, lateral bracing members, and wind bracing. In some bridge types, secondary members enhance the load-carrying capacity by increasing the stiffness of the bridge. Advantage can be taken of the effects of secondary members provided that it can be shown that they are effective at the designated service load level.

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8.2.5—Participation of Nonstructural Members

Load distribution, stresses, and deflections may be affected by the stiffness contribution from nonstructural members such as railings, parapets, and barriers, and to a lesser extent by the curbs and utilities on the bridge. Since the stiffness contribution from such members cannot be relied upon at the ultimate load condition, it is important that their contributions be considered in comparing the bridge-test-load response with the calculated response.

8.2.6—Portion of Load Carried by Deck

Depending on the bridge span and the thickness of the deck, there may be a portion of the load carried directly by the deck slab spanning between end supports of the bridge. The deck may, however, not be able to carry significant amounts of load at higher load levels so that any portion carried during the diagnostic test should be determined and transferred back, if necessary, into the main load-carrying members.

8.3—BENEFITS OF NONDESTRUCTIVE LOAD TESTS

8.3.1—Unknown or Low-Rated Components

Load tests may provide sufficient data to establish safe live-load levels for older bridges. In some instances, the make-up of the bridge members, the members' response to loading, or both cannot be determined because of lack of existing as-built information. In other cases, theoretical rating calculations may result in a low live load requiring posting of the rated bridge, and nondestructive load tests may provide a more realistic safe service live-load capacity. In some instances, the test results may indicate that the actual safe service live-load capacity is less than computed, thus alerting the Bridge Owners to speedy action to reinforce or close the bridge.

Existing bridges that have been strengthened over the years may not be accurately load rated due to the unknown interaction of the various elements of the repaired structure in supporting live loads.

Nondestructive load tests can help evaluate the performance of such a bridge, and generally improve its load rating.

8.3.2—Load Distribution

An important part of the rating equation concerns the distribution of the live loads to the main load-carrying members of the bridge and to the individual components of a multicomponent member. Typically, in design and rating, the load distribution to main supporting members is based on design distribution factors. These factors are known to generally result in conservative approximations of the actual distribution. A major aim of diagnostic testing is to confirm the

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precise nature of the load distribution. In a multicomponent member, such as truss chords, test results could reveal if the components share the load equally as is assumed in the analysis.

8.3.3—Deteriorated or Damaged Members

It is often difficult to analyze the effects of observed deterioration or damage on the load-carrying capacity of the bridge and on load distribution, especially in the case of heavily deteriorated bridges. In such cases, field load testing serves as a powerful tool to identify existing behavior.

8.3.4—Fatigue Evaluation

In assessing the remaining fatigue life of steel bridges, both the range of stress and the number of stress cycles acting on a member need to be evaluated. Field load testing can provide data for both of these parameters. The range of live-load stress is influenced by the enhanced section modulus evidenced by most beam and slab sections. Measured stresses can be used in place of computed stresses in making remaining life assessments. In addition, stress spectra may be obtained for distortion-induced stresses, which have been found to be a major cause of distress in steel bridges and can lead to cracking of components and eventual failure.

8.3.5—Dynamic Load Allowance

Design dynamic load allowance is generally conservative for most spans. Dynamic load allowance is influenced primarily by the surface roughness of the deck and approaches. The use of full-scale dynamic testing under controlled or normal traffic conditions remains the most reliable and cost-effective way of obtaining the dynamic load allowance for a specific bridge. Measured dynamic load allowance may be used in place of code-specified value in load-rating calculations.

8.4—TYPES OF NONDESTRUCTIVE LOAD TESTS

8.4.1—Static Tests

8.4.1.1—Diagnostic Tests

Diagnostic load tests are employed to improve the Engineer's understanding of the behavior of a bridge and to reduce uncertainties related to material properties, boundary conditions, cross-section contributions, effectiveness of repair, influence of damage and deterioration, and other similar variables. Diagnostic load tests include the measurement of load effects in one or more critical bridge members and comparison of the measured load effects with that computed using an analytical model (theory). Diagnostic tests serve to verify and adjust the predictions of an analytical model. The

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calibrated analytical models are then used to calculate the load-rating factors. During a diagnostic load test, the applied load should be sufficiently high to properly model the physical behavior of the bridge at the rating load level.

Bridges for which analytical methods of strength evaluation may significantly underestimate the actual strength (e.g., redundant spans, spans with boundary conditions different from assumed idealized behavior, etc.) are candidates for diagnostic load testing. Thus, candidate bridges are limited to those bridges for which an analytical load-rating model can be developed.

8.4.1.2—Proof Tests

In this form of field load testing, a bridge is subjected to specific loads, and observations are made to determine if the bridge carries these loads without damage. Loads should be applied in increments and the bridge monitored to provide early warning of possible distress or nonlinear behavior. The proof test is terminated when:

1. a predetermined maximum load has been reached, or
2. the bridge exhibits the onset of nonlinear behavior or other visible signs of distress.

Although simple in concept, proof testing will in fact require careful preparation and experienced personnel for implementation. Caution is required to avoid causing damage to the structure or injury to personnel or the public.

Bridges that are candidates for proof load testing may be separated into two groups. The first group consists of those bridges whose make-up is known and which can be load rated analytically. Proof load testing of "known" bridges is called for when the calculated load ratings are low and the field testing may provide realistic results and higher ratings. Bridges with large dead loads compared with the live loads are also suitable candidates for proof load testing.

The second group consists of "hidden" bridges, those bridges which cannot be load rated by computations because of insufficient information on their internal details and configuration. Many older reinforced concrete and prestressed concrete beam and slab bridges whose construction plans, design plans, or both are not available need proof testing to determine a realistic live-load capacity. Bridges that are difficult to model analytically because of uncertainties associated with their construction and the effectiveness of repairs are also potential candidates and beneficiaries of proof load testing.

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8.4.2—Dynamic Tests

8.4.2.1—Weigh-In-Motion Testing

The actual site survey of truck weight spectra and volume can be determined by weigh-in-motion (WIM) systems. WIM systems utilize axle sensors and other measurement systems which make use of the bridge as the scale. Such WIM techniques could provide data on vehicle arrivals; and determine axle and gross loads, axle configurations, and speeds of passing vehicles. The WIM data can be utilized to provide a precise site-specific load model and can also be utilized in fatigue evaluation.

8.4.2.2—Dynamic Response Tests

Dynamic response tests, under normal traffic or controlled conditions using test vehicles, can be performed to obtain realistic estimates of the dynamic load allowance and live-load stress ranges that can be used in load rating and fatigue evaluation calculations. Dynamic load allowance is influenced primarily by the surface roughness of the deck and the bridge approach, and to a lesser extent by the bridge frequency and the weight and dynamic characteristics of the vehicle. Many of these parameters are difficult to quantify without the use of full-scale dynamic testing.

The dynamic load allowance may be estimated from the peak dynamic strain and the corresponding peak static strain for vehicles on the same path or transverse position on the bridge. A variety of vehicle types, speeds, weights, and positions should be considered in estimating the appropriate dynamic load allowance. A representative estimate of the dynamic load allowance can be obtained from statistical analyses of measured values.

C8.4.2.2

Dynamic tests preferably should use heavy test vehicles since load rating is governed by heavy vehicles with much lower dynamic impact effects.

8.4.2.3—Vibration Tests

Vibration tests are used to determine bridge dynamic characteristics such as frequencies of vibration, mode shapes, and damping. Earthquake response is strongly influenced by bridge frequency and damping. Vibration testing can sometimes be used to evaluate defects and deterioration as they affect the vibration characteristics. The principal results of a dynamic response test may be the bridge natural frequencies and corresponding mode shapes as well as damping values. Vibration tests may be conducted by means of portable sinusoidal shakers, sudden release of applied deflections, sudden stopping of vehicles by braking, and impulse devices such as hammers.

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8.5—LOAD TEST MEASUREMENTS

Load test instrumentation is used to measure the following: 1) strain (stresses) in bridge components, 2) relative or absolute displacement of bridge components, 3) relative or absolute rotation of bridge components, and 4) dynamic characteristics of the bridge.

Prior to conducting a field test, the Engineer must determine the goals of the test and the types and magnitude of the measurements to be made. Preliminary calculations may be needed to estimate the range of the measurements as well as the best locations for the instrumentation.

8.6—WHEN NOT TO LOAD TEST

The following conditions could render a bridge an unsuitable candidate for load testing:

- The cost of testing reaches or exceeds the cost of bridge strengthening.
- Pretest evaluation shows that the load test is unlikely to show the prospect of improvement in load-carrying capacity.

C8.5

Strain Measurements

Strain sensors are usually attached on critical members to monitor response. Different types of gages are available for steel and concrete structures. The locations should be selected so that the analytical model can be validated. The most common sensors for field measurement of strains are electrical resistance gages (bonded or welded), strain transducers (clamped or anchored), and acoustic strain gages. Careful selection of gage characteristics is required to optimize gage performance for specified environmental and operating conditions.

Displacement Measurements

Three methods of monitoring displacements are mechanical, optical, and electrical. Dial gages are mechanical devices that are easy to set up and monitor, and their accuracy is usually sufficient for load tests. Optical methods include laser methods and other surveying tools that can be used when higher accuracy is required.

Electrical methods include displacement transducers such as Linear Variable Differential Transformers (LVDT) that transform displacement to a proportional change of electrical voltage. They can be used to monitor both static and dynamic displacements.

Rotation Measurements

Mechanical tiltmeters can be installed on beam webs to monitor beam rotations. The measurement of end rotations can establish the extent of end restraint at bearings. The elastic curve for a bending member can be developed by measuring rotations along the length of the member.

Measurement of Dynamic Characteristics

Accelerometers are used if the modal frequencies, mode shapes, and damping ratios are to be obtained. Accelerometers are usually placed at midspan and quarter-span points to determine first and second longitudinal mode shapes, and on either side of the bridge to determine torsional mode shapes.

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- According to calculations, the bridge cannot sustain even the lowest level of load.
- There is a possibility of sudden failure (shear or fracture).
- Load tests may be impractical because of access difficulties or site traffic conditions.

8.7—BRIDGE SAFETY DURING LOAD TESTS

An element of risk is inherent in all load testing. The Bridge Owner and evaluators must be aware of the risks and their consequences. In assessing the risks, consideration should be given to safety of the public, safety of personnel, possible structural damage, traffic disruption, and possible load posting. Bridge load testing should not be attempted by inexperienced personnel. Common sense, good engineering judgment, and sound analytical principles are not to be ignored.

8.8—LOAD RATING THROUGH LOAD TESTING

8.8.1—Introduction

Diagnostic and proof load tests can be employed to improve the Evaluator's understanding of the behavior of the bridges being tested and to identify and quantify in a scientific manner their true inherent reserve capacity. A major part of the Evaluator's responsibility is in determining how much of any potentially enhanced load-carrying capacity observed during the load test, as compared to the values predicted analytically, could be reliably utilized in establishing the bridge load rating. Article 8.8 outlines methods and procedures for the application of nondestructive load tests in the load rating process and translating the results of the bridge load tests into bridge load ratings.

8.8.2—Diagnostic Load Tests

8.8.2.1—Introduction

Prior to initiating a diagnostic load test, the bridge should be rated analytically using procedures contained in this Manual. The procedures outlined in Article 8.8.2 will enable the Engineer to re-examine the theoretical values and adjust these ratings to reflect the actual performance of the bridge obtained from the diagnostic test results.

8.8.2.2—Approach

As long as a bridge exhibits linear behavior, a diagnostic load test can be used to validate an updated analytical model. It is thus important that the test load be placed at various positions on the bridge to determine the response in all critical bridge members. Further, the magnitude of the test load must be sufficiently high so that there is little likelihood of nonlinear behavior at the

C8.8.1

General load testing procedures are contained in Appendix A8. For additional guidance, Evaluators should consult *NCHRP Research Results Digest* No. 234.

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anticipated service-load levels. If the Engineer is satisfied that the model is valid, then an extrapolation to load levels higher than those placed on the bridge during the test may be feasible. The following Articles present a method for extrapolating the results of a diagnostic load test.

8.8.2.3—Application of Diagnostic Test Results

A major part of diagnostic testing is the assessment of the differences between predicted and measured responses for subsequent use in determining the load rating of the bridge. Article 8.8.2.3 provides guidelines for modifying the calculated load rating for a bridge based on the results of a diagnostic load test.

The following equation should be used to modify the calculated load rating following a diagnostic load test:

$$RF_T = RF_c K \qquad (8.8.2.3\text{-}1)$$

RF_T = load-rating factor for the live-load capacity based on the load test result

RF_c = rating factor based on calculations prior to incorporating test results (Eq. A6.4.2.1-1 should be used).

K = adjustment factor resulting from the comparison of measured test behavior with the analytical model (represents the benefits of the field load test, if any)

8.8.2.3.1—Determining K

The Adjustment Factor K is given by:

$$K = 1 + K_a K_b \qquad (8.8.2.3.1\text{-}1)$$

where:

K_a = accounts for both the benefit derived from the load test, if any, and consideration of the section factor (area, section modulus, etc.) resisting the applied test load

K_b = accounts for the understanding of the load test results when compared with those predicted by theory

Without a load test, $K = 1$. If the load test results agree exactly with theory, then $K = 1$ also. Generally, after a load test K is not equal to one. If $K > 1$, then response of the bridge is more favorable than predicted by theory and the bridge load capacity may be enhanced. On the other hand, if $K < 1$, then actual response of the bridge is more severe than that predicted and the theoretical bridge load capacity may have to be reduced.

The following general expression should be used in determining K_a:

C8.8.2.3

The appropriate section factor (area, section modulus) to be used in calculating RF_c should be determined after evaluation of the load test results, including observations made during the placement of the test vehicle on the bridge. Observed enhancement to the section factor resulting from unintended composite action needs to be critically evaluated. Analytical evaluation of composite action in slab-and-girder bridges without mechanical shear connection and the reliability of composite action found by a diagnostic test is discussed in *NCHRP Research Results Digest* No. 234.

For composite structures with shear connectors, the full composite section as defined by the *AASHTO LRFD Bridge Design Specifications* should be used unless observations during the test indicate slippage at the deck-girder interface. Noncomposite structures which show no evidence of composite action under the test load should be evaluated based on noncomposite section factors.

C8.8.2.3.1

The intent of "Can member behavior be extrapolated to 1.33*W*?" in Table 8.8.2.3.1-1 is to provide some assurance that the structure has adequate reserve capacity beyond its rating load level *W*. Normally this would be established by calculation, but proof testing would also be acceptable.

Examples of typical calculations which could be performed to check this criterion include:

1. Load the analytical model with 1.33*W* and determine whether there is linear behavior of the components of the structure. The model could be based on the LRFD specifications or a three-dimensional computer model.
2. Using the procedures given in *NCHRP Research Results Digest* No. 234, determine whether there is composite action at 1.33*W* where none was intended.

Diagnostic load test does not specifically address the fatigue limit state. However, at the time of the test it may be necessary to measure stresses at fatigue sensitive details to determine if fatigue cracking is possible.

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$$K_a = \frac{\varepsilon_c}{\varepsilon_T} - 1 \qquad (8.8.2.3.1\text{-}2)$$

where:

ε_T = maximum member strain measured during load test

ε_C = corresponding calculated strain due to the test vehicle, at its position on the bridge which produced ε_T

K_a may be positive or negative depending on the results of the load test.

In general:

$$\varepsilon_c = \frac{L_T}{(SF)E} \qquad (8.8.2.3.1\text{-}3)$$

where:

L_T = calculated theoretical load effect in member corresponding to the measured strain ε_T

SF = member appropriate section factor (area, section modulus, etc.); see C8.8.2.3

E = member modulus of elasticity

The theoretical strain ε_C resulting from the test load should be calculated using a section factor which most closely approximates the member's actual resistance during the test. (See example in *NCHRP Research Results Digest* No. 234, pages 46–47.) For noncomposite sections, the factor K_a represents the test benefit without the effect of unintended composite action.

K_b takes into account the analysis performed by the load test team and their understanding and explanations of the possible enhancements to the load capacity observed during the test. In particular, the load test team should consider the items below and reduce K_b to account for those contributions that cannot be depended on at the rating load level. Table 8.8.2.3.1-1 provides guidance based on the anticipated behavior of the bridge members at the rating load level, and the relationship between the unfactored test vehicle effect T and the unfactored gross rating load effect W.

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Table 8.8.2.3.1-1—Values for K_b

Can member behavior be extrapolated to 1.33W?		Magnitude of Test Load			
Yes	No	$\frac{T}{W} < 0.4$	$0.4 < \frac{T}{W} \le 0.7$	$\frac{T}{W} > 0.7$	K_b
√		√			0
√			√		0.8
√				√	1.0
	√	√			0
	√		√		0
	√			√	0.5

The factor K_b should be assigned a value between 0 and 1.0 to indicate the level of test benefit that is expected at the rating load level. $K_b = 0$ reflects the inability of the test team to explain the test behavior or validate the test results, whereas $K_b = 1$ means that the test measurements can be directly extrapolated to performance at higher loads corresponding to the rating levels.

8.8.3—Proof Load Tests

8.8.3.1—Introduction

Proof load testing provides an alternative to analytically computing the load rating of a bridge. A proof test "proves" the ability of the bridge to carry its full dead load plus some "magnified" live load. A larger load than the live load the bridge is expected to carry is placed on the bridge. This is done to provide a margin of safety in the event of an occasional overload during the normal operation of the bridge.

The proof loads provide a lower bound on the true strength capacity of the components and hence leads to a lower bound on the load-rating capacity. A satisfactory proof load test usually provides higher confidence in the load capacity than a calculated capacity.

8.8.3.2—Approach

During a proof load test, the loads must be incremented and the response measured until the desired load is reached or until the test is stopped for reasons cited below. Loads must also be moved to different positions to properly check all load path components. Upon load removal, the structure should again be inspected to see that no damage has occurred and that there are no residual movements or distress.

Usually, the loads are applied in steps so that the response of the bridge under each load increment can be monitored for linear-elastic behavior and to limit distress due to cracking or other physical damage. The proof load test is usually terminated when either of the following occurs:

1. The desired live load plus the appropriate margin of safety is reached.

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2. The bridge response exhibits the start of nonlinear behavior or other visible signs of distress, such as buckle patterns appearing in compressive zones in steel or cracking in concrete.

The test loads must provide for both the rating vehicles, including the dynamic load allowance, and a load factor for the required margins of safety. The load factor may be as described in Article 8.8.3.3 or as specified by the Bridge Agency.

8.8.3.3—Target Proof Loads

8.8.3.3.1—Selection of Target Live-Load Factor

C8.8.3.3.1

A proof test provides information about the bridge capacity, including dead-load effect, live-load distributions, and component strengths. However, other uncertainties, in particular the possibility of bridge overloads during normal operations as well as the impact allowance, are not measured during the test. These remaining uncertainties should be considered in establishing a target proof load.

X_p represents the target live-load factor (applied to the test load) needed to bring the bridge to a rating factor of 1.0. If the test safely reaches this level of load, namely the legal rating plus impact allowance magnified by the factor X_p, then the rating factor is 1.0. The proof test load factors are calibrated to provide the same safety targets implicit in the calculated ratings using load and resistance factor rating procedures. Only the live load is factored during the proof test. The dead load is assumed to be the mean value.

Higher proof loads may also be warranted to incorporate ratings for permit vehicles, and in this instance the permit load vehicle plus dynamic load allowance should be magnified by X_p.

Several site conditions may have an influence on the load rating. These factors are included herein by making adjustments to X_p to account for such conditions. Each of these adjustment quantities is presented below. After X_{pA} (the adjusted X_p) is obtained, this value is multiplied by the rating load plus dynamic load allowance to get the proof-load magnitude that is needed to reach a rating factor of 1.0.

The recommended base value for X_p before any adjustments are applied is 1.40. This value was calibrated to give the same overall reliability as the level inherent in the calculated load capacity. The 1.40 factor on live loads may be reduced if the purpose of the test is solely to verify a rating for a permit load. In this case the corresponding permit load factors given in Table 6A.4.5.4.2a-1 should be used.

For strength based on test:

$$R_n = 1.40\left(L + I\right) + D \qquad (8.8.3.3.1\text{-}1)$$

For strength based on calculation:

$$R_n = \gamma_L\left(L + I\right) + \gamma_D D \qquad (8.8.3.3.1\text{-}2)$$

The reliability levels associated with Eqs. 8.8.2.3.1-1 and 8.8.2.3.1-2 are equivalent because the strength value obtained from a proof test is more reliable than that obtained solely by analytical methods.

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The following are some of the adjustments to X_p that should be considered in selecting a live-load test magnitude to achieve a rating factor of 1.0, as given in Table 8.8.2.3.1-1. Any of these adjustments may be neglected, however, if the posting and permit policies of the agency already include allowances for these factors.

1. For most situations, the live-load factor applies to a test with loads in two lanes. If one-lane load controls response, then increase X_p by 15 percent. This increase is consistent with overload statistics generated for the *AASHTO LRFD Bridge Design Specifications*.
2. For spans with fracture-critical details, the live load factor X_p shall be increased by ten percent in order to raise the reliability level to a safer level. A similar increase in test load shall be considered for any structure without redundant load paths.
3. Increase X_p by ten percent for structures in poor condition (NBI Code 4 or less) to account for increased uncertainties in resistance and future deterioration. A five-percent reduction in test load may be taken if an in-depth inspection is performed.
4. If the structure is rateable, that is, there are no hidden details, and if the calculated rating factor exceeds 1.0, X_p can be reduced by five percent. The test in this instance is performed to confirm calculations.
5. Reduction in test load is warranted for bridges with reduced traffic intensity.

Table 8.8.3.3.1-1—Adjustments to X_p

Consideration	Adjustment
One-Lane Load Controls	+15%
Nonredundant Structure	+10%
Fracture-Critical Details Present	+10%
Bridges in Poor Condition	+10%
In-Depth Inspection Performed	–5%
Rateable, Existing $RF \geq 1.0$	–5%
$ADTT \leq 1{,}000$	–10%
$ADTT \leq 100$	–15%

The adjustments described above should be considered as minimum values; larger values may be selected by the Engineer as deemed appropriate.

8.8.3.3.2—Application of Target Live-Load Factor, X_{pA}

Applying the adjustments recommended above leads to the target live-load factor X_{pA}. The net percent increase in X_p (Σ percent) is found by summing the appropriate adjustments given above. Then:

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$$X_{pA} = X_p\left(1+\frac{\sum\%}{100}\right) \qquad (8.8.3.3.2\text{-}1)$$

The target proof load L_T is then:

$$L_T = X_{pA}L_R\left(1+IM\right) \qquad (8.8.3.3.2\text{-}2)$$

where:

L_R = comparable unfactored live load due to the rating vehicle for the lanes loaded

IM = dynamic load allowance

X_{pA} = target adjusted live-load factor

In no case should a proof test load be applied that does not envelop the rating vehicle plus dynamic load allowance. For multiple-lane bridges, a minimum of two lanes should be loaded concurrently.

X_{pA} should not be less than 1.3 or more than 2.2.

The target proof load L_T should be placed on the bridge in stages, with the response of the bridge to the applied loads carefully monitored. The first-stage loading should not exceed $0.25L_T$ and the second stage loading should not exceed $0.5L_T$. Smaller increments of loading between load stages may be warranted, particularly when the applied proof load approaches the target load.

8.8.3.3.3—Load Capacity and Rating

At the conclusion of the proof load test, the actual maximum proof live load L_p applied to the bridge is known. The Operating level capacity OP is found as follows:

$$OP = \frac{k_O L_p}{X_{pA}} \qquad (8.8.3.3.3\text{-}1)$$

where:

X_{pA} = target live load factor resulting from the adjustments described in Article 8.8.3.3.2

k_O = factor which takes into consideration how the proof load test was terminated and is found from Table 8.8.3.3.3-1

Table 8.8.3.3.3-1—Values for *ko*

Terminated	k_O
Reached Target Load	1.00
Reached Distress Level	0.88

If the test is terminated prior to reaching the target load, the load L_P to be used in Eq. 8.8.3.3.3-1 should be the load just prior to reaching the load causing the distress which resulted in the termination of the test.

C8.8.3.3.3

If there are observed signs of distress prior to reaching the target proof load and the test must be stopped, then the actual maximum proof live load must be reduced by 12 percent by means of the factor k_O. This reduction is consistent with observations that show that nominal material properties used in calculations are typically 12 percent below observed material properties from tests.

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The rating factor at the operating level RF_o is:

$$RF_o = \frac{OP}{L_R\left(1+IM\right)} \qquad (8.8.3.3.3\text{-}2)$$

The Operating capacity, in tons, is the rating factor times the rating vehicle weight in tons.

8.9—USE OF LOAD TEST RESULTS IN PERMIT DECISIONS

Load tests may be used to predict load capacity for purposes of reviewing special permit loads which exceed the normal legal levels. These tests should be carried out using a load pattern similar to the effects of the permit vehicle. Special consideration should be given in the interpretation of the tests and the review of the permit load calculations to the following:

1. Will other traffic be permitted on the bridge when the permit load crosses the structure?
2. Will the load path of the vehicle crossing the bridge be known in advance, and can it be assured?
3. Will the speed of the vehicle be controlled to limit dynamic impact?
4. Will the bridge be inspected after the movement to ensure that the bridge is structurally sound?

Based on these considerations, the results of the bridge load test, whether diagnostic or proof, can be extrapolated to provide a basis for the review of requests for permit vehicles. If a diagnostic test has been performed, then test results should be used to predict the response of the bridge to the permit vehicle. The same modifications and reduced use of any enhancements in capacity observed during the test shall apply to the permit evaluation in the same way as discussed with the rating computation. Similarly, if the test is a proof load, it is necessary that the load effects of the test vehicles exceed the permit effects. A safety margin will also be needed to account for variations in weight of the permit trucks, the position of the loading, possible dynamic effects, and the possible presence of random traffic on the bridge when the permit vehicle crosses the bridge.

8.10—SERVICEABILITY CONSIDERATIONS

Load testing is primarily geared to evaluating the strength and safety of existing bridges. Load testing could also provide live-load stresses, stress ranges, and live-load deflections that could assist in the evaluation of fatigue and service limit states when these limit states may have been deemed to be of consequence by the evaluator. Careful pretest planning should be used to establish the needed response measurements for the purpose of evaluating the serviceability of an existing bridge.

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8.11—REFERENCES

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APPENDIX A8—GENERAL LOAD-TESTING PROCEDURES

A8.1—GENERAL

The steps required for load rating of bridges through load testing include the following:

Step 1. Inspection and theoretical load rating

Step 2. Development of load test program

Step 3. Planning and preparation for load test

Step 4. Execution of load test

Step 5. Evaluation of load test results

Step 6. Determination of final load rating

Step 7. Reporting

A8.2—STEP 1: INSPECTION AND THEORETICAL LOAD RATING

Prior to load testing, a thorough evaluation of the physical condition of the bridge by a field inspection should be carried out, followed by a theoretical load rating (where feasible) in accordance with the procedures described in Section 6. These are necessary for use as the base condition for planning and conducting the load test and to ensure the safety of the bridge under the test load. At this stage, a determination should be made as to whether load testing is a feasible alternative to establishing the load rating of the bridge.

The analytical model developed for the theoretical rating will also be used in establishing the target test loading required, predicting the response of the bridge to the test loading, evaluating the results of the load test, and establishing the final load rating for the bridge. The procedure to interpret the test results should be determined before the tests are commenced so that the instrumentation can be arranged to provide the relevant data.

A8.3—STEP 2: DEVELOPMENT OF LOAD TEST PROGRAM

A test program should be prepared prior to commencing with a load test and should include the test objectives, the type of test(s) to be performed, and related criteria. The choice of either the diagnostic or proof load test method depends on several factors including type of bridge, availability of design and as-built details, bridge condition, results of preliminary inspection and rating, availability of equipment and funds, level of risk involved, and test objectives.

A8.4—STEP 3: PLANNING AND PREPARATION FOR LOAD TEST

Careful planning and preparation of test activities are required to ensure that the test objectives are realized. At this stage, the load effects to be measured are identified, instrumentation is selected, personnel requirements are established, and test loadings are defined, all with due regard to safety considerations. The magnitude, configuration, and position of the test loading are selected based on the type of bridge and the type of test to be conducted.

A8.5—STEP 4: EXECUTION OF LOAD TEST

The first step in the execution of a load test is to install and check the instrumentation, which could usually be done without closing the bridge to traffic. The actual load test may then be conducted, preferably with the bridge closed to all vehicular and pedestrian traffic. The loads should be applied in several increments while observing structural behavior. Measurements of strains, displacements, and rotations should be taken at the start of the bridge load test and at the end of each increment. To ensure that accurate and reliable data is obtained during the test, it is important to assess the response of the bridge to repeated load positions and to account for temperature variations during the load test. Load-deformation response and deflection recovery at critical locations should be monitored to determine the onset of nonlinear behavior. Once any nonlinearity is observed, the bridge should be unloaded immediately and the deflection recovery recorded.

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A8.6—STEP 5: EVALUATION OF LOAD TEST RESULTS

At the completion of the field load test and prior to using the load test results in establishing a load rating for the bridge, the reliability of the load test results should be considered in evaluating the overall acceptability of the test results. It is important to understand any differences between measured load effects and those predicted by theory. This evaluation is generally performed in the office after the completion of the load test.

A8.7—STEP 6: DETERMINATION OF FINAL LOAD RATING

The determination of a revised load rating based on field testing should be done in accordance with Article 8.8.2 for Diagnostic Tests and Article 8.8.3 for Proof Tests. The rating established should be consistent with the structural behavior observed during the load test and good engineering judgment, and should also consider factors which cannot be determined by load testing, but are known to influence bridge safety.

A8.8—STEP 7: REPORTING

A comprehensive report should be prepared describing the results of field investigations and testing, description of test loads and testing procedures, types and location of instrumentation, theoretical rating, and final load rating calculations. The report should include the final assessment of the bridge according to the results of the load test and rating calculations, and may also contain recommendations for remedial actions.

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APPENDIX A:

ILLUSTRATIVE EXAMPLES

A1—SIMPLE SPAN COMPOSITE STEEL STRINGER BRIDGE

PART A—LOAD AND RESISTANCE FACTOR RATING METHOD

A1A.1—Evaluation of an Interior Stringer

A1A.1.1—Bridge Data

Span:	65 ft
Year Built:	1964
Material:	A36 Steel
	F_y = 36 ksi
	f'_c = 3 ksi
Condition:	No deterioration (NBI Item 59 = 7)
	Member is in good condition
Riding Surface:	Minor surface deviations (Field verified and documented)
ADTT (one direction):	1,000
Skew:	0°
Additional Information:	Diaphragms spaced at 16 ft 3 in.

A1A.1.2—Section Properties

In unshored construction, the noncomposite steel stringer must support its own weight plus the weight of the concrete slab. For the composite section, the concrete is transformed into an equivalent area of steel by dividing the area of the slab by the modular ratio. Live load plus impact stresses are carried by the composite section using a modular ratio of n. To account for the effect of creep, superimposed dead-load stresses are carried by the composite section using a modular ratio of $3n$ (LRFD Design 6.10.1.1.1.b). The as-built section properties are used in this analysis as there is no deterioration.

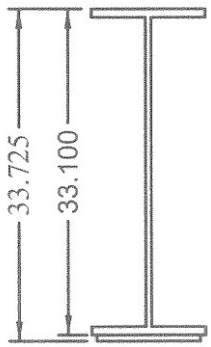
A1A.1.2.1—Noncomposite Section Properties

Section properties of rolled shapes are subject to change with changes in rolling practices of the steel industry. Identify steel components from available records, construction date, and field measurements. The section properties for this beam were determined from *AISC Manual of Steel Construction*, Sixth Edition, printed during the period from July 1963 to March 1967, which is consistent with the "Year Built" date for this bridge.

W 33 × 130
t_f = 0.855 in.
b_f = 11.51 in.
t_w = 0.58 in.
A = 38.26 in.^{2}
I = 6,699 in.^{4}

PL $^5/_8$ in. × 10 $^1/_2$ in.
t = 0.625 in.
b = 10.5 in.

$A = t \times b$ = 6.56 in.^{2}
I ~0 in.^{4} (negligible)



$$\bar{y} = \frac{\left(\frac{D_{W33\times130}}{2} + t_{PL}\right)\left(A_{W33\times130}\right) + \left(\frac{t_{PL}}{2}\right)\left(t_{PL} \times b_{PL}\right)}{A_{W33\times130} + \left(t_{PL} \times b_{PL}\right)}$$

$$\bar{y} = \frac{(17.175)(38.26) + (0.313)(6.56)}{38.26 + 6.56} \text{ Distance to } C.G.$$

$$\bar{y} = 14.71 \text{ in. from bottom of section to centroid}$$

$$I_x = 6{,}699 + 38.26(2.47)^2 + 6.56(14.40)^2$$

$$I_x = 8{,}293 \text{ in.}^4$$

$$S_t = \frac{8{,}293}{19.02} = 436.0 \text{ in.}^3 \quad \text{Section Modulus at top of steel}$$

$$S_b = \frac{8{,}293}{14.71} = 563.7 \text{ in.}^3 \quad \text{Section Modulus at bottom of steel}$$

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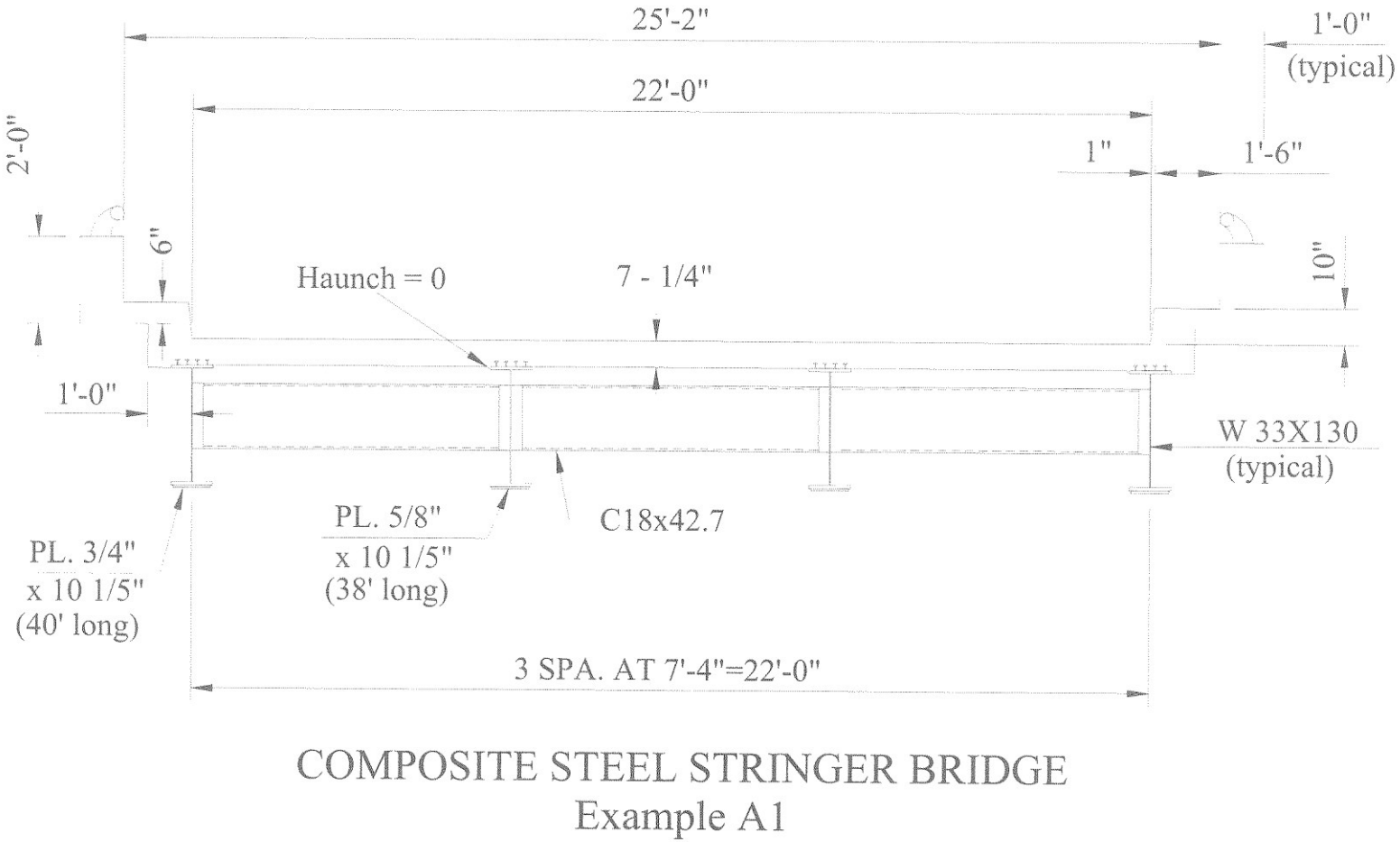

Figure A1A.1.2.1-1—Composite Steel Stringer Bridge

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A1A.1.2.2—Composite Section Properties (LRFD Design 4.6.2.6.1)

Effective Flange Width, b_e, may be taken as the tributary width perpendicular to the axis of the member.

b_e = 7'–4" = 88 in.

Modular Ratio, n LRFD Design 6.10.1.1.1b

f'_c = 3 ksi

For $2.9 < f'_c < 3.6$, $n = 9$ LRFD Design C6.10.1.1.1b

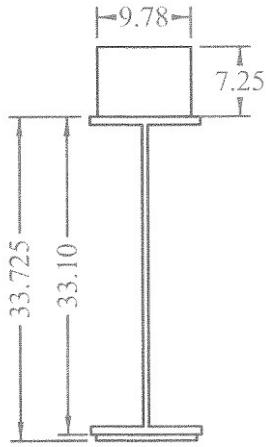
Typical Interior Stringer:

Short-Term Composite, (n):

$W33 \times 130$, *PL* $^5/_8$ in. × $10^1/_2$ in. and *Conc.* $7^1/_4$ in. × 88 in.

Effective Flange Width, $b_e = \dfrac{88}{n} = 9.78$ in.

Transformed Slab



$$\overline{y} = \frac{(17.175)(38.26) + (0.313)(6.56) + \left(\frac{88}{9} \times 7.25\right)(37.35)}{38.26 + 6.56 + \left(\frac{88}{9} \times 7.25\right)}$$

$\overline{y} = 28.58$ in. from bottom of section to centroid

$$I_x = (6,699) + (38.26)(11.40)^2 + (6.56)(28.27)^2 + \frac{\left(\frac{88}{9}\right)(7.25)^3}{12} + \left(\frac{88}{9} \times 7.25\right) \times (8.77)^2$$

$I_x = 22,677$ in.^{4}

$$S_t = \frac{22,677}{5.14} = 4,412 \text{ in.}^3$$ Section Modulus at top of steel

$$S_b = \frac{22,677}{28.58} = 793 \text{ in.}^3$$ Section Modulus at bottom of steel

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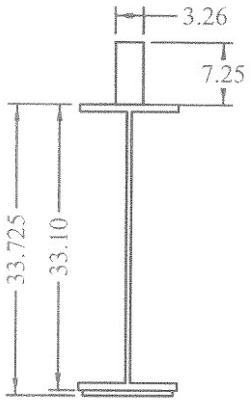

Long-Term Composite, 3*n*:

*W*33 × 130, *PL* 5/8 in. × 10 1/2 in. and *Conc.* 7 1/4 in. × 88 in.

Effective Flange Width, $b_e = \frac{88}{(3\times 9)} = 3.26$ in.

$$\bar{y} = \frac{(17.175)(38.26)+(0.313)(6.56)+\left(\frac{88}{27}\times 7.25\right)(37.35)}{38.26+6.56+\left(\frac{88}{27}\times 7.25\right)}$$

$\bar{y} = 22.52$ in. from bottom of section to centroid

$$I_x = (6,699)+(38.26)(5.34)^2+(6.56)(22.21)^2+\frac{\left(\frac{88}{27}\right)(7.25)^3}{12}+\left(\frac{88\times 7.25}{27}\right)\times(14.83)^2$$

$I_x = 16,326$ in.^{4}

$S_t = \frac{16,326}{11.20} = 1,458$ in.^{3} Section Modulus at top of steel

$S_b = \frac{16,326}{22.52} = 725$ in.^{3} Section Modulus at bottom of steel

A1A.1.2.3—Summary of Section Properties at Midspan

A1A.1.2.3a—Steel Section Only

S_{TOP} = 436 in.^{3}

S_{BOT} = 563.7 in.^{3}

A1A.1.2.3b—Composite Section—Short Term, n = 9

$S_{TOPsteel}$ = 4,412 in.^{3}

S_{BOT} = 793 in.^{3}

A1A.1.2.3c—Composite Section—Long Term, 3n = 27

$S_{TOPsteel}$ = 1,458 in.^{3}

S_{BOT} = 725 in.^{3}

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A1A.1.3—Dead-Load Analysis—Interior Stringer

A1A.1.3.1—Components and Attachments, DC

In general, attachments may include connection plates, stiffeners, diaphragms, bracing, and other miscellaneous components. A refined rating calculation accounts for major weight components; alternatively, a percentage of stringer weight can be used as an estimate. For this example, three interior diaphragms were taken into account and end diaphragms that are directly over the supports were neglected when estimating uniform span loads.

A1A.1.3.1a—Noncomposite Dead Loads, DC_1

Deck: $(7.33\text{ ft})\dfrac{(7.25\text{ in.})}{12}(0.150\text{ kcf})$ = 0.664 kip/ft

Stringer: (0.130 kip/ft) (1.06) = 0.138 kip/ft
(six percent increase for connections)

Cover Plate:

$$\frac{(0.625\text{ in.})(10.5\text{ in.})\left(\dfrac{0.490\text{ kcf}}{144}\right)(1.06)(38\text{ ft})}{65\text{ ft}} = 0.014\text{ kip/ft}$$

Diaphragms: $\dfrac{(3)(0.0427\text{ kip/ft})(7.33\text{ ft})(1.06)}{65\text{ ft}}$ = 0.015 kip/ft

Total per stringer = 0.831 kip/ft

$$M_{DC1} = \frac{0.831(65)^2}{8} = 439\text{ kip-ft at midspan}$$

$$V_{DC1} = 0.831\left(\frac{65}{2}\right) = 27\text{ kips at bearing}$$

A1A.1.3.1b—Composite Dead Loads, DC_2

All permanent loads on the deck are uniformly distributed among the beams. LRFD Design 4.6.2.2.1

The unit weight of reinforced concrete is generally taken as .005 kcf greater than the unit weight of plain concrete, hence for estimating concrete loads 0.150 kcf was assumed. LRFD Design C3.5.1

Curb: $(1\text{ ft})\left(\dfrac{10\text{ in.}}{12}\right)(0.150\text{ kcf})\left(\dfrac{2\text{ curbs}}{4\text{ beams}}\right)$ = 0.062 kip/ft

Parapet:

$$\left[\left(\frac{6\text{ in.}\times 19\text{ in.}}{144}\right)+\left(\frac{18\text{ in.}\times 12\text{ in.}}{144}\right)\right](0.150\text{ kcf})\left(\frac{2\text{ parapets}}{4\text{ beams}}\right) = 0.172\text{ kip/ft}$$

Railing: Assume $0.020\,\text{kip/ft}\left(\dfrac{2\text{ railings}}{4\text{ beams}}\right)$ = 0.010 kip/ft

Total per stringer = 0.244 kip/ft

$$M_{DC2} = \frac{0.244(65)^2}{8} = 129\text{ kip-ft at midspan}$$

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$$V_{DC2} = 0.244\left(\frac{65}{2}\right) = 8 \text{ kips at bearing}$$

A1A.1.3.2—Wearing Surface

$DW = 0$

A1A.1.4—Live Load Analysis—Interior Stringer (LRFD Design Table 4.6.2.2.1-1)

A1A.1.4.1—Compute Live Load Distribution Factors (Type (a) cross section)

Longitudinal Stiffness Parameter, K_g LRFD Design 4.6.2.2.1

$$K_g = n\left(I + Ae_g^2\right)$$ LRFD Design Eq. 4.6.2.2.1-1

in which $n = \frac{E_B}{E_D}$ LRFD Design Eq. 4.6.2.2.1-2

$$E_D = 33,000\left(w_c\right)^{1.5}\sqrt{f_c'}$$ LRFD Design Eq. C5.4.2.4-2

$$= 33,000\left(0.145\right)^{1.5}\sqrt{3}$$

$$= 3,155.9 \text{ ksi}$$

$E_B = 29,000$ ksi

Beam + Cov. *PL*

I = 8,293 in.^{4}

A = 44.82 in.^{2}

e_g = $\frac{1}{2}$ (7.25) + 19.02 = 22.65 in.

$$K_g = \frac{29,000}{3,155.9}\left(8,293 + 44.82 \times 22.65^2\right)$$

K_g = 287,498 in.^{4}

A1A.1.4.1a—Distribution Factor for Moment, g_m (LRFD Design Table 4.6.2.2.2b-1)

$$\frac{K_g}{12.0Lt_s^3} = \frac{287,498}{12 \times 65 \times 7.25^3} = 0.967$$

One Lane Loaded:

$$g_{m1} = 0.06 + \left(\frac{S}{14}\right)^{0.4}\left(\frac{S}{L}\right)^{0.3}\left(\frac{K_g}{12.0Lt_s^3}\right)^{0.1}$$

$$= 0.06 + \left(\frac{7.33}{14}\right)^{0.4}\left(\frac{7.33}{65}\right)^{0.3}\left(0.967\right)^{0.1}$$

$$= 0.460$$

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Two or More Lanes Loaded:

$$g_{m2} = 0.075 + \left(\frac{S}{9.5}\right)^{0.6}\left(\frac{S}{L}\right)^{0.2}\left(\frac{K_g}{12.0Lt_s^3}\right)^{0.1}$$

$$= 0.075 + \left(\frac{7.33}{9.5}\right)^{0.6}\left(\frac{7.33}{65}\right)^{0.2}(0.967)^{0.1}$$

$$= 0.626 > 0.460$$

$\therefore$ use $g_m = 0.626$

A1A.1.4.1b—Distribution Factor for Shear, g_v (LRFD Design 4.6.2.2.3a)

One Lane Loaded:

$$g_{v_1} = 0.36 + \frac{S}{25.0}$$ LRFD Design Table 4.6.2.2.3a-1

$$= 0.36 + \frac{7.33}{25.0}$$

$$= 0.653$$

Two or More Lanes Loaded:

$$g_{v_2} = 0.2 + \frac{S}{12} - \left(\frac{S}{35}\right)^{2.0}$$ LRFD Design Table 4.6.2.2.3a-1

$$= 0.2 + \frac{7.33}{12} - \left(\frac{7.33}{35}\right)^{2.0}$$

$$= 0.767 > 0.653$$

$\therefore$ use $g_v = 0.767$

A1A.1.4.2—Compute Maximum Live Load Effects

A1A.1.4.2a—Maximum Design Live Load (HL-93) Moment at Midspan

The maximum moment effects are estimated to occur with the design live load centered on the span. Calculate moments by statics.

$$\text{Design Lane Load Moment} = \frac{wl^2}{8} = \frac{0.640\,\text{klf}\,(65\,\text{ft})^2}{8} = 338\,\text{kip-ft}$$

Design Truck Moment with the middle axle located at midspan:

$$\text{Design Truck Moment} = \frac{P_{32}\ell}{4} + \frac{\text{D}_{32})xb}{\ell}$$

$$= \frac{32^k \times 65\,\text{ft}}{4} + \frac{\left(8^k + 32^k\right)32.5\,\text{ft} \times 18.5\,\text{ft}}{65\,\text{ft}}$$

Design Truck Moment = 890 kip-ft Governs

Tandem Axles Moment with tandem axles located equidistant from midspan:

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Tandem Axles Moment $= P_{25}a = 25^k \times 30.5\,\text{ft} = 762.5$ kip-ft

IM = 33 percent LRFD Design Table 3.6.2.1-1

M_{LL+IM} = 338 + 890 × 1.33

= 1,521.7 kip-ft

A1A.1.4.2b—Maximum Design Live Load Shear at Beam Ends

The maximum shear effects occur with the heaviest axle located to create the maximum end reaction. Calculate shears by statics.

Design Lane Load Shear $= \frac{wl}{2} = \frac{0.640\,\text{klf}\left(65\,\text{ft}\right)}{2} = 20.8\,\text{kips}$

Design Truck Shear $= P_{32} + P_{32}\left(\frac{1 - x_{32}}{l}\right) + P_8\left(\frac{1 - x_8}{l}\right)$

$= 32^k + 32^k\left(\frac{65\,\text{ft} - 14\,\text{ft}}{65\,\text{ft}}\right) + 8^k\left(\frac{65\,\text{ft} - 28\,\text{ft}}{65\,\text{ft}}\right)$

Design Truck Shear = 61.7 kips Governs

Tandem Axles Shear $= P_{25} + P_{25}\left(\frac{1 - x_{25}}{l}\right) = 25^k + 25^k\left(\frac{65\text{ ft} - 4\text{ ft}}{65\text{ ft}}\right) = 48.5$ kips

V_{LL+IM} = 20.8 kips + 61.7 kips × 1.33

= 102.9 kips

A1A.1.4.2c—Distributed Live Load Moments and Shears

Design Live-Load HL-93:

M_{LL+IM} = 1,521.7 × g_m

= 1,521.7 × 0.626

= 952.6 kip-ft

V_{LL+IM} = 102.9 × g_v

= 102.9 × 0.767

= 78.9 kips

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A1A.1.5—Compute Nominal Resistance of Section at Midspan

Locate Plastic Neutral Axis PNA:

t_f = 0.855 in.

t_w = 0.58 in.

b_f = 11.51 in.

Cov. *PL* Area A_p

= 6.56 in.^{2}

(*PL* $^5/_8$ in. × 10$^1/_2$ in.)

Web Depth:

D = 33.10 in. – 2 (0.855 in.)

= 31.39 in.

Treat the bottom flange and the cover plate as one element.

$$A_t = (11.51)(0.855)+(10.5)(0.625)=16.40 \text{ in.}^2$$

$$y = \frac{(11.51)(0.855)\frac{(0.855)}{2}+(10.5)(0.625)\left(0.855+\frac{0.625}{2}\right)}{(11.51)(0.855)+(10.5)(0.625)}$$

= 0.724 in. (from top of tension flange to centroid of flange and cover plate)

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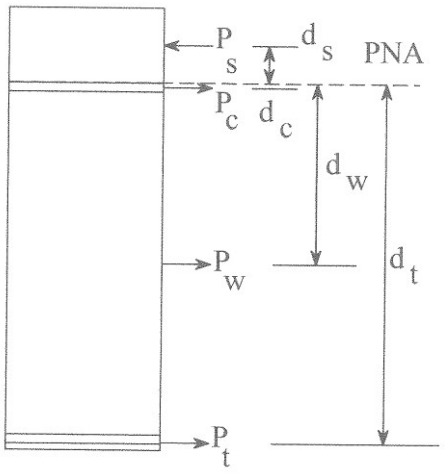

Plastic Forces LRFD Design Appendix D6.1

Note the forces in longitudinal reinforcement may be conservatively neglected.

Set P_{rb} and P_{rt} = 0

P_s = $0.85 f'_c b_{eff} t_s$

= 0.85 × 3.0 × 88 × 7.25

= 1626.9 kips

$\frac{c_{rb}}{t_s} = \frac{5.25}{7.25}$

where c_{rb} is the distance from the top of the concrete slab to the center of the bottom layer of the longitudinal concrete deck reinforcement and t_s is the thickness of the concrete deck. Asssume cover + $^1/_2$ bar diameter = 2 in., then c_{rb} equals 5.25 in.

P_c = $F_y A_c$ where $A_c = b_f t_f$

= 36 × 11.51 × 0.855

= 354.3 kips

P_w = $F_y D t_w$

D = 33.10 – 2 x 8.55 = 31.39

= 36 × 31.39 × 0.58

= 655.4 kips

P_t = $F_y A_t$ where $A_t = b_f t_f + A_p$

= 36(11.51 × 0.855 + 6.56)

= 590.4 kips

$$P_t + P_w + P_c = 590.4 + 655.4 + 354.3 = 1,600.1 \text{ kips}$$

$$\frac{c_{rb}}{t_s} P_s + P_{rb} + P_{rt} = \frac{5.25}{7.25} 1,626.9 + 0.0 + 0.0 \text{ kips} = 1,178.1 \text{ kips}$$

$$P_c + P_w + P_t \geq \frac{c_{rb}}{t_s} P_s + P_{rb} + P_{rt}$$

$$1,600.1 \geq 1,178.1$$

The PNA lies in the slab; only a portion of the slab (depth = $\overline{y}$) is required to balance the plastic forces in the steel beam.

$$\overline{Y} = (t_s)\left[\frac{P_c + P_w + P_t - P_{rt} - P_{rb}}{P_s}\right]$$ LRFD Design Appendix D6.1

$$\overline{Y} = (7.25)\frac{1,600.1}{1,626.9}$$

$\overline{Y}$ = 7.13 in. from the top of the concrete deck slab

A1A.1.5.1—Classify Section (LRFD Design 6.10.7 and Figure C6.4.5-1)

Following the I-Sections in Positive Flexure Flowchart
(Section is considered to be Constant Depth)

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A1A.1.5.1a—Check Web Slenderness (LRFD Design 6.10.6.2.2)

Since PNA is in the slab, the web slenderness requirement is automatically satisfied.

For composite sections in positive bending, the remaining stability criteria are automatically satisfied. The section is compact.

A1A.1.5.1b—Check Ductility Requirement (LRFD Design 6.10.7.1.2)

$D_p = \overline{Y} = 7.13$ in.

D_t = Depth of Composite Section
$= d + t_s = 33.725 + 7.25$
$= 40.98$ in.

If $D_p \le 0.1D_t$, then $M_n = M_p$ LRFD Design Eq. 6.10.7.1.2-1

Otherwise, $M_n = M_p\left(1.07 - 0.7\frac{D_p}{D_t}\right)$ LRFD Design Eq. 6.10.7.1.2-2

$0.1D_t = 0.1 \times 40.98 = 4.098$ in.

7.13 in. $\not\le$ 4.098 in. therefore calculate $M_n < M_p$

A1A.1.5.2—Plastic Moment, M_p

Moment arms about the PNA:

Compression Flange: $d_c = \left(t_s - \overline{Y}\right) + \frac{t_c}{2}$

$= \left(7.25 - 7.13\right) + \frac{0.855}{2}$

$= 0.55$ in.

$d_w = \left(t_s - \overline{Y}\right) + t_c + \frac{D}{2}$

$= \left(7.25 - 7.13\right) + 0.855 + \frac{31.39}{2}$

$= 16.67$ in.

Tension Flange: $d_t = \left(t_s - \overline{Y}\right) + t_c + D + \frac{t_t}{2}$

$= \left(7.25 - 7.13\right) + 0.855 + 31.39 + 0.724$

(0.724 in. is the distance to the centroid of the bottom flange and cover plate from the top of the flange)

$= 33.09$ in.

The plastic moment, M_p, is the sum of the moments of the plastic forces about the PNA.

$M_p = \left(\frac{\overline{Y}^2 P_s}{2t_s}\right) + [P_{rt}d_{rt} + P_{rb}d_{rb} + P_c d_c + P_w d_w + P_t d_t]$ LRFD Design Table D6.1-1

$= \left(\frac{7.13^2 \times 1,626.9}{2 \times 7.25}\right) + \left[0 + 0 + 354.3 \times 0.55 + 655.4 \times 16.67 + 590.4 \times 33.09\right]$

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= 36,361 kip-in. or 3,030 kip-ft

A1A.1.5.3—Nominal Flexural Resistance, M_n (LRFD Design 6.10.7.1.2)

$$D_p \not\leq 0.1D_t$$ LRFD Design Eq. 6.10.7.1.2-1

$$\text{Therefore, } M_n = M_p\left(1.07 - 0.7\frac{D_p}{D_t}\right)$$ LRFD Design Eq. 6.10.7.1.2-2

Flange lateral bending stress: $f_l = 0$.

A1A.1.5.4—Nominal Shear Resistance, V_n (LRFD Design 6.10.9.2)

*W*33 × 130 Rolled section, no stiffeners.

D = *d* – 2*tf* (Clear distance between flanges)
= 33.1 – 2 × 0.855
= 31.39 in.

$$\text{If } \frac{D}{t_w} \leq 1.12\sqrt{\frac{Ek}{F_{yw}}} \text{ then } C = 1.0$$

where *k* = 5 for unstiffened web

$$\frac{D}{t_w} = \frac{31.39}{0.580} = 54.12$$

$$1.12\sqrt{\frac{Ek}{F_{yw}}} = 1.12\sqrt{\frac{29,000 \times 5}{36}} = 71.1$$ LRFD Design Eq. 6.10.9.3.2-4

$$54.12 \leq 71.1, \text{therefore } C = 1.0$$

then:

$$V_n = V_{cr} = CV_p$$ LRFD Design Eq. 6.10.9.2-1

$$\text{where } V_p = 0.58F_{yw}Dt_w$$ LRFD Design Eq. 6.10.9.2-2

= 1.0 x 0.58 × 36 × 31.39 × 0.580

= 380.15 kips

A1A.1.5.5—Summary for Interior Stringer

	Dead Load DC_1	Dead Load DC_2	Live Load Distribution Factor	Dist. Live Load + Impact	Nominal Capacity
Moment, kip-ft	439.0	129.0	*gm* = 0.626	952.6	2,873.0
Shear, kips	27.0	8.0	*gv* = 78.9	78.9	380.15

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A1A.1.6—General Load-Rating Equation

$$RF = \frac{C - (\gamma_{DC})(DC) - (\gamma_{DN})(DW) \pm (\gamma_P)(P)}{(\gamma_L)(LL + IM)}$$ Eq. 6A.4.2.1-1

A1A.1.7—Evaluation Factors (for Strength Limit States)

1. Resistance Factor, φ LRFD Design 6.5.4.2
   φ = 1.0 for flexure and shear

2. Condition Factor, φ_c 6A4.2.3
   φ_c = 1.0 Member is in good condition. NBI Item 59 = 7.

3. System Factor, φ_s 6A.4.2.4
   φ_s = 1.0 4-girder bridge, spacing > 4 ft (for flexure and shear).

A1A.1.8—Design Load Rating (6A.4.3)

A1A.1.8.1—Strength I Limit State (6A.6.4.1)

Capacity $C = (\varphi_c)(\varphi_s)(\varphi)R_n$

$$RF = \frac{(\varphi_c)(\varphi_s)(\varphi)R_n - (\gamma_{DC})(DC) - (\gamma_{DW})(DW)}{(\gamma_L)(LL + IM)}$$

A1A.1.8.1a—Inventory Level

Load Factors Table 6A.4.2.2-1

γ_{DC}	1.25
γ_{DW}	1.50
γ_{LL}	1.75

The dead load demands established for load cases DC_1 and DC_2 are permanent loads and therefore the load factor for these loads will be taken from the load case *DC*.

$$RF = \frac{(1.0)(1.0)(1.0)(2{,}873) - (1.25)(439 + 129) - (1.50)(0)}{(1.75)(952.6)}$$

Flexure: $$RF = \frac{(1.0)(1.0)(1.0)(2{,}873) - (1.25)(439) - (1.25)(129)}{(1.75)(952.6)}$$

$$= 1.2975$$

Note: The general rule for simple spans carrying moving concentrated loads states: the maximum bending moment produced by moving concentrated loads occurs under one of the loads when that load is as far from one support as the center of gravity of all the moving loads on the beam is from the other support. In a refined analysis with the HL-93 truck located in such a manner, the resulting rating factor for flexure is *RF* = 1.2922 for this stringer. It should be understood that locating the precise critical section and load position for rating depends on the combined influence of dead load, live load, member capacity, and load factors that make up the general rating factor equation.

Shear: $$RF = \frac{(1.0)(1.0)(1.0)(360.3) - (1.25)(27 + 8)}{(1.75)(78.9)}$$

$$= 2.29$$

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A1A.1.8.1b—Operating Level

Load	Load Factor γ	Table 6A.4.2.2-1
DC	1.25	
LL	1.35	

For Strength I Operating Level, only the live-load factor changes; therefore, the rating factor can be calculated by direct proportions.

Flexure: $RF = 1.29 \times \frac{1.75}{1.35}$

$= 1.67$

Shear: $RF = 2.29 \times \frac{1.75}{1.35}$

$= 2.97$

A1A.1.8.2—Service II Limit State (6A.6.4.1)

Capacity $C = f_R$

$$RF = \frac{f_R - (\gamma_{DC})(f_{DC}) - (\gamma_{DW})(f_{DW}) \pm (\gamma_P)(f_P)}{(\gamma_{LL})(f_{LL+IM})}$$ Eq. 6A.6.4.2.1-1

For this example, the terms:

$(\gamma_{DW})(f_{DW}) \pm (\gamma_P)(f_P)$

do not contribute and the general equation reduces to:

$$RF = \frac{f_R - (\gamma_{DC})(f_{DC})}{(\gamma_{LL})(f_{LL+IM})}$$

A1A.1.8.2a—Inventory Level

Allowable Flange Stress for tension flange $f_R = 0.95 R_h F_{yf}$ $(f_\ell = 0)$ LRFD Design Eq. 6.10.4.2.2-2

Checking the tension flange as compression flanges typically do not govern for composite sections.

R_h = 1.0 for non-hybrid sections LRFD Design 6.10.1.10.1

$f_R = 0.95 \times 1.0 \times 36$

$= 34.2$ ksi

$f_D = f_{DC_1} + f_{DC_2}$

$= \frac{439 \times 12}{563.7} + \frac{129 \times 12}{725}$

$= 9.35 + 2.14 = 11.49$ ksi

$f_{LL+IM} = \frac{952.6 \times 12}{793} = 14.42$ ksi

$\gamma_{LL} = 1.30$ $\gamma_{DC} = 1.0$ Table 6A.4.2.2-1

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$$RF = \frac{34.2-(1.0)(11.49)}{(1.3)(14.42)}$$

$$= 1.21$$

A1A.1.8.2b—Operating Level

$\gamma_{LL} = 1.0 \qquad \gamma_{DC} = 1.0$ Table 6A.4.2.2-1

$$RF = \frac{34.2-(1.0)(11.49)}{(1.0)(14.42)}$$

$$= 1.57$$

A1A.1.8.3—Fatigue Limit State (6A.6.4.1)

Determine if the bridge has any fatigue-prone details (Category C or lower).

The transverse welds detail connecting the ends of cover plates to the flange are fatigue-prone details. Use Category E' details because the flange thicknesss = 0.855 in. is greater than 0.8 in. LRFD Design Table 6.6.1.2.3-1

If $2.2(\Delta f)_{tension} > f_{dead\text{-}load\ compression}$, the detail may be prone to fatigue. Eq. 7.2.3-1

$f_{dead\text{-}load\ compression}$ 7.2.3

= 0 at cover plate at all locations because beam is a simple span and cover plate is located in the tension zone

$\therefore$ must consider fatigue; compute *RF* for fatigue load for infinite life.

$$RF = \frac{f_R-(\gamma_{DC})(f_{DC})}{(\gamma_{LL})(\Delta f_{LL+IM})_{max}}$$

$f_R = (\Delta F)_{TH}$

$\gamma_{LL} = 0.80 \qquad \gamma_{DC} = 0.00$ Table 6A.4.2.2-1

Composite section properties without cover plate:

$$\bar{y} = \frac{\sum A \times \bar{y}}{\sum A} = \frac{(38.26)(16.55)+\left(\frac{88}{9}\times 7.25\right)(36.725)}{(38.26)+\left(\frac{88}{9}\times 7.25\right)}$$

= 29.65 in. from bottom of flange

$$I_x = 6,699+(38.26)(13.10)^2+\frac{\left(\frac{88}{9}\right)(7.25)^3}{12}+\frac{88}{9}(7.25)(7.07)^2$$

$$= 17{,}119 \text{ in.}^4$$

$$S_b = \frac{17{,}119}{29.65} = 577 \text{ in.}^3$$

Live Load at Cover Plate Cut-Off (13.5 ft. from centerline of bearing)

Fatigue Load: Design truck with a spacing of 30 ft between 32 kip axles. LRFD Design 3.6.1.4.1 and Figure 3.6.1.2.2-1

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M_{LL} = (32 kips) (10.69 ft) + (32 kips) (4.46 ft) + (8 kips) (1.56 ft)

= 497 kip-ft = 5,967 kip-in. Using influence lines.

IM = 15 percent LRFD Design Table 3.6.2.1-1

M_{LL+IM} = (1.15) (5,967) = 6,862 kip-in.

A1A.1.8.3a—Load Distribution for Fatigue LRFD Design 3.6.1.4.3b

The single-lane distribution factor will be used for fatigue. LRFD Design 3.6.1.1.2

Remove multiple presence factor from the single-lane distribution. LRFD Design C3.6.1.1.2

$$g_{Fatigue} = \frac{1}{1.2}(g_{m1})$$

$$= \frac{1}{1.2}(0.46)$$

$$= 0.383$$

Distributed Live-Load Moment:

gM_{LL+IM} = (0.383) (6862)

= 2,628 kip-in.

Fatigue Load Stress Range:

$$\Delta f_{LL+IM} = \frac{2,628}{577}$$

= 4.56 ksi at the cover plate weld

Nominal fatigue resistance for infinite life:

$(\Delta F)_{TH}$ = 2.6 ksi for Detail Category E′ LRFD Design Table 6.6.1.2.5-3

Infinite-Life Fatigue Check: 7.2.4

$ADTT$ = 1,000
Span Length = 65 ft
Number of lanes = 2

R_p = 0.988 + 6.87x10^{-5} Span Length + 4.01x10^{-6} $ADTT$ + 0.0107 / Number of lanes 7.2.2.1
= 0.988 + 6.87x10^{-5}*65 + 4.01x10^{-6}*1,000+0.0107/2
= 1.0018

$(\Delta f)_{max} = (R_p)(\Delta f_{FATIGUE\ I})$ = (1.0018)(1.75)(4.56) 7.2.4
= 7.99 ksi > 2.6 ksi
Thus, $(\Delta f)_{max} > (\Delta F)_{TH}$.

The detail does not possess infinite fatigue life.
Evaluate fatigue life using procedures given in Section 7.

A1A.1.8.3b—Calculation of Fatigue Life

Fatigue life determination will be based upon the finite fatigue life.

7.2.5.1

$$Y = \frac{\log\left[\frac{R_R A}{365n\left[(ADTT)_{SL}\right]_{PRESENT}((\Delta f)_{eff})^3} g(1+g)^{a-1}+1\right]}{\log(1+g)}$$

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$ADTT$ (One Direction) = 1,000 (present value) — LRFD Design
$[ADTT_{SL}]_{PRESENT}$ = 0.85(1,000) = 850 — Table 3.6.1.4.2-1

Traffic Growth Rate g: 2 percent is applied over the life of the bridge
Bridge Age a: 48 years

Assume Evaluation 1 Life to be used for bridge assessment.
Hence, R_R = 1.3 — Table 7.2.5.1-1

Calculate effective stress range:

R_p = 1.0018

R_{sa} = 1.0 — Table 7.2.2.1-1
R_{st} = 1.0
$R_s = R_{sa} \times R_{st}$ = 1.0
$\Delta f_{eff} = (R_p)(R_s)(\Delta f_{FATIGUE\ II})$ = (1.0018)(1.0)(0.80)(4.56) = 3.65 ksi — 7.2.2
$A = 3.9 \times 10^8$ ksi^3 — LRFD Design Table 6.6.1.2.5-1

n = 1.0 simple span girders with L > 40 ft — LRFD Design Table 6.6.1.2.5-2

$$Y = \frac{\log\left[\frac{(1.3)(3.9\times10^8)}{(365)(1.0)(850)(3.65^3)}(0.02)(1+0.02)^{48-1}+1\right]}{\log(1+0.02)} = 50 \text{ years}$$

A1A.1.8.3c—Calculation of Fatigue Serviceability Index

Fatigue Serviceability Index $Q = \left(\frac{Y-a}{N}\right)GRI$ — 7.2.6.1

No. of load paths (in this case, girders) = 4
G = 1.0 — Table 7.2.6.1-1
No. of Spans = 1 (Simple Span)
R = 0.90 — Table 7.2.6.1-2
N = (larger of 100 or Y) = 100
Assuming that the bridge is on an Interstate Highway, I = 0.9 — Table 7.2.6.1-3

$$Q = \left(\frac{50-48}{100}\right)(1)(0.9)(0.9) = 0.02$$

Based on the value of the Fatigue Serviceability Index, the bridge owner will need to define the inspection frequency based upon the importance of the structure. Note, however, that the Fatigue Serviceability Index value could be increased if the owner decided to accept a greater risk of fatigue cracking and use an Evaluation 2 Life estimate instead of the Evaluation 1 Life estimate. This is illustrated below for the same example.

Assume that Evaluation 2 Life is used for the bridge fatigue assessment.
Hence, R_R = 1.6 — Table 7.2.5.1-1
$(\Delta f)_{eff}$ = 3.65 ksi
$A = 3.9 \times 10^8$ ksi^3 — LRFD Design Table 6.6.1.2.5-1

n = 1.0 simple span girders with L > 40 ft — LRFD Design Table 6.6.1.2.5-2

$$Y = \frac{\log\left[\frac{(1.6)(3.9\times10^8)}{(365)(1.0)(850)(3.65^3)}(0.02)(1+0.02)^{48-1}+1\right]}{\log(1+0.02)} = 57 \text{ years}$$

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CALCULATION OF FATIGUE SERVICEABILITY INDEX

Fatigue Serviceability Index $Q = \left(\frac{Y-a}{N}\right)GRI$ 7.2.6.1

No. of load paths (in this case, girders) = 4

G = 1.0 Table 7.2.6.1-1

No. of Spans = 1 (Simple Span)

R = 0.90 Table 7.2.6.1-2

N = (larger of 100 or Y) = 100

Assuming that the bridge is on an Interstate Highway, I = 0.9 Table 7.2.6.1-3

$$Q = \left(\frac{57-48}{100}\right)(1)(0.9)(0.9) = 0.07$$

Note that the Fatigue Serviceability Index, Q, has increased from 0.06 to 0.13 as a result of accepting a greater risk of fatigue cracking at the critical detail.

A1A.1.9—Legal Load Rating 6A.6.4.2

Note: The Inventory Design Load Rating produced rating factors greater than 1.0 (with the exception of fatigue). This indicates that the bridge has adequate load capacity to carry all legal loads within LRFD exclusion limits and need not be subject to legal load ratings. Appendix A6A

The load rating computations that follow have been done for illustrative purposes. Shear ratings have not been illustrated.

Live Load: AASHTO Legal Loads—Type 3, 3S2, 3-3 (rate for all three)

g_m = 0.626

IM = 20 percent The standard dynamic load allowance of 33 percent is decreased based on a field evaluation verifying that the approach and bridge riding surfaces have only minor surface deviations or depressions. Table C6A.4.4.3-1

Table E6A-1

The following table compares interpolating to determine M_{LL} without impact for 65 ft span with exact values determined by statics. Note that for the Type 3-3, interpolating M_{LL} results in a value that is 1.5 percent greater than the true value. Judgement should be exercised whether to interpolate tabulated values.

	Type 3	Type 3S2	Type 3-3	
M_{LL} interpolated	660.7	707.2	654.5	kip-ft
M_{LL} statics	660.77	707.03	644.68	kip-ft
gM_{LL+IM}	496.3	531.2	484.3	kip-ft

Live Load: AASHTO Legal Loads—Specialized Hauling Units and Notional Rating Load—SU4, SU5, SU6, SU7, and NRL

Interpolated values shall be used for the Specialized Hauling Units in this example for illustrative purposes and to familiarize the reader with the Appendix A tables. Table E6A-2

Interpolating to determine M_{LL} without impact for 65 ft span

	SU4	SU5	SU6	SU7	NRL	
M_{LL} interpolated	744.7	821.2	913.5	994.1	1037.0	kip-ft
gM_{LL+IM}	559.4	616.9	686.2	746.8	779.0	kip-ft

A1A.1.9.1—Strength I Limit State 6A.6.4.2.1

For Types 3, 3S2, and 3-3

Dead Load DC: γ_{DC} = 1.25 Table 6A.4.2.2-1

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$ADTT$ = 1,000

Generalized Live-Load Factor for Legal Loads, γ_{LL} = 1.30 Table 6A.4.4.2.3a-1

Flexure: $RF = \dfrac{(1.0)(1.0)(1.0)(2873)-(1.25)(439)-(1.25)(129)}{(1.30)(M_{LL+IM})}$

	Type 3	Type 3S2	Type 3-3
RF	3.35	3.12	3.43

For Specialized Hauling Units and NRL

Dead Load *DC*: γ_{DC} = 1.25 Table 6A.4.2.2-1

$ADTT$ = 1,000 Assumed

Generalized Live Load Factor for Legal Loads γ_{LL} = 1.30 Table 6A.4.4.2.3b-1

Flexure: $RF = \dfrac{(1.0)(1.0)(1.0)(2,873)-(1.25)(439)-(1.25)(129)}{(1.30)(M_{LL+IM})}$

	SU4	SU5	SU6	SU7	NRL
RF	2.97	2.69	2.42	2.23	2.13

A1A.1.9.2—Service II Limit State 6A.6.4.2.2

For Types 3, 3S2, and 3-3, and for Specialized Hauling Units and NRL

γ_{LL} = 1.3 γ_D = 1.0 Table 6A.4.2.2-1

f_R = 34.2 ksi

$f_D = f_{DC_1} + f_{DC_2}$

$= \dfrac{439\times12}{563.7} + \dfrac{129\times12}{725} = 11.49$ ksi

$f_{LL+IM} = \dfrac{M_{LL+IM}\times12}{793}$

$RF = \dfrac{34.2-11.49}{1.3(f_{LL+IM})}$

	Type 3	Type 3S2	Type 3-3	
f_{LL+IM}	7.51	8.04	7.33	ksi
RF	2.33	2.17	2.38	

	SU4	SU5	SU6	SU7	NRL	
f_{LL+IM}	8.47	9.34	10.38	11.30	11.79	ksi
RF	2.06	1.87	1.68	1.55	1.48	

No posting required as $RF > 1.0$.

A1A.1.9.3—Summary

Truck	Type 3	Type 3S2	Type 3-3
Weight (tons)	25	36	40
RF (Service II Controlling)	2.33	2.17	2.38
Safe Load Capacity (tons)	58	78	95

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Truck	SU4	SU5	SU6	SU7	NRL
Weight (tons)	27	31	34.8	38.8	40
RF (Service II Controlling)	2.06	1.87	1.68	1.55	1.48
Safe Load Capacity (tons)	55	58	58	60	59

The NRL rating demonstrates Article C6A.4.4.2.1b: "Bridges that rate for the NRL loading will have adequate load capacity for all legal Formula B truck configurations up to 80 kips."
Example A1 shows this holding true NRL $RF > 1$ and all SU $RF > 1$, while Example A2 shows when NRL $RF < 1$, RF for the SUs may or may not be >1 and need to be checked on an individual basis.

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A1A.1.10—Permit Load Rating 6A.6.4.2

Permit Type: Special (Single-Trip, Escorted)
Permit Weight: 220 kips
Permit Vehicle: Shown in Figure A1A.1.10-1
ADTT (one direction): 1,000

From Live Load Analysis by Computer Program:

Undistributed Maximum M_{LL} = 2,127.9 kip-ft

Undistributed Maximum V_{LL} = 143.5 kips

A1A.1.10.1—Strength II Limit State 6A.6.4.2.1

γ_{LL} = 1.10 (Single-Trip, Escorted) Table 6A.4.5.4.2a-1

Use one-lane distribution factor and divide out the 1.2 multiple presence factor. 6A.4.5.4.2b

$$g_{m1} = \frac{0.46}{1.2} = 0.383$$

$$g_{v1} = \frac{0.653}{1.2} = 0.544$$

IM = 20 percent (no speed control, minor surface deviations) 6A.4.5.5

Distributed Live-Load Effects:

M_{LL+IM} = (2,127.9) (0.383) (1.20)
= 978.0 kip-ft

V_{LL+IM} = (143.5) (0.544) (1.20)
= 93.7 kips

Flexure: $$RF = \frac{(1.0)(1.0)(1.0)(2873)-(1.25)(439)-(1.25)(129)}{(1.10)(978.0)}$$

= 2.01 > 1.0 OK

Shear: $$RF = \frac{(1.0)(1.0)(1.0)(360.3)-(1.25)(27+8)}{(1.10)(93.7)}$$

= 3.07 > 1.0 OK

A1A.1.10.2—Service II Limit State (Optional) 6A.6.4.2.2

$$RF = \frac{f_R - f_D}{\gamma_L (f_{LL+IM})}$$

IM = 20 percent (no speed control, minor surface deviations)

γ_L = 1.0 γ_D = 1.0 Table 6A.4.2.2-1

f_R = 34.2 ksi

f_D = 11.49 ksi

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Live-load effects for the Service II permit rating of vehicles that mix with traffic are calculated using the LRFD distribution analysis methods. This check is based on past practice and does not use the one-lane distribution with permit load factors that have been calibrated for the Strength II permit rating. For escorted permits, a one-lane distribution factor can be used as the permit crosses the bridge with no other vehicles allowed on the bridge at the same time. C6A.6.4.2.2

g_m = 0.383 (m = 1.2 has been divided out)

M_{LL+IM} = (2,127.9) (0.383) (1.2) = 978.0 kip-ft. = 11,736 kip-in.

$$f_{LL+IM} = \frac{M_{LL+IM}}{S_b} = \frac{11,736}{793} = 14.8 \text{ ksi}$$

$$RF = \frac{34.2 - (1.0)(11.49)}{(1.0)(14.8)} = 1.53$$

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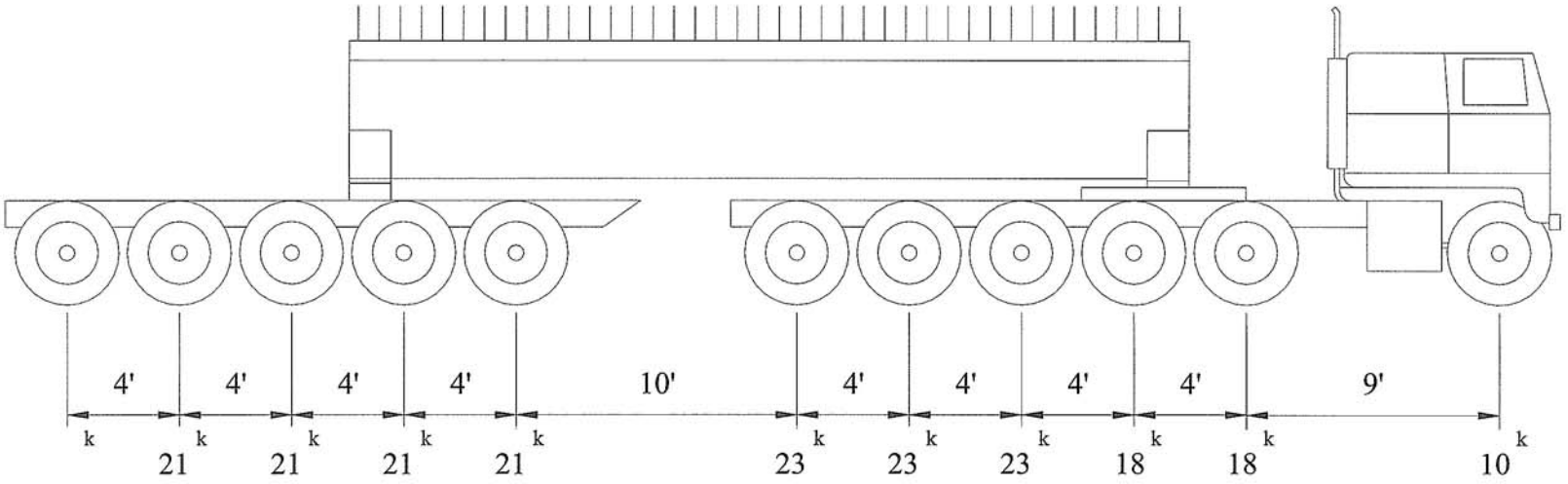

Figure A1A.1.10-1—Permit Truck Loading Configuration

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A1A.2—Evaluation of an Exterior Stringer

The same given bridge data as for interior stringers applies.

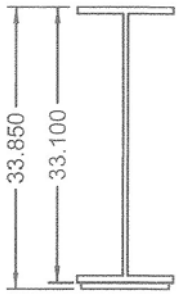
A1A.2.1—Section Properties

A1A.2.1.1—Noncomposite Section Properties

W 33 × 130 and *PL* 3/4 in. × 10 1/2 in.

The section properties for this beam were determined from the *AISC Manual of Steel Construction*, Sixth Edition, printed during the period from July 1963 to March 1967, which is consistent with the "Year Built" date for this bridge.

W 33 × 130			*PL* 3/4 in. × 10 1/2 in.	
t_f	=	0.855 in.	t	= 0.750 in.
b_f	=	11.51 in.	b	= 10.5
t_w	=	0.58 in.		
A	=	38.26 in.^{2}	$A = t \times b$	= 7.875 in.^{2}
I	=	6,699 in.^{4}	I	~ 0 in.^{4} (negligible)



$$\bar{y} = \frac{(17.30)(38.26) + (0.375)(7.875)}{38.26 + 7.875}$$ Distance to *C.G.*

$\bar{y} = 14.41$ in. from bottom of section to centroid

$$I_x = 6,699 + 38.26(2.89)^2 + 7.875(14.04)^2$$

$I_x = 8,570.9$ in.^{4}

$$S_t = \frac{8,570.9}{19.44} = 440.8 \text{ in.}^3$$ Section Modulus at top of steel

$$S_b = \frac{8,570.9}{14.41} = 594.7 \text{ in.}^3$$ Section Modulus at bottom of steel

A1A.2.1.2—Composite Section Properties

Barrier is not known to be structurally continuous.

Effective Flange Width, b_e, may be taken as one-half the distance to the adjacent stringer or girder plus the full overhang width. LRFD Design 4.6.2.6.1

Effective Flange Width $b_e = \frac{1}{2}(88 \text{ in.}) + 12 \text{ in.} = 56 \text{ in.}$

Modular Ratio, *n* LRFD Design 6.10.1.1.1b

f'_c = 3 ksi

For $2.9 < f'_c < 3.6$, $n = 9$ LRFD Design C6.10.1.1.1b

Short-Term Composite, *n*:

W 33 × 130, *PL* 3/4 in. × 10 1/2 in. and *Conc.* 7 1/4 in. × 56 in.

$$\frac{56}{9} = 6.22 \text{ in.}$$

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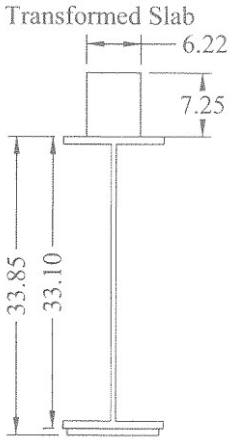


$$\bar{y} = \frac{(17.30)(38.26)+(0.375)(7.875)+\left(\frac{56}{9}\right)(7.25)(37.475)}{38.26+7.875+\left(\frac{56}{9}\right)(7.25)}$$

$\bar{y}$ = 25.81 in. from bottom of section to centroid

$$I_x = 6,699+38.26(8.51)^2+(7.875)(25.43)^2 + \frac{\left(\frac{56}{9}\right)(7.25)^3}{12}+\left(\frac{56}{9}\right)(7.25)(11.66)^2$$

I_x = 20,893 in.^{4}

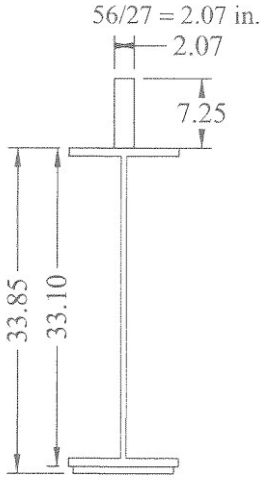
$S_t = \frac{20,893}{8.04} = 2,599$ in.^{3} Section Modulus at top of steel

$S_b = \frac{20,893}{25.81} = 809$ in.^{3} Section Modulus at bottom of steel

Long-Term Composite, 3*n*:

$3n = 3 \times 9 = 27$

W 33 × 130, *PL* $^3/_4$ in. × 10$^1/_2$ in. and *Conc.* 7$^1/_4$ in. × 56 in.



$$\bar{y} = \frac{(17.30)(38.26)+(0.375)(7.875)+\left(\frac{56}{27}\times 7.25\right)(37.475)}{38.26+7.875+\left(\frac{56}{27}\times 7.25\right)}$$

$\bar{y}$ = 20.08 in. from bottom of section to centroid

$$I_x = 6,699+38.26(2.78)^2+(7.875)(19.70)^2 + \frac{\left(\frac{56}{27}\right)(7.25)^3}{12}+\left(\frac{56}{27}\right)(7.25)(17.39)^2$$

I_x = 14,664 in.^{4}

$S_t = \frac{14,664}{13.77} = 1,065$ in.^{3} Section Modulus at top of steel

$S_b = \frac{14,664}{20.08} = 730$ in.^{3} Section Modulus at bottom of steel

A1A.2.1.3—Summary of Section Properties at Midspan

1. Steel Section Only

S_{TOP} = 440.8 in.^{3}

S_{BOT} = 594.7 in.^{3}

2. Composite Section—Short Term, $n = 9$

$S_{TOP\ steel}$ = 2,599 in.^{3}

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S_{BOT} = 809 in.^{3}

3. Composite Section—Long Term, $3n = 27$

$S_{TOP\ steel}$ = 1,065 in.^{3}

S_{BOT} = 730 in.^{3}

A1A.2.2—Dead Load Analysis—Exterior Stringer

A1A.2.2.1—Components and Attachments, DC

A1A.2.2.1a—Noncomposite Dead Loads, DC_1

Deck: $\left(1+\frac{7.33}{2}\right)\left(\frac{7.25}{12}\right)(0.150\ \text{kip/ft})$ = 0.423 kip/ft

Stringer: (same as interior) = 0.138 kip/ft

Cover Plate: $\frac{0.75\times10.5}{144\,\text{in.}^2/\text{ft}^2}\times0.490\,\text{klf}\times1.06\times\frac{40\,\text{ft}}{65\,\text{ft}}$ = 0.017 kip/ft

Diaphragms: $\frac{(3)(0.0427)\left(\frac{7.33}{2}\right)(1.06)}{65\ \text{ft}}$ = 0.008 kip/ft

Total per stringer = 0.586 kip/ft

$M_{DC_1} = \frac{(0.586)(65)^2}{8}$ = 309.5 kip-ft at midspan

$V_{DC_1} = (0.586)\left(\frac{65}{2}\right)$ = 19.0 kips at bearing

A1A.2.2.1b—Composite Dead Loads, DC_2 (same as interior)

M_{DC_2} = 129 kip-ft

V_{DC_2} = 8 kips

A1A.2.2.2—Wearing Surface

DW = 0

A1A.2.3—Live Load Analysis—Exterior Stringer

A1A.2.3.1—Compute Live Load Distribution Factors

A1A.2.3.1a—Distribution Factor for Moment, g_m (LRFD Design Table 4.6.2.2.2d-1)

One Lane Loaded:

Lever Rule

For one lane loaded, the multiple presence factor, $m = 1.20$ LRFD Design Table 3.6.1.1.2-1

For:

$S + d_e$ = 7.33 ft + 0 ft < 8 ft one wheel acting upon the girder

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$$g_{m1} = m\left(\frac{S+d-2\text{ ft}}{2S}\right) = 1.2\left(\frac{7.33+0-2}{2(7.33)}\right) = 0.436$$

Two or More Lanes Loaded:

$$g_{m2} = eg_{interior} \qquad e = 0.77+\frac{d_e}{9.1} = 0.77$$

$$g_{m2} = (0.77)\,(0.626) = 0.482 > 0.436$$

A1A.2.3.1b—Distribution Factor for Shear, g_v (LRFD Design Table 4.6.2.2.3b-1)

One Lane Loaded:

Lever Rule

$g_{v1} = g_{m1} = 0.436$

Two or More Lanes Loaded:

$$g = eg_{interior} \qquad e = 0.6+\frac{d_e}{10} = 0.6$$

$$g_{v2} = (0.6)\,(0.767) = 0.460 > 0.436$$

A1A.2.3.1c—Special Analysis for Exterior Girders with Diaphragms or Cross-Frames (LRFD Design 4.6.2.2.2d)

Roadway Layout: two 11-ft wide lanes

$$R = \frac{N_L}{N_b}+\frac{X_{ext}\sum^{N_L} e}{\sum^{N_b} x^2}$$ LRFD Design Eq. C4.6.2.2.2d-1

$$g_{special} = (m)\,(R)$$

One Lane Loaded:

$$R = \frac{1}{4}+\frac{(11)(6)}{\left[11^2+3.67^2+(-3.67)^2+(-11)^2\right]} = 0.495$$

$$g_{special1} = 1.2\,(0.495) = 0.595$$

Two Lanes Loaded:

$$R = \frac{2}{4}+\frac{(11)[6+(-5)]}{\left[11^2+3.67^2+(-3.67)^2+(-11)^2\right]} = 0.541$$

$$g_{special2} = 1.0\,(0.541) = 0.541$$

A1A.2.3.1d—Summary of Distribution Factors for the Exterior Girders

Moment, g_m

1 Lane	=	0.436	
2 or More Lanes	=	0.482	
Special Analysis (1 Lane)	=	0.595	Governs
Special Analysis (2 Lanes)	=	0.541	
g_m	=	0.595	

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Shear, g_v

1 Lane	=	0.436	
2 or More Lanes	=	0.460	
Special Analysis (1 Lane)	=	0.595	Governs
Special Analysis (2 Lanes)	=	0.541	
g_v	=	0.595	

A1A.2.3.2—Compute Maximum Live Load Effects for HL-93

Same as for interior girder

Midspan: M_{LL+IM} = 1,521.7 kip-ft

Bearing: V_{LL+IM} = 102.9 kips

A1A.2.3.2a—Distributed Live Load Moments and Shears

Design Live Load HL-93

M_{LL+IM} = 1,521.7 × g_m = (1,521.7) (0.595)

= 905.4 kip-ft

V_{LL+IM} = 102.9 kips × g_v = (102.9) (0.595)

= 61.2 kips

A1A.2.4—Compute Nominal Resistance of Section at Midspan

Locate PNA:

D = 31.39 in.

t_f = 0.855 in.

t_w = 0.58 in.

b_f = 11.51 in.

Cov. *PL* A_p = 7.875 in.^{2}

(*PL* $^3/_4$ in. × $10^1/_2$ in.)

Treat the bottom flange and the cover plate as one component.

A_t = (11.51) (0.855) + (10.5) (0.75) = 17.72 in.^{2}

$$y = \frac{(11.51)(0.855)\frac{(0.855)}{2} + (10.5)(0.75)\left(0.855 + \frac{0.75}{2}\right)}{(11.51)(0.855) + (10.5)(0.75)}$$

= 0.784 in. (from top of tension flange to centroid of flange and cover plate)

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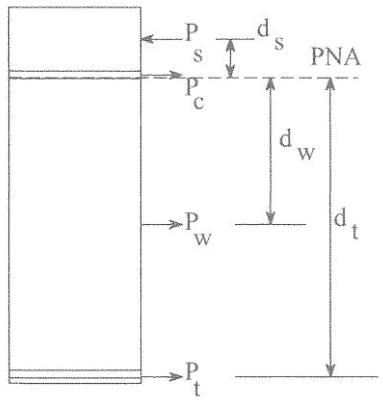

Plastic Forces LRFD Design Article D6.1

Note the forces in longitudinal reinforcement may be conservatively neglected.

Set P_{rb} and $P_{rt} = 0$

$P_s = 0.85f'_c b_{eff} t_s$

$= 0.85\ (3.0)\ (56)\ (7.25)$

$= 1035.3$ kips

$P_c = F_y b_f t_f$

$= (36)\ (11.51)\ (0.855)$

$= 354.3$ kips

$P_w = F_y D t_w$

$= (36)\ (31.39)\ (0.58)$

$= 655.4$ kips

$P_t = F_y\ (b_f t_f + A_p)$

$= 36\ (11.51 \times 0.855 + 7.875)$

$= 637.8$ kips

$P_t + P_w < P_c + P_s + P_{rb} + P_{rt}$ $\therefore$ Conditions for Case I are not met

$P_t + P_w + P_c \geq P_s + P_{rb} + P_{rt}$ $\therefore$ The PNA lies in the top flange

$$\overline{Y} = \left(\frac{t_c}{2}\right)\left(\frac{P_w + P_t - P_s}{P_c} + 1\right) = \left(\frac{0.855}{2}\right)\left(\frac{655.4 + 637.8 - 1035.3}{354.3} + 1\right)$$ LRFD Design Table D6.1-1

$= 0.739$ in. from top of flange

A1A.2.4.1—Classify Section

Following the I-Sections in Flexure Flowchart (section is considered to be constant depth). LRFD Design Figure C6.4.5-1

A1A.2.4.1a—Check Web Slenderness

Since PNA is in the top flange, the web slenderness requirement is automatically satisfied.

For composite sections in positive bending, the remaining stability criteria are automatically satisfied. The section is compact.

A1A.2.4.1b—Check Ductility (LRFD Design 6.10.7.1.2)

$D_p = t_s + \overline{Y} = 7.25 + 0.739$

$= 7.99$ in.

$D_t = 33.85 + 7.25 = 41.1$ in.

If $D_p \leq 0.1 D_t$, then $M_n = M_p$

Otherwise, $M_n = M_p\left(1.07 - 0.7\frac{D_p}{D_t}\right)$ LRFD Design Eq. 6.10.7.1.2-1

$0.1 D_t = 0.1 \times 41.1 = 4.11$ in. LRFD Design Eq. 6.10.7.1.2-2

7.99 in. $\not\leq 4.11$ in. therefore calculate $M_n < M_p$

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A1A.2.4.2—Plastic Moment, M_p

Moment arms about the PNA.

Slab:

$$d_s = \frac{t_s}{2} + \overline{Y}$$

$$= \frac{7.25}{2} + 0.739$$

$$= 4.36 \text{ in.}$$

Web:

$$d_w = \frac{D}{2} + t - \overline{Y}$$

$$= \frac{31.39}{2} + 0.855 - 0.739$$

$$= 15.81 \text{ in.}$$

Tension Flange:

$$d_t = t_c - \overline{Y} + D + 0.784$$

$$= 0.855 - 0.739 + 31.39 + 0.784$$

$$= 32.29 \text{ in.}$$

$$M_p = \frac{P_c}{2t_c}\left[\left(\overline{Y}\right)^2 + \left(t_c - \overline{Y}\right)^2\right] + P_s d_s + P_{rt} d_{rt} + P_{rb} d_{rb} + P_w d_w + P_t d_t \qquad \text{LRFD Design Table D6.1-1}$$

$$= \left\{\begin{array}{l} \dfrac{354.3}{2(0.855)}\left[(0.739)^2 + (0.855 - 0.739)^2\right] \\ +(1035.3)(4.36) + 0 + 0 + (655.4)(15.81) + (637.8)(32.29) \end{array}\right\}$$

$$= 35586 \text{ kip-in.} = 2965 \text{kip-ft}$$

A1A.2.4.3—Nominal Flexural Resistance, M_n (LRFD Design 6.10.7.1.2)

$$D_p \not\leq 0.1 D_t \qquad \text{LRFD Design Eq. 6.10.7.1.2-1}$$

Therefore, $M_n = M_p(1.07 - 0.7\frac{D_p}{D_t})$ LRFD Design Eq. 6.10.7.1.2-2

$= 2965(1.07 - 0.7 \times 0.194)$

$= 2770.0$ kip-ft

A1A.2.4.4—Nominal Shear Resistance, V_n

Classification and Resistance same as for interior.

V_n = 360.3 kips

A1A.2.4.5—Summary for Exterior Stringer

	Dead Load DC_1	Dead Load DC_2	LiveLoad Distribution Factor	Dist. Live Load + Impact	Nominal Capacity
Moment kip-ft	309.5	129.0	$gm = 0.595$	905.4	2770.0
Shear kips	19.0	8.0	$gm = 0.595$	61.2	360.3

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A1A.2.5—General Load-Rating Equation

$$RF = \frac{C - (\gamma_{DC})(DC) - (\gamma_{DW})(DW) \pm (\gamma_P)(P)}{(\gamma_{LL})(LL + IM)}$$ Eq. 6A.4.2.1-1

A1A.2.6—Evaluation Factors (for Strength Limit State)

1. Resistance Factor, φ LRFD Design 6.5.4.2

φ = 1.0 for flexure and shear

Condition Factor, φ_c 6A.4.2.3

Member is in good condition. NBI Item 59 = 7.

φ_c = 1.0

System Factor, φ_s 6A.4.2.4

φ_s = 1.0 Multi-girder bridge.

A1A.2.7—Design Load Rating (6A.4.3)

A1A.2.7.1—Strength I Limit State (6A.6.4.1)

$$RF = \frac{(\varphi_c)(\varphi_s)(\varphi)R_n - (\gamma_{DC})(DC) - (\gamma_{DW})(DW)}{(\gamma_{LL})(LL + IM)}$$

A1A.2.7.1a—Inventory Level

Load	Load Factor γ
DC	1.25
LL	1.75

Table 6A.4.2.2-1

Flexure:

$$RF = \frac{(1.0)(1.0)(1.0)(2770) - (1.25)(309.5 + 129)}{(1.75)(905.4)}$$

= 1.40

Shear:

$$RF = \frac{(1.0)(1.0)(1.0)(360.3) - (1.25)(19 + 8)}{(1.75)(61.2)}$$

= 3.05

A1A.2.7.1b—Operating Level

Load	Load Factor γ
DC	1.25
LL	1.35

Table 6A.4.2.2-1

For Strength I Operating Level, only the live load factor changes; therefore the rating factor can be calculated by direct proportions.

Flexure:

$$RF = 1.40 \times \frac{1.75}{1.35}$$

= 1.81

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Shear:

$$RF = 3.05 \times \frac{1.75}{1.35}$$

$$= 3.95$$

A1A.2.7.2—Service II Limit State (6A.6.4.1)

For Service Limit States, Capacity C = f_R

$$RF = \frac{f_R - (\gamma_D)(f_D)}{(\gamma_L)(f_{LL+IM})}$$

A1A.2.7.2a—Inventory Level

Allowable Flange Stress for tension flange:

$f_R = 0.95 R_h F_{yf}$ ($f_\ell = 0$) LRFD Design Eq. 6.10.4.2.2-2

Checking the tension flange as a compression flange typically does not govern for composite sections.

R_h = 1.0 for non-hybrid sections LRFD Design 6.10.1.10.1

$f_R = 0.95 \times 1.0 \times 36$

$= 34.2$ ksi

$f_D = f_{DC_1} + f_{DC_2}$

$$f_D = \frac{(309.5)(12)}{594.7} + \frac{(129)(12)}{730}$$

$= 6.24 + 2.12 = 8.36$ ksi

$$f_{LL+IM} = \frac{(905.4)(12)}{809} = 13.43 \text{ ksi}$$

$\gamma_{LL} = 1.30$ $\gamma_{DC} = 1.0$ Table 6A.4.2.2-1

$$RF = \frac{34.2 - (1.0)(8.36)}{1.3(13.43)}$$

$= 1.48$

A1A.2.7.2b—Operating Level

$\gamma_{LL} = 1.0$ $\gamma_{DC} = 1.0$ Table 6A.4.2.2-1

$$RF = \frac{34.2 - (1.0)(8.36)}{1.0(13.43)}$$

$= 1.92$

A1A.2.7.3—Fatigue Limit State

The calculations are not shown. See the calculations for interior stringers.

A1A.2.8—Legal Load Rating (6A.6.4.2)

Note: The design load check produced a rating factor greater than 1.0 for the Inventory Design Load Rating. This indicates that the bridge has adequate load capacity to carry all legal loads and need not be subject to load ratings for legal loads. The load rating

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computations that follow have been done for illustrative purposes. Shear ratings have not been illustrated.

Live Load: AASHTO Legal Loads—Types 3, 3S2, and 3-3 (Rate for all three) Appendix D6A

g_m = 0.595
IM = 20 percent Table C6A.4.4.3-1

The standard dynamic load allowance of 33 percent is decreased based on a field evaluation certifying that the approach and bridge riding surfaces have only minor surface deviations or depressions.

	Type 3	Type 3S2	Type 3-3	
M_{LL}	660.7	707.2	644.7	kip-ft
gM_{LL+IM}	471.7	504.9	460.3	kip-ft

Live Load: AASHTO Legal Loads—Specialized Hauling Units and Notional Rating Load —SU4, SU5, SU6, SU7, and NRL

Interpolating to determine M_{LL} without impact for 65 ft span Table E6A-2

	SU4	SU5	SU6	SU7	NRL	
M_{LL}	744.7	821.2	913.5	994.1	1037.0	kip-ft
gM_{LL+IM}	531.7	586.3	652.2	709.8	740.4	kip-ft

A1A.2.8.1—Strength I Limit State (6A.6.4.2.1)

Dead load and capacity remain the same

For Types 3, 3S2, and 3-3

Dead Load *DC*: γ_{DC} = 1.25 Table 6A.4.2.2-1

$ADTT$ = 1,000

Generalized Live-Load Factor for Legal Loads:

γ_{LL} = 1.30 Table 6A.4.4.2.3a-1

Flexure:

$$RF = \frac{(1.0)(1.0)(1.0)(2,770)-(1.25)(309.5+129)}{(1.30)(M_{LL+IM})}$$

	Type 3	Type 3S2	Type 3-3
RF	3.62	3.38	3.72

For Specialized Hauling Units and NRL

Dead Load *DC*: $\gamma_{DC} = 1.25$ Table 6A.4.2.2-1

$ADTT$ = 1,000 Assumed

Generalized Live-Load Factor for Legal Loads $\gamma_{LL} = 1.30$ Table 6A.4.4.2.3b-1

Flexure: RF = $$\frac{(1.0)(1.0)(1.0)(2,770)-(1.25)(309.5+129)}{(1.30)(M_{LL+IM})}$$

	SU4	SU5	SU6	SU7	NRL
RF	3.06	2.70	2.50	2.29	2.21

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A1A.2.8.2—Service II Limit State (6A.6.4.2.2)

For Types 3, 3S2, and 3-3, and for Specialized Hauling Units and NRL

γ_{LL} = 1.3 γ_{DC} = 1.0 Table 6A.4.2.2-1

f_R = 34.2 ksi

f_D = 8.36 ksi

$f_{LL+IM} = \dfrac{M_{LL+IM} \times 12}{809}$

Service II: $RF = \dfrac{34.2 - (1.0)(8.36)}{(1.3)(f_{LL+IM})}$

	Type 3	Type 3S2	Type 3-3	
f_{LL+IM}	7.00	7.49	6.82	ksi
RF	2.84	2.65	2.91	

	SU4	SU5	SU6	SU7	NRL	
f_{LL+IM}	7.89	8.70	9.67	10.53	10.98	ksi
RF	2.52	2.29	2.05	1.89	1.81	

No posting is required as for all legal loads, $RF > 1.0$.

A1A.2.8.3—Summary (6A.4.4.4)

Safe Load Capacity (tons), $RT = RF \times W$ Eq. 6A.4.4.4-1

Truck	Type 3	Type 3S2	Type 3-3
Weight (tons)	25	36	40
RF (Service II Controlling)	2.84	2.65	2.91
Safe Load Capacity (tons)	71	95	116

Truck	SU4	SU5	SU6	SU7	NRL
Weight (tons)	27	31	34.8	38.8	40
RF (Service II Controlling)	2.52	2.29	2.05	1.89	1.81
Safe Load Capacity (tons)	68	70	71	73	72

A1A.2.9—Permit Load Rating (6A.6.4.2)

Permit Type: Special (Single-Trip, Escorted)

Permit Weight: 220 kips

Permit Vehicle: Shown in Figure A1A.1.10-1.

ADTT: 1,000

From Live-Load Analysis by Computer Program:

Undistributed Maximum:

M_{LL} = 2,127.9 kip-ft

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V_{LL} = 143.5 kips

A1A.2.9.1—Strength II Limit State (6A.6.4.2.1)

Dead load and capacity remain the same as that calculated for the design load rating

γ_{LL} = 1.10 (Single Trip, Escorted) Table 6A.4.5.4.2a-1

γ_{DC} = 1.25

Use the One-Lane Loaded Distribution Factor and divide out the 1.2 multiple presence factor. 6A.4.5.4.2b

$g_{special1}$ = 0.595 (Special method for rigid torsional behavior governs.) LRFD Design 4.6.2.2.2d

g_{m1} = $g_{v1} = \frac{g_{special1}}{1.2} = 0.496$

Distributed Live-Load Effects:

IM = 20 percent (no speed control, minor surface deviations)

M_{LL+IM} = (2,127.9) (0.496) (1.2)

= 1,266.5 kip-ft

V_{LL+IM} = (143.5) (0.496) (1.2)

= 85.4 kips

Flexure: $RF = \frac{(1.0)(1.0)(1.0)(2,770)-(1.25)(309.5+129)}{(1.10)(1,266.5)}$

= 1.60 > 1.0 OK

Shear: $RF = \frac{(1.0)(1.0)(1.0)(360.3)-(1.25)(19+8)}{(1.10)(85.4)}$

= 3.48 > 1.0 OK

A1A.2.9.2—Service II Limit State (Optional)

$RF = \frac{f_R - \gamma_{DC} f_D}{\gamma_{LL}(f_{LL+IM})}$

IM = 20 percent (no speed control, minor deviations)

γ_{LL} = 1.0 γ_{DC} = 1.0 Table 6A.4.2.2-1

Dead load and capacity expressed in terms of stresses remain the same as that calculated for the design load rating

f_R = 34.2 ksi

f_D = 8.36 ksi

Live-load effects for the Service II permit rating of an escorted permit are calculated using the same one-lane-loaded procedures as for the Strength II rating. C6A.6.4.2.2

g_{m1} = 0.496

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M_{LL+IM} = (2,127.9) (0.496) (1.2) = 1,266.5 kip-ft

= 15,198 kip-in.

$f_{LL+IM} = \frac{M_{LL+IM}}{S_b} = \frac{15,192}{809} = 18.8$ ksi

$RF = \frac{34.2 - (1.0)8.36}{1.0(18.8)} = 1.37 > 1.0$ OK

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A1A.3—Summary of Rating Factors for Load and Resistance Factor Rating Method

Table A1A.3-1—Summary of Rating Factors for Load and Resistance Factor Rating Method—Interior Stringer

Limit State		Design Load Rating		Legal Load Rating								Permit Load Rating
		Inventory	Operating	Type 3	Type 3S2	Type 3-3	SU4	SU5	SU6	SU7	NRL	
Strength I	Flexure	1.29	1.67	3.35	3.12	3.44	2.97	2.70	2.43	2.23	2.13	—
	Shear	2.29	2.97	—	—	—	—	—	—	—	—	—
Strength II	Flexure	—	—	—	—	—	—	—	—	—	—	2.01
	Shear	—	—	—	—	—	—	—	—	—	—	3.07
Service II		1.21	1.57	2.33	2.17	2.38	2.06	1.87	1.68	1.55	1.48	1.53
Fatigue		0.38	—	—	—	—	—	—	—	—	—	—

Table A1A.3-2—Summary of Rating Factors for Load and Resistance Factor Rating Method—Exterior Stringer

Limit State		Design Load Rating		Legal Load Rating								Permit Load Rating
		Inventory	Operating	Type 3	Type 3S2	Type 3-3	SU4	SU5	SU6	SU7	NRL	
Strength I	Flexure	1.40	1.81	3.62	3.39	3.71	3.21	2.92	2.62	2.41	2.31	—
	Shear	3.05	3.95	—	—	—	—	—	—	—	—	—
Strength II	Flexure	—	—	—	—	—	—	—	—	—	—	1.60
	Shear	—	—	—	—	—	—	—	—	—	—	3.48
Service II		1.48	1.92	2.84	2.65	2.91	2.52	2.29	2.05	1.89	1.81	1.37

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PART B—ALLOWABLE STRESS AND LOAD FACTOR RATING METHODS

A1B.1—EVALUATION OF AN INTERIOR STRINGER

Note: When reference is given as "AASHTO," e.g., "AASHTO 10.38.1," it refers to that Article in the AASHTO *Standard Specifications for Highway Bridges*, 17th Edition.

A1B.1.1—Bridge Data

Refer to Article A1A.1.1, Simple Span Composite Steel Stringer Bridge Data.

A1B.1.2—Section Properties

In unshored construction, the steel stringer must support its own weight plus the weight of the concrete slab. For the composite section, the concrete is transformed into an equivalent area of steel by dividing the area of the slab by the modular ratio. Live load plus impact stresses are carried by the composite section using a modular ratio of n. To account for the effect of creep, superimposed dead load stresses are carried by the composite section using a modular ratio of $3n$ (AASHTO 10.38.1). The as-built section properties are used in this analysis.

A1B.1.2.1—Noncomposite Section Properties

Section properties of rolled shapes are subject to change with changes in rolling practices of the steel industry. Identify steel components from available records, construction date, and field measurements. The section properties for this beam were determined from the *AISC Manual of Steel Construction*, Sixth Edition, printed during the period from July 1963 to March 1967, which is consistent with the "Year Built" date for this bridge.

W 33 × 130 and PL $^5/_8$ in. × $10^1/_2$ in.
t_f = 0.855 in.; b_f = 11.51 in.; t_w = 0.58 in.
A = 38.26 in.^{2}

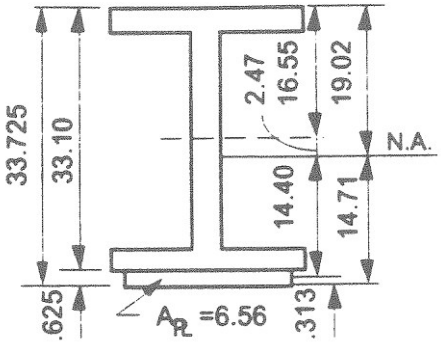

Figure A1B.1.2.1-1 Cross Section—Interior Stringer, Noncomposite

$$\bar{y} = \frac{(17.175)(38.26)_{W} + (0.313)(6.56)_{PL}}{38.26 + 6.56}$$

$$\bar{y} = 14.71 \text{ in.}$$

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$$I_x = \underset{W}{6,699} + \underset{W}{38.26(2.47)^2} + \underset{PL}{6.56(14.40)^2}$$
$$= 8,293 \text{ in.}^4$$

$$S_t = \frac{8,293}{19.02} = 436.0 \text{ in.}^3 = S_t^{DL}$$

$$S_b = \frac{8,293}{14.71} = 563.7 \text{ in.}^3 = S_b^{DL}$$

A1B.1.2.2—Composite Section Properties

Effective Flange Width AASHTO 10.38.3.1

1/4 (65)(12)	=	195 in.	
(7.33)(12)	=	88 in.	
(7.25)(12)	=	87 in.	⇐ Controls

Modular Ratio *n* 6B.5.2.4

$$\text{for } f_c' = 3,000 \text{ psi} - n = 10$$

Composite $n = n$: *W* 33 × 130, *PL* 5/8 in. × 10 1/2 in. and *Conc.* 7 1/4 in. × 87 in.

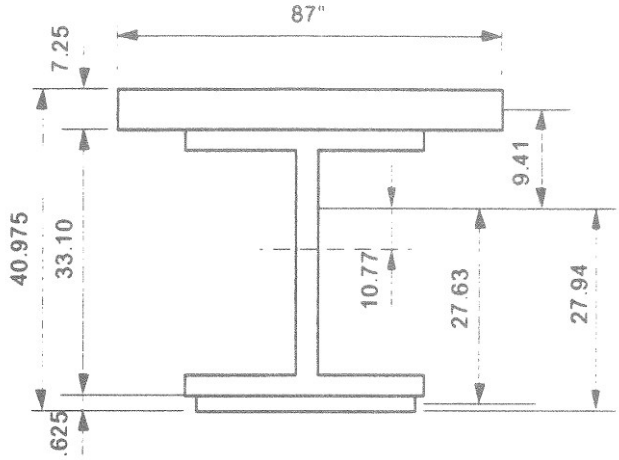

Figure A1B.1.2.2-1—Cross Section—Interior Stringer, Composite *n* = *n*

$$\bar{y} = \frac{\underset{W}{(17.175)(38.26)} + \underset{PL}{(0.313)(6.56)} + \underset{Conc.}{(87 \times 7.25 \div 10)(37.35)}}{38.26 + 6.56 + (87 \times 7.25) \div 10}$$

$$\bar{y} = 27.94 \text{ in.}$$

$$I_x = \underset{W}{6,699} + \underset{W}{(38.26)(10.77)^2} + \underset{PL}{(6.56)(27.63)^2} + \underset{Conc.}{\frac{(87 \div 10)(7.25)^3}{12}} + \underset{Conc.}{(87 \times 7.25) \div 10 (9.41)^2}$$
$$= 22,007 \text{ in.}^4$$

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Note: I_x for the bottom cover plate is negligible, however, its Ad^2 term makes a significant contribution.

$$S_t = \frac{22,007}{5.79} = 3,801 \text{ in.}^3 \text{ Section modulus at top of steel}$$

$$S_b = \frac{22,007}{27.94} = 787.7 \text{ in.}^3 = S_b^L$$

Use with Live Load.

Composite $n = 3n$: *W* 33 × 130, *PL* $^5/_8$ in. × $10^1/_2$ in. and *Conc.* $7^1/_4$ in. × 87 in.

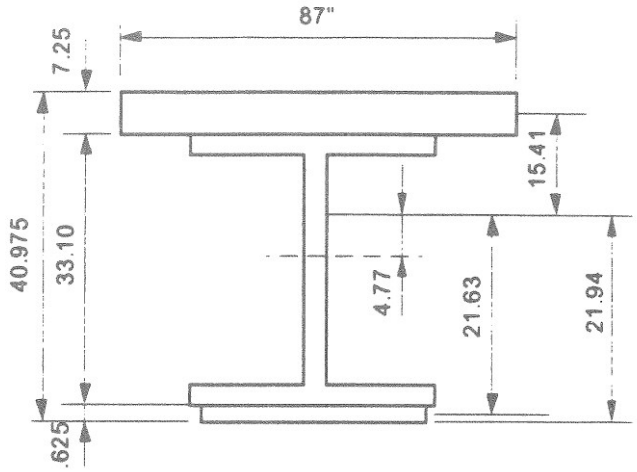

Figure A1B.1.2.2-2—Cross Section—Interior Stringer, Composite *n* = 3*n*

$$\bar{y} = \frac{\underset{W}{(17.175)(38.26)} + \underset{PL}{(0.313)(6.56)} + \underset{Conc.}{(87 \times 7.25 \div 30)(37.35)}}{38.26 + 6.56 + (87 \times 7.25) \div 30}$$

$$\bar{y} = 21.94 \text{ in.}$$

$$I_x = \underset{W}{6,699} + \underset{W}{(38.26)(4.77)^2} + \underset{PL}{(6.56)(21.63)^2} + \underset{Conc.}{\frac{(87 \div 30)(7.25)^3}{12} + \left(\frac{87 \times 7.25}{30}\right)(15.41)^2}$$

$$I_x = 15,725 \text{ in.}^4$$

$$S_t = \frac{15,725}{11.79} = 1,333.8 \text{ in.}^3 \text{ (Section modulus at top of steel)}$$

$$S_b = \frac{15,725}{21.94} = 716.7 \text{ in.}^3 = S_b^{SDL}$$

Use with Superimposed Dead Load (*SDL*).

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A1B.1.3—Dead Load Analysis—Interior Stringer

A1B.1.3.1—Dead Loads (Includes an Allowance of Six Percent of Steel Weight for Connections)

Deck $(7.33)\left(\frac{7.25}{12}\right)(150 \text{ pcf})$	=	664.3 lb/ft
Stringer $(130)(1.06)$	=	137.8 lb/ft
Cover *PL* $(0.625)(10.5)(490 \div 144)(1.06)(38) \div 65$	=	13.8 lb/ft
Diaphragms $(3)(42.7)(7.33)(1.06) \div 65$	=	15.4 lb/ft
Total per stringer	=	831.3 lb/ft

A1B.1.3.2—Superimposed Dead Loads (AASHTO 3.23.2.3.1.1)

Curb $(1)\left(\frac{10}{12}\right)(150 \text{ pcf}) \times \left(\frac{2 \text{ curbs}}{4 \text{ beams}}\right)$	=	62.5 lb/ft
Parapet $\left[\left(\frac{6 \times 19}{144}\right) + \left(\frac{18 \times 12}{144}\right)\right](150 \text{ pcf}) \times \left(\frac{2 \text{ parapets}}{4 \text{ beams}}\right)$	=	171.9 lb/ft
Railing $(\text{assume } 20 \text{ plf}) \times \left(\frac{2 \text{ railings}}{4 \text{ beams}}\right)$	=	10.0 lb/ft
Wearing Surface	=	0.0 lb/ft
Total per stringer	=	244.4 lb/ft

A1B.1.4—Live Load Analysis—Interior Stringer

Live Load: Rate for HS-20

Moments:

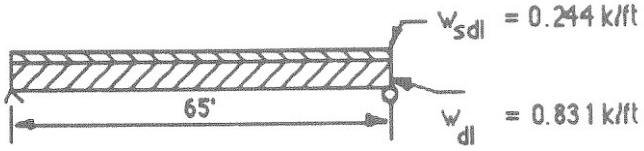

Figure A1B.1.4-1—Load Diagram—Interior Stringer, Dead Load, and Superimposed Dead Load

$$M_{DL} = \frac{w_{DL}L^2}{8} = \frac{0.831(65)^2}{8} = 439 \text{ kip-ft}$$

$$M_{SDL} = \frac{w_{SDL}L^2}{8} = \frac{0.244(65)^2}{8} = 129 \text{ kip-ft}$$

M_{LL} Appendix C6B[a]

Span	M_{LL}	
60	403.3	$M_{LL} = \frac{403.3 + 492.8}{2}$
	$\Leftarrow$65 ft	
70	492.8	$M_{LL} = 448$ kip-ft[b]
		(without Impact, without Distribution)

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a Note the moments given in the MBE are for one line of wheels. The values given in AASHTO are for the entire axle and are therefore twice the MBE value.

b Maximum M_{LL} without impact for 65 ft span, with exact values determined by statics, is 448.02 kip-ft. Nevertheless, judgment should be exercised whether to interpolate tabulated values. The general rule for simple spans carrying moving concentrated loads states that the maximum bending moment produced by moving concentrated loads occurs under one of the loads when that load is as far from one support as the center of gravity of all the moving loads on the beam is from the other support. It should be understood that locating the precise critical section and load position for rating depends on the combined influence of dead load, live load, and member capacity that make up the general Rating Factor equation.

A1B.1.5—Allowable Stress Rating (6B.3.1, 6B.4.2, and 6B.5.2)

Consider Maximum Moment Section only for this example.

A1B.1.5.1—Impact (Use Standard AASHTO) (6B.6.4, AASHTO 3.8.2.1)

$$I = \frac{50}{L+125} \le 0.3$$

$$I = \frac{50}{65+125} = 0.26$$

A1B.1.5.2—Distribution (Use Standard AASHTO) (6B.6.3, AASHTO 3.23.2.2, and Table 3.23.1)

Thus:

$$DF = \frac{S_s}{5.5} = \frac{7.33 \text{ ft}}{5.5} = 1.33$$

$$M_{LL+I} = M_{LL}\left(1+I\right) \times DF = 448\left(1+0.26\right)\left(1.33\right)$$

$$M_{LL+I} = 751 \text{ kip-ft}$$

A1B.1.5.3—Inventory Level (Bottom Tension Controls) (6B.5.2.1, Table 6B.5.2.1-1)

For steel with $F_y = 36$ ksi $\rightarrow f_I = 0.55 f_y$

Thus:

$$f_I = 0.55\left(36\right) = 20 \text{ ksi}$$

The Resisting Capacity $\left(M_{RI}\right) = f_I S_x^L$

$$M_{RI} = 20 \text{ ksi } \left(787.7 \text{ in.}\right)^3 = 15754 \text{ kip-in.} = 1{,}313 \text{ kip-ft}$$

Then:

$$RF_I = \frac{M_{RI} - M_{DL}\dfrac{S_b^L}{S_b^{DL}} - M_{SDL}\dfrac{S_b^L}{S_b^{SDL}}}{M_{LL+I}}$$

$$= \frac{1313 - 439\dfrac{787.7}{563.7} - 129\dfrac{787.7}{716.7}}{751} = \frac{557.8}{751}$$

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= 0.74 or 0.74 × 36 tons = 26.7 tons

Alternatively, in terms of stress:

$$RF_I = \frac{f_s - \frac{M_{DL}}{S_b^{DL}} - \frac{M_{SDL}}{S_b^{SDL}}}{\frac{M_{LL+I}}{S_b^{LL+I}}}$$

$$= \frac{20 \text{ ksi} - \frac{439 \text{ ft-kips} \times 12 \text{ in./ft}}{563.7 \text{ in.}^3} - \frac{129 \text{ ft-kips} \times 12 \text{ ft-kips}}{716.7 \text{ in.}^3}}{\frac{751 \text{ ft-kips} \times 12 \text{ in./ft}}{787.7 \text{ in.}^3}}$$

$$= \frac{20 - 9.345 - 2.160}{11.441}$$

$$= \frac{8.495}{11.441} = 0.74 \text{ as above}$$

A1B.1.5.4—Operating Level (6B.5.2.1, Table 6B.5.2.1-2)

For steel with $F_y = 36$ ksi $\rightarrow f_O = 0.75 f_y$

Thus:

$$f_O = 0.75(36) = 27 \text{ ksi}$$

and

$$M_{RO} = 27(787.7) = 21268 \text{ kip-in.} = 1772 \text{ kip-ft}$$

and:

$$RF_O = \frac{1772 - 439\frac{787.7}{563.7} - 129\frac{787.7}{716.7}}{751} = \frac{1016.8}{751}$$

$RF_O = 1.35$ or 1.35 × 36 tons = 48.7 tons

A1B.1.5.5—Summary of Ratings for Allowable Stress Rating Method

Table A1B.1.5.5-1—Summary of Ratings for Allowable Stess Rating Method—Interior Stringer

	RF	Tons
Inventory	0.74	26.7
Operating	1.35	48.7

A1B.1.6—Load Factor Rating (6B.3.2, 6B.4.3, and 6B.5.3)

Consider maximum moment section only for this example. See general notes.

A1B.1.6.1—Impact (Use Standard AASHTO) (6B.6.4)

From Allowable Stress Rating $I = 0.26$

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A1B.1.6.2—Distribution (Use Standard AASHTO) (6B.6.3)

From Allowable Stress Rating $DF = 1.33$

$$M_{LL+I} = M_{LL}(1+I)DF = 448(1+0.26)(1.33)$$

$$= 751 \text{ kip-ft (as for AS Rating)}$$

A1B.1.6.3—Capacity of Section, M_R (6B.5.3.1)

For braced, compact, composite sections:

$$M_R = M_u \qquad \text{AASHTO 10.50.1.1}$$

where M_u is found in accordance with applicable load factor provisions of AASHTO.

Check assumptions:

1. Section is fully braced along top flange by composite deck (for Live Load and *SDL*).
2. To check if section is compact, need to apply provisions of AASHTO 10.50.1.1.1. These checks follow.

The compressive force in the slab C is equal to the smallest value given by the following equations: AASHTO 10.50.1.1.1(a)

$$C = 0.85 f'_c b t_s + (AF_y)_c \qquad \text{AASHTO Eq. 10-123}$$

Neglecting that part of the reinforcement that lies in the compressive zone the equation reduces to:

$$C_{CONC} = 0.85 f'_c b_{eff} t_s = 0.85(3 \text{ ksi})(87 \text{ in.})(7.25 \text{ in.}) = 1608 \text{ kips}^*$$

$$C = (AF_y)_{bf} + (AF_y)_{tf} + (AF_y)_w \qquad \text{AASHTO Eq. 10-124}$$

where $(AF_y)_{bf}$ includes cover plate, this equation reduces to:

$$C_{STL} = A_s f_y = (38.26 \text{ in.}^2 + 6.56 \text{ in.}^2)(36 \text{ ksi}) = 1613.5 \text{ kips}$$

$$C_{CONC} < C_{STL} \therefore C_{CONC} = 1,608 \text{ controls}$$

Capacity:

$$C' = \frac{\sum(AF_y) - C}{2} = \frac{1,613.5 - 1,608}{2} = 2.75 \text{ kips} \qquad \text{AASHTO Eq. 10-126}$$

$$(AF_y)_{TF} = (11.51 \times 0.855)(36) = 354 \text{ kips} >>> 2.75 \text{ kips} \therefore \text{ NA in top flange} \qquad \text{AASHTO 10.50.1.1.1(d)}$$

$$\bar{y} = \frac{C'}{(AF_y)_{TF}} t_{TF} = \frac{2.75}{354}(0.855) = 0.007 \text{ in. neglect. Say NA at top of steel.} \qquad \text{AASHTO Eq. 10-127}$$

Since the PNA is at the top of the flange, the depth of the web in compression at the plastic moment, D_{cp}, is equal to zero. Hence, the web slenderness requirement given by Eq. 10-129 in AASHTO Article 10.50.1.1.2 is automatically satisfied.

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Check the ductility requirement given by Eq. 10-129a in AASHTO Article 10.50.1.1.2:

$$\left(\frac{D_p}{D'}\right) \le 5 \qquad \text{AASHTO Eq. 10.129a}$$

$$D' = \beta\frac{\left(d + t_s + t_h\right)}{7.5} \qquad \beta = 0.9 \text{ for } F_y = 36{,}000 \text{ psi}$$

$$D' = 0.9\frac{\left(33.725 + 7.25 + 0.0\right)}{7.5} = 4.92$$

$$D_p = 7.25 \text{ in.}$$

$$\left(\frac{D_p}{D'}\right) = \frac{7.25}{4.92} = 1.47 < 5 \text{ OK}$$

Since the top flange is braced by the hardened concrete deck, local and lateral buckling requirements need not be checked. The capacity of composite beams in simple spans satisfying the preceding web slenderness and ductility requirements is given by Eq. 10-129c in AASHTO 10.50.1.1.2 when D_p exceeds D':

$D' < D_p \le 5D'$

4.92 in. < 7.25 in. ≤ 5 x 4.92 in. = 24.6 in.

Therefore:

$$C = M_R = M_U = \frac{5M_p - 0.85M_y}{4} + \frac{0.85M_y - M_p}{4}\left(\frac{D_p}{D'}\right) \qquad \text{AASHTO Eq. 10.129c}$$

$$M_y = F_y S = \left(36\right)\frac{787.7}{12} = 2{,}363 \text{ kip-ft}$$

Compute the plastic moment capacity, M_p

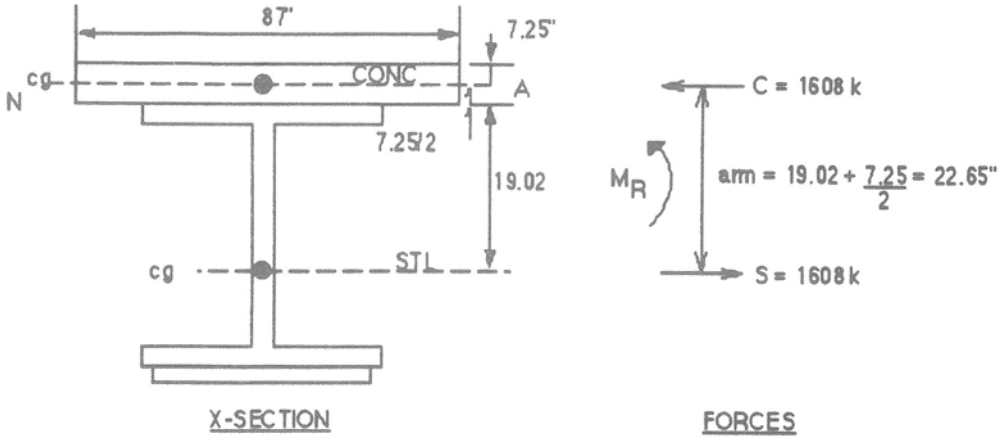

Figure A1B.1.6.3-1—Cross Section—Interior Stringer, for Determining Plastic Moment Capacity, M_p

$$M_p = C \times arm = 1{,}608\left(22.65\right) = 36{,}421 \text{ kip-in.} = 3{,}035 \text{ kip-ft}$$

$$M_R = \frac{5\left(3{,}035\right) - 0.85\left(2{,}363\right)}{4} + \frac{0.85\left(2{,}363\right) - 3{,}035}{4}\left(1.47\right) = 2{,}914 \text{ kip-ft}$$

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A1B.1.6.4—Inventory Level (6B.4.1 and 6B.5.3)

$$RF_I^{LF} = \frac{M_R - A_1 M_D}{A_2 M_{L+I}}$$ Eq. 6B.4.1-1

where: 6B.4.3

$A_1 = 1.3$

$A_2 = 2.17$

Thus:

$$RF_I^{LF} = \frac{(2914) - 1.3(439 + 129)}{2.17(751)}$$

RF_I^{LF} = 1.33 or 1.33 × 36 tons = 47.9 tons

A1B.1.6.5—Operating Level (6B.4.3)

Only change is A_2 = 1.3

Thus:

$$RF_O^{LF} = \frac{2.17}{1.3} RF_I^{LF} = \frac{2.17}{1.3}(1.33)$$

RF_O^{LF} = 2.22 or 2.22 × 36 tons = 79.9 tons

A1B.1.6.6—Check Serviceability Criteria

For HS loadings overload is defined as $D + 5(L + I)/3$ AASHTO 10.57

A1B.1.6.6a—At Inventory Level (Bottom Steel in Tension Controls)

$$f_{DL} + f_{SDL} + 1.67(f_{LL+I}) \le \text{Serv. Strength} = 0.95F_y$$ AASHTO 10.57.2

Thus A_1 = 1.0 and A_2 = 1.67 for service rating:

$$RF_I^{LF} = \frac{0.95F_y - (1.0) f_{DL} - (1.0) f_{SDL}}{(1.67) f_{LL+I}}$$

$$= \frac{0.95(36 \text{ ksi}) - \frac{439(12)}{563.7} - \frac{129(12)}{716.7}}{1.67 \frac{751(12)}{787.7}}$$

= RF_I^{LF} = 1.19 or 1.19 × 36 tons = 42.8 tons

Check the web compressive stress:

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$$C = F_{cr} = \frac{26,200,000\alpha k}{\left(\frac{D}{t_w}\right)^2} \le F_{yw}$$ AASHTO Eq. 10-173

where:

$k = 9\left(D \div D_c\right)^2$

$\alpha = 1.3$

Since D_c is a function of the dead-to-live load stress ratio according to the provisions of AASHTO 10.50(b), an iterative procedure may be necessary to determine the rating factor:

Compute the compressive stresses at the top of the web:

$$f_{DL} = \frac{439(12)(18.165)}{8,293} = 11.5 \text{ ksi}$$

$$f_{SDL} = \frac{129(12)(10.935)}{15,725} = 1.1 \text{ ksi}$$

$$f_{LL+I} = \frac{(751)(12)(4.935)}{22,007} = 2.02 \text{ ksi}$$

$$\sum = 14.62 \text{ksi}$$

Compute the tensile stresses at the bottom of the web:

$$f_{DL} = \frac{439(12)(13.23)}{8,293} = 8.4 \text{ ksi}$$

$$f_{SDL} = \frac{129(12)(20.46)}{15,725} = 2.0 \text{ ksi}$$

$$f_{L+I} = \frac{(751)(12)(26.46)}{22,007} = 10.84 \text{ksi}$$

$$\sum = 21.24 \text{ksi}$$

$$D_c = 31.39\left(\frac{14.62}{14.62+21.24}\right) = 12.80 \text{ in.}$$

$$k = 9\left(D \div D_c\right)^2 = 9\left(31.39 \div 12.80\right)^2 = 54.1$$

$$C = F_{cr} = \frac{26,200,000(1.3)(54.1)}{\left(\frac{31.39}{0.58}\right)^2 (1,000)} = 629 \text{ ksi} > F_{yw}$$

$\therefore F_{cr} = F_{yw} = 36$ ksi

$$RF_I^{LF} = \frac{36 - 11.5 - 1.1}{1.67(2.02)} = \underline{6.9} \text{ or } 6.9 \times 36 \text{ tons} = \underline{248.4 \text{ tons}}$$

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Since the computed rating factor would cause the total stresses in the tension flange to far exceed F_y (causing the neutral axis to be higher on the web), further iterations are not necessary in this case. The web compressive stress does not govern the serviceability rating.

A1B.1.6.6b—At Operating Level

$$f_D = RF_O^{LF}\left(f_{L+I}\right) \leq \text{ Service Strength}$$

Thus A_1 = 1.0 and A_2 = 1.0 for service rating:

$$RF_O^{LF} = RF_I^{LF} \times 1.67 = 1.19 \times 1.67$$

RF_O^{LF} = 1.98 or 1.98 × 36 tons = 71.3 tons

A1B.1.6.7—Summary of Ratings for Load Factor Rating Method

Table A1B.1.6.7-1—Summary of Ratings for Load Factor Rating Method—Interior Stringer

	RF	Tons	Controlled
Inventory	1.19	42.8	AASHTO 10.57.2
Operating	1.98	71.3	AASHTO 10.57.2

A1B.1.7—Load Factor Rating—Rate for Single-Unit Formula B Loads

M_{LL+I} from Appendix C6B:

Span	HS-20	NRL	SU4	SU5	SU6	SU7	
60 ft	512.2	595.1	430.2	472.5	525.0	569.9	kip-ft
70 ft	619.2	714.2	510.2	564.4	628.3	685.4	kip-ft

By interpolation:

65 ft	565.7	654.7	470.2	518.5	576.7	627.7	kip-ft

Apply distribution factor DF = 1.33

65 ft	751.0	870.8	625.4	689.6	767.0	834.8	kip-ft

Capacity of Section M_R = 2,914 kip-ft

Dead Load M_{DL} = 439 kip-ft

Superimposed Dead Loads M_{SDL} = 129 kip-ft

$$\text{Inv. } RF = \frac{2,914 - 1.3(439 + 129)}{2.17\left(M_{L+I}\right)}$$

$$\text{Opr. } RF = \frac{2,914 - 1.3(439 + 129)}{1.3\left(M_{L+I}\right)}$$

Strength Rating Factors:

	HS-20	NRL	SU4	SU5	SU6	SU7
Inventory	1.33	1.15	1.60	1.45	1.31	1.20
Operating	2.22	1.92	2.67	2.42	2.19	2.00

Check Serviceability Criteria:

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$$RF = \frac{0.95F_y - f_{DL} - f_{SDL}}{1.67 f_{LL+I}}$$

$$RF = \frac{34.2 - 9.35 - 2.16}{1.67\left(M_L + I \times 12 \times 1.0 / 787.7\right)}$$

Serviceability Rating Factors (Controls):

HS-20	NRL	SU4	SU5	SU6	SU7
1.19	1.03	1.43	1.29	1.16	1.07

As the Notional Rating Load NRL $RF > 1.0$ for strength and serviceability, the bridge has adequate capacity for all legal loads, including the single-unit Formula B trucks.

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PART C—SUMMARY

A1C.1—Summary of All Ratings for Example A1

Table A1C.1-1—Summary of Rating Factors for All Rating Methods—Interior Stringer

			Design Load Rating (HL-93)		Legal Load Rating								Permit Load Rating	HS-20 Rating	
			Inventory	Operating	Type 3	Type 3S2	Type 3-3	SU4	SU5	SU6	SU7	NRL		Inventory	Operating
LRFR Method	Strength I	Flexure	1.29	1.67	2.64	2.46	2.71	2.76	2.50	2.25	2.07	1.98	—		
		Shear	2.29	2.97	—	—	—	—	—	—	—	—	—	—	—
Limit State	Strength II	Flexure	—	—	—	—	—	—	—	—	—	—	1.92	—	—
		Shear	—	—	—	—	—	—	—	—	—	—	2.94	—	—
	Service II		1.21	1.57	2.33	2.17	2.38	2.06	1.87	1.68	1.55	1.48	1.53	—	—
	Fatigue		0.38	—	—	—	—	—	—	—	—	—	—	—	—
Allowable Stress Method			—	—	—	—	—	—	—	—	—	—	—	0.74	1.35
Load Factor Method	Strength		—	—	—	—	—	1.60	1.45	1.31	1.20	1.15	—	1.33	2.22
	Serviceability		—	—	—	—	—	1.43	1.29	1.16	1.07	1.03	—	1.19	1.98

Table A1C.1-2—Summary of Rating Factors for Load and Resistance Factor Rating Method—Exterior Stringer

Limit State		Design Load Rating		Legal Load Rating								Permit Load Rating
		Inventory	Operating	Type 3	Type 3S2	Type 3-3	SU4	SU5	SU6	SU7	NRL	
Strength I	Flexure	1.40	1.81	2.85	2.66	2.93	2.85	2.58	2.32	2.13	2.05	—
	Shear	3.05	3.95	—	—	—	—	—	—	—	—	—
Strength II	Flexure	—	—	—	—	—	—	—	—	—	—	1.53
	Shear	—	—	—	—	—	—	—	—	—	—	3.33
Service II		1.48	1.92	2.84	2.65	2.91	2.52	2.29	2.05	1.89	1.81	1.37

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A1C.2—References

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A2—REINFORCED CONCRETE T-BEAM BRIDGE: EVALUATION OF AN INTERIOR BEAM

PART A—LOAD AND RESISTANCE FACTOR RATING METHOD

A2A.1—Bridge Data

Span:	26 ft
Year Built	1925
Materials:	
Concrete:	f'_c = 3 ksi
Reinforcing Steel:	Unknown f_y
Condition:	Minor deterioration has been observed, but no section loss. NBI Item 59 = 6
Riding Surface:	Field verified and documented: Smooth approach and deck
ADTT (one direction):	1,850
Skew:	0°

A2A.2—Dead-Load Analysis—Interior Beam

Permanent loads on the deck are distributed uniformly among the beams. LRFD Design 4.6.2.2.1

A2A.2.1—Components and Attachments, *DC*

Structural Concrete:

Consisting of deck + stem + haunches (conservatively, $2^1/_2$-in. chamfers were not deducted)

$$\left[\frac{6 \text{ in.}}{12}\times 6.52 \text{ ft} + 1.25 \text{ ft}\times 2 \text{ ft} + 2\left(\frac{1}{2}\times\frac{6 \text{ in.}}{12}\times\frac{6 \text{ in.}}{12}\right)\right]\times(0.150 \text{ kcf}) = 0.902 \text{ kip/ft}$$

$$\text{Railing and curb } 0.200 \text{ kip/ft}\times\frac{1}{2} = 0.100 \text{ kip/ft}$$

$$\text{Total per beam, } DC = \underline{\quad\quad} 1.002 \text{ kip/ft}$$

$$M_{DC} = \frac{1}{8}\times 1.002\times 26^2 = 84.7 \text{ kip-ft}$$

$$V_{DCmax} = 1.002(0.5\times 26) = 13.0 \text{ kips}$$

A2A.2.2—Wearing Surface, *DW*

Thickness was field measured: 6A.2.2.3

Asphalt Overlay:

$$\left(\frac{5 \text{ in.}}{12}\right)(22 \text{ ft})(0.144 \text{ kcf})\left(\frac{1}{4}\right) = 0.330 \text{ kip/ft}$$

$$M_{DW} = \frac{1}{8}\times 0.330\times 26^2 = 27.9 \text{ kip-ft}$$

$$V_{DWmax} = 0.33(0.5\times 26) = 4.3 \text{ kips}$$

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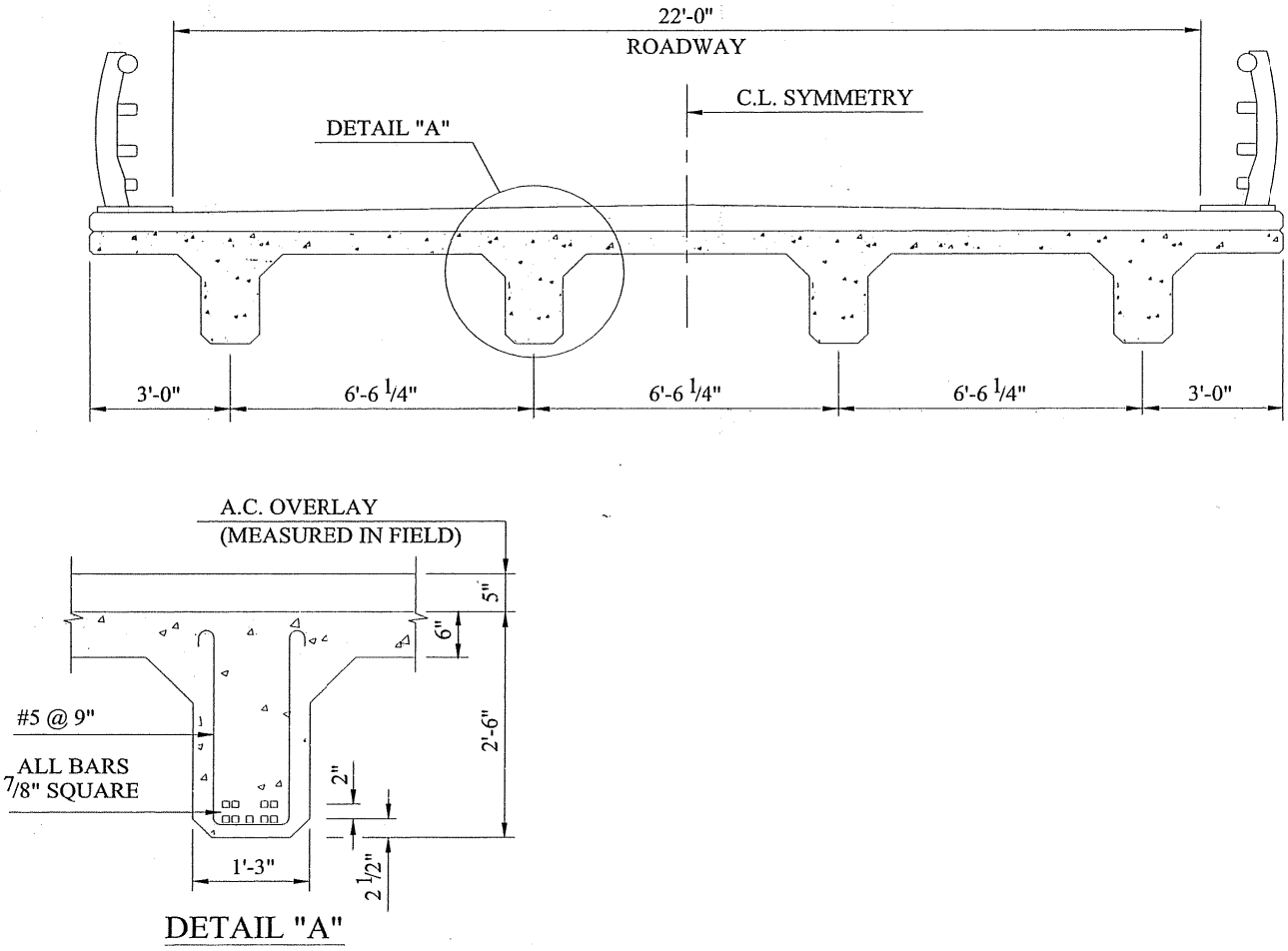

Figure A2A.2.2-1—Reinforced Concrete T-Beam Bridge

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A2A.3—Live-Load Analysis—Interior Beam

A2A.3.1—Compute Live-Load Distribution Factor

AASHTO LRFD Type (e) cross section

LRFD Design Table 4.6.2.2.1-1

Longitudinal Stiffness Parameter, K_g

LRFD Design Eq. 4.6.2.2.1-1

$$K_g = n\left(I + Ae_g^2\right)$$

$n = 1.0$

$$I = \frac{1}{12}\times 15\times 24^3 = 17{,}280 \text{ in.}^4$$

$$A = 15 \times 24 = 360 \text{ in.}^2$$

$$e_g = \frac{1}{2}(24+6) = 15 \text{ in.}$$

$$K_g = 1.0\ (17{,}280 + 360 \times 15^2)$$

$$= 98{,}280 \text{ in.}^4$$

$$\frac{K_g}{12Lt_s^3} = \frac{98{,}280}{12\times 26\times 6^3} = 1.46$$

A2A.3.1.1—Distribution Factor for Moment, g_m (LRFD Design Table 4.6.2.2.2b-1)

One Lane Loaded:

$$g_{m1} = 0.06 + \left(\frac{S}{14}\right)^{0.4}\left(\frac{S}{L}\right)^{0.3}\left(\frac{K_g}{12Lt_s^3}\right)^{0.1}$$

$$= 0.06 + \left(\frac{6.52}{14}\right)^{0.4}\left(\frac{6.52}{26}\right)^{0.3}(1.46)^{0.1}$$

$$= 0.565$$

Two or More Lanes Loaded:

$$g_{m2} = 0.075 + \left(\frac{S}{9.5}\right)^{0.6}\left(\frac{S}{L}\right)^{0.2}\left(\frac{K_g}{12Lt_s^3}\right)^{0.1}$$

$$= 0.075 + \left(\frac{6.52}{9.5}\right)^{0.6}\left(\frac{6.52}{26}\right)^{0.2}(1.46)^{0.1}$$

$$= 0.703 > 0.565$$

$\therefore$ use $g_m = 0.703$

A2A.3.1.2—Distribution Factor for Shear, g_v (LRFD Design Table 4.6.2.2.3a-1)

One Lane Loaded:

$$g_{v1} = 0.36 + \frac{S}{25.0}$$

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$$= \quad 0.36 + \frac{6.52}{25.0}$$

$$= \quad 0.621$$

Two or More Lanes Loaded:

$$g_{v2} = \quad 0.2 + \frac{S}{12} - \left(\frac{S}{35}\right)^{2.0}$$

$$= \quad 0.2 + \frac{6.52}{12} - \left(\frac{6.52}{35}\right)^{2.0}$$

$$= \quad 0.709 > 0.62$$

$\therefore$ use $g_v =$ 0.709

A2A.3.2—Compute Maximum Live Load Effects

A2A.3.2.1—Maximum Design Live Load (HL-93) Moment at Midspan

Design Lane Load Moment = 54.1 kip-ft

Design Truck Moment = 208.0 kip-ft

Tandem Axles Moment = 275.0 kip-ft Governs

IM = 33 percent LRFD Design Table 3.6.2.1-1

M_{LL+IM} = 54.1 + 275.0 × 1.33

= 419.9 kip-ft

A2A.3.2.2—Maximum Design Live Load Shear (HL-93) at Critical Section

See Article A2A.7.

A2A.3.2.3—Distributed Live Load Moments

Design Live Load HL-93:

M_{LL+IM} = 419.9 × 0.703

= 295.2 kip-ft

A2A.4—Compute Nominal Flexural Resistance

A2A.4.1—Compute Effective Flange Width, b_e (LRFD Design 4.6.2.6.1)

Effective Flange Width, b_e, may be taken as the tributary width perpendicular to the axis of the member.

$\therefore$ use $b_e = 78.25$ in.

A2A.4.2—Compute Distance to Neutral Axis, *c* LRFD Design 5.6.3.1.1

Assume rectangular section behavior.

β_1 = 0.85 for $f'_c = 3{,}000$ psi LRFD Design 5.6.2.2

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$$c = \frac{A_s f_y}{0.85 f_c' \beta_1 b}$$ LRFD Design Eq. 5.6.3.1.1-4

$A_s = 9\left(\frac{7}{8}\right)^2 = 6.89$ in.^{2} (nine $^7/_8$-in.^{2} bars)

$b = 78.25$ in.

$f_y = 33$ ksi (unknown steel) Table 6A.5.2.2-1

$$c = \frac{6.89 \times 33}{0.85 \times 3.0 \times 0.85 \times 78.25}$$

$= 1.34$ in. < 6 in.

The neutral axis is within the slab. Therefore, there will be rectangular section behavior.

$a = c\beta_1$

$= 1.34 \times 0.85$

$= 1.14$ in.

Distance from bottom of section to *CG* of reinforcement, $\bar{y}$

$$\bar{y} = \frac{4 \times 4.5 + 5 \times 2.5}{9}$$

$\bar{y} = 3.39$ in.

$d_s = h - \bar{y}$

$h = 30$ in.

$d_s = 30$ in. $- 3.39$ in.

$= 26.61$ in.

$$M_n = A_s f_y \left(d_s - \frac{a}{2}\right)$$ LRFD Design 5.6.3.2.3, LRFD Design Eq. 5.6.3.2.2-1

$$= 6.89 \times 33\left(26.61 - \frac{1.14}{2}\right)\frac{1}{12}$$

$= 493.4$ kip-ft

A2A.5—Maximum Reinforcement (6A.5.5)

The factored resistance (φ factor) of compression controlled sections shall be reduced in accordance with LRFD Design Article 5.5.4.2. This approach limits the capacity of over-reinforced (compression controlled) sections. C6A.5.5

The net tensile strain, ε_t, is the tensile strain at nominal strength and determined by strain compatibility using similar triangles. LRFD Design C5.6.2.1

Given an allowable concrete strain of 0.003 and depth to neutral axis c = 1.34 in.

$$\frac{\varepsilon_c}{c} = \frac{\varepsilon_t}{d - c}$$

$$\frac{0.003}{1.34 \text{ in.}} = \frac{\varepsilon_t}{26.61 \text{ in.} - 1.34 \text{ in.}}$$

Solving for ε_t, $\varepsilon_t = 0.0566$.

For $\varepsilon_t = 0.0566 > 0.005$, the section is tension controlled. LRFD Design 5.6.2.1

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For conventional construction and tension controlled reinforced concrete sections, resistance factor φ shall be taken as 0.90. LRFD Design 5.5.4.2

A2A.6—Minimum Reinforcement (6A.5.6)

The amount of reinforcement must be sufficient to develop M_r equal to the lesser of: LRFD Design 5.6.3.3

$1.2M_{cr}$ or $1.33M_u$

M_r = $\varphi_f M_n$ = 0.90 × 493.4 kip-ft

= 444.1 kip-ft

1. $1.33M_u$ = 1.33 (1.75 × 295.2 + 1.25 × 84.7 + 1.25 × 27.9)

= 874.3 kip-ft > 444.1 kip-ft No Good

2. $M_{cr} = \gamma_3 \left[\left(\gamma_1 f_r + \gamma_2 f_{cpe} \right) S_c - M_{dnc} \left(\frac{S_c}{S_{nc}} - 1 \right) \right]$ LRFD Design Eq. 5.6.3.3-1

M_{dnc} = 0 Total unfactored dead load moment acting on the monolithic or noncomposite section

f_{cpe} = 0 Compressive stress in concrete due to effective prestress forces only at extreme fiber of section where tensile stress is caused by externally applied loads

S_{nc} = $\frac{I}{y_t}$ Uncracked section modulus (neglect steel)

γ_1 = flexural cracking variability factor = 1.6

γ_2 = prestress variability factor = 0

γ_3 = ratio of specified minimum yield strength to ultimate tensile strength of nonprestressed reinforcement = 0.67

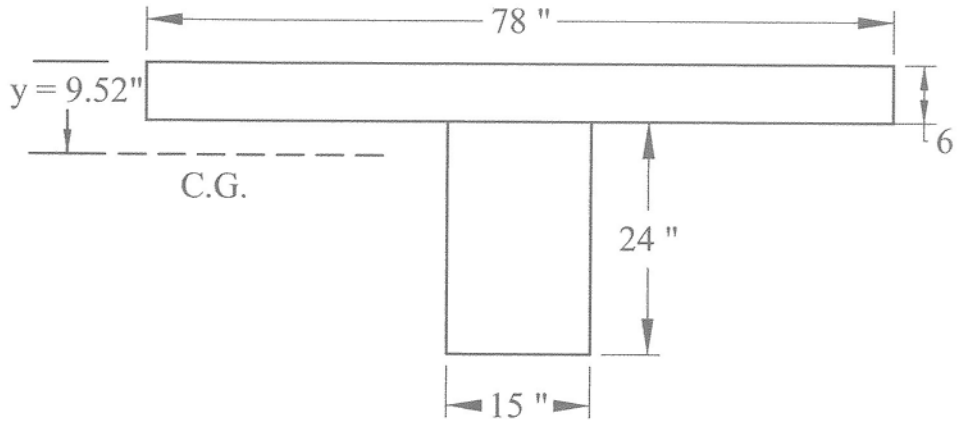

Figure A2A.6-1 Cross Section of Concete T-Beam—Depth to Centroid of Uncracked Section

$$y = \frac{\sum (A_i \times y_i)}{\sum A_i}$$

$$y = \frac{(78 \times 6 \times 3) + (24 \times 15 \times 18)}{(78 \times 6) + (24 \times 15)} = 9.52 \text{ in.}$$

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from top of slab to centroid of uncracked section

$I = \sum\left(I_o + A_c d^2\right)$ where $I_0 = bh^3/12$

	y	A_c	$A_c y$	d	Ad^2	I_0
slab	3	468	1,404	6.52	19,895	1,404
stem	18	360	6,480	8.48	25,888	17,280
		828	7,884		45,783	18,684

$I = \left(18,684 + 45,783\right) = 64,467$

y_b = 30 in. – 9.52 in. = 20.48 in.

$S_{bc} = \dfrac{64,467}{20.48} = 3,148 \text{ in.}^3$

$f_r = 0.24\sqrt{f'_c} = 0.24\sqrt{3.0} = 0.416$ ksi

$M_{cr} = (0.67)\left[1.6\left(0.416 + 0\right)3,148 - 0\right] = 116.99$ kip-ft LRFD Design 5.4.2.6

$M_r = \varphi M_n = 0.9\ (493.4)$

$M_r = 444.1$ kip-ft $> M_{cr} = 116.99$ kip-ft OK

The section meets the requirements for minimum reinforcement.

A2A.7—Compute Nominal Shear Resistance

Stirrups: #5 bars at 9 in.

$$A_v = 2 \times \frac{\pi}{4}\left(\frac{5}{8}\right)^2 = 0.6136 \text{ in.}^2$$

Unknown $f_y \rightarrow 33$ ksi

Critical section for shear: LRFD Design 5.7.3.2

Effective Shear Depth: d_v LRFD Design 5.7.2.8

Distance, meassured perpendicular to the neutral axis, between resultants of the tensile and compressive forces. It need not be taken to be less than the greater of:

$0.9d_e$

$0.72h$

1. $d_v = \dfrac{M_n}{A_s f_y + A_{ps} f_{ps}}$ LRFD Design Eq. C5.7.2.8-1

This quantity depends upon the transfer and development of the reinforcement. Conservatively, we will take d_v as the greater of the remaining criteria to reduce required calculations.

2. 0.9 (26.61) = 23.95 in.

3. 0.72 (30.0) = 21.60 in.

d_v = 23.95 in.

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Assume θ = 45°

$0.5d_v \cot \theta$ = (0.5) (26.04) (cot 45) = $0.5d_v < d_v$ Use d_v

Critical section for shear at 23.95 in. from face of support.

Bearing pad width = 4 in.

Calculate shear at $23.95 + \frac{4}{2} = 25.95$ in. from centerline of bearing.

Maximum Shear at Critical Section Near Support (25.95 in.) calculated by statics:

V_{TANDEM} = 41.9 kips Governs
V_{TRUCK} = 41.4 kips
V_{LANE} = 7.0 kips

Total Live-Load Shear = (1.33) (41.9) + 7.0 = 62.7 kips LRFD Design Table 3.6.2.1-1

(including 33 percent increase for dynamic load allowance)

Distributed Shear, V_{LL+IM} = (62.7) (0.709) = 44.5 kips

Dead-Load Shears:

$$V_{DC} = 1.002\left(0.5 \times 26 - \frac{25.95}{12}\right) = 10.8 \text{ kips}$$

$$V_{DW} = 0.33\left(0.5 \times 26 - \frac{25.95}{12}\right) = 3.6 \text{ kips}$$

Resistance:

The lesser of :

$V_n = V_c + V_s + V_p$ LRFD Design Eq. 5.7.3.3-1

$V_n = 0.25 f'_c b_v d_v + V_p$ LRFD Design Eq. 5.7.3.3-2

In this case there is no V_p contribution, and:

Effective shear depth, d_v = 23.95 in. LRFD Design Eq. 5.7.2.8-1

Minimum web width within the depth d_v, b_v = 15 in. LRFD Design Eq. 5.7.2.8-1

$V_c = 0.0316 \beta \lambda \sqrt{f'_c} b_v d_v$ LRFD Design Eq. 5.7.3.3-3

$V_s = \frac{A_v f_y d_v \cot \theta}{s}$ (for $\alpha = 90°$) LRFD Design Eq. 5.7.3.3-4

Simplified Approach: LRFD Design 5.7.3.4.1

β = 2.0
θ = 45

$$V_c = (0.0316)(2)\sqrt{3.0}(15)(23.95) = 39.3 \text{ kips}$$

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$$V_s = \frac{(0.6136)(33)(23.95)\cot 45}{9} = 53.9 \text{ kips}$$

$V_n = 39.3 + 53.9 = 93.2$ kips

$V_n = 0.25 \times 3.0 \times 15 \times 23.95 = 269.4$ kips

93.2 kips < 269.4 kips, therefore $V_n = 93.2$ kips

A2A.8—Summary for Interior Concrete T-Beam

	Dead Load *DC*	Dead Load *DW*	Live Load Distribution Factor	Dist. Live Load + Impact	Nominal Capacity
Moment, kip-ft	84.7	27.9	$g_m = 0.703$	295.2	493.4
Shear, kips	10.8	3.6	$g_v = 0.709$	44.5	93.2

A2A.9—General Load Rating Equation

$$RF = \frac{C - (\gamma_{DC})(DC) - (\gamma_{DW})(DW) \pm (\gamma_P)(P)}{(\gamma_L)(LL + IM)} \text{ s}$$ Eq. 6A.4.2.1-1

For Strength Limit States $C = (\varphi_c)(\varphi_s)(\varphi)R_n$

A2A.10—Evaluation Factors (for Strength Limit States)

1. Resistance Factor, φ LRFD Design 5.5.4.2

φ = ~~1.0~~ 0.90 for flexure and shear of normal weight concrete

2. Condition Factor, φ_c 6A.4.2.3

No member condition information available. NBI Item 59 = 6.

$\varphi_c = 1.0$

3. System Factor, φ_s 6A.4.2.4

$\varphi_s = 1.0$ 4-girder bridge with $S > 4$ ft (for flexure and shear)

A2A.11—Design Load Rating (6A.4.3)

A2A.11.1—Strength I Limit State

$$RF = \frac{(\varphi_c)(\varphi_s)(\varphi)R_n - (\gamma_{DC})(DC) - (\gamma_{DW})(DW)}{(\gamma_L)(LL + IM)}$$

A2A.11.2—Inventory Level (6A.5.4.1)

Load	Load Factor
DC	1.25
DW	1.25
LL	1.75

Table 6A.4.2.2-1

Thickness was field verified

Flexure:

$$RF = \frac{(1.0)(1.0)(0.90)(493.4) - [(1.25)(84.7) + (1.25)(27.9)]}{(1.75)(295.2)}$$

$= 0.59$

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Shear:

$$RF = \frac{(1.0)(1.0)(0.90)(93.2)-\left[(1.25)(10.8)+(1.25)(3.6)\right]}{(1.75)(44.5)}$$

$$= 0.85$$

The shear ratings factors for Design Load Rating are calculated for illustration purposes only. In-service concrete bridges that show no visible signs of shear distress need not be checked for shear during design load or legal load ratings. 6A.5.8

A2A.11.3—Operating Level

Load	Load Factor γ
DC	1.25
DW	1.25
LL	1.35

Table 6A.4.2.2-1

For Strength I Operating Level only the live load factor changes; therefore the rating factor can be calculated by direct proportions.

Flexure:

$$RF = 0.59\times\frac{1.75}{1.35}$$

$$= 0.76$$

Shear:

$$RF = 0.85\times\frac{1.75}{1.35}$$

$$= 1.10$$

Note: The shear resistance using MCFT varies along the length. The simplified assumptions of $\beta = 2.0$ and $\theta = 45°$ in this example are conservative for high shear–low moment regions. Example A3 demonstrates a case where the shear rating must be performed at multiple locations along the length of the member. Tension in the longitudinal reinforcement caused by moment-shear interaction (LRFD Design Article 5.7.3.5) has not been checked in this example. Example A3 includes demonstrations of this check.

No service limit states apply to reinforced concrete bridge members at the design load check.

A2A.12—Legal Load Rating (6A.5.4.2)

Note: Since the Operating Level Design Load Rating produced $RF < 1.0$ for flexure, load ratings for legal loads should be performed to determine the need for posting.

Live Load: AASHTO Legal Loads—Types 3, 3S2, and 3-3 (Rate for all three) 6A.4.4.2.1

g_m = 0.703

L = 26 ft ($L < 40$ ft)

IM = 33 percent

Even though the condition of the wearing surface has been field evaluated as smooth, the length of the flexure members prevents the use of a reduced *IM*. C6A.4.4.3

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	Type 3	Type 3S2	Type 3-3		
M_{LL+IM} =	250.6	240.7	206.2	kip-ft	Table E6A-1
gM_{LL+IM}=	176.2	169.2	145.0	kip-ft	

Live Load: AASHTO Legal Loads—Specialized Hauling Units and Notional Rating Load—SU4, SU5, SU6, SU7, and NRL 6A.4.4.2.1b

As before:

g_m = 0.703

L = 26 ft (L <40 ft)

IM = 33 percent C6A.4.4.3

	SU4	SU5	SU6	SU7	NRL		
M_{LL+IM} =	296.9	323.2	350.1	358.6	360.4	kip-ft	Table E6A-2
gM_{LL+IM}=	208.7	227.2	246.1	252.1	253.4	kip-ft	

A2A.12.1—Strength I Limit State (6A.5.4.2.1)

$ADTT$ = 1850

For AASHTO Legal Loads—Types 3, 3S2, and 3-3

Generalized Live-Load Factor:

Linear interpolation is permitted for other $ADTT$. Therefore: Table 6A.4.4.2.3a-1

$$\gamma_L = 1.30 + \frac{1{,}850 - 1{,}000}{5{,}000 - 1{,}000}(1.45 - 1.30) = 1.33$$

γ_L = 1.33

Flexure:

$$RF = \frac{(1.0)(1.0)(0.90)(493.4) - \left[(1.25)(84.7) + (1.25)(27.9)\right]}{(1.33)(M_{LL+IM})}$$

Truck	Type 3	Type 3S2	Type 3-3
RF =	1.29	1.35	1.57
Vehicle Weight (tons)	25	36	40
Safe Load Capacity (tons)	32	48	62

For Specialized Hauling Units and Notional Rating Load—SU4, SU5, SU6, SU7, and NRL

Generalized Live-Load Factor:

γ_L = 1.33 by interpolation Table 6A.4.4.2.3b-1

Flexure:

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$$RF = \frac{(1.0)(1.0)(0.90)(493.4)-\left[(1.25)(84.7)+(1.25)(27.9)\right]}{(1.33)(M_{LL+IM})}$$

Truck	SU4	SU5	SU6	SU7	NRL
RF	1.09	1.00	0.95	0.90	0.90
Vehicle Weight, tons	27	31	34.8	38.8	40
Safe Load Capacity, tons	29	31	32	34	36

No posting is required for the Types 3, 3S2, and 3-3.

Comparison of the above safe capacities for the SU4, SU5, SU6, and SU7 to the NRL Safe Load Capacity demonstrates that for bridges that do not rate the NRL Load, a posting analysis should be performed to resolve posting requirements for single unit multiaxle trucks. The above results show that the Safe Load Capacity for the SU4 and SU5 vehicle is adequate; however, posting may be required for SU6 and SU7 vehicles. 6A.8.2 and C6A.8.2

The descision to post a bridge should be made by the Bridge Owner. When for any legal truck the Rating Factor *RF* is between 0.3 and 1.0 then the following formula should be used to establish the safe posting load for that vehicle type. 6A.8.3

$$\text{Safe Posting Load} = \frac{W}{0.7}\left[(RF)-0.3\right] \quad \text{Eq. 6A.8.3-1}$$

Therefore, for SU6 and SU7, the recommended safe posting loads are:

	SU6	SU7
Safe Posting Load	31	33

No service limit states apply to reinforced concrete bridge members at the legal load check.
This example focused on the interior stringer for illustrative purposes only. Before a final posting descision can be made the exterior beam should be analyzed.

A2A.12.2—Summary

Truck	Type 3	Type 3S2	Type 3-3	SU4	SU5	SU6	SU7	NRL
Weight, tons	25	36	40	27	31	34.8	38.8	40
RF	1.29	1.35	1.57	1.09	1.00	0.98	0.90	0.90
Safe Load Capacity, tons	32	48	62	29	31	32	34	36
Safe Posting Load (tons)	—	—	—	—	—	31	33	

A2A.13—Permit Load Rating (6A.4.5)

Permit Type: Special, Multiple-Trips, no speed control

Permit Weight: 175 kips

Permit Vehicle: Shown in Figure A2A.13-1.

ADTT (one direction): 1,850

IM = 33 percent ($L < 40$ ft) C6A.4.4.3

Undistributed Maximum:

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M_{LL} = 347.3 kip-ft at midspan
V_{LL} = 52.6 kips at 26 in.

A2A.13.1—Strength II Limit State (6A.5.4.2.1)

ADTT (one direction): 1850
Load Factor, γ_L: 1.40 Table 6A.4.5.4.2a-1

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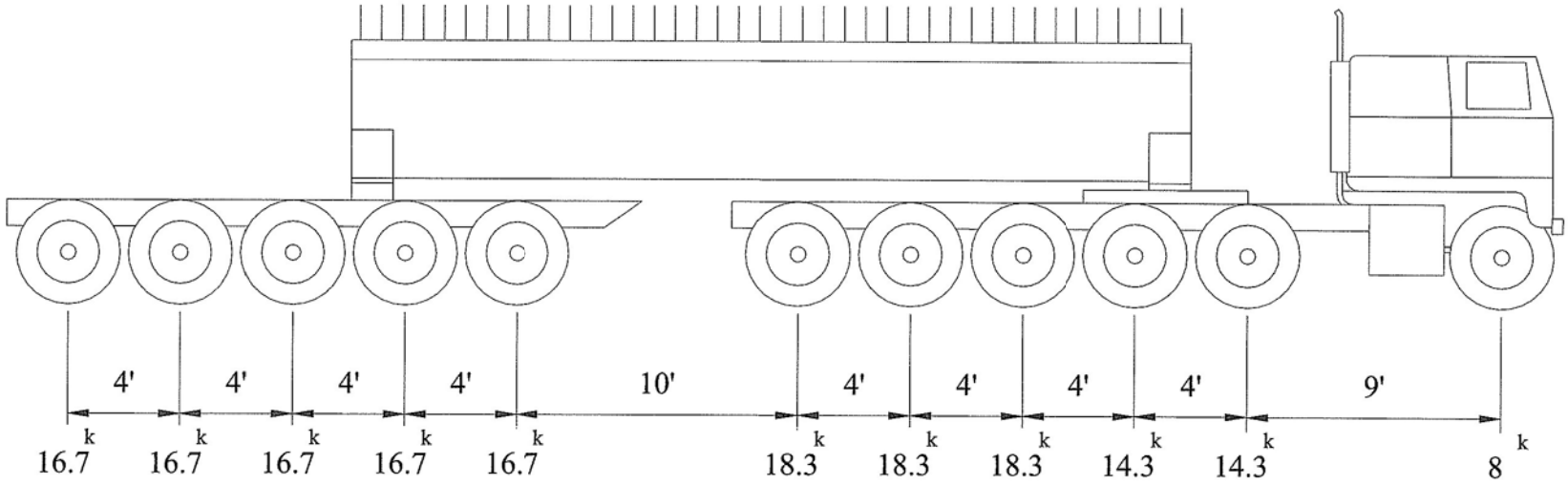

Figure A2A.13-1—Permit Truck Loading Configuration

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Use One-Lane Distribution Factor and divide out the 1.2 multiple presence factor. 6A.4.5.4.2b

$$g_{m1} = 0.565 \times \frac{1}{1.2} = 0.471$$

$$g_{v1} = 0.621 \times \frac{1}{1.2} = 0.518$$

Distributed Live-Load Effect:

M_{LL+IM} = (347.3) (0.471) (1.33) = 217.6 kip-ft

V_{LL+IM} = (52.6) (0.518) (1.33) = 36.2 kips

$$RF = \frac{C - (\gamma_{DC})(DC) - (\gamma_{DW})(DW) \pm (\gamma_P)(P)}{(\gamma_L)(LL+IM)}$$ Eq. 6A.4.2.1-1

For Strength Limit States: $C = (\varphi_c)(\varphi_s)(\varphi)R_n$

Flexure:

$$RF_M = \frac{(1.0)(1.0)(0.9)(493.4) - (1.25)(84.7) - (1.25)(27.9)}{(1.40)(217.6)}$$

= 1.0 ≤ 1.0 No Good

Shear: Shear evaluation may be considered for Permit Load Ratings. 6A.5.8

Since V_n was determined by the simplified approach, it is not dependent upon the vehicle.

$$RF_V = \frac{(1.0)(1.0)(0.9)(93.2) - (1.25)(10.8+3.6)}{(1.40)(36.2)}$$

= 1.30 > 1.0 OK

A2A.13.2—Service I Limit State (Optional) (6A.5.4.2.2b)

$\gamma_L = \gamma_{DC} = \gamma_{DW} =$ 1.0 Table 6A.4.2.2-1

Use the distribution factors that were used for the design and for legal loads. C6A.5.4.2.2b

g_m = 0.703

Distributed Live-Load Effect

M_{LL+IM} = (347.3) (0.703) (1.33) = 324.7 kip-ft

M_{DC} = 84.7 kip-ft

M_{DW} = 27.9 kip-ft

A2A.13.2.1—Simplified Check Using 0.75M_n (C6A.5.4.2.2b)

Unfactored Moments:

$M_{DC} + M_{DW} + M_{LL+IM}$ = 437.3 kip-ft

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Nominal flexural resistance:

M_n = 493.4 kip-ft

(Use nominal resistance, not factored.)

$0.75M_n$ = 0.75×493.4 = 370.1 kip-ft < 437.3 kip-ft No Good

Moment Ratio = $\dfrac{0.75M_n}{M_{DC}+M_{DW}+M_{LL+IM}} = \dfrac{370.1}{437.3} = 0.86 < 1.0$ No Good

A2A.13.2.2—Refined Check Using $0.9f_y$ C6A.5.4.2.2b

$M_{DC} + M_{DW}$ = 112.6 kip-ft

The Service I moments act upon the cracked section to produce stress in the reinforcement. An elastic model of the cracked concrete section with transformed steel is used to calculate the stress in the reinforcement due to the Service I loads.

$E_c = 1{,}820\sqrt{f_c'}$ LRFD Design Eq. C5.4.2.4-3

$= 1{,}820\sqrt{3.0}$

= 3,152 ksi

E_s = 29,000 ksi

$n = \dfrac{29{,}000}{3{,}152} = 9.2$ Use $n = 9$

For permanent loads at the Service limit states, use an effective modular ratio of n. LRFD Design 5.6.1

b_e = 78 in.

t_s = 6 in.

t_w = 15 in.

A_s = 6.89 in.^{2}

d_s = 26.61 in.

Assume neutral axis is within the slab.

$$\bar{y} = \frac{(b_e \times \bar{y})\frac{\bar{y}}{2} + (n \times A_s)(d_s)}{(b_e \times \bar{y}) + (n \times A_s)}$$

For $n = 9$:

$\bar{y}$ = 5.76 in. (within the slab)

$$I = \frac{1}{12} b_e \times \bar{y}^3 + (b_e \times \bar{y})\left(\frac{\bar{y}}{2}\right)^2 + (n \times A_s)(d_s - \bar{y})^2$$

$$= \frac{1}{12} \times 78 \times 5.76^3 + (78 \times 5.76)\left(\frac{5.76}{2}\right)^2 + (9 \times 6.89)(26.61 - 5.76)^2$$

$$= 31{,}926 \text{ in.}^3$$

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Stress in the extreme tension reinforcement:

$$\text{bending stress, } f = n \times \frac{M \times 12 \times \left(h - cov. - \bar{y}\right)}{I}$$

$$f_{LL+IM} = 9 \times \frac{324.7 \times 12 \times (30 - 2.5 - 5.76)}{31,926} = 23.88 \text{ ksi}$$

$$f_D = 9 \times \frac{112.6 \times 12 \times (30 - 2.5 - 5.76)}{31,926} = 8.28 \text{ ksi}$$

$f_s = f_{LL+IM} + f_D = 23.88 + 8.28 = 32.2$ ksi

$f_R = 0.90 f_y = 0.90 \times 33$ ksi $= 29.7$ ksi 6A.5.4.2.2b

$29.7 < 32.2$ No Good

Stress Ratio:

$$\frac{f_R - f_{DC} - f_{DW}}{f_{LL+IM}} = \frac{29.7 - 8.28}{23.88} = 0.9 \text{ No Good}$$

Some improvement versus the simplified check, but not enough to allow the permit if this optional check is applied. The truck also has an $RF \leq 1.0$ under flexure.

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A2A.14—Summary of Rating Factors for Load and Resistance Factor Rating Method

Table A2A.14-1—Summary of Rating Factors for Load and Resistance Factor Rating Method—Interior Beam

Limit State		Design Load Rating		Legal Load Rating								Permit Load Rating
		Inventory	Operating	Type 3	Type 3S2	Type 3-3	SU4	SU5	SU6	SU7	NRL	
Strength I	Flexure	0.59	0.76	1.29	1.35	1.57	1.69	1.00	0.93	0.90	0.90	—
	Shear	0.85	1.10	—	—	—	—	—	—	—	—	—
Strength II	Flexure	—	—	—	—	—	—	—	—	—	—	1.00
	Shear	—	—	—	—	—	—	—	—	—	—	1.30
Service I		—	—	—	—	—	—	—	—	—	—	Stress Ratio = 0.90

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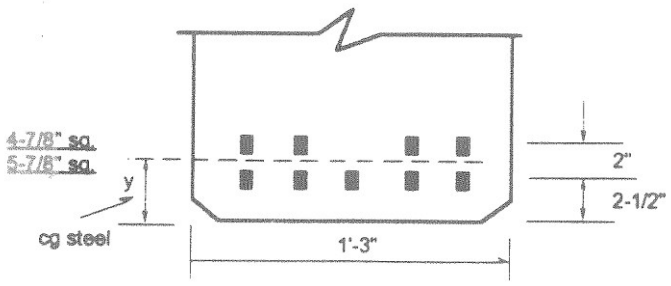
PART B—ALLOWABLE STRESS AND LOAD FACTOR RATING METHODS

A2B.1—Bridge Data

Refer to Article A2A.1 for Bridge Data.

A2B.2—Section Properties

Find cg steel:



$$y = \frac{4(.766)(2 + 2\text{-}1/2) + 5(.766)(2\text{-}1/2)}{4(.766) + 5(.766)}$$

$y = 3.39"$

$d = 30" - 3.39 = 26.61"$

$A_{1BAR} = 7/8" \times 7/8" = 0.766 \text{ in}^2$

$A_s = 9 \times A_{1BAR} = 6.89 \text{ in}^2$

Figure A2B.2-1—Steel Reinforcement Arrangment

Effective Slab Width (for T-Girder): AASHTO 8.10.1.1

$$\frac{1}{4}L = \frac{26 \text{ ft} \times 12 \text{ in./ft}}{4} = 78 \text{ in.}$$

or:

$$CC\ SPCG = 6 \text{ ft} - 6\ \frac{1}{4} \text{ in.} = 78.25 \text{ in.}$$

or:

$$12t_s = 12 \times 6 \text{ in.} = 72 \text{ in.} \Leftarrow \text{Controls}$$

$$\rho_{act} = \frac{As}{b_{eff}d} = \frac{6.89 \text{ in}^2}{78 \text{ in.} \times 26.61 \text{ in.}} = 0.0036$$

(if compression within flange)

A2B.3—Dead-Load Analysis—Interior Beam

Structural Concrete:

$$0.15 \text{ kip/ft}^3 \left[\left(\frac{6 \text{ in.}}{12 \text{ in./ft}} \times 6.52 \text{ ft} \right) + (1.25 \text{ ft} \times 2.0 \text{ ft}) + 2\left(\frac{1}{2}\frac{6}{12}\frac{6}{12} \right) \right] = 0.92 \text{ kip/ft}$$

AC Overlay:

$$0.144 \text{ kip/ft}^3 \left(\frac{5 \text{ in.}}{12 \text{ in./ft}} \times 6.52 \text{ ft} \right) = 0.39 \text{ kip/ft}$$

$W_{DL} = 0.902 + 0.39 = 1.292 \text{ kip/ft}$ say 1.3 kip/ft

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Midspan Moments:

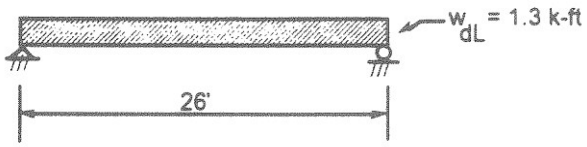

Figure A2B.3-1—Load Diagram for Uniform Dead Load

$$M_{DL} = \frac{W_{DL}L^2}{8} = \frac{1.3 \text{ kip/ft} \times 26^2 \text{ ft}^2}{8} = 109.9 \text{ kip-ft}$$

A2B.4—Live-Load Analysis—Interior Beam

Rate for HS-20 vehicle. Figure 6B.6.2-1

For HS-20—Using Table C6B-1, select from column "Without Impact." Appendix C6B

$M_L = 111.1$ kip-ft (without impact and without distribution)

A2B.5—Allowable Stress Rating (6B.3.1, 6B.4.2, and 6B.5.2)

For this example, we consider only the maximum moment section.

A2B.5.1—Impact (Use standard AASHTO) (6B.6.4, AASHTO 3.8.2.1)

$$I = \frac{50}{L+125} \le 0.30$$

$$I = \frac{50}{26+125} = 0.33 \text{ use } \underline{0.30}$$

A2B.5.2—Distribution (Use standard AASHTO) (6B.6.3, AASHTO 3.23.2.2 and Table 3.23.1)

$DF = \frac{S_G}{6.0}$ Concrete T-Beam

$$DF = \frac{6 \text{ ft} - 6 \text{ 1/4 in.}}{6.0} = \frac{6.52 \text{ ft}}{6} = 1.087$$

Thus:

$$M_{L+I} = M_L(1+I)(DF) = 111.1(1+0.30)(1.087) = 157 \text{ kip-ft}$$

A2B.5.3—Inventory Level (6B.4.2, 6B.5.2.4)

The inventory unit stresses are determined in accordance with AASHTO Article 8.15, "Service Load Design Method," or taken from 6B.5.2.4[a].

Inventory allowable stresses: AASHTO 8.15.2.1.1

$f_c^I = 1200 \text{ psi} = 1.2 \text{ ksi}$ 6B.5.2.4.1

[a] Note the moments given in the MBE are for one line of wheels. The values given in AASHTO are for the entire axle and are therefore twice the MBE values.

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For Reinforcing Steel, 6B.5.2.3 controls:

$f_s^I = 18,000 \text{ psi} = 18 \text{ ksi}$ (unknown steel prior to 1954) 6B.5.2.3

Capacity (Traditional Approach):

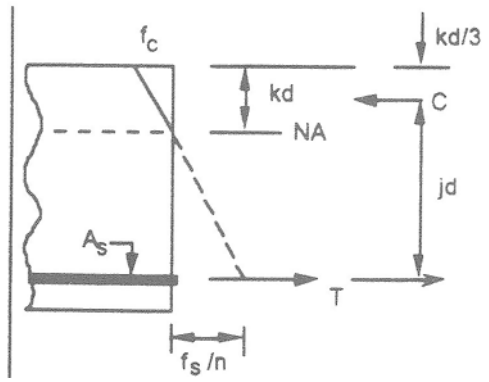

Figure A2B.5.3-1—Stress and Force Diagram, *nts*

The actual steel and concrete stresses are not known and must be found. Since this is a T-beam, assume neutral axis *NA* is within slab. Thus, rectangular beam formulas apply. Check this assumption later.

The following formulas for the Traditional Approach were referenced from *Reinforced Concrete Design Handbook* Working Stresss Method in accordance with ACI 318-63, ACI Publication SP-3.

Position of Neutral Axis:

$$k = \sqrt{2\rho n + (\rho n)^2} - \rho n$$ SP-3 Eq. (2)

where:

$$\rho = \frac{A_s}{bd} = \frac{6.89 \text{ in.}^2}{(72 \text{ in.})(26.61 \text{ in.})}$$ SP-3 Table 1

$\rho = 0.0036$

$$n = \frac{E_s}{E_c}$$

$n = 10$ 6B.5.2.4

$$k = \sqrt{2(0.0036)(10) + [(0.0036)(10)]^2} - (0.0036)(10)$$

$k = 0.235$

$$j = 1 - \frac{k}{3} = 1 - \frac{0.235}{3} = 0.922$$ SP-3 Table 1

then:

Capacity if concrete allowable stress controls:

$$M_c = \frac{1}{2} f_c jkbd^2$$

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$$= \frac{1}{2}(1.2 \text{ ksi})(0.922)(0.235)(72 \text{ in.})(26.61 \text{ in.})^2$$

$$= 6,622.8 \text{ kip-in.} = 552 \text{ kip-ft}$$

Capacity if steel reinforcement allowable stress controls:

$$M_s = A_s f_s jd$$

$$M_s = (6.89 \text{ in.}^2)(18 \text{ ksi})(0.922)(26.61 \text{ in.})$$

$$M_s = 3,042.8 \text{ kip-in.} = 253 \text{ kip-ft} \Leftarrow \text{Controls since } M_s < M_c$$

Check neutral axis assumption:

k_d = (0.235)(26.61 in.) = 6.25 in. > 6 in. the slab thickness $\therefore NA$ is below bottom of slab and slightly into web. This could be ignored in this case. However, for the sake of completeness, capacity will be figured below based on the *NA* below the slab and ignoring the compression in the stem concrete.

$$kd = \frac{2ndA_s + bt^2}{2nA_s + 2bt}$$

$$kd = \frac{2(10)(26.61 \text{ in.})(6.89 \text{ in.}) + (72 \text{ in.})(6 \text{ in.})^2}{2(10)(6.89 \text{ in.}) + 2(72 \text{ in.})(6 \text{ in.})} = \frac{6258.9}{1001.8}$$

$$kd = 6.25 \text{ in.} \rightarrow k = \frac{kd}{d} = \frac{6.25 \text{ in.}}{26.61 \text{ in.}} = 0.235$$

$$Z = \left(\frac{3kd - 2t}{2kd - t}\right)\frac{t}{3}$$

$$Z = \left(\frac{3(6.25 \text{ in.}) - 2(6 \text{ in.})}{2(6.25 \text{ in.}) - (6 \text{ in.})}\right)\frac{6 \text{ in.}}{3} = \frac{6.75 \text{ in.}}{6.5 \text{ in.}}(2 \text{ in.})$$

$$Z = 2.077 \text{ in.}$$

$$jd = d - Z$$

$$jd = 26.61 \text{ in.} - 2.077 \text{ in.} = 24.53 \text{ in.}$$

$$M_s = A_s f_s jd$$

$$M_s = (6.89 \text{ in.}^2)(18 \text{ ksi})(24.53 \text{ in.}) = 3042.2 \text{ kip-in.}$$

$$M_s = 253 \text{ kip-ft as before}$$

(Note concrete was not checked since capacity of section is limited by steel allowable stress.)

$$RF_I^A = \frac{M_{RI} - M_D}{M_{L+I}} \qquad \text{Eq. 6B.4.1-1}$$

$$RF_I^A = \frac{253 \text{ kip-ft} - 109.9 \text{ kip-ft}}{157 \text{ kip-ft}} = 0.91$$

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A2B.5.4—Operating Level (6B.5.2)

The Operating allowable stresses for concrete with f'_c = 3,000 psi:

$f_c^o = 1900 \text{ psi} = 1.9 \text{ ksi}$ 6B.5.2.4.1

For reinforcing steel:

$f_s^o = 25{,}000 \text{ psi} = 25 \text{ ksi}$ (unknown steel, prior to 1954) 6B.5.2.3

The basic relationships defined previously apply:

Since ρ and n do not change, the neutral axis, k, j, and Z terms do not change.

Thus:

$$M_s = A_s f_s jd$$

$$= \left(6.89 \text{ in.}^2\right)\left(25 \text{ ksi}\right)\left(24.53 \text{ in.}\right)$$

$$= 4225.3 \text{ kip-in.} = 352 \text{ kip-ft}$$

and checking concrete stress to ensure that concrete does not control:

$$f_c = \frac{f_s}{n}\left(\frac{k}{1-k}\right)$$ SP-3 Table 1

$$f_c = \left(\frac{25 \text{ ksi}}{10}\right)\left(\frac{0.235}{1-0.235}\right) = 0.77 \text{ ksi} << 1.9 \text{ ksi allowable}$$

Therefore, capacity of section is controlled by allowable steel stress.

$$M_{R_O} = 352 \text{ kip-ft}$$

$$RF_O^A = \frac{M_{R_O} - M_{DL}}{M_{L+I}} = \frac{352 \text{ kip-ft} - 109.9 \text{ kip-ft}}{157 \text{ kip-ft}}$$

$$RF_O^A = 1.54$$

A2B.6—Load Capacity Based on Allowable Stress

Inventory: $0.91 \times 36^T = 32.8^T \, HS$

Operating: $1.54 \times 36^T = 55.4^T \, HS$

To transform HS rating to H rating, multiply HS rating factor by ratio of HS moment to H moment:

For 26-ft span:

$M_L^{\text{HS-20}} = 111.1 \text{ kip-ft}$

$\rightarrow M_L^{\text{H-15}} = 78 \text{ kip-ft}$ Table C6B-1

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Then:

$$M_L^{\text{H-20}} = \frac{20T}{15T} \times 78 \text{ kip-ft} = 104 \text{ kip-ft}$$

and:

$$\text{Ratio} = \frac{M_L^{\text{HS-20}}}{M_L^{\text{H-20}}} = \frac{111.1}{104} = 1.068$$

Thus for H-20 Truck:

Inventory: $0.91 \times 1.068 \times 20^T = 19.4^T H$

Operating: $1.54 \times 1.068 \times 20^T = 32.9^T H$

A2B.7—Capacity (Alternate Approach)

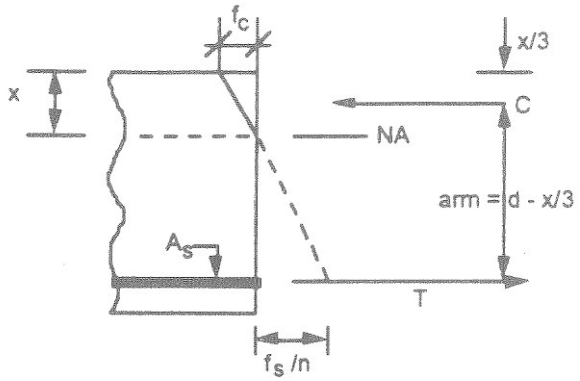

Figure A2B.7-1—Stress and Force Diagram, *nts*

Since the location of the neutral axis *NA* and the corresponding stresses in the steel and concrete are not known, these must be determined consistent with the principles of equilibrium of the cross section.

1. From the stresses on the cross section using similar triangles:

$$\frac{f_c}{x} = \frac{f_s \div n}{d - x} \rightarrow f_c = \frac{f_s}{n}\left(\frac{x}{d - x}\right) \qquad \text{(Eq. A2B.7-1)}$$

2. Assume the steel allowable stress controls the capacity of the section. This will be checked later. Then:

$$T = A_s f_s = \left(6.89 \text{ in.}^2\right)\left(18 \text{ ksi}\right) = 124 \text{ kips}$$

and:

$$C = \frac{1}{2} f_c b x$$

but:

$$C = T$$

thus:

$$\frac{1}{2} f_c b x = A_s f_s$$

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$$x = \frac{A_s f_s}{\frac{1}{2} f_c b} \qquad \text{(Eq. A2B.7-2)}$$

Solve Eqs. A2B.7-1 and A2B.7-2 to find location of neutral axis. This may be done by trial and error as follows.

Assume f_s = 18 ksi, i.e., steel allowable stress controls.

Try x = 6.0 in. Then by Eq. A2B.7-1:

$$f_c = \frac{f_s}{n}\left(\frac{x}{d-x}\right) = \frac{18 \text{ ksi}}{10}\left(\frac{6.0 \text{ in.}}{26.61 \text{ in.} - 6.0 \text{ in.}}\right) = 0.524 \text{ ksi} < 1.2 \text{ ksi} \qquad \text{allowable OK}$$

and by Eq. A2B.7-2:

$$x = \frac{A_s f_s}{\frac{1}{2} f_c b} = \frac{\left(6.89 \text{ in.}^2\right)\left(18 \text{ ksi}\right)}{\frac{1}{2}\left(0.524 \text{ ksi}\right)\left(72 \text{ in.}\right)} = 6.57 \text{ in.} > 6.0 \qquad \text{assumed. Try again}$$

Try x = 6.25 in.

$$f_c = \frac{18}{10}\left(\frac{6.25}{26.61 - 6.25}\right) = 0.552 < 1.2 \text{ ksi} \qquad \text{allowable OK}$$

and:

$$x = \frac{(6.89)(18)}{\frac{1}{2}(0.552)(72)} = 6.24 \approx 6.25 \qquad \text{assumed OK}$$

3. Since x = 6.24 > t = 6.0, *NA* is below bottom of slab and slightly into web. If web concrete in compression is neglected:

$arm \approx d - \frac{x}{3}$ for this example.

$$arm \approx 26.61 - \frac{6.24}{3} = 24.53 \text{ in.}$$

and capacity is:

$$M = A_s f_s \left(arm\right) = \left(6.89\right)\left(18\right)\left(24.53\right) = 3,042.2 \text{ kip-in.} = 253 \text{ kip-ft} \qquad \text{as before}$$

The exact *arm* may be determined from the concrete stress diagram as follows:

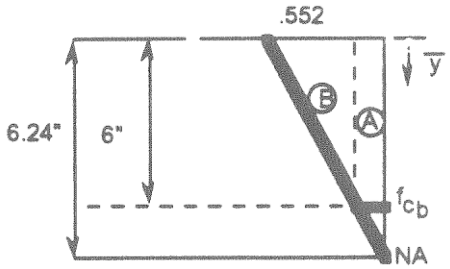

Figure A2B.7-2—Concrete Stress Diagram for Slab Portion of T-Beam, *nts*

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at bottom of slab:

$$f_{cb} = 0.552\left(\frac{0.24}{6.24}\right) = 0.021$$

Next find centroid of stress diagram from top of slab.

$$\bar{y} = \frac{\sum A_y}{\sum A} = \frac{(0.021)(6)\left(\frac{6}{2}\right) + (0.552 - 0.021)(6)\left(\frac{1}{2}\right)\left(\frac{6}{3}\right)}{(0.021)(6) + (0.552 - 0.021)(6)\left(\frac{1}{2}\right)}$$

$$\bar{y} = \frac{3.576}{1.722} = 2.08 \text{ in.}$$

$\therefore arm = 26.61 - 2.08 = 24.53$ in. as found previously

4. The operating capacity may be found as above and will be the same as for the "traditional method." The rating calculations are not shown here since they too will be the same as for the traditional method.

A2B.8—Allowable Stress Rating—Rate for AASHTO Legal Loads

M_{L+I} from Appendix C6B (all values have 30 percent impact):

Span	Type 3	Type 3S2	Type 3-3	
26 ft	122.4	117.7	100.8	kip-ft

Span	NRL	SU4	SU5	SU6	SU7	
26 ft	176.2	145.1	158.0	171.1	175.2	kip-ft

Apply distribution factor $DF = 1.087$

Span	Type 3	Type 3S2	Type 3-3	
26 ft	133.0	127.9	109.6	kip-ft

Span	NRL	SU4	SU5	SU6	SU7	
26 ft	191.5	157.7	171.7	186.0	190.4	kip-ft

Capacity of section as previously determined in A2B.5.3 and A2B.5.4 respectively:

Inventory Level $M_{RI} = 253$ kip-ft Operating Level $M_{RO} = 352$ kip-ft.

Dead Load $M_{DL} = 109.9$ kip-ft.

For Allowable Stress Method $A_1 = 1.0$ and $A_2 = 1.0$ 6B.4.2

$$RF_I^A = \frac{M_{RI} - A_1 M_D}{A_2 M_{L+I}} = \frac{253 \text{ kip-ft} - (1.0)109.9 \text{ kip-ft}}{(1.0)M_{L+I}}$$ Eq. 6B.4.1-1

$$RF_O^A = \frac{M_{R_O} - A_1 M_{DL}}{A_2 M_{L+I}} = \frac{352 \text{ kip-ft} - (1.0)109.9 \text{ kip-ft}}{(1.0)M_{L+I}}$$

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Alowable Stress Method Rating Factors:

	Type 3	Type 3S2	Type 3-3
Inventory	1.08	1.12	1.31
Operating	1.82	1.89	2.21

	NRL	SU4	SU5	SU6	SU7
Inventory	0.75	0.91	0.83	0.77	0.75
Operating	1.26	1.53	1.41	1.30	1.27

Load Capacity in Tons:

Inventory: $RF_I^A \times \text{vehicle weight} = \text{Inv.Cap.}$

Operating: $RF_O^A \times \text{vehicle weight} = \text{Opr.Cap.}$

Load	Type 3	Type 3S2	Type 3-3
Vehicle Weight	25	36	40
Inv. Cap.	27.0	40.3	52.4
Opr. Cap.	45.5	68.0	88.4

Load	NRL	SU4	SU5	SU6	SU7
Vehicle Weight	40	27	31	34.8	38.8
Inv. Cap.	30.0	24.6	25.7	26.8	29.1
Opr. Cap.	50.4	41.3	43.7	45.2	49.3

A2B.9—Summary of Ratings for Allowable Stress Rating Method

Table A2B.9-1—Summary of Ratings for Allowable Stess Rating Method—Interior Beam

Load	HS-20	H-20	Type 3	Type 3S2	Type 3-3
Vehicle Weight (tons)	36	20	25	36	40
Inventory RF	0.91	0.91	1.08	1.12	1.31
Inv. Cap.	32.8	19.4	27.0	40.3	52.4
Operating RF	1.54	1.54	1.82	1.89	2.21
Opr. Cap.	55.4	32.9	45.5	68.0	88.4

Load	NRL	SU4	SU5	SU6	SU7
Vehicle Weight (tons)	40	27	31	34.8	38.8
Inventory RF	0.75	0.91	0.83	0.77	0.75
Inv. Cap.	30.0	24.6	25.7	26.8	29.1
Operating RF	1.26	1.53	1.41	1.30	1.27
Opr. Cap.	50.4	41.3	43.7	45.2	49.3

A2B.10—Load Factor Rating (6B.3.2, 6B.4.3, 6B.5.3)

For this example, we consider only the maximum moment section.

A2B.10.1—Impact (Use standard AASHTO) (6B.6.4, AASHTO 3.8.2.1)

$$I = \frac{50}{L+125} \le 0.30$$

$$I = \frac{50}{26+125} = 0.33 \text{ use } \underline{0.30}$$

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A2B.10.2—Distribution (Use standard AASHTO) (6B.6.3, AASHTO 3.23.2.2 and Table 3.23.1)

$$DF = \frac{S_G}{6.0} = \frac{6.52 \text{ ft}}{6} = 1.087$$

Thus:

$$M_{LL+I} = M_L(1+I) \times DF = 111.1(1+0.30)(1.087) = 157 \text{ kip-ft}$$

A2B.10.3—Capacity of Section (6B.5.3.2)

For unknown steel prior to 1954, f_y = 33,000 psi = 33 ksi

M_u is found in accordance with applicable strength requirements of AASHTO Article 8.16.

Consider a rectangular section with compression limited to top slab. Then check 6B.5.3.2 requirement for 75 percent of balanced condition.

$$\rho_{max} = 0.75\rho_{bal} = 0.75\frac{0.85\beta_1 f_c'}{f_y}\frac{87000}{87000 + f_y}$$ AASHTO Eq. 8-18

$$\rho_{max} = 0.75\frac{0.85(0.85)(3000)}{33000}\left(\frac{87000}{87000+33000}\right)$$

$\rho_{act} = 0.0036 << \rho_{max}$ OK

Then:

$$a = \frac{A_s f_y}{0.85 f_c' b_{eff}}$$ AASHTO Eq. 8-17

$$a = \frac{6.89 \text{ in.}^2 (33 \text{ ksi})}{0.85(3 \text{ ksi})72 \text{ in.}} = 1.24 \text{ in.} < 6 \text{ in.}$$ OK within slab

$$M_R = A_s f_y\left(d - \frac{a}{2}\right)$$ AASHTO Eq. 8-16

$$M_R = \left(6.89 \text{ in.}^2\right)(33 \text{ ksi})\left(26.61 \text{ in.} - \frac{1.24}{2}\right)$$

$M_R = 5909$ kip-in. = <u>492 kip-ft</u>

$M_u = \phi M_R$

where $\phi = 0.90$ AASHTO 8.16.1.2.2

$M_u = 0.90 \times 492 = 443$ kip-ft.

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A2B.10.4—Inventory Level (6B.4.1, 6B.5.3)

M_{DL} is the same as what was estimated for the ASD rating calculation:

$$R_I^{LF} = \frac{M_u - A_1 M_{DL}}{A_2 M_{L+I}} \qquad \text{Eq. 6B.4.1-1}$$

where in accordance with 6B.4.3:

$A_1 = 1.3$

$A_2 = 2.17$

Thus:

$$RF_I^{LF} = \frac{443 - 1.3(109.9)}{2.17(157)} = 0.88$$

A2B.10.5—Operating Level (6B.4.1, 6B.5.3)

$$R_O^{LF} = \frac{M_u - A_1 M_{DL}}{A_2 M_{L+I}} \qquad \text{Eq. 6B.4.1-1}$$

where in accordance with 6B.4.3:

$A_1 = 1.3$

$A_2 = 1.3$

Thus:

$$RF_O^{LF} = \frac{443 - 1.3(109.9)}{1.3(157)} = 1.47$$

Load capacity based on Load Factor Method, HS-20 truck:

Inventory: $0.88 \times 36^T = 31^T HS$

Operating: $1.47 \times 36^T = 52^T HS$

Load capacity based on Load Factor Method, H-20 truck, where the ratio of HS moment to H moment has been determined in A2B.6 as 1.068:

Inventory: $0.88 \times 1.068 \times 20^T = 18.8^T H$

Operating: $1.47 \times 1.068 \times 20^T = 31.4^T H$

A2B.10.6—Summary of Ratings for Load Factor Rating Method

Table A2B.10.6-1—Summary of Ratings for Load Factor Rating Method—Interior Beam

	RF	HS-20 Rating, tons	H-20 Rating, tons
Inventory	0.88	31.7	18.8
Operating	1.47	52.9	31.4

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A2B.10.7—Load Factor Rating—Rate for AASHTO Legal Loads

M_{L+I} from Appendix C6B (all values have 30 percent impact)

Span	Type 3	Type 3S2	Type 3-3	
26 ft	122.4	117.7	100.8	kip-ft

Apply distribution factor $DF = 1.087$

26 ft	133.0	127.9	109.6	kip-ft

Capacity of Section $M_U = 443$ kip-ft

Dead Load $M_{DL} = 109.9$ kip-ft

For Inventory level, $A_1 = 1.3$ and $A_2 = 2.17$ 6B.4.3

$$\text{Inv. } RF = \frac{443 - 1.3(109.9)}{2.17(M_{L+I})}$$

For Operating level, $A_1 = 1.3$ and $A_2 = 2.17$ 6B.4.3

$$\text{Opr. } RF = \frac{443 - 1.3(109.9)}{1.3(M_{L+I})}$$

Strength Rating Factors:

	Type 3	Type 3S2	Type 3-3
Inventory	1.01	1.05	1.22
Operating	1.74	1.81	2.11

Load Capacity in Tons:

Load	Type 3	Type 3S2	Type 3-3
Vehicle Weight	25	36	40
Inv. Cap.	25.3	37.8	48.8
Opr. Cap.	43.5	65.2	84.4

The bridge has adequate Inventory load capacity for Types 3, 3S2, and 3-3 Legal Loads.

A2B.10.8—Load Factor Rating—Rate for Single-Unit Formula B Loads

M_{L+I} from Appendix C6B (all values have 30 percent impact)

Span	NRL	SU4	SU5	SU6	SU7	
26 ft	176.2	145.1	158.0	171.1	175.2	kip-ft

Apply distribution factor $DF = 1.087$

Span	NRL	SU4	SU5	SU6	SU7	
26 ft	191.5	157.7	171.7	186.0	190.4	kip-ft

Capacity of Section $M_U = 443$ kip-ft

Dead Load $M_{DL} = 109.9$ kip-ft

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For Inventory level, $A_1 = 1.3$ and $A_2 = 2.17$ 6B.4.3

$$\text{Inv. } RF = \frac{443 - 1.3(109.9)}{2.17(M_{L+I})}$$

For Operating level, $A_1 = 1.3$ and $A_2 = 2.17$ 6B.4.3

$$\text{Opr. } RF = \frac{443 - 1.3(109.9)}{1.3(M_{L+I})}$$

Strength Rating Factors:

	NRL	SU4	SU5	SU6	SU7
Inventory	0.72	0.88	0.81	0.74	0.73
Operating	1.20	1.47	1.35	1.24	1.22

Load Capacity in Tons:

Load	NRL	SU4	SU5	SU6	SU7
Vehicle Weight	40	27	31	34.8	38.8
Inv. Cap.	28.8	23.8	25.1	25.8	28.3
Opr. Cap.	48.0	39.7	41.9	43.2	47.3

The bridge has inadequate Inventory load capacity for the notional rating load NRL, and the posting loads SU4, SU5, SU6, and SU7.

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PART C—SUMMARY

A2C.1—Summary of All Ratings for Example A2

Table A2C.1-1—Summary of Rating Factors for All Rating Methods—Interior Beam

Limit State		Design Load Rating				Legal Load Rating								Permit Load Rating
		Inventory	Operating	HS-20 Rating	H-20 Rating	Type 3	Type 3S2	Type 3-3	SU4	SU5	SU6	SU7	NRL	
Strength I	Flexure	0.59	0.76			1.29	1.35	1.57	1.09	1.00	0.93	0.90	0.90	—
	Shear	0.85	1.10			—	—	—	—	—	—	—	—	—
Strength II	Flexure	—	—			—	—	—	—	—	—	—	—	0.996
	Shear	—	—			—	—	—	—	—	—	—	—	1.30
Service I		—	—	—	—	—	—	—	—	—	—	—	—	Stress Ratio = 0.90
Allowable Stress Method	Inv.	—	—	0.91	0.97	1.08	1.12	1.31	0.91	0.83	0.77	0.75	0.75	—
	Opr.	—	—	1.54	1.64	1.82	1.89	2.21	1.53	1.41	1.30	1.27	1.26	
Load Factor Method	Inv.	—	—	0.88	0.94	1.01	1.05	1.22	0.88	0.81	0.74	0.73	0.72	—
	Opr.	—	—	1.47	1.57	1.74	1.81	2.11	1.47	1.35	1.24	1.22	1.20	—

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A2C.2—References

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A3—SIMPLE SPAN PRESTRESSED CONCRETE: I-GIRDER BRIDGE EVALUATION OF AN INTERIOR GIRDER (LRFR ONLY)

Note: This example illustrates rating an interior prestressed concrete girder at midspan for moment, at the critical section for shear, and at a change in stirrup spacing for shear. The example member contains debonded tendons to illustrate how this affects the rating at the two shear locations.

A3.1—Bridge Data

Span:	80 ft (Total Length = 81 ft)
Year Built:	1985
Materials:	
Concrete:	$f'_c = 4$ ksi (Deck)
	$f'_c = 5$ ksi (P/S Beam)
	$f_{ci} = 4$ ksi (P/S Beam at transfer)
Prestressing Steel:	$^1/_2$ in. diameter, 270 ksi, Low-Relaxation Strands
	$A_{ps} = 0.153$ in.^{2} per strand
	32 prestressing strands; ten are debonded over the last 12 ft on each end
Stirrups:	#4 at 9 in. over end 20 ft
	#3 at 12 in. over center 40 ft
Compression Steel:	six #6 Grade 60
Condition:	No Deterioration, NBI Item 59 Code = 6
Riding Surface:	Minor surface deviations (Field verified and documented)
ADTT (one direction)	5,000
Skew:	0°
Effective Flange Width	b_e may be taken as the tributary width perpendicular to the axis of the member LFRD Design 4.6.2.6.1

Effective Flange Width $b_e = 8.5$ ft × 12 in. = 102 in.

$$E_c = 33{,}000\, W_c^{1.5} \sqrt{f'_c}$$ LRFD Design Eq. C5.4.2.4-2

For deck, $E_c = 33{,}000 \times (0.145)^{1.5} \sqrt{4.0} = 3.64 \times 10^3$ ksi

For P/S Beam, $E_c = 33{,}000 \times (0.145)^{1.5} \sqrt{5.0} = 4.07 \times 10^3$ ksi

Modular Ratio, $n = \dfrac{E_{deck}}{E_{beam}} = \dfrac{3.64 \times 10^3}{4.07 \times 10^3} = 0.89$

Transformed Width, $b_{trans} = 102 \text{ in.} \times 0.89 = 90.8 \text{ in.}$

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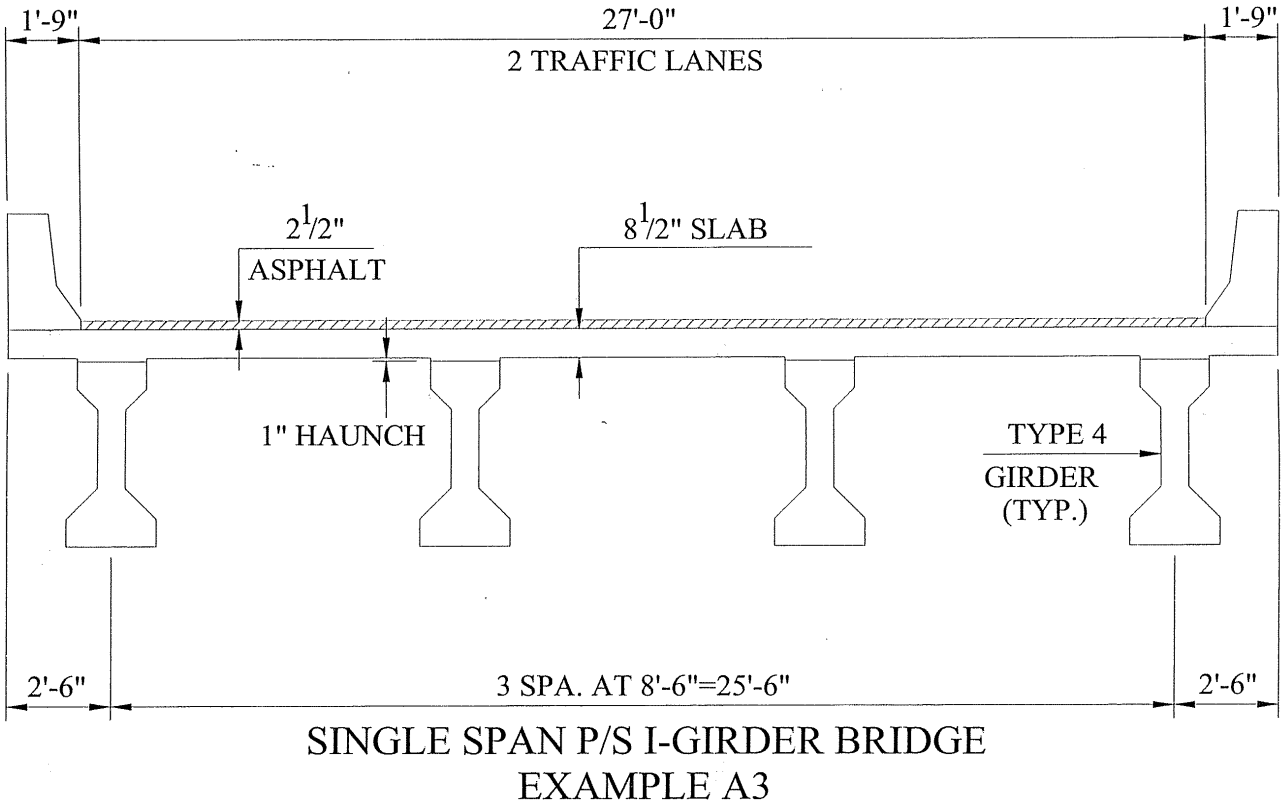

Figure A3.1-1—Cross Section—Single Span Prestressed I-Girder Bridge

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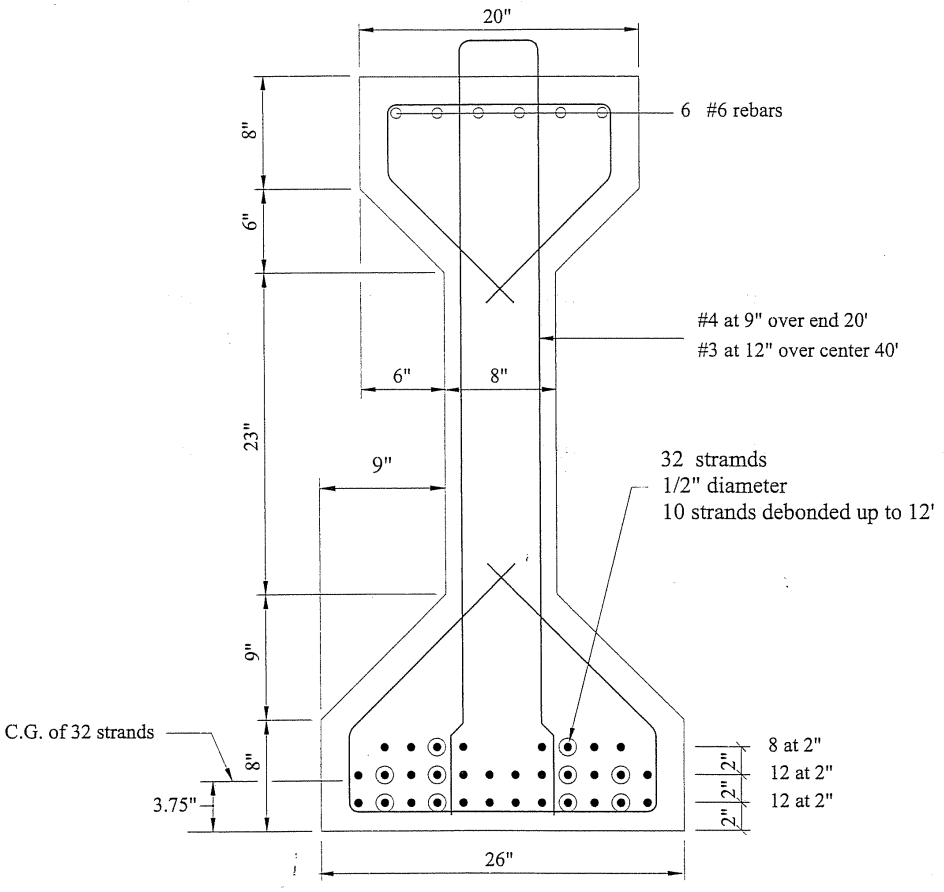

Figure A3.1-2 Cross Section—Interior Girder AASHTO Type 4 I-Girder

A3.2—Summary of Section Properties

Type 4 Girder:

h = 54 in.

A = 789 in.^{2}

I = 260,730 in.^{4}

Y_b = 24.73 in.

S_b = 10,543 in.^{3}

S_t = 8,908 in.^{3}

Composite Section

	Area, in.^{2}	y, in.	Ay	d	Ad^2 in.^{4}	I_0 in.^{4}
P/S beam	789	24.73	19,512	17.07	229,903	260,730
Slab	772	59.25	45,741	17.45	235,076	4,647
Totals	1,561		65,253		464,979	265,377

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Area slab = 8.5 in. × 90.8 in. = 772 in.^{2} (uses full slab thickness of deck)

y slab = 54 in. + 1 in. + ½ × 8.5 in. = 59.25 in. (includes 1-in. haunch)

$\bar{y}$ = 65,253 ÷ 1,561 = 41.80 in.

d = $y - \bar{y}$

y_{bot} = $\bar{y}$ = 41.80 in. y_{top} = $h - \bar{y}$ = 54 in. – 41.80 in. = 12.20 in.

I_0 slab = $\frac{bh^3}{12} = \frac{90.8\times(8.5)^3}{12}$ = 4,647 in.^{4}

I_{comp} = $\sum I_0 + \sum Ad^2$ = 464,979 + 265,377 = 730,356 in.^{4}

S_b = $\frac{I}{y_{bot}}$ = 730,356 ÷ 41.80 = 17,473 in.^{3} (Bottom of Beam)

S_t = $\frac{I}{y_{top}}$ = 730,356 ÷ 12.20 = 59,865 in.^{3} (Top of Beam)

A3.3—Dead Load Analysis—Interior Girder

A3.3.1—Components and Attachments, *DC*

A3.3.1.1—Noncomposite Dead Loads, DC_1

Girder Self Weight: = 0.822 kip/ft *PCI Design Manual*

Diaphragms: = 0.150 kip/ft

Slab + haunch:

$$\left[\frac{8.5\,\text{in.}}{12\,\text{in./ft}}\times 8.5\,\text{ft}+\frac{1\,\text{in.}\times 20\,\text{in.}}{144\,\text{in.}^2/\text{ft}^2}\right]\times 0.15\,\text{kcf} = 0.925\ \text{kip/ft}$$

Total per Girder DC_1 = 1.90 kip/ft

$$V_{DC_1} = 1.90\,\text{kip/ft}\times\frac{80\,\text{ft}}{2} = 76\ \text{kip}$$ At support

$$M_{DC_1} = \frac{1}{8}\times 1.90\,\text{kip/ft}\times(80\,\text{ft})^2 = 1520\ \text{kip-ft}$$ At midspan

A3.3.1.2—Composite Dead Load, DC_2

Concrete Barriers: LRFD Design 4.6.2.2.1

Assuming equal distribution among 4 beams

$(2\times 0.500\ \text{kip/ft}) \div 4 = 0.25\ \text{kip/ft}$

$$V_{DC_2} = 0.25\,\text{kip/ft}\times\frac{80\,\text{ft}}{2} = 10\ \text{kips}$$ At support

$$M_{DC_2} = \frac{1}{8}\times 0.25\,\text{kip/ft}\times(80\,\text{ft})^2 = 200\ \text{kip-ft}$$ At midspan

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A3.3.2—Wearing Surface, *DW*

Asphalt Overlay: $\frac{2.5\text{in.}}{12\text{in./ft}} \times 27\,\text{ft} \times 0.144\,\text{kcf} \div 4\,\text{beams} = 0.203$ kip/ft

Overlay thickness was not field measured. Use $\gamma_{DW} = 1.5$

$V_{DW} = 0.203\,\text{kip/ft} \times \frac{80\text{ ft}}{2} = 8.12$ kips — At support

$M_{DW} = \frac{1}{8} \times 0.203$ kip/ft $\times 80^2 = 162$ kip-ft — At midspan

A3.4—Live Load Analysis—Interior Girder

A3.4.1—Compute Live Load Distribution Factors, *g*

AASHTO LRFD Type (k) cross-section — LRFD Design Table 4.6.2.2.1-1

Longitudinal Stiffness Parameter, K_g:

$K_g = n\,(I + Ae_g^2)$ — LRFD Design Eq. 4.6.2.2.1-1

$n = \frac{E_B}{E_D} = \frac{4.07 \times 10^3\text{ ksi}}{3.64 \times 10^3\text{ ksi}} = 1.12$

A = 789 in.^{4}

I = 260,730 in.^{4}

L = 80 ft

t_s = 8.5 in.

e_g = girder depth – Y_b + haunch + t_s/2

= $(54 - 24.73) + 1 + \frac{8.5}{2}$

= 34.52 in.

K_g = $1.12\,(260,730 + 789 \times 34.52^2)$

= 1,345,038 in.

$\frac{K_g}{12\,Lt_s^3} = \frac{1{,}345{,}038}{12 \times 80 \times 8.5^3} = 2.28$

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A3.4.1.1—Distribution Factor for Moment, g_m (LRFD Design Table 4.6.2.2.2b-1)

One Lane Loaded:

$$g_{m1} = 0.06 + \left(\frac{S}{14}\right)^{0.4}\left(\frac{S}{L}\right)^{0.3}\left(\frac{K_g}{12.0Lt_s^3}\right)^{0.1}$$

$$= 0.06 + \left(\frac{8.5}{14}\right)^{0.4}\left(\frac{8.5}{80}\right)^{0.3}(2.28)^{0.1}$$

$$= 0.514$$

Two or More Lanes Loaded:

$$g_{m2} = 0.075 + \left(\frac{S}{9.5}\right)^{0.6}\left(\frac{S}{L}\right)^{0.2}\left(\frac{K_g}{12.0Lt_s^3}\right)^{0.1}$$

$$= 0.075 + \left(\frac{8.5}{9.5}\right)^{0.6}\left(\frac{8.5}{80}\right)^{0.2}(2.28)^{0.1}$$

$$= 0.724 > 0.514$$

Use $g_m = 0.724$

A3.4.1.2—Distribution Factor for Shear, gv (LRFD Design Table 4.6.2.2.3a-1)

One Lane Loaded:

$$g_{v1} = 0.36 + \frac{S}{25}$$

$$= 0.36 + \frac{8.50}{25}$$

$$= 0.70$$

Two or More Lanes Loaded:

$$g_{v2} = 0.2 + \left(\frac{S}{12}\right) - \left(\frac{S}{35}\right)^2$$

$$= 0.2 + \left(\frac{8.5}{12}\right) - \left(\frac{8.5}{35}\right)^2$$

$$= 0.849 > 0.70$$

use $g_v = 0.849$

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A3.4.2—Compute Maximum Live Load Effects

A3.4.2.1—Maximum Design Live Load (HL-93)—Moment at Midspan

Note: The general rule for simple spans carrying moving concentrated loads states that the maximum bending moment produced by moving concentrated loads occurs under one of the loads when that load is as far from one support as the center of gravity of all the moving loads on the beam is from the other support. It should be understood that locating the precise critical section and load position for rating depends on the combined influence of dead load, live load, member capacity, and load factors that make up the general Rating Factor equation. For simplicity and illustrative purposes only, the moment at midspan is used to closely approximate the maximum moment. See also Example A1, which illustrates that for a beam with a constant section capacity throughout the maximum moment region and a long span, the resulting rating factor obtained by a refined analysis yields only a small difference compared to the rating factor obtained from the maximum moment approximated at midspan.

Calculated by statics with the load centered at midspan:

$$\text{Design Lane Load Moment} = 0.64 \text{ klf} \times \frac{(80 \text{ ft})^2}{8} = 512 \text{ kip-ft}$$

$$\text{Design Truck Moment} = \frac{\left(8^k + 32^k\right) \times 40 \text{ ft} \times 26 \text{ ft}}{80 \text{ ft}} + \frac{32^k \times 80 \text{ ft}}{4} = 1{,}160 \text{ kip-ft} \quad \text{Governs}$$

$$\text{Tandem Axles Moment} = 25^k \times 38 \text{ ft} = 950 \text{ kip-ft}$$

IM = 33 percent — LRFD Design Table 3.6.2.1-1

M_{LL+IM} = 512 + 1,160 × 1.33

= 2,054.8 kip-ft

Distributed Live Load Moment at Midspan:

$$M_{LL+IM} = 2{,}054.8 \times g_m$$
$$= 2{,}054.8 \times 0.724$$
$$= 1{,}487.7 \text{ kip-ft}$$

A3.5—Compute Nominal Flexural Resistance at Midspan

$$f_{ps} = f_{pu}\left(1 - k\frac{c}{d_p}\right)$$ — LRFD Design Eq. 5.6.3.1.1-1

k = 0.28 for low-relaxation strands — LRFD Design Table 5.6.3.1.1-1

f_{pu} = 270 ksi

d_p = distance from extreme compression fiber to the centroid of the prestressing tendons

	Strands	y	Strands × y
Layer 1	12	2	24
Layer 2	12	4	48
Layer 3	8	6	48
Total	32		120

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$\bar{y} = \frac{strands \times y}{strands} = \frac{120}{32}$

$\bar{y}$ = 3.75 in. distance from bottom of girder to centroid of prestressing strands

$d_p = (54+1+8.5) - 3.75 = 59.75$ in.

c = distance from the neutral axis to the compressive face

To compute c, assume rectangular section behavior. (Neglect any nonprestressed reinforcement.)

Given $A_{ps} = 0.153$ in.^{2} for $^1/_2$-in. diameter Low-Relaxation strands:

$$c = \frac{A_{ps} f_{pu}}{\alpha_1 f_c' \beta_1 b + k A_{ps} \frac{f_{pu}}{d_p}}$$ LRFD Design Eq. 5.6.3.1.1-4

A_{ps} = 4.896 in^2

b = be = 102 in (Effective Flange Width of Deck)

f'_c = 4.0 ksi (Deck Concrete Strength)

β_1 = 0.85

α_1 = 0.85

$$c = \frac{4.896 \text{ in.}^2 \times 270 \text{ksi}}{0.85 \times 4.0 \text{ksi} \times 0.85 \times 102 \text{in.} + 0.28 \times 4.896 \text{in.}^2 \times \frac{270 \text{ksi}}{59.75 \text{in.}}}$$ LRFD Design 5.6.2.2

$a = \beta_1 c = 0.85 \times 4.39 = 3.73 \text{ in.} < ts = 8.5 \text{ in.}$ LRFD Design 5.6.2.2

Therefore, the rectangular section behavior assumption is valid.

$$f_{ps} = 270\left(1 - 0.28 \times \frac{4.39}{59.75}\right)$$
$$= 264.4 \text{ ksi}$$

Nominal Flexural Resistance (Midspan): LRFD Design Eq. 5.6.3.2.2-1

$$M_n = A_{ps} f_{ps}\left(d_p - \frac{a}{2}\right)$$
$$= 4.896 \times 264.4\left(59.75 - \frac{3.73}{2}\right)\frac{1}{12}$$
$$= 6{,}244.4 \text{ kip-ft}$$

A3.6—Maximum Reinforcement 6A.5.5

The factored resistance (φ factor) of compression controlled sections shall be limited in accordance with LRFD Design Article 5.6.2.1. This approach limits the capacity of over-reinforced (compression controlled) sections. C6A.5.5

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The net tensile strain, ε_t, is the tensile strain at nominal strength and determined by strain compatibility using similar triangles. LRFD Design C5.6.2.1

Given an allowable concrete strain of 0.003 and depth to neutral axis c = 4.39 in. and a depth from the extreme concrete compression fiber to the center of gravity of the prestressing strands d_p = 59.75 in. LRFD Design C5.6.2.1

$$\frac{\varepsilon_c}{c} = \frac{\varepsilon_t}{d-c}$$

$$\frac{0.003}{4.39 \text{ in.}} = \frac{\varepsilon_t}{59.75 \text{ in.} - 4.39 \text{ in.}}$$

$$\varepsilon_t = 0.0378$$

For ε_t = 0.0378 > 0.005, the section is tension controlled and Resistance Factor φ shall be taken as 1.0. LRFD Design 5.6.2.1, 5.5.4.2

A3.7—Minimum Reinforcement 6A.5.6

Amount of reinforcement must be sufficient to develop M_r equal to the lesser of: LRFD Design 5.6.3.3

M_{cr} or 1.33 M_u

$$M_{cr} = \gamma_3 \left[\left(\gamma_1 f_r + \gamma_2 f_{cpe} \right) S_c - M_{dnc} \left(\frac{S_c}{S_{nc}} - 1 \right) \right]$$

Load	Load Factor, γ
DC	1.25
DW	1.50
LL	1.75

LRFD Design Tables 3.4.1-1, 3.4.1-2

$M_R = \varphi M_n = (1.0)\ (6,244.4) = 6,244.4$ kip-ft

1. $1.33 M_u =$
   $1.33\ [1.75\ (1,487.7) + 1.25\ (1,520 + 200) + 1.5\ (162)]$
   $= 6,645.3$ kip-ft > 6,244.4 kip-ft
   No Good

2. $M_{cr} = \gamma_3 \left[\left(\gamma_1 f_r + \gamma_2 f_{cpe} \right) S_c - M_{dnc} \left(\frac{S_c}{S_{nc}} - 1 \right) \right]$ 6A.5.6, LRFD Design Eq. 5.6.3.3-1

   $M_{dnc} = M_{DC1} = 1,520$ kip-ft

   $S_c = 17,473$ in.^{3}

   $S_{nc} = 10,543$ in.^{3}

Modulus of Rupture $f_r = 0.24\lambda\sqrt{f_c'}$ LRFD Design 5.4.2.6
$= 0.24(1)\sqrt{5} = 0.537$ ksi

$\lambda = 1.0$

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f_{cpe} = compressive stress in concrete due to effective prestress force (after allowance for all prestress losses) at extreme fiber of section where tensile stress is caused by externally applied loads.

$$f_{cpe} = \frac{P_{pe}}{A} + \frac{P_{pe}e}{S_b}$$

where P_{pe} = Effective Prestress Force

A3.7.1—Determine Effective Prestress Force, P_{pe}

$$P_{pe} = A_{ps} f_{pe}$$

Total Prestress Losses $\Delta f_{pT} = \Delta f_{pES} + \Delta f_{pLT}$ immediately before transfer — LRFD Design Eq. 5.9.3.1-1

Effective Prestress f_{pe} = Initial Prestress – Total Prestress Losses

A3.7.1.1—Loss Due to Elastic Shortening and/or External Loads, Δf_{pES} — LRFD Design 5.9.3.2.3a

$$\Delta f_{pES} = \frac{E_p}{E_{ct}} f_{cgp}$$

$$f_{cgp} = \frac{P_i}{A} + \frac{P_i e^2}{I} - \frac{M_D e}{I}$$

Initial Prestress immediately prior to transfer = $0.75 f_{pu}$ for low-relaxation prestressing strands — LRFD Design Table 5.9.2.2-1

For estimating P_i immediately after transfer, use $0.90(0.75 f_{pu})$ — LRFD Design C5.9.8.2.3a

P_i = $0.90 \times (0.75 \times 270 \text{ ksi}) \, 32 \times 0.153$

= 892.3 kips

M_D = Moment due to Self-Weight of Member at Section of Maximum Moment (Midspan)

= $\frac{1}{8} \times 0.822 \times 80^2$

= 657.6 kip-ft

Y_b = 24.73 basic beam section

$\bar{y}$ = 3.75 in. distance from bottom of girder to centroid of prestressing strands

e = 24.73 – 3.75 = 20.98 in.

eccentricity of P/S strands from *CG* of beam

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LRFD Design 5.9.3.2.3a

$$f_{cgp} = \frac{892.3}{789} + \frac{892.3 \times 20.98^2}{260,741} - \frac{657.6 \times 12 \times 20.98}{260,741}$$

$$= 1.131 + 1.506 - 0.635$$

$$= 2.002 \text{ ksi}$$

$$E_p = 28,500 \text{ ksi}$$

LRFD Design Eq. C5.4.2.4-2

$$E_{ct} = 33,000(w_c)^{1.5}\sqrt{f'_{ci}}$$

$$= 33,000\ (0.145)^{1.5}\sqrt{4.0}$$

$$= 3,644 \text{ ksi}$$

LRFD Design Eq. C5.9.3.2.3a-1

$$\Delta f_{pES} = \frac{28,500}{3,644} \times 2.002$$

$$= 15.658 \text{ ksi}$$

A3.7.1.2—Approximate Lump Sum Estimate of Time-Dependent Losses, Δf_{pLT}

Time-dependent losses include shrinkage of concrete, creep of concrete, and relaxation of steel. For refined estimates:

LRFD Design Eq. 5.9.3.4.1-1

$$\Delta f_{pLT} = \left(\Delta f_{pSR} + \Delta f_{pCR} + \Delta f_{pR1}\right)_{id} + \left(\Delta f_{pSD} + \Delta f_{pCD} + \Delta f_{pR2} - \Delta f_{pSS}\right)_{df}$$

for I-Girders, time-dependent losses can be approximated by: LRFD Design C5.9.3.3

LRFD Design Eq. 5.9.3.3-1

$$\Delta f_{pLT} = 10.0\frac{f_{pi}A_{ps}}{A_g}\gamma_h\gamma_{st} + 12.0\gamma_h\gamma_{st} + \Delta f_{pR}$$

where $\gamma_h = 1.7 - 0.01H$ LRFD Design Eq. 5.9.3.3-2

Assuming a relative humidity H ranging between 40 to 100 percent.

For this example, assume H = 70% or refer to LRFD Design Figure 5.4.2.3.3-1

$$\gamma_h = 1.7 - 0.01(70) = 1.0$$

and: LRFD Design Eq. 5.9.3.3-3

$$\gamma_{st} = \frac{5}{1 + f'_{ci}}$$

$$\gamma_{st} = \frac{5}{1+4} = 1.0$$

and:

Δf_{pR} = an estimate of relaxation loss

LRFD Design 5.9.3.3

$$\Delta f_{pR} = 2.4 \text{ ksi}$$

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and:

$f_{pi} = 0.75 \times 270 \text{ ksi} = 202.5 \text{ ksi}$

then:

$$\Delta f_{pLT} = 10.0 \times \frac{202.5 \times (32 \times 0.153)}{789} \times 1.0 \times 1.0 + 12.0 \times 1.0 \times 1.0 + 2.4$$

$\Delta f_{pLT} = 26.97 \text{ ksi}$

A3.7.1.3—Total Prestress Losses, Δf_{pT}

$\Delta f_{pT} = \Delta f_{pES} + \Delta f_{pLT}$ LRFD Design Eq. 5.9.3.1-1

$= 15.658 + 26.97$

$= 42.63 \text{ ksi}$

$f_{pe} = \text{Initial Prestress} - \text{Total Prestress Losses}$

$= 0.75 \times 270 - 42.63 = 159.87 \text{ ksi}$

$P_{pe} = 159.87 \times 32 \times 0.153$

$= 782.7 \text{ kips}$

Substitute in:

$$f_{pb} = \frac{P_{pe}}{A} + \frac{P_{pe} e}{S_b}$$

$$= \frac{782.7}{789} + \frac{(782.7)(20.98)}{10,543}$$

$= 2.550 \text{ ksi}$

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$$M_{cr} = \gamma_3 \left[\left(\gamma_1 f_r + \gamma_2 f_{cpe} \right) S_c - M_{dnc} \left(\frac{S_c}{S_{nc}} - 1 \right) \right]$$

LRFD Design Eq. 5.6.3.3.-1

$\gamma_1 = 1.6$

$\gamma_2 = 1.1$ (for bonded tendons)

$\gamma_3 = 1.0$ (for prestressing steel)

$$M_{cr} = 1.0 \left[\left(1.6(0.537) + 1.1(2.550) \right) \frac{17,473}{12} - 1,520 \left(\frac{17,473}{10,543} - 1 \right) \right]$$

$$M_{cr} = 1.0 \left[\left(0.859 + 2.803 \right) \frac{17,473}{12} - 1,520 \left(0.657 \right) \right]$$

Therefore, $M_{cr} = 4,333.5$ kip-ft and:

$1.33 M_u = 6{,}645.3$ kip-ft (previously calculated)

$M_{cr} < 1.33\ M_u$ therefore, M_{cr} governs

$M_r = 6{,}244.4$ kip-ft (previously calculated)

$M_r = 6{,}244.4$ kip-ft > 4,333.5 kip-ft OK

6A.5.6

The minimum reinforcement check is satisfied.

A3.8—Compute Nominal Shear Resistance at First Critical Section

Note: Article 6A.5.8 of this Manual does not require a shear evaluation for the design load and legal loads if the bridge shows no visible sign of shear distress. Shear calculations shown here for HL-93 are for illustrative purposes only.

Shear Location:

Critical section for shear near the supports is the greater of d_v or $0.5 d_v \cot \theta$ from the face of support.

LRFD Design 5.7.3.2

Effective Shear Depth, d_v:

LRFD Design 5.7.2.8

Maximum of:

i) distance between resultants of the tensile and compressive forces

ii) $0.9 d_e$

iii) $0.72 h$

The first critical section will, by inspection, be within the 12-ft debonded end region. Ten strands have been debonded at the ends.

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LRFD Design Eq. 5.6.3.1.1-4

$$c = \frac{A_{ps} f_{pu}}{\alpha_1 f_c' \beta_1 b + k A_{ps} \frac{f_{pu}}{d_p}}$$

A_{ps} = $(32-10)(0.153) = 3.366$ in^2

b = $b_e = 102$ in. (Effective flange width of deck)

α_1 = 0.85

β_1 = 0.85

f_c' = 4.0 ksi (Deck concrete strength)

k = 0.28 for low-relaxation strands

f_{pu} = 270 ksi

	Strands	y	Strands × y
Layer 1	8	2	16
Layer 2	8	4	32
Layer 3	6	6	36
Total	22		84

$$\bar{y} = \frac{\text{strands} \times y}{\text{strands}} = \frac{84}{22}$$

distance from bottom of beam to 22 strand centroid = 3.82 in.

d_p = $(54 + 1 + 8.5) - 3.82 = 59.68$ in.

$$c = \frac{3.366 \text{ in.}^2 \times 270 \text{ ksi}}{0.85 \times 4.0 \text{ ksi} \times 0.85 \times 102 \text{ in.} + 0.28 \times 3.366 \text{ in.}^2 \times \frac{270 \text{ ksi}}{59.68 \text{ in.}}}$$

c = 3.04 in. $a = \beta_1 c = 2.58$ in.

d_v = $59.68 - \frac{2.58}{2} = 58.4$ in.

For establishing the critical shear section assume: θ = 30°, a high assumption is conservative.

$$0.5 d_v \cot\theta = (0.5)\ (d_v)\ (\cot 30°)$$
$$= 0.87 d_v < d_v$$

Distance from face of support to centerline of bearing = 6 in. (12-in. bearing pads)

Distance from centerline of bearing to critical shear section:

= 58.4 in. + 6 in. LRFD Design 5.7.3.2

= 64.4 in.

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= 5.37 ft

A3.9—Maximum Shear at Critical Section Near Supports

Calculated by statics with the loads applied no closer than 5.37 ft from the supports

$$V_{TANDEM} = 25^k \times \frac{(74.63 \text{ ft} + 70.63 \text{ ft})}{80 \text{ ft}} = 45.4 \text{ kips}$$

$$V_{TRUCK} = \frac{32^k (74.63 \text{ ft} + 60.63 \text{ ft}) + 8^k (46.63 \text{ ft})}{80 \text{ ft}} = 58.8 \text{ kips (Governs)}$$

$$V_{LANE} = \frac{0.64 \text{ klf} (74.63 \text{ ft})^2}{2 \times 80 \text{ ft}} = 22.3 \text{ kips}$$

$$IM = 33 \text{ percent}$$

$$= 45.4 \text{ kips}$$

Total Shear = $V_{LANE} + V_{TRUCK} \times 1.33$ = 100.5 kips

Distributed V_{LL+IM} = 100.5 kips × 0.849 = 85.3 kips

Dead Load Shears:

From A3.3.1, DC_1 = 1.90 kip/ft and DC_2 = 0.25 kip/ft

From A3.3.2, DW = 0.203 kip/ft

V_{DC} = (1.90 klf + 0.25 klf)(0.5 × 80 ft − 5.37 ft) = 74.5 kips

V_{DW} = (0.203 klf) (0.5 × 80 ft − 5.37 ft) = 7.03 kips

A3.10—Compute Nominal Shear Resistance

The nominal shear resistance V_n shall be the lesser of:

$$V_n = V_s + V_c + V_p$$

LRFD Design Eqs. 5.7.3.3-1, 5.7.3.3-2

$$V_n = 0.25 f_c' b_v d_v + V_p$$

$V_p = 0.0$ as straight tendons are provided

Critical section for shear near the support is at 64.4 in. from centerline of bearing (within the debonded length). Transverse reinforcement provided at critical section: #4 vertical stirrups at 9-in. spacings.

Minimum Transverse Reinforcement — LRFD Design 5.7.2.5

effective web width, b_v = 8 in.

stirrup spacing, s = 9 in.

Grade 60 rebar, f_y = 60 ksi

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$A_v = 0.0316\sqrt{f'_c}\frac{b_v s}{f_y}$ LRFD Design Eq. 5.7.2.5-1

$= 0.0316\sqrt{5}\frac{(8)(9)}{60}$

$= 0.0815$ in.²

Area provided 2 legs × 0.20 in.^{2} = 0.40 in.^{2} > 0.0815 in.^{2} OK

$V_c = 0.0316\beta\sqrt{f'_c}b_v d_v$ LRFD Design Eq. 5.7.3.3-3

$V_s = \frac{A_v f_y d_v \cot\theta}{s}$ for α= 90° LRFD Design Eq. 5.7.3.3-4

$V_n = 0.25 f'_c b_v d_v + V_p = 0.25\times 5.0\times 8\text{in.}\times 58.4\text{in.} + 0.0 = 584$ kips LRFD Design Eq. 5.7.3.3-2

These equations are based on the Modified Compression Field Theory (MCFT) and require the determination of β and θ by detailed analysis. A simplified analysis using θ = 45° and β = 2.0 may be utilized for an initial evaluation before resorting to the MCFT, if necessary, for likely improved shear capacity. C6A.5.8

A3.10.1—Simplified Approach

θ = 45° β = 2.0

Concrete: $V_c = 0.0316\beta\sqrt{f'_c}b_v d_v$

Effective Web Width: $b_v = 8$ in.

Effective Shear Depth: $d_v = 58.4$ in.

$V_c = (0.0316)(2.0)\sqrt{5.0}(8)(58.4)$

$= 66.0$ kips

Steel:

$V_s = \frac{A_v f_y d_v \cot\theta}{s}$

#4 at 9 in.

$A_v = 2\times 0.20 = 0.40$ in.^{2}

$V_s = \frac{(0.40)(60)(58.4)(\cot 45)}{9}$

$= 155.7$ kips

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Total Nominal Shear Resistance, V_n:

$V_c + V_s + V_p = 66.0 + 155.7 + 0.0 = 221.7 \text{ kips}$

221.7 kips < 584 kips, $\therefore\ V_n = 221.7$ kips

Φ = 0.9 LRFD Design 5.5.4.2

φV_n = 0.9 × 221.7= 199.5 kips

Maximum distributed shears at critical section (HL-93 Inventory Loading):

V_{LL+IM} = 85.3 kips

V_{DC} = 74.5 kips

V_{DW} = 7.03 kips

Load	Load Factor γ
DC	1.25
DW	1.50
LL	1.75

LRFD Design Tables 3.4.1-1, 3.4.1-2

Factored Shear:

$V_u = (1.75)(85.3) + (1.25)(74.5) + (1.5)(7.03)$
$= 252.9 \text{ kips} > 199.5 \text{ kips}$ No Good

Try MCFT approach.

A3.10.2—MCFT Approach

LRFD Design Eq. 5.7.2.8-1

Shear stress on the concrete:

$$v = \frac{V_u - \varphi V_p}{\varphi b_v d_v}$$

$$= \frac{252.9}{(0.9)(8)(58.4)} = 0.601 \text{ ksi}$$

$$\frac{v}{f_c'} = \frac{0.601}{5.0} = 0.12 < 0.25$$

OK, Therefore Appendix B5, Table B5.2-1 can be used.

At First Critical Section for Shear (64.4 in. from centerline of bearing)

Live Load Moments at first critical section determined by statics:

$$M_{TRUCK} = \frac{32^k \times 5.37 \text{ ft}(74.63 \text{ ft} + 60.63 \text{ ft}) + 8^k \times 5.37 \text{ ft}(46.63 \text{ ft})}{80 \text{ ft}} = 315.6 \text{ kip-ft}$$

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$M_{LANE} = 0.64\text{klf} \times \frac{(74.63\text{ ft})^2}{2 \times 80\text{ ft}} \times 5.37\text{ ft} = 119.6\text{ kip-ft}$

$M_{LL+IM} = 119.6\text{ kip-ft} + 1.33 \times 315.6\text{ kip-ft} = 539.6\text{ kip-ft}$

Distributed Moment:

$g_m \times M_{LL+IM} = 0.724 \times 539.3 = 390.5\text{ kip-ft}$

Dead Load Moments at First Critical Section for Shear:

$M_{DC} = 0.5\ (1.90\text{ klf} + 0.25\text{ klf})\ (5.37\text{ ft})\ (80\text{ ft} - 5.37\text{ ft}) = 430.8\text{ kip-ft}$

$M_{DW} = 0.5\ (0.203\text{ klf})\ (5.37\text{ ft})\ (80\text{ ft} - 5.37\text{ ft}) = 40.7\text{ kip-ft}$

Load	Load Factor, γ
DC	1.25
DW	1.50
LL	1.75

LRFD Design Tables 3.4.1-1, 3.4.1-2

Factored Moment:

$M_u = (1.75)\ (390.5) + (1.25)\ (430.8) + (1.50)\ (40.7)$

$= 1{,}282.9\text{ kip-ft}$

Following the approach in the LRFD Shear Design Flowchart in Appendix B5 and LRFD Design Table B5.2-1 in Appendix B5.
Transfer Length 60 strand diameters = 30 in. < 64.4 in.

LRFD Design Figure CB5.2-5

As the section is outside the transfer length, the full value of f_{po} is used in calculating the shear resistance.

The Modified Compression Field Theory (MCFT) follows an iterative process in Appendix B5:

$\frac{v}{f_c'} = \frac{0.601}{5.0} = 0.12$ (≤ 0.125, row 3 of LRFD Design Table B5.2-1

LRFD Design Figure CB5.2-5, LRFD Design Table B5.2-1

Assume $\varepsilon_x \leq -0.10 \times 10^{-3}$ $(\varepsilon_s \times 1{,}000 \leq -0.10)$

From LRFD Design Table B5.2-1 (row 3, column 2):

$\theta = 21.9°$ $\beta = 2.99$

Calculate ε_x:

LRFD Design Figure B5.2-3

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LRFD Design Appendix B5.2

$$\varepsilon_x = \frac{\frac{|M_u|}{d_v} + 0.5N_u + 0.5|V_u - V_p|\cot\theta - A_{ps}f_{po}}{2\left(E_sA_s + E_pA_{ps}\right)} \le 0.001$$

A_{ps} = $22 \times 0.153 = 3.366$ in.^{2}

f_{po} = $0.7f_{pu} = 0.7 \times 270 = 189$ ksi

$$\varepsilon_x = \frac{\frac{|12 \times 1,282.9|}{58.4} + 0.5|252.9|(\cot 21.9°) - (3.366)(189)}{2(0 + 28,500 \times 3.366)}$$

$= -0.302 \times 10^{-3}$

If ε_x is negative, it must be recalculated including concrete stiffness.

LRFD Design Figure CB5.2-5

A_c = Area below $h/2$
$= (8)(26) + 1/2\ (8\ +\ 26)(9) + (10)(8)$
$= 441$ in.^{2}

LRFD Design Eq. B5.2-5

$$\varepsilon_x = \frac{\frac{|M_u|}{d_v} + 0.5N_u + 0.5|V_u - V_p|\cot\theta - A_{ps}f_{po}}{2\left(E_cA_c + E_sA_s + E_pA_{ps}\right)}$$

$$\varepsilon_x = \frac{\frac{12 \times 1,282.9}{58.4} + (0.5)(252.9)(\cot 21.9°) - (3.366)(189)}{2\left[(4,030)(441) + 0 + (28,500)(3.366)\right]}$$

$= -0.015 \times 10^{-3} >$ assumed $\varepsilon_x \le -0.10 \times 10^{-3}$

Assume $\varepsilon_x \le 0$

From LRFD Design Table B5.2-1 (row 3, column 4):

$\theta = 23.7°$ $\beta = 2.87$

LRFD Design Eq. 5.7.3.4.2-1, 5.7.3.4.2-2, 5.7.3.4.2-3

Calculate ε_x:

$$\varepsilon_x = \frac{\frac{12 \times 1,282.9}{58.4} + (0.5)(252.8)(\cot 23.7°) - (3.366)(189)}{2\left[(4,030)(441) + 0 + (28,500)(3.366)\right]}$$

$= -0.023 \times 10^{-3} <$ assumed $\varepsilon_x \le 0$ OK

Note $-0.023 \times 10^{-3} > -0.05 \times 10^{-3}$ (adjacent column), ∴ no further interactions

Calculate V_n:

LRFD Design Eq. 5.7.3.3-3

$V_c = 0.0316\ \beta\sqrt{f'_c}b_vd_v$
$= (0.0316)(2.87)\sqrt{5}(8)(58.4)$
$= 94.75$ kips

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$$V_s = \frac{A_v f_y d_v \cot\theta}{s}$$

LRFD Design Eq. C5.7.3.3-4

$$= \frac{(0.39)(60)(58.4)(\cot 23.7°)}{9}$$

$$= 345.9 \text{ kips}$$

Total Nominal Shear Resistance:

$$V_n = V_c + V_s$$

LRFD Design Eq. 5.7.3.3-1

$$= 94.75 + 345.9 = 440.7 \text{ kips}$$

(versus 221.7 by simplified method)

$0.25 f_c' b_v d_v + V_p = 584$ kips (previously calculated)

LRFD Design Eq. 5.7.3.3-2

440.7 kips < 584 kips, $\therefore$ $V_n = 440.7$ kips

$\varphi V_n = 0.9 \times 440.7 = 396.6$ kips

Alternatively, LRFD Design 5.7.3.4.2 allows for solving for β and θ directly:

$$\varepsilon_s = \frac{\left(\frac{|M_u|}{d_v} + 0.5N_u + |V_u - V_p| - A_{ps} f_{po}\right)}{E_s A_s + E_p A_{ps}}$$

LRFD Design Eq. 5.7.3.4.2-4

$$= \frac{\frac{|12 \times 1{,}282.9|}{58.4} + |252.9| - (3.366)(189)}{(0 + 28{,}500 \times 3.366)}$$

$$= -0.0012$$

use 0

LRFD Design 5.7.3.4.2

$$\beta = \frac{4.8}{(1+750\varepsilon_s)}$$

LRFD Design Eq. 5.7.3.4.2-1

$$= \frac{4.8}{(1+750(0))}$$

$$= 4.8$$

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$\theta = 29 + 3{,}500\,\varepsilon_s$ LRFD Design Eq. 5.7.3.4.2-3

$= 29 + 3{,}500\,(0)$
$= 29$

Calculate V_n:

$V_c = 0.0316\beta\sqrt{f_c'}\,b_v d_v$

$= 0.0316(4.8)\sqrt{5}(8)(58.4)$

$= 158.46$ kips

$V_s = \dfrac{A_v f_y d_v \cot\theta}{s}$

$= \dfrac{(0.39)(60)(58.4)\cot(29)}{9}$

$= 273.9$ kips

Total Nominal Shear Resistance:

$V_n = V_c + V_s$

$= 158.5 + 273.9$

$= 432.40$ kips

(versus 440.7 kips by LRFD Design Appendix B5)

A3.10.3—Check Longitudinal Reinforcement (LRFD Design 5.7.3.5)

Tensile capacity of the longitudinal reinforcement on the flexural tension side of the member shall be proportioned to satisfy LRFD Design Eq. 5.7.3.5-1. "Any lack of full development shall be accounted for."

$$A_{ps}f_{ps} + A_s f_y \geq \frac{|M_u|}{d_v\varphi_f} + 0.5\frac{N_u}{\varphi_c} + \left[\left|\frac{V_u}{\varphi_v} - V_p\right| - 0.5V_s\right]\cot\theta$$ LRFD Design Eq. 5.7.3.5-1

Calculate minimum required tensile capacity:

$V_s = 345.9 \text{ kips} > \dfrac{V_u}{\varphi} = \dfrac{252.9}{0.9} = 281.0$ use 281.0 kips

The right side of LRFD Design Eq. 5.7.3.5-1 yields:

$= \dfrac{|(1,282.9)(12)|}{(58.4)(1.0)} + \left(\left|\dfrac{252.9}{0.9}\right| - 0.5(281.0)\right)\cot 23.7°$

$= 583.7$ kips

Transfer Length: LRFD Design 5.9.4.3.1

ℓ_t = 60 strand diameters

= 60 × 0.5 in.= 30 in.

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Development Length: LRFD Design Eq. 5.9.4.3.2-1

$$\ell \quad \left(f_{ps} - \frac{2}{3} f_{pe} \right) d_b$$

where $\kappa = 1.6$ for pretensioned members with a depth greater than 24.0 in.

$$\ell \quad (264.4 - \frac{2}{3} \times 159.87) \times 0.5 = 126.3 \text{ in}$$

The 22 effective strands at the critical shear section are bonded over the full length of the beam. The section at 64.4 in. from centerline of the bearing is between the transfer length (30 in. from end of beam, 26 in. from centerline of bearing) and the development length (126.3 in. from end of beam, 120.3 in. from centerline of bearing). Use a linear growth in strand capacity from f_{pe} at the transfer length to f_{ps} at the development length. LRFD Design 5.9.4.3.2 LRFD Design 5.9.4.3.1

ℓ_{px} = 64.4 in. to critical section LRFD Design Eq. 5.9.4.3.2-3

$$f_{px} = f_{pe} + \frac{\ell}{\ell} (f_{ps} - f_{pe})$$

$$f_{px} = 159.87 + \frac{64.4 - 30}{126.3 - 30} (264.4 - 159.87) = 197.21$$

The left side of LRFD Design Eq. 5.7.3.5-1 yields:

$$= f_{px} \times A_{ps} = 197.21 \text{ ksi} \times 3.366 \text{ in.}^2 = 663.8 \text{ kips}$$

$A_{ps} f_{ps} + A_s f_y \geq \frac{|M_u|}{d_v \varphi_f} + 0.5 \frac{N_u}{\varphi_c} + \left[\left| \frac{V_u}{\varphi_v} - V_p \right| - 0.5 V_s \right] \cot\theta$ reduces to

663.8 kips ≥ 583.7 kips OK

A3.11—Compute Nominal Shear Resistance at Stirrup Change/ Quarter Point (6A.5.8) C6A.5.8

Multiple locations need to be checked for shear. Typically, locations near the quarter point could be critical because the corresponding moment may be quite low.

(20 ft from centerline of bearing)

Effective Shear Depth, d_v, is based upon 32 strands.

→ check transfer length LRFD Design 5.9.4.3

60 strand diameters = 30 in.

debonded length = 12 ft

All 32 strands are bonded at: 12 ft + 30 in. = 14.5 ft < 20 ft OK

$d_v = d_e - \frac{a}{2}$ LRFD Design 5.7.2.8

$d_e = h - \bar{y} = 63.5 - 3.75 = 59.75$ in.

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$d_v = d_e - \frac{a}{2}$

a = 3.73 in. (from Article A3.5 of this example)

d_v need not be less than the greater of minimum effective shear depth limits $0.9d_e$ or $0.72h$.

d_v = 57.89 in. > $0.9d_e = 53.78$ in.

> $0.72h = 45.72$ in.

If we base d_v on: LRFD Design Eq. C5.7.2.8-1

$$d_v = \frac{M_n}{A_s f_y + A_{ps} f_{ps}}$$

including the effects of development, then:

$$d_v = \frac{6244.4 \text{ kip-ft} \times 12 \text{ in./ft}}{0 + \left(32 \times 0.153 \text{ in.}^2\right) 264.4 \text{ ksi}} = 57.89 \text{ in.}$$

A3.12—Maximum Shear at Stirrup Change

(20 ft from centerline of bearing)

for HL-93 loading

Calculated by statics with the loads applied no closer than 5.37 ft from the support

V_{TANDEM} = $25^k \times \frac{(60 \text{ ft} + 56 \text{ ft})}{80 \text{ ft}} = 36.25$ kips = 36.25 kips

V_{TRUCK} = $\frac{32^k (60 \text{ ft} + 46 \text{ ft}) + 8^k (32 \text{ ft})}{80 \text{ ft}}$ = 45.6 kips Governs

V_{LANE} = $\frac{0.64 \text{ klf} (60 \text{ ft})^2}{2 \times 80 \text{ ft}}$

IM = 33 percent

V_{LL+IM} = 14.4 + 1.33 × 45.6 kips = 75.05 kips

Distributed $g_v = 0.849$

$g_v V_{LL+IM} = 0.849 \times 75.05 = 63.7$ kips

Dead Load Shears:

From A3.3.1, DC_1 = 1.90 kip/ft and DC_2 = 0.25 kip/ft

From A3.3.2, DW = 0.203 kip/ft

V_{DC} = (1.90 klf + 0.25 klf)(0.5 × 80 ft − 20 ft)

V_{DC} = 38 + 5 = 43 kips

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V_{DW} = (0.203klf) (0.5 ×80 ft – 20 ft)

V_{DW} = 4.1 kips

Minimum Transverse Reinforcement:

Effective Web Width: LRFD Design Eq. 5.7.2.5-1

b_v = 8 in.

Spacing of Transverse Reinforcement:

s = 12 in.

$$A_v = 0.0316\sqrt{f_c'}\frac{b_v s}{f_y}$$

$$A_v = 0.0316\sqrt{5}\frac{(8)(12)}{60} = 0.113 \text{ in.}^2$$

Area provided 2 # 3 = 2 (0.11) = 0.22 in.² > 0.113 in.² OK

A3.12.1—Simplifed Approach

θ = 45°

β = 2.0

Concrete:

$$V_c = 0.0316\beta\sqrt{f_c'}b_v d_v$$ LRFD Design Eq. 5.7.3.3-3

Effective Shear Depth:

d_v = 57.89 in.

$$V_c = (0.0316)(2.0)\sqrt{5.0}(8)(57.89)$$
$$= 66.0 \text{ kips}$$

Steel:

$$V_s = \frac{A_v f_y d_v \cot\theta}{s}$$

#3 at 12 in.

A_v = 2 (0.110) = 0.22 in.^{2}

$$V_s = \frac{(0.22)(60)(57.89)\cot 45°}{12}$$

V_s = 63.7 kips

Total Nominal Shear Resistance:

$$V_n = V_c + V_s$$
$$= 65.4 + 63.7 = 129.1 \text{ kips}$$

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$0.25 f_c' b_v d_v + V_p = 0.25 \times 5.0 \times 8 \text{ in.} \times 57.89 \text{ in.} + 0.0 = 578.9 \text{ kips}$ LRFD Design Eq. 5.7.3.3-2

129.1 kips < 578.9 kips ∴ V_n = 129.1 kips

φV_n = 0.9x129.1 = 116.2 kips

Factored Shear V_u:

$V_u = 1.75\ (63.7) + 1.25\ (43) + 1.5\ (4.1) = 171.4 \text{ kips}$

116.2 kips < 171.4 kips No Good

Try MCFT Approach.

A3.12.2—MCFT Approach

Shear stress on the concrete:

$$v = \frac{V_u - \varphi V_p}{\varphi b_v d_v} = \frac{172.6}{(0.9)(8)(57.89)} = 0.414 \text{ ksi}$$ LRFD Design Eq. 5.7.2.8-1

$$\frac{v}{f_c'} = \frac{0.414}{5} = 0.0828 < 0.25$$ OK

At Stirrup Change:

$$M_{TRUCK} = \frac{32^k \times 20 \text{ ft.} (60 \text{ ft} + 46 \text{ ft}) + 8^k \times 20 \text{ ft} (32 \text{ ft})}{80 \text{ ft}} = 912.0 \text{ kip-ft}$$

$$M_{LANE} = 0.64 \text{ klf} \times \frac{(60 \text{ ft})^2}{2 \times 80 \text{ ft}} \times 20 \text{ ft} = 288 \text{ kip-ft}$$

M_{LL+IM} = 288 kip-ft + 1.33 × 912 kip-ft = 1,501 kip-ft

$g_m M_{LL+IM}$ = (0.724) (1,501) = 1,087 kip-ft

M_{DC} = 0.5 (1.90 klf + 0.25 klf) (20 ft) (80 ft – 20 ft) = 1,290 kip-ft

M_{DW} = 0.5 (0.203 klf) (20 ft) (80 ft – 20 ft) =121.8 kip-ft

M_u = 1.75 (1087) + 1.25 (1,290) + 1.5 (121.8)

= 3,697.5 kip-ft

Following the approach in the LRFD Shear Design Flowchart in Appendix B5 and LRFD Design Table B5.2-1 in Appendix B5. LRFD Design Appendix B5

Check upper limit of shear V_n

$0.25 f_c' b_v d_v + V_p = 0.25 \times 5.0 \times 8 \text{ in.} \times 57.89 \text{ in.} + 0.0 = 578.9 \text{ kips}$ LRFD Design Eq. 5.7.3.3-2

$\frac{v}{f_c'} = 0.0828 \leq 0.100$ (2nd row) LRFD Design Figure CB5.2-5, Table B5.2-1

$A_{ps} = 32 \times 0.153 = 4.896 \text{ in.}^2$

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$f_{po} = 0.7 f_{pu} = 0.7 \times 270 = 189$ ksi

LRFD Design Eq. B5.2-3

$$\varepsilon_x = \frac{\frac{|M_u|}{d_v} + 0.5N_u + 0.5|V_u - V_p|\cot\theta - A_{ps}f_{po}}{2\left(E_s A_s + E_p A_{ps}\right)} \le 0.001$$

$$\varepsilon_x = \frac{\frac{(12)(3,697.5)}{57.89} + 0.5(171.4)\cot\theta - (4.896)(189)}{2(0 + 28,500 \times 4.896)} \le 0.001$$

$$\varepsilon_x = 0.3071 \times 10^{-3} \cot\theta - 0.5694 \times 10^{-3}$$

Assume $\varepsilon_x \le 0.125 \times 10^{-3}$ ($\varepsilon_x \times 1{,}000 \le 0.125$)

From LRFD Design Table B5.2-1 (row 2, column 5):

$\theta = 24.9°$ $\beta = 2.91$

$\varepsilon_x = 0.3071 \times 10^{-3} \cot 24.9° - 0.5694 \times 10^{-3} = 0.0922 \times 10^{-3}$

The calculated ε_x is less than the assumed but not less than the adjacent ε_x value 0.0, $\therefore$ the assumption was not too conservative — OK

Calculate V_n:

LRFD Design Eq. 5.7.3.3-3

$$V_c = 0.0316\,\beta\sqrt{f_c'}\,b_v d_v$$

$$V_c = (0.0316)(2.91)\sqrt{5}(8)(57.89) = 95.2 \text{ kips}$$

LRFD Design Eq. 5.7.3.3-4

$$V_s = \frac{A_v f_y d_v \cot\theta}{s}$$

$$V_s = \frac{(0.22)(60)(57.89)(\cot 24.9°)}{12} = 137 \text{ kips}$$

LRFD Design Eq. 5.7.3.3-1

$$V_n = V_c + V_s = 95.2 + 137$$

$$V_n = 232.2 \text{ kips}$$

$0.25 f_c' b_v d_v + V_p = 578.9$ kips (previously calculated)

232.2 kips < 578.9 kips therefore $V_n = 232.2$ kips

$\varphi V_n = 0.9 \times 232.2 = 209.0$ kips

A3.12.3—Check Longitudinal Reinforcement (LRFD Design 5.7.3.5)

LRFD Design Eq. 5.7.3.5-1

$$A_{ps}f_{ps} + A_s f_y \ge \frac{|M_u|}{d_v \varphi_f} + 0.5\frac{N_u}{\varphi_c} + \left[\left|\frac{V_u}{\varphi_v} - V_p\right| - 0.5V_s\right]\cot\theta$$

$$V_s = 137 \text{ kips} < \frac{V_u}{\varphi} = \frac{171.4}{0.9} = 190.4 \text{ use } 137 \text{ kips}$$

The right side of LRFD Design Eq. 5.7.3.5-1 yields:

$$= \frac{(3,697.5)(12)}{(57.89)(1.0)} + \left(\frac{171.4}{0.9} - 0.5(137)\right)\cot 24.9° = 1,029 \text{ kips}$$

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The 20 fully bonded strands are fully developed at this location (f_{ps} = 264.4 ksi). As a portion of the remaining ten strands are debonded, their development length from the end of the debonded zone is calculated by LRFD Design Eq. 5.9.4.3.2-1 with k = 2.0. LRFD Design 5.9.4.3

$$\ell \quad {}_{,s} - 2/3 f_{pe})d_b \qquad \text{LRFD Design Eq. 5.9.4.3.2-1}$$

$$\ell \quad 264.4 - \frac{2}{3} \times 159.87)0.5 = 157.8 \text{ in.}$$

Check to see that the debonded strands are fully developed at the stirrup change location.

157.8 in. + 12 ft = 25.2 ft > 20 ft

Therefore, the strands are not fully developed and f_{px} must be determined.

Using a linear increase from f_{pe} at the transfer length to f_{ps} at the development length

From end of debonded zones

$$\ell \quad \text{ft} - 12 \text{ ft}) \times 12 \text{ in./ft} = 96 \text{ in.}$$

$$f_{px} = 159.87 + \frac{96 - 30}{157.8 - 30}(264.4 - 159.87) = 213.9 \text{ ksi}$$

Then, the left side of LRFD Design Eq. 5.7.3.5-1 yields:

$$= \quad 264.4 \times 22 \times 0.153 + 213.9 \times 10 \times 0.153 = 1,217 \text{ kips} \qquad \text{OK}$$

$$A_{ps} f_{ps} + A_s f_y \geq \frac{|M_u|}{d_v \varphi_f} + 0.5 \frac{N_u}{\varphi_c} + \left[\left| \frac{V_u}{\varphi_v} - V_p \right| - 0.5 V_s \right] \cot\theta \qquad \text{LRFD Design Eq. 5.7.3.5-1}$$

reduces to: 1,217 kips ≥ 1,029 kips OK

A3.12.4—Summary

Table A3.12.4-1—Summary of Moments and Shears

Location	Support	Critical Shear	Stirrup Change	Midspan
x/L	0.0	0.067	0.25	0.5
X, ft	0.0	5.37	20	40
V_{DC1}, kips	76	65.8	38	—
V_{DC2}, kips	10	8.7	5	—
V_{DW}, kips	8.12	7.03	4.1	—
$g_m V_{LL+IM}$, kips	—	85.3	63.7	—
V_n, kips, simplified	—	221.7	129.1	—
V_n, kips, MCFT	—	440.7	232.2	—
M_{DC1}, kip-ft	—	380.7	1140	1520
M_{DC2}, kip-ft	—	50.1	150	200
M_{DW}, kip-ft	—	40.7	121.8	162
$g_m M_{LL+IM}$, kip-ft	—	390.5	1087	1487.7
M_n, kip-ft	—	—	—	6244.4

A3.13—General Load Rating Equation (6A.4.2)

$$RF = \frac{C - (\gamma_{DC})(DC) - (\gamma_{DW})(DW) \pm (\gamma_P)(P)}{(\gamma_L)(LL + IM)} \qquad \text{Eq. 6A.4.2.1-1}$$

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A3.13.1 Evaluation Factors (for Strength Limit State)

A3.13.1.1—Resistance Factor, φ (LRFD Design 5.5.4.2)

φ = 1.0 for flexure (previously determined to be a tension-controlled section; see Article A3.6)

φ = 0.9 for shear

A3.13.1.2—Condition Factor, φ_c (6A.4.2.3)

φ_c = 1.0 No member deterioration, NBI Item 59 Code = 6

A3.13.1.3—System Factor, φ_s (6A.4.2.4)

φ_s = 1.0 4-girder bridge with spacing > 4 ft

A3.13.2—Design Load Rating (6A.4.3)

A3.13.2.1—Strength I Limit State (6A.5.4.1)

$$RF = \frac{(\varphi_c)(\varphi_s)(\varphi)R_n - (\gamma_{DC})(DC) - (\gamma_{DW})(DW)}{(\gamma_L)(LL + IM)}$$

A3.13.2.1a—Inventory Level

Table 6A.4.2.2-1

Load	Load Factor
DC	1.25
DW	1.50 Overlay thickness was not field measured.
LL	1.75

Flexure at Midspan:

$$RF = \frac{(1.0)(1.0)(1.0)(6244.4) - \left[(1.25)(1520 + 200) + (1.5)(162)\right]}{(1.75)(1,487.7)}$$

$= 1.48$

The shear rating factors for Design Load Rating are calculated for illustration purposes only. In-service concrete bridges that show no visible signs of shear distress need not be checked for shear during design load or legal load ratings. 6A.5.8

Shear at First Critical Shear Section (64.4 in. from centerline of bearing):

1. Simplified Approach

$$RF = \frac{(1.0)(1.0)(0.9)(221.7) - \left[(1.25)(65.8 + 8.7) + (1.50)(7.03)\right]}{(1.75)(85.3)}$$

$= 0.64$

2. MCFT

$$RF = \frac{(1.0)(1.0)(0.9)(440.7) - \left[(1.25)(65.8 + 8.7) + (1.50)(7.03)\right]}{(1.75)(85.3)}$$

$= 1.96$

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3. General Procedure

$$RF = \frac{(1.0)(1.0)(0.9)(432.4) - \left[(1.25)(65.8+8.7) + (1.50)(7.03)\right]}{(1.75)(85.3)}$$

$$= 1.91$$

Shear at Stirrup Change (20 ft from centerline of bearing):

1. Simplified Approach

$$RF = \frac{(1.0)(1.0)(0.9)(129.1) - [(1.25)(38+5) + (1.50)(4.1)]}{(1.75)(63.7)}$$

$$= 0.51$$

2. MCFT

$$RF = \frac{(1.0)(1.0)(0.9)(232.2) - [(1.25)(38+5) + (1.50)(4.1)]}{(1.75)(63.7)}$$

$$= 1.34$$

A3.13.2.1b—Operating Level

For Strength I Operating Level only the live load factor changes; therefore the rating factor can be calculated by direct proportions.

Table 6A.4.2.2-1

Load	Load Factor, γ
DC	1.25
DW	1.50
LL	1.35

Flexure at Midspan:

$$RF = 1.48 \times \frac{1.75}{1.35}$$

$$= 1.92$$

Shear: Prestressed concrete shear capacity is load-dependent. Therefore, the change in the rating factor using MCFT will not be linear with the change in the live-load factor. The Operating Design Load Rating for shear is not illustrated here.

This example has illustrated the calculation for the shear rating factor with the longitudinal yield check at the first critical section for shear and at a stirrup change. Due to the variation of resistances for shear along the length of this prestressed concrete I-beam, it is not certain that these two locations govern for the Strength I limit state. A systematic evaluation of the shear and longitudinal yield criteria based on shear-moment interaction should be performed along the length of the beam.

Flexure rating should be checked at maximum moment sections and at sections where there are changes in flexural resistance.

The checks performed for minimum and maximum reinforcement will also vary along the length; these checks are required to be satisfied at each cross section in the LRFD Design specification.

A3.13.2.2—Service III Limit State (Inventory Level) (6A.5.4.1)

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$$RF = \frac{f_R - (\gamma_D)(f_D)}{(\gamma_L)(f_{LL+IM})}$$

Flexural Resistance $f_R = f_{pb}$ + Allowable tensile stress

f_{pb} = Compressive stress due to effective prestress

= 2.550 ksi (from Article A3.7.1.3 of this example)

Allowable Tensile Stress $= 0.19\sqrt{f'_c}$ LRFD Design 5.9.2.3.2b

$= 0.19\sqrt{5}$

$= 0.425$ ksi

$f_R = 2.550 + 0.425$

$= 2.975$ ksi

Determine Dead Load Stresses at Midspan:

From A3.3.1, M_{DC1} = 1,520 kip-ft and M_{DC2} = 200 kip-ft

From A3.3.2, M_{DW} = 162 kip-ft

From A3.2, $S_{b\,(nc)}$ = 10,543 in.^{3} $S_{b\,(comp)}$ = 17,473 in.

$$f_{DC} = \frac{1,520 \times 12}{10,543} + \frac{200 \times 12}{17,473} = 1.87 \text{ ksi}$$

$$f_{DW} = \frac{162 \times 12}{17,473} = 0.11 \text{ ksi}$$

Total f_D = 1.98 ksi

Live Load Stress at Midspan:

From A3.4.2, M_{LL+IM} = 1,487.7 kip-ft

From A3.2, $S_{b\,(comp)}$ =17,473 in.^{3}

$$f_{LL+IM} = \frac{1,487.7 \times 12}{17,473} = 1.02 \text{ ksi}$$

$$RF = \frac{2.973 - (1.0)(1.98)}{(0.8)(1.02)}$$

$$= 1.22$$

A3.13.3—Legal Load Rating (6A.4.4)

Inventory Design Load Rating $RF > 1.0$, therefore the legal load ratings do not need to be performed and no posting is required. 6A.4.3.1

A3.13.4—Permit Load Rating (6A.4.5)

Permit Type: Special, single-trip, mix with traffic, no escort

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Permit Weight: 220 kips

The permit vehicle is shown in Example A1, Figure A1A.1.10-1.

ADTT (one direction): 5,000

From Live-Load Analysis by Computer Program:

Undistributed Maximum M_{LL} = 2,950.5 kip-ft

Undistributed Maximum V_{LL} = 157.9 kips

A3.13.4.1—Strength II Limit State (6A.5.4.2.1)

Load	Load Factor, γ
DC	1.25
DW	1.5
LL	1.2

LRFD Design Tables 3.4.1-1, 3.4.1-2; Table 6A.4.5.4.2a-1

Use One-Lane Distribution Factor and divide out the 1.2 multiple presence factor. 6A.4.5.4.2b

6A.4.5.5

$g_{m1} = 0.514 \times \frac{1}{1.2} = 0.428$

$g_{v1} = 0.70 \times \frac{1}{1.2} = 0.583$

IM = 20% (Riding surface condition verified by inspection: Minor Deviations)

Maximum Live Load Effect:

M_{LL+IM} = (2950.5) (0.428) (1.20)

= 1515.4 kip-ft at midspan

V_{LL+IM} = (157.9) (0.583) (1.20)

= 110.5 kips

φ factors are the same as those for the design calculations. See Article A3.13.1.

A3.13.4.1a—Flexure

$$RF = \frac{(1.0)(1.00)(1.0)(6244.4) - [(1.25)(1,520 + 200) + (1.5)(162)]}{(1.2)(1515.4)}$$

$= 2.11 > 1.0$ OK

Shear evaluation is required for Permit Load Rating. 6A.5.8

A3.13.4.1b—Shear (Using MCFT)

$$RF = \frac{(1.0)(1.0)(0.9)(440.7) - [(1.25)(74.5) + (1.50)(7.03)]}{(1.2)(110.5)}$$

$= 2.21 > 1.0$ OK

Shear resistance taken from HL-93. Acceptable and conservative as long as M_u and V_u for HL-93 are both ≥ M_u and V_u for permit. Must be recalculated if permit values are greater.

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A3.13.4.2—Service I Limit State (Optional) (6A.5.4.2.2b)

$\gamma_L = \gamma_{DC} = \gamma_{DW} = 1.0$ Table 6A.4.2.2-1

LRFD distribution analysis methods as described in LRFD Design Article 4.6.2 should be used. 6A.4.5.4.2a

$g_m = 0.724$ A3.4.1.1

Distributed Live-Load Effect:

Dead Load Moments at Midspan:

From A3.3.1, M_{DC1} = 1520 kip-ft and M_{DC2} = 200 kip-ft

From A3.3.2, M_{DW} = 162 kip-ft

$M_{LL+IM} = (2,950.5)(0.724)(1.2) = 2,563.4$ kip-ft

$M_{DC} + M_{DW} + M_{LL+IM} = (1,520 + 200) + 162 + 2,563.4 = 4,445.4$ kip-ft

A3.13.4.2a—Simplified Check Using 0.75M_n (C6A.4.2.2)

Nominal flexural resistance: M_n = 6,244.4 kip-ft
(use nominal, not factored resistance)

$0.75M_n = 0.75 \times 6,244.4 = 4,683.3$ kip-ft > 4,445.4 kip-ft OK

$$\text{Moment Ratio} = \frac{0.75M_n}{M_{DC} + M_{DW} + M_{LL+IM}} = \frac{4,683.3}{4,445.4} = 1.05 > 1.0$$ OK

A3.13.4.2b—Refined Check Using 0.9f_y

Calculate stress in outer reinforcement at Midspan. Stress due to moments in excess of the cracking moment acts upon the cracked section. The moments up to the cracking moment cause stress in the reinforcement equal to the effective prestress.

$f_R = 0.9F_y = 0.9\left(0.9F_{pu}\right) = 0.9\left(0.9 \times 270\right) = 218.7$ ksi 6A.5.4.2.2b, Table 6A.5.4.2.2b-1

M_{cr} = 4,333.5 kip-ft (previously calculated; see Article A3.7.1.3)

Effective prestress: (0.75 × 270 – 42.63) = 159.87 ksi (previously calculated; see Article A3.7.1.3)

$M_{DC} + M_{DW} + M_{LL+IM} - M_{cr} = 4,445.4 - 4,333.5 = 111.9$

Section Properties for the Cracked Composite Section:

b_{trans} = 102 in. × 0.89 = 90.8 in. (see Article A3.2)

h = 54 in. + 1 in. + 8.5 in. = 63.5 in.

A_{ps} = 32 × 0.153 in.^{2} = 4.896 in.^{2}

Modular ratio, n:

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$n = \frac{E_{ps}}{E_{beam}} = \frac{28.5\times10^3}{4.07\times10^3} = 7$

A_{trans} = 4.896 in.^{2} × 7 = 34.3 in.^{2}

y = 3.75 in. (see Article A3.5)

Outer strand y = 2 in.

Assume neutral axis is in the slab.

$$c = \frac{\left(\frac{c}{2}\right)\left(b_{trans}\times c\right)+(h-y)(A_{trans})}{\left(b_{trans}\times c\right)+(A_{trans})}$$

$$c = \frac{\frac{c}{2}(90.8)c+(63.5-3.75)(34.3)}{(90.8)c+34.3}$$

$$45.4c^2+34.3c-2,049.4=0$$

Solving for c:

$$c = \frac{-34.3\pm\sqrt{34.3^2-4(45.4)(-2049.4)}}{2(45.4)}$$

$$c = 6.35\,\text{in.}$$

$$I_{cr} = \frac{1}{12}(90.8)(6.35)^3+(90.8)(6.35)\left(\frac{6.35}{2}\right)^2+(34.3)(63.5-3.75-6.35)^2$$

= 105,558 in.^{4}

Stress beyond the effective prestress (increase in stress after cracking):

$$f_s = n\frac{M_y}{I} = 7\frac{(111.9)(12)(63.5-2.0-6.35)}{105,558} = 4.9\text{ ksi}$$

Stress in the reinforcement at Permit crossing Service I:

$f_s = 159.87+4.9 = 164.8$ ksi $< f_R = 0.9F_y = 218.7$ ksi OK

Stress Ratio $= \frac{0.9f_y}{f_s} = \frac{218.7}{164.8} = 1.33 > 1.0$ OK

All permit checks for an interior girder are satisfied.

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A3.14—Summary of Rating Factors

Table A3.14-1—Summary of Rating Factors—Interior Girder

Limit State	Design Load Rating (HL-93)		Permit Load Rating
	Inventory	Operating	
Strength I	—	—	—
Flexure (at midspan)	1.48	1.92	—
Shear (at 64 in.)	1.96	—	—
Shear (at 20 ft)	1.34	—	—
Strength II	—	—	—
Flexure (at midspan)	—	—	2.11
Shear	—	—	2.21
Service III	—	—	—
Flexure (at midspan)	1.22	—	—
Service I	—	—	Stress Ratio = 1.33

A3.15—References

AASHTO. *AASHTO LRFD Bridge Design Specifications*, Eighth Edition, LRFD-8. American Association of State Highway and Transportation Officials, Washington, DC, 2017.

NCHRP. *Legal Truck Loads and AASHTO Legal Loads for Posting*, NCHRP Report 575. Transportation Research Board, National Research Council, Washington, DC, 2007.

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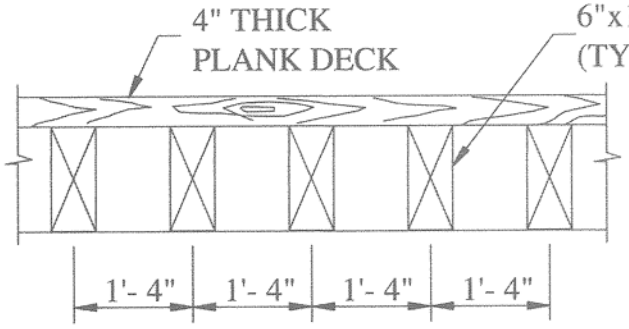
A4—TIMBER STRINGER BRIDGE: EVALUATION OF AN INTERIOR STRINGER

PART A—LOAD AND RESISTANCE FACTOR RATING METHOD

A4A.1—Bridge Data

Span:	17 ft 10 in.
Year Built:	1930
Year Reconstructed:	1967
Material:	Southern Pine No. 2
Condition:	No deterioration. NBI Item 59 Code = 6
Riding Surface:	Unknown condition
Traffic:	Two Lanes
ADTT (one direction):	150
Skew:	0°



Actual

Figure A4A.1-1—Partial Cross Section of Deck

A4A.2—Dead Load Analysis—Interior Stringer in Flexure

A4A.2.1—Components and Attachments, *DC*

Deck: $\frac{16}{12}\times\frac{4}{12}\times 0.050$ = 0.022 kip/ft LRFD Design Table 3.5.1-1

Stringer: $\frac{6\times 14}{144}\times 0.050$ = 0.029 kip/ft

Total per stringer = 0.051 kip/ft

$$M_{DC} = \frac{1}{8}\times 0.051\times 17.83^2$$
$$= 2.03 \text{ kip-ft}$$

A4A.2.2—Wearing Surface

$DW = 0$

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A4A.3—Live Load Analysis—Interior Stringer in Flexure

A4A.3.1—Distribution Factor for Moment and Shear

LRFD Design Type ℓ cross section — LRFD Design Table 4.6.2.2.1-1

One Lane Loaded: — LRFD Design Table 4.6.2.2.2a-1

$$g_1 = \frac{S}{6.7}$$

$$= \frac{\frac{16}{12}}{6.7} = 0.20$$

Two or More Lanes Loaded:

$$g_2 = \frac{S}{7.5}$$

$$= \frac{\frac{16}{12}}{7.5} = 0.18 < 0.20$$

One Lane Loaded Governs

$$g = 0.20$$

A4A.3.2—Compute Maximum Live Load Effects

A4A.3.2.1—Maximum Design Live Load (HL-93) Moment at Midspan

Design Lane Load Moment	=	25.4 kip-ft		
Design Truck Moment	=	142.6 kip-ft		
Design Tandem Moment	=	175.7 kip-ft	Governs	
IM	=	0 percent		6A.7.5
M_{LL}	=	25.4 + 175.7		
	=	201.1 kip-ft		

A4A.3.2.2—Distributed Live-Load Moments

Design Live Load HL-93:

$$g \times M_{LL} = 0.20 \times 201.1$$

$$= 40.2 \text{ kip-ft}$$

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A4A.4—Compute Nominal Flexural Resistance

Section Properties for Stringers (based on actual dimensions):

$I_x = \frac{bh^3}{12} = \frac{6 \times 14^3}{12} = 1372 \text{ in.}^4$

$S_x = \frac{I_x}{\frac{h}{2}} = \frac{1,372}{\frac{14}{2}} = 196 \text{ in.}^3$

$A = bh = 6 \times 14 = 84 \text{ in.}^2$

A4A.4.1—LRFD Design

$F_b = F_{bo} C_{KF} C_M \left(C_F \text{ or } C_V\right) C_{fu} C_i C_d C_\lambda$ LRFD Design Eq. 8.4.4.1-1

F_{bo} = 0.85 ksi Reference Design Value LRFD Design Table 8.4.1.1.4-1

C_{KF} = 2.5/ϕ = 2.5 / 0.85 = 2.94 Format Conversion Factor LRFD Design 8.4.4.2

C_M = 1.0 Wet Service Factor LRFD Design 8.4.4.3

(reduction for wet use not required due to species and member size)

C_F = Size Effect Factor for sawn lumber $\left(\frac{12}{d}\right)^{\frac{1}{9}} = \left(\frac{12}{14}\right)^{\frac{1}{9}} = 0.98 \le 1.0$ LRFD Design Eq. 8.4.4.4-2

C_{fu} = 1.0 Flat Use Factor LRFD Design 8.4.4.6

C_i = 1.0 Incising Factor (only apply to dimension lumber) LRFD Design 8.4.4.7

C_d = 1.0 Deck Factor LRFD Design 8.4.4.8

C_λ = 0.8 Time Effect Factor for Strength I LRFD Design 8.4.4.9

F_b = 0.85 × 2.94 × 1.0 × 0.98 × 1.0 ×1.0 × 1.0 × 0.8 = 1.96

Adjusted Design Value = F_b = 1.96 ksi

Nominal Resistance $M_n = F_b S C_L$ LRFD Design Eq. 8.6.2-1

C_L = 1.0

M_n = 1.96 ksi × 196 in.^{3} × 1.0 × 1ft/12 in. = 32.01 kip-ft

A4A.5—General Load-Rating Equation (6A.4.2)

$$RF = \frac{C - \left(\gamma_{DC}\right)\left(DC\right) - \left(\gamma_{DW}\right)\left(DW\right) \pm \left(\gamma_P\right)\left(P\right)}{\left(\gamma_L\right)\left(LL + IM\right)}$$ Eq. 6A.4.2.1-1

A4A.6—Evaluation Factors (for Strength Limit State)

1. Resistance Factor, φ LRFD Design 8.5.2.2

   φ = 0.85 for Flexure
   φ = 0.75 for Shear

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2. Condition Factor, φ_c 6A.4.2.3

   $\varphi_c = 1.0$ Good Condition

3. System Factor φ_s 6A.4.2.4

   $\varphi_s = 1.0$ for flexure and shear in timber bridges

A4A.7—Design Load Rating (6A.4.3)

A4A.7.1—Strength I Limit State (6A.7.4.1)

$$RF = \frac{(\varphi_c)(\varphi_s)(\varphi)R_n - (\gamma_{DC})(DC) - (\gamma_{DW})(DW)}{(\gamma_L)(LL + IM)}$$

A4A.7.1.1—Inventory Level

Load	Load Factor
DC	1.25
LL	1.75

Table 6A.4.2.2-1

Flexure:

$$RF = \frac{(1.0)(1.0)(0.85)(32.0) - (1.25)(2.03)}{(1.75)(40.2)}$$

$= 0.35$

A4A.7.1.2—Operating Level

Load	Load Factor
DC	1.25
LL	1.35

Table 6A.4.2.2-1

Flexure:

$$RF = 0.35 \times \frac{1.75}{1.35}$$

$= 0.45$

A4A.7.1.3—Shear (Horizontal Shear) (LRFD Design 8.7)

Critical Section for Live Load Shear is at a distance d = 14 in. = 1.17 ft from face of support

Place live load to cause maximum shear at lesser of:

1. Three times the depth = 3 × 14 = 42 in. = 3.5 ft Governs
2. $\frac{1}{4}$ of span length = $\frac{1}{4} \times 17.83$ = 4.46 ft

A4A.7.1.4—Compute Maximum Shear at Critical Section (14 in. = 1.17 ft)

A4A.7.1.4a—Dead Load Shear

$$V_{DC} = \frac{1}{2}(0.051)(17.83) - (0.051)(1.17)$$

$= 0.395$ kips

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A4A.7.1.4b—Live Load Shear (HL-93)

Live load placed at 3.5 ft from face of support:

V_{TANDEM} = 34.6 kips Governs

V_{TRUCK} = 26.3 kips

V_{LANE} = 3.7 kips

Undistributed Shear:

V_{LU} = 3.7 + 34.6

= 38.3 kips

Distributed:

V_{LD} = 38.3 × 0.20

= 7.7 kips

For Horizontal Shear:

$V_{LL} = 0.50\left[\left(0.60V_{LU}\right)+V_{LD}\right]$ LRFD Design Eq. 4.6.2.2.2a-1

V_{LU} = Maximum vertical shear at 3*d* or *L*/4 due to undistributed wheel loads (kips)

= For undistributed wheel loads, one line of wheels is assumed to be carried by one bending member. LRFD Design 4.6.2.2.2a

$= \frac{V_{LU}}{2} = \frac{(38.3)}{2} = 19.1$ kips

V_{LD} = Maximum vertical shear at 3*d* or *L*/4 due to wheel loads distributed laterally as specified herein (kips)

= 7.7 kips

V_{LL} = 0.50[(0.60×19.1) + 7.7] = 9.58 kips

A4A.7.1.5—Compute Nominal Shear Resistance

A4A.7.1.5a—LRFD Design

$V_n = \frac{F_v bd}{1.5}$ LRFD Design Eq. 8.7-2

$F_v = F_{vo}C_{KF}C_M C_i C_\lambda$ LRFD Design Eq. 8.4.4.1-2

F_{vo} = 0.165 ksi Reference Design Value LRFD Design Table 8.4.1.1.4-1

C_{KF} = 2.5/φ = 2.5 / 0.75 = 3.33 Format Conversion Factor LRFD Design 8.4.4.2

C_M = 1.0 Wet Service Factor LRFD Design 8.4.4.3

(reduction for wet use not required due to species and member size)

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C_i = 1.0 Incising Factor LRFD Design 8.4.4.7

C_λ = 0.8 Time Effect Factor for Strength I LRFD Design 8.4.4.9

F_v = $0.165 \times 3.33 \times 1.0 \times 1.0 \times 0.8$

Adjusted Design Value:

F_v = 0.440 ksi

V_n = $\frac{(0.440)(6)(14)}{1.5} = 24.6$ kips

A4A.7.1.5b—Inventory Level

Load	Load Factor
DC	1.25
LL	1.75

Shear:

$$RF = \frac{(1.0)(1.0)(0.75)(24.6) - (1.25)(0.395)}{(1.75)(9.58)}$$

$$= 1.07$$

A4A.7.1.5c—Operating Level

Shear:

$$RF = 1.07 \times \frac{1.75}{1.35} = 1.39$$

No service limit states apply.

A4A.8—Legal Load Rating (6A.4.4)

Live Load: AASHTO Legal Loads—Types 3, 3S2, and 3-3 (Rate for all three) 6A.4.4.2.1

g = 0.20

IM = 0 percent 6A.7.5

	Type 3	Type 3S2	Type 3-3	
M_{LL}	119.5	108.9	98.4	kip-ft
gM_{LL}	23.9	21.8	19.7	kip-ft

A4A.8.1—Strength I Limit State (6A.7.4.2)

Dead Load *DC*:

Load Factor = 1.25 Table 6A.4.2.2-1

ADTT = 150

Live-Load Factor = 1.30 Table 6A.4.4.2.3a-1

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Flexure:

$$RF = \frac{(1.0)(1.0)(0.85)(32.0)-(1.25)(2.03)}{(1.30)(M_{LL})}$$

A4A.8.1.1—Shear Capacity

Live load shear at critical section (14 in.) with live load placed to cause maximum shear effect at 3.5 ft (3*d*).

g = 0.20

IM = 0 percent 6A.7.5

The distributed live load is calculated in the same manner as demonstrated for the design load check.

$$V_{LL} = 0.50\left[(0.60V_{LU})+V_{LD}\right]$$ LRFD Design Eq. 4.6.2.2.2a-1

	Type 3	Type 3S2	Type 3-3	
V_{LU}	11.76	10.72	9.68	kips
V_{LD}	4.70	4.29	3.87	kips
V_{LL}	5.87	5.35	4.83	kips

Shear:

$$RF = \frac{(1.0)(1.0)(0.75)(24.6)-(1.25)(0.395)}{(1.30)(V_{LL})}$$

RF	Type 3	Type 3S2	Type 3-3
Shear	2.35	2.58	2.86
Flexure	0.79	0.87	0.96

A4A.8.2—Summary

Truck	Type 3	Type 3S2	Type 3-3
Weight, tons	25	36	40
RF	0.73	0.80	0.88
Safe Load Capacity, tons	18	28	35

A4A.9—Summary of Rating Factors for Load and Resistance Factor Rating Method

Table A4A.9-1—Summary of Rating Factors for Load and Resistance Factor Method—Interior Stringer

Limit State		Design Load Rating		Legal Load Rating		
		Inventory	Operating	Type 3	Type 3S2	Type 3-3
Strength I	Flexure	0.35	0.45	0.79	0.887	0.96
	Shear	1.07	1.39	2.35	2.58	2.864

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PART B—ALLOWABLE STRESS RATING METHOD

A4B.1—Bridge Data

Refer to Article A4A.1 for bridge data.

A4B.2—Section Properties

$I_x = \frac{bh^3}{12} = \frac{6 \times 14^3}{12} = 1{,}372 \text{ in.}^4$

$S_x = \frac{I_x}{h \div 2} = \frac{1{,}372}{14 \div 2} = 196 \text{ in.}^3$

$A = bh = 6 \times 14 = 84 \text{ in.}^2$

A4B.3—Dead Load Analysis—Interior Stringer

Deck:

$\frac{(1 \text{ ft} - 4 \text{ in.})4 \text{ in.}}{144 \text{ in.}^2/\text{ft}^2} \times 50 \text{ lb./ft}^3 = 22.2 \text{ lb/ft}$

Stringer:

$\frac{6 \text{ in.} \times 14 \text{ in.}}{144} \times 50 = \frac{29.2 \text{ lb/ft}}{51.4 \text{ lb/ft}}$ say 0.051 kip/ft

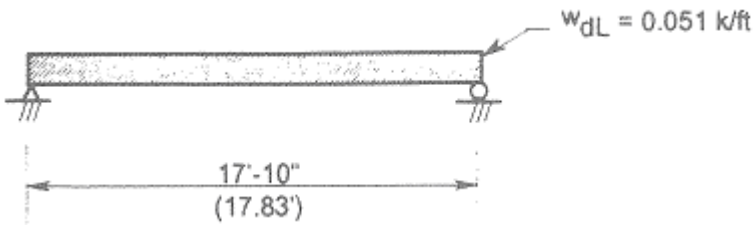

Figure A4B.3-1—Load Diagram for Interior Stringer—Uniform Dead Load

$M_{DL} = \frac{w_{DL}L^2}{8} = \frac{0.051(17.83)^2}{8}$

$M_{DL} = 2.03$ kip-ft

A4B.4—Live Load Analysis—Interior Stringer

Live Load: Rate for H-15 truck

Determine the maximum live load moment by statics. For small spans, verify that the maximum moment will occur at midspan with the heaviest wheel positioned at midspan.

$M_L = PL/4$

$M_L = (12 \text{ kips} \times 17.83 \text{ ft})/4 = 53.49$ kip-ft

Alternatively, interpolation could be used for estimating. Note that for longer spans and for interpolation between span increments greater than 1 ft, interpolated values yield approximate results. Appendix C6B

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Span	M_L	
17 ft	51 kip-ft	
		← For 17.83-ft span, interpolate
18 ft	54 kip-ft	
$M_L = 51 + \frac{17.83-17}{18-17}(54-51) = 53.5$ kip-ft		

A4B.5—Allowable Stress Rating (6B.3.1, 6B.4.2, 6B.5.2)

Consider stringer only; consider maximum moment and shear sections only for this example.

A4B.5.1—Impact (Use standard AASHTO) (6B.6.4)

No impact for timber members: AASHTO 3.8.1.2

$I = 0$

A4B.5.2—Distribution (Use standard AASHTO) (6B.6.3)

For two lanes and plank deck [a]: AASHTO 3.23.2.2, Table 3.23.1

$$DF = \frac{S}{3.75} = \frac{16 \text{ in.}/12 \text{ in.}/\text{ft}}{3.75} = 0.36$$

a Note that the moments given in MBE are for one line of wheels. The values given in AASHTO are for the entire rear axle and are therefore twice the MBE values.

Thus:

$M_{LL} = M_L \times DF = 53.5 \text{kip-ft} \times 0.36$

$M_{LL} = 19.26 \text{kip-ft}$

A4B.5.3—Stresses to be Used (Use NDS, *National Design Specification for Wood Construction*, 2005 Edition)

The general equations for adjusted Reference Design Values are:

$F_b' = F_b \times C_D C_M C_t C_L C_F C_{fu} C_i C_r$

$F_V' = F_V \times C_D C_M C_t C_i$

F_b = 850 psi Reference Design Value, NDS Table 4D

F_V = 165 psi Reference Design Value, NDS Table 4D

C_D = 1.15 Load Duration Factor for two months is assumed as cumulative effect of live load. Wood bridges are typically located on low-volume roads; therefore, the accumulated live load duration is lower than 30 days. It is assumed that the live load duration is two months in the reliability analysis.

C_M = 1.0 Wet Service Factor is in NDS Table 4D for Sothern Pine

C_t = 1.0 Temperature Factor

C_L = 1.0 Beam Stability Factor

C_F = 0.98 Size Factor = $(12/d)^{1/9}$ for beam depth exceeding 12 in.

C_{fu} = 1.0 Flat Use Factor; not applicable

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C_i = 1.0 Incising Factor

C_r = 1.0 Repetitive Use Factor, not applicable

A4B.5.3.1—Inventory Level Stresses (6B.5.2.7)

$F_b^I = 850 \times 1.15 \times 0.98 \times 1.0 = 958 \text{ psi} = 0.96 \text{ ksi}$

$C_D = 1.15$

$C_F = 0.98$

$C_i = 1.0$

and:

$F_V^I = 165 \times 1.15 \times 1.0 = 190 \text{ psi} = 0.19 \text{ ksi}$

A4B.5.3.2—Operating Level Stresses (Use standard AASHTO) (6B.5.2.7)

$F_b^O = F_b^I \times 1.33 = 950 \times 1.33$

$F_b^O = 1274 \text{ psi} = 1.27 \text{ ksi}$

and:

$F_V^O = 1.33 F_V^I = 1.33 \times 190 \text{ psi} = 253 \text{ psi}$

A4B.5.4—Inventory Level Rating for Flexure

Capacity:

$M_{R_I} = F_b^I S_x = 0.96 \text{ ksi} \times 196 \text{ in.}^3 = 188 \text{ kip-in.}$

$M_{R_I} = 15.68 \text{ kip-ft}$

then:

$$RF_I^M = \frac{M_{R_I} - M_{DL}}{M_{LL}} = \frac{15.68 \text{ kip-ft} - 2.03 \text{ kip-ft}}{19.26 \text{ kip-ft}} \qquad \text{Eq. 6B.4.1-1}$$

RF_I^M = 0.71 or 0.71 × 15 tons = 10.7 tons H truck

A4B.5.5—Operating Level Rating for Flexure

Capacity:

$M_{R_O} = F_b^O S_x = 1.27 \text{ ksi} \times 196 \text{ in.}^3 = 248.9 \text{ kip-in.}$

$M_{R_O} = 20.74 \text{ kip-ft}$

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then:

$$RF_O^M = \frac{M_{R_O} - M_{DL}}{M_{LL}} = \frac{20.74\,\text{kip-ft} - 2.03\,\text{kip-ft}}{19.26\,\text{kip-ft}}$$ Eq. 6B.4.1-1

RF_O^M = 0.97 or 0.97 × 15 tons = 14.6 tons H truck

A4B.5.6—Check Horizontal Shear

Computed shear at: AASHTO 13.6.5.2

1. A distance from the support equal to three times the depth of the stringer, or
2. At the quarter point, whichever is less.

Thus by:

1. $3(14\text{ in.}) = 42\text{ in.} \leftarrow$ Controls $= 3.5$ ft
2. $\frac{17.83\text{ ft} \times 12\text{ in./ft}}{4} = 53.5\text{ in.}$

For H-15 Truck:

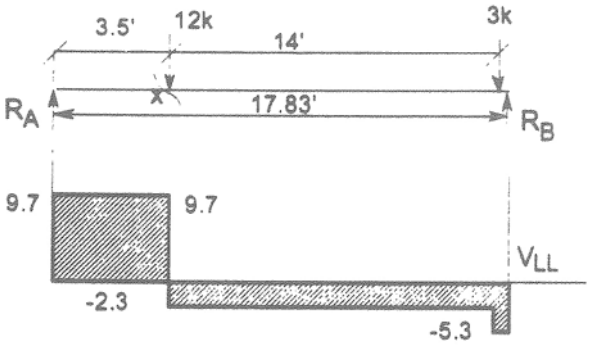

Figure A4B.5.6-1—Shear Diagram for Interior Stringer—H-15 Live Load

$$V_x = \frac{15(x - 2.8)}{L}$$ Appendix H6B

where $L = 17.83$ ft

$x = 17.83 - 3.5 = 14.33$ ft

$V_x = \frac{15(14.33 - 2.8)}{17.83} = 9.7$ kips per wheel line without distribution

$$V_{L_x} = \frac{1}{2}\left(0.6V_x^{L\ no\ dist.} + DFV_x^{L\ no\ dist.}\right)$$ AASHTO 13.6.5.2, Eq. 13-10

$V_{L_x} = \frac{1}{2}\left[0.6(9.7) + 0.36(9.7)\right]$

$V_{L_x} = 4.7$ kips

For $w_{DL} = 0.051$ kip/ft

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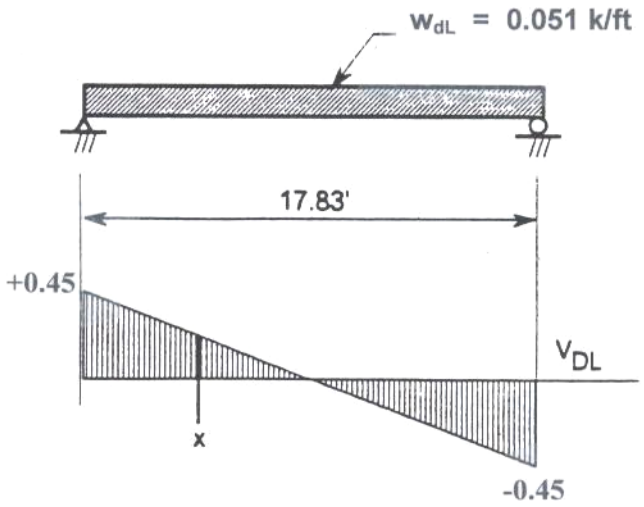

Figure A4B.5.6-2—Load and Shear Diagrams—Uniform Dead Load

$$R_A = R_B = \frac{1}{2} w_{DL} L$$

$$= \frac{1}{2}(0.051) \times 17.83$$

$$= 0.45 \text{ kips}$$

$$V_{D_x} = 0.45 - 0.051 \times 14/12$$

$$V_{D_x} = 0.4 \text{ kips}$$

A4B.5.7—Inventory Level Rating for Shear

Capacity:

$$V_R = \frac{2}{3} b d f_v \qquad \text{AASHTO Eq. 13-9}$$

then:

$$V_{R_I} = \frac{2}{3}(6)(14)(190) \text{ psi} = 10640 \text{ lbs} = 10.64 \text{ kips}$$

$$RF_I^V = \frac{V_{R_I} - V_{D_x}}{V_{L_x}} = \frac{10.64 \text{ kips} - 0.4 \text{ kips}}{4.7 \text{ kips}} \qquad \text{Eq. 6B.4.1-1}$$

RF_I^V = 2.18 or 2.18 × 15 tons = 32.7 tons H truck

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A4B.5.8—Operating Level Rating for Shear

Capacity:

$$V_{R_O} = \frac{2}{3}(6)(14)(253)\text{ psi} = 14168\text{ lbs} = 14.17\text{ kips}$$

$$RF_O^V = \frac{V_{R_O} - V_{D_x}}{V_{L_x}} = \frac{14.17\text{ kips} - 0.4\text{ kips}}{4.7\text{ kips}} \qquad \text{Eq. 6B.4.1-1}$$

RF_O^V = 2.93 or 2.93 × 15 tons = 43.95 tons H truck

A4B.5.9—Summary of Ratings for Allowable Stress Rating Method

Table A4B.5.9-1—Summary of Ratings for Allowable Stress Rating Method—Interior Stringer

Method/Force	*RF*	H Truck Max. Load, tons
Allowable Stress Moment:		
Inventory	0.71	10.7
Operating	0.97	14.6
Allowable Stress Shear:		
Inventory	2.18	32.7
Operating	2.93	43.9

∴ Rating governed by moment rather than shear.

A4B.6—Load Factor Rating

Not currently available for timber.

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PART C—SUMMARY

A4C.1—Summary of All Ratings for Example A4

Table A4C-1—Summary of Rating Factors for All Rating Methods—Interior Stringer

		Design Load Rating		Legal Load Rating			H-15 Rating			
							Flexure		Shear	
LRFR Method		Inventory	Operating	Type 3	Type 3S2	Type 3-3	Inv.	Opr.	Inv.	Opr.
Strength I Limit State	Flexure	0.35	0.45	0.79	0.87	0.96	—	—	—	—
	Shear	1.07	1.39	2.35	2.58	2.86	—	—	—	—
Allowable Stress Method		—	—	—	—	—	0.71	0.97	2.18	2.93
Load Factor Method		—	—	—	—	—	—	—	—	—

A4C.2—References

AASHTO. *Standard Specifications for Highway Bridges*, 17th Edition, HB-17. American Association of State Highway and Transportation Officials, Washington, DC, 2002.

AASHTO. *AASHTO LRFD Bridge Design Specifications*, Eighth Edition, LRFD-8. American Association of State Highway and Transportation Officials, Washington, DC, 2017.

NFPA. *National Design Specification for Wood Construction*. National Forest Products Association, Washington, DC, 2005.

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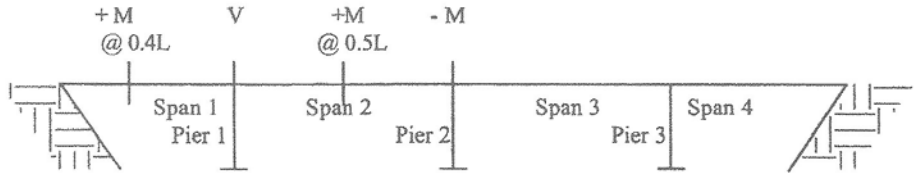
A5—FOUR-SPAN CONTINUOUS STRAIGHT WELDED PLATE GIRDER BRIDGE: EVALUATION OF AN INTERIOR GIRDER

Note: This example demonstrates the rating calculations for a straight, continuous plate girder for the design load, legal loads, and a permit load. Ratings have been performed only at critical moment and shear locations.

A5.1—Bridge Data

Span Lengths: 112 ft—140 ft—140 ft—112 ft
Year Built: 1965 (HS20 Design Load)
Noncomposite construction LRFD Design C6.10.1.6
Top flange is considered to be continuously braced by encasement in concrete Haunches
Material: F_y = 32 ksi
$f_c^{'}$ = 3 ksi
Condition: No deterioration
Riding Surface: Not field verified and documented
ADTT (one direction): 5,500
Skew: 0°



BRIDGE ELEVATION SKETCH WITH RATING LOCATIONS

Figure A5.1-1—Bridge Elevation

A5.1.1—Girder Bracing

1. Cross Frames

   Spaced at 18 ft 2 in. at piers.

   Spaced at 24 ft 4 in. elsewhere.

2. Stiffeners

   Welded vertical intermediate stiffeners at 5 ft spacing.

A5.1.2—Girder Section Properties

See Figure A5.2.1-1.

	Region	Area (in.^{2})	I (in.^{4})	S (in.^{3})
	A	54.63	42,540	1,189.9
*	B	66.63	58,038	1,606.6
	C	54.63	42,540	1,189.9
	D	74.63	68,550	1,884.6
*	E	98.63	100,965	2,719.6
	F	74.63	68,550	1,884.6
	G	54.63	42,540	1,189.9
*	H	66.63	58,038	1,606.6

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A5.1.3—Girder Sections

Region	Web Depth	Web Thickness	Top Flange Width	Top Flange Thickness	Bottom Flange Width	Bottom Flange Thickness
B	70 in.	0.4375 in.	16 in.	1.125 in.	16 in.	1.125 in.
C	70 in.	0.4375 in.	16 in.	0.75 in.	16 in.	0.75 in.
D	70 in.	0.4375 in.	16 in.	1.375 in.	16 in.	1.375 in.
E	70 in.	0.4375 in.	16 in.	2.125 in.	16 in.	2.125 in.
H	70 in.	0.4375 in.	16 in.	1.125 in.	16 in.	1.125 in.

A5.2—Dead Load Analysis—Interior Girder

Since the girders are noncomposite, all dead loads act upon the steel section.

A5.2.1—Components and Attachments, *DC*

Permanent loads on the deck are distributed uniformly among the beams.

Deck $\left(\frac{7.5}{12}\right)(7.833)(0.150) = 0.734$ kip/ft

Haunch = 0.066 kip/ft

Stay-in-place forms = 0.098 kip/ft

Average Girder Self Weight: $\left(\frac{66}{144}\right)(0.490) = 0.224$ kip/ft

Web Stiffeners = 0.011 kip/ft

Diaphragms = 0.015 kip/ft

Parapet Weight per girder = 0.310 kip/ft

Total per girder = 1.458 kip/ft

Say *DC* = 1.50 kip/ft

A5.2.2—Wearing Surface, *DW*

Overlay thickness was not field measured.

1.5 in. LMC Overlay: $\left(\frac{1.5}{12}\right)(32.7)(0.150)\times\frac{1}{5} = 0.122$ kip/ft

Say *DW* = 0.12 kip/ft

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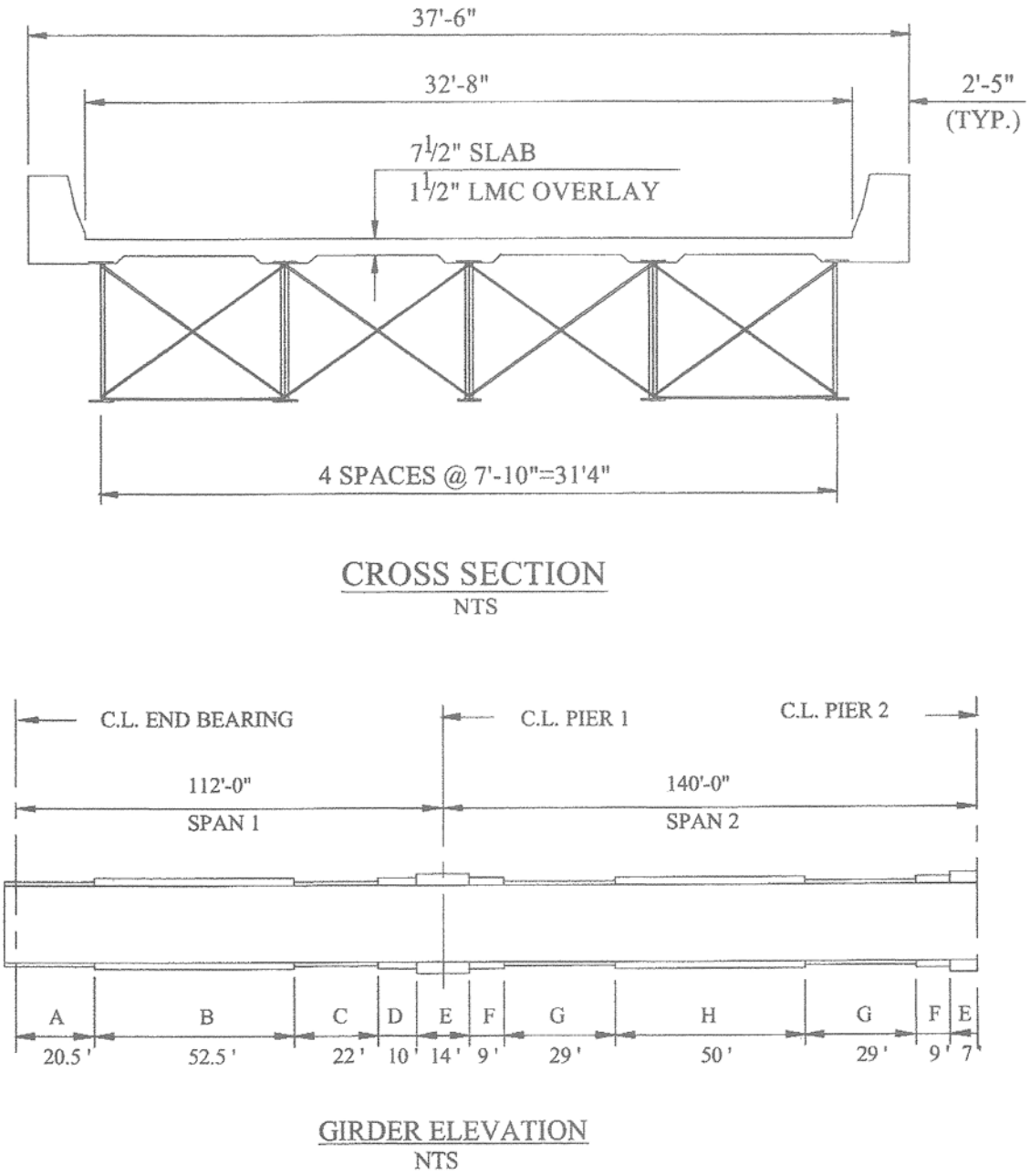

Figure A5.2.1-1—Bridge Cross-Section and Plate Girder Elevation

5.3—Dead Load Effects

Continuous beam analysis results:

A5.3.1—Maximum Positive Moment at Span 1 (at 0.4*L* = 44.8 ft)

M_{DC} = 1,236.6 kip-ft

M_{DW} = 98.9 kip-ft

A5.3.2—Maximum Positive Moment at Span 2 (at 0.5*L* = 182 ft)

M_{DC} = 1,119.8 kip-ft

M_{DW} = 89.6 kip-ft

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A5.3.3—Maximum Negative Moment at Pier 2 (252 ft)

M_{DC} = 2,558.0 kip-ft

M_{DW} = 204.6 kip-ft

A5.3.4—Maximum Shear left of Pier 1 (112 ft)

V_{DC} = –106.8 kips

V_{DW} = –8.5 kips

A5.3.5—Negative Moments at Pier 1

M_{DC} = –2,557.2 kip-ft

M_{DW} = –204.6 kip-ft

A5.4—Live Load Distribution Factors

AASHTO Type (a) cross section — LRFD Design Table 4.6.2.2.1-1

A5.4.1—Positive Flexure and Shear to the Left of Pier 1

Span 1 (same for Span 4)

$K_g = n\left(I + Ae_g^2\right)$ — LRFD Design Eq. 4.6.2.2.1-1

n = 9

For noncomposite construction, $e_g = 0$

I = 58,037.9 in.^{4} (Region B and Region H)

K_g = 9 × 58,037.9

= 522341 in.^{4}

$$\frac{K_g}{12Lt_s^3} = \frac{522341}{12\times112\left(7.5\right)^3}$$

= 0.92

Weighted Average of K_g may also be used, but distribution factor is not overly sensitive to K_g

A5.4.1.1—Interior Girder (LRFD Design Table 4.6.2.2.2b-1)

$$g_{m1} = 0.06 + \left(\frac{S}{14}\right)^{0.4}\left(\frac{S}{L}\right)^{0.3}\left(\frac{K_g}{12Lt_s^3}\right)^{0.1}$$

$$= 0.06 + \left(\frac{7.833}{14}\right)^{0.4}\left(\frac{7.833}{112}\right)^{0.3}\left(0.92\right)^{0.1}$$

$g_{m1} = 0.414$

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$$g_{m2} = 0.075 + \left(\frac{S}{9.5}\right)^{0.6}\left(\frac{S}{L}\right)^{0.2}\left(\frac{K_g}{12Lt_s^3}\right)^{0.1}$$

$$= 0.075 + \left(\frac{7.833}{9.5}\right)^{0.6}\left(\frac{7.833}{112}\right)^{0.2}(0.92)^{0.1}$$

$$= 0.594 > 0.414$$

$g_m = g_{m2} = 0.594$ For checking + M at 44.8 ft (0.4L of Span 1)

LRFD Design Table 4.6.2.2.3a-1

$$g_{V1} = 0.36 + \frac{S}{25}$$

$$= 0.36 + \frac{7.833}{25}$$

$$= 0.673$$

$$g_{V2} = 0.2 + \frac{S}{12} - \left(\frac{S}{35}\right)^{2.0}$$

$$= 0.2 + \frac{7.833}{12} - \left(\frac{7.833}{35}\right)^{2.0}$$

$$= 0.803 > 0.673$$

$g_V = g_{V2} = 0.803$ For checking V left of Pier 1 (112 ft)

Span 2 and Span 3:

Substitute $L = 126_{ave}$ ft into the distribution factor equations.

$g_m = 0.560$ For checking +M at 182 ft

(0.5L of Span 2)

A5.4.2—Negative Flexure

Use K_g based on the section properties of the Pier section.

L = 140 ft for center pier (Pier 2) as adjacent spans are both 140 ft

LRFD Design Table 4.6.2.2.1-2

L = (140 + 112)/2 = 126 ft for Pier 1

Pier 2:

$K_g = n\left(I + Ae_g^2\right)$

n = 9

e_g = 0 noncomposite section

I = 100,965.1 in.^{4} (Region E)

K_g = 9 × 100,965.1 = 908,686

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$$\frac{K_g}{12Lt_s^3} = \frac{908,686}{12(140)(7.5)^3} = 1.282$$

A5.4.2.1—Interior Girder

$$g_{m1} = 0.06 + \left(\frac{S}{14}\right)^{0.4}\left(\frac{S}{L}\right)^{0.3}\left(\frac{K_g}{12Lt_s^3}\right)^{0.1}$$

$$= 0.06 + \left(\frac{7.833}{14}\right)^{0.4}\left(\frac{7.833}{140}\right)^{0.3}(1.282)^{0.1}$$

$$g_{m1} = 0.402$$

$$g_{m2} = 0.075 + \left(\frac{S}{9.5}\right)^{0.6}\left(\frac{S}{L}\right)^{0.2}\left(\frac{K_g}{12Lt_s^3}\right)^{0.1}$$

$$= 0.075 + \left(\frac{7.833}{9.5}\right)^{0.6}\left(\frac{7.833}{140}\right)^{0.2}(1.282)^{0.1}$$

$$= 0.588 > 0.402$$

$g_m = g_{m2} = 0.588$ For checking $-M$ at Pier 2.

Pier 1:

Substitute L = 140 ft into the distribution factor equations.

g_m = 0.604 For checking $-M$ at Pier 1.

A5.5—Live Load Effects

Continuous beam analysis results are described in Articles A5.5.1 through A5.5.4 below.

A5.5.1—Maximum Positive Moment at Span 1 (at 0.4*L*)

A5.5.1.1—Design Live Load (HL-93)

Design Lane Load = 841.0 kip-ft

Design Truck = 1,404.0 kip-ft Governs

Design Tandem = 1,108.0 kip-ft

IM = 33 percent 6A.4.3.3

$M_{LL+IM} = 841.0 + 1,404.0 \times 1.33 = 2,708.3$ kip-ft

$g_m \times M_{LL+IM} = (0.594)(2,708.3) = 1,608.7$ kip-ft

A5.5.1.2—Legal Loads

Use only truck loads as span length < 200 ft

1. Type 3 = 1,011.1 kip-ft
2. Type 3S2 = 1,230.1 kip-ft

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3. Type 3-3 = 1,232.6 kip-ft Governs

IM = 33 percent 6A.4.4.3

M_{LL+IM} = $1,232.6 \times 1.33$ unknown riding surface condition = $1,639.4$ kip-ft

$g_m \times M_{LL+IM}$ = $(0.594)(1639.4) = 973.8$ kip-ft

A5.5.2—Maximum Positive Moment at Span 2 (at 0.5*L*)

A5.5.2.1—Design Live Load (HL-93)

Design Lane Load = 903.5 kip-ft

Design Truck = 1,405.2 kip-ft Governs

Design Tandem = 1,109.2 kip-ft

IM = 33 percent

M_{LL+IM} = $903.5 + 1,405.2 \times 1.33 = 2,772.4$ kip-ft

$g_m \times M_{LL+IM}$ = $(0.560)(2,772.4) = 1,552.5$ kip-ft

A5.5.2.2—Legal Loads (Use Only Truck Loads)

4. Type 3 = 1,012.8 kip-ft

5. Type 3S2 = 1,234.7 kip-ft

6. Type 3-3 = 1,259.1 kip-ft Governs

IM = 33 percent

M_{LL+IM} = $1,259.1 \times 1.33 = 1,674.6$ kip-ft

$g_m \times M_{LL+IM}$ = $(0.560)(1674.6) = 937.8$ kip-ft

A5.5.3—Maximum Negative Moment at Pier 2

Live-load analysis for negative moment and reactions at interior piers in a continuous bridge requires the consideration of an additional lane-type load model. LRFD and LRFR recognize the possibility of more than one truck in a lane causing the maximum force effect. The influence line for moment at Pier 2 is shown in Figure A5.5.3-1 along with the governing load placement for the design load case, the legal load case, and the permit load case.

Live Load Models and Placement:

Design Load 6A.4.3.2.1

Legal Load 6A.4.4.2.1

Permit Load 6A.4.5.4.1

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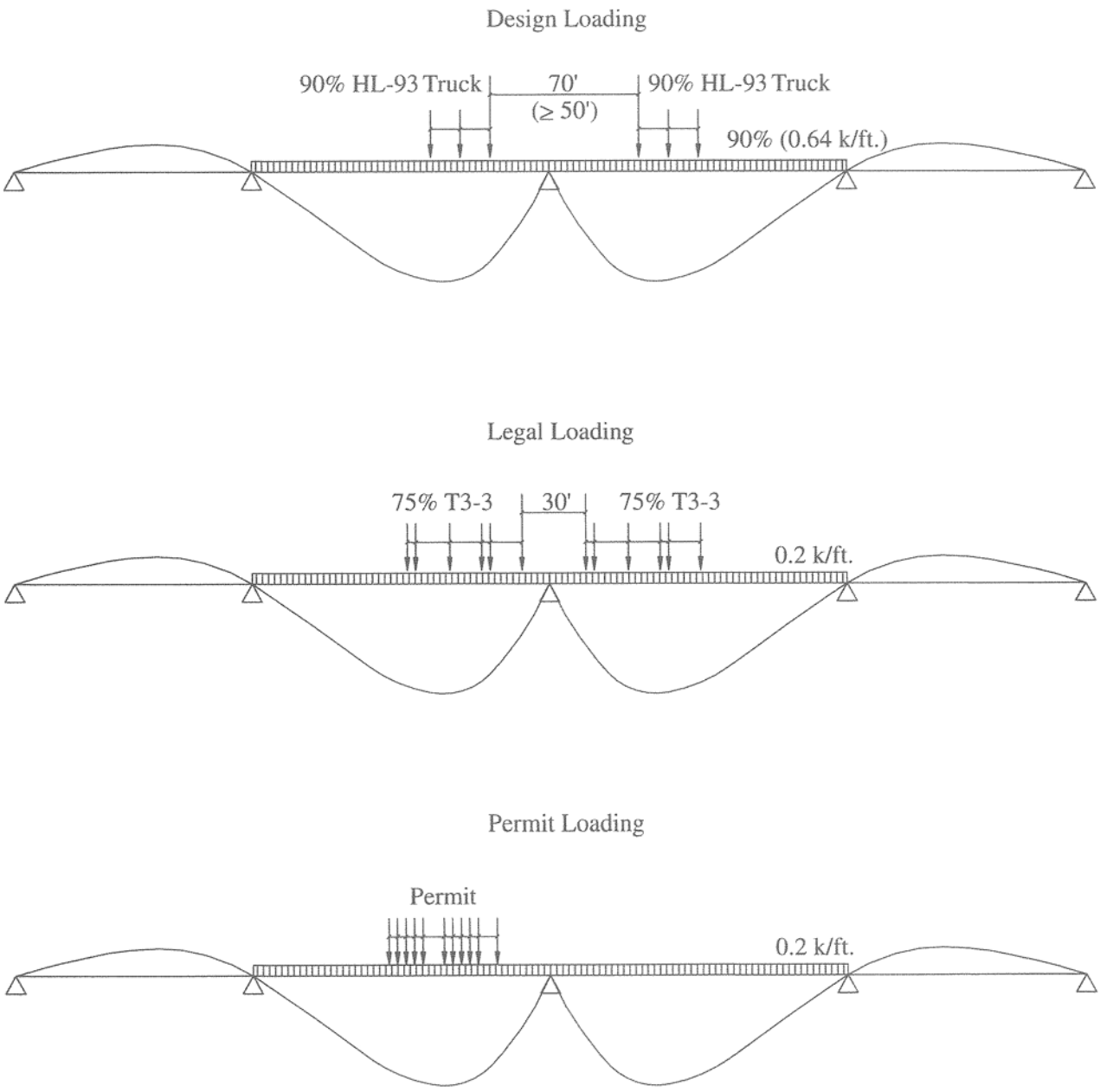

Figure A5.5.3-1—Influence Line for Moment Over Center Pier (Pier 2) with Design, Legal, and Permit Loading

(shows lane-type loading)

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A5.5.3.1—Calculate Maximum Negative Moment at Pier 2

A5.5.3.1a—Design Live Load (HL-93)

Design Lane Load = –1,388 kip-ft

Design Truck = –895.5 kip-ft

Design Tandem = –612.8 kip-ft

Double Trucks = –1,790.1 kip-ft

IM = 33 percent

Lane Load + Design Truck = $-1{,}388 - 895.5 \times 1.33 = -2{,}579$ kip-ft

Lane Load + Tandem Axles = $-1{,}388 - 612.8 \times 1.33 = -2{,}203$ kip-ft

0.9 (Lane Load + Double Trucks) = $0.9(-1{,}388 - 1{,}790.1 \times 1.33) = -3{,}392$ kip-ft
Governs

M_{LL+IM} = –3,392 kip-ft

$g_m \times M_{LL+IM}$ = $(0.588)(-3{,}392) = -1{,}994.5$ kip-ft

A5.5.3.1b—Legal Loads (Truck Loads and Lane-Type Load)

1. Type 3 = –582.0 kip-ft
2. Type 3S2 = –800.6 kip-ft
3. Type 3-3 = –858.9 kip-ft Governs
4. Lane Type Load

   Axle Loads = –1,291.0 kip-ft

   Uniform Load = –433.9 kip-ft

IM = 33 percent

is applied to axle loads only.

Type 3 = (-582.0×1.33)

= –774.1 kip-ft

Type 3S2 = (-800.6×1.33)

= –1,065 kip-ft

Type 3-3 = (-858.9×1.33)

= –1,142 kip-ft

Lane-Type Load = $(-1{,}291.0 \times 1.33) + (-433.9)$

= –2,150.9 kip-ft Governs

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M_{LL+IM} = –2,150.9 kip-ft

$g_m \times M_{LL+IM}$ = (0.588)(–2,150.9)

= 1,264.7 kip-ft

Table A5.5.3.1b-1—Girder Bending Stresses at Critical Sections

Location	S (in.^{3})	Live Load	$g_m M_{LL+IM}$ (kip-ft)	f_{LL+IM} (ksi)	M_{DC} (kip-ft)	M_{DW} (kip-ft)	f_{DC} (ksi)	f_{DW} (ksi)
Span 1 at 0.4L	1,606.6	HL-93 Legal Load	1,608.7 973.8	12.02 7.27	1,236.6	98.9	9.24	0.74
Pier 2	2,719.6	HL-93 Legal Load	–1,994.5 –1,264.7	–8.80 –5.58	–2,558.0	–204.6	–11.29	–0.90
Span 2 at 0.5L	1,606.6	HL-93 Legal Load	1,552.5 937.8	11.60 7.00	1,119.8	89.6	8.36	0.67

A5.5.4—Maximum Shear at Pier 1 (Left of Support)

A5.5.4.1—Design Live Load (HL-93)

Design Lane Load = –53.9 kips

Design Truck = –68.3 kips Governs

Design Tandem = –49.5 kips

IM = 33 percent

= –53.9 – 68.3 × 1.33

V_{LL+IM} = –144.7 kips

$g_v \times V_{LL+IM}$ = (0.803)(–144.7)

= –116.2 kips

A5.5.4.2—Legal Loads

1. Type 3 = –48.0 kips 6A.4.4.2.1a
2. Type 3S2 = –63.9 kips
3. Type 3-3 = –67.7 kips Governs

Note: Lane-type load is not required when checking shear.

IM = 33 percent

V_{LL+IM} = (–67.7)(1.33)

= –90.0 kips

$g_v \times V_{LL+IM}$ = (0.803)(–90.0)

= –72.3 kips

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A5.6—Compute Nominal Flexural Resistance of Section (Positive and Negative Moment)

A5.6.1—Noncomposite Symmetric Section

A5.6.1.1—Check Web for Noncompact Slenderness Limit

$$\frac{2D_c}{t_w} < 5.7\sqrt{\frac{E}{F_{yc}}}$$ LRFD Design Eq. 6.10.6.2.3-1

$$\frac{2D_c}{t_w} = \frac{70}{0.4375} = 160$$

$$5.7\sqrt{\frac{E}{F_{yc}}} = 5.7\sqrt{\frac{29{,}000}{32}} = 171.6 > \frac{2D_c}{t_w}$$

And check that flanges satisfy the ratio:

$$\frac{I_{yc}}{I_{yt}} \geq 0.3$$ LRFD Design Eq. 6.10.6.2.3-2

in this case:

$$\frac{I_{yc}}{I_{yt}} = 1.0 \geq 0.3$$

Because the bridge is straight and F_y of the flanges does not exceed 70 ksi, the optional provisions of LRFD Design Appendix A may be applied to determine the nominal flexural resistance of noncomposite sections. LRFD Design C6.10.6.2.3

A5.6.2—Regions B and H—Positive Moment Sections with Continuously Braced Compression Flanges

$M_u \leq \varphi_f R_{pc} M_{yc}$ where R_{pc} = Web Plastification Factor LRFD Design Eq. A6.1.3-1

For rating $R_n = R_{pc}M_{yc}$

Noncomposite sections that satisfy the following shall qualify as compact web sections:

$$\frac{2D_{cp}}{t_w} \leq \lambda_{pw(D_{cp})}$$ LRFD Design Eq. A6.2.1-1

$$\frac{2D_{cp}}{t_w} = 160$$

$$\lambda_{pw(D_{cp})} = \frac{\sqrt{\frac{E}{F_{yc}}}}{\left[0.54\frac{M_p}{R_h M_y} - 0.09\right]^2} \leq \lambda_{rw}\left(\frac{D_{cp}}{D_c}\right)$$ LRFD Design Eq. A6.2.1-2

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where: LRFD Design Eq. A6.2.1-3

$$\lambda_{rw}\left(\frac{D_{cp}}{D_c}\right)=5.7\sqrt{\frac{29{,}000}{32}}\left(\frac{35}{35}\right)=171.59$$

A5.6.2.1—Calculate Plastic Moment, M_p (LRFD Design D6.1)

Top flange:

P_c = 16 in. × 1.125 in. × 32 ksi = 576 kips

Bottom flange:

P_t = 16 in. × 1.125 in. × 32 ksi = 576 kips

Web:

P_w = 70 in. × 0.4375 in. × 32 ksi = 980 kips

d_t = $d_c=\frac{70}{2}+\frac{1.125}{2}=35.56$ in.

D = 70 in.

Referring to LRFD Design Appendix D6.1, Table D6.1-1, Case I:

$\bar{y}$ = $\frac{D}{2}=35$ in.

M_p = $\frac{P_w}{2D}\left[\bar{y}^2+(D-\bar{y})^2\right]+(P_c d_c+P_t d_t)$

= $\frac{980}{2\times 70}\left[35^2+(70-35)^2\right]+2\times 576\times 35.56$

= $(17{,}150+40{,}965.1)\times\frac{1}{12\ \text{in./ft}}$

= 4,842.9 kip-ft

M_y = $F_y S$ LRFD Design D6.2.1

= $32\times 1{,}606.6\times\frac{1}{12}$

= 4,284.3 kip-ft

R_h = 1.0

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$$\lambda_{pw(D_{cp})} = \frac{\sqrt{\frac{29{,}000}{32}}}{\left[0.54\times\frac{4{,}842.9}{1.0\times 4{,}284.3}-0.09\right]^2}$$

$$= \quad 111.16 < \frac{2D_{cp}}{t_w} = 160$$

Therefore, the web section is not compact.

Check if section satisfies the requirements for noncompact web sections. LRFD Design A6.2.2

$\lambda_w < \lambda_{rw}$ LRFD Design Eq. A6.2.2-1

$\lambda_w = \frac{2D_c}{t_w} = 160$ LRFD Design Eq. A6.2.2-2

$\lambda_{rw} = 5.7\sqrt{\frac{E}{F_{yc}}} = 171.6$ LRFD Design Eq. A6.2.2-3

$\lambda_w = 160 < \lambda_{rw} = 171.6$

Therefore, the section qualifies as a noncompact web section.

R_{pc} shall be taken as:

$$R_{pc} = \left[1-\left(1-\frac{R_h M_{yc}}{M_p}\right)\left(\frac{\lambda_w-\lambda_{pw(D_c)}}{\lambda_{rw}-\lambda_{pw(D_c)}}\right)\right]\frac{M_p}{M_{yc}} \le \frac{M_p}{M_{yc}}$$ LRFD Design Eq. A6.2.2-4

$$\lambda_{pw(D_c)} = \lambda_{pw(D_{cp})}\left(\frac{D_c}{D_{cp}}\right) \le \lambda_{rw}$$ LRFD Design Eq. A6.2.2-6

$$111.16\left(\frac{35}{35}\right) = 111.16 \le 171.6$$

$$= \left[1-\left(1-\frac{1.0\times 4{,}284.3}{4{,}842.9}\right)\left(\frac{160-111.16}{171.6-111.16}\right)\right]\frac{4{,}842.9}{4{,}284.3}$$

$$= 0.9068\frac{M_p}{M_{yc}} \le \frac{M_p}{M_{yc}}$$

$$= 1.025 \le 1.13$$

$R_{pc} = 1.025$

$M_n = R_{pc} M_{yc} = 1.025 \times 4{,}284.3 \text{ kip-ft} = 4{,}391.5 \text{ kip-ft}$

Because f_ℓ is equal to zero in this case and M_{yc} is equal to M_{yt}, the flexural resistance based on the discretely braced tension flange at this section does not control and need not be checked (LRFD Design CA6.1.2).

$\therefore R_n = M_n = 4{,}391.5 \text{ kip-ft}$

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A5.6.3—Region E—Negative Moment Sections with Discretely Braced Compression Flange (LRFD Design A6.1.1)

$$M_u + \frac{1}{3} f_\ell S_{xc} \le \phi M_{nc}$$ LRFD Design Eq. A6.1.1-1

For rating:

$$R_n = M_{nc} - \frac{1}{3} f_\ell S_{xc}$$

where:

M_{nc} = nominal flexural resistance specified in LRFD Design Appendix A6.3 and based on the compression flange. M_{nc} is to be determined as the smaller of the local buckling resistance and the lateral torsional buckling resistance.

A5.6.3.1—Calculate Local Buckling Resistance (LRFD Design A6.3.2)

$$\lambda_f = \frac{b_{fc}}{2t_{fc}}$$ LRFD Design Eq. A6.3.2-3

$$= \frac{16}{2 \times 2.125} = 3.76$$

$$\lambda_{pf} = 0.38\sqrt{\frac{E}{F_{yc}}}$$ LRFD Design Eq. A6.3.2-4

$$= 0.38\sqrt{\frac{29{,}000}{32}}$$

$$= 11.4 > \lambda_f$$

As $\lambda_f \le \lambda_{pf}$, then: LRFD Design Eq. A6.3.2-1

$$M_{nc} = R_{pc} M_{yc}$$

Recalculating $M_n = R_{pc}M_{yc}$ for Region E:

$$M_{yc} = F_y S$$

$$= \frac{32 \text{ ksi} \times 2{,}719.6 \text{ in.}^3}{12 \text{ in./ft}}$$

$$= 7{,}252.3 \text{ kip-ft}$$

P_c = 16 in. × 2.125 in. × 32 ksi = 1,088 kips

$P_t = P_c$ = 1,088 kips

P_w = 980 kips

$d_t = d_c = \frac{70}{2} + \frac{2.125}{2}$ = 36.06 in.

D = 70 in.

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$\bar{y}$ = 35 in.

$$M_p = \frac{980}{2\times 70}\left[35^2+(70-35)^2\right]+2\times 1{,}088\times 36.06$$

= 17,150 + 78,467

= 95,617 kip-in.

= 7,968 kip-ft

Then:

$$R_{pc} = \left[1-\left(1-\frac{R_h M_{yc}}{M_p}\right)\left(\frac{\lambda_w-\lambda_{pw(D_c)}}{\lambda_{rw}-\lambda_{pw(D_c)}}\right)\right]\frac{M_p}{M_{yc}} \le \frac{M_p}{M_{yc}}$$

where:

$$\lambda_{pw(D_c)} = \lambda_{pw(D_{cp})}\left[\frac{D_c}{D_{cp}}\right] \le \lambda_{rw}$$

$$\frac{\sqrt{\frac{29{,}000\,\text{ksi}}{32\,\text{ksi}}}}{\left(0.54\times\frac{7{,}968\,\text{kip-ft}}{1.0\times 7{,}252.3}-0.09\right)^2}\times\frac{35}{35}=118.85\le\lambda_{rw}$$

λ_{rw} = 171.6 λ_w = 160

$$R_{pc} = \left[1-\left(1-\frac{1.0\times 7{,}252.3}{7{,}968}\right)\left(\frac{160-118.85}{171.6-118.85}\right)\right]\frac{7{,}968}{7{,}252.3}\le\frac{7{,}968}{7{,}252.3}$$

= 0.9299 × 1.0987

= 1.0217 ≤ 1.0987

M_{nc} = 1.0217 × 7,252.3 = 7,409.7 kip-ft

A5.6.3.2—Calculate Lateral Torsional Buckling Resistance (LRFD Design A6.3.3)

L_b = Unbraced length = 18 ft 2 in. = 218 in.

In this example, the unbraced length encompasses three cross-section regions, C, D, and E (or E, F, and G). LRFD Design Article C6.10.8.2.3 states that for unbraced lengths containing one or more transitions, only transitions located within 20 percent of the unbraced length from the brace point with the smaller moment may be ignored and the lateral torsional buckling resistance of the remaining nonprismatic unbraced length may be computed as the smallest resistance based on the remaining sections. Because only the transition between Regions C and D is located within 20 percent of the unbraced length from the brace point with the smaller moment, that particular transition may be ignored. The lateral torsional buckling must be computed based on the section in Region D.

Determine L_p and L_r for Section D:

$b_{fc} = b_{ft}$ = 16 in., $t_{fc} = t_{ft}$ = 1.375 in., web depth D = 70 in. t_w = 0.4375 in.

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Calculate effective radius of gyration r_t:

$$r_t = \frac{b_{fc}}{\sqrt{12\left(1+\frac{1}{3}\frac{D_c t_w}{b_{fc} t_{fc}}\right)}}$$ LRFD Design Eq. A6.3.3-10

$$= \frac{16}{\sqrt{12\left(1+\frac{1}{3}\times\frac{35\times 0.4375}{16\times 1.375}\right)}}$$

$$= 4.16 \text{ in.}$$

$$L_p = 1.0 r_t \sqrt{\frac{E}{F_{yc}}}$$ LRFD Design Eq. A6.3.3-4

$$L_p = 1.0\times 4.16\sqrt{\frac{29{,}000}{32}} = 125.23 \text{ in.}$$

Calculate St. Venant torsional constant J

$$J = \frac{D t_w^3}{3} + \frac{b_{fc} t_{fc}^3}{3}\left(1-0.63\frac{t_{fc}}{b_{fc}}\right) + \frac{b_{ft} t_{ft}^3}{3}\left(1-0.63\frac{t_{ft}}{b_{ft}}\right)$$ LRFD Design Eq. A6.3.3-9

$$= \frac{70\times 0.4375^3}{3} + \frac{16\times 1.375^3}{3}\left(1-0.63\times\frac{1.375}{16}\right) + \frac{16\times 1.375^3}{3}\left(1-0.63\times\frac{1.375}{16}\right)$$

$$= 28.18 \text{ in.}^4$$

Depth between centerline of flanges, h = 70 in. + 1.375 in. = 71.375 in.

Calculate F_{yr} in order to compute L_r, where F_{yr} is the smaller of: LRFD Design A6.3.3

$0.7F_{yc} = 0.7\times 32$ ksi = 22.4 ksi

and:

$$R_h F_{yt}\frac{S_{xt}}{S_{xc}} = 1.0\times 32 \text{ ksi}\times\frac{1{,}884.6 \text{ in.}^3}{1{,}884.6 \text{ in.}^3} = 32 \text{ ksi}$$

but not less than $0.5F_{yc} = 0.5\times 32$ ksi = 16 ksi

Therefore, 22.4 ksi governs.

$$L_r = 1.95 r_t \frac{E}{F_{yr}}\sqrt{\frac{J}{S_{xc} h}}\sqrt{1+\sqrt{1+6.76\left(\frac{F_{yr}}{E}\times\frac{S_{xc} h}{J}\right)^2}}$$ LRFD Design Eq. A6.3.3-5

$$= 1.95\times 4.16\times\frac{29{,}000}{22.4}\sqrt{\frac{28.18}{1{,}884.6\times 71.375}}\sqrt{1+\sqrt{1+6.76\left(\frac{22.4}{29{,}000}\cdot\frac{1{,}884.6\times 71.375}{28.18}\right)^2}}$$

$$= 495.8 \text{ in.}$$

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The moment gradient modifier C_b can be taken equal to 1.0 in this case according to LRFD Design Article A6.3.3.

Note: If all transitions had been located within 20 percent of the unbraced length from the brace point with the smaller moment, C_b would not have to be taken equal to 1.0 and the lateral torsional buckling resistance could be based on the larger flange. Under those circumstances, C_b should be calculated because the results would lead to a larger rating.

Determine R_{pc} in accordance with LRFD Design Articles A6.2.1 or A6.2.2 as applicable and determine M_{yc}:

Top flange P_c = 16 in. × 1.375 in. × 32 ksi = 704 kips

Bottom flange P_t = 16 in. × 1.375 in. × 32 ksi = 704 kips

Web P_w = 70 in. × 0.4375 in. × 32 ksi = 980 kips

$d_t = d_c = \frac{70}{2} + \frac{1.375}{2} = 36.375$ in.

D = 70 in.

Referring to LRFD Design Appendix D6.1, Table D6.1-1, Case I:

$$\bar{y} = \frac{D}{2} = 35 \text{ in.}$$

$$M_p = \frac{P_w}{2D}\left[\bar{y}^2 + (D-\bar{y})^2\right] + (P_c d_c + P_t d_t)$$

$$= \frac{980}{2\times 70}\left[35^2 + (70-35)^2\right] + 2\times 704 \times 36.375$$

$$= (17{,}150 + 51{,}216) \times \frac{1}{12 \text{ in./ft}}$$

$$= 5{,}697.2 \text{ kip-ft}$$

$M_y = F_y S$ LRFD Design D6.2.1

$$= 32 \times 1{,}884.6 \times \frac{1}{12}$$

$$= 5{,}025.6 \text{ kip-ft}$$

R_h = 1.0

$$\lambda_{pw(D_{cp})} = \frac{\sqrt{\frac{29{,}000}{32}}}{\left[0.54 \times \frac{5{,}697.2}{1.0 \times 5{,}025.6} - 0.09\right]^2} = 110.4$$

$$\frac{2D_{cp}}{t_w} = \frac{2\times 35}{0.4375} = 160$$

$$= 110.4 < 160$$

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Therefore, the web section is not compact.

Check if section satisfies the requirements for noncompact web sections: LRFD Design A6.2.2

$$\lambda_w < \lambda_{rw}$$ LRFD Design Eq. A6.2.2-1

$$\lambda_w = \frac{2D_c}{t_w} = 160$$ LRFD Design Eq. A6.2.2-2

$$\lambda_{rw} = 5.7\sqrt{\frac{E}{F_{yc}}} = 171.6$$ LRFD Design Eq. A6.2.2-3

$$\lambda_w = 160 < \lambda_{rw} = 171.6$$

R_{pc} shall be taken as:

$$R_{pc} = \left[1-\left(1-\frac{R_h M_{yc}}{M_p}\right)\left(\frac{\lambda_w-\lambda_{pw(D_c)}}{\lambda_{rw}-\lambda_{pw(D_c)}}\right)\right]\frac{M_p}{M_{yc}} \le \frac{M_p}{M_{yc}}$$ LRFD Design Eq. A6.2.2-4

$$\lambda_{pw(D_c)} = \lambda_{pw(D_{cp})}\left(\frac{D_c}{D_{cp}}\right) \le \lambda_{rw}$$ LRFD Design Eq. A6.2.2-6

$$110.4\left(\frac{35}{35}\right) = 110.4 \le 171.6$$

$$= \left[1-\left(1-\frac{1.0\times 5,025.6}{5,697.2}\right)\left(\frac{160-110.4}{171.6-110.4}\right)\right]\frac{5,697.2}{5,025.6}$$

$$= 0.9045\frac{M_p}{M_{yc}} \le \frac{M_p}{M_{yc}}$$

$$= 1.025 \le 1.13$$

R_{pc} = 1.025 and M_{yc} = 5,025.6 kip-ft = 60,307 kip-in.

Then:

$$M_{nc}\,(\text{for Region D}) = C_b\left[1-\left[1-\frac{F_{yr}S_{xc}}{R_{pc}M_{yc}}\right]\left[\frac{L_b-L_p}{L_r-L_p}\right]\right]R_{pc}M_{yc} \le R_{pc}M_{yc}$$

where F_{yr} was previously determined by $0.7F_{yc}$ = 0.7×32 ksi = 22.4 ksi LRFD Design A6.3.3

$$M_{nc} = 1.0\left[1-\left[1-\frac{22.4\,\text{ksi}\times 1,884.6\text{ in.}^3}{1.025\times 60,307\text{ kip-in.}}\right]\left[\frac{218\text{ in.}-125.23\text{ in.}}{495.8\text{ in.}-125.23\text{ in.}}\right]\right]R_{pc}M_{yc} \le R_{pc}M_{yc}$$

$$M_{nc} = 1.0\times\left[1-(1-0.6829)(0.2503)\right]\times 1.025\times 5025.6\text{ kip-ft} = 4742.4\text{ kip-ft}$$

$$M_{nc(pier)} = M_{nc}\times\frac{S_{xc(Region\,E)}}{S_{xc(Region\,D)}} = 4,742.4\times\frac{2,719.6}{1,884.6} = 6,843.6\text{ kip-ft}$$

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$6{,}843.6 \text{ kip-ft} \le M_{nc}$ for local buckling $= 7{,}409.7$ kip-ft

Because M_{yc} is equal to M_{yt}, the flexural resistance based on the continuously braced tension flange at this section does not control and need not be checked.

Therefore, $R_n = M_{nc(pier)} = 6{,}943.6$ kip-ft

A5.7—General Load Rating Equation (6A.4.2)

$$RF = \frac{C - (\gamma_{DC})(DC) - (\gamma_{DW})(DW) \pm (\gamma_P)(P)}{(\gamma_L)(LL + IM)}$$ Eq. 6A.4.2.1-1

Evaluation Factors (for Strength Limit State)

- Resistance Factor, φ LRFD Design 6.5.4.2

φ = 1.0 for flexure and shear

- Condition Factor, φ_c 6A.4.2.3

φ_c = 1.0 No deterioration

- System Factor, φ_s 6A.4.2.4

φ_s = 1.0 Multi-girder bridge

A5.8—Design Load Rating 6A.4.3

A5.8.1—Strength I Limit State 6A.6.4.1

$$RF = \frac{(\varphi_c)(\varphi_s)(\varphi) R_n - (\gamma_{DC})(DC) - (\gamma_{DW})(DW)}{(\gamma_L)(LL + IM)}$$

A5.8.1.1—Flexure at Span 1, 0.4L

$$\text{Inventory } RF = \frac{(1.0)(1.0)(1.0)(4{,}391.5) - (1.25)(1{,}236.6) - (1.5)(98.9)}{(1.75)(1{,}608.7)}$$

$= 0.96$ Governs

$$\text{Operating } RF = 0.96 \times \frac{1.75}{1.35}$$

$= 1.24$ Governs

A5.8.1.2—Flexure at Span 2, 0.5L

$$\text{Inventory } RF = \frac{(1.0)(1.0)(1.0)(4{,}391.5) - (1.25)(1{,}119.8) - (1.5)(89.6)}{(1.75)(1{,}552.5)}$$

$= 1.05$

$$\text{Operating } RF = 1.05 \times \frac{1.75}{1.35}$$

$= 1.36$

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A5.8.1.3—Flexure at Pier 2

$$\text{Inventory } RF = \frac{(1.0)(1.0)(1.0)(6,943.7)-(1.25)(2,558.0)-(1.5)(204.6)}{(1.75)(1,994.5)}$$

$$= 0.99$$

$$\text{Operating } RF = 0.99\times\frac{1.75}{1.35}$$

$$= 1.28$$

A5.8.2—Service II Limit State (6A.6.4.1)

Calculated for illustration; does not govern for noncomposite, noncompact sections as discussed later.

For Service Limit States, $C = f_R$ 6A.4.2.1

$$RF = \frac{f_R-(\gamma_D)(f_D)}{(\gamma_L)(f_{LL+IM})}$$

f_R = $0.80R_hF_{yf}$ for noncomposite sections LRFD Design 6.10.4.2.2

R_h was previously determined to be 1.0

f_R = $0.80 \times 1.0 \times 32$

= 25.6 ksi

γ_D = $\gamma_{DC} = \gamma_{DW} = 1.0$ Table 6A.4.2.2-1

γ_L = 1.3 for Inventory

= 1.0 for Operating

A5.8.2.1—At Span 1, 0.4L

$$\text{Inventory } RF = \frac{25.6-(1.0)(9.24+0.74)}{(1.3)(12.02)} = 1.00$$

$$\text{Operating } RF = 1.00\times\frac{1.30}{1.00} = 1.30$$

A5.8.2.2—At Span 2, 0.5L

$$\text{Inventory } RF = \frac{25.6-(1.0)(8.36+0.67)}{(1.3)(11.60)} = 1.10$$

$$\text{Operating } RF = 1.10\times\frac{1.30}{1.00} = 1.43$$

A5.8.2.3—At Pier 2

$$\text{Inventory } RF = \frac{25.6-(1.0)(11.29+0.90)}{(1.3)(8.80)} = 1.17$$

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Operating RF = $1.17 \times \frac{1.30}{1.00} = 1.52$

As seen here the Strength I rating factors govern over the corresponding Service II rating factors. This is a true statement for all noncomposite, noncompact steel beams. During normal ratings the Service II rating factors do not need to be calculated for the Design Load Rating when the steel beam is both noncomposite and noncompact. This is true in both LRFD and LRFR.

A5.8.3—Legal Load Rating (6A.4.4)

The Design Load Ratings at the inventory level were not all > 1.0. The Design Load Ratings at operating level were all > 1.0. If a state (or Owner) allows legal vehicles that exceed the AASHTO legal loads, then load ratings with the state legal vehicles will be necessary. Legal Load Ratings using the AASHTO legal loads are demonstrated for illustration.

Type 3-3 is governed for the positive moment locations and the Lane-Type Loading is governed for the negative moment location. The rating factors will be demonstrated using only the governing loadings. (See Table A5.5.3.1b-1 for girder bending stresses.)

A5.8.3.1—Strength I Limit State (6A.6.4.2.1)

$ADTT$ = 5,500 Table 6A.4.4.2.3a-1

γ_L = 1.45

A5.8.3.1a—Flexure at Span 1, 0.4L

Type 3-3 + $g_m M_{LL+IM}$ = 973.8 kip-ft

$$RF = \frac{(1.0)(1.0)(1.0)(4,391.5) - (1.25)(1,236.6) - (1.5)(98.9)}{(1.45)(973.8)}$$

= 1.91

A5.8.3.1b—Flexure at Span 2, 0.5L

Type 3-3 + $g_m M_{LL+IM}$ = 937.8 kip-ft

$$RF = \frac{(1.0)(1.0)(1.0)(4,391.5) - (1.25)(1,119.8) - (1.5)(89.6)}{(1.45)(937.8)}$$

= 2.10

A5.8.3.1c—Flexure at Pier 2

Lane-Type Load – $g_m M_{LL+IM}$ = 1,264.7 kip-ft

$$RF = \frac{(1.0)(1.0)(1.0)(6,943.6) - (1.25)(2,558.0) - (1.5)(204.6)}{(1.45)(1,264.7)}$$

= 1.88 Governs

A5.8.3.2—Service II Limit State (6A.6.4.2.2)

$f_R = 0.80 R_h F_{yf}$ for noncomposite sections

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R_h was previously determined to be 1.0 LRFD Design 6.10.4.2.2

f_R = 0.80 × 1.0 × 32

= 25.6 ksi

γ_D = $\gamma_{DC} = \gamma_{DW} = 1.0$ Table 6A.4.2.2-1

γ_{LL} = 1.3

A5.8.3.2a—At Span 1, 0.4L (Type 3-3 Truck Governs)

$$RF = \frac{25.6 - (1.0)(9.24 + 0.74)}{(1.3)(7.27)}$$

= 1.65

A5.8.3.2b—At Span 2, 0.5L (Type 3-3 Truck Governs)

$$RF = \frac{25.6 - (1.0)(8.36 + 0.67)}{(1.3)(7.00)}$$

= 1.82

A5.8.3.2c—At Pier 2 (Lane-Type Load Governs)

$$RF = \frac{25.6 - (1.0)(11.29 + 0.90)}{(1.3)(5.58)}$$

= 1.85

A5.9—Shear Evaluation 6A.6.10

Maximum shear at Pier 1 (see previous calculations):

V_{DC} = 106.8 kips

V_{DW} = 8.5 kips

$g_v V_{LL+IM}$ = 116.2 kips (HL-93)

$g_v V_{LL+IM}$ = 72.3 kips (Type 3-3)

A5.9.1—Shear Resistance at Pier 1

Spacing of vertical stiffeners = 5 ft c/c

Web depth:

D = 70 in. = 5.83 ft

$3D$ = 3 × 70 in. = 210 in. = 17.5 ft

As transverse stiffener spacing is less than $3D$, the interior web panels are considered stiffened. LRFD Design 6.10.9

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A5.9.2—Shear Resistance for Interior Panel

Check: LRFD Design Eq. 6.10.9.3.2-1

$$\frac{2Dt_w}{(b_{fc}t_{fc}+b_{ft}t_{ft})} \le 2.5$$

$$\frac{2Dt_w}{(b_{fc}t_{fc}+b_{ft}t_{ft})} = \frac{2\times 70\times 0.4375}{(16\times 2.125+16\times 2.125)} = 0.9 \le 2.5$$

Then:

LRFD Design Eq. 6.10.9.3.2-2

$$V_n = V_p\left[C+\frac{0.87(1-C)}{\sqrt{1+\left(\frac{d_o}{D}\right)^2}}\right]$$

LRFD Design Eq. 6.10.9.3.2-3

$$V_n = 0.58F_{yw}Dt_w$$

$$= 0.58 \times 32 \times 70 \times 0.4375$$

$$= 568.4 \text{ kips}$$

Determine C:

LRFD Design Eq. 6.10.9.3.2-7

$$k = 5+\frac{5}{\left(\frac{d_o}{D}\right)^2}$$

where d_o = stiffener spacing = 60 in.

$$k = 5+\frac{5}{\left(\frac{60}{70}\right)^2} = 11.81$$

If: LRFD Design Eq. 6.10.9.3.2-4

$$\frac{D}{t_w} < 1.12\sqrt{\frac{Ek}{F_{yw}}}$$

then:

$$C = 1.0$$

$$\frac{D}{t_w} = \frac{70}{0.4375} = 160$$

$$1.12\sqrt{\frac{Ek}{F_{yw}}} = 1.12\sqrt{\frac{29{,}000\times 11.81}{32}} = 115.9$$

160 > 115.9 FAIL

If: LRFD Design Eq. 6.10.9.3.2-5

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$$1.12\sqrt{\frac{Ek}{F_{yw}}} \quad \leq \quad \frac{D}{t_w} \leq 1.40\sqrt{\frac{Ek}{F_{yw}}}$$

then:

$$C = \frac{1.12}{\left(\frac{D}{t_w}\right)^2}\left(\frac{Ek}{F_{yw}}\right)$$

$$1.40\sqrt{\frac{Ek}{F_{yw}}} = 144.9$$

$160 > 144.9$ FAIL

If: LRFD Design Eq. 6.10.9.3.2-6

$$\frac{D}{t_w} > 1.40\sqrt{\frac{Ek}{F_{yw}}} \quad \text{TRUE}$$

then:

$$C = \frac{1.57}{\left(\frac{D}{t_w}\right)^2}\left(\frac{Ek}{F_{yw}}\right)$$

$$C = \frac{1.57}{160^2}\left(\frac{29{,}000 \times 11.81}{32}\right) = 0.656$$

$$V_n = V_p\left[C + \frac{0.87(1-C)}{\sqrt{1+\left(\frac{d_o}{D}\right)^2}}\right]$$

$$= 568.4\left[0.656 + \frac{0.87(1-0.656)}{\sqrt{1+\left(\frac{60}{70}\right)^2}}\right]$$

$$= 502.0 \text{ kips}$$

$$V_r = \varphi_v V_n$$

$$= 1.0 \times 502.0 = 502.0 \text{ kips}$$

A5.10—Shear Rating at Pier 1

$\varphi_v = 1.00$ LRFD Design 6.5.4.2

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φ_c = 1.00 6A.4.2.3

φ_s = 1.00 6A.4.2.4

A5.10.1—Design Load Rating 6A.4.3

Strength I Limit State: 6A.6.4.1

Inventory Shear:

$$RF = \frac{(1.0)(1.0)(1.0)(502.0)-(1.25)(106.8)-(1.50)(8.5)}{(1.75)(116.2)}$$

= 1.75

Operating Shear:

$$RF = 1.75\times\frac{1.75}{1.35} = 2.27$$

using same R as inventory.

A5.10.2—Legal Load Rating (Type 3-3 Governs) 6A.4.4

Strength I Limit State: 6A.6.4.2.1

Shear:

$$RF = \frac{(1.0)(1.0)(1.0)(502.0)-(1.25)(106.8)-(1.50)(8.5)}{(1.45)(72.3)}$$

= 3.39

Using same shear resistance as for HL-93.

Note: R could be recalculated for legal loads resulting in a higher resistance and rating.

A5.10.3—Permit Load Rating (6A.4.5)

Permit Type: Routine 6A.4.5.2
Legal Load $RF > 1.0$ ∴ bridge may be evaluated for permits
Permit Weight: 220 kips

The permit vehicle is shown in Example A1A, Figure A1A.1.10-1

ADTT (one direction): 5,500

Routine Permit, Unlimited Crossings

$$\frac{GVW}{AL} = \frac{220}{51} = 4.3 > 3.0$$
$$\therefore \gamma_L = 1.3$$

Strength II Limit State: 6A.6.4.2

Load Factor, γ_L = 1.30 Table 6A.4.5.4.2a-1

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IM = 33 percent (riding surface condition is unknown) 6A.4.5.5

Use the Multi-Lane Loaded Live Load Distribution Factors. 6A.4.5.4.2a

Span 1: $+M$ g_m = 0.594

Span 2: $+M$ g_m = 0.560

Pier 2: $-M$ g_m = 0.588

Pier 1: Max V g_v = 0.803

g_m = 0.604

6A.4.5.4.1

		Permit	Lane Load, 0.2 kip/ft
Max $+M$	Span 1	3,775.3	NA
Max $+M$	Span 2	3,884.8	NA
–M at	Pier 2	2,621.8	433.9 No *IM* for lane load
Max V left of	Pier 1	190.6	NA

Distributed Load Effects with *IM*:

Span 1: $+M_{LL+IM}$ = (3,775.3)(1.33)(0.594) = 2,982.6 kip-ft

Span 2: $+M_{LL+IM}$ = (3,884.8)(1.33)(0.560) = 2,893.4 kip-ft

Pier 2: $-M_{LL+IM}$ = [(2,621.8)(1.33) + 433.0](0.588) = 2,305.0 kip-ft

Pier 1: V_{LL+IM} = (190.6)(1.33)(0.803) = 203.6 kips

Flexure	S, in.^{3}	$g_m M_{LL+IM}$, kip-ft	f_{LL+IM}, ksi	f_{DC}, ksi	f_{DW}, ksi
Span 1 at 0.4L	1,606.6	2,982.6	22.3	9.24	0.74
Pier 2	2,719.6	2,305.0	10.2	11.29	0.90
Span 2 at 0.5L	1,606.6	2,893.4	21.6	8.36	0.67

The nominal flexure resistance of each section was previously determined. See Article A5.6.

For positive moment Regions B and H, M_n = 4,391.5 kip-ft

For negative moment Region E, M_{nc}= 6,943.6 kip-ft

Flexural Rating Factors

A5.10.3.1—Flexure at Span 1, 0.4L

$$RF = \frac{(1.0)(1.0)(1.0)(4{,}391.5)-(1.25)(1{,}236.6)-(1.5)(98.9)}{(1.3)(2{,}982.6)}$$

= 0.70 < 1.0 Governs

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A5.10.3.2—Flexure at Span 2, 0.5L

$$RF = \frac{(1.0)(1.0)(1.0)(4{,}391.5)-(1.25)(1{,}119.8)-(1.5)(89.6)}{(1.3)(2{,}893.4)}$$

$$= 0.76 < 1.0$$

A5.10.3.3—Flexure at Pier 2

$$RF = \frac{(1.0)(1.0)(1.0)(6{,}943.6)-(1.25)(2{,}558.0)-(1.5)(204.6)}{(1.3)(2{,}305.0)}$$

$$= 1.15 > 1.0$$

As the governing flexure:

$$RF = 0.70 < 1.0$$

The permit check fails in flexure.

If the flexural Strength II rating factors were greater than 1.0, the shear Strength II and Service II rating factors should also be evaluated prior to permit approval.

A5.11—Summary of Rating Factors

Table A5.11-1—Summary of Rating Factors—Interior Girder

Limit State	Design Load Rating (HL-93)		Legal Load Rating		Permit Load Rating
	Inventory	Operating	Governing Load		
Strength I					
Flexure at 0.4*L* (+*M*)	0.96	1.24	Type 3-3	1.91	
Flexure at 0.5*L* (+*M*)	1.05	1.36	Type 3-3	2.10	
Flexure at pier 2 (–*M*)	0.99	1.28	Lane	1.88	
Shear at pier 1	1.75	2.27	Type 3-3	3.39	
Service II					
Flexure at 0.4*L* (+*M*)	1.00	1.30	Type 3-3	1.65	
Flexure at 0.5*L* (+*M*)	1.10	1.43	Type 3-3	1.82	
Flexure at pier 2 (–*M*)	1.17	1.52	Lane	1.85	
Strength II					
Flexure at 0.4*L* (+*M*)					0.70
Flexure at 0.5*L* (+*M*)					0.76
Flexure at pier 2 (–*M*)					1.15

A5.12—Reference

AASHTO. *AASHTO LRFD Bridge Design Specifications*, Eighth Edition, LRFD-8. American Association of State Highway and Transportation Officials, Washington, DC, 2017.

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A6—THROUGH PRATT TRUSS BRIDGE: DESIGN LOAD CHECK OF SELECTED TRUSS MEMBERS

A6.1—Bridge Data

Span Length: 175 ft (single span, pin-connected truss)
Year Built: 1909
Material: Steel F_y = 36 ksi (nominal yield by testing)
F_u = 65.4 ksi (nominal ultimate by testing)
Condition: No deterioration. NBI Item 59 Code = 7
Riding Surface: Not field verified and documented
ADTT (one direction): Unknown
Skew: 0°

A6.2—Member Properties

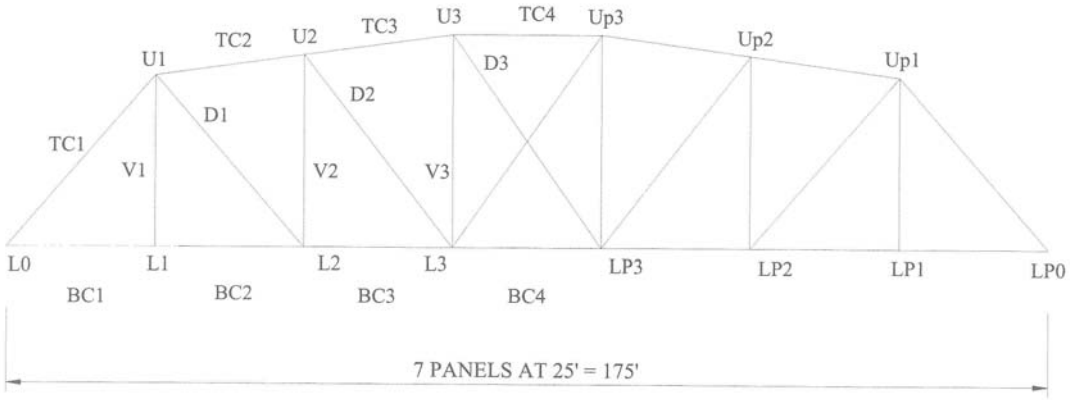
Member	Section	A, in.^{2}	r, in.
Top Chord TC4 Riveted	Built-up Section 2 Web Pl. 21 × ½	55.3	9.1
	2 Bottom Angle 5 × 3½ × $^5/_8$ with 6" × ½" bar		
	2 Top Angle 3$^1/_2$ × 3½ × $^3/_8$		
	Top Cover Plate 27 × ½		
Bottom Chord BC4	6 Eyebars 8 × 1	48.0	—
Diagonal D1	2 Eyebars 8 × 1½	24.0	—
Vertical V1 Riveted	2 Channels—15C33.9#	19.92	—

A6.3—Dead Load Analysis

Asphalt Thickness = 3 in. (field measured)

Dead Load Force Effects (*DC* = Component, *DW* = Wearing Surface)

Member	P_{DC}	P_{DW}
TC4 (Top Chord)	–558.1 kips	–39.4 kips
BC4 (Bottom Chord)	535.1 kips	37.7 kips
D1 (Diagonal)	253.2 kips	17.8 kips
V1 (Vertical)	106.2 kips	9.2 kips



TRUSS ELEVATION

Figure A6.3-1—Truss Elevation with Joint and Member Designations

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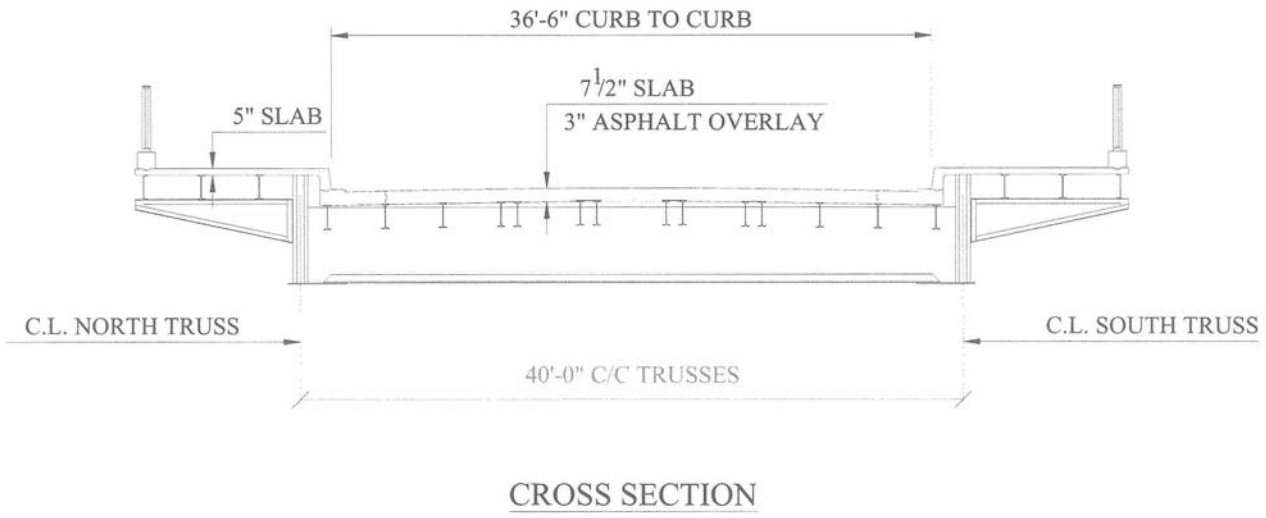

Figure A6.3-2—Bridge Cross Section

A6.4—Live Load Analysis (Design Load Check)

Use lever rule for the distribution of live loads to the North truss. LRFD Design 4.6.2.4

Analyzing as a planar structure.

Application of HL-93 Loading within a Lane: LRFD Design 3.6.1.3.1

R represents the resultant of lane and wheel loads.

W = lane load

P = wheel loads

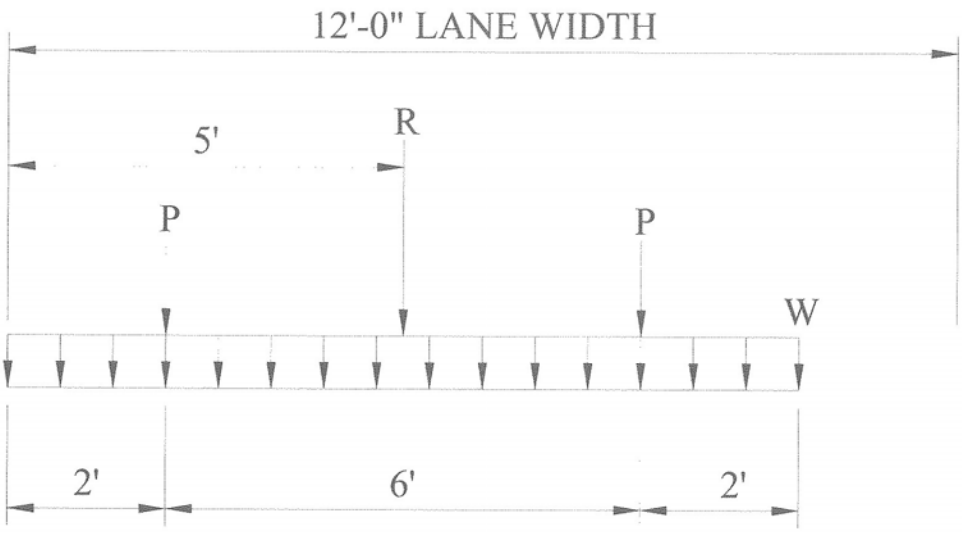

Figure A6.4-1—Typical Load Placement within a Lane

Road width = 36.5 ft

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Distance between trusses = 40 ft

Edge distances = 1.75 ft

A6.4.1—Live Load Distribution Factors

A6.4.1.1*—One Lane Loaded (See Figure A6.4.1-1)*

Multiple Presence Factor = 1.2 LRFD Design Table 3.6.1.1.2-1

Distribution Factor $= \left(\frac{40 - 1.75 - 5}{40}\right) \times 1.2$

$= 0.998$

A6.4.1.2*—Two Lanes Loaded (See Figure A6.4.1-1)*

Multiple Presence Factor = 1.0

Distribution Factor $= (33.25 + 21.25)\frac{1}{40.00} \times 1.0$

= 1.363 Governs

A6.4.1.3*—Three Lanes Loaded (See Figure A6.4.1-1)*

Multiple Presence Factor = 0.85

Distribution Factor $= (33.25 + 21.25 + 9.25)\frac{1}{40.00} \times 0.85$

= 1.355

A6.4.2—Live Load Force Effects (Due to HL-93)

Distribution Factor g = 1.363

Dynamic Load Allowance IM = 33 percent

The following member forces were computed using influence lines. Undistributed, no impact.

A6.4.2.1—Member TC4 (See Figure A6.3-1)

Design Lane Load = –68.1 kips

Design Truck = –76.3 kips Governs

Design Tandem = –53.2 kips

P_{LL+IM} = –68.1 – 76.3 × 1.33

= –169.6 kips

$g \times P_{LL+IM}$ = (1.363) (–169.6 kips)

= –231.1 kips

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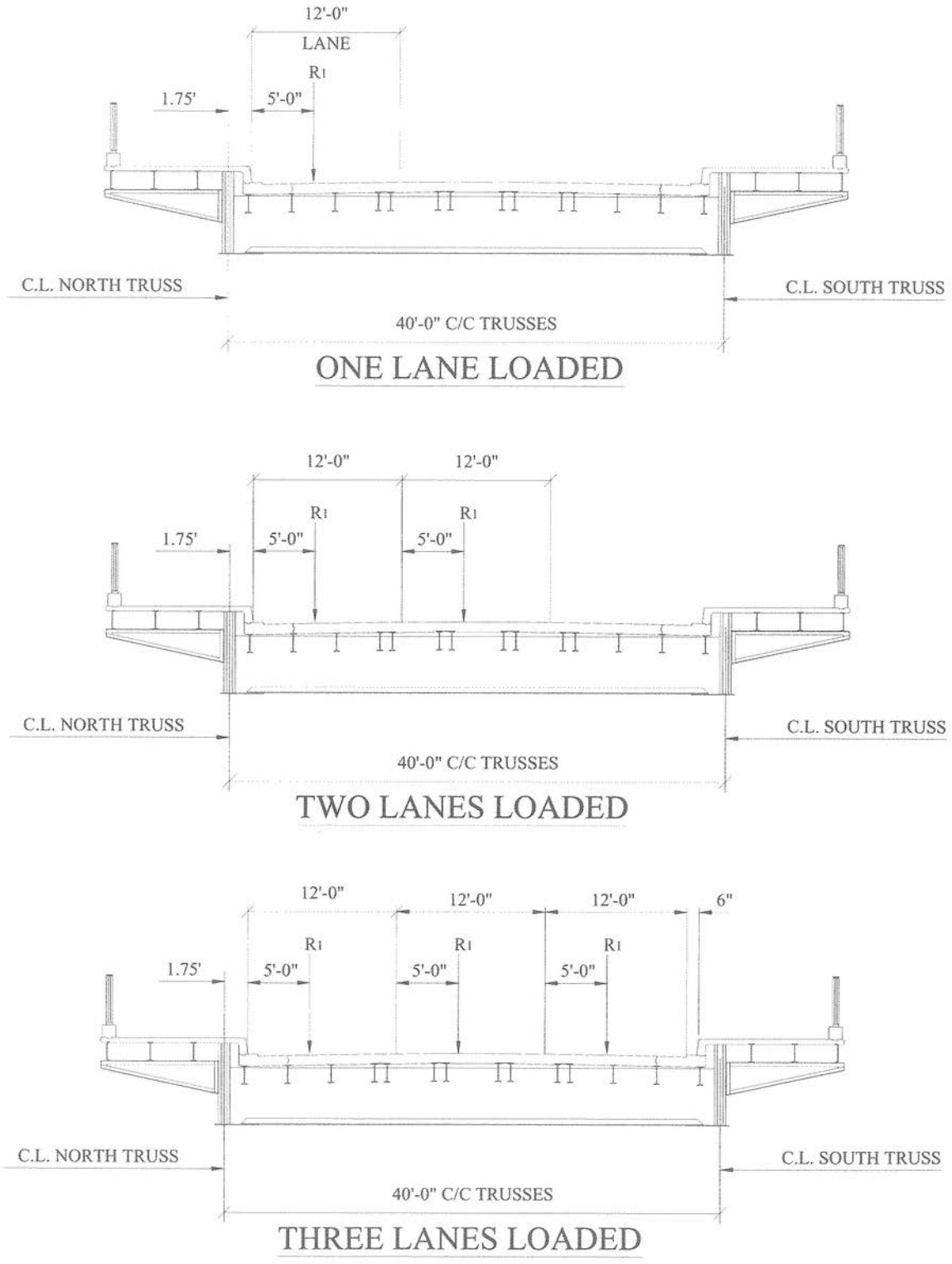

Figure A6.4.1-1—Load Placement for Distribution to the North Truss

A6.4.2.2—Member BC4

Design Lane Load	= 65.3 kips	
Design Truck	= 73.1 kips	Governs
Design Tandem	= 51.0 kips	

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P_{LL+IM} = 65.3 kips + 73.1 × 1.33

= 162.5 kips

$g \times P_{LL+IM}$ = (1.363) (162.5 kips)

= 221.5 kips

A6.4.2.3—Member D1

Design Lane Load = 33.9 kips

Design Truck = 49.3 kips Governs

Design Tandem = 36.4 kips

P_{LL+IM} = 33.9 kips + 49.3 × 1.33

= 99.5 kips

$g \times P_{LL+IM}$ = (1.363) (99.5 kips)

= 135.6 kips

A6.4.2.4—Member V1

Design Lane Load = 16.0 kips

Design Truck = 49.6 kips Governs

Design Tandem = 46.0 kips

P_{LL+IM} = 16.0 kips + 49.6 × 1.33

= 82.0 kips

$g \times P_{LL+IM}$ = (1.363) (82.0 kips)

= 111.7 kips

A6.5—Compute Nominal Resistance of Members

A6.5.1—Top Chord TC4 (Compression Member)

Area = 55.30 in.^{2} r = 9.1 in.

Length = 25 ft

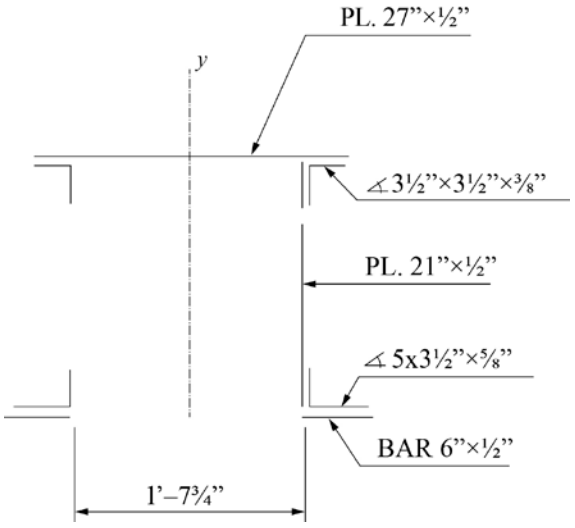

Figure A6.5.1-1—Cross Section of Top Chord

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Member TC4:

Area = 55.30 in.^{2}

I_y = 5,716.8 in.^{4}

I_z = 4,541.3 in.^{4}

The gravity axis of the top chord coincides with the working line connecting the pins.

The top chord is therefore evaluated as a concentrically loaded column.

Appendix I6A illustrates an example where the pins are eccentric. The method is simple and conservative.

Article A6.10 shows how the load rating of a member with an eccentric connection can be established using the interaction equation for the design of members with combined axial loads and concurrent moments.

Limiting Slenderness Ratio: LRFD Design 6.9.3

$$\frac{K\ell}{r} = \frac{0.875\times 25\times 12}{9.1} = 28.8 < 120 \text{ for main members} \quad \text{OK}$$

K = 0.875 for pinned ends LRFD Design 4.6.2.5

Nominal Compressive Resistance: LRFD Design 6.9.5.1

Column slenderness term λ is defined as:

$$\lambda = \left(\frac{k\ell}{r_s\pi}\right)^2 \frac{F_y}{E}$$

$$= \left(\frac{k\ell/r}{\pi}\right)^2 \frac{F_y}{E}$$

$$= \left(\frac{28.8}{\pi}\right)^2 \frac{36}{29,000}$$

= 0.104 < 2.25 Intermediate length column

Check Limiting Width/Thickness Ratios: LRFD Design 6.9.4.2

$$\frac{b}{t} \le \lambda_r$$ LRFD Design Eq. 6.9.4.2.1-1

λ_r = width to thickness ratio limit as specified in LRFD Design Eq. 6.9.4.2.1-1.

$$\lambda_r = 1.4\sqrt{\frac{E}{F_y}}$$ LRFD Design Table 6.9.4.2.1-1

$$\frac{b}{t} \le 1.40\sqrt{\frac{E}{F_y}}$$

b = 19.75 in. (back-to-back angles)

$$\frac{b}{t} = \frac{19.75}{½} = 39.5$$

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$$1.40\sqrt{\frac{E}{F_y}} = 1.40\sqrt{\frac{29{,}000}{36}} = 39.7$$

$$\frac{b}{t} = 39.5 < 39.7 \quad \text{OK}$$

Web Plates, $\lambda_r = 1.49\sqrt{\frac{E}{F_y}}$ LRFD Design Table 6.9.4.2.1-1

$$\frac{h}{t_w} \leq 1.49\lambda_r$$

$$\frac{h}{t_w} = \frac{21}{0.5} = 42$$

$$1.49\sqrt{\frac{E}{F_y}} = 1.49\sqrt{\frac{29{,}000}{36}}$$

$$= 42.3 > \frac{h}{t} = 42 \quad \text{OK}$$

Bottom Flange, $\lambda_r = 0.45\sqrt{\frac{E}{F_y}}$ LRFD Design Table 6.9.4.2.1-1

$$\frac{b}{t} \leq 0.45\sqrt{\frac{E}{F_y}}$$

$$\frac{b}{t} = \frac{6}{\left(\frac{5}{8} + 0.5\right)} = 5.33$$

$$0.45\sqrt{\frac{E}{F_y}} = 0.45\sqrt{\frac{29{,}000}{36}}$$

$$= 12.8 > \frac{b}{t} = 5.33 \quad \text{OK}$$

The built-up section meets limiting width/thickness ratios; local buckling prior to yielding will not occur.

$\lambda < 2.25$ (See previous calculations)

$P_n = -0.66^{\lambda} F_y A_s$ LRFD Design Eq. 6.9.5.1-1

$= -0.66^{(0.104)} \times 36 \times 55.30$

$= -0.957 \times 36 \times 55.30$

$= -1{,}906.6$ kips

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P_r = $\varphi_c P_n$ LRFD Design Eq. 6.9.2.1-1

φ_c = 0.90 LRFD Design 6.5.4.2

P_r = $0.9 \times (-1{,}906.6) = 1{,}715.9$ kips

A6.5.2—Bottom Chord Member BC4 (Tension Member)

6 Eyebars 8 in. $\times$ 1 in.

Total Area = 48 in.^{2}

A6.5.2.1—Limit State: Yielding over Gross Area (in the Shank of the Eyebar)

P_r = $\varphi_y F_y A_g$ LRFD Design Eq. 6.8.2.1-1

φ_y = 0.95 LRFD Design 6.5.4.2

P_r = $0.95 \times 36 \times 48 = 0.95(1{,}728)$

= 1,641.6 kips Governs

A6.5.2.2—Limit State: Fracture at the Eyebar Head

P_r = $\varphi_u F_u A_n R_p U$ LRFD Design Eq. 6.8.2.1-2

U = 1.0

φ_u = 0.80 (tension, fracture in net section) LRFD Design 6.5.4.2

R_p = 1.0

Width of eyebar head at centerline of pin = 18 in.

Size of pin hole = $6\,^1/_2$ in. + $^1/_{32}$ in.

A_n = (18 in. – $6\,^1/_2$ in. – $^1/_{32}$ in.) $\times$ 1 in.

= 11.53 in.^{2} per eyebar

$\frac{A_n}{A_{shank}} = \frac{11.53}{8 \times 1} = 1.43 > 1.35$ OK 6A.6.6.2

P_r = $0.80 \times 65.4 \times 11.53 \times 6 = 0.80(4{,}524.4)$

= $3{,}619.5$ kips > $1{,}641.6$ kips

Lesser value of P_r governs:

P_r = 1,641.6 kips

A6.5.3—Diagonal Member D1

2 Eyebars 8 in. $\times$ 1 $^1/_2$ in.

Total Area = 24 in.^{2}

A6.5.3.1—Limit State: Yielding over Gross Area (in the Shank of the Eyebar)

P_r = $\varphi_y F_y A_g$ LRFD Design

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$= 0.95 \times 36 \times 24 = 0.95(864)$ Eq. 6.8.2.1-1

$= 820.8$ kips

A6.5.3.2—Limit State: Fracture at the Eyebar Head

$P_r = \varphi_u F_u A_n R_p U$ LRFD Design Eq. 6.8.2.1-2

$U = 1.0$

$\varphi_u = 0.80$

$R_p = 1.0$

Width of eyebar head at centerline of pin = 18 in.

Size of pin hole = $6^1/_2$ in. + $^1/_{32}$ in.

A_n = (18 in. – $6^1/_2$ in. – $^1/_{32}$ in.) × 1.5 in. 6A.6.6.2

$= 17.20$ in.^{2}

$\frac{A_n}{A_{shank}} = \frac{17.20}{8 \times 1.5} = 1.43 > 1.35$ OK

$P_r = 0.80 \times 65.4 \times 17.2 \times 2 = 0.80(2,249.8)$

$= 1{,}799.8$ kips > 820.8 kips

Lesser value of P_r governs

$P_r = 820.8$ kips

A6.5.4—Vertical Member V1

$2 - 15$ C $33.9^{\#}$

Total Area A_g = 19.92 in.^{2}

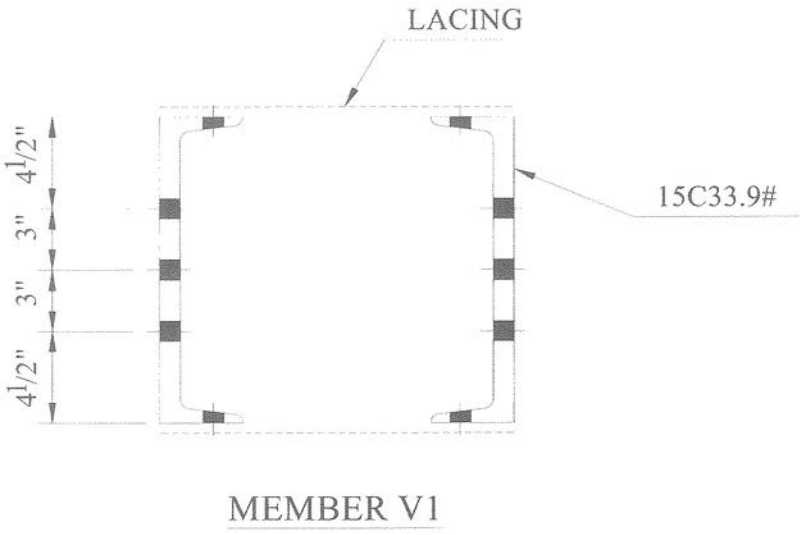

Figure A6.5.4-1—Cross Section of Vertical Member

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A6.5.4.1—Limit State: Yielding over Gross Area

$P_r = \varphi_y F_y A_n R_p U$ LRFD Design Eq. 6.8.2.1-1

$= 0.95 \times 36 \times 19.92 = 0.95\ (717.1)$

$= 681.3$ kips

A6.5.4.2—Limit State: Fracture at Net Area (at Rivet Holes)

$P_r = \varphi_u F_u A_n U$ LRFD Design Eq. 6.8.2.1-2

$\varphi_u = 0.80$

$U = 1 - \frac{\overline{x}}{L}$ LRFD Design Table 6.8.2.2-1

$\overline{x} = 0.788$ in.

$L = 6$ in.

$U = 1 - \frac{0.788}{6} = 0.87$

$R_p = 1.0$ LRFD Design 6.8.2.2

Loads transmitted through webs only; three or more rivets per line

Net Area:

Gross Area per channel = 9.96 in.^{2} LRFD Design 6.8.3

Web thickness = 0.4 in.

Flange thickness = 0.6 in.

Rivet hole = $^{15}/_{16}$ in.

s = $1^1/_2$ in.

g = $4^1/_2$ in.

$\frac{s^2}{4g} = \frac{1.5^2}{4 \times 4.5} = 0.125$

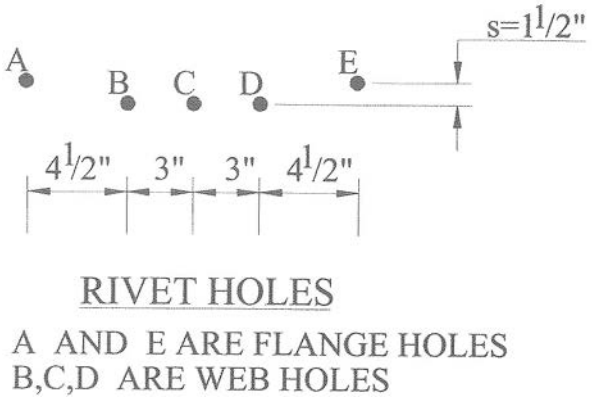

Figure A6.5.4.2-1—Rivet Hole Spacing for Net Area

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B-C-D:

A_{net} = $9.96 - 3 \times \frac{15}{16} \times 0.4$

= 8.84 in.^{2} per channel

A-B-C-D-E:

A_{net} = A_{gross} – hole areas + (# of diagonals) $\left(\frac{s^2}{4g}\right)$ (thickness)

= $9.96 - \left(3 \times \frac{15}{16} \times 0.4 + 2 \times \frac{15}{16} \times 0.6\right) + 2 \times 0.125 \times 0.40$

= 9.96 – 2.25 + 0.1

= 7.81 in.^{2} per channel < 8.84 in.^{2}

A_{net} = 7.81 in.^{2} per channel

Total A_{net} = 2 × 7.81 =15.62 in.^{2}

P_r = $0.80 \times 65.4 \times 15.62 \times 0.87 = 0.80\ (888.7)$

= 711.0 kips > 681.3 kips

Lesser value of P_r governs

P_r = 681.3 kips

A6.6—General Load Rating Equation

$$RF = \frac{C - (\gamma_{DC})(DC) - (\gamma_{DW})(DW) \pm (\gamma_P)(P)}{(\gamma_L)(LL + IM)}$$ Eq. 6A.4.2.1-1

A6.7—Evaluation Factors (for Strength Limit States)

A6.7.1—Resistance Factor, φ

Included in previous calculations of factored axial resistances and not used in *RF* equations that follow.

A6.7.2—Condition Factor, φ_c 6A.4.2.3

φ_c = 1.0 no deterioration

A6.7.3—System Factor, φ_s 6A.4.2.4

φ_s = 0.90 for riveted truss members and multiple eyebars

A6.8—Design Load Rating (6A.4.3)

Strength I Limit State: 6A.6.4.1

Load	Inventory	Operating	
DC, *DW*	1.25	1.25	Asphalt thickness was field measured
LL + *IM*	1.75	1.35	

Table 6A.4.2.2-1

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A6.8.1—Top Chord TC4

P_{DC} = –558.1 kips

P_{DW} = –39.4 kips

P_{LL+IM} = –231.1 kips

P_r = –1,715.9 kips

Inventory: $RF = \dfrac{(1.0)(0.90)(-1,715.9) - (1.25)(-558.1) - (1.25)(-39.4)}{(1.75)(-231.1)}$

$= 1.97$

Operating: $RF = 1.97 \times \dfrac{1.75}{1.35}$

$= 2.55$

A6.8.2—Bottom Chord BC4

P_{DC} = 535.1 kips

P_{DW} = 37.7 kips

P_{LL+IM} = 221.5 kips

P_r = 1,641.6 kips

Inventory: $RF = \dfrac{(1.0)(0.90)(1,641.6) - (1.25)(535.1) - (1.25)(37.3)}{(1.75)(221.5)}$

$= 1.97$

Operating: $RF = 1.97 \times \dfrac{1.75}{1.35}$

$= 2.55$

A6.8.3—Diagonal D1

P_{DC} = 253.2 kips

P_{DW} = 17.8 kips

P_{LL+IM} = 135.6 kips

P_r = 820.8 kips

Inventory: $RF = \dfrac{(1.0)(0.90)(820.8) - (1.25)(253.2) - (1.25)(17.8)}{(1.75)(135.6)}$

$= 1.69$

Operating: $RF = 1.69 \times \dfrac{1.75}{1.35}$

$= 2.18$

A6.8.4—Vertical V1

P_{DC} = 106.2 kips

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P_{DW} = 9.2 kips

P_{LL+IM} = 111.7 kips

P_r = 681.3 kips

$$\text{Inventory: } RF = \frac{(1.0)(0.90)(681.3)-(1.25)(106.2)-(1.25)(9.2)}{(1.75)(111.7)}$$

$$= 2.40$$

$$\text{Operating: } RF = 2.40 \times \frac{1.75}{1.35}$$

$$= 3.11$$

Service II limits will be satisfied if Strength I limits are satisfied for axial members.

A6.9—Summary of Rating Factors

Table A6.9-1—Summary of Rating Factors —Truss Members

Limit State	Member	Design Load Rating (HL-93)	
		Inventory	Operating
Strength I	Top Chord TC4	1.97	2.55
	Bottom Chord BC4	1.97	2.55
	Diagonal D1	1.69	2.18
	Vertical V1	2.40	3.11

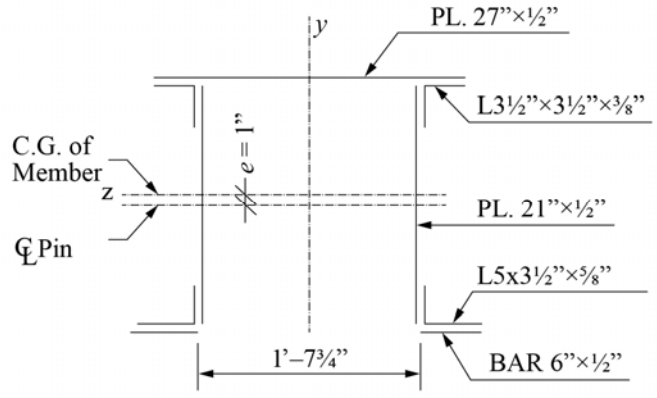
A6.10—Rating of Steel Compression Member (TC4) with Eccentric Connections

The following is an example to illustrate the approach that could be utilized to load rate compression members that have eccentric connections. As stated in Article C6A.6.8, in compression members with ***a symmetrical section*** *(such as TC4 in this example),* the gravity axis of the section may not coincide with the working lines, resulting in an eccentric connection.

Axially loaded members with eccentric connections could be treated as concentrically loaded members, with an equivalent applied moment of P_e at both ends of the member, where e is the eccentricity at the end of the member. The following example illustrates a rating analysis based on the method described in Appendix H6A for the compression chord member TC4, where the pins are assumed to be 1 inch eccentric in the negative y-coordinate.

e = 1 in.

$S_{z\text{-}bottom}$ = 376.0 in.^{3}



A6.10.1—Cross Section of Top Chord

The axial compression capacity calculated in the previous section:

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ϕP_n = 1,715.9 kip (See Section A6.5.1)

P_{DL} = –558.1 kip; P_{DW} = -39.4 kip; P_{LL+I} = –231.1 kip (HL-93 Demand)

γ_{DL} = 1.25; γ_{DW} = 1.25 (field measured AC); γ_{LL} = 1.75 (Inventory);

Denote the inventory rating as *RF*. The total axial compression load can be expressed as:

$$P_u = \gamma_D P_{DL} + RF \times \gamma_{LL} \times P_{LL}$$

$$P_u = 1.25 \times 558.1 + 1.25 \times 39.4 + RF \times 1.75 \times 231.1$$

$$P_u = \left(746.9 + 404.4 \times RF\right) \text{ kip}$$

The moment due to the eccentricity can be expressed as:

$$M_{uz} = P_u \times e$$

$$M_{uz} = \left(P_u\right) \times 1$$

$$M_{uz} = \left(746.9 + 404.4 \times RF\right) \times 1 \qquad \text{(since eccentricity } e = 1.0 \text{ in)}$$

$$M_{uz} = \left(746.9 + 404.4 \times RF\right) \text{ kip-in.}$$

As shown in Appendix H6A, the moment magnifier is:

$$\delta_b = \frac{C_m}{1 - \left[\dfrac{\gamma_D P_D + RF \times \gamma_{LL} P_{LL+I}}{\varphi_K P_e}\right]}$$

For the constant member moment:

$$C_m = 0.6 + 0.4\left(\frac{M_{1b}}{M_{2b}}\right)$$ LRFD Design Eq. 4.5.3.2.2b-6

Moment demand at one end will be $M_{1b} = P_u e_1 = P_u \times 1 = P_u$

Moment demand at the other end will be $M_{2b} = P_u e_2 = P_u \times 1 = P_u$

$$C_m = 0.6 + 0.4\left(\frac{M_{1b}}{M_{2b}}\right) = 0.6 + 0.4 \times \left(\frac{P_u}{P_u}\right) = 0.6 + 0.4 = 1.0$$

The Euler buckling load is:

$$P_e = \frac{\pi^2 EI}{(KL)^2}$$ LRFD Design Eq. 4.5.3.2.2b-5

$$P_e = \frac{\pi^2 \times 29{,}000 \times 4{,}541.3}{(0.875 \times 25 \times 12)^2} = 18{,}863 \text{ kip}$$

$\varphi = 1.0$ for steel member

$$\delta_b = \frac{1}{1 - \dfrac{746.9 + 404.4 \times RF}{1.0 \times 18{,}863}}$$

Although the top chord is not a true box section, it behaves similarly to a box section with the assumed bottom lacing (see Note). The moment capacity may be determined with the general approach listed in LRFD Design Article C6.12.2.2.2, but requires a different method to calculate the torsion constant for the nontrue box shape. However, the

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effect due to the lateral buckling capacity typically has a minimal effect on the bending capacity for such a section and the elastic bending capacity may be used.

$$\varphi_f M_{nz} = \varphi_f F_y S_{z-bottom} = 1.0 \times 36 \times 376.0 = 13,536 \text{ kip-in.}$$

Note: For double lacing with $2^1/_2$ in. × $^7/_8$ in. bars at 25 in. spacing, the torsion constant (from 3D FEM analysis) is:

J = 1,399 in.^{4}

A similar result is obtained using the Effective Section Properties formula for J for a laced compression member from Appendix B of the *Caltrans Guide Specifications for Seismic Design of Steel Bridges*, First Edition.

With

$$M_{CR} = \frac{\pi}{l}\sqrt{EI_y GJ}$$ LRFD Design (C6.12.2.2.2-1)

where $G = 0.385E$ LRFD Design (C6.12.2.2.2-2)

$$M_{CR} = \frac{\pi \times 29,000}{25 \times 12}\sqrt{0.385 \times 4,541.3 \times 1,399} = 474,958 \text{ kip-in.}$$

The nominal bending capacity can be calculated as:

$$M_n = F_y S_z \left[1 - \frac{F_y S_z}{4M_{CR}}\right]$$ LRFD Design (C6.12.2.2.2-5)

$$= 36 \times 376 \times \left[1 - \frac{36 \times 376}{4 \times 474,958}\right]$$

$$= 36 \times 376 \times 0.9929$$

$$= 13,440 \text{ kip-in.}$$

Since the reduction due to torsion is insignificant, a reduction factor of 0.9929, or 0.7 percent reduction for this case, using the elastic moment capacity will eliminate the need for the complicated torsion constant calculation and still produce similar results.

Setting the left side of Eq. 6.9.2.2-2 equal to the right side as:

$$\frac{P_u}{P_r} + \frac{8.0}{9.0}\left(\frac{\delta_b M_{uz}}{M_{rz}}\right) = 1.0$$

$M_{rz} = \varphi_s \varphi_c \varphi_f M_{nz}$, where $\varphi_f M_{nz} = 13536$ kip-in.

$P_r = \varphi_s \varphi_c \varphi_c P_n$, where $\varphi_c P_n = 1,715.9$ kip

with

$\varphi_c = 1.0,\ \varphi_s = 0.9$

$$\frac{746.9 + 404.4 \times RF}{1.0 \times 0.9 \times 1,715.9} + \frac{8.0}{9.0}\left[\frac{1}{1 - \left(\frac{746.9 + 404.4 \times RF}{1.0 \times 18,863}\right)}\right]\left(\frac{746.9 + 404.4 \times RF}{1.0 \times 0.9 \times 13,536}\right) = 1.0$$

With iteration, the rating factor can be determined.

Resulting inventory rating factor RF = 1.558.

Check axial force condition:

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$$\frac{P_u}{P_r} = \frac{746.9 + 404.4 \times 1.558}{1.0 \times 0.9 \times 1715.9} = 0.89 > 0.2 \; OK$$

The rating factors calculated with this method are usually larger than those calculated with methods listed in Appendix I6A. The differences could be significant for bridges with low rating factors.

A6.11—References

AASHTO. *AASHTO LRFD Bridge Design Specifications*, Eighth Edition, LRFD-8. American Association of State Highway and Transportation Officials, Washington, DC, 2017.

California Department of Transportation. *Caltrans Guide Specifications for Seismic Design of Steel Bridges*, First Edition, Sacramento, CA, 2001.

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A7—REINFORCED CONCRETE SLAB BRIDGE DESIGN AND LEGAL LOAD CHECK

A7.1—Bridge Data

Span Length:	21.5 ft (simple span)
Year Built:	1963
Material:	Concrete $f_c' = 3$ ksi
	Reinforced Steel $f_y = 40$ ksi
Condition:	No deterioration. NBI Item 59 Code = 6
Riding Surface:	Not field verified and documented
ADTT (one direction):	Unknown
Skew:	0°

A7.2—Dead Load Analysis

A7.2.1—Interior Strip—Unit Width

A7.2.1.1—Components, DC

Concrete slab:

$$\left(\frac{14}{12}\right)(1.0)(0.150) = 0.175 \text{ kip/ft}$$

Parapet and curb:

$$\frac{2\left[(1.5)(1.5)+(2.33)(1.0)\right](1.0)(0.150)}{43} = \underline{0.032 \text{ kip/ft}}$$

$$DC = 0.207 \text{ kip/ft}$$

$$M_{DC} = \frac{1}{8}\times 0.207 \times 21.5^2$$

$$= 12.0 \text{ kip-ft}$$

A7.2.1.2—Wearing Surface, DW

Asphalt Thickness = $3^1/_2$ in. (field measured)

Asphalt Overlay $= \left(\frac{3.5}{12}\right)(1.0)(0.144) = 0.042$ kip/ft

$$M_{DW} = \frac{1}{8}\times 0.042 \times 21.5^2$$

$$= 2.4 \text{ kip-ft}$$

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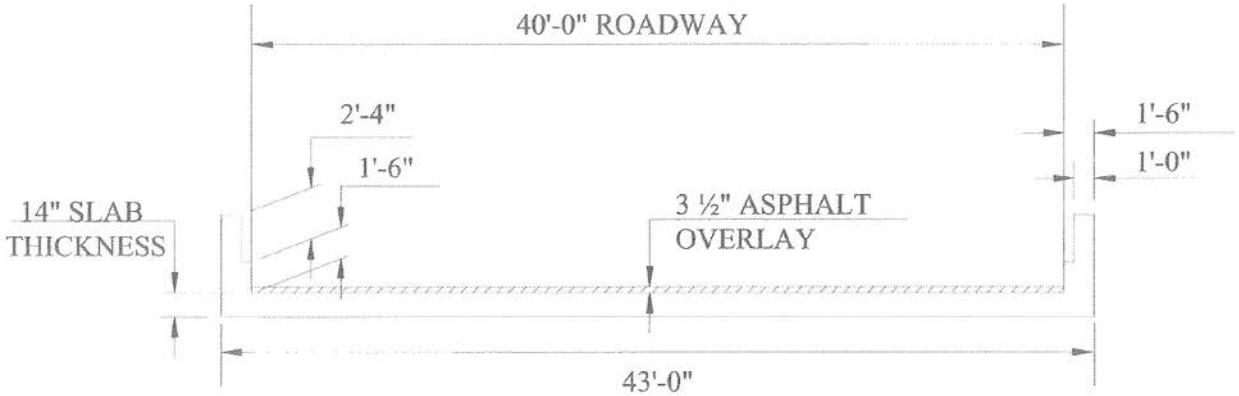
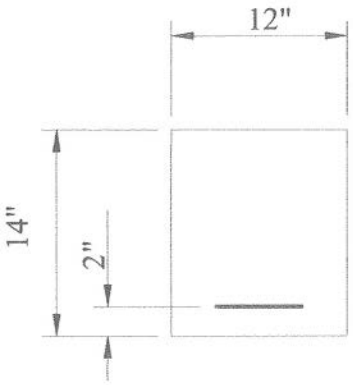


CROSS SECTION



MAIN REINFORCEMENT
#8 AT 6" CC

$A_S = 1.58$ in.^{2}

SLAB REINFORCEMENT
(UNIT WIDTH)

Figure A7.1-1—Reinforced Concrete Slab Bridge

A7.3—Live Load Analysis (Design Load Check)

Equivalent strip width for slab type bridges (Interior Strip) — LRFD Design 4.6.2.3

A7.3.1—One Lane Loaded

$E = 10.0 + 5.0\sqrt{L_1 W_1}$ — LRFD Design Eq. 4.6.2.3-1

L_1 = 21.5 ft < 60 ft

W_1 = Lesser of 43.0 ft or 30.0 ft

= 30.0 ft

$E = 10.0 + 5.0\sqrt{21.5 \times 30}$

= 137.0 in.

= 11.41 ft

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A7.3.2—More than One Lane Loaded

$$E = 84.0 + 1.44\sqrt{L_1 W_1} \le \frac{12.0\,W}{N_L}$$

L_1 = 21.5 ft < 60.0 ft

W_1 = Lesser of 43.0 ft or 60.0 ft

= 43.0 ft

$E = 84 + 1.44\sqrt{21.5 \times 43}$ LRFD Design Eq. 4.6.2.3-2

= 127.8 in. = 10.65 ft < 11.41 ft

$N_L = \frac{40.0}{12} = 3$ Design Lanes

$\frac{12.0\,W}{N_L} = \frac{12 \times 43}{3} = 172$ in. > 127.8 in. OK

Use E = 10.65 ft

For longitudinal edge strips, the effective strip width is:

Sum of: LRFD Design 4.6.2.1.4b

the distance between the edge of the deck and the inside face of the barrier

+ one-quarter the strip width specified in LRFD Design Article 4.6.2.1.3, 4.6.2.3, or 4.6.2.10, as appropriate

+ 12.0 in.

The effective edge strip width shall not exceed either one-half the full strip width or 72.0 in.

E_2 = 18.0 in. + 0.25 x 137.0 in. + 12.0 in. = 64.25 in.

E_2 = 0.5 x 137.0 in. = 68.5 in.

E_2 = 72 in.

64.25 in ≤ 68.5 in.

∴ use E_2 = 64.25 in.

LRFD Design Article 4.6.2.1.4b assumes the longitudinal edge strip supports one wheel line and a tributary portion of the design lane load where appropriate.

By comparison of the ratios of the tributary design lane load width to effective slab width, the edge strip is estimated not to govern for this bridge. Note that parapet dead load was assumed to be uniformly distributed across the full bridge width and that parapet width can play an influential role when determining the governing case.

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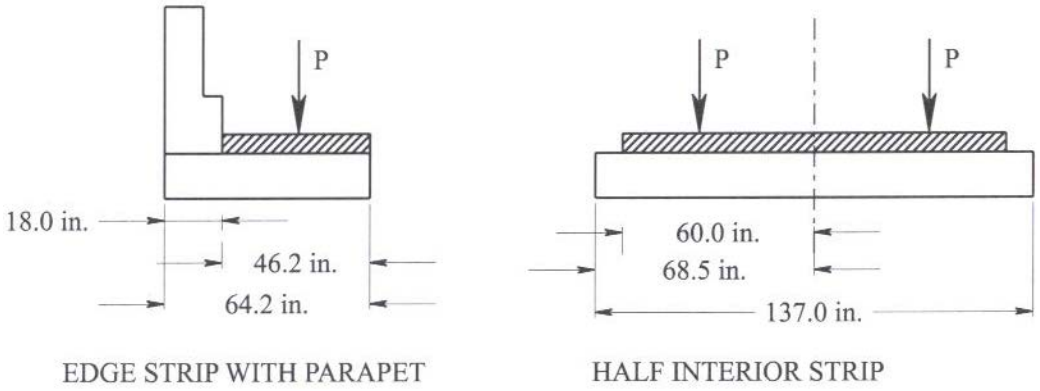

Figure A7.3.2-1—Longitudinal Edge Strip Comparison

Ratio edge strip:

46.25/64.25 = 0.72

Ratio half interior strip:

60.0/68.5 = 0.88 Governs

The rating will consider only the interior strip width.

A7.3.2.1—Midspan Live Load Force Effects (HL-93)

Dynamic Load Allowance = 33%

Equivalent Strip Width = 10.65 ft

Design-Lane Load Moment = 37.0 kip-ft

Design Truck Moment = 172.0 kip-ft

Design Tandem Moment = 219.4 kip-ft Governs

M_{LL+IM} = 37.0 + 319.4 × 1.33

= 328.8 kip-ft

Live Load Moment per unit width of slab:

$$M_{LL+IM} = \frac{328.8}{10.65} = 30.9 \text{ kip-ft/ft}$$

A7.4—Compute Nominal Resistance

Flexural Resistance:

Rectangular Section = $b_w = b = 12$ in. LRFD Design 5.6.3.1

$$c = \frac{A_s f_y}{\alpha_1 f_c' \beta_1 b}$$ LRFD Design Eq. 5.6.3.1.1-4

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A_s = 0.79×2 #8 bars at 6 in.

= 1.58 in.^{2}/ft

α_1 = 0.85 LRFD Design 5.6.2.2

β_1 = 0.85 LRFD Design 5.6.2.2

b = 12 in.

$c = \dfrac{1.58 \times 40}{0.85 \times 3 \times 0.85 \times 12}$

= 2.43 in.

a = $c\beta_1$ LRFD Design 5.6.2.2

= 2.43×0.85

= 2.07 in.

d_s = 14 – 2 = 12 in. Distance to *C.G.* of steel

$M_n = A_s f_y \left(d_s - \dfrac{a}{2} \right)$

$= 1.58 \times 40 \left(12 - \dfrac{2.07}{2} \right) \times \dfrac{1}{12}$

= 57.75 kip-ft/ft

A7.5—Minimum Reinforcement (6A.5.6)

Amount of reinforcement must be sufficient to develop M_r equal to the lesser of: 6A.5.6

M_{cr} or $1.33M_u$

$M_r = \varphi M_n = 0.90 \times 57.75$ kip-ft=51.98 kip-ft

1. $1.33M_u = 1.33\ M_u = 1.33 \times (1.75 \times 30.9 + 1.25 \times 12 + 1.25 \times 2.4)$

= 95.9 kip-ft > 51.98 kip-ft No Good

2. $M_{cr} = \gamma_3 \left[(\gamma_1 f_r + \gamma_2 f_{cpc}) S_c - M_{dnc} \left(\dfrac{S_c}{S_{nc}} - 1 \right) \right]$ LRFD Design Eq. 5.6.3.3-1

Where a monolithic or non-composite section is designed to resist all the loads, S_{nc} is substituted for S_c. In this case, $f_{cpe} = 0$, therefore:

$M_{cr} = \gamma_3 (\gamma_1 f_r) S_c$

$\gamma_1 = 1.6$

$\gamma_3 = \dfrac{F_y}{F_n} = \dfrac{40 \text{ ksi}}{70 \text{ ksi}} = 0.57$

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$$S_{nc} = \frac{I}{y_t}$$

where:

I = moment of inertia of uncracked section (neglecting reinforcement steel)

y_t = distance from the neutral axis of the uncracked section to the extreme tension fiber

$$= \frac{14}{2} = 7 \text{ in.}$$

$$I = \frac{1}{12} \times 12 \times 14^3 = 2744 \text{ in.}^4$$

$$S_{nc} = \frac{2,744}{7} = 392 \text{ in.}^3$$

$$f_r = 0.24\lambda\sqrt{f'_c}$$ LRFD Design 5.4.2.6

$\lambda = 1.0$

$f_r = 0.24\sqrt{3'} = 0.416$ ksi

$M_{cr} = 0.57 \times 1.6 \times 0.416 \text{ ksi} \times 392 \text{ in}^3 = 148.7 \text{ kip-in} = 12.4 \text{ kip-ft}$

$1.2M_{cr} = 1.2 \times 12.4 = 14.88 \text{ kip-ft} < 51.98 \text{ kip-ft}$ OK

The section meets the requirements for minimum reinforcement.

A7.6—Maximum Reinforcement (6A.5.5)

Current provisions of the LRFD specification have eliminated the check for maximum reinforcement. Instead, the factored resistance (φ factor) of compression controlled sections shall be reduced in accordance with LRFD Design Article 5.5.4.2. This approach limits the capacity of over-reinforced (compression controlled) sections. C6A.5.5

The net tensile strain, ε_t, is the tensile strain at nominal strength and determined by strain compatibility using similar triangles. LRFD Design C5.6.2.1

Given an allowable concrete strain of 0.003 and depth to neutral axis c = 2.43 in.

$$\frac{\varepsilon_c}{c} = \frac{\varepsilon_t}{d-c}$$

$$\frac{0.003}{2.43 \text{ in.}} = \frac{\varepsilon_t}{12 \text{ in.} - 2.43 \text{ in.}}$$

$\varepsilon_t = 0.0118$

For $\varepsilon_t = 0.0118 > 0.005$, the section is tension controlled and Resistance Factor, φ, shall be taken as 0.90. LRFD Design 5.6.2.1, 5.5.4.2

A7.7—Shear

Concrete slabs and slab bridges designed in conformance with AASHTO specifications may be considered satisfactory for shear. LRFD Design 5.12.2.1

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Also shear need not be checked for design load and legal load ratings of concrete members. 6A.5.8

A7.8—General Load-Rating Equation (6A.4.2)

$$RF = \frac{C - (\gamma_{DC})(DC) - (\gamma_{DW})(DW) \pm (\gamma_P)(P)}{(\gamma_L)(LL + IM)}$$ Eq. 6A.4.2.1-1

A7.9—Evaluation Factors (for Strength Limit States)

A7.9.1—Resistance Factor, φ (LRFD Design 5.5.4.2)

φ = 0.90 For flexure

A7.9.2—Condition Factor, φ_c (6A.4.2.3)

φ_c = 1.0 No deterioration

A7.9.3—System Factor, φ_s (6A.4.2.4)

φ_s = 1.0 Slab bridge

A7.10—Design Load Rating (6A.4.3)

A7.10.1—Strength I Limit State (6A.5.4.1)

$$RF = \frac{(\varphi_c)(\varphi_s)(\varphi) R_n - (\gamma_{DC})(DC) - (\gamma_{DW})(DW)}{(\gamma_L)(LL + IM)}$$

Load	Inventory	Operating	
DC, *DW*	1.25	1.25	Asphalt thickness
LL + *IM*	1.75	1.35	was field measured

Table 6A.4.2.2-1

Inventory:

$$RF = \frac{(1.0)(1.0)(0.9)(57.75) - (1.25)(12.0) - (1.25)(2.4)}{(1.75)(30.9)}$$

= 0.63

Operating:

$$RF = 0.63 \times \frac{1.75}{1.35}$$

= 0.82

A7.10.2—Service Limit State

No service limit states apply to reinforced concrete bridge members. As *RF* < 1.0 for HL-93, evaluate the bridge for legal loads.

A7.11—Legal Load Rating (6A.4.4)

Live Load: AASHTO Legal Loads—Type 3, 3S2, 3-3 (Rate for all 3) 6A.4.4.2.1

E = 10.65 ft 6A.4.4.3

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IM = 33 percent Unknown riding surface conditions

	Type 3	Type 3S2	Type 3–3	
M_{LL}	150.4	137.1	123.8	kip-ft
$\frac{M_{LL+IM}}{E}$	18.8	17.1	15.5	kip-ft/ft

A7.11.1—Strength I Limit State 6A.5.4.2.1

Generalized Live-Load Factor: Table 6A.4.4.2.3a-1

γ_L = 1.45

$ADTT$ = Unknown

Flexure:

$$RF = \frac{(1.0)(1.0)(0.90)(57.75)-\left[(1.25)(12.0)+(1.25)(2.4)\right]}{(1.45)(M_{LL+IM})}$$

	Type 3	Type 3S2	Type 3-3
RF	1.25	1.37	1.51

No posting required as $RF > 1.0$ for all AASHTO Legal Loads.

A7.11.2—Service Limit State

No service limit states apply to reinforced concrete bridge members at the legal load rating.

A7.11.3—Shear LRFD Design 5.12.2.1

Concrete slabs and slab bridges designed in conformance with AASHTO Specifications may be considered satisfactory for shear.

Shear need not be checked for legal loads. 6A.5.8

A7.11.4—Summary

Truck	Type 3	Type 3S2	Type 3-3
Weight, tons	25	36	40
RF	1.25	1.37	1.51
Safe Load Capacity, tons	31	49	60

A7.12—Summary of Rating Factors

Table A7.12-1 Summary of Rating Factors—Concrete Slab Interior Strip

Limit State		Design Load Rating		Legal Load Rating		
		Inventory	Operating	Type 3	Type 3S2	Type 3-3
Strength I	Flexure	0.63	0.82	1.25	1.37	1.51

A7.13—Reference

AASHTO. *AASHTO LRFD Bridge Design Specifications*, Eighth Edition, LRFD-8. American Association of State Highway and Transportation Officials, Washington, DC, 2017.

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A8—TWO-GIRDER STEEL BRIDGE: DESIGN LOAD RATING OF GIRDER AND FLOORBEAM

A8.1—Bridge Data

Span Length: 94 ft $8^1/_4$ in. (simple span)
Year Built: 1934
Material: Concrete f'_c = 3 ksi
Steel F_y = 33 ksi
Condition: No deterioration. NBI Item 59 Code = 6
Main girders are built-up, riveted plate girders
ADTT (one direction): Unknown
Skew: 0°

A8.2—Rating of Intermediate Floorbeam

Rolled Section:

*W*24 × 70# Noncomposite

A = 20.44 in.^{2}

I_z = 1,905.48 in.^{4}

S_z = 159.59 in.^{3}

Floorbeam Spacing: 9 ft $5^5/_8$ in. (9.47 ft)

(11 floorbeams counting ends)

Overlay Thickness: 2 in. (field measured)

As the overlay thickness was field measured, the load effects for *DC* and *DW* have been combined as the same load factor will apply for both loadings.

The cross section, Figure A8.2-1, shows all of the appurtenances contributing dead loads. The point loads and distributed loads due to the tributary areas of the appurtenances on an interior (intermediate) floorbeam are shown in Figure A8.2-2.

Rating factors are calculated for the maximum positive moment, maximum negative moment, and the maximum shear.

A8.3—Dead Load Force Effects

See Figure A8.2-2.

Table A8-1 Dead Load Force Effects

Location on Floorbeam	M_{DC+DW}	V_{DC+DW}	Effect
At East Girder	42.8 kip-ft	13.1 kips	*M*, *V* (left of G1)
At West Girder	33.4 kip-ft	12.1 kips	
Max M_D (8.63 ft from West Girder)	18.7 kip-ft	0 kips	
At 8.17 ft from West Girder	18.5 kip-ft	0.64 kips	+*M*

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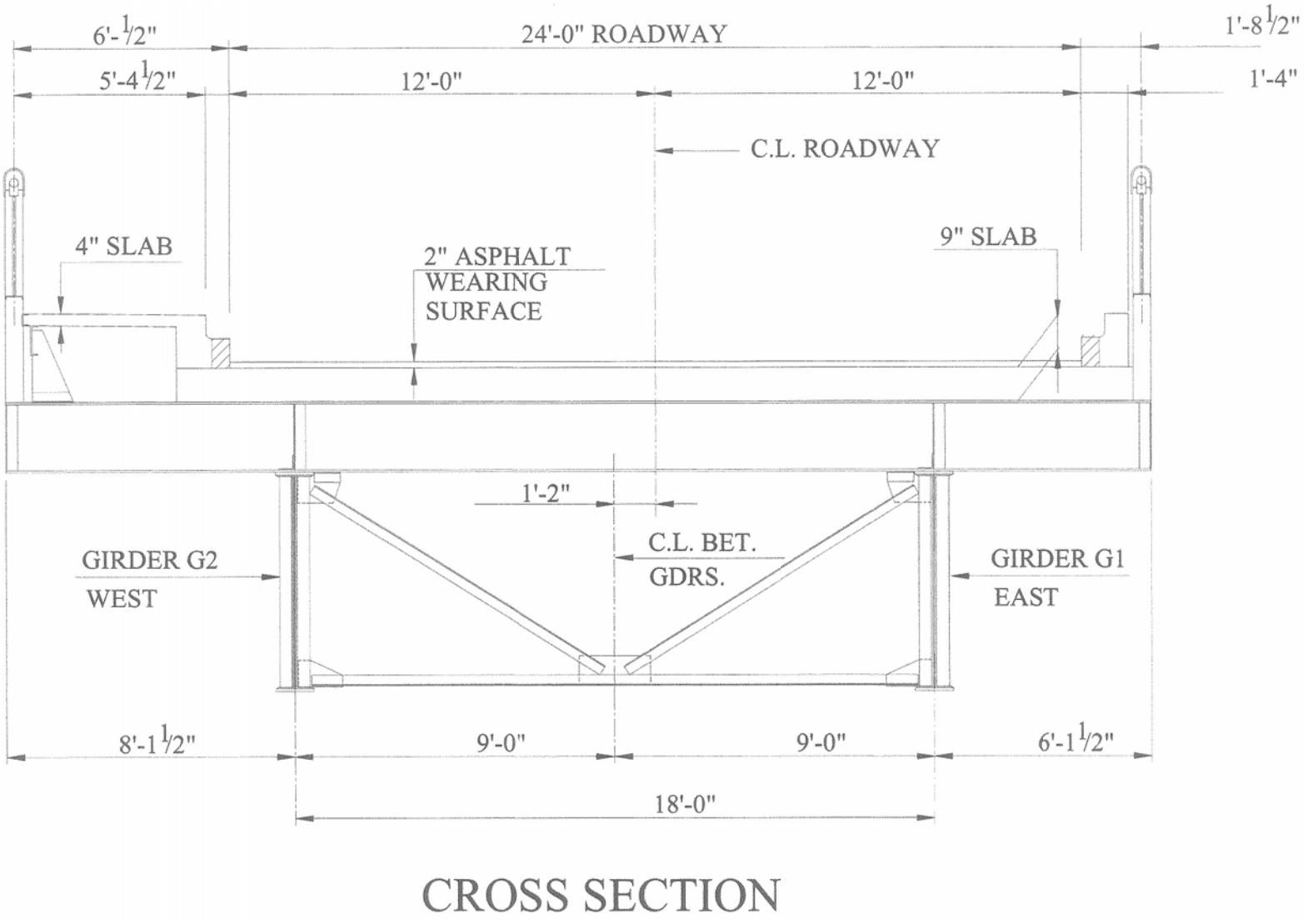

Figure A8.2-1—Cross Section—Two-Girder Steel Bridge

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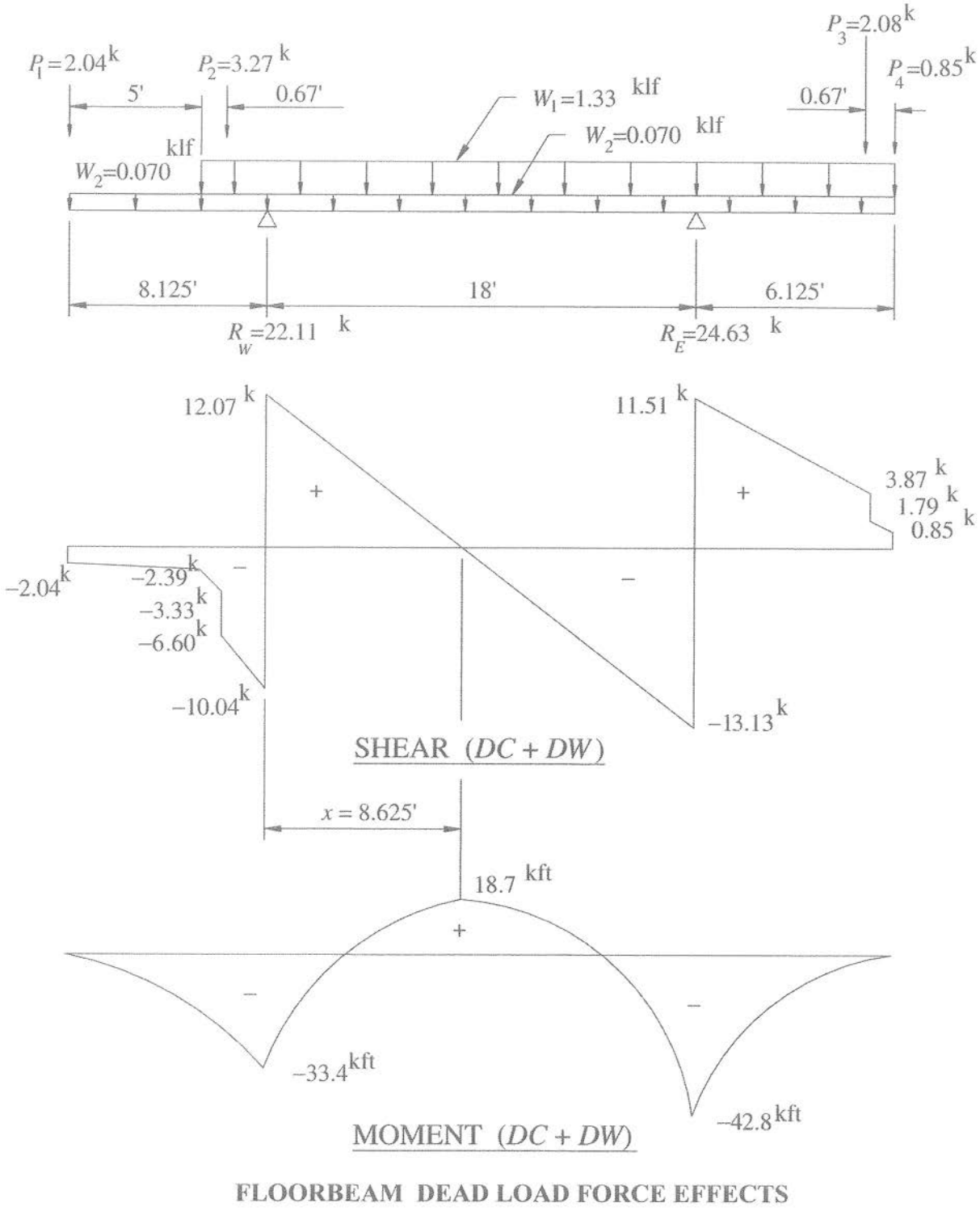

Figure A8.2-2—Intermediate Floorbeam Dead Load Force Effects

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A8.4—Live Load (HL-93) Force Effects

A8.4.1—Live Load (HL-93) Reactions on Intermediate Floorbeam

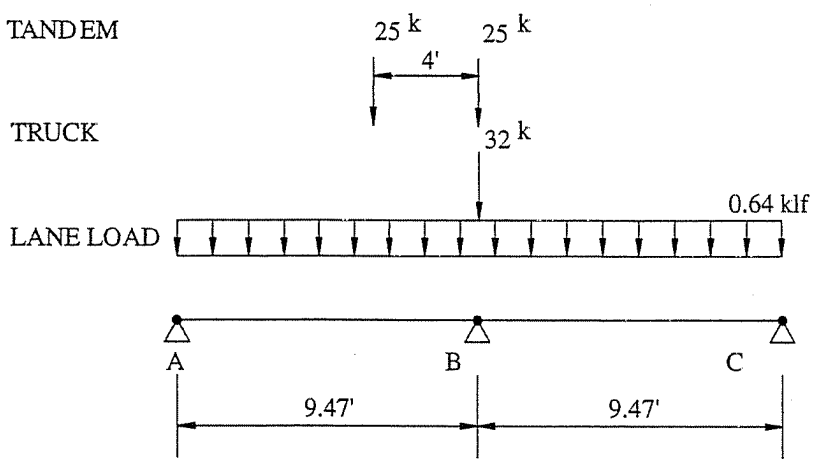

Figure A8.4.1-1—Critical Live Load Position for Reactions on Intermediate Floorbeam

Modeling deck as hinged at the floorbeams.

Reaction at Floorbeam B: LRFD Design Table 3.6.2.1-1

IM = 33 percent

Truck + Lane:

R_{LL+IM} = 32 kips × 1.33 + 0.64 × 9.47 ft

= 48.62 kips

Tandem + Lane:

$$R_{LL+IM} = \left(25 + 25 \times \frac{9.47 - 4}{9.47}\right) \times 1.33 + 0.64 \times 9.47$$

= 58.62 kips > 48.62 kips Governs

$$R_{Tandem} = \left(25 + 25 \times \frac{9.47 - 4}{9.47}\right) \times 1.33 = 52.46 \text{ kips}$$

$$R_{TandemWheel} = \frac{52.46}{2} = 26.23 \text{ kips} = P$$

R_{Lane} = 0.64 × 9.47 = 6.06 kips

$$R_{Lane\ per\ foot\ width} = \frac{6.06}{10} = 0.606 \text{ kip/ft} = W$$

A8.4.2—Live Load (HL-93) Maximum Positive Moment

Critical positions of the two lanes to produce maximum positive moment in the floorbeam

Multiple presence factor, m = 1.0 LFRD Design Table 3.6.1.1.2-1

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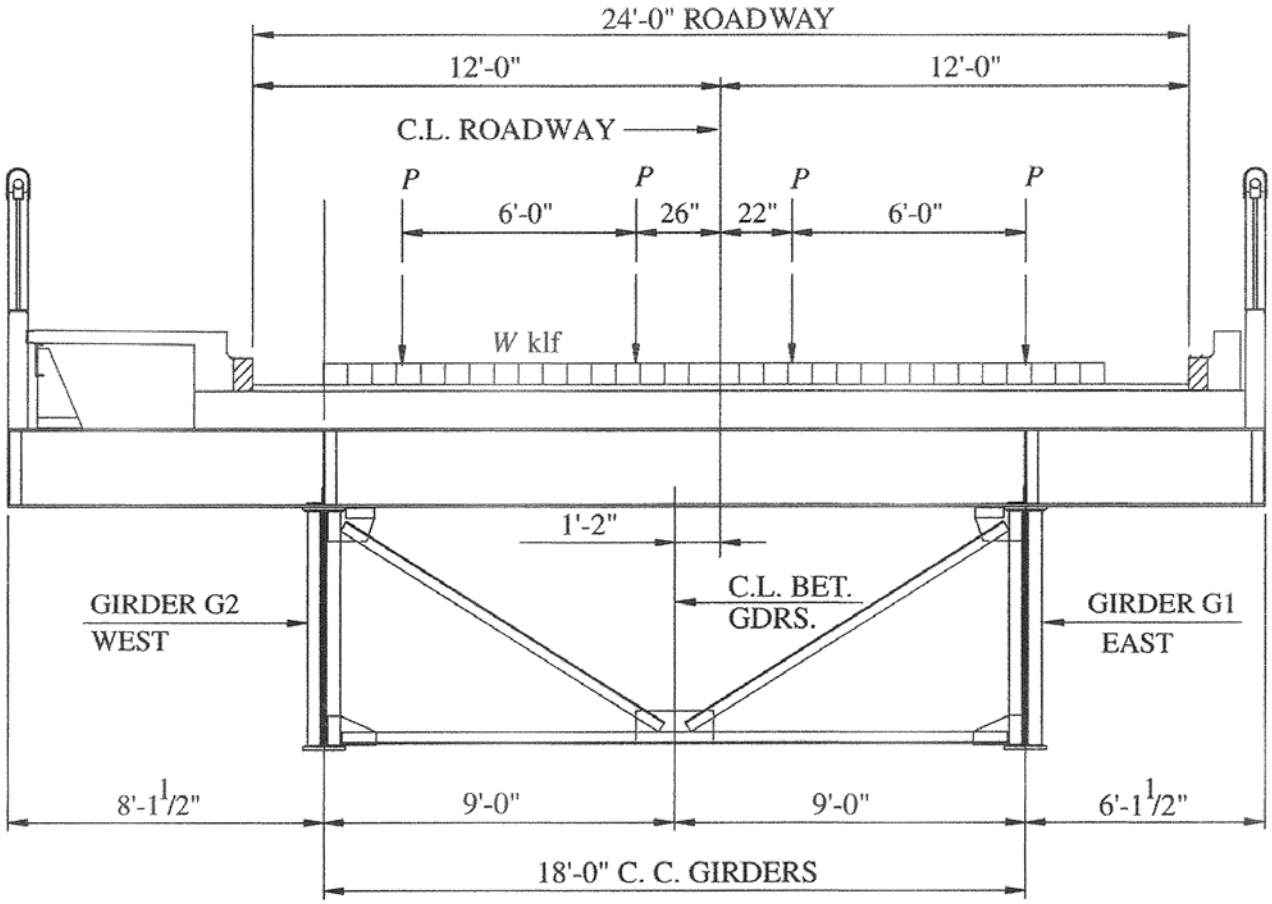

Figure A8.4.2-1—Critical Lane Positions for Maximum Positive Moment in Floorbeam

Maximum positive live load moment in floorbeam is at 8.17 ft from G2.

Two lanes occupying 12 ft each:

P = 26.23 kips

W = 0.606 kip/ft over two 10-ft adjacent sections.

Neglect the farthest east wheel load and the lane load overhanging G1 for the maximum floorbeam moment calculation.

The moment at 8.17 ft from G2 is calculated by statics. Each main girder is treated as a pinned support.

M_{LL+IM} = 242 kip-ft at 8.17 ft from West Girder (G2)

A8.4.3—Live Load (HL-93) Maximum Shear

Critical position of one loaded lane to produce maximum shear in the floorbeam

Multiple presence factor, m = 1.2 LFRD Design Table 3.6.1.1.2-1

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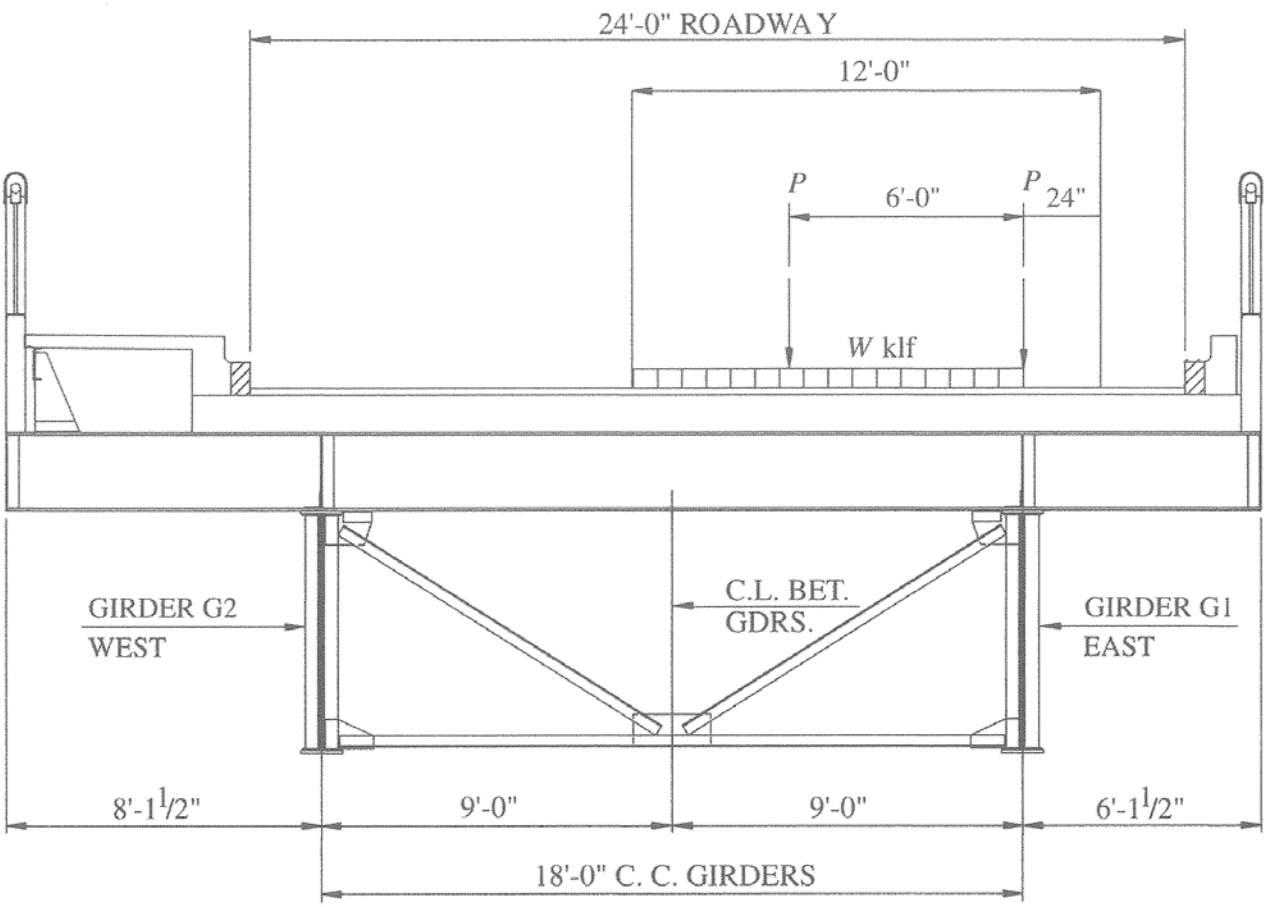

Figure A8.4.3-1—Critical Position to Produce Maximum Shear in the Floorbeam

Maximum live load shear in floorbeam is to the left of G1.

One lane loaded, wheel load just left of G1 is the governing case.

P = 26.23 kips

W = 0.606 kip/ft over one 10-ft section

The shear left of G1 is calculated by statics. Each main girder is treated as a pinned support. The multiple presence factor m for one lane loaded is 1.2.

The loading in Figure A8.4.3-1 results in a shear of 48.2 kips. Multiply by the multiple presence factor.

V_{LL+IM} = 48.2 × 1.2 = 57.8 kips at floorbeam section above and to the left of the east girder (G1)

A8.4.4—Live Load (HL-93) Maximum Negative Moment

Critical position of east lane to produce maximum negative moment in the floorbeam

Multiple presence factor, m = 1.2

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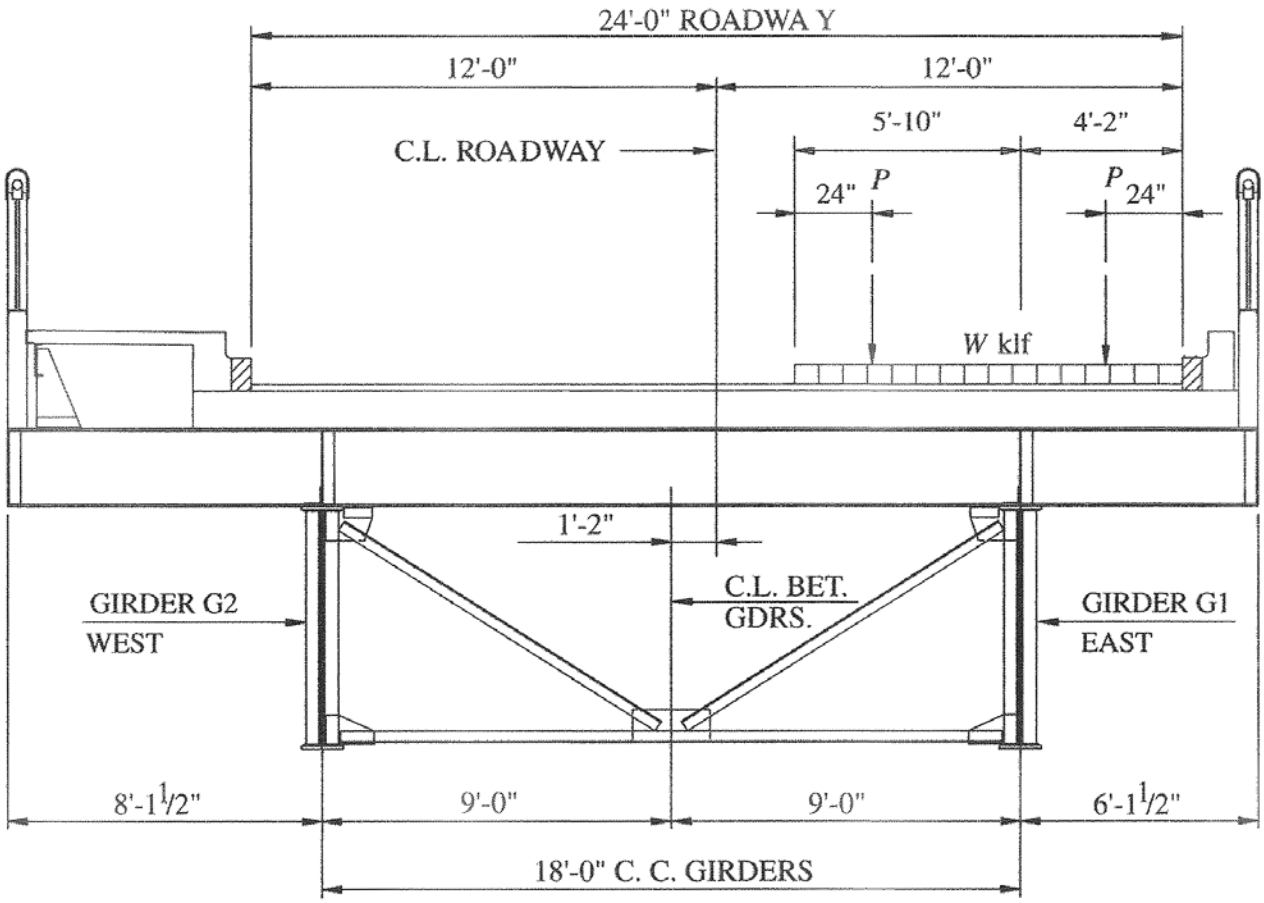

Figure A8.4.4-1—Critical Position to Produce Maximum Negative Moment in the Floorbeam

Maximum negative live load moment in floorbeam is at G1.

One lane loaded, loads positioned as far to the right as permitted in LRFD Design Article 3.6.1.3.1.

P = 26.23 kips

W = 0.606 kip/ft over one 10-ft section

The moment at G1 is calculated by statics. Each main girder is treated as a pinned support.

The loading in Figure A8.4.4-1 results in a moment of 62.2 kip-ft. Multiply by the multiple presence factor.

M_{LL+IM} = –62.2 × 1.2 = –74.7 kip-ft at the floorbeam section above the east girder (G1)

A8.5—Summary of Live Load (HL-93) Force Effects in Floorbeam

Location	M_{LL+IM}	V_{LL+IM}	Loading
At East Support	–74.7 kip-ft	–57.8 kips	one lane
8.17 ft from West Girder	242.0 kip-ft	0 kips	two lanes

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A8.6—Compute Nominal Resistance of Floorbeam

A8.6.1—Positive Moment Section—Noncomposite Construction

*W*24×70#, no deterioration

The following dimensions were assumed for the purpose of calculating this example:

t_w = 0.41 in.

b_f = 8.995 in.

D_w = 22.64 in.

t_f = 0.62 in.

Web slenderness check: LRFD Design 6.10.6.2.3

Minimum yield strength of flanges is less than 70 ksi, and:

$$\frac{2D_c}{t_w} < 5.7\sqrt{\frac{E}{F_{yc}}}$$ LRFD Design Eq. 6.10.6.2.3-1

LRFD Design D6.3.2

$$D_{cp} = \frac{D_w}{2} = \frac{22.64}{2} = 11.32 \text{ in.}$$

$$\frac{2D_c}{t_w} = \frac{D_w}{t_w} = \frac{22.64}{0.41} = 55.22$$

$$5.7\sqrt{\frac{E}{F_{yc}}} = 5.7\sqrt{\frac{29{,}000}{33}} = 169 > 55.22 \text{ OK}$$

and:

$$\frac{I_{yc}}{I_{yt}} = 1.0 > 0.3$$ LRFD Design Eq. 6.10.6.2.3-2

Compression flange is taken to be continuously braced by the concrete deck.

The optional provisions of LRFD Appendix A6 may be applied to determine the norminal flexural resistance of non-composite sections. LRFD Design C6.10.6.2.3

LRFD Design Article A6.2.1.

Sections that satisfy the following requirement shall qualify as compact web sections:

$$\frac{2D_{cp}}{t_w} \le \lambda_{pw(D_{cp})}$$ LRFD Design Eq. A6.2.1-1

$$\lambda_{pw(D_{cp})} = \frac{\sqrt{\frac{E}{F_{yc}}}}{\left(0.54\frac{M_p}{R_h M_y} - 0.09\right)^2} \le \lambda_{rw}\left(\frac{D_{cp}}{D_c}\right)$$ LRFD Design Eq. A6.2.1-2

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Plastic Moment: M_p LRFD Design D6.1

Flange Width:

b_c = 8.995 in.

Flange Thickness:

t_c = 0.62 in.

Top Flange:

$P_c = F_{yc} b_c t_c$

= 33 ksi × 8.995 in. × 0.62 in.

= 184.0 kips

Bottom Flange:

$P_t = P_c$ = 184.0 kips

Web:

$P_{wc} = P_{wt}$ = 33 ksi × 11.32 in. × 0.41 in.

= 153.2 kips

LRFD Design Article D6.1, Case I:

$$\bar{y} = \frac{D}{2} = \frac{22.64}{2} = 11.32 \text{ in.}$$

$$d_t = d_c = 11.32 + \frac{0.62}{2} = 11.63 \text{ in.}$$

$$M_p = \frac{P_w}{2D}\left[\left(\bar{y}\right)^2 + \left(D - \bar{y}\right)^2\right] + \left[P_c d_c + P_t d_t\right]$$

$$= \frac{2 \times 153.2}{2 \times 22.64}\left[11.32^2 + \left(22.64 - 11.32\right)^2\right] + \left[2 \times 184.0 \times 11.63\right]$$

= 6,014.1 kip-in.

= 501.2 kip-ft

Yield Moment, M_y:

$M_y = F_y S_z$

= 33 × 159.59

= 5,266.5 kip-in. = 438.9 kip-ft

R_h = 1.0 LRFD Design 6.10.1.10.1

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$$\lambda_{pw(D_{cp})} = \frac{\sqrt{\frac{29{,}000}{33}}}{(0.54\times\frac{501.2}{1.0\times438.9}-0.09)^2} = 106.9$$

$$\lambda_{rw}\left(\frac{D_{cp}}{D_c}\right) = 5.7\sqrt{\frac{E}{F_{yc}}}\left(\frac{D_{cp}}{D_c}\right) = 5.7\sqrt{\frac{29{,}000}{33}}(1.0) = 169$$

$$\lambda_{pw(D_{cp})} \le \lambda_{rw}\left(\frac{D_{cp}}{D_c}\right) \therefore \text{ use } 106.9$$

$$\frac{2D_{cp}}{t_w} = 55.22 \le \lambda_{pw(D_{cp})} = 106.9$$

The section qualifies as compact web section.

$$R_{pc} = \frac{M_p}{M_{yc}}$$ LRFD Design Eq. A6.2.1-4

$$R_{pc} = \frac{501.2}{438.9} = 1.14$$

Sections with continuously braced compression flanges LRFD Design A6.1.3

$$M_u = \varphi_f R_{pc} M_{yc}$$ LRFD Design Eq. A6.1.3-1

$$= \varphi_f \left(\frac{M_p}{M_{yc}}\right) M_{yc}$$

$= \varphi_f M_p$ where $\varphi_f = 1.0$ LRFD Design 6.5.4.2

$= 1.0 \times 501.2$

$= 501.2$ kip-ft

A8.6.2—Negative Moment Section

Sections with discretely braced compression flanges LRFD Design A6.1.1

$$M_u + \frac{1}{3} f_\ell S_{xc} \le \varphi_f M_{nc}$$ LRFD Design Eq. A6.1.1-1

M_{nc} = nomimal flexural resistance determined as specified in LRFD Design Article A6.3 (smaller of the local buckling resistance and lateral torsional buckling resistance)

Local buckling resistance LRFD Design A6.3.2

$$\lambda_f = \frac{b_{fc}}{2t_{fc}}$$ LRFD Design Eq. A6.3.2-3

$$= \frac{8.995}{2\times0.62} = 7.3$$

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$$\lambda_{pf} = 0.38\sqrt{\frac{E}{F_{yc}}}$$ LRFD Design Eq. A6.3.2-4

$$= 0.38\sqrt{\frac{29,000}{33}} = 11.3$$

$$\lambda_f \leq \lambda_{pf}$$

then:

$$M_{nc} = R_{pc}M_{yc}$$ LRFD Design Eq. A6.3.2-1

$$= \frac{M_p}{M_{yc}} \times M_{yc}$$

$$= M_p$$

Lateral torsional buckling resistance LRFD Design A6.3.3

The unbraced length, L_b, is taken as the distance between cross sections braced against twist and lateral displacement. While it is assumed that the deck continuously braces the top flange within this regon, there is no indication in the bridge data that intermediate stiffeners or bracing are present to prevent torsion of the section. Therefore, girders G1 and G2 are taken as brace points for the full beam cross section.

$$L_b = 18 \text{ ft} = 216 \text{ in.}$$

$$L_p = 1.0r_t\sqrt{\frac{E}{F_{yc}}}$$ LRFD Design Eq. A6.3.3-4

$$r_t = \frac{b_{fc}}{\sqrt{12\left(1+\frac{1}{3}\frac{D_c t_w}{b_{fc}t_{fc}}\right)}}$$ LRFD Design Eq. A6.3.3-10

$$r_t = \frac{8.995}{\sqrt{12\left[1+\frac{1}{3}\left(\frac{11.32\times 0.41}{8.995\times 0.62}\right)\right]}} = 2.3 \text{ in.}$$

$$L_p = 1.0\times 2.3\times\sqrt{\frac{29,000}{33}} = 68.2 \text{ in.}$$

$$L_r = 1.95r_t\frac{E}{F_{yr}}\sqrt{\frac{J}{S_{xc}h}}\sqrt{1+\sqrt{1+6.76\left(\frac{F_{yr}}{E}\frac{S_{xc}h}{J}\right)^2}}$$ LRFD Design Eq. A6.3.3-5

where J, the St. Venant torsional constant, is:

$$J = \frac{Dt_w^3}{3} + \frac{b_{fc}t_{fc}^3}{3}\left(1-0.63\frac{t_{fc}}{b_{fc}}\right) + \frac{b_{ft}t_{ft}^3}{3}\left(1-0.63\frac{t_{ft}}{b_{ft}}\right)$$ LRFD Design Eq. A6.3.3-9

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$$J = \frac{22.64\times 0.41^3}{3}+\frac{8.995\times(0.62)^3}{3}\left(1-0.63\frac{0.62}{8.995}\right)+\frac{8.995\times(0.62)^3}{3}\left(1-0.63\frac{0.62}{8.995}\right)$$

$$= 0.520 + 0.684 + 0.684$$

$$= 1.889$$

where h, distance between centerline of flanges, is:

h = 22.64 + 0.62 = 23.26 in.

and:

$F_{yr} = 0.7F_{yc} = 0.7\times 33$ ksi = 23.1 ksi

then:

$$L_r = 1.95\times 2.3\times\frac{29000}{23.1}\sqrt{\frac{1.889}{159.58\times 23.26}}\sqrt{1+\sqrt{1+6.76\left(\frac{23.1}{29000}\times\frac{159.58\times 23.26}{1.889}\right)^2}}$$

$$= 5630.52 \times 0.0226 \times 2.2783$$

$$= 289.9 \text{ in.} > L_b = 216 \text{ in.}$$

If $L_p < L_b \le L_r$, then $M_{nc} = C_b\left[1-\left(1-\frac{F_{yr}S_{xc}}{R_{pc}M_{yc}}\right)\left(\frac{L_b-L_p}{L_r-L_p}\right)\right]R_{pc}M_{yc} \le R_{pc}M_{yc}$ — LRFD Design Eq. A6.3.3-2

C_b is taken as 1.0 where $M_{mid}/M_2 > 1$ — LRFD Design Eq. A6.3.3-6

$$M_{nc} = 1.0\left[1-\left(1-\frac{23.1\times 159.59}{1.14\times 438.9\times 12}\right)\left(\frac{216-68.2}{289.9-68.2}\right)\right]1.14\times 438.9$$

$$= 0.74 \times 1.14 \times 438.9 = 370.3 \text{ kip-ft}$$

In general, the lateral torsional buckling resistance of the cantilever portion of the beam should also be checked.

In this bridge, the floorbeam's cross section is uniform:

$L_p = 68.2 \text{ in.} \le L_b = 73.5 \text{ in.} \le L_r = 289.9 \text{ in.}$

$C_b = 1.0$ — LRFD Design Eq. A6.3.3-6

By comparison, the unbraced length between girders determines the critical lateral torsional buckling resistance.

For negative moment section, compare:

Local buckling resistance:

$M_{nc} = M_p = 501.2$ kip-ft

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Lateral torsional buckling resistance:

M_{nc} = 370.3 kip-ft Governs

Therefore:

M_{nc} = 370.3 kip-ft

A8.6.3—Nominal Shear Resistance (unstiffened web) LRFD Design 6.10.9.2

$V_n = V_{cr} = CV_p$ LRFD Design Eq. 6.10.9.2-1

$V_p = 0.58F_{yw}Dt_w$ LRFD Design Eq. 6.10.9.2-2

Determine C, the ratio of shear buckling resistance to shear yield strength with k taken equal to 5.0.

$$\frac{D}{t_w} = \frac{22.64}{0.41} = 55.2$$

$$1.12\sqrt{\frac{Ek}{F_{yw}}} = 1.12\sqrt{\frac{29{,}000 \times 5.0}{33}} = 74.24$$ LRFD Design Eq. 6.10.9.3.2-4

$$\frac{D}{t_w} = 55.2 \le 74.24$$

$\therefore C = 1.0$

then: LRFD Design Eq. 6.10.9.2-2

$V_n = V_{cr} = CV_p = 1.0 \times 0.58F_{yw}Dt_w$

$= 1.0 \times 0.58 \times 33 \times 22.64 \times 0.41$

$= 177.7$ kips

A8.7—General Load-Rating Equation (6A.4.2)

$$RF = \frac{C - (\gamma_{DC})(DC) - (\gamma_{DW})(DW) \pm (\gamma_P)(P)}{(\gamma_L)(LL + IM)}$$ Eq. 6A.4.2.1-1

A8.7.1—Evaluation Factors (for Strength Limit States)

A8.7.1.1—Resistance Factor, φ (LRFD Design 6.5.4.2)

φ = 1.0 for flexure and shear

A8.7.1.2—Condition Factor, φ_c (6A.4.2.3)

φ_c = 1.0 No deterioration

A8.7.1.3—System Factor, φ_s (6A.4.2.4)

φ_s = 1.0 for floorbeams, floorbeam spacing < 12 ft

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A8.7.2—Design Load Rating (6A.4.3)

A8.7.2.1—Strength I Limit State (6A.6.4.1)

$$RF = \frac{(\varphi_c)(\varphi_s)(\varphi)R_n - (\gamma_{DC})(DC) - (\gamma_{DW})(DW)}{(\gamma_L)(LL+IM)}$$

Load	Inventory	Operating	
DC, *DW*	1.25	1.25	Asphalt thickness was field measured
LL + *IM*	1.75	1.35	

Table 6A.4.2.2-1

A8.7.2.1a—Flexure at 8.17 ft from West Girder (Max. Positive Live Load Moment)

Inventory:

$$RF = \frac{(1.0)(1.0)(1.0)(501.2) - (1.25)(18.5)}{(1.75)(242)} = 1.13$$

Operating:

$$RF = 1.13 \times \frac{1.75}{1.35} = 1.46$$

A8.7.2.1b—Flexure at East Girder (Max. Negative Moment)

Inventory:

$$RF = \frac{(1.0)(1.0)(1.0)(370.3) - (1.25)(42.8)}{(1.75)(74.7)} = 2.42$$

Operating:

$$RF = 2.42 \times \frac{1.75}{1.35} = 3.14$$

A8.7.2.1c—Shear at East Girder

Inventory:

$$RF = \frac{(1.0)(1.0)(1.0)(177.7) - (1.25)(13.1)}{(1.75)(57.8)} = 1.60$$

Operating:

$$RF = 1.60 \times \frac{1.75}{1.35} = 2.07$$

A8.7.2.2—Service II Limit State 6A.6.4.1

$$RF = \frac{f_R - (\gamma_D)(f_D)}{(\gamma_L)(f_{LL+IM})}$$

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A8.7.2.2a—At 8.17 ft from West Girder

$$f_f + \frac{f_\ell}{2} \le 0.80 R_h F_{yf}$$ LRFD Design Eq. 6.10.4.2.2-3

For homogeneous sections, R_h shall be taken as 1.0. LRFD Design 6.10.1.10.1

f_ℓ = 0.0 ksi LRFD Design 6.10.1.6

f_f = $0.80 \times 1.0 \times 33 = 26.4$ ksi

γ_D = $\gamma_{DC} = \gamma_{DW} = 1.0$ Table 6A.4.2.2-1

γ_L = 1.3 for Inventory

= 1.0 for Operating

$$f_D = \frac{(18.5)(12)}{159.59} = 1.39 \text{ ksi}$$

$$f_{LL+IM} = \frac{(242)(12)}{159.59} = 18.20 \text{ ksi}$$

Inventory:

$$RF = \frac{26.4 - 1.0(1.39)}{1.3(18.20)} = 1.06$$

Operating:

$$RF = 1.06 \times \frac{1.30}{1.00} = 1.38$$

A8.7.2.2b—At East Girder

$$f_D = \frac{(42.8)(12)}{159.59} = 3.22 \text{ ksi}$$

$$f_{LL+IM} = \frac{(74.7)(12)}{159.59} = 5.62 \text{ ksi}$$

Inventory:

$$RF = \frac{26.4 - 1.0(3.22)}{1.3(5.62)} = 3.17$$

Operating:

$$RF = 3.17 \times \frac{1.30}{1.00} = 4.12$$

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A8.8—Rating of East Girder (G1)

Section at midspan:

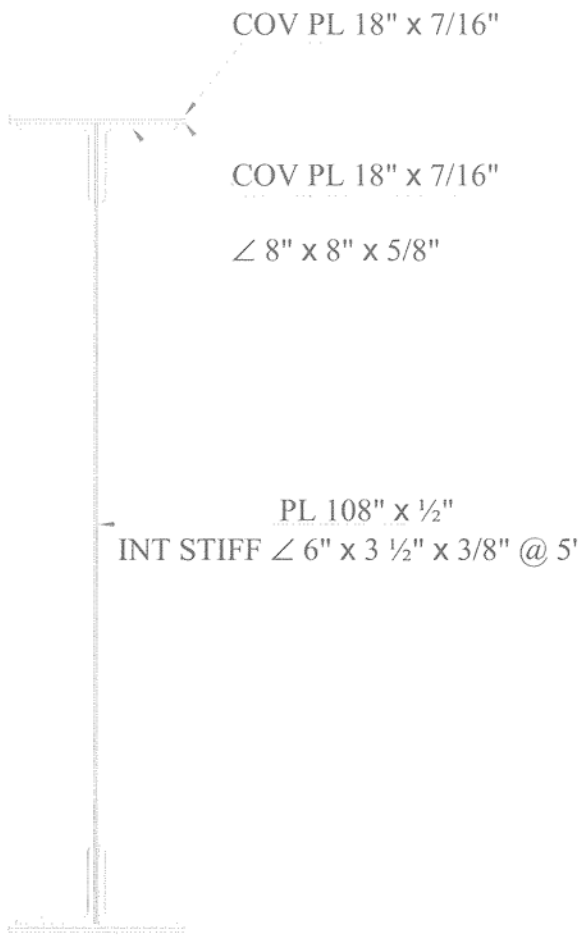

Figure A8.8-1—Girder Cross Section at Midspan

A8.9—Dead Load Force Effects

Each floorbeam transmits a concentrated load of 24.63 kips due to dead loads to the east girder. The built-up girder has a self weight of 0.49 kip/ft.

At Midspan:

M_{DC+DW} = 3,512.2 kip-ft

At Midspan:

S = 4,556 in.3 for the net section

At Girder End:

V_{DC+DW} = 136.0 kips

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A8.10—Live Load Analysis

Compute distribution factors for east girder:

Application of Live Load (HL-93)

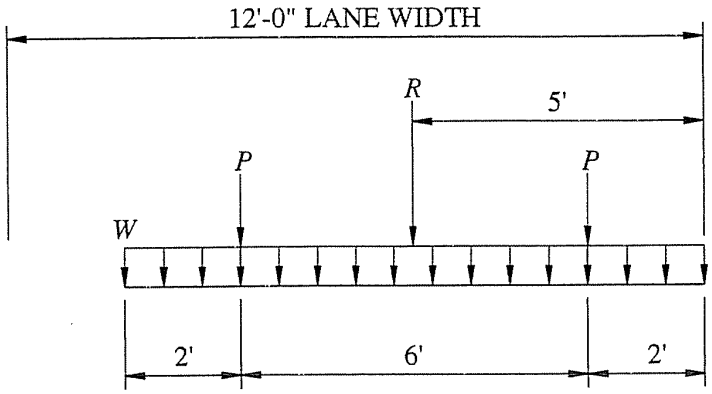

Figure A8.10-1—HL-93 Live Load Position within a Lane

R = Resultant Live Load — For calculating reactions, the resultant of each lane may be used instead of the wheel loads and distributed load.

Case of Only East Lane Loaded:

Multiple Presence Factor: LRFD Design 3.6.1.1.2

$m = 1.2$

Distribution Factor:

$$g_1 = \frac{18-\frac{10}{12}}{18}\times 1.2$$

$$= 1.14$$

Case of Both Lanes Loaded (see Figure A8.10-2):

Multiple Presence Factor:

$m = 1.0$

Distribution Factor:

$$g_2 = \left[\frac{18-\frac{10}{12}}{18}+\frac{18-\frac{10}{12}-12}{18}\right]\times 1.0$$

$$= 1.24 > 1.14$$

$g = g_2 = 1.24$

Axle loads are distributed between adjacent floorbeams assuming the deck acts as hinged at the floorbeams. The lane load imposes 6.06 kips per floorbeam as previously determined. Live loads are applied to the main girders as concentrated forces at the floorbeam locations.

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At Midspan: Moments due to HL-93

IM	=	33 percent	LRFD Design Table 3.6.2.1-1
Design Lane Load	=	717.4 kip-ft	
Design Truck	=	1425.0 kip-ft	Governs over Tandem
M_{LL+IM}	=	717.4 + 1425.0 × 1.33	
	=	2612.7 kip-ft	
$g \times M_{LL+IM}$	=	1.24 × 2612.7	
	=	3239.7 kip-ft	

At Girder End: Shear due to HL-93

IM	=	33 percent	
Design Lane Load	=	30.3 kips	
Design Truck	=	64.8 kips	Governs over Tandem
Design Tandem	=	48.9 kips	
V_{LL+IM}	=	30.3 kips + 64.8 kips × 1.33	
	=	116.5 kips	
$g \times V_{LL+IM}$	=	1.24 ×116.5 kips	
	=	144.4 kips	

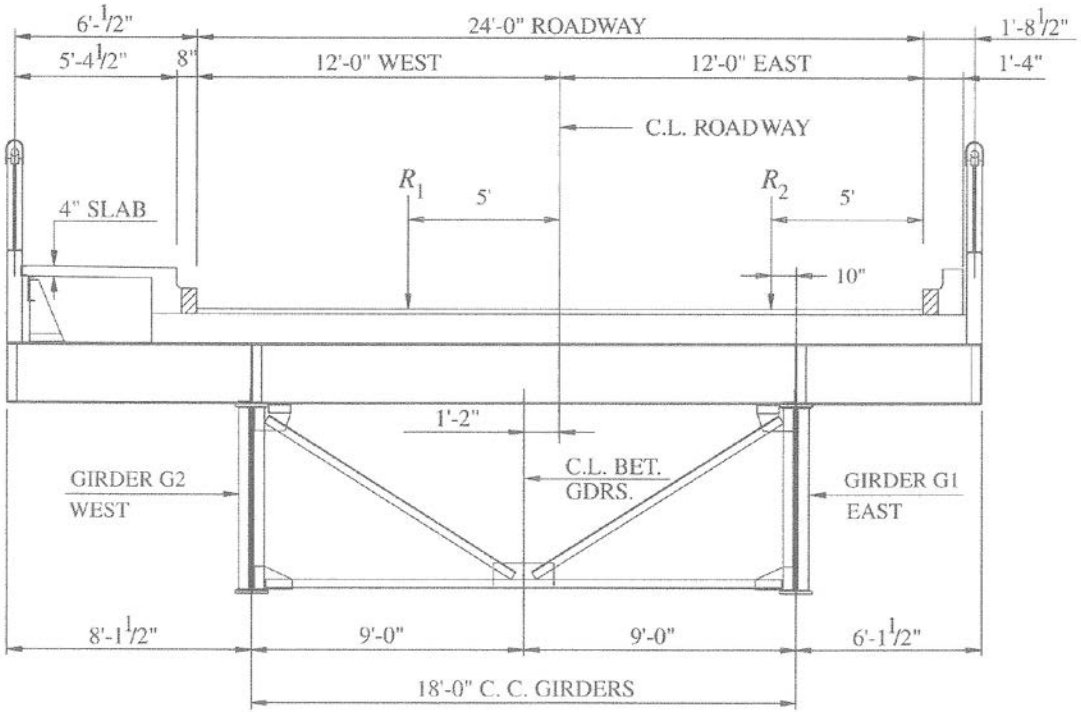

Figure A8.10-2—Critical Position of the Two Lanes to Produce Maximum Load on the East Girder G1

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A8.11—Compute Norminal Flexural Resistance of Section

Check web for noncompact slenderness limit:

$$\frac{2D_c}{t_w} < 5.7\sqrt{\frac{E}{F_{yc}}}$$ LRFD Design Eq. 6.10.6.2.3-1

$$\frac{2D_c}{t_w} = \frac{D_w}{t_w} = \frac{108}{0.5} = 216$$

$$5.7\sqrt{\frac{E}{F_{yc}}} = 5.7\sqrt{\frac{29,000}{33}} = 169 < 216$$

Provision specified in LRFD Design Article 6.10.8 shall apply.

For discretely braced flanges in compression:

$$f_{bu} + \frac{1}{3}f_\ell \le \varphi_f F_{nc}$$ LRFD Design Eq. 6.10.8.1.1-1

F_{nc} = Smaller of the local buckling and the lateral torsional buckling resistance as specified in LRFD Design Articles 6.10.8.2.2 and 6.10.8.2.3.

A8.11.1—Local Buckling Resistance

Slenderness ratio of the compression flange:

$$\lambda_f = \frac{b_{fc}}{2t_{fc}}$$ LRFD Design Eq. 6.10.8.2.2-3

At mid span:

$$\lambda_f = \frac{18}{2\left(2\times\frac{7}{16}+\frac{5}{8}\right)} = 6$$

$$\lambda_{pf} = 0.38\sqrt{\frac{E}{F_{yc}}}$$ LRFD Design Eq. 6.10.8.2.2-4

$$= 0.38\sqrt{\frac{29,000}{33}} = 11.2$$

$$\lambda_f = 6 \le \lambda_{pf} = 11.2$$

Then:

$$F_{nc} = R_b R_h F_{yc}$$ LRFD Design Eq. 6.10.8.2.2-1

$$R_h = 1.0$$ LRFD Design Eq. 6.10.1.10.1

Determine Load Shedding Factor R_b: LRFD Design 6.10.1.10.2

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$$\frac{2D_c}{t_w} \leq \lambda_{rw}$$ LRFD Design Eq. 6.10.1.10.2-2

$$\frac{2D_c}{t_w} = \frac{2\times 54}{0.5} = 216$$

$$\lambda_{rw} = 5.7\sqrt{\frac{E}{F_{yc}}} = 5.7\sqrt{\frac{29{,}000}{33}} = 169$$ LRFD Design Eq. 6.10.1.10.2-4

$$\frac{2D_c}{t_w} = 216 > \lambda_{rw} = 169$$

Therefore:

$$R_b = 1-\left(\frac{a_{wc}}{1{,}200+300a_{wc}}\right)\left(\frac{2D_c}{t_w}-\lambda_{rw}\right) \leq 1.0$$ LRFD Design Eq. 6.10.1.10.2-3

where:

$$\lambda_w = 5.7\sqrt{\frac{E}{F_{yc}}} = 169$$

$$a_{wc} = \frac{2D_c t_w}{A_c}$$ LRFD Design Eq. 6.10.1.10.2-5

A_c = compression flange area at midspan

= $b_{fc}t_{fc}$

= $(2 \times 8 \times {}^{5}/_{8} + 2 \times 18 \times {}^{7}/_{16}) = 25.75$

$$a_{wc} = \frac{2\times 54\times 0.5}{25.75} = 2.097$$

$$R_b = 1-\left(\frac{2.097}{1{,}200+300\times 2.097}\right)(216-169) = 0.946$$

$F_{nc} = 1.0\times 0.946\times 33 \text{ ksi} = 31.2 \text{ ksi}$

A8.11.2—Lateral Torsional Buckling Resistance (LRFD Design 6.10.8.2.3)

L_b = Unbraced length (in.)

= Spacing of floorbeams

= 9 ft $5^{5}/_{8}$ in. = 113.6 in.

$$L_p = 1.0r_t\sqrt{\frac{E}{F_{yc}}}$$ LRFD Design Eq. 6.10.8.2.3-4

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$$r_t = \frac{b_{fc}}{\sqrt{12\left(1+\frac{1}{3}\frac{D_c t_w}{b_{fc} t_{fc}}\right)}}$$ LRFD Design Eq. 6.10.8.2.3-9

$$= \frac{18}{\sqrt{12\left(1+\frac{1}{3}\times\frac{54\times 0.5}{2\times 8\times\frac{5}{8}+2\times 18\times\frac{7}{16}}\right)}}$$

$$= 4.5 \text{ in.}$$

$$L_p = 1.0\times 4.5\sqrt{\frac{29{,}000}{33}} = 133.4 \text{ in.}$$

$$L_b = 113.6 \text{ in.} \le L_p = 133.4 \text{ in.}$$

Then: LRFD Design Eq. 6.10.8.2.3-1

$$F_{nc} = R_b R_h F_{yc}$$

$$= 0.946\times 1.0\times 33 = 31.2 \text{ ksi}$$

A8.12—General Load-Rating Equation (6A.4.2)

$$RF = \frac{C-(\gamma_{DC})(DC)-(\gamma_{DW})(DW)\pm(\gamma_P)(P)}{(\gamma_L)(LL+IM)}$$ Eq. 6A.4.2.1-1

A8.12.1—Evaluation Factors (for Strength Limit States)

A8.12.1.1—Resistance Factor, φ LRFD Design 6.5.4.2

$\varphi_f = \varphi_v = 1.0$ For flexure and shear

A8.12.1.2—Condition Factor, φ_c 6A.4.2.3

$\varphi_c = 1.0$ No deterioration NBI Item 59 Code = 6

A8.12.1.3—System Factor, φ_s 6A.4.2.4

$\varphi_s = 0.90$ For flexure, riveted two-girder system

$\varphi_s = 1.00$ For shear

A8.12.2—Design Load Rating (6A.4.3)

Table 6A.4.2.2-1

Load	Inventory	Operating	
DC, *DW*	1.25	1.25	Asphalt thickness was field measured
LL + *IM*	1.75	1.35	

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A8.12.2.1—Flexure

A8.12.2.1a—Strength I Limit State 6A.6.4.1

Flexural stresses at midspan (unfactored):

$$f_{DC+DW} = \frac{M_{DC+DW}}{S} = \frac{3{,}512.2 \times 12}{4{,}556} = 9.25 \text{ ksi}$$

$$f_{LL+IM} = \frac{M_{LL+IM}}{S} = \frac{3{,}239.7 \times 12}{4{,}556} = 8.53 \text{ ksi}$$

Resistance at midspan:

F_n = 31.2 ksi

Inventory:

$$RF = \frac{(1.0)(0.90)(1.0)(31.2) - (1.25)(9.25)}{(1.75)(8.53)}$$

$$= 1.11$$

Operating:

$$RF = 1.11 \times \frac{1.75}{1.35}$$

$$= 1.44$$

A8.12.2.1b—Service II Limit State 6A.6.4.1

Since the section is non-composite and noncompact, the Service II limit state does not need to be checked for the Design Load Rating as discussed in Example A5 (will not govern load ratings).

A8.12.2.2—Shear 6A.6.4.1

A8.12.2.2a—Strength I Limit State

Shear forces at girder ends:

V_{DC+DW} = 136.0 kips

V_{LL+IM} = 144.4 kips

Girder Web:

D = 108 in. = 9 ft

t_w = $^1/_2$ in.

Transverse stiffener spacing = 5 ft

Required end panel transverse stiffener spacing (for stiffened girders) < 1.5D LRFD Design 6.10.9.3.3

1.5D = 13.5 ft > 9 ft OK

Interior panel ratings have not been illustrated here.

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Shear Resistance of End Panel: LRFD Design 6.10.9.3.3

$V_n = CV_p$

Determine C:

$$\frac{D}{t_w} = \frac{108}{0.5} = 216$$ LRFD Design Eq. 6.10.9.3.2-7

$$k = 5 + \frac{5}{\left(\frac{d_o}{D}\right)^2} = 5 + \frac{5}{\left(\frac{60}{108}\right)^2} = 21.2$$

$$1.12\sqrt{\frac{Ek}{F_{yw}}} = 1.12\sqrt{\frac{29{,}000 \times 21.2}{33}}$$ FAIL

$$= 153 < \frac{D}{t_w} = 216$$

$$1.40\sqrt{\frac{Ek}{F_{yw}}} = 1.40\sqrt{\frac{29{,}000 \times 21.2}{33}}$$

$$= 191 < \frac{D}{t_w} = 216$$ FAIL

If:

$$\frac{D}{t_w} > 1.40\sqrt{\frac{Ek}{Fyw}}$$

Then:

$$C = \frac{1.57}{\left(\frac{D}{t_w}\right)^2}\left(\frac{Ek}{F_{yw}}\right)$$ LRFD Design Eq. 6.10.9.3.2-6

$$= \frac{1.57}{216^2}\left(\frac{29{,}000 \times 21.2}{33}\right)$$

$$= 0.627$$

$$V_p = 0.58F_{yw}Dt_w$$ LRFD Design Eq. 6.10.9.3.3-2

$$= 0.58 \times 33 \times 108 \times 0.5$$

$$= 1{,}033.6 \text{ kips}$$

$$V_n = CV_p$$ LRFD Design Eq. 6.10.9.3.3-1

$$= 0.627 \times 1{,}033.6 \text{ kips}$$

$$= 648.0 \text{ kips}$$

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Inventory:

$$RF = \frac{(1.0)(1.0)(1.0)(648.0)-(1.25)(136.0)}{(1.75)(144.4)}$$

$$= 1.89$$

Operating:

$$RF = 1.89 \times \frac{1.75}{1.35}$$

$$= 2.45$$

As the bridge has sufficient capacity ($RF > 1.0$) for the HL-93 loading, further evaluation for legal loads is not required.

A8.13—Summary of Rating Factors

Table A8.13-1—Summary of Rating Factors—Floorbeam

Limit State			Design Load Rating (HL-93)	
			Inventory	Operating
Strength I	Flexure	Max $+M$	1.13	1.46
		Max $-M$	2.42	3.14
	Shear		1.60	2.07
Service II	Flexure Max $+M$ (Governs)		1.06	1.38

Table A8.13-2—Summary of Rating Factors—Girder

Limit State		Design Load Rating (HL-93)	
		Inventory	Operating
Strength I	Flexure	1.11	1.44
	Shear	1.89	2.45

A8.14—Reference

AASHTO. *AASHTO LRFD Bridge Design Specifications*, Eighth Edition, LRFD-8. American Association of State Highway and Transportation Officials, Washington, DC, 2017.

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A9—P/S CONCRETE ADJACENT BOX-BEAM BRIDGE: DESIGN LOAD AND PERMIT LOAD RATING OF AN INTERIOR BEAM

Note: This example demonstrates the rating calculations for moment at the centerline of a prestressed concrete adjacent box beam bridge.

A9.1—Bridge Data

Span Length: 70 ft (simple span)
Year Built: 1988
Concrete: f'_c = 5 ksi (P/S beam)
f'_{ci} = 4 ksi (P/S beam at transfer)
Prestressing Steel: $^1/_2$ in. diameter, 270 ksi stress-relieved strand
Reinforcing Steel: Grade 60
Condition: No deterioration. NBI Item 59 Code = 7
Riding Surface: Field verified and documented: Smooth approach and deck
ADTT (one direction): 4600
Skew: 0°

A9.1.1—Section Properties

48 in. × 33 in. Box Beams

A = 753 in.^{2}

I_x = 110,499 in.^{4}

S_{bot} = 6,767 in.^{3}

S_{top} = 6,629 in.^{3}

A9.2—Dead Load Analysis—Interior Beam

The beams are sufficiently transversley post-tensioned to act as a unit. Conditions given in LRFD Design Article 4.6.2.2.1 are also satisfied. Therefore, permanent loads due to barrier, wearing surface, and utilities may be uniformly distributed among the beams.

A9.2.1—Components and Attachments, *DC*

Beam Self Weight (including diaphragms) = 0.815 kip/ft

Sidewalks:

$$2\left(\frac{10.25}{12}\times 7\times 0.150\right)\frac{1}{12} = 0.150 \text{ kip/ft}$$

Parapets:

$$2\left(1.0\times 2.25\times 0.150\right)\frac{1}{12} = 0.056 \text{ kip/ft}$$

Railing:

$$2\times 0.02 \text{ kip/ft}\times\frac{1}{12} = 0.003 \text{ kip/ft}$$

Total DC = 1.024 kip/ft

M_{DC} = $M_{DC} = \frac{1}{8}\times 1.024\times 70^2$

= 627.2 kip-ft

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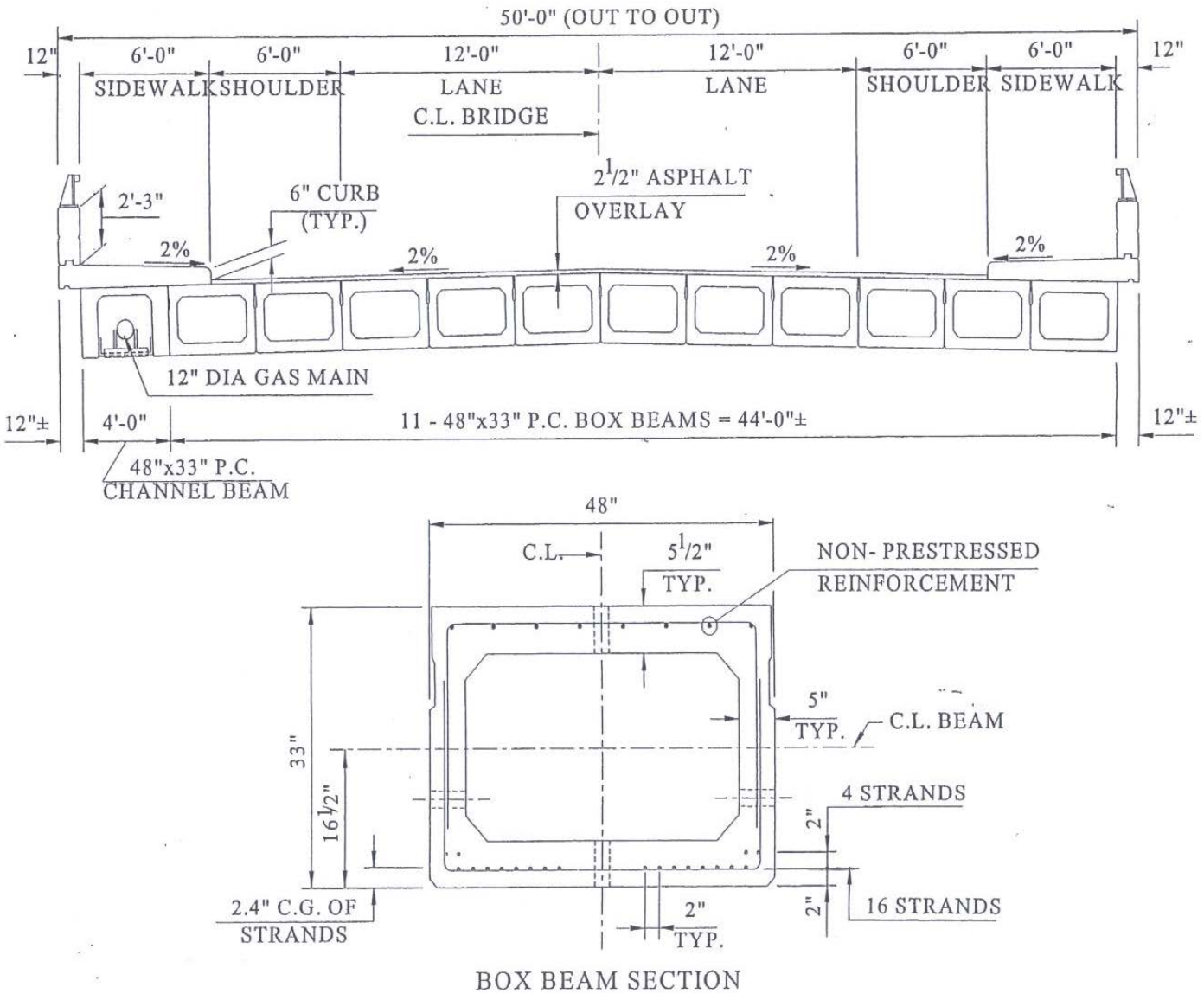

Figure A9.2.1-1—Bridge Cross Section and Interior Box Beam Cross Section

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Asphalt thickness = $2^1/_2$ in. (not field measured)

A9.2.2—Wearing Surface and Utilities, *DW*

Asphalt Overlay:

$$\frac{2.5}{12}\times 36.0\times 0.144\times\frac{1}{12}=0.09\,\text{kip/ft}$$

12-in. Gas Main:

$$0.05\text{ kip/ft}\times\frac{1}{12}=0.005\text{ kip/ft}$$

Total *DW* = 0.095 kip/ft

$$M_{DW} = \frac{1}{8}\times 0.095\times 70^2$$

= 58.2 kip-ft

A9.3—Live Load Analysis—Interior Girder

Type (g) cross section. LRFD Design Table 4.6.2.2.1-1

The beams are transversely post-tensioned to act as a unit.

A9.3.1—Compute Live Load Distribution Factors for an Interior Beam (LRFD Design Table 4.6.2.2.2b-1)

N_b = 12

$k = 2.5\left(N_b\right)^{-0.2} \geq 1.5$

= $(2.5)(12)^{-0.2} = 1.52$ Say 1.5

I = 110,499 in.^{4}

b = 48 in.

For closed thin-walled shapes: LRFD Design Eq. C.4.6.2.2.1-3

$$J = \frac{4A_o^2}{\Sigma\frac{s}{t}}$$

A_o = Area enclosed by the centerlines of elements

= $(48-5)(33-5\ ^1/_2) = 1{,}182.5$ in.^{2}

s = Length of a side element

$$J = \frac{4\times 1{,}182.5^2}{\frac{2(48-5)}{5.5}+\frac{2(33-5.5)}{5}}$$

= 209,985 in.^{4}

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A9.3.1.1—Distribution Factor for Moment

One Lane Loaded:

$$g_{m1} = k\left(\frac{b}{33.3L}\right)^{0.5}\left(\frac{I}{J}\right)^{0.25}$$

$$= 1.50\left(\frac{48}{33.3\times 70}\right)^{0.50}\left(\frac{110,499}{209,985}\right)^{0.25}$$

$$= 0.183$$

Two or More Lanes Loaded:

$$g_{m2} = k\left(\frac{b}{305}\right)^{0.6}\left(\frac{b}{12L}\right)^{0.2}\left(\frac{I}{J}\right)^{0.06}$$

$$= 1.50\left(\frac{48}{305}\right)^{0.6}\left(\frac{48}{12\times 70}\right)^{0.2}\left(\frac{110,499}{209,985}\right)^{0.06}$$

$$= 0.268 > 0.183$$

$$g_m = g_{m2} = 0.268$$

A9.3.2—Maximum Live Load (HL-93) Moment at Midspan

Design Lane Load:

$$0.64\,\text{klf} \times \frac{(70\ \text{ft})^2}{8} = 392.0\ \text{kip-ft}$$

Design Truck (with the middle axle positioned at midspan):

$$\frac{32^K \times 70\,\text{ft}}{4} + \frac{\left(8^K + 32^K\right)\times 21\,\text{ft}\times 35\,\text{ft}}{70} = 980.0\ \text{kip-ft} \quad \text{Governs}$$

Design Tandem (with tandem centered on midspan):

$$25^K \times 33\,\text{ft} = 825.0\ \text{kip-ft}$$

IM = 33 percent

M_{LL+IM} = 392.0 + 980.0 × 1.33

= 1,695.4 kip-ft

g × M_{LL+IM} = (0.268)(1,695.4)

= 454.4 kip-ft

A9.4—Compute Nominal Flexural Resistance

$$f_{ps} = f_{pu}\left(1 - k\frac{c}{d_p}\right)$$

LRFD Design Eq. 5.6.3.1.1-1

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$k = 2(1.04 - \frac{f_{py}}{f_{pu}}) = 2(1.04 - \frac{229.5}{270}) = 0.38$ — LRFD Design Eq. 5.6.3.1.1-1, Table 6A.5.4.2.2b-1

f_{pu} = 270 ksi

d_p = distance from extreme compression fiber to the *C.G.* of prestressing tendons

= 33 in. – 2.4 in.

= 30.6 in.

For rectangular section:

$$c = \frac{A_{ps} f_{pu}}{\alpha_1 f_c' \beta_1 b + k A_{ps} \frac{f_{pu}}{d_p}}$$ — LRFD Design Eq. 5.6.3.1.1-4

Neglects nonprestressed reinforcement.

A_{ps} = 20×0.153

= 3.06 in.^{2}

b = 48 in.

f'_c = 5 ksi

α_1 = 0.85 — LRFD Design 5.6.2.2

β_1 = $0.85 - 0.05(f_c' - 4) = 0.85 - 0.05(5-4) = 0.80$ — LRFD Design 5.6.2.2

$$c = \frac{3.06 \times 270}{0.85 \times 5 \times 0.80 \times 48 \times 0.38 \times 3.06 \times \frac{270}{30.6}}$$ — LRFD Design 5.6.3.1.1

= 4.76 in.

a = $\beta_1 c$ — LRFD Design 5.6.2.2

= 0.80×4.76

= 3.81 in. < 5.5 in.

Therefore, the rectangular section behavior assumption is valid.

$$f_{ps} = 270\left(1 - 0.38 \times \frac{4.76}{30.6}\right)$$

= 254.0 ksi

$M_n = A_{ps} f_{ps}\left(d_p - \frac{a}{2}\right)$ — LRFD Design Eq. 5.6.3.2.2-1

$$= 3.06 \times 254.0\left(30.6 - \frac{3.81}{2}\right)\frac{1}{12}$$

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= 1,858.6 kip-ft

A9.5—Maximum Reinforcement (C6A.5.5)

The factored resistance (φ factor) of compression controlled sections shall be limited in accordance with LRFD Design Article 5.6.2.1. This approach limits the capacity of over-reinforced (compression controlled) sections. C6A.5.5

The net tensile strain, ε_t, is the tensile strain at nominal strength and determined by strain compatibility using similar triangles. LRFD Design C5.6.2.1

Given an allowable concrete strain of 0.003 and depth to neutral axis $c = 4.76$ in. and a depth from the extreme concrete compression fiber to the center of gravity of the prestressing strands, $d_p = 30.6$ in.

$$\frac{\varepsilon_c}{c} = \frac{\varepsilon_t}{d-c}$$

$$\frac{0.003}{4.76 \text{ in.}} = \frac{\varepsilon_t}{30.6 \text{ in.} - 4.76 \text{ in.}}$$

$$\varepsilon_t = 0.0163$$

For $\varepsilon_t = 0.0163 > 0.005$, the section is tension controlled and Resistance Factor, φ, shall be taken as 1.0. LRFD Design 5.6.2.1, 5.5.4.2

A9.6—Minimum Reinforcement 6A.5.6

Amount of reinforcement must be sufficient to develop M_r equal to the lesser of: LRFD Design 5.6.3.3

$1.33M_u$ or M_{cr}

$$M_r = \varphi M_n = (1.0)(1,858.6) = 1,858.6$$

$$M_u = 1.75(454.4) + 1.25(627.2) + 1.5(58.2) = 1,666.5$$

$1.33M_u = 2,216.4 > M_r$ check $M_r \geq M_{cr}$

$$M_{cr} = \gamma_3 \left[\left(\gamma_1 f_r + \gamma_2 f_{cpe} \right) S_c - M_{dnc} \left(\frac{S_c}{S_{nc}} - 1 \right) \right]$$ LRFD Design Eq. 5.6.3.3-1

Where a monolithic or noncomposite section is designed to resist all the loads, S_{nc} is substituted for S_c. Therefore:

$$M_{cr} = \gamma_3 \left[\left(\gamma_1 f_r + \gamma_2 f_{cpe} \right) S_{nc} \right]$$

γ_3 = 1.0 for prestressing steel LRFD Design 5.6.3.3

γ_1 = 1.6

γ_2 = 1.1 for bonded tendons

S_{nc} = $S_b = 6,767$ in.^{3}

f_{cpe} = compressive stress in concrete due to effective prestress force (after allowance for all prestress losses) at extreme fiber of section where tensile stress is caused by externally applied loads

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$$f_{cpe} = \frac{P_{pe}}{A} + \frac{P_{pe}e}{S_b}$$

where:

P_{pe} = effective prestress force

Modulus of Rupture: LRFD Design 5.4.2.6

$$f_r = 0.24\lambda\sqrt{f_c'}$$

λ = 1.0

$f_r = 0.24(1.0)\sqrt{5}$ LRFD Design 5.4.2.8

= 0.537 ksi

A9.6.1—Determine Effective Prestress Force, P_{pe}

$P_{pe} = A_{ps}f_{pe}$

Total Prestress Losses: LRFD Design Eq. 5.9.3.1-1

Δf_{pT} = $\Delta f_{pES} + \Delta f_{pLT}$ immediately before transfer

Effective Prestress:

f_{pe} = Initial Prestress – Total Prestress Losses

A9.6.1.1—Loss Due to Elastic Shortening, Δf_{pES} (LRFD Design 5.9.3.2.3a)

$$\Delta f_{pES} = \frac{E_p}{E_{ct}} f_{cgp}$$ LRFD Design Eq. 5.9.3.2.3a-1

$$f_{cgp} = \frac{P_i}{A} + \frac{P_i e^2}{I} - \frac{M_D e}{I}$$

Initial Prestress immediately prior to transfer = $0.7f_{pu}$ if not available in plans. (This value assumes stress-relieved strand.) 6A.5.2.3

For estimating P_i immediately after transfer, use $0.90(0.7f_{pu})$. LRFD Design C5.9.3.2.3a

P_i = $0.90 \times (0.7 \times 270)\ 20 \times 0.153$

= 520.5 kips

A = 753 in.^{2}

I = 110,499 in.^{4}

e = 16.5 in. – 2.4 in.

= 14.1 in.

M_D = Moment due to self-weight of the member

$$= \frac{1}{8} \times 0.815 \times 70^2 = 499.2 \text{ kip-ft}$$

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f_{cgp} = $\frac{520.5}{753}+\frac{520.5\times14.1^2}{110,499}-\frac{499.2\times14.1\times12}{110,499}$

= 0.691 + 0.936 – 0.764

= 0.863 ksi

E_{ct} = $33,000\,K_1\,(w_c)^{1.5}\,\sqrt{f'_{ct}}$ LRFD Design Eq. C5.4.2.4-2

= $33,000\,(1.0)(0.145)^{1.5}\,\sqrt{4.0}$

= 3,644 ksi

E_p = 28,500 ksi LRFD Design 5.4.4.2

Δf_{pES} = $\frac{28,500}{3,644}\times0.863$ LRFD Design Eq. 5.9.3.2.3a-1

= 6.750 ksi

A9.6.1.2—Refined Estimates of Time-Dependent Losses (LRFD Design 5.9.3.4)

Δf_{pLT} = $(\Delta f_{pSR}+\Delta f_{pCR}+\Delta f_{pR1})id+(\Delta f_{pSD}+\Delta f_{pCD}+\Delta f_{pR2}-\Delta f_{pSS})df$ LRFD Design Eq. 5.9.3.4-1

$(\Delta f_{pSD}+\Delta f_{pCD}+\Delta f_{pR2}-\Delta f_{pSS})$ = 0 (No composite deck)

Shrinkage of Girder Concrete

Δf_{pSR} = $E_{bid}E_pK_{id}$ LRFD Design Eq. 5.9.3.4.2a-1

E_{bid} = $K_sK_{hs}K_fK_{td}0.48\times10^{-3}$ LRFD Design Eq. 5.4.2.3.3-1

K_s = $1.45–0.13\left(\frac{V}{S}\right)\geq1.0$ LRFD Design Eq. 5.4.2.3.2-2

$\frac{V}{S}$ = $\frac{753}{(48+48+33+33+81)}$ = 3.10 in.

K_s = 1.45–0.13(3.10) = 1.05

K_{hs} = $2.00–0.014H$ LRFD Design Eq. 5.4.2.3.3-2

H = average annual humidity; assume 70 percent

K_{hs} = 2.00–0.014(70) = 1.02

K_f = $\frac{5}{1+f'_{ci}}$ LRFD Design Eq. 5.4.2.3.2-4

K_f = $\frac{5}{1+4}=1.0$

K_{td} = $\frac{t}{12\left(\frac{100-4(f'_{ci})}{f'_{ci}+20}\right)+t}$ LRFD Design Eq. 5.4.2.3.2-5

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For time development factor at time of transfer of prestressing force, assume 0.75 days.

$$K_{td1} = \frac{0.75}{12\left(\frac{100-4(4)}{4+20}\right)+0.75} = 0.018$$

For time development factor at final design time, assume 30 years.

t_f = 30 × 365 = 10,950 days

$$K_{td2} = \frac{10,950}{12\left(\frac{100-4(4)}{4+20}\right)+10,950} = 0.996$$

E_{bid} = $1.05(1.02)(1.0)(0.996–0.018)(0.48\times10^{-3}) = 5.02\times10^{-4}$

$$K_{id} = \frac{1}{1+\frac{E_p}{E_{ci}}\frac{A_{ps}}{A_g}\left(1+\frac{A_g e_{pg}^2}{I_g}\right)\left[1+0.7\psi_b\left(t_f,t_i\right)\right]}$$

LRFD Design Eq. 5.9.3.4.2-2

$$\psi_b\left(t_f,t_i\right) = 1.9K_sK_{hc}K_fK_{td}t_i^{-0.118}$$

LRFD Design Eq. 5.4.2.3.2-1

K_{hc} = $1.56 - 0.008H$

LRFD Design Eq. 5.4.2.3.2-3

= $1.56 - 0.008(70) = 1.0$

$$\psi_b\left(t_f,t_i\right) = 1.9(1.05)(1.0)(1.0)(0.996-0.018)(0.75)^{-0.118}$$

= 2.02

$$K_{id} = \frac{1}{1+\left(\frac{28,500}{3,644}\right)\left(\frac{3.06}{753}\right)\left(1+\frac{753(14.1)^2}{110,499}\right)\left[1+0.7(2.02)\right]}$$

= 0.847

f_{pSR} = $(5.02\times10^{-4})(28,500)(0.847) = 12.12$ ksi

Creep of Girder Concrete

$$\Delta f_{pCR} = \frac{E_p}{E_{ci}} f_{cgp}\psi_b\left(t_d,t_i\right)K_{id}$$

LRFD Design Eq. 5.9.3.4.2b-1

$$= \frac{28,500}{3,644}(0.863)(2.02)(0.847)$$

= 11.55 ksi

Relaxation of Prestressing Strands

$$\Delta f_{pR1} = \frac{f_{pt}}{K_L}\left(\frac{f_{pt}}{f_{py}}-0.55\right)$$

LRFD Design Eq. 5.9.3.4.2c-1

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f_{pt} = $\max(0.70(f_{pn}) - \Delta f_{pES}, 0.55f_{py})$

$0.70(f_{pn}) - \Delta f_{pES} = 0.70(270) - 6.750 = 182.25 \leftarrow$ Governs

$0.55f_{py}$ = $0.55(0.85)(270) = 126.23$

K_L = 7.0 (Stress Relieved Strands)

$$\Delta f_{pR1} = \frac{182.25}{7}\left(\frac{182.25}{0.85(270)} - 0.55\right) = 6.36 \text{ ksi}$$

Total Time Dependent Losses

Δf_{pLT} = $\Delta f_{pSR} + \Delta f_{pCR} + \Delta f_{pR1}$

= $12.12 + 11.55 + 6.36 = 30.03$ ksi

A9.6.1.3—Total Prestress Losses, Δf_{pT}

Δf_{pT} = $\Delta f_{pES} + \Delta f_{pLT}$ LRFD Design Eq. 5.9.3.1-1

= $6.75 + 30.03$

= 36.78 ksi

f_{pe} = Initial Prestress – Total Prestress Losses

= $(0.7 \times 270) - 36.78$

= 152.22 ksi

P_{pe} = $152.22 \times 20 \times 0.153$

= 465.79 kips

$$f_{cpe} = \frac{P_{pe}}{A} + \frac{P_{pe}e}{S_b}$$

$$= \frac{465.79}{753} + \frac{465.79(16.5 - 2.4)}{6,767}$$

= 1.589 ksi

$$M_{cr} = \gamma_3\left[\left(\gamma_1 f_r + \gamma_2 f_{cpe}\right)S_{nc}\right]$$

$$= (1.0)\left[\left(1.6(0.537) + 1.1(1.589)\right)6,767 \times \frac{1}{12}\right]$$

= 1,470.2 kip-ft

M_r = φM_n

= $1.0 \times 1,858.6 = 1,858.6$ kip-ft

M_r = $1,858.6 > M_{cr} = 1,470.2$ OK

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Minimum reinforcement check is satisfied. 6A.5.6

A9.7—General Load-Rating Equation (6A.4.2)

$$RF = \frac{C - (\gamma_{DC})(DC) - (\gamma_{DW})(DW) \pm (\gamma_P)(P)}{(\gamma_L)(LL + IM)}$$ Eq. 6A.4.2.1-1

A9.7.1—Evaluation Factors for Strength Limit States

A9.7.1.1—Resistance Factor, φ LRFD Design 5.5.4.2

φ = 1.0 for flexure

A9.7.1.2—Condition Factor, φ_c 6A.4.2.3

φ_c = 1.0 no deterioriation

A9.7.1.3—System Factor, φ_s 6A.4.2.4

φ_s = 1.0

A9.7.2—Design Load Rating (6A.4.3)

A9.7.2.1—Strength I Limit State (6A.5.4.1)

$$RF = \frac{(\varphi_c)(\varphi_s)(\varphi)R_n - (\gamma_{DC})(DC) - (\gamma_{DW})(DW)}{(\gamma_L)(LL + IM)}$$

Table 6A.4.2.2-1

Load	Inventory	Operating	
DC	1.25	1.25	
DW	1.5	1.50	Asphalt thickness was not field measured
LL + IM	1.75	1.35	

A9.7.2.1a—Flexure at Midspan

Inventory:

$$RF = \frac{(1.0)(1.0)(1.0)(1,858.6) - (1.25)(627.2) - (1.50)(58.2)}{(1.75)(454.4)}$$

= 1.24

Operating:

$$RF = 1.24 \times \frac{1.75}{1.35}$$

= 1.61

Shear need not be checked for the design load as the bridge does not exhibit signs of shear distress.

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A9.7.2.2—Service III Limit State for Inventory Level (6A.5.4.1)

$$RF = \frac{f_R - (\gamma_D)(f_D)}{(\gamma_L)(f_{LL+IM})}$$

Flexural Resistance:

f_R = f_{pb} + Allowable tensile stress

f_{pb} = compressive stress due to effective prestress

= 1.589 (See previous calculation, A9.6.1.3)

Allowable Tensile Stress $= 0.19\sqrt{f'_c}$ LRFD Design Table 5.9.2.3.2b-1

= $0.19\sqrt{5}$

= 0.425 ksi

f_R = 1.589 + 0.425

= 2.014 ksi

Dead Load Stress:

f_{DC} = $\frac{627.2 \times 12}{6,767} = 1.112$ ksi

f_{DW} = $\frac{58.2 \times 12}{6,767} = 0.103$ ksi

Total f_D = 1.215 ksi

Live Load Stress:

f_{LL+IM} = $\frac{454.4 \times 12}{6,767} = 0.806$ ksi

γ_L = 0.80

γ_D = 1.0

RF = $\frac{2.014 - (1.0)(1.215)}{(0.80)(0.806)}$ Table 6A.4.2.2-1

= 1.24 > 1.0 OK

A9.7.3—Legal Load Rating (6A.4.4)

Inventory design load rating RF > 1.0, therefore the legal load ratings do not need to be performed and no posting is required. 6A.4.3.1

A9.7.4—Permit Load Rating (6A.4.5)

Permit Type: Routine

Permit Weight: 220 kips

Permit Vehicle: Shown in Example A1, Figure A1A.1.10-1

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ADTT (one direction): 4,600

From Live Load Analysis by Computer Program:

Undistributed maximum:

M_{LL} = 2,592 kip-ft

A9.7.4.1—Strength II Limit State 6A.4.5.4.2a

$$\frac{GVW}{AL} = \frac{220}{51} = 4.3 > 3.0$$ Table 6A.4.5.4.2a-1

$$\gamma_L = \frac{1.30 - 1.20}{5,000 - 1,000} = \frac{\gamma_L - 1.20}{4,600 - 1,000}$$

$$= 1.29$$

For a routine permit, use a multi-lane loaded distribution factor. 6A.4.5.4.2a

g_m = 0.268 (two lanes loaded distribution factor)

IM = 10 percent Field inspection verified: Smooth Riding Surface Table C6A.4.4.3-1

Distributed Live Load Effects:

M_{LL+LL} = $(2,592)(0.268)(1.10) = 764.1$ kip-ft

Flexure:

$$RF = \frac{(1.0)(1.0)(1.0)(1,858.6) - 1.25(627.2) - 1.5(58.2)}{(1.29)(764.1)}$$

RF = 1.00 = 1.0 OK

Note: Permit trucks should be checked for shear incrementally along the length of the member. 6A.5.8
Not illustrated here; see Example A3.

A9.7.4.2—Service I Limit State 6A.6.4.2.2

$\gamma_L = \gamma_{DC} = \gamma_{DW} = 1.0$ Table 6A.4.2.2-1

LRFD distribution analysis methods as described in LRFD 4.6.2 should be used. C6A.6.4.2.2

g_m = 0.268

M_{LL+LL} = (2,592)(0.268)(1.10) = 764.1 kip-ft

M_{DC} = 627.2 kip-ft

M_{DW} = 58.2 kip-ft

M_{cr} = 1,470.2 kip-ft (previously calculated)

f_{pe} = 159.3 ksi (previously calculated)

$M_{DC} + M_{DW} + M_{LL+IM} - M_{cr} = 627.2 + 58.2 + 764.1 - 1470.2 = -20.70$ kip-ft

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A9.7.4.2a—Simplified check using 0.75M_n C6A.5.4.2.2b

$M_{DC} + M_{DW} + M_{LL+IM}$ = *1449.5 kip-ft*

$0.75M_n$ = 0.75 × 1,858.6 kip-ft

= 1,394.0 kip-ft < 1,449.5 kip-ft NO GOOD

Moment Ratio:

$$\frac{0.75\,M_n}{M_{DC}+M_{DW}+M_{LL+IM}} = \frac{1,394.0}{1,449.5} = 0.96 < 1.0 \qquad \text{NO GOOD}$$

A9.7.4.2b—Refined check using 0.90f_y

Calculate stress in outer reinforcement at midspan. Stress due to moments in excess of the cracking moment acts upon the cracked section. The moments up to the cracking moment cause stress in the reinforcement equal to the effective prestress.

f_R = $0.9 f_y = 0.9(0.85 f_{pu}) = 0.9(0.85 \times 270) = 206.6$ ksi Table 6A.5.4.2.2b-1

Section Properties for the Cracked Section:

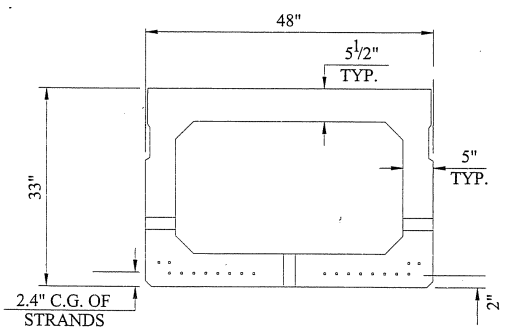

Figure A9.7.4.2b-1—Box Beam Cross Section

Assume neutral axis is in the top flange.

A_{ps} = 3.06 in.^{2} LRFD Design 5.6.1

f'_c = 5 ksi

Effective modular ratio of 2*n* is applicable

$$n = \frac{E_p}{E_c} = \frac{28,500}{4,000}$$

n = 7; therefore, $2n$ = 14

A_{trans} = $A_{ps} 2n = 3.06 \times 14 =$ 42.8in.^{2}

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$$c = \frac{\frac{c}{2}(b)(c) + (d_e)(A_{trans})}{(b)(c) + A_{trans}}$$

$$c = \frac{\frac{c}{2}(48)(c) + (33 - 2.4)(42.8)}{(48)(c) + 42.8}$$

$$24c^2 + 42.8c - 1,309.7 = 0$$

Solving for c:

c = 6.55 in. > 5.50 in. assumed; therefore, find neutral axis depth by trial and adjustment

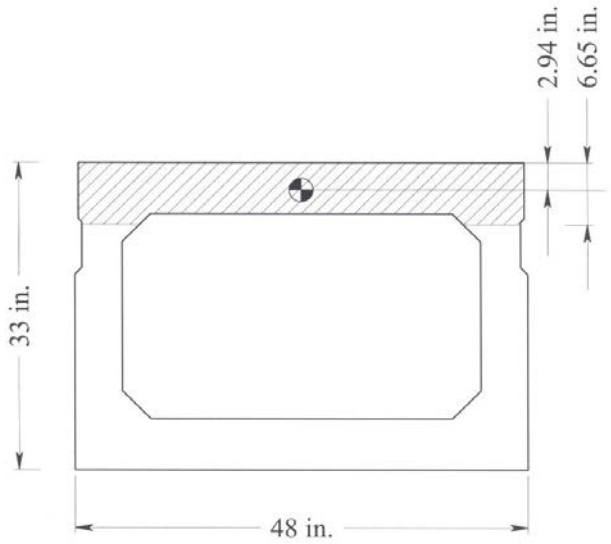

Figure A9.7.4.2b-2—Box Beam Cross Section for Determining *c*

Table A9.7.4.2b-1—Trial and Adjustment Values for *c*

Trial *c*	Centroid	Area Concrete	Calculated *c*	Difference Trial – Calculated
5.5	2.75	264.0000	6.6383	–1.138
5.8294	2.8041	264.7872	6.6749	–0.846
6.3	2.8827	271.3392	6.6621	–0.362
6.6	2.9318	275.1264	6.6596	–0.060
6.65	2.9400	275.7456	6.6594	–0.009
6.7	2.9482	276.3504	6.6595	+0.041

By trial and adjustment, c approximately equals 6.65:

$$c = \frac{2.94(275.7456) + \left[(16\times.153 \text{ in.}^2\times31 \text{ in.}) + (4\times.153 \text{ in.}^2\times29 \text{ in.})\right]\times14}{(275.7456) + (20\times.153 \text{ in.}^2\times14)} = 6.65$$

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$$I_{cr} = \begin{bmatrix} \frac{1}{12}(47.25)(5.5)^3 + (47.25)(5.5)\left(6.65 - \frac{5.5}{2}\right)^2 + \\ 2\times\left[\frac{1}{12}(6.9)(1.15)^3 + (6.9)(1.15)(0.58)^2\right] + \\ (42.8)(33 - 2.4 - 6.65)^2 \end{bmatrix} = 29165 \text{ in.}^4$$

Stress beyond the effective prestress (increase in stress after cracking):

$$f = n\frac{M_y}{I} = 7\frac{(-20.7)(12)(33-2-6.65)}{29{,}165} = -1.45 \text{ ksi}$$

Stress in the reinforcement at permit crossing Service I:

$$f_s = 152.2 - 1.45 = 150.75 \text{ ksi} < f_R = 0.9F_y = 206.6 \text{ ksi} \quad \text{OK}$$

Stress Ratio:

$$\frac{0.9f_y}{f_s} = \frac{206.6}{150.75} = 1.37 > 1.0 \text{ Allowable steel stress; actual on cracked section}$$

OK

For this bridge, the simplified check indicates that the Service I condition is violated for the permit truck; the more detailed check indicates that the condition is acceptable.

A9.8—Summary of Rating Factors

Table A9.8-1—Summary of Rating Factors—Interior Box Beam

Limit State		Design Load Rating (HL-93)		Permit Load Rating
		Inventory	Operating	
Strength I	Flexure	1.24	1.61	—
Strength II	Flexure	—	—	1.00
Service III		1.24	—	—
Service I	Approximate	—	—	Stress Ratio = 0.96
	Refined	—	—	Stress Ratio = 1.37

A9.9—Reference

AASHTO. *AASHTO LRFD Bridge Design Specifications*, Eighth Edition, LRFD-8. American Association of State Highway and Transportation Officials, Washington, DC, 2017.

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A10—LRFR RATING OF A REINFORCED CONCRETE BOX CULVERT

A10.1—Culvert Data

Materials:

Concrete: f'_c = 2.5 ksi
Reinforcing Steel: f_y = 33 ksi

The culvert is in satisfactory physical condition.
Cross Section: Dimensions are given in feet.

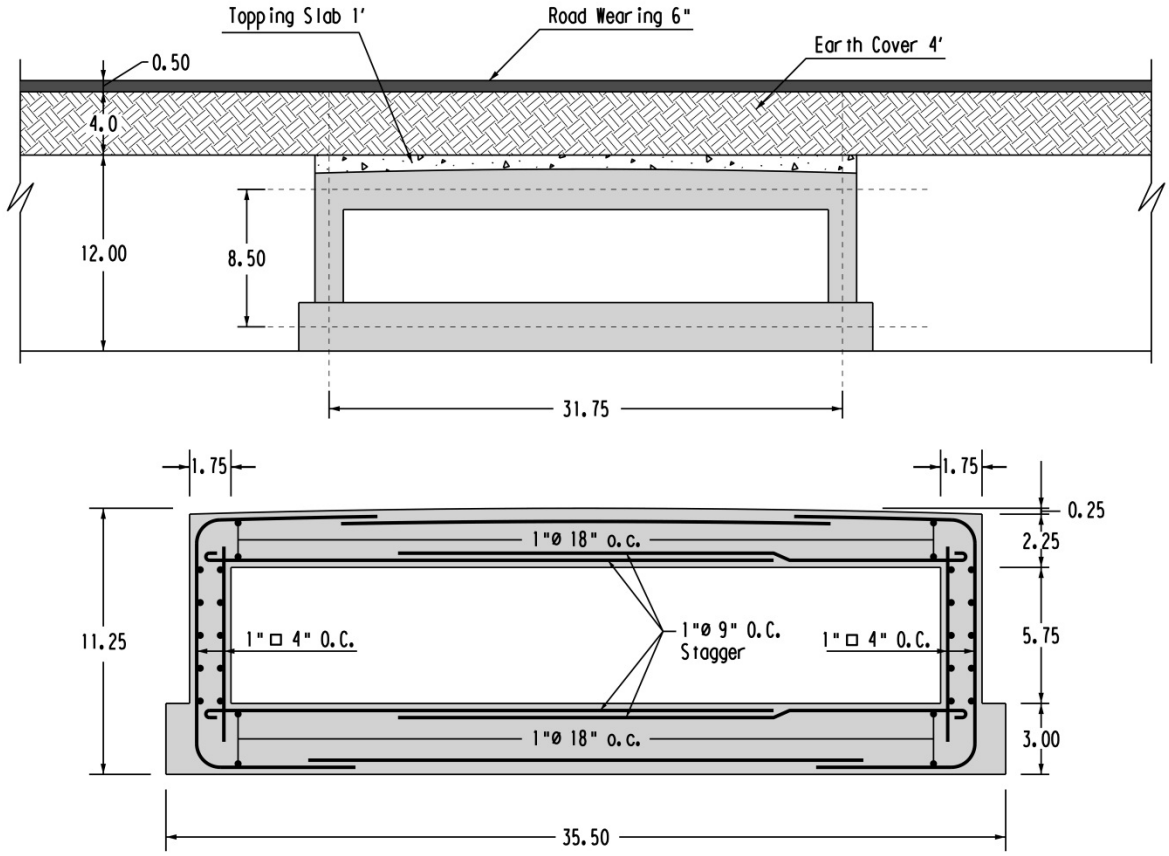

Figure A10.1-1—Reinforced Concrete Culvert (a) Cross Section and (b) Reinforcing Steel Details

Rating Approach

- LRFR evaluation is performed for a 1-ft strip of the culvert structure.
- The concrete topping slab (1 ft) is considered only in the dead load calculations (no resistance contribution). The 6-in. asphalt wearing surface thickness was field verified.
- Culvert is reinforced with 1 in. × 1 in. square-shape bars that are continuous at the connections. All connections are considered as capable of transferring moment.
- LRFR live load ratings are evaluated for HL-93 design loading (Tandem, Truck) and legal loads (Type 3, SU-4) at Strength I Level.

A10.2—Dead Load Analysis, *DC* (6A.5.12.10.1)

Normal weight concrete: w_c = 0.145 kcf LRFD Design Table 3.5.1-1

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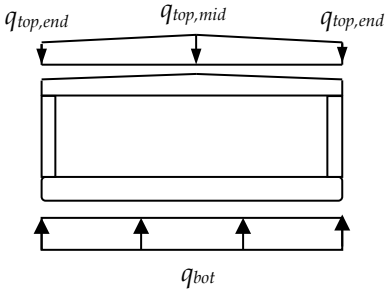
Top slab weight:

$q_{top,\,end}$ = 2.25 ft × 1 ft × 0.145 kcf = 0.326 kip/ft

$q_{top,\,mid}$ = 2.50 ft × 1 ft × 0.145 kcf = 0.363 kip/ft



1 ft topping slab weight is added to $q_{top,end}$ and $q_{top,mid}$:

$q_{top,add}$ = 1.0 ft × 1 ft × 0.145 kcf = 0.145 kip/ft

Sidewall weight = 5.75 ft × 1.75 ft × 1 ft × 0.145 kip/ft + …
1.0 ft × 1.75 ft × 1.0 ft × 0.145 kip/ft × $^1/_2$ +…
2.25 ft × 1.75/2 ft × 1 ft × 0.145 kip/ft = 1.87 kips

Sidewall weight is applied as a point load to consider thrust in sidewalls.

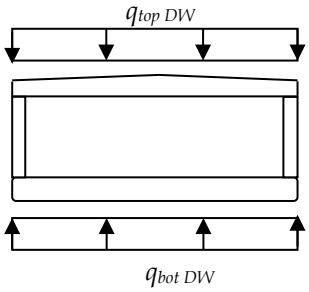
Bottom slab: Uniform uplift reaction on bottom slab

q_{bot} = [(0.326 + 0.363)/2 kip/ft × 31.75ft + 2 × 1.87 kips + 0.145 kip/ft × 31.75 ft] / 31.75 ft
= 0.607 kip/ft

Total *DL* in vertical direction shall be in balance in box culvert model so that element length (31.75 ft) is used.

A10.3—Wearing Surface, *DW*

Asphalt wearing surfaces: w_w = 0.140 kcf LRFD Design Table 3.5.1-1

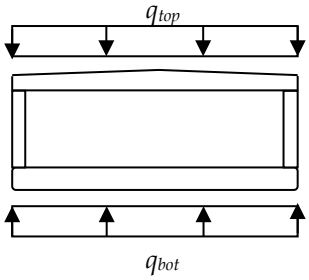


Thickness is measured as 6 in.
(6/12) ft × 1 ft × 0.140 kcf = 0.07 kip/ft

Top slab: Uniform gravity loading, $q_{top\ DW}$

Bottom slab: Uniform uplift loading, $q_{bot\ DW}$
$q_{top\ DW}$ = $q_{bot\ DW}$ = 0.07 kip/ft

A10.4—Earth Load: Vertical, *EV* (6A.5.12.10.2a)



Earth fill over the top slab: 4 ft 0 in. (Wearing surface will be considered as a uniform surcharge load on the surface in *ES* estimations.) The amount of earth load carried by the structure is dependent on the construction method. The rating of the existing culvert is based on construction in embankment conditions.

EV load factor *Fe* = 1+ 0.20 (*H*/B_c): *H* = 4 ft depth of earthfill, B_c: out-to-out width of the culvert: *Fe* = 1 + 0.20 (4 ft/33.5 ft) = 1.02 LRFD Design 12.11.2.2

$q_{top\ EV}$ = $q_{bot\ EV}$ = (*Fe* = 1.02) × 4.0 ft × 1 ft × 0.120 kcf = 0.49 kip/ft
Top slab: Uniform gravity loading, $q_{top\ EV}$

Bottom slab: Uniform uplift loading, $q_{bot\ EV}$

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A10.5—Earth Load: Horizontal, *EH* (6A.5.12.10.2b)

Depth of fill at the top slab = 4 ft

$p = k\gamma_s z$ LRFD Design Eq. 3.11.5.1-1

$k = k_o = 0.55$: a typical value is assumed LRFD Design C3.11.5.2

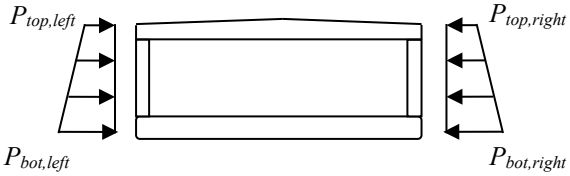
$\gamma_s = 0.120$ kcf: soil density LRFD Design Table 3.5.1-1

Effective soil height at top and bottom of the culvert:

$z_{top} = 5.0$ ft (including 1-ft topping), $z_{bot} = 16.00$ ft

$P_{top,left} = 0.33$ kip/ft , $P_{top,right} = -0.33$ kip/ft

$P_{bot,left} = 1.06$ kip/ft , $P_{bot,right} = -1.06$ kip/ft



A10.6—Surcharge Earth Load: Horizontal, *ES* (6A.5.12.10.2c)

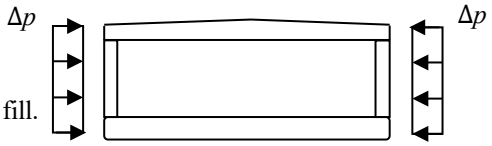
Sidewalls are subjected to a uniform surcharge load due to continuous roadway fill.

$\Delta p = k_s q_s$ LRFD Design Eq. 3.11.6.1-1

$k_s = k_o = 0.55$

$q_s = q_{top\ DW} = 0.07$ kip/ft^2:

Uniform surcharge, q_s, applied to the upper surface by continuous roadway fill.



$q_{ES} = (k_s = 0.55) \times 1$ ft $\times (q_s = 0.07$ kip/ft$) = 0.04$ kip/ft

A10.7—Live Load Analysis, *LL* (6A.5.12.10.3)

Live Load Distribution: 6A.5.12.10.3a

Transmission of wheel loads to the buried structure through the earth fill is calculated per LRFD Design Article 3.6.1.2.6. Instead of distributed point live loads, the patch loading is considered with a contact area of 20 in. × 10 in. For the earth cover higher than 2 ft, the strip pressure is applied to the culvert top slab, including the impact loading for buried structures. For a 6-in.-thick asphalt wearing surface, the total depth of the projection is taken as 4.5 ft ($H = 4.5$ ft). Per LRFD Design Article 3.6.1.2.6, load distribution and projected axle loads are calculated through LRFD Design Eqs. 3.6.1.2.6a-1 to 3.6.1.2.6b-7.

Live Load Distribution Factor (LLDF) = 1.15 LRFD Design Table 3.6.1.2.6a-1

Traffic Parallel to the Culvert Span LRFD Design 3.6.1.2.6b

a. For live load distribution transverse to culvert spans, axle load interaction depth: H_{int_t}

$H_{int_t} = (s_w - w_t / 12 - 0.06 D_i / 12) / LLDF$ LRFD Design Eq. 3.6.1.2.6b-1

s_w = wheel spacing in the transverse direction, 6 ft

w_t = tire patch width, 20 in.

D_i = inside diameter or clear span of the culvert, 30 ft

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H_{int_t} = (6 ft – 20 in. / 12 – 0.06 × 30 ft) / (*LLDF* = 1.15) = 2.20 ft, $H > H_{int_t}$: Use LRFD Design Eq. 3.6.1.2.6b-3

$w_w = (w_t /12 + s_w + LLDF \times H + 0.06\ D_i / 12)$ LRFD Design Eq. 3.6.1.2.6b-3

w_w = (20 in. / 12 +6 ft + 1.15 × 4.5 ft + 0.06 × 30 ft) = 14.64 ft

(This width is valid for all truck types in transverse direction (HL-93 Truck, HL-93 Tandem, Type 3, SU-4)

b. For live load distribution parallel to culvert span, axle load interaction depth: H_{int_p}

$H_{int_p} = (s_a – l_t / 12) / LLDF$ LRFD Design Eq. 3.6.1.2.6b-4

s_a = axle spacing (ft)
HL-93 Truck: s_a = 14 ft (minimum)
HL-93 Tandem, Type 3, SU-4: s_a = 4 ft (minimum)

l_t = tire patch-length, 10 in., in longitudinal direction

HL-93 Truck

H_{int_p} = (14 ft – 10 in. / 12) / (*LLDF* = 1.15) = 11.45 ft, $H < H_{int_p}$: Use LRFD Design Eq. 3.6.1.2.6b-5

$l_w = lt /12 + LLDF \times H$ LRFD Design Eq. 3.6.1.2.6b-5

l_w = 10 in. / 12 + 1.15 × 4.5 ft = 6.01 ft

HL-93 Tandem, Type 3

H_{int_p} = (4 ft – 10 in. / 12) / (*LLDF* = 1.15) = 2.75 ft, $H > H_{int_p}$: Use LRFD Design Eq. 3.6.1.2.6b-6

$l_w = lt /12 + s_a + LLDF \times H$ LRFD Design Eq. 3.6.1.2.6b-6

l_w = 10 in. / 12 + 4 ft + 1.15 × 4.5 ft = 10.01 ft

SU-4: See for three-axle load distribution calculations in following pages.

Impact Loading (*IM*) 6A.5.12.10.3b

Impact Factor for buried structures is calculated using the earth fill over the structure:

IM = 33 (1.0 – 0.125 × 4.5 ft) = 14.4 percent LRFD Design Eq. 3.6.2.2-1

Uniform Live Load on Culvert Surface: P_L

Rectangular Area $A_{LL} = l_w w_w$ LRFD Design Eq. 3.6.1.2.6a-1
$P_L = P\ (1 + IM/100)\ (m)/A_{LL}$ LRFD Design Eq. 3.6.1.2.6b-7

(m) = Multiple Presence Factor 6A.5.12.10.3

P = Live load applied at surface on all interacting wheels (kips)

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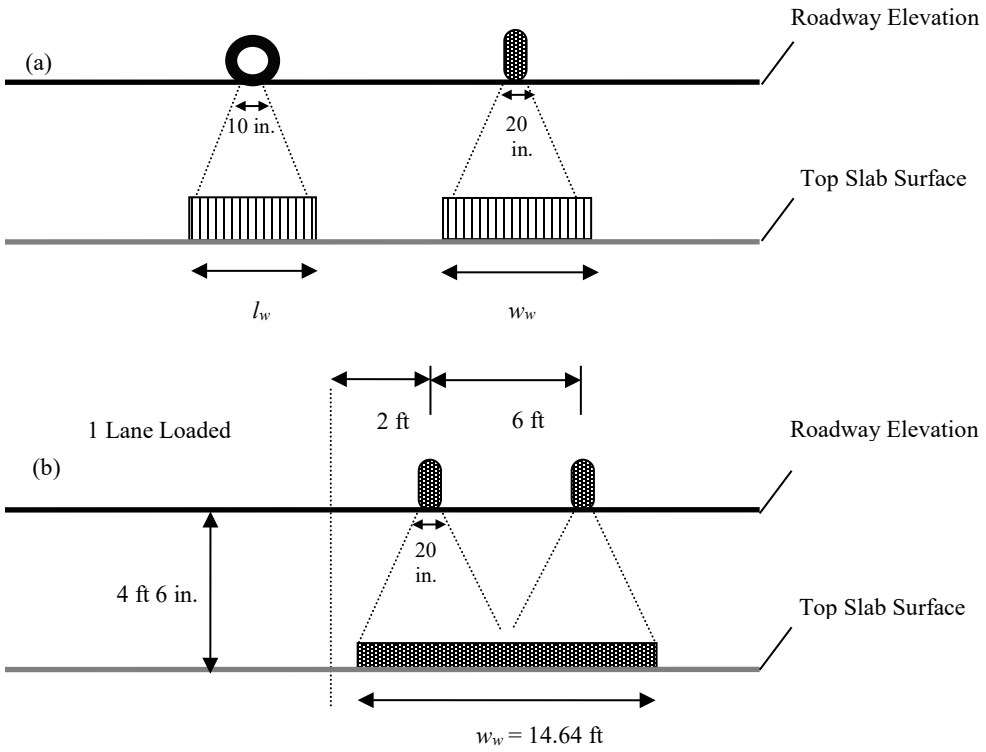

Figure A10.7-1—(a) Contact Area Projection of One Wheel in Transverse and Longitudinal Directions; (b) Transverse Load Distribution of 6-ft Wheel Spacing for All Vehicle Types

A10.7.1—Design Load: HL-93 Tandem

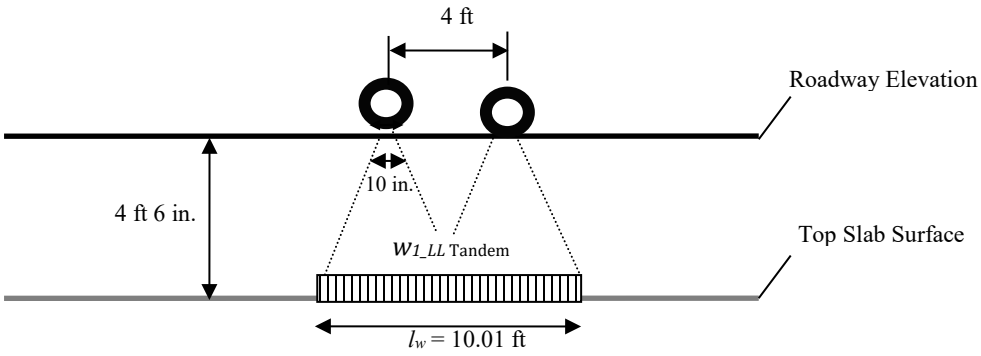

Figure A10.7.1-1—Tandem Axle Projections in Longitudinal Direction

HL-93 Design Tandem:

For Design Load Rating, Single Lane Loading with *MPF* is applied:

Multiple Presence Factor (*MPF*) = 1.2 6A.5.12.10.3

$w_{LL,Tandem}$ = [4 × 12.5 kips × (*MPF* = 1.20) × (1+ *IM* = 14.4 percent)] / (14.64 ft × 10.01 ft) = 0.47 kip/ft^2

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A10.7.2—Design Load: HL-93 Truck

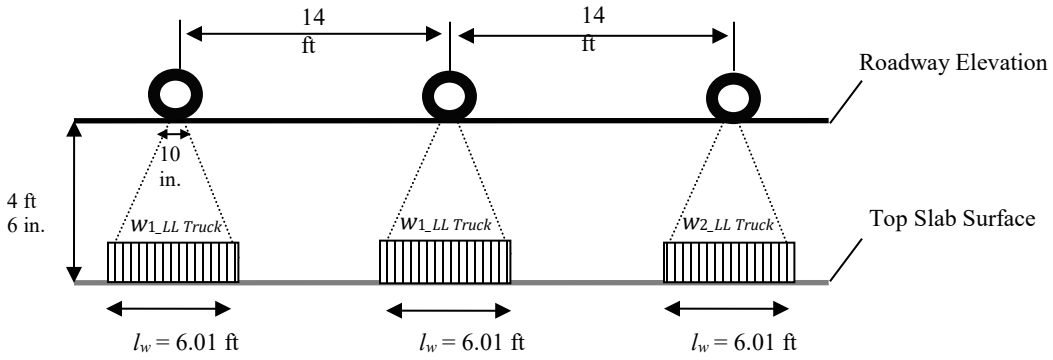

Figure A10.7.2-1—Design Truck Axle Projections in Longitudinal Direction

HL-93 Design Truck:

For Design Load Rating, Single-Lane Loading with *MPF* is applied:

Multiple Presence Factor (*MPF*) = 1.2 6A.5.12.10.3

$w_{1_LL,\ HL93}$ = [2 × 16 kips × (*MPF* =1.20) × (1+ *IM* =14.4 percent)] / (14.64 ft × 6.01 ft) = 0.50 kip/ft^2

$w_{2_LL,\ HL93}$ = [2 × 4 kips × (*MPF* =1.20) × (1+ *IM* =14.4 percent)] / (14.64 ft × 6.01 ft) = 0.125 kip/ft^2

A10.7.3—Legal Load: Type 3

Live load projection in the perpendicular direction is the same for design and tandem trucks as given in the earlier sections and the width of projection is 14.64 ft. The uniform pressure on the culvert from the 16-kip wheel and 17-kip tandem is distributed on the projected area and calculated as $w_{1_LL\ T3}$ and $w_{2_LL\ T3}$, respectively. *MPF* = 1.0.

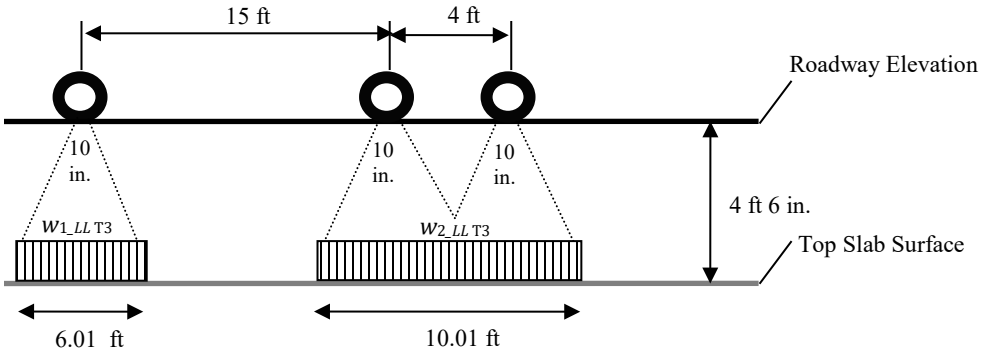

Figure A10.7.3-1—Legal Type 3 Axle Projections in Longitudinal Direction

Type 3: $w_{1_LL\ T3}$ = [16 kips × (*MPF* = 1.00) × (1 + *IM* = 14.4 percent)] / (14.64 ft × 6.01 ft) = 0.21 kip/ft^2

$w_{2_LL\ T3}$ = [2 × 17 kips × (*MPF* = 1.00) × (1 + *IM* = 14.4 percent)] / (14.64 ft × 10.01 ft) = 0.27 kip/ft^2

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A10.7.4—Legal Load: SU-4

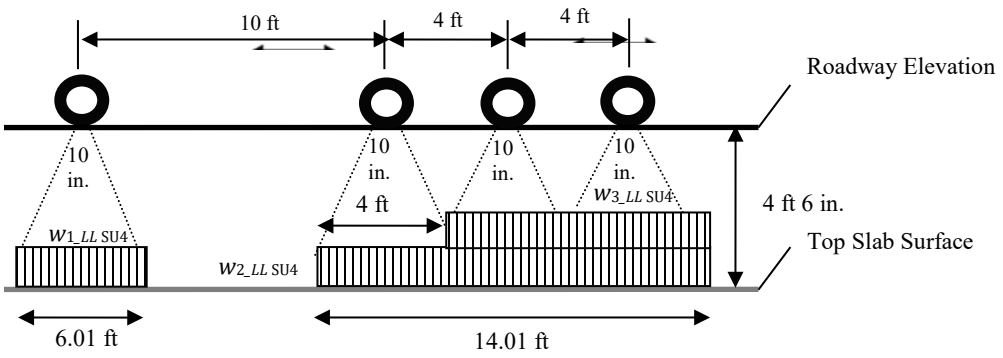

Figure A10.7.4-1—Legal SU-4 Axle Projections in Longitudinal Direction

SU-4: $w_{1_LL\ SU4}$ = [12 kips × (*MPF* = 1.00) × (1 + *IM* = 14.4 percent)] / (6.01ft × 14.64 ft) = 0.16 kip/ft^2

$w_{2_LL\ SU4}$ = [3 × 8 kips × (*MPF* = 1.00) × (1 + *IM* = 14.4 percent)] / (14.01 ft × 14.64 ft) = 0.14 kip/ft^2

$w_{3_LL\ SU4}$ = $w_{2_LL\ SU4}$ + [2 × (17 – 8) kips × (*MPF* = 1.00) × (1 + *IM* = 14.4 percent)] / (10.01 ft × 14.64 ft)
= 0.28 kip/ft^2

Due to the axle weight difference between the rear tandem axles, distributing this 9-kip difference in tandem load over its corresponding footprint length is more realistic than distributing the total load over the entire length. For the three rear tandem axles of the SU-4 truck, 8 kips of the load in all three is distributed over 14.01 ft and calculated as $w_{2_LL\ SU4}$. The remaining 9 kips in the rear two axles are projected on the area with the typical size of 10.01 ft × 14.64 ft.

A10.7.5—Surcharge Live Load, *LS* (6A.5.12.10.3c)

The increase in horizontal pressure due to live load surcharge:

$\Delta p = k\gamma_s h_{eq}$ LRFD Design Eq. 3.11.6.4-1

h_{eq} = equivalent height of soil for vehicular load (ft) LRFD Design Table 3.11.6.4-1

Height = 16.5 ft; obtained value by interpolation = 2.35 ft

q_{LS} = 1 ft × (*k* = 0.55) × 2.35 ft × 0.120 kcf = 0.155 kip/ft

A10.8—Structural Analysis of Box Culvert (6A.5.12.3)

Finite element analysis procedure is applied for the culvert model with beam-column elements. The widths of slabs and sidewalls are considered as a unit length of 1 ft for the two-dimensional representation of the continuous structure. All the load effects and member resistances are calculated using this 1-ft strip representation. Structural analysis of the culvert is based on a box frame model with moment-resisting connections between both top and bottom slabs and the sidewall joints, and the box frame is fixed against lateral movement at the bottom. The statically indeterminate structure is pinned at the ends of the bottom slab member and free to side-sway at the top slab. One pinned support is fixed in the horizontal direction and the opposite end is a roller. This simplification yields demand on the structure that may be somewhat conservative when compared to other modeling approaches. For more precise approximations regarding support conditions, alternate analysis methods can be employed such as the application of linear elastic or non-linear elastic soil foundation springs, or considering the box culvert as a continuous beam on an elastic foundation.

As per the reinforcing steel details in the reference drawings, bars are continuous between the slab and sidewalls; therefore, moment-transferring connection is assumed.

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Vertical loading is balanced by the bottom slab subjected to uplift reaction forces so that the total external loads are zero and total reaction forces will be zero. For dead loads, vertical dead loads, and live loads, supporting soil pressure is applied as the total load applied downward and uniformly distributed along the length of the bottom slab. Hydrostatic pressure does not exist in the culvert and the soil is assumed to be drained so that no hydrostatic pressure is applied on the outside. In addition, soil–structure interaction is not considered.

The critical failure modes for the box culvert model are due to flexure, shear, and axial forces. The shear demand is checked at the critical shear section (d_v away from the sidewall surface), whereas the moment demand is checked at the inner surface of the sidewalls. For seven critical sections, the unfactored ultimate section forces are listed below. The factored loads are superposed using the minimum and maximum load factors to construct an envelope of ultimate force occurrences. Tables A10.8-1 through A10.8-4 present the analysis results for dead load, earth load, design, and legal load maximum section forces.

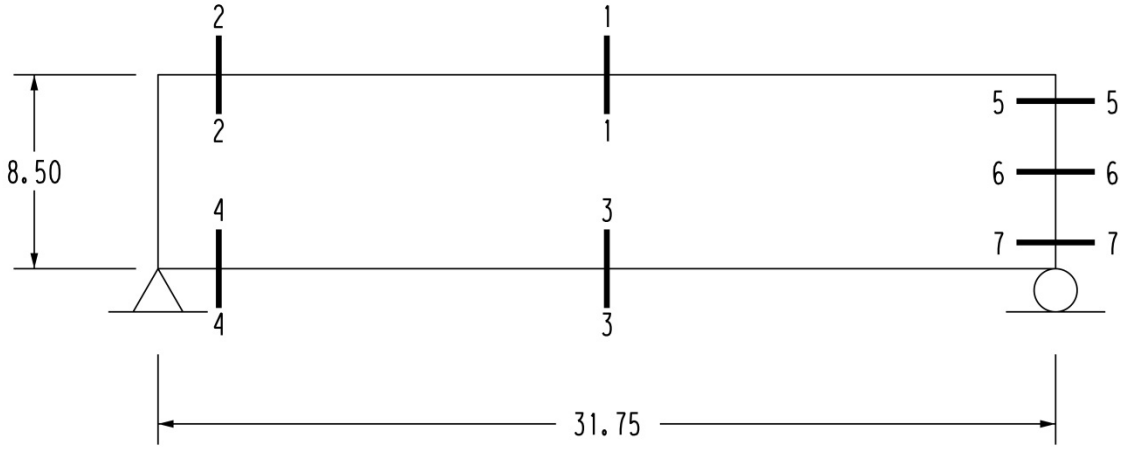

Figure A10.8-1—Seven Critical Sections Evaluated in Box Culvert Ratings

Table A10.8-1—Analysis Results: Moment for *DL*, *EL*, and Design *LL*

		Moment (kip-ft/ft)							
	Load	*DC*	*DW*	*EV*	*EH*	*ES*	*LL + IM* (Tandem)	*LL + IM* (Truck)	*LS*
1	Top Slab Center	34.78	4.87	34.07	–1.29	–0.08	18.89	15.43	–0.30
2	Top Slab Corner	–26.97	–2.19	–27.62	–1.29	–0.08	–10.43	–9.93	–0.30
3	Bottom Slab Center	–55.93	–6.64	–46.41	2.71	0.15	–14.61	–19.10	0.60
4	Bottom Slab Corner	20.35	2.19	15.29	2.71	0.15	4.18	3.44	0.60
5	Sidewall Top	–26.97	–3.95	–27.62	–1.00	–0.08	–11.32	–11.88	–0.31
6	Sidewall Center	–23.76	–3.07	–21.46	4.21	0.25	–8.06	–9.13	0.95
7	Sidewall Bottom	–20.35	–2.19	–15.29	1.57	–0.08	–6.40	–7.65	–0.27

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Table A10.8-2—Analysis Results: Shear for *DL*, *EL*, and Design *LL*

		Shear (kip/ft)—Unfactored							
	Load	*DC*	*DW*	*EV*	*EH*	*ES*	*LL + IM* (Tandem)	*LL + IM* (Truck)	*LS*
1	Top Slab Center	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2	Top Slab Corner	7.35	1.05	6.58	0.00	0.00	3.62	3.15	0.00
3	Bottom Slab Center	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
4	Bottom Slab Corner	–7.51	–1.05	–6.06	0.00	0.00	–2.29	–2.85	0.00
5	Sidewall Top	–0.75	–0.21	–1.45	–1.02	–0.12	–1.03	–0.88	–0.45
6	Sidewall Center	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7	Sidewall Bottom	–0.75	–0.21	–1.45	1.28	0.12	–1.03	–0.88	0.46

Table A10.8-3—Analysis Results: Axial Thrust for *DL*, *EL*, and Design *LL*

		Thrust (kip/ft)—Unfactored							
	Load	*DC*	*DW*	*EV*	*EH*	*ES*	*LL + IM* (Tandem)	*LL + IM* (Truck)	*LS*
1	Top Slab Center	–0.75	–0.21	–1.45	–2.24	–0.28	–1.03	–0.88	–0.62
2	Top Slab Corner	–0.75	–0.21	–1.45	–2.24	–0.28	–1.03	–0.88	–0.62
3	Bottom Slab Center	0.75	0.21	1.45	–3.62	–0.31	1.03	0.88	–0.69
4	Bottom Slab Corner	0.75	0.21	1.45	–3.62	–0.31	1.03	0.88	–0.69
5	Sidewall Top	–9.65	–1.11	–7.77	0.00	0.00	–4.02	–4.15	0.00
6	Sidewall Center	–9.65	–1.11	–7.77	0.00	0.00	–4.02	–4.15	0.00
7	Sidewall Bottom	–9.65	–1.11	–7.77	0.00	0.00	–4.02	–4.15	0.00

Table A10.8-4 Analysis Results: Moment, Shear, Thrust for Legal Loads Type 3 (T3) and SU-4

		Moment (kip-ft/ft)		Shear (kip/ft)		Thrust (kip/ft)	
	Load	*LL + IM* (T3)	*LL + IM* (SU4)	*LL + IM* (T3)	*LL + IM* (SU4)	*LL + IM* (T3)	*LL + IM* (SU4)
1	Top Slab Center	10.89	12.64	0.00	0.00	–1.03	–0.88
2	Top Slab Corner	–5.90	–6.98	2.11	2.45	–1.03	–0.88
3	Bottom Slab Center	–11.89	–12.69	0.00	0.00	1.03	0.88
4	Bottom Slab Corner	2.05	1.92	–1.83	–1.97	1.03	0.88
5	Sidewall Top	–7.30	–8.11	–0.55	–0.63	–4.02	–4.15
6	Sidewall Center	–5.89	–6.42	0.00	0.00	–4.02	–4.15
7	Sidewall Bottom	–4.72	–5.25	–0.55	–0.63	–4.02	–4.15

A10.9—Nominal Flexural Resistance

A10.9.1—Top Slab Center

Rectangular Section height, h = 2.5 ft

Rectangular Section width, b = 1.0 ft

β_t = 0.85 for f'_c = 2.5 ksi LRFD Design 5.6.2.2

$c = (A_s \times f_y) / (0.85 f'_c\ 0.85b)$ LRFD Design Eq. 5.6.3.1.1-4

A_s + = 2.11 in.^{2}/ft (stagger 2 × 1 in. φ / 9 in.)

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$c = (2.11 \times 33) / (0.85 \times 2.5 \times 0.85 \times 12) = 3.21$ in.

$a = c\beta_t = 3.21 \times 0.85 = 2.73$ in.

Distance from bottom edge to center of rebar, y

$y = 2$ in. (clean cover) + 0.5 in. = 2.5 in.

$d_e = h - y = 30$ in. – 2.5 in. = 27.5 in.

Nominal Moment Resistance LRFD Design Eq. 5.6.3.2.2-1

$M_n = A_s f_y (d_e - a / 2) = 2.11 \times 33 \times (27.5 - 2.73/2) \times (1/12)$

$M_n = 151.65$ kip-ft

A10.9.2—Top Slab Corner

Rectangular Section height, $h = 2.25$ ft

Rectangular Section width, $b = 1.0$ ft

$\beta_t = 0.85$ for $f'_c =$ 2.5 ksi LRFD Design 5.6.2.2

$c = (A_s \times f_y) / (0.85 f'_c\ 0.85 b)$ LRFD Design Eq. 5.6.3.1.1-4

A_s- = 3.00 in.^{2}/ft (1 in. □ / 4 in., minimum 3 1 in. × 1 in. square shape rebar in 1 ft)

$c = (3.00 \times 33) / (0.85 \times 2.5 \times 0.85 \times 12) = 4.57$ in

$a = c\ \beta_t = 4.57 \times 0.85 = 3.88$ in
Distance from bottom edge to center of rebar, y

$y = 3$ in. (clean cover) + 0.5 in. = 3.5 in.

$d_e = h - y = 27$ in. – 3.5 in. = 23.5 in.

Nominal Moment Resistance LRFD Design Eq. 5.6.3.2.2-1

$M_n = A_s f_y (d_e - a / 2) = 3.00 \times 33 \times (23.5 - 3.88/2) \times (1/12)$

M_n- = 177.87 kip-ft

A10.9.3—Bottom Slab Center

Rectangular Section height, $h = 3.0$ ft

Rectangular Section width, $b = 1.0$ ft

$\beta_t = 0.85$ for $f'_c =$ 2.5 ksi LRFD Design 5.6.2.2

$c = (A_s \times f_y) / (0.85 f'_c\ 0.85 b)$ LRFD Design Eq. 5.6.3.1.1-4

A_s- = 2.11 in.^{2}/ft (stagger 2 × 1 in. φ /9 in.)

$c = (2.11 \times 33) / (0.85 \times 2.5 \times 0.85 \times 12) = 3.21$ in.

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$a = c\beta_t = 3.21 \times 0.85 = 2.73$ in.

Distance from bottom edge to center of rebar, y

$y = 2$ in. (clean cover) + 0.5 in. = 2.5 in.

$d_e = h - y = 36$ in. – 2.5 in. = 33.5 in.

Nominal Moment Resistance LRFD Design Eq. 5.6.3.2.2-1

$M_n = A_s f_y (d_e - a / 2) = 2.11 \times 33 \times (33.5 - 2.73/2) \times (1/12)$

M_n- = 186.46 kip-ft

A10.9.4—Bottom Slab Corner

Rectangular Section height, $h = 3.0$ ft

Rectangular Section width, $b = 1.0$ ft

$\beta_t = 0.85$ for $f'_c =$ 2.5 ksi LRFD Design 5.6.2.2

$c = (A_s \times f_y) / (0.85 f'_c\ 0.85b)$ LRFD Design Eq. 5.6.3.1.1-4

$A_s = 3.0$ in.^{2}/ft (1 in. □ / 4 in., minimum 3 1in. × 1 in. square shape rebar in 1 ft)

$c = (3.00 \times 33) / (0.85 \times 2.5 \times 0.85 \times 12) = 4.57$ in.

$a = c\beta_t = 4.57 \times 0.85 = 3.88$ in.

Distance from bottom edge to center of rebar, y

$y = 3$ in. (clean cover) + 0.5 in. = 3.5 in.

$d_e = h - y = 36$ in. – 2.5 in. = 33.5 in.

Nominal Moment Resistance LRFD Design Eq. 5.6.3.2.2-1

$M_n = A_s f_y (d_e - a / 2) = 3.0 \times 33 \times (33.5 - 3.88/2) \times (1/12)$

M_n- = 260.37 kip-ft

A10.9.5—Sidewall: Top–Center–Bottom

Rectangular Section height, $h = 1.75$ ft

Rectangular Section width, $b = 1.0$ ft

$\beta_t = 0.85$ for $f'_c =$ 2.5 ksi LRFD Design 5.6.2.2

$c = (A_s \times f_y) / (0.85 f'_c\ 0.85b)$ LRFD Design Eq. 5.6.3.1.1-4

$A_s = 3.00$ in.^{2}/ft (1 in. □ / 4 in., minimum 3 1 in. × 1 in. square shape rebar in 1 ft)

$c = (3 \times 33) / (0.85 \times 2.5 \times 0.85 \times 12) = 4.57$ in.

$a = c\beta_t = 3.81 \times 0.85 = 3.88$ in

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Distance from bottom edge to center of rebar, y

y = 3 in. (clean cover) + 0.5 in. = 3.5 in.

$d_e = h$ – y = 21 in. – 3.5 in. = 17.5 in.

Nominal Moment Resistance LRFD Design Eq. 5.6.3.2.2-1

$M_n = A_s f_y (d_e - a / 2) = 3.00 \times 33 \times (17.5 - 3.88/2) \times (1/12)$

M_n = 128.37 kip-ft

A10.10—Nominal Shear Resistance

A10.10.1—Top Slab Corner

Rectangular Section height, h = 2.25 ft

Rectangular Section width, b = 1.0 ft

A_v = 0.79 in.^{2}/ 18 ft

Critical section for shear: LRFD Design 5.7.3.2

Effective Shear Depth: $d_v = M_n / (A_s f_y)$

$d_v = 177.87/(3.00 \times 33) \times 12 = 21.56$ in.

Nominal shear resistance in slabs of the box culvert is calculated per LRFD Design 5.12.7.3.

$V_c = [0.0676\sqrt{f'_c} + 4.6 (A_s / bd_e) (V_u d_e / M_u)] bd_e$ LRFD Design Eq. 5.12.7.3-1

For each vehicle, the V_u / M_u ratio shall be calculated using concurrent shear and moment demand. Note that for single-cell box culverts, minimum shear capacity shall not be taken to be less than:

$V_c = 0.0948\sqrt{f'_c}\ bd_e$ LRFD Design 5.12.7.3

Whereas nominal capacity shall not be greater than:

$V_c = 0.126\sqrt{f'_c}\ bd_e$ LRFD Design Eq. 5.12.7.3-2

For conservative simplification of calculations, longitudinal reinforcing steel contribution is not considered.

$V_c = 0.0676\sqrt{f'_c}\ b\ d_e < V_c = 0.0948\ \sqrt{f'_c}\ bd_e$

$V_c = 0.0948 \times \sqrt{2.5} \times 12 \times 23.5 = 42.27$ kips

Critical section for shear is at 21.56 in. from face of support. Calculate at 21.56 in. + 21 / 2 in. = 32.06 in.

Shear Resistance is the lesser of LRFD Design Eqs. 5.7.3.3-1 and 5.7.3.3-2:

$V_n = V_c + V_s$ LRFD Design Eq. 5.7.3.3-1

$V_n = 0.25 f'_c b_v d_v$ LRFD Design Eq. 5.7.3.3-2

$V_s = A_v f_y d_v \cot\theta / s$ ($\alpha = 90°$ and θ is assumed as 45°) LRFD Design Eq. 5.7.3.3-4

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$V_s = 0.79 \times 33 \times \cot 45 \times 21.56 / 18 = 31.23$ kips

$V_n = 0.25 \times 2.5 \times 12 \times 21.56 = 161.7$ kips: Use $V_n = 42.27 + 31.23 = 57.08$ kips

A10.10.2—Bottom Slab Corner

Rectangular Section height, $h = 3.0$ ft

Rectangular Section width, $b = 1.0$ ft

$A_v = 0.79$ in.^{2}/ 18 ft

Critical section for shear: LRFD Design 5.7.3.2

Effective Shear Depth: $d_v = M_n / (A_s f_y)$

$d_v = 260.37/(3.00 \times 33) \times 12 = 31.56$ in.

Nominal shear resistance in slabs of the box culvert is calculated per LRFD Design 5.12.7.3.

$V_c = [0.0676 \sqrt{f'_c} + 4.6 (A_s / bd_e) (V_u d_e / M_u)] bd_e$ LRFD Design Eq. 5.12.7.3-1

For each vehicle, V_u / M_u ratio shall be calculated using concurrent shear and moment demand. Note that for the single-cell box culverts, minimum shear capacity shall not be taken to be less than:

$V_c = 0.0948 \sqrt{f'_c} bd_e$ LRFD Design 5.12.7.3

Whereas nominal capacity shall not be greater than:

$V_c = 0.126 \sqrt{f'_c} bd_e$ LRFD Design Eq. 5.12.7.3-2

For conservative simplification of calculations, the contribution of longitudinal reinforcing steel is not considered.

$V_c = 0.0676 \sqrt{f'_c} bd_e < V_c = 0.0948 \sqrt{f'_c} bd_e$

$V_c = 0.0948 \times \sqrt{2.5} \times 12 \times 33.5 = 60.26$ kips

Critical section for shear: 31.56 in. from face of support. Calculate at 31.56 in. + 21 / 2 in. = 42.1 in.

Shear Resistance is the lesser of:

$V_n = V_c + V_s$ LRFD Design Eq. 5.7.3.3-1

$V_n = 0.25 f'_c b_v d_v$ LRFD Design Eq. 5.7.3.3-2

$V_s = A_v f_y d_v \cot \theta / s$ ($\alpha = 90°$ and θ is assumed as 45°) LRFD Design Eq. 5.7.3.3-4

$V_s = 0.79 \times 33 \times \cot 45 \times 31.56 / 18 = 45.71$ kips

$V_n = 0.25 \times 2.5 \times 37.84 \times 12 = 283.8$: Use $V_n = 60.26 + 45.71 = 105.97$ kips

A10.10.3—Sidewall Top–Center–Bottom

Rectangular Section height, $h = 3.0$ ft

Rectangular Section width, $b = 1.0$ ft

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$A_v = 0.0$ in.^{2}

Critical section for shear: LRFD Design 5.7.3.2

Effective Shear Depth: $d_v = M_n / (A_s f_y)$

$d_v = 128.37/(3.00 \times 33) \times 12 = 15.56$ in.

Nominal shear resistance of sidewalls is calculated per LRFD Design 5.7.3.3. The general procedure in LRFD Design 5.7.3.4.2 is applied for concurrent moment, shear, and axial demand by HL-93 Truck, HL-93 Tandem, Type 3, and SU-4 loadings. For all load cases, the following values are obtained: $\theta = 47.8°$, $\beta = 1.53$ at critical sections at the tops and bottoms of sidewalls.

Check $0.5 d_v \cot \theta < d_v$: Use d_v

Critical section for shear: 15.56 in. from face of support. Calculate at 15.56 in. + 27/2 in. = 29.1 in. (Top)

Critical section for shear: 15.56 in. from face of support. Calculate at 15.56 in. + 36/2 in. = 33.6 in. (Bottom)

Shear Resistance is the lesser of:

$V_n = V_c + V_s$ LRFD Design Eq. 5.7.3.3-1

$V_n = 0.25 f'_c b_v d_v$ LRFD Design Eq. 5.7.3.3-2

$V_c = 0.0316 \beta \sqrt{f'_c} b_v d_v$ LRFD Design Eq. 5.7.3.3-3

$V_c = 0.0316 \times 1.53 \times \sqrt{2.5} \times 15.56 \times 12 = 14.27$ kips

$V_s = A_v f_y d_v \cot \theta / s$ ($\alpha = 90°$) = 0.0 LRFD Design Eq. 5.7.3.3-4

$V_n = 0.25 \times 2.5 \times 15.56 \times 12 = 116.7$ kips: Use $V_n = 14.27$ kips

A10.11—Axial Thrust Resistance

Axial capacity of the culvert is calculated per LRFD Design 5.6.4.5. The maximum factored demand load is checked for the ten percent of the nominal compressive capacity of culvert sections. LRFD Design 5.6.4.5 states that if factored axial demand is lower than the 10 percent of factored nominal compressive strength as given below then factored bending moment load can be checked for the factored flexural resistance without considering the axial–flexural interaction.

$\varphi P = 0.1 \ \varphi f_k A_g$

$\varphi = 0.70$ (since no spirals or ties) LRFD Design 5.5.4.2

For the uncoupled axial–flexural check, a rating factor (RF_{thrust}) is calculated while ten percent of factored nominal compressive strength is accounted for. For the case where RF_{thrust} is lower than 1.0, an additional load rating needs to be performed per Article 6A.5.8 and LRFD Design Eq. 5.6.4.5-1.

Factored compressive resistance for each section of the culvert is given below.

Top Slab: $P_{top} = 0.1 \times 2.5$ ksi $\times 2.25$ ft $\times 12 \times 1$ ft $\times 12 = 81$ kips

Bottom Slab: $P_{bottom} = 0.1 \times 2.5$ ksi $\times 3$ ft $\times 12 \times 1$ ft $\times 12 = 108$ kips

Sidewall: $P_{side} = 0.1 \times 2.5$ ksi $\times 1.75$ ft $\times 12 \times 1$ ft $\times 12 = 63$ kips

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A10.12—Calculate Rating Factors, *RF*

For Strength Limit State, Rating Factor (*RF*):

$$RF = \frac{C \pm \gamma_{DC} DC \pm \gamma_{DW} DW \pm \gamma_{EV} EV \pm \gamma_{EH} EH \pm \gamma_{ES} ES}{\gamma_{ll} (LL + IM) \pm \gamma_{LS} LS}$$

$C = \varphi_C \varphi_S \varphi R_n$

$\varphi_C = 1.0$: Condition Factor (satisfactory condition) 6A.4.2.3

$\varphi_S = 1.0$: System Factor 6A.4.2.4

Resistance Factor for Flexural and Shear for Reinforced Concrete Cast-in-Place Box Structures:

Flexure: $\varphi_{flexure} = 0.90$ LRFD Design Table 12.5.5-1

Shear: $\varphi_{shear} = 0.85$ LRFD Design Table 12.5.5-1

A10.12.1—Load Factors for Culvert Load Rating (Table 6A.5.12.5-1) MBE Table 6A.5.12.5-1

Given the unfactored load analysis results, load factors with their minimum and maximum values define a load envelope that combines the load cases to produce the largest load demand. Considering the direction of the unfactored moment and shear forces in the sections, the minimum or the maximum load factor is applied in the next section.

Article 6A.5.12.10 indicates the effect of horizontal loads, i.e., *EH*, *ES*, and *LS*, may reduce effects caused by other loads and recommends 50 percent reduction in the horizontal earth load. Moreover, the minimum rating factor value of live load surcharge is equal to zero and its effect on the positive moments on the top slab can be neglected to generate highest load effect.

The factored load tables given below were generated using maximum–minimum load rating factors for the design load case at Strength I. The ultimate moment shear values are compared between the HL-93 tandem and the HL-93 truck and the higher load effect is accounted for in the RF calculations. For flexure, shear, and thrust, live load rating equations are given where minimum load factors are positively affecting the capacity and where maximum load factors have a negative effect on capacity—that is, they reduce capacity. For each force effect, formulations are explicitly provided for each section. Additionally, numerical examples at critical loads demonstrate the use of min–max load factors.

A10.12.2—Strength I Limit State: Design Load Rating Factor Calculation for Flexure

The use of min–max rating factors generates the highest demand on the sections. At seven sections, the rating factors are evaluated and the *RF* equation for each critical case is as given below.

Design Truck Loads: Single Lane Loads with *MPF* = 1.2.

See Figure A10.8-1 for locations of numbered sections.

Section 1: Top Slab Center Moment

Use maximum factors for *DC*, *DW*, *EV*, and minimum factors for *LS*.

$RF = [C - \gamma_{DC_max} DC - \gamma_{DW_max} DW - \gamma_{EV_max} EV - EH^* - ES^*] / [\gamma_{LL_} (LL + IM) + \gamma_{LS_min} LS]$ LRFD Design 3.11.7

*EH**, *ES** are 50 percent reduced and not combined with load factors.

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Section 2: Top Slab Corner Moment

Use maximum factors for all load cases.

$$RF = [C - \gamma_{DC_max}DC - \gamma_{DW_max}DW - \gamma_{EV_max}EV - \gamma_{EH_max}EH - \gamma_{ES_max}ES] / [\gamma_{LL_}(LL + IM) + \gamma_{LS_max}LS]$$

Section 3: Bottom Slab Center Moment

Use maximum factors for *DC*, *DW*, *EV*, and minimum factors for *LS*.

$$RF = [C - \gamma_{DC\,max}DC - \gamma_{DW_max}DW - \gamma_{EV_max}EV - \gamma_{EH_min}EH - \gamma_{ES_min}ES] / [\gamma_{LL}(LL + IM) + \gamma_{LS_min}LS]$$

Section 4: Bottom Slab Corner Moment

Use maximum factors for all load cases.

$$RF = [C - \gamma_{DC\,max}DC - \gamma_{DW_max}DW - \gamma_{EV_max}EV - \gamma_{EH_min}EH - \gamma_{ES_min}ES] / [\gamma_{LL}(LL + IM) + \gamma_{LS_min}LS]$$

Section 5: Sidewall Top Corner Moment

Use maximum factors for all load cases.

$$RF = [C - \gamma_{DC\,max}DC - \gamma_{DW_max}DW - \gamma_{EV_max}EV - \gamma_{EH_min}EH - \gamma_{ES_min}ES] / [\gamma_{LL}(LL + IM) + \gamma_{LS_min}LS]$$

Section 6: Sidewall Center Moment

Use maximum factors for *DC*, *DW*, *EV*, and minimum factors for *LS*.

$$RF = [C - \gamma_{DC\,max}DC - \gamma_{DW_max}DW - \gamma_{EV_max}EV - \gamma_{EH_min}EH - \gamma_{ES_min}ES] / [\gamma_{LL}(LL + IM) + \gamma_{LS_min}LS]$$

Section 7: Sidewall Bottom Corner Moment

Use maximum factors for all load cases.

$$RF = [C - \gamma_{DC\,max}DC - \gamma_{DW_max}DW - \gamma_{EV_max}EV - \gamma_{EH_min}EH - \gamma_{ES_min}ES] / [\gamma_{LL}(LL + IM) + \gamma_{LS_min}LS]$$

Table A10.12.2-1—Load Factor Selection: Moment Ratings for Design Loads

		Summary for *R*-Factor Selection (Moment)					
	Load	γ_{DC}	γ_{DW}	γ_{EV}	γ_{EH}	γ_{ES}	γ_{LS}
1	Top Slab Center	max	max	max	[a]	[a]	min [b]
2	Top Slab Corner	max	max	max	max	max	max
3	Bottom Slab Center	max	max	max	min [b]	min [b]	min [b]
4	Bottom Slab Corner	max	max	max	max	max	max
5	Sidewall Top	max	max	max	max	max	max
6	Sidewall Center	max	max	max	min [b]	min [b]	min [b]
7	Sidewall Bottom	max	max	max	max	max	max
	Design γ, max	1.25	1.50	1.30	1.35	1.50	1.75
	Strength I, min	0.90	0.65	0.90	0.90	0.75	0.00

[a] *EH* and *ES* are reduced 50 percent and not combined with load factors as indicated in LRFD Design 3.11.7.

[b] Minimum factor indicates *EH*, *ES*, *LS* are acting in the opposite direction with respect to *LL*.

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Table A10.12.2-2—Design Loads: Factored Moment

		Moment (kip-ft/ft) Factored (*LL* Inventory Factor)							
	Load	*DC*	*DW*	*EV*	*EH*	*ES*	*LL* + *IM* (Tandem)	*LL* + *IM* (Truck)	*LS*
1	Top Slab Center	43.48	7.31	44.29	–0.65 [a, b]	–0.04 [a, b]	33.06	27.00	0.00 [b]
2	Top Slab Corner	–33.71	–3.28	–35.91	–1.74	–0.12	–18.25	–17.38	–0.52
3	Bottom Slab Center	–69.91	–9.95	–60.33	2.44 [b]	0.12 [b]	–26.09	–33.43	0.00 [b]
4	Bottom Slab Corner	25.44	3.28	19.87	3.66	0.23	7.32	6.02	1.05
5	Sidewall Top	–33.71	–5.93	–35.91	–1.35	–0.12	–19.81	–20.78	–0.53
6	Sidewall Center	–29.70	–4.60	–27.89	3.79 [b]	0.18 [b]	–14.11	–15.98	0.00 [b]
7	Sidewall Bottom	–25.44	–3.28	–19.87	2.12	–0.12	–11.19	–13.39	–0.47
	Design γ, max	1.25	1.50	1.30	1.35	1.50	1.75	1.75	1.75
	Strength I, min	0.90	0.65	0.90	0.90 [b]	0.75 [b]	—	—	0.00 [b]

[a] Values are 50 percent reduced and unfactored moment values of *EH* and *ES*.

[b] The force is in the opposite direction with respect to *LL* and is factored by the minimum load factor.

Table A10.12.2-3—Strength I Design Load Moment Ratings

				Inventory	Operating
	Resistance	Moment (kip-ft/ft)	$\varphi_c\varphi_s\varphi R$	*RF*	*RF*
1	Top Slab Center	151.65	136.49	1.27	1.65
2	Top Slab Corner	–177.87	–160.08	4.55	5.89
3	Bottom Slab Center	–186.46	–167.81	0.90	1.17
4	Bottom Slab Corner	260.37	234.33	25.73	33.35
5	Sidewall Top	–128.37	–115.53	1.89	2.45
6	Sidewall Center	–128.37	–115.53	4.06	5.27
7	Sidewall Bottom	–128.37	–115.53	5.91	7.66

Moment Critical Sections

Top Slab Center: Same *RF* equation in A10.12.2 Section 1.

$$RF = [C \pm \gamma_{DC_max}DC \pm \gamma_{DW_max}DW \pm \gamma_{EV_max}EV \pm EH^* \pm ES^*] / [\gamma_{LL_}(LL + IM) + \gamma_{LS_min}LS]$$

$$RF = [1 \times 1 \times 0.9 \times 151.65 – 1.25 \times 34.73 – 1.50 \times 4.87 – 1.30 \times 34.07 – (–1.08) – (–0.07)] / [1.75 \times 23.18 + 0.0 \times (–0.67)] = 1.05$$

Bottom Slab Center: Same *RF* equation in A10.12.2 Section 3.

$$RF = [C \pm \gamma_{DC_max}DC \pm \gamma_{DW_max}DW \pm \gamma_{EV_max}EV \pm \gamma_{EH_min}EH \pm \gamma_{ES_min}ES] / [\gamma_{LL}(LL + IM) + \gamma_{LS_min}LS]$$

$$RF = [1 \times 1 \times 0.9 \times (–186.46) – 1.25 \times (–55.00) – 1.50 \times (–6.64) – 1.30 \times (–46.41) – 0.90 \times 4.55 – 0.75 \times 0.27] / [1.75 \times (–20.19) + 0.0 \times 1.31] = 0.94$$

A10.12.3—Strength I Limit State: Design Load Rating Factor Calculation for Shear

Section 2: Top Slab Corner Shear

Use maximum factors for all load cases.

EH, *ES*, *LS* do not induce shear force at Top Slab Corner

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$$RF = [C - \gamma_{DC_max}DC - \gamma_{DW_max}DW - \gamma_{EV_max}EV] / \gamma_{LL_}(LL + IM)$$

Section 4: Bottom Slab Corner Shear

Use maximum factors for all load cases.

EH, *ES*, *LS* do not induce shear force at Bottom Slab Corner

$$RF = [C - \gamma_{DC_max}DC - \gamma_{DW_max}DW - \gamma_{EV_max}EV] / \gamma_{LL}(LL + IM)$$

Section 5: Sidewall Top Corner Shear

Use maximum factors for *DW*, *EV*, *EH*, *ES*, *LS* and minimum factors for *DC*.

$$RF = [C - \gamma_{DC_min}DC - \gamma_{DW_max}DW - \gamma_{EV_max}EV - \gamma_{EH_max}EH - \gamma_{ES_max}ES] / [\gamma_{LL}(LL + IM) + \gamma_{LS_max}LS]$$

Section 7: Sidewall Bottom Corner Shear

Use maximum factors for all load cases.

$$RF = [C - \gamma_{DC_min}DC - \gamma_{DW_max}DW - \gamma_{EV_max}EV - \gamma_{EH_min}EH - \gamma_{ES_min}ES] / [\gamma_{LL}(LL + IM) + \gamma_{LS_min}LS]$$

Table A10.12.3-1—Load Factor Selection: Shear Ratings for Design Loads

		Summary for *R*-Factor Selection (Shear)					
	Load	γ_{DC}	γ_{DW}	γ_{EV}	γ_{EH}	γ_{ES}	γ_{LS}
1	Top Slab Center	—	—	—	—	—	—
2	Top Slab Corner	max	max	max	—	—	—
3	Bottom Slab Center	—	—	—	—	—	—
4	Bottom Slab Corner	max	max	max	—	—	—
5	Sidewall Top	max	max	max	max	max	max
6	Sidewall Center	—	—	—	—	—	—
7	Sidewall Bottom	max	max	max	min	min	min
	Design γ, max	1.25	1.50	1.30	1.35	1.50	1.75
	Strength I, min	0.90	0.65	0.90	0.90	0.75	0.00

Table A10.12.3-2—Design Loads: Factored Shear

		Moment (kip-ft/ft) Factored (*LL* Inventory Factor)							
	Load	*DC*	*DW*	*EV*	*EH*	*ES*	*LL* + *IM* (Tandem)	*LL* + *IM* (Truck)	*LS*
1	Top Slab Center	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2	Top Slab Corner	9.19	1.58	8.55	0.00	0.00	6.33	5.51	0.00
3	Bottom Slab Center	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
4	Bottom Slab Corner	–9.39	–1.58	–7.88	0.00	0.00	–4.00	–4.99	0.00
5	Sidewall Top	–0.94	–0.32	–1.88	–1.38	–0.17	–1.81	–1.54	–0.79
6	Sidewall Center	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7	Sidewall Bottom	–0.94	–0.32	–1.88	1.15 [a]	0.09 [a]	–1.81	–1.54	0.00 [a]
	Design γ, max	1.25	1.50	1.30	1.35	1.50	1.75	1.75	1.75
	Strength I, min	0.90	0.65	0.90	0.90 [a]	0.75 [a]	—	—	0.00 [a]

[a] The force is in the opposite direction with respect to *LL* and factored by the minimum load factor.

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Table A10.12.3-3–Strength I Design Load Shear Ratings

				Inventory	Operating
	Resistance	Shear (kip/ft)	$\varphi_c\varphi_s\varphi R$	*RF*	*RF*
1	Top Slab Center	—	—	—	—
2	Top Slab Corner	57.08	48.52	4.62 [a]	5.98 [a]
3	Bottom Slab Center	—	—	—	—
4	Bottom Slab Corner	–105.97	–90.07	14.29 [a]	18.52 [a]
5	Sidewall Top	–14.27	–12.13	2.87 [a]	3.72 [a]
6	Sidewall Center	—	—	—	—
7	Sidewall Bottom	–14.27	–12.13	5.66 [a]	7.33 [a]

Shear Critical Section

Side Wall Top: Same *RF* equation as in Article A10.12.3, Section 5.

$$RF = [C - \gamma_{DC_min}DC - \gamma_{DW_max}DW - \gamma_{EV_max}EV - \gamma_{EH_max}EH - \gamma_{ES_max}ES] / [\gamma_{LL}(LL + IM) + \gamma_{LS_max}LS]$$

$$RF = [1 \times 1 \times 0.9 \times (-18.66) - 1.25 \times (-0.80) - 1.50 \times (-0.21) - 1.30 \times (-1.45) - 1.35 \times (-3.03) - 1.50 \times (-0.20)] / [1.75 \times (-1.25) + 1.75 \times (-0.56)] = 2.91$$

A10.12.4—Strength I Limit State: Design Load Rating Factor Calculation for Axial Thrust

The sidewall sections are subjected to the maximum live load demand while providing the minimum axial capacity compared to top and bottom slabs. Therefore, the thrust checks shown in Article A10.11 are performed for these critical sections, RF_{thrust} for all sidewall sections are greater than 1.0 so that no further rating considering flexure–axial interaction is needed.

Section 5, 6, 7: Sidewall Top Corner, Center, and Bottom Corner Thrust

Use maximum factors for *DC*, *DW*, *EV*. Horizontal loads *EH*, *ES*, and *LS* have no influence on sidewall axial thrust.

$$RF = [C - \gamma_{DC_max}DC - \gamma_{DW_max}DW - \gamma_{EV_max}EV] / [\gamma_{LL}(LL + IM)]$$

Table A10.12.4-1—Load Factor Selection: Axial Thrust Ratings for Design Loads

		Summary for *R*-Factor Selection (Thrust)						
	Load	γ_{DC}	γ_{DW}	γ_{EV}	γ_{EH}	γ_{ES}	γ_{LL}	γ_{LS}
5	Sidewall Top	max	max	max	—	—	max	—
6	Sidewall Center	max	max	max	—	—	max	—
7	Sidewall Bottom	max	max	max	—	—	max	—
	Design γ, max	1.25	1.50	1.30	1.35	1.50	1.75	1.75
	Strength I, min	0.90	0.65	0.90	0.90	0.75	—	0.00

Note: Thrust is not induced on the vertical sidewalls due to *EH*, *ES*, and *LS*.

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Table A10.12.4-2—Design Loads: Factored Thrust

		Thrust (kip/ft) Factored							
	Load	*DC*	*DW*	*EV*	*EH*	*ES*	*LL + IM* (Tandem)	*LL + IM* (Truck)	*LS*
5	Sidewall Top	–12.06	–1.67	–10.10	0.00	0.00	–7.04	–7.26	0.00
6	Sidewall Center	–12.06	–1.67	–10.10	0.00	0.00	–7.04	–7.26	0.00
7	Sidewall Bottom	–12.06	–1.67	–10.10	0.00	0.00	–7.04	–7.26	0.00
	Design γ, max	1.25	1.50	1.30	1.35	1.50	1.75	1.75	1.75
	Strength I, min	0.90	0.65	0.90	0.90	0.75	—	—	0.00

Table A10.12.4-3—Strength I Design Load Thrust Ratings

				Inventory	Operating
	Resistance	Thrust (kip/ft)	$\varphi_c\varphi_s\varphi R$	*RF*	*RF*
5	Sidewall Top	–63.00	–44.10	2.88 [a]	3.73 [a]
6	Sidewall Center	–63.00	–44.10	2.88 [a]	3.73 [a]
7	Sidewall Bottom	–63.00	–44.10	2.88 [a]	3.73 [a]

Axial Thrust Critical Section

Side Wall:

$$RF = [C - \gamma_{DC_max}DC - \gamma_{DW_max}DW - \gamma_{EV_max}EV] \ / \ [\gamma_{LL}(LL + IM)]$$

$$RF = [1 \times 1 \times 0.7 \times (-63.0) - 1.25 \times (-9.52) - 1.50 \times (-1.11) - 1.30 \times (-7.77)] / [1.75 \times (-4.91)] = 2.38$$

A10.12.5—Strength I Limit State: Legal Load Rating for Flexure

The load factors selected in Article A10.12.2 are used in legal load ratings for flexure. Load ratings for Type 3 and SU-4 are given below. *MPF* = 1.2 is omitted for legal load rating per Article 6A.5.12.10.3.

Table A10.12.5-1—Legal Loads: Factored Moment

		Moment (kip-ft/ft) Factored							
	Load	*DC*	*DW*	*EV*	*EH*	*ES*	*LL + IM* (Type 3)	*LL + IM* (SU-4)	*LS*
1	Top Slab Center	43.48	7.31	44.29	–0.65 [a, b]	–0.04 [a, b]	21.78	25.27	0.00 [b]
2	Top Slab Corner	–33.71	–3.28	–35.91	–1.74	–0.12	–11.80	–13.96	–0.60
3	Bottom Slab Center	–69.91	–9.95	–60.33	2.44 [b]	0.12 [b]	–23.78	–25.37	0.00 [b]
4	Bottom Slab Corner	25.44	3.28	19.87	3.66	0.23	4.10	3.84	1.20
5	Sidewall Top	–33.71	–5.93	–35.91	–1.35	–0.12	–14.61	–16.21	–0.61
6	Sidewall Center	–29.70	–4.60	–27.89	3.79 [b]	0.18 [b]	–11.78	–12.83	0.00 [b]
7	Sidewall Bottom	–25.44	–3.28	–19.87	2.12	–0.12	–9.45	–10.50	–0.54
	Design γ, max	1.25	1.50	1.30	1.35	1.50	2.00	2.00	2.00
	Strength I, min	0.90	0.65	0.90	0.90 [b]	0.75 [b]	—	—	0.00 [b]

[a] Values are 50 percent reduced and unfactored moment values due to *EH* and *ES*.

[b] The force is in the opposite direction with respect to *LL*.

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Table A10.12.5-2—Strength I Legal Load Moment Ratings

				Type 3	SU-4
	Resistance	Moment (kip-ft/ft)	$\varphi_c\varphi_s\varphi R$	*RF*	*RF*
1	Top Slab Center	151.65	136.49	1.93 [a]	1.67 [a]
2	Top Slab Corner	–143.17	–128.85	6.88 [a]	5.86 [a]
3	Bottom Slab Center	–186.46	–167.81	1.27 [a]	1.19 [a]
4	Bottom Slab Corner	208.33	187.50	34.31 [a]	36.14 [a]
5	Sidewall Top	–128.37	–115.53	2.53 [a]	2.29 [a]
6	Sidewall Center	–128.37	–115.53	4.86 [a]	4.47 [a]
7	Sidewall Bottom	–128.37	–115.53	6.90 [a]	6.25 [a]

A10.12.6—Strength I Limit State: Legal Load Rating for Shear

The load factors selected in Article A10.12.3 are used in legal load ratings for flexure. Load ratings for Type 3 and SU-4 are given below.

Table A10.12.6-1—Legal Loads: Factored Shear

		Shear (kip/ft) Factored							
	Load	*DC*	*DW*	*EV*	*EH*	*ES*	*LL* + *IM* (Type 3)	*LL* + *IM* (SU-4)	*LS*
1	Top Slab Center	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2	Top Slab Corner	9.19	1.58	8.55	0.00	0.00	4.22	4.91	0.00
3	Bottom Slab Center	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
4	Bottom Slab Corner	–9.39	–1.58	–7.88	0.00	0.00	–3.21	–3.93	0.00
5	Sidewall Top	–0.94	–0.32	–1.88	–1.38	–0.17	–1.10	–1.26	–0.90
6	Sidewall Center	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7	Sidewall Bottom	–0.94	–0.32	–1.88	1.15 [a]	0.09 [a]	–1.10	–1.26	0.00
	Design γ, max	1.25	1.50	1.30	1.35	1.50	2.00	2.00	2.00
	Strength I, min	0.90	0.65	0.90	0.90 [a]	0.75 [a]	—	—	0.00

[a] The force is in the opposite direction with respect to *LL* and factored by the minimum load factor.

Table A10.12.6-2—Strength I Legal Load Shear Ratings

				Type 3	SU-4
	Resistance	Shear (kip/ft)	$\varphi_c\varphi_s\varphi R$	*RF*	*RF*
1	Top Slab Center	—	—	—	—
2	Top Slab Corner	57.08	51.37	6.93 [a]	5.86 [a]
3	Bottom Slab Center	—	—	—	—
4	Bottom Slab Corner	–92.64	–83.38	22.19 [a]	18.11 [a]
5	Sidewall Top	–18.66	–16.79	3.73 [a]	3.46 [a]
6	Sidewall Center	—	—	—	—
7	Sidewall Bottom	–18.66	–16.79	9.30 [a]	8.15 [a]

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A10.12.7—Strength I Limit State: Legal Load Rating for Axial Thrust

The load factors selected in Article A10.12.4 are used in legal load ratings for flexure. Load ratings for Type 3 and SU-4 are given below.

Table A10.12.7-1—Legal Loads: Factored Axial Thrust

		Thrust (kip/ft) Factored							
	Load	*DC*	*DW*	*EV*	*EH*	*ES*	*LL + IM* (Type 3)	*LL + IM* (SU-4)	*LS*
5	Sidewall Top	–12.06	–1.67	–10.10	0.00	0.00	–6.44	–7.34	0.00
6	Sidewall Center	–12.06	–1.67	–10.10	0.00	0.00	–6.44	–7.34	0.00
7	Sidewall Bottom	–12.06	–1.67	–10.10	0.00	0.00	–6.44	–7.34	0.00
	Design γ, max	1.25	1.50	1.30	1.35	1.50	2.00	2.00	2.00
	Strength I, min	0.90	0.65	0.90	0.90	0.75	—	—	0.00

Table A10.12.7-2—Strength I Legal Load Axial Thrust Ratings

				Type 3	SU-4
	Resistance	Thrust (kip/ft)	$\varphi_c\varphi_s\varphi R$	*RF*	*RF*
5	Sidewall Top	–63.00	–44.10	3.15 [a]	2.76 [a]
6	Sidewall Center	–63.00	–44.10	3.15 [a]	2.76 [a]
7	Sidewall Bottom	–63.00	–44.10	3.15 [a]	2.76 [a]

A10.13—Rating Summary

Rating summaries for Strength I design and legal loads are below. Note that the design load ratings are controlled by Tandem loading at sidewalls and top slab, and by truck load at the bottom slab. For the legal load ratings, SU-4 ratings are lower than Type 3 rating for all members.

Table A10.13-1—Strength I Design Load Ratings Summary

		Moment	Shear	Thrust		
	Design—Inventory	*RF*	*RF*	*RF*	Controlling Load	
1	Top Slab Center	1.27	—	—	Tandem	Moment
2	Top Slab Corner	4.55	4.62	—	Tandem	Moment
3	Bottom Slab Center	0.90	—	—	Truck	Moment
4	Bottom Slab Corner	25.73	14.29	—	Truck	Shear
5	Sidewall Top	1.89	2.87	2.88	Tandem	Moment
6	Sidewall Center	4.06	—	2.88	Tandem	Thrust
7	Sidewall Bottom	5.91	5.66	2.88	Tandem	Thrust

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Table A10.13-2 Strength I Legal Load Ratings Summary

		Moment	Shear	Thrust		
	Design—Inventory	*RF*	*RF*	*RF*	Controlling Load	
1	Top Slab Center	1.67	—	—	SU-4	Moment
2	Top Slab Corner	5.86	5.95	—	SU-4	Moment
3	Bottom Slab Center	1.19	—	—	SU-4	Moment
4	Bottom Slab Corner	36.14	18.11	—	SU-4	Shear
5	Sidewall Top	2.29	3.46	2.76	SU-4	Moment
6	Sidewall Center	4.47	—	2.76	SU-4	Thrust
7	Sidewall Bottom	6.25	8.15	2.76	SU-4	Thrust

A10.14—Capacity Check for Permanent Loads

This check investigates the capacity of the structure for permanent loads (without any live load) for the possible ultimate demand obtained by the max-min load factors. Moment, shear, and axial thrust factored *C* / *D* ratios are presented below where *D* also considers the effect of vertical and horizontal earth loads.

Table A10.14-1—Factored Capacity/Demand Ratio: Moment

		Moment (kip-ft/ft) Factored					Capacity	*C* / *D*
	Load	*DC*	*DW*	*EV*	*EH*	*ES*	*C*	*C* / *f* (*DL* + *EL*)
1	Top Slab Center	43.48	7.31	44.29	–0.65 [a]	–0.04 [a]	136.49	1.45
2	Top Slab Corner	–33.71	–3.28	–35.91	–1.74	–0.12	–160.08	2.14
3	Bottom Slab Center	–69.91	–9.95	–60.33	2.44 [a]	0.12 [a]	–167.81	1.22
4	Bottom Slab Corner	25.44	3.28	19.87	3.66	0.23	234.33	4.47
5	Sidewall Top	–33.71	–5.93	–35.91	–1.35	–0.12	–115.33	1.50
6	Sidewall Center	–29.70	–4.60	–27.89	3.79 [a]	0.18 [a]	–115.33	1.98
7	Sidewall Bottom	–25.44	–3.28	–19.87	2.12	–0.12	–115.33	2.48
	Design γ, max	1.25	1.50	1.30	1.35	1.50		
	Strength I, min	0.90	0.65	0.90	0.90 [a]	0.75 [a]		

[a] Numbers are *DL* and *EL* factored by minimum load factor to obtain the maximum demand.

Table A10.14-2—Factored Capacity / Demand Ratio: Shear

		Shear (kip/ft) Factored					Capacity	*C* / *D*
	Load	*DC*	*DW*	*EV*	*EH*	*ES*	*C*	*C* / *f* (*DL* + *EL*)
1	Top Slab Center	0.00	0.00	0.00	0.00	0.00	—	—
2	Top Slab Corner	9.19	1.67	9.91	0.00	0.00	51.37	2.47
3	Bottom Slab Center	0.00	0.00	0.00	0.00	0.00	—	—
4	Bottom Slab Corner	–9.39	–1.58	–7.88	0.00	0.00	–83.38	4.43
5	Sidewall Top	–0.94	–0.32	–1.88	–1.38	–0.17	–16.79	3.58
6	Sidewall Center	0.00	0.00	0.00	0.00	0.00	—	—
7	Sidewall Bottom	–0.94	–0.14 [a]	–1.30 [a]	1.73	0.18	16.79	–35.65
	Design γ, max	1.25	1.50	1.30	1.35	1.50		
	Strength I, min	0.90	0.65 [a]	0.90 [a]	0.90	0.75		

[a] Numbers are *DL* and *EL* factored by minimum load factor to obtain the maximum demand.

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Table A10.14-3—Factored Capacity / Demand Ratio: Axial Thrust

		Shear (kip/ft) Factored					Capacity	*C* / *D*
	Load	*DC*	*DW*	*EV*	*EH*	*ES*	*C*	*C* / *f* (*DL* + *EL*)
1	Top Slab Center	–0.94	–0.32	–1.88	–4.95	–0.42	–56.70	6.66
2	Top Slab Corner	–0.94	–0.32	–1.88	–4.95	–0.42	–56.70	6.66
3	Bottom Slab Center	0.68 [a]	0.14 [a]	1.30 [a]	–8.29	–0.47	–75.60	11.39
4	Bottom Slab Corner	0.68 [a]	0.14 [a]	1.30 [a]	–8.29	–0.47	–75.60	11.39
5	Sidewall Top	–12.06	–1.67	–10.10	0.00	0.00	–44.10	1.85
6	Sidewall Center	–12.06	–1.67	–10.10	0.00	0.00	–44.10	1.85
7	Sidewall Bottom	–12.06	–1.67	–10.10	0.00	0.00	–44.10	1.85
	Design γ, max	1.25	1.50	1.30	1.35	1.50		
	Strength I, min	0.90 [a]	0.65 [a]	0.90 [a]	0.90	0.75		

[a] Numbers are *DL* and *EL* factored by minimum load factor to obtain the maximum demand.

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A11—THROUGH TRUSS BRIDGE: GUSSET PLATE RATING

These gusset plate checks for design loads illustrate the application of the LRFR and LFR specifications, using the L2 joint of the truss depicted in Figure A11.1-1. Articles A11A.1 through A11A.9 provide an LRFR gusset plate rating example; Articles A11B.1 through A11B.9 provide an LFR gusset plate rating example.

A11.1—Bridge and Member Data

The truss bridge on I-94 was constructed in 1991. All members and plates were made with AASHTO M 223 Grade 50 material; all fasteners are $^{7}/_{8}$-in. diameter, A325 fully-tensioned bolts in standard holes that were subpunched and reamed to size. For the purposes of this design example, a center-to-center distance of 3 in. is assumed for all bolts. The truss is striped for four lanes of traffic and carries an *ADTT* of 10,000/day. There is no wearing surface on the deck (i.e. *DW* = 0). The superstructure has a current NBI Condition Rating of 7.

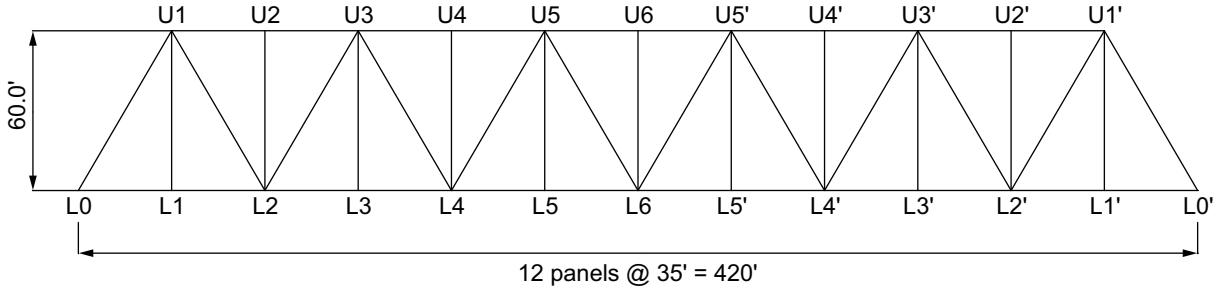

Figure A11.1-1—Truss Elevation

Detailing of the joint is shown in Figure A11.1-2. The loads shown are envelope loads developed using HS20-44 and alternate loading. Impact loading was considered to be 28.5 percent of the live load.

Note: For the purposes of this example, this LRFR rating will assume these loads were from HL-93 loading considering 33 percent impact, though a separate structural analysis would have to be performed to truly understand the differences in live load effects.

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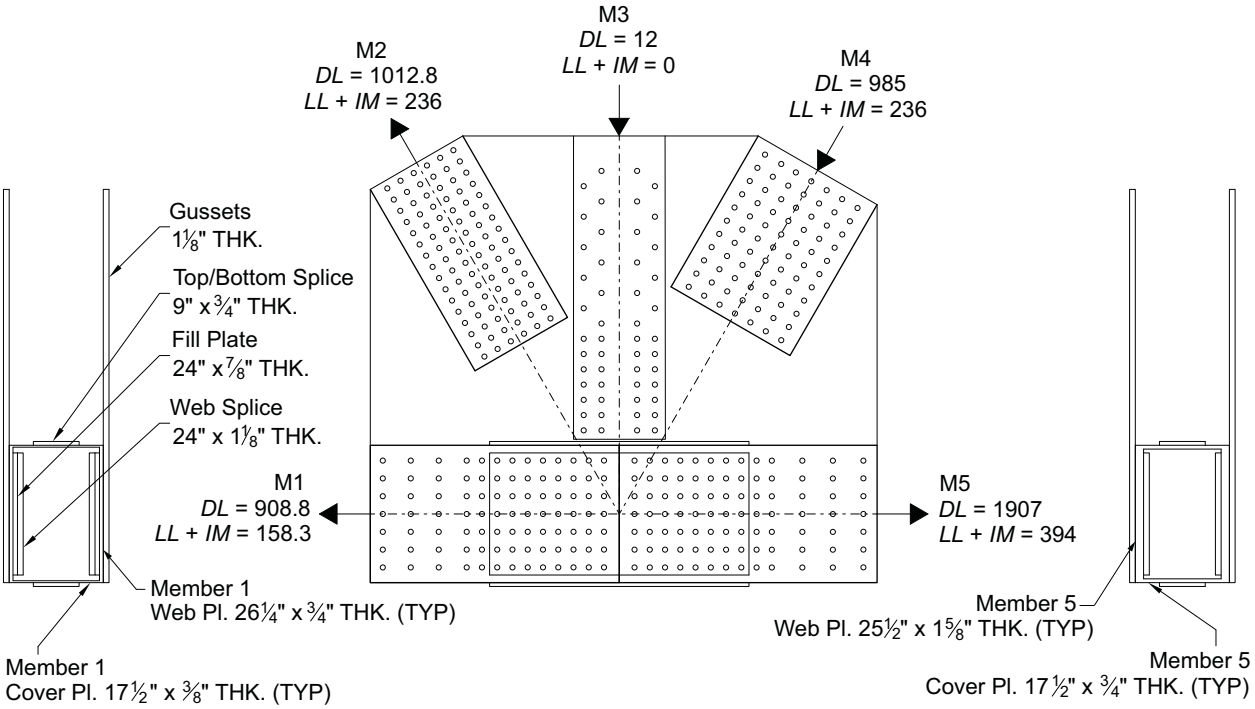

Figure A11.1-2—Panel Point L2 Details

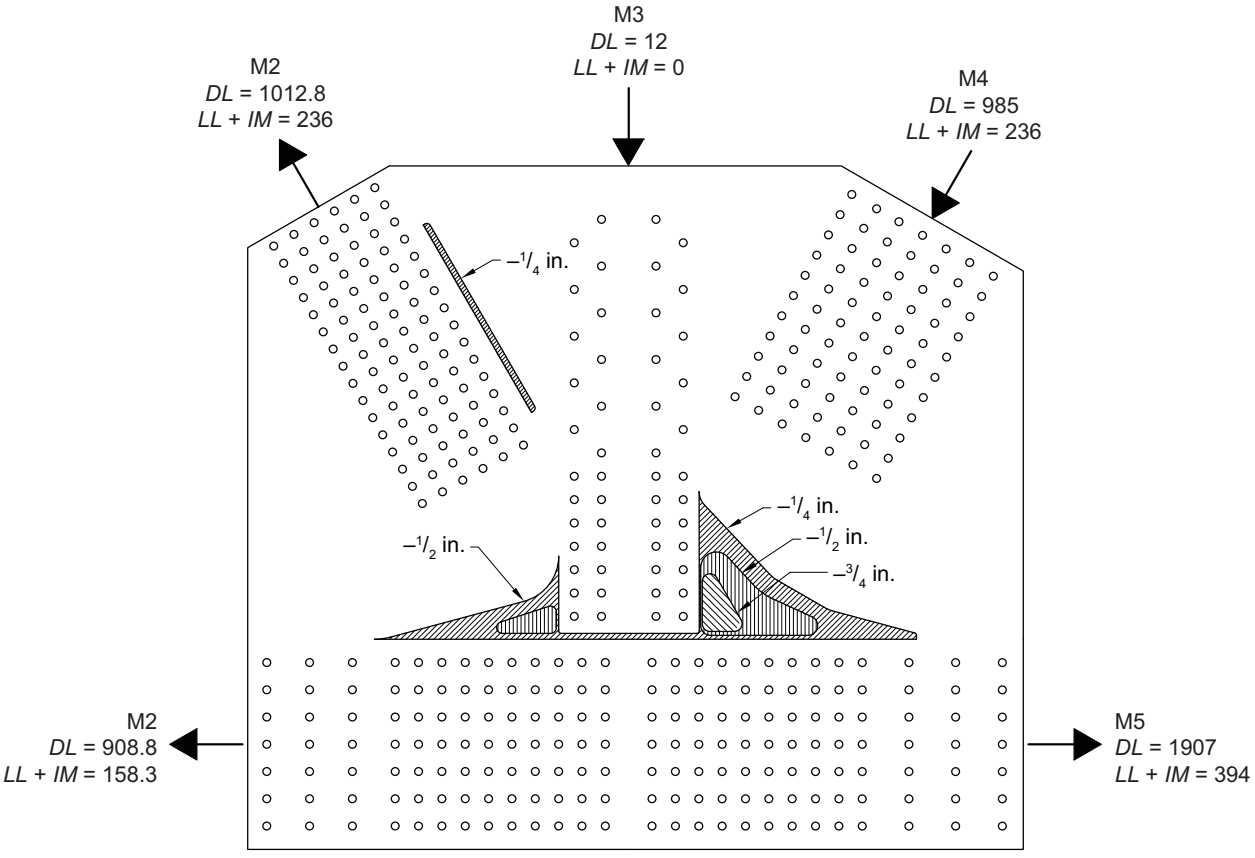

Figure A11.1-3—Losses to One Gusset Plates at L2

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PART A—LOAD AND RESISTANCE FACTOR RATING METHOD

A11A.1—Load and Resistance Factor Rating of Gusset Plates

The various evaluation factors used throughout the example are shown in Table A11A.1-1.

Table A11A.1-1—LRFR Load Factors for Gusset Plates

	Strength (Table 6A.4.2.2-1)	Service (Table 6A.4.2.2-1)
Inventory Live Load Factor (γ_{LL_INV})	1.75	1.3
Operating Live Load Factor (γ_{LL_OPR})	1.35	1.00
Dead Load Factor (γ_{DL})	1.25	1.00
System Factor (φ_s) (Table 6A.4.2.4-1)	0.90	0.90
Condition Factor (φ_c) (Table 6A.4.2.3-1)	1.00	1.00

In addition, detailed inspection reports of the L2 joint noted the discrete section loss on only one of the two gusset plates. The contour image in Figure A11.1-3 shows the pattern of section loss, again, only on one of two gusset plates. This section loss will be used in the relevant resistance checks.

The following rating checks will be performed:

1. Tension resistance of M1 (see Article A11A.2),
2. Chord splice resistance between M1 and M5 (see Article A11A.3),
3. Tension resistance of M2 (see Article A11A.4),
4. Compression resistance of M3 (see Article A11A.5),
5. Compression resistance of M4 (see Article A11A.6),
6. Horizontal shear resistance of the gusset plate (see Article A11A.7), and
7. Vertical shear resistance of the gusset plate (see Article A11A.8).

A11A.2—Check the Tension Resistance of M1

Only block shear resistance of the gusset plate, shear resistance of the bolts, and bearing resistance of the gusset must be checked. The checks for M5 are very similar in nature and will not be shown. Additionally, the bolt shear and bearing resistance checks will only be demonstrated for the gusset plate at M1; these checks would normally be performed at each fastener pattern and for the splice plates, too, for a complete rating.

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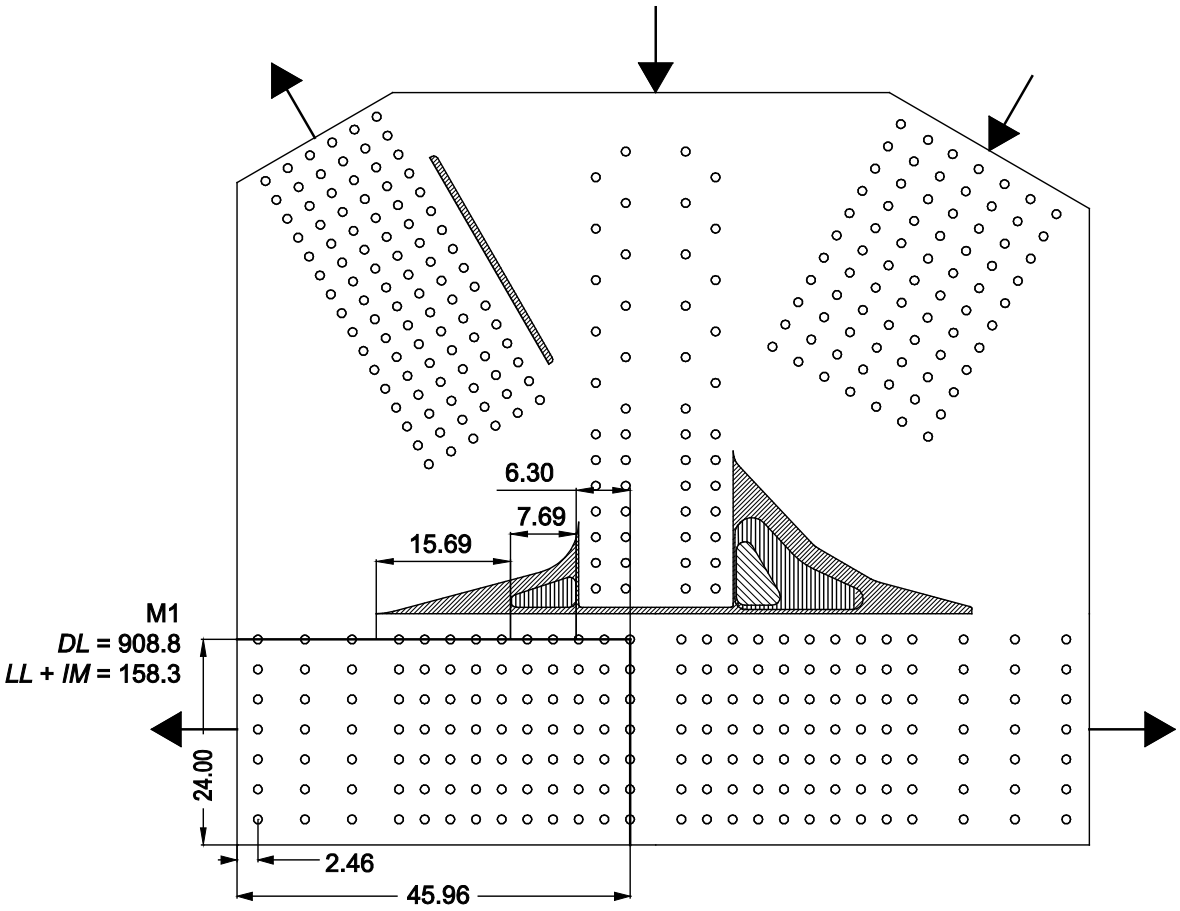

Figure A11A.2-1—Block Shear Failure Plane for Member M1 at L2

First, determine the shear resistance of the fasteners using Article 6A.6.12.6.2. Because web splice plates are used, some fasteners will be in double shear, but conservatively assume just single shear. The connection is not greater than 50 in. in length; therefore, a connection length reduction factor does not apply. Since the web splice plates have fill plates greater than 0.25 in., the bolt shear must be reduced by the fill plate factor.

$$\varphi R_n = \varphi_s 0.48 A_b F_{ub} N_s R \qquad \text{Eq. 6A.6.12.6.2-1}$$

$$R = \left[\frac{1+\gamma}{1+2\gamma}\right] \qquad \text{Eq. 6A.6.12.6.2-3}$$

$$\gamma = \left[\frac{A_f}{A_p}\right]$$

In this case, A_p is determined as the lesser of area of the lower chord web plate ($^3/_4$ in. × 26$^1/_4$ in.) and the sum of the areas of the web splice plate (1$^1/_8$ in. × 24 in.) and gusset plate (1$^1/_4$ in. × 87.8 in.). Given the large area of the gusset plate, it is clear that the area of the member web plate will control the calculation of A_p.

$$\gamma = \frac{(24)(0.875)}{(26.25)(0.75)} = 1.067$$

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$$R = \left[\frac{1+\gamma}{1+2\gamma}\right] = \left[\frac{1+1.067}{1+(2)(1.067)}\right] = 0.66$$

$$\varphi R_n = 0.80(0.48)\left(\frac{\pi(0.875^2)}{4}\right)120(1)(0.66) = 18.3 \text{ kip/bolt}$$

Check the bearing resistance of the gusset plate. All holes are at distances greater than 2 bolt diameters from each other and the edges; therefore, only one bearing resistance equation needs to be used.

$$\varphi Rn = \varphi_{bb} 2.4 dtF_u$$

$$\varphi R_n = 0.80(2.4)\left(\frac{7}{8}\right)(1.125)(65) = 122.8 \text{ kip/bolt} \qquad \text{Eq. 6A.6.12.6.4-1}$$

Note that the chord web plate thickness is 3/4 in. and the bearing resistance of the web plate will govern over the bearing resistance of the gusset plate. Therefore, a similar bearing resistance computation for the web plate should be performed as part of the member rating. However, as this example is for gusset plates alone, the bearing resistance computation for the chord member will not be shown herein.

The resistance is the lesser of the fastener shear resistance and plate bearing resistance. In this case, the fastener resistance controls. For this member, there are 182 bolts. Conservatively, the fill plate reduced bolt shear resistance is assumed over all the bolts, despite the fill plate not covering the entire bolt pattern. The total resistance is:

$$R_r = 182 \times 18.3 \text{ kip/bolt} = 3{,}330 \text{ kips}$$

Next, determine the resistance due to block shear. (See Figure A11A.2-1.) First, calculate the shear and tensile areas.

$$A_{tg} = 2 \times 24.00 \times 1.125 = 54.00 \text{ in.}^2$$

$$A_{tn} = 54.00 - 13\left(\frac{7}{8} + \frac{1}{8}\right)1.125 = 39.38 \text{ in.}^2$$

Gross shear plane of the gusset plate without corrosion damage:

$$A_{vg1} = 45.96 \times 1.125 = 51.71 \text{ in.}^2$$

Gross shear plane of the gusset plate with corrosion damage:

$$A_{vg2} = (45.96)(1.125) - (6.30 + 15.69)(0.25) - (7.69)(0.50) = 42.36 \text{ in.}^2$$

Net shear plane:

$$A_{vn} = (51.71 + 42.36) - 25\left(\frac{7}{8} + \frac{1}{8}\right)1.125 = 65.95 \text{ in.}^2$$

Calculate the DL/LL ratio for this limit state to account for further reduction according to Article 6A.6.12.6.1.

$\frac{P_{1DL}}{P_{1LL}} = \frac{908.8}{158.3} = 5.74$ which is greater than 1.0; therefore, the additional reduction is:

$$R_{DL_LL} = 1 - 0.1\left(\frac{5.74 - 1.0}{5}\right) = 0.90 \qquad \text{6A.6.12.6.1}$$

Calculate the block shear resistance of each gusset plate.

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Is $F_u A_{vn} \le F_y \left(A_{vg1} + A_{vg2} \right)$?

$65(65.95) \le 50(51.71+42.36) \rightarrow 4,287 \le 4,703,$ yes. Therefore:

$$P_{bs1} = \varphi_{bs} R_p \left(0.58 F_u A_{vn} + F_u A_{tn} \right)$$ Eq. 6A.6.12.6.8-1

$$P_{bs1} = (1.00)(1.0)\left[\left(0.58 \right)(65)(65.95) + \left(65 \right)(39.38) \right] = 5,046$$

Since the resistance from the bolts is less than the block shear resistance, then the capacity is controlled by the bolt resistance, C = 3,330 kips.

Calculate the rating factors.

$$RF_{LRFR} = \frac{\varphi_s \varphi_c C(R_{DL_LL}) - \gamma_{DL} P_{1DL}}{\gamma_{LL} P_{1LL}}$$ Eq. 6A.4.2.1-1

$$RF_{LRFRinv} = \frac{0.90(1.00)(3,330)(0.90) - (1.25)(908.8)}{(1.75)(158.3)} = 5.63$$

$$RF_{LRFRopr} = RF_{LRFRinv} \left(\frac{\gamma_{LLinv}}{\gamma_{LLopr}} \right) = 5.63 \left(\frac{1.75}{1.35} \right) = 7.30$$

The gusset plate was also originally designed to be slip-critical with the members. Therefore rating factors must also be calculated for the slip resistance. Slip resistance is checked with the Service II load combination.

The slip resistance of the bolt group is calculated as:

$$R_r = R_n = K_h K_s N_s P_t (\text{\# of bolts})$$ Eq. 6A.6.12.6.3-1

For standard holes, K_h = 1.0 6A.6.12.6.3

A Class B surface preparation was specified, K_s = 0.50 Table 6A.6.12.6.3-2

Conservatively, one shear plane is assumed for all bolts, N_s = 1

P_t = 39 kips Table 6A.6.12.6.3-1

$R_r = 1.0(0.50)(1)(39)(182) = 3549$ kips

Per Article 6A.6.12.6.3, slip only needs to be calculated at the operating level under Service II loads.

$$RF_{LRFRopr} = \frac{\varphi_s \varphi_c C - \gamma_{DL} P_{1DL}}{\gamma_{LL} P_{1LL}} = \frac{0.90(1.00)(3,549) - (1.00)(908.8)}{(1.00)(158.3)} = 14.44$$ Eq. 6A.4.2.1-1

Note that in the above equation, the dead-to-live load ratio reduction factor is not applied to the capacity per Article 6A.6.12.6.1 since it is a service limit state.

A11A.3—Check the Chord Splice Resistance

Since the gusset plate is also performing as a tension chord splice, the resistance of the spliced section must be evaluated according to the provisions of Article 6A.6.12.6.9. To begin, the net and gross section modulus need to be calculated for the composite section of gusset and splice plates (see Figure A11A.3-1).

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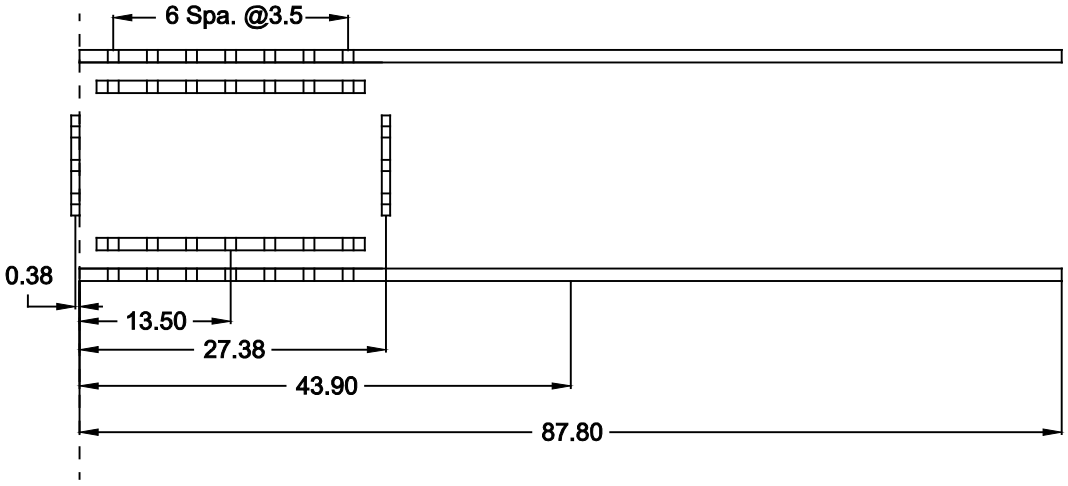

Figure A11A.3-1—Section of Gusset and Splice Plates at L2

Start by calculating the gross and net section properties of the combined section of gusset and web splice plates. For brevity, the calculations will not be shown. In this case, they were calculated using a drawing program; alternatively, the calculations can be completed using a spreadsheet or long-hand equations.

The gross and net section areas are:

$A_g = 265.1\,\text{in.}^2$

$A_n = 229.0\,\text{in.}^2$

The heights of the gross and net section centroids, using the bottom of the gusset as the datum, are:

$h_g = 36.2\,\text{in.}$

$h_n = 39.7\,\text{in.}$

Next, calculate the gross and net moments of inertia.

$I_g = 178{,}592.7\,\text{in.}^4$

$I_n = 154{,}792.4\,\text{in.}^4$

The maximum bending stress will be at the bottom of the gusset plate. Therefore, calculate the gross and net section moduli relative to the bottom of the gusset.

$$S_g = \frac{178{,}592.7}{36.2} = 4{,}933.5\,\text{in.}^3$$

$$S_n = \frac{154{,}792.4}{39.7} = 3{,}899.1\,\text{in.}^3$$

Since the chord splice section intersects the workpoint of the joint, the force resultant of the chord is at the center of the chord depth, so the gross and net section eccentricities are:

$$e_{p_\text{gross}} = 36.2 - \frac{27}{2} = 22.7\ \text{in.}$$

$$e_{p_\text{net}} = 39.7 - \frac{27}{2} = 26.2\ \text{in.}$$

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The resistance of the chord splice can be calculated as the minimum of Eqs. 6A.6.12.6.9-3 and 6A.6.12.6.9-4.

$$P_{r_\text{gross}} = \varphi_{cs} F_y \left(\frac{S_g A_g}{S_g + e_{p_\text{gross}} A_g} \right) \qquad \text{Eq. 6A.6.12.6.9-3}$$

$$P_{r_\text{gross}} = 0.85(50)\left(\frac{(4{,}933.5)(265.1)}{4{,}933.5 + (22.7)(265.1)} \right) = 5{,}076 \text{ kips}$$

$$P_{r_\text{net}} = \varphi_{cs} F_u \left(\frac{S_n A_n}{S_n + e_{p_\text{net}} A_n} \right) \qquad \text{Eq. 6A.6.12.6.9-4}$$

$$P_{r_\text{net}} = 0.85(65)\left(\frac{(3{,}899.1)(229.0)}{3{,}899.1 + (26.2)(229.0)} \right) = 4{,}984 \text{ kips}$$

The lesser resistance controls; in this case, it is controlled by fracture on the net section, $C = 4984$ kips.

Next, determine the factored load in the chord splice. Look at the loads on both sides of the splice and use the more severe.

Left Side

$$1.25\left\{908.8 + (1{,}012.8)\left[\cos(60°)\right]\right\} + 1.75\left\{158.3 + (236)\left[\cos(60°)\right]\right\} = 2{,}252 \text{ kips}$$

Right Side

$$1.25\left\{1907 - (985)\left[\cos(60°)\right]\right\} + 1.75\left\{394 - (236)\left[\cos(60°)\right]\right\} = 2{,}251 \text{ kips}$$

Note: This example does not explore the potential unconservatism of the use of nonconcurrent envelope forces in the determination of the resolved forces, as discussed in Article C6A.6.12.6.9.

Since the loads are nearly the same on each side of the splice, determine the most severe *DL*/*LL* ratio reduction.

$$\frac{P_{DL_\text{left}}}{P_{LL_\text{left}}} = \frac{908.8 + (1{,}012.8)\left[\cos(60°)\right]}{158.3 + (236)\left[\cos(60°)\right]} = 5.12$$

$$\frac{P_{DL_\text{right}}}{P_{LL_\text{right}}} = \frac{1{,}907 - (985)\left[\cos(60°)\right]}{394 - (236)\left[\cos(60°)\right]} = 5.12$$

Since the dead load-to-live load ratio reduction factor is the same for both sides, use the controlling loads from the left side of the splice.

$$\text{R}_{\text{DL_LL}} = 1 - 0.1\left(\frac{5.12 - 1.0}{5} \right) = 0.92 \qquad \text{6A.6.12.6.1}$$

Calculate the LRFR rating factors.

$$RF_{LRFR} = \frac{\varphi_s \varphi_c C\left(R_{DL_LL}\right) - \gamma_{DL} P_{DL}}{\gamma_{LL} P_{LL}} \qquad \text{Eq. 6A.4.2.1-1}$$

$$RF_{LRFRinv} = \frac{0.90(1.00)(4{,}984)(0.92) - (1.25)\left\{908.8 + (1{,}012.8)\left[\cos(60°)\right]\right\}}{(1.75)\left\{158.3 + (236)\left[\cos(60°)\right]\right\}} = 4.87$$

$$RF_{LRFRopr} = RF_{LRFRinv} \frac{\gamma_{LLinv}}{\gamma_{LLopr}} = 4.87 \frac{1.75}{1.35} = 6.32$$

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A11A.4—Check Tension Resistance of M2

The tension capacity will be governed by the lesser of the gross yield and net fracture on the Whitmore section (see Figure A11A.4-1), and block shear (See Figure A11A.4-2). Consider the section loss in the one gusset plate.

Define the gross and net section areas on the Whitmore section.

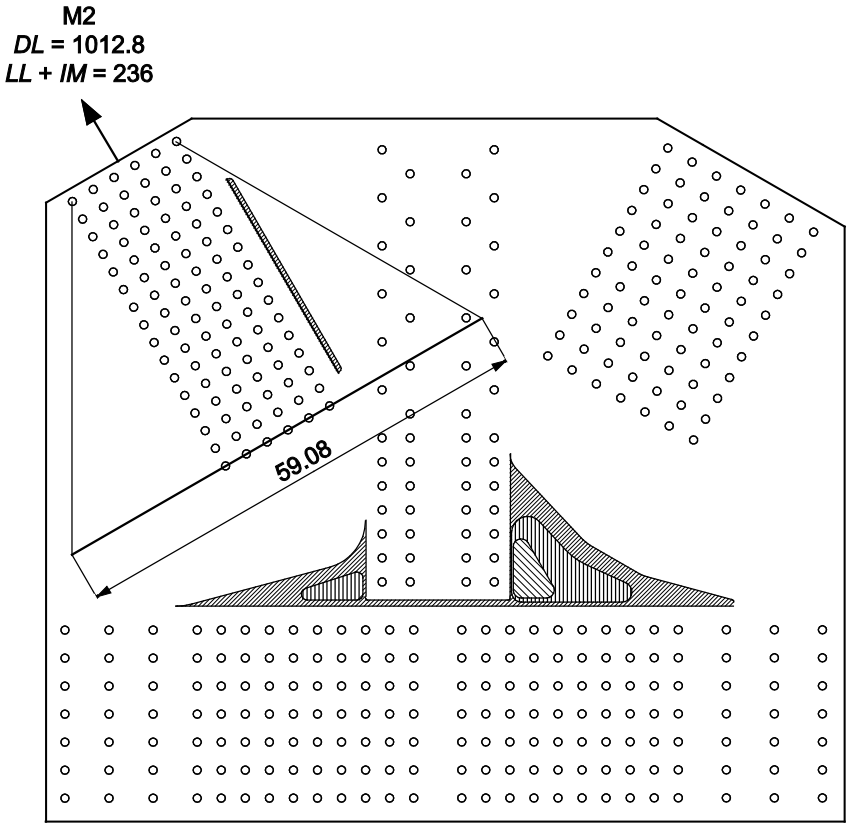

Figure A11A.4-1—Member M2 Whitmore Section at L2

$$A_g = 2 \times 59.08 \times 1.125 = 132.93 \text{ in.}^2$$

$$A_n = 132.93 - 12\left(\frac{7}{8} + \frac{1}{8}\right)1.125 = 119.43 \text{ in.}^2$$

Note: There are no reductions due to corrosion for the Whitmore tension resistance because no section loss intersects the Whitmore section.

Define the gross and net sections for block shear check.

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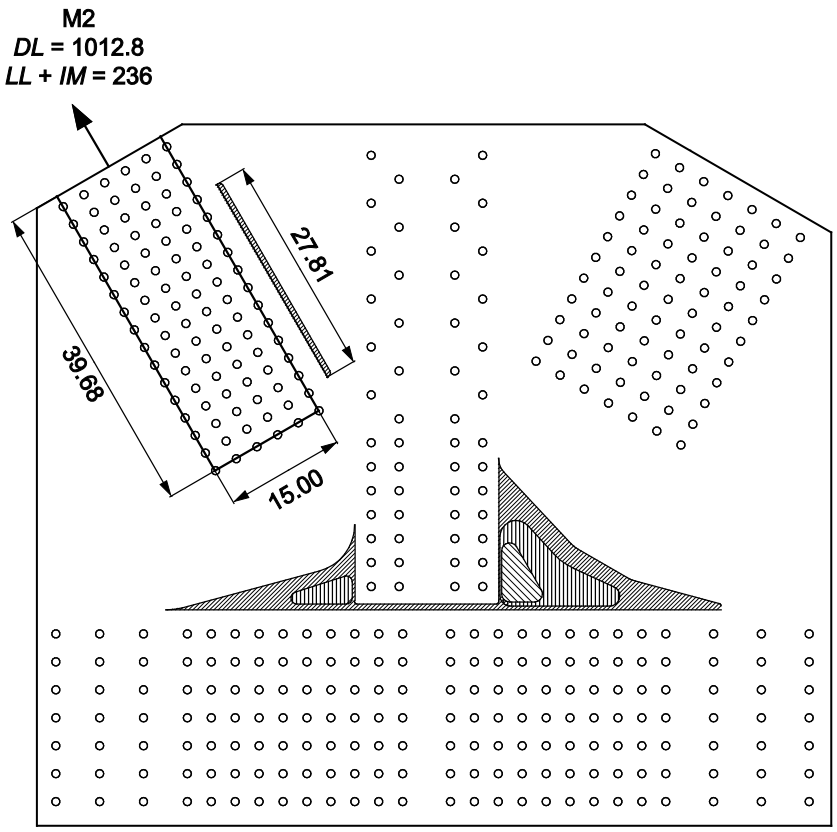

Figure A11A.4-2—Member M2 Block Shear Plane at L2

$$A_{tg} = 2 \times 15.00 \times 1.125 = 33.75 \text{ in.}^2$$

$$A_{tn} = 33.75 - 10\left(\frac{7}{8} + \frac{1}{8}\right)1.125 = 22.5 \text{ in.}^2$$

Gross shear plane of the gusset plate without corrosion damage:

$$A_{vg1} = 2 \times 39.68 \times 1.125 = 89.28 \text{ in.}^2$$

Gross shear plane of the gusset plate with corrosion damage:

$$A_{vg2} = 39.68 \times 1.125 + 27.81(1.125 - 0.25) + (39.68 - 27.81)1.125 = 82.33 \text{ in.}^2$$

Net shear plane:

$$A_{vn} = (2 \times 89.28) - 62\left(\frac{7}{8} + \frac{1}{8}\right)1.125 = 108.81 \text{ in.}^2$$

Calculate the LRFR rating factors.

Calculate the *DL*/*LL* ratio for this limit state to account for further reduction according to Article 6A.6.12.6.1.

$\frac{P_{2DL}}{P_{2LL}} = \frac{1{,}012.8}{236} = 4.29,$ which is greater than 1.0; therefore, the additional reduction is:

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$$R_{DL_LL} = 1 - 0.1\left(\frac{4.29 - 1.0}{5}\right) = 0.93 \qquad \text{6A.6.12.6.1}$$

Calculate the yield on gross and fracture on net capacities on the Whitmore section.

$$P_y = \varphi_y F_y A_g = 0.95 \times 50 \times 132.93 = 6,314 \qquad \text{Eq. 6A.6.12.6.8-2}$$

$$P_r = \varphi_u F_u A_n R_p U = 0.80 \times 65 \times 119.43 \times 1.0 \times 1.0 = 6,210 \qquad \text{Eq. 6A.6.12.6.8-3}$$

Calculate the block shear resistance of each gusset plate.

Is $F_u A_{vn} \le F_y \left(A_{vg1} + A_{vg2}\right)$? Eq. 6A.6.12.6.8-1

$65(108.81) \le 50(89.28 + 82.33) \rightarrow 7,073 \le 8,580$, yes. Therefore:

$$P_{bs1} = \varphi_{bs} R_p \left(0.58 F_u A_{vn} + F_u A_{tn}\right) = 1.00 \times 1.0\left[\left(0.58 \times 65 \times 108.81\right) + \left(65 \times 22.5\right)\right] = 5,564.6$$

The overall tension capacity, *C*, is the lesser of the Whitmore yield on gross, Whitmore fracture on net, and block shear; in this case, it is controlled by block shear so *C* = 5564.6 kips.

Calculate the rating factors.

$$RF_{LRFR} = \frac{\varphi_s \varphi_c C(R_{DL_LL}) - \gamma_{DL} P_{2DL}}{\gamma_{LL} P_{2LL}} \qquad \text{Eq. 6A.4.2.1-1}$$

$$RF_{LRFRinv} = \frac{0.90 \times 1.00 \times 5,564.6 \times 0.93 - 1.25 \times 1,012.8}{1.75 \times 236} = 8.21$$

$$RF_{LRFRopr} = RF_{LRFRinv} \frac{\gamma_{LLinv}}{\gamma_{LLopr}} = 8.21 \frac{1.75}{1.35} = 10.64$$

A11A.5—Check the Compression Resistance of M3

It is clear that this member will not control with only 12 kips of dead load and, since there is no live load, the rating factor cannot be calculated. For illustration, only the capacity calculations will be shown.

For the Whitmore buckling check, project the worst-case section loss onto the Whitmore section (see Figure A11A.5-1) in the direction of the member to calculate the equivalent gusset plate thickness.

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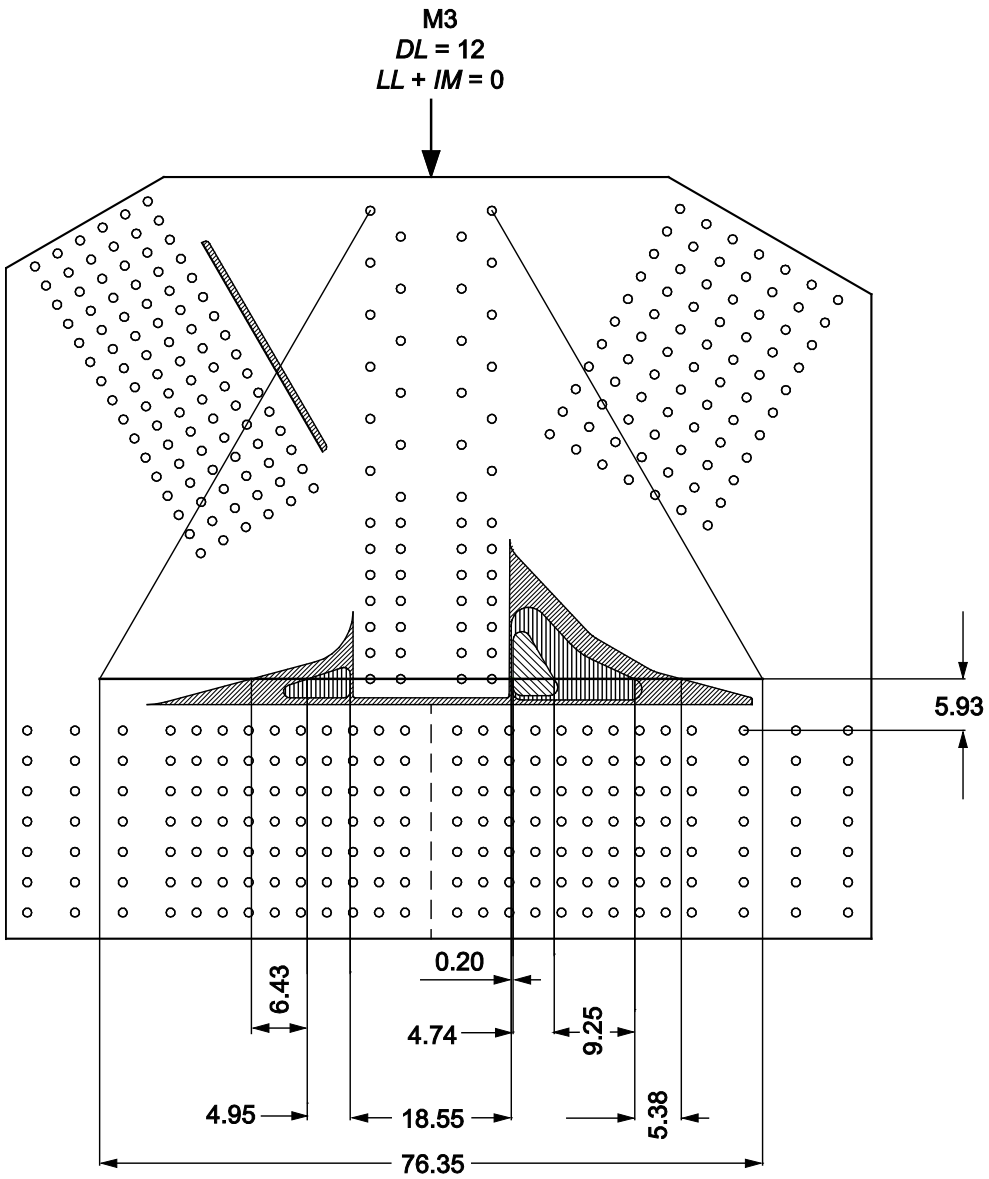

Figure A11A.5-1—Member M3 Whitmore Section at L2

$$t_{eq_whit} = \frac{30.36(1.125-0.25)+14.40(1.125-0.5)+4.74(1.125-0.75)+1.125(76.35-30.36-14.40-4.74)}{76.35}$$

$$t_{eq_whit} = 0.88$$

This member has no live load so it has no extra dead-to-live load reduction.

Calculate the factored Whitmore buckling strength per gusset plate because only one of the plates has section loss.

For the gusset plate with section loss:

$$P_{el} = \frac{3.29E}{\left(\frac{L_{mid}}{t_{eq_whit}}\right)^2} A_g = \frac{3.29(29{,}000)}{\left(\frac{5.93}{0.88}\right)^2}(76.35\times 0.88) = 141{,}928 \qquad \text{Eq. 6A.6.12.6.7-4}$$

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$$P_{o1} = F_y A_g = 50\left(76.35 \times 0.88\right) = 3,377.4$$ 6A.6.12.6.7

$$\frac{P_{e1}}{P_{o1}} = \frac{141,928}{3,377.4} = 42.0 \text{ is greater than } 0.44\text{; therefore :}$$

$$\text{Whit}_1 = 0.658^{P_{o1}/P_{e1}} P_{o1} = 0.658^{3,377.4/141,928} 3,377.4 = 3,343.9$$ Eq. 6A.6.12.6.7-2

For the gusset plate without section loss:

$$P_{e2} = \frac{3.29E}{\left(\frac{L_{mid}}{t_g}\right)^2} A_g = \frac{3.29\left(29,000\right)}{\left(\frac{5.93}{1.125}\right)^2}\left(76.35 \times 1.125\right) = 294,952$$ Eq. 6A.6.12.6.7-4

$$P_{o2} = F_y A_g = 50\left(76.35 \times 1.125\right) = 4,294.7$$ 6A.6.12.6.7

$$\frac{P_{e2}}{P_{o2}} = \frac{294,952}{4,294.7} = 68.68 \text{ is greater than } 0.44\text{; therefore :}$$

$$Whit_2 = 0.658^{P_{o2}/P_{e2}} P_{o2} = 0.658^{4,294.7/294,952} 4,294.7 = 4,268.6$$ Eq. 6A.6.12.6.7-2

Therefore, the total Whitmore buckling capacity is the sum of the two individual gusset plates:

$$Whit = \varphi_{cg}\left(Whit_1 + Whit_2\right) = 0.95 \times \left(3,343.9 + 4,268.6\right) = 7,231.9$$ Eq. 6A.6.12.6.7-1

This vertical member does not have any admissible partial shear planes that would control the compression resistance. Therefore, the capacity, *C*, is the Whitmore buckling strength.

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A11A.6—Check the Compression Resistance of M4

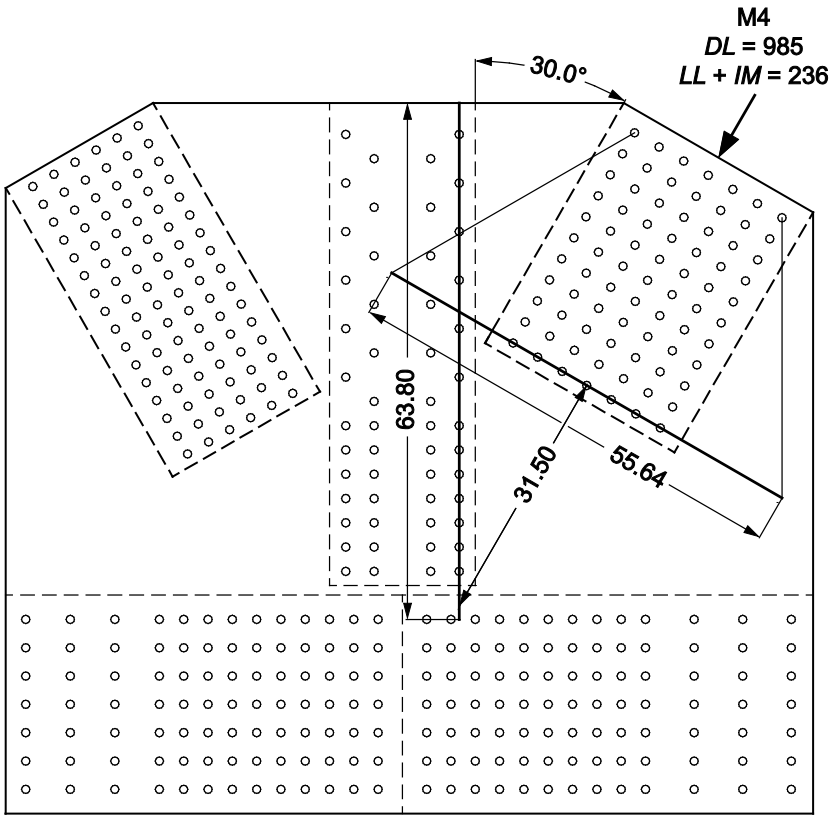

Figure A11A.6-1—Member M4 Compression Checks at L2

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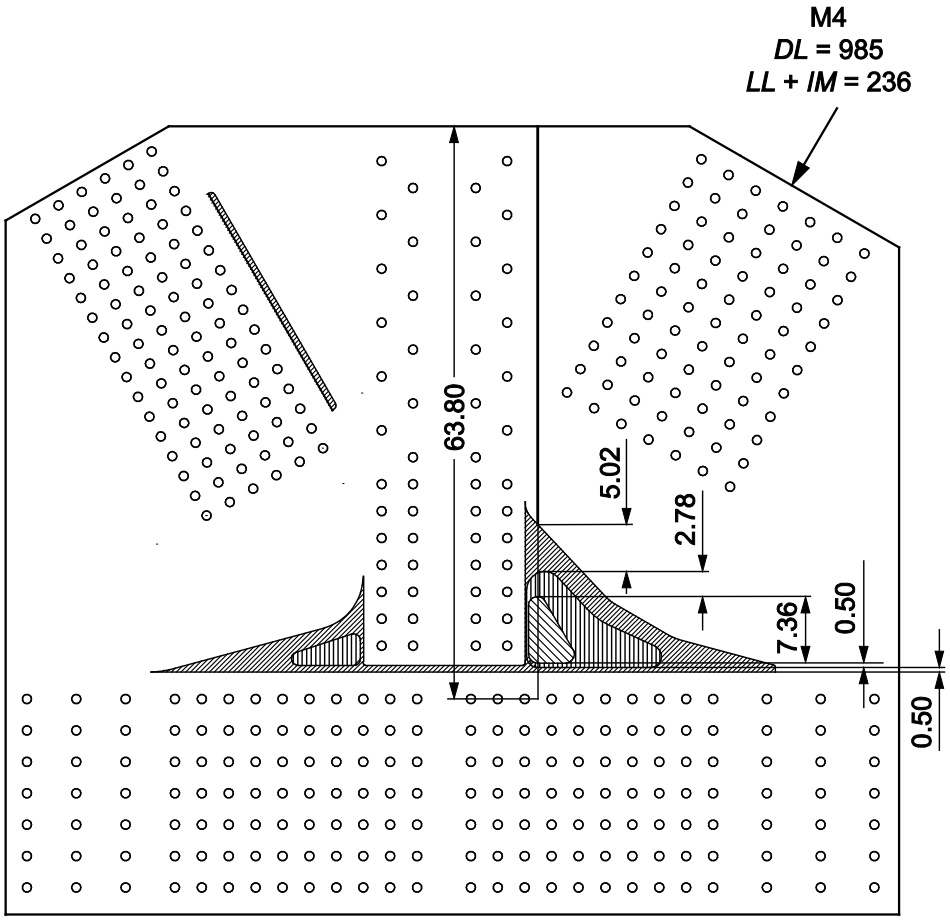

Figure A11A.6-2—Member M4 Partial Shear Plane at L2

Calculate the equivalent gusset plate thickness on the partial shear plane for the gusset plate with section loss (see Figure A11A.6-2).

$$t_{eq_shr} = \frac{5.52(1.125-0.25)+3.28(1.125-0.5)+7.36(1.125-0.75)+1.125(63.80-5.52-3.28-7.36)}{63.80}$$

$$t_{eq_shr} = 0.99$$

For the Whitmore buckling check, project the worst case section loss onto the Whitmore section in the direction of the member to calculate the equivalent gusset plate thickness (see Figure A11A.6-3).

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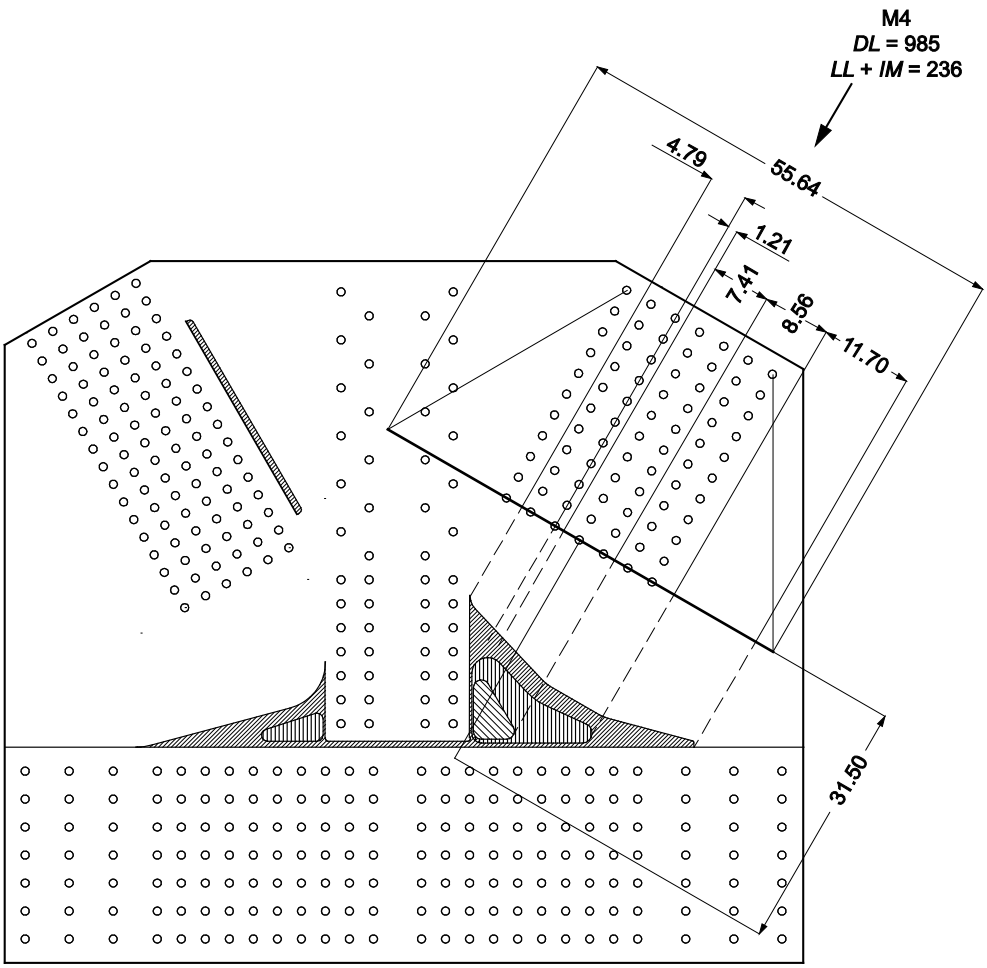

Figure A11A.6-3—Member M4 Whitmore Section at L2

$$t_{eq_whit} = \frac{16.49(1.125-0.25)+9.77(1.125-0.5)+7.41(1.125-0.75)+1.125(55.64-16.49-9.77-7.41)}{55.64}$$

$$t_{eq_whit} = 0.86$$

For LRFR, calculate the DL/LL ratio reduction according to Article 6A.6.12.6.1.

$\frac{P_{4DL}}{P_{4LL}} = \frac{985}{236} = 4.17$ which is greater than 1.0; therefore, the additional reduction is:

$$R_{DL_LL} = 1-0.1\left(\frac{4.17-1.0}{5}\right) = 0.94 \qquad \text{6A.6.12.6.1}$$

Calculate the factored Whitmore buckling strength per gusset plate, because only one of the plates has section loss.

For the gusset plate with section loss:

$$P_{e1} = \frac{3.29E}{\left(\frac{L_{mid}}{t_{eq_whit}}\right)^2} A_g = \frac{3.29(29,000)}{\left(\frac{31.50}{0.86}\right)^2}(55.64\times 0.86) = 3,441.4 \qquad \text{Eq. 6A.6.12.6.7-4}$$

$$P_{o1} = F_y A_g = 50(55.64\times 0.86) = 2,401.5 \qquad \text{6A.6.12.6.7}$$

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$\frac{P_{e1}}{P_{o1}} = \frac{3{,}441.4}{2{,}401.5} = 1.43$ is greater than 0.44; therefore :

$$Whit_1 = 0.658^{P_{o1}/P_{e1}} P_{o1} = 0.658^{2{,}401.5/3{,}441.4} 2{,}401.5 = 1{,}793.2 \qquad \text{Eq. 6A.6.12.6.7-2}$$

For the gusset plate without section loss:

$$P_{e2} = \frac{3.29E}{\left(\frac{L_{mid}}{t_g}\right)^2} A_g = \frac{3.29(29{,}000)}{\left(\frac{31.50}{1.125}\right)^2} (55.64 \times 1.125) = 7{,}617.6 \qquad \text{Eq. 6A.6.12.6.7-4}$$

$$P_{o2} = F_y A_g = 50(55.64 \times 1.125) = 3{,}129.8 \qquad \text{6A.6.12.6.7}$$

$\frac{P_{e2}}{P_{o2}} = \frac{7{,}617.6}{3{,}129.8} = 2.43$ is greater than 0.44; therefore:

$$Whit_2 = 0.658^{P_{o2}/P_{e2}} P_{o2} = 0.658^{3{,}129.8/7{,}617.64} 3{,}129.8 = 2{,}635.3 \qquad \text{Eq. 6A.6.12.6.7-2}$$

Therefore, the total Whitmore buckling capacity is the sum of the two individual gusset plates.

$$Whit = \varphi_{cg}\left(Whit_1 + Whit_2\right) = 0.95 \times \left(1{,}793.2 + 2{,}635.3\right) = 4{,}207.1 \qquad \text{Eq. 6A.6.12.6.7-1}$$

Since this is a compression member, the partial plane shear yield capacity also needs to be calculated.

For the gusset plate with section loss:

$$PS_1 = \frac{\Omega\left(0.58F_y\right) A_{\text{shear}}}{\cos\left(\theta\right)} = \frac{0.88\left(0.58 \times 50\right)\left(63.80 \times 0.99\right)}{\cos\left(30\right)} = 1{,}863.4 \qquad \text{Eq. 6A.6.12.6.6-1}$$

For the gusset plate without section loss:

$$PS_2 = \frac{\Omega\left(0.58F_y\right) A_{\text{shear}}}{\cos\left(\theta\right)} = \frac{0.88\left(0.58 \times 50\right)\left(63.80 \times 1.125\right)}{\cos\left(30\right)} = 2{,}115.1 \qquad \text{Eq. 6A.6.12.6.6-1}$$

Therefore, the total partial plane shear capacity is the sum of the two individual gusset plates.

$$PS = \varphi_{vg}\left(PS_1 + PS_2\right) = 1.00 \times \left(1{,}863.4 + 2{,}115.1\right) = 3{,}978.5$$

The overall buckling capacity, *C*, is the lesser of the Whitmore buckling strength and that from the partial plane shear yield. In this case, partial plane shear yielding controls, and *C* = 3978.5 kips.

Calculate the rating factors.

$$RF_{LRFR} = \frac{\varphi_s \varphi_c C\left(R_{DL_LL}\right) - \gamma_{DL} P_{4DL}}{\gamma_{LL} P_{4LL}} \qquad \text{Eq. 6A.4.2.1-1}$$

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$$RF_{LRFRinv} = \frac{0.9 \times 1.00 \times 3978.5 \times 0.94 - 1.25 \times 985}{1.75 \times 236} = 5.17$$

$$RF_{LRFRopr} = RF_{LRFRinv} \frac{\gamma_{LLinv}}{\gamma_{LLopr}} = 5.17 \frac{1.75}{1.35} = 6.70$$

A11A.7—Check Horizontal Shear Capacity of the Gusset Plate

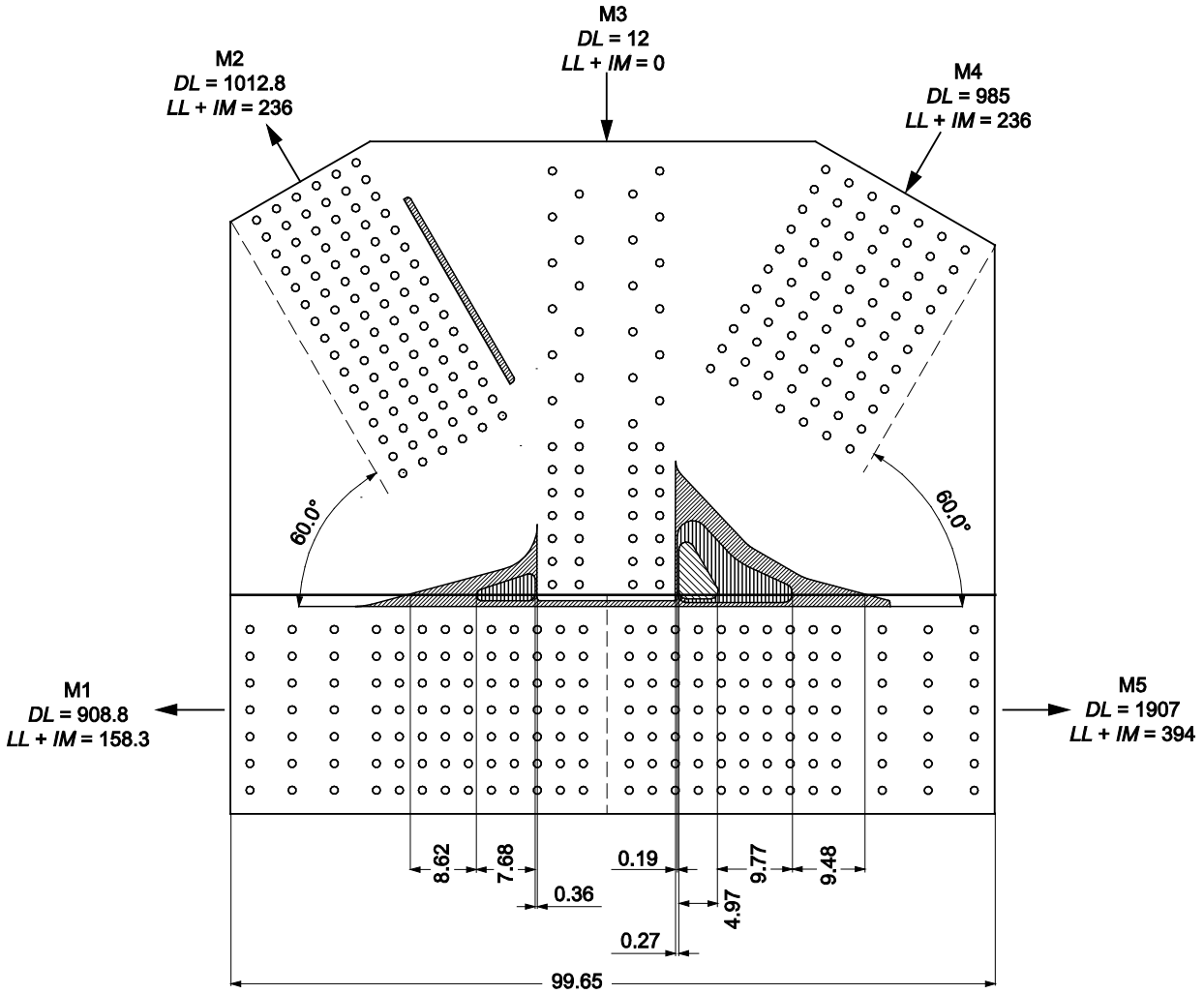

Figure A11A.7-1—L2 Horizontal Shear Plane

First, calculate the horizontal shear loads acting on the horizontal shear plane (see Figure A11A.7-1). Note that the loads represented on the joint are based on envelope loads, not coincident loads. This will result in a conservative rating, which could be refined based on analysis of coincident shear loads.

$$P_{HDL} = P_{2HDL} + P_{4HDL} = (1{,}012.8)\left[\cos(60°)\right] + (985)\left[\cos(60°)\right] = 998.9 \text{ kips}$$

$$P_{HLL} = P_{2HLL} + P_{4HLL} = (236)\left[\cos(60°)\right] + (236)\left[\cos(60°)\right] = 236 \text{ kips}$$

Calculate the DL/LL ratio for this limit state to account for further reduction according to Article 6A.6.12.6.1.

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$\frac{P_{HDL}}{P_{HLL}} = \frac{998.9}{236} = 4.23$, which is greater than 1.0; therefore, the additional reduction is:

$$R_{DL_LL} = 1 - 0.1\left(\frac{4.23 - 1.0}{5}\right) = 0.94 \qquad \text{6A.6.12.6.1}$$

Calculate the equivalent gusset plate thickness on the horizontal shear plane for the gusset plate with section loss.

$$t_{eq_shr} = \frac{99.65(1.125) - (8.62 + 0.36 + 0.19 + 9.48)(0.25) - (7.68 + 0.27 + 9.77)(0.5) - 4.97(0.75)}{99.65}$$

$$t_{eq_shr} = 0.95$$

Calculate the resistance to gross section shear yielding.

For the gusset plate with section loss:

$$V_{ny1} = \varphi_{vg}\left(0.58F_y\right)A_g\Omega = 1.00(0.58 \times 50)(99.65 \times 0.95)(0.88) = 2{,}416 \qquad \text{Eq. 6A.6.12.6.6-1}$$

For the gusset plate without section loss:

$$V_{ny2} = \varphi_{vg}\left(0.58F_y\right)A_g\Omega = 1.00(0.58 \times 50)(99.65 \times 1.125)(0.88) = 2{,}861 \qquad \text{Eq. 6A.6.12.6.6-1}$$

Therefore, the total resistance to shear yielding is:

$$V_{ny} = V_{ny1} + V_{ny2} = 2{,}416 + 2{,}861 = 5{,}277$$

Calculate the resistance to shear fracture. Note that the section loss in the one gusset plate does not affect the net section.

$$V_{nu} = \varphi_{vu}\left(0.58F_u\right)A_{vn} = 0.80(0.58 \times 65)\left[2 \times 99.65 - 52\left(\frac{7}{8} + \frac{1}{8}\right)\right](1.125) = 4{,}998 \qquad \text{Eq. 6A.6.12.6.6-2}$$

The capacity is determined by the lesser of the shear yield and shear fracture capacities. In this case, the shear fracture of the net section governs, and C = 4,998 kips.

Calculate the LRFR rating factors.

$$RF_{LRFR} = \frac{\varphi_s\varphi_c C\left(R_{DL_LL}\right) - \gamma_{DL}P_{HDL}}{\gamma_{LL}P_{HLL}} \qquad \text{Eq. 6A.4.2.1-1}$$

$$RF_{LRFRinv} = \frac{0.9 \times 1.00 \times 4{,}998 \times 0.94 - 1.25 \times 998.9}{1.75 \times 236} = 7.21$$

$$RF_{LRFRopr} = RF_{LRFRinv}\frac{\gamma_{LLinv}}{\gamma_{LLopr}} = 7.21\frac{1.75}{1.35} = 9.35$$

A11A.8—Check Vertical Shear Capacity of the Gusset Plate

This check will not control because a valid shear plane cannot develop through the chord member.

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A11A.9—SUMMARY OF LRFR LOAD RATING FACTORS

Table A11A.9-1—Gusset Plate L2 LRFR Summary

	Inventory	Operating
Member 1 (bolt shear and slip resistance control)	5.63	7.30
Member 2 (block shear controls)	8.21	10.64
Member 3	Rating factors were not calculated	
Member 4 (partial plane shear controls)	5.17	6.70
Member 5	Rating factors were not demonstrated; procedures are similar to those for Member 1.	
Chord Splice (net fracture controls)	4.87	6.32
Horizontal Shear (fracture controls)	7.21	9.35
Vertical Shear	This is not a relevant failure mode for this joint	

Note that the summary of rating factors presented in Table A11A.9-1 considers solely the checks performed in this rating example. For a complete rating, the rating engineer must also consider the capacity of the bolts to ensure that the bolts do not govern the ratings for members 2 and 4 and for the chord splice.

PART B—LOAD FACTOR RATING METHOD

A11B.1—LOAD FACTOR RATING OF GUSSET PLATES

These gusset plate checks for design loads illustrate the application of the LFR specifications, using the L2 joint of the truss depicted in Figure A11.1-1. See Figure A11.1-2 for detailing of Joint L2. For overall bridge and member data, including a description of live loading, see Section A11.1. For section losses to the gusset plates at L2 considered in this rating, see Article A11A.1 and Figure A11.1-3.

The loads shown are envelope loads developed using HS20-44 and alternate loading. Impact loading was considered to be 28.5 percent of the live load.

The various evaluation factors used throughout the example are shown in Table A11B.1-1.

Table A11B.1-1—LFR Load Factors for Gusset Plates

	Strength (Article 6B.4.3)	Service (Article 6B.5.3.1)
Inventory Live Load Factor (γ_{LL_INV})	2.17	N/A
Operating Live Load Factor (γ_{LL_OPR})	1.30	1.00
Dead Load Factor (γ_{DL})	1.30	1.00

The following rating checks will be performed:

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1. Tension resistance of M1 (see Article A11B.2),
2. Chord splice resistance between M1 and M5 (see Article A11B.3),
3. Tension resistance of M2 (see Article A11B.4),
4. Compression resistance of M3 (see Article A11B.5),
5. Compression resistance of M4 (see Article A11B.6),
6. Horizontal shear resistance of the gusset plate (see Article A11B.7), and
7. Vertical shear resistance of the gusset plate (see Article A11B.8).

A11B.2—Check the Tension Resistance of M1

Only block shear resistance of the gusset plate, shear resistance of the bolts, and bearing resistance of the gusset must be checked. The checks for M5 are very similar in nature and will not be shown. Additionally, the bolt shear and bearing resistance checks will only be demonstrated for the gusset plate at M1; these checks would normally be performed at each fastener pattern and for the splice plates, too, for a complete rating.

First, determine the shear resistance of the fasteners. Because web splice plates are used, some fasteners will be in double shear but conservatively assume just single shear. The connection is not greater than 50 in. in length; therefore, a connection length reduction factor does not apply. Since the web splice plates have fill plates greater than 0.25 in., the bolt shear must be reduced by the fill plate factor.

$$\varphi R = (\varphi F) m A R_{\text{fill}} \qquad \text{L6B.2.6.1}$$

$$\gamma = \left(\frac{A_{\text{fill}}}{A_p} \right)$$

In this case, A_p is determined as the lesser of area of the lower chord web plate ($^3/_4$ in. × 26$^1/_4$ in.) and the sum of the areas of the web splice plate (1$^1/_8$ in. × 24 in.) and gusset plate (1$^1/_8$ in. × 87.8 in.). Given the large area of the gusset plate, it is clear that the area of the member web plate will control the calculation of A_p.

$$\gamma = \frac{(24)(0.875)}{(26.25)(0.75)} = 1.067$$

$$R_{\text{fill}} = \left(\frac{1+\gamma}{1+2\gamma} \right) = \left(\frac{1+1.067}{1+(2)(1.067)} \right) = 0.66 \qquad \text{L6B.2.6.1}$$

$$\varphi R = (43)(1)\left(\frac{\pi(0.875^2)}{4} \right)(0.66) = 17.1 \text{ kip/bolt}$$

Check the bearing resistance of the gusset plate. Since all holes are centered a minimum of 3 in. from each other, assume a clear edge distance of 2.46 in. controls for all bolts.

$$\varphi R = 0.9 L_c t F_u \le 1.8 d t F_u \qquad \text{Standard Specs (10-166b)}$$

$$\varphi R = 0.9(2.46)(1.125)(65) \le 1.8(0.875)(1.125)(65) \rightarrow 161.9 \le 115.2$$

Since the inequality does not hold true, the bearing resistance is capped at 115.2 kips/bolt. This is much larger than the shear resistance of the fastener; therefore, the shear resistance controls the capacity.

$$\varphi R = 182 \times 17.1 \text{ kip/bolt} = 3{,}112 \text{ kips}$$

Note that the chord web plate thickness is $^3/_4$ in. and the bearing resistance of the web plate will govern over the bearing resistance of the gusset plate. Therefore, a similar bearing resistance computation for the web plate should be

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performed as part of the member rating. However, as this example is for gusset plates alone, the bearing resistance computation for the chord member will not be shown herein.

Next, determine the resistance due to block shear. See Article A11A.2 for the gusset plate shear and tensile areas to consider for block shear.

Calculate the block shear resistance of each gusset plate (see Figure A11A.2-1).

Is $F_u A_{vn} \le F_y \left(A_{vg1} + A_{vg2} \right)$?

$65\left(65.95\right) \le 50\left(51.71+42.36\right) \rightarrow 4,287 \le 4,703$, yes; therefore: L6B.2.6.5

$C = \varphi_{bs} R_p \left(0.58 F_u A_{vn} + F_u A_{tn}\right)$

$C = \left(0.85\right)\left(1.0\right)\left[\left(0.58\right)(65)(65.95) + \left(65\right)(39.38)\right] = 4289$

Since the resistance from the bolts is less than the block shear resistance, the capacity is controlled by the bolt resistance, C = 3112 kips.

Calculate the rating factors.

$$RF_{LFR} = \frac{C - \gamma_{DL} P_{1DL}}{\gamma_{LL} P_{1LL}} \qquad \text{Eq. 6B.4.1-1}$$

$$RF_{LFRinv} = \frac{3112 - 1.30 * 908.8}{2.17 * 158.3} = 5.62$$

$$RF_{LFRopr} = RF_{LFRinv} \frac{\gamma_{LLinv}}{\gamma_{LLopr}} = 5.62 \frac{2.17}{1.30} = 9.38$$

The gusset plate was also originally designed to be slip-critical with the members; therefore, rating factors must also be calculated for the slip resistance. Slip resistance is checked with the Service II load combination.

The slip resistance per bolt is calculated as:

$$R_n = \left(\varphi F_s\right) A_b (\text{\# of bolts}) = 21\left(\frac{\pi(0.875^2)}{4}\right)(182) = 2298 \qquad \text{Standard Specs Eq. 10-172a}$$

φF_s is taken as 21 ksi according to Article 6B.5.3.1.

Per Article 6B.5.3.1, slip only needs to be calculated at the operating level under Service II loads.

$$RF_{LFRopr} = \frac{C - \gamma_{DL} P_{1DL}}{\gamma_{LL} P_{1LL}} = \frac{2298 - (1.00)(908.8)}{(1.00)(158.3)} = 8.78 \qquad \text{Eq. 6B.4.1-1}$$

A11B.3—Check the Chord Splice Resistance

Since the gusset plate is also performing as a tension chord splice, the resistance of the spliced section must be evaluated according to the provisions of Article L6B.2.6.6. For the net and gross section properties for the composite section of gusset and splice plates, see Article A11A.3 and Figure A11A.3-1.

The resistance of the chord splice can be calculated according to Article L6B.2.6.6.

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$$\frac{Kl\sqrt{12}}{t_g} < 25$$

$$\frac{(0.50)(6.0)\sqrt{12}}{(1.125)} = 9.24 < 25$$

$$C_{\text{gross}} = \varphi_{cs} F_{cr}\left(\frac{S_g A_g}{S_g + e_{p_\text{gross}} A_g}\right)$$

$$C_{\text{gross}} = 1.00(50)\left(\frac{(4{,}933.5)(265.1)}{4{,}933.5 + (22.7)(265.1)}\right) = 5{,}971 \text{ kips}$$

$$C_{\text{net}} = \varphi_{cs} F_u\left(\frac{S_n A_n}{S_n + e_{p_\text{net}} A_n}\right)$$

$$C_{\text{net}} = 1.00(65)\left(\frac{(3{,}899.1)(229.0)}{3{,}899.1 + (26.2)(229.0)}\right) = 5{,}863 \text{ kips}$$

The lesser resistance controls; in this case, it is controlled by fracture on the net section, C = 5,863 kips.

Next, determine the factored load in the chord splice. Look at the loads on both sides of the splice and use the more severe.

Left Side

$$1.3\left\{908.8 + (1{,}012.8)\left[\cos(60°)\right]\right\} + 2.17\left\{158.3 + (236)\left[\cos(60°)\right]\right\} = 2{,}439 \text{ kips}$$

Right Side

$$1.3\left\{1{,}907 - (985)\left[\cos(60°)\right]\right\} + 2.17\left\{394 - (236)\left[\cos(60°)\right]\right\} = 2{,}438 \text{ kips}$$

Use the controlling loads from the left side of the splice.

Calculate the LFR rating factors per Eq. 6B.4.1-1.

$$RF_{LFRinv} = \frac{C - \gamma_{DL} P_{DL}}{\gamma_{LL} P_{LL}} = \frac{5{,}863 - 1.30\left\{908.8 + (1{,}012.8)\left[\cos(60°)\right]\right\}}{2.17\left\{158.3 + (236)\left[\cos(60°)\right]\right\}} = 6.71$$

$$RF_{LFRopr} = RF_{LFRinv}\frac{\gamma_{LLinv}}{\gamma_{LLopr}} = 6.71\frac{2.17}{1.30} = 11.20$$

A11B.4—Check Tension Resistance of M2

The tension capacity will be governed by the lesser of the gross yield and net fracture on the Whitmore section (see Figure A11A.4-1), and block shear (see Figure A11A.4-2). Consider the section loss in the one gusset plate.

Calculate the effective area for calculating capacity of the Whitmore section in tension.

$$A_e = A_n + \beta A_g \le A_g$$ Standard Specs Eq. 10-4w

$A_e = 119.43 + (0.15)(132.93) \le 132.93 = 139.37 \le 132.93$ is not true, so $A_e = 132.93$

$$P_n = \varphi F_y A_e = 1.00 \times 50 \times 132.93 = 6646$$ L6B.2.6.5

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For the gross and net sections for block shear check, see Article A11A.4. Calculate the block shear resistance of each gusset plate.

Is $F_u A_{vn} \le F_y \left(A_{vg1} + A_{vg2} \right)$?

$65(108.81) \le 50(89.28 + 82.33) \rightarrow 7,073 \le 8,580$, yes; therefore: L6B.2.6.5

$C = \varphi_{bs} R_p \left(0.58 F_u A_{vn} + F_u A_{tn} \right)$

$C = (0.85)(1.0)\left[(0.58)(65)(108.81) + (65)(22.5) \right] = 4,729.9$

The tension capacity is controlled by the minimum capacity based on the Whitmore yield and block shear. In this case, block shear controls, so C = 4729.9 kips.

Calculate the rating factors.

$$RF_{LFR} = \frac{C - \gamma_{DL} P_{2DL}}{\gamma_{LL} P_{2LL}} = \frac{4,729.9 - 1.30 \times 1,012.8}{2.17 \times 236} = 6.66 \quad \text{Eq. 6B.4.1-1}$$

$$RF_{LFRopr} = RF_{LFRinv} \frac{\gamma_{LLinv}}{\gamma_{LLopr}} = 6.66 \frac{2.17}{1.30} = 11.12$$

A11B.5—Check the Compression Resistance of M3

It is clear that this member will not control with only 12 kips of dead load and, since there is no live load, the rating factor cannot be calculated. For illustration, only the capacity calculations will be shown.

For the Whitmore buckling check, project the worst case section loss onto the Whitmore section (see Figure A11A.5-1) in the direction of the member to calculate the equivalent gusset plate thickness.

Calculate the factored Whitmore buckling strength per gusset plate because only one of the plates has section loss. For the equivalent gusset plate thickness for the plate with section loss, see Article A11A.5.

For the gusset plate with section loss:

$K = 0.5$ L6B.2.6.4

$$r_{s1} = \frac{t_{eq_whit}}{\sqrt{12}} = \frac{0.88}{\sqrt{12}} = 0.25$$

Is $\frac{KL_{mid}}{r_{s1}} \le \sqrt{\frac{2\pi^2 E}{F_y}} \rightarrow \left(\frac{0.5 \times 5.93}{0.25} \right) \le \sqrt{\frac{2\pi^2 (29,000)}{50}} \rightarrow 11.67 \le 107$? L6B.2.6.4

Yes. Therefore:

$$F_{cr1} = F_y \left[1 - \frac{F_y}{4\pi^2 E} \left(\frac{KL_{mid}}{r_{s1}} \right)^2 \right] = 50 \left[1 - \frac{50}{4\pi^2 (29,000)} \left(11.67^2 \right) \right] = 49.7 \quad \text{L6B.2.6.4}$$

$$Whit_1 = 0.85 F_{cr} A_g = 0.85 \times 49.7 (76.35 \times 0.88) = 2,838.5 \quad \text{L6B.2.6.4}$$

For the gusset plate without section loss:

$K = 0.5$ L6B.2.6.4

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$$r_{s2} = \frac{t_g}{\sqrt{12}} = \frac{1.125}{\sqrt{12}} = 0.32$$

$$\text{Is } \frac{KL_{mid}}{r_{s2}} \le \sqrt{\frac{2\pi^2 E}{F_y}} \rightarrow \left(\frac{0.5\times 5.93}{0.32}\right) \le \sqrt{\frac{2\pi^2\,(29{,}000)}{50}} \rightarrow 9.13 \le 107\,? \qquad \text{L6B.2.6.4}$$

Yes. Therefore:

$$F_{cr2} = F_y\left[1 - \frac{F_y}{4\pi^2 E}\left(\frac{KL_{mid}}{r_{s2}}\right)^2\right] = 50\left[1 - \frac{50}{4\pi^2(29{,}000)}\left(9.13^2\right)\right] = 49.8 \qquad \text{L6B.2.6.4}$$

$$Whit_2 = 0.85F_{cr}A_g = 0.85\times 49.8\left(76.35\times 1.125\right) = 3{,}637.2 \qquad \text{L6B.2.6.4}$$

Therefore, the total Whitmore buckling capacity is the sum of the two individual gusset plates.

$$Whit = \varphi_{cg}\left(Whit_1 + Whit_2\right) = 1.00\times\left(2{,}838.5 + 3{,}637.2\right) = 6{,}475.7 \qquad \text{L6B.2.6.4}$$

This vertical member does not have any admissible partial shear planes that would control the compression resistance. Therefore, the capacity, *C*, is the Whitmore buckling strength.

A11B.6—Check the Compression Resistance of M4

For the Whitmore buckling check, project the worst case section loss onto the Whitmore section in the direction of the member to calculate the equivalent gusset plate thickness. For the equivalent gusset plate thickness for the gusset plate with section loss, see Article A11A.6 and Figure A11A.6-3. Calculate the factored Whitmore buckling strength per gusset plate, because only one of the plates has section loss.

For the gusset plate with section loss:

$$K = 0.5 \qquad \text{L6B.2.6.4}$$

$$r_{s1} = \frac{t_{eq_whit}}{\sqrt{12}} = \frac{0.86}{\sqrt{12}} = 0.25$$

$$\text{Is } \frac{KL_{mid}}{r_{s1}} \le \sqrt{\frac{2\pi^2 E}{F_y}} \rightarrow \left(\frac{0.5\times 31.5}{0.25}\right) \le \sqrt{\frac{2\pi^2(29{,}000)}{50}} \rightarrow 63.2 \le 107\,? \qquad \text{L6B.2.6.4}$$

Yes. Therefore:

$$F_{cr1} = F_y\left[1 - \frac{F_y}{4\pi^2 E}\left(\frac{KL_{mid}}{r_{s1}}\right)^2\right] = 50\left[1 - \frac{50}{4\pi^2(29{,}000)}\left(63.2^2\right)\right] = 41.3 \qquad \text{L6B.2.6.4}$$

$$Whit_1 = 0.85F_{cr}A_g = 0.85\times 41.3(55.64\times 0.86) = 1{,}685.2 \qquad \text{L6B.2.6.4}$$

For the gusset plate without section loss:

$$K = 0.5 \qquad \text{L6B.2.6.4}$$

$$r_{s2} = \frac{t_g}{\sqrt{12}} = \frac{1.125}{\sqrt{12}} = 0.32$$

$$\text{Is } \frac{KL_{mid}}{r_{s2}} \le \sqrt{\frac{2\pi^2 E}{F_y}} \rightarrow \left(\frac{0.5\times 31.5}{0.32}\right) \le \sqrt{\frac{2\pi^2\,(29{,}000)}{50}} \rightarrow 48.5 \le 107\,? \qquad \text{L6B.2.6.4}$$

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Yes. Therefore:

$$F_{cr2} = F_y\left[1-\frac{F_y}{4\pi^2 E}\left(\frac{KL_{mid}}{r_{s2}}\right)^2\right] = 50\left[1-\frac{50}{4\pi^2\,(29{,}000)}\left(48.5^2\right)\right] = 44.9 \qquad \text{L6B.2.6.4}$$

$$Whit_2 = 0.85F_{cr}A_g = 0.85\times 44.9\left(55.64\times 1.125\right) = 2{,}387.0 \qquad \text{L6B.2.6.4}$$

Therefore, the total Whitmore buckling capacity is the sum of the two individual gusset plates.

$$Whit = \varphi_{cg}\left(Whit_1 + Whit_2\right) = 1.00\times\left(1{,}685.2 + 2{,}387.0\right) = 4{,}072.2 \qquad \text{L6B.2.6.4}$$

Since this is a compression member, the partial plane shear yield capacity also needs to be calculated. For the equivalent gusset plate thickness on the partial shear plane for the gusset plate with section loss, see Article A11A.6 and Figure A11A.6-2.

For the gusset plate with section loss:

$$PS_1 = \frac{\Omega\left(0.58F_y\right)A_{\text{shear}}}{\cos\left(\theta\right)} = \frac{0.88\left(0.58\times 50\right)\left(63.80\times 0.99\right)}{\cos\left(30\right)} = 1{,}863.4 \qquad \text{L6B.2.6.3}$$

For the gusset plate without section loss:

$$PS_2 = \frac{\Omega\left(0.58F_y\right)A_{\text{shear}}}{\cos\left(\theta\right)} = \frac{0.88\left(0.58\times 50\right)\left(63.80\times 1.125\right)}{\cos\left(30\right)} = 2{,}115.1 \qquad \text{L6B.2.6.3}$$

Therefore, the total partial plane shear capacity is the sum of the two individual gusset plates.

$$PS = \varphi_{vy}\left(PS_1 + PS_2\right) = 1.00\times\left(1{,}863.4 + 2{,}115.1\right) = 3{,}978.5$$

The overall buckling capacity, C, is the lesser of the Whitmore buckling strength and that from the partial plane shear yield. In this case, partial plane shear yielding controls, and C = 3978.5 kips.

Calculate the rating factors.

$$RF_{LFRinv} = \frac{C-\gamma_{DL}P_{4DL}}{\gamma_{LL}P_{4LL}} = \frac{3{,}978.5-1.30\times 985}{2.17\times 236} = 5.27 \qquad \text{Eq. 6B.4.1-1}$$

$$RF_{LFRopr} = RF_{LFRinv}\frac{\gamma_{LLinv}}{\gamma_{LLopr}} = 5.27\frac{2.17}{1.30} = 8.79$$

A11B.7—Check Horizontal Shear Capacity of the Gusset Plate

First calculate the horizontal shear loads acting on the horizontal shear plan (see Figure A11.7-1). Note that the loads represented on the joint are based on envelope loads, not coincident loads. This will result in a conservative rating, which could be refined based on analysis of coincident shear loads.

$$P_{HDL} = P_{2HDL} + P_{4HDL} = (1012.8)\left[\cos(60°)\right] + (985)\left[\cos(60°)\right] = 998.9 \text{ kips}$$

$$P_{HLL} = P_{2HLL} + P_{4HLL} = (236)\left[\cos(60°)\right] + (236)\left[\cos(60°)\right] = 236 \text{ kips}$$

For the equivalent gusset plate thickness on the horizontal shear plane for the gusset plate with section loss, see Article A11A.7.

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Calculate the resistance to gross section shear yielding.

For the gusset plate with section loss:

$$V_{ny1} = \varphi_{vy}\left(0.58F_y\right)A_g\Omega = 1.00\left(0.58\times 50\right)\left(99.65\times 0.95\right)\left(0.88\right) = 2{,}416 \qquad \text{L6B.2.6.3}$$

For the gusset plate without section loss:

$$V_{ny2} = \varphi_{vy}\left(0.58F_y\right)A_g\Omega = 1.00\left(0.58\times 50\right)\left(99.65\times 1.125\right)\left(0.88\right) = 2{,}861 \qquad \text{L6B.2.6.3}$$

Therefore the total resistance to shear yielding is:

$$V_{ny} = V_{ny1} + V_{ny2} = 2{,}416 + 2{,}861 = 5{,}277$$

Calculate the resistance to shear fracture. Note that the section loss in the one gusset plate does not affect the net section.

$$V_{nu} = \varphi_{vu}\left(0.58F_u\right)A_{vn} = 0.85(0.58\times 65)\left[2\times 99.65 - 52\left(\frac{7}{8}+\frac{1}{8}\right)\right]1.125 = 5{,}310 \qquad \text{L6B.2.6.3}$$

The capacity is determined by the lesser of the shear yield and shear fracture capacities. In this case, the shear yielding of the gross section governs, and C = 5,277 kips.

Calculate the rating factors.

$$RF_{LFRinv} = \frac{C - \gamma_{DL}P_{HDL}}{\gamma_{LL}P_{HLL}} = \frac{5{,}277 - 1.30\times 998.9}{2.17\times 236} = 7.77 \qquad \text{Eq. 6B.4.1-1}$$

$$RF_{LFRopr} = RF_{LFRinv}\frac{\gamma_{LLinv}}{\gamma_{LLopr}} = 7.77\frac{2.17}{1.30} = 12.97$$

A11B.8—Check Vertical Shear Capacity of the Gusset Plate

This check will not control because a valid shear plane cannot develop through the chord member.

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A11B.9—Summary of LFR Load Rating Factors

Table A11B.9-1—Gusset Plate L2 LFR Summary

	Inventory	Operating
Member 1 (bolt shear and slip resistance control)	5.62	8.78
Member 2 (block shear controls)	6.66	11.12
Member 3	Rating factors were not calculated.	
Member 4 (partial plane shear controls)	5.27	8.79
Member 5	Rating factors were not demonstrated; procedures are similar to those for Member 1.	
Chord Splice (net fracture controls)	6.71	11.20
Horizontal Shear (yielding controls)	7.77	12.97
Vertical Shear	This is not a relevant failure mode for this joint.	

Note that the summary of rating factors presented in Table A11B.9-1 considers solely the checks performed in this rating example. For a complete rating, the rating engineer must also consider the capacity of the bolts to ensure that the bolts do not govern the ratings for members 2 and 4 and for the chord splice.

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